open62541

open62541 Documentation

Release 1.5.9

Oct 27, 2026

CONTENTS

1 Introduction **1**
- 1.1 OPC Unified Architecture . . . 1
- 1.2 open62541 Features . . . 2
- 1.3 Getting Help . . . 3
- 1.4 Contributing . . . 3

2 OPC UA Basics **5**
- 2.1 Information Modelling . . . 6
- 2.2 Protocol . . . 11
- 2.3 Services . . . 14

3 Building open62541 **19**
- 3.1 Building the Library . . . 19
- 3.2 Build Options . . . 22
- 3.3 Prebuilt packages . . . 26
- 3.4 Building the Examples . . . 27

4 Data Types **29**
- 4.1 Builtin Types . . . 29
- 4.2 Generic Type Handling . . . 46
- 4.3 Namespace Mapping . . . 50
- 4.4 Binary Encoding/Decoding . . . 51
- 4.5 JSON En/Decoding . . . 53
- 4.6 XML En/Decoding . . . 54
- 4.7 Array handling . . . 55
- 4.8 Generated Data Type Definitions . . . 56

5 Server **133**
- 5.1 Server Lifecycle . . . 133
- 5.2 Timed Callbacks . . . 135
- 5.3 Application Notification . . . 136
- 5.4 SecureChannel Handling . . . 137
- 5.5 Session Handling . . . 138
- 5.6 Attribute Service Set . . . 139
- 5.7 Method Service Set . . . 144
- 5.8 View Service Set . . . 145
- 5.9 MonitoredItem Service Set . . . 146
- 5.10 Node Management Service Set . . . 149
- 5.11 Async Operations . . . 161

5.12 Events . . . 163
5.13 Remote Discovery Server Registration . . . 166
5.14 Local Discovery Server Records . . . 166
5.15 Drivers . . . 168
5.16 Statistics . . . 170
5.17 Reverse Connect . . . 170
5.18 Utility Functions . . . 171
5.19 Role-Based Access Control (RBAC) . . . 172
5.20 Server Configuration . . . 174
5.21 Configuration from File . . . 190

6 Client **193**
6.1 Client Lifecycle . . . 193
6.2 Connect to a Server . . . 194
6.3 Discovery . . . 197
6.4 Services . . . 198
6.5 Client Utility Functions . . . 200
6.6 Application Notification . . . 203
6.7 Client Configuration . . . 203
6.8 Configuration from File . . . 210
6.9 Subscriptions . . . 211
6.10 Highlevel Client Functionality . . . 217
6.11 Asynchronous Services . . . 226

7 PubSub **239**
7.1 PubSub Information Model Representation . . . 240
7.2 PublisherId . . . 240
7.3 PubSub Components . . . 241
7.4 Global PubSub Configuration . . . 243
7.5 PubSub Custom State Machine . . . 244
7.6 PubSubConnection . . . 245
7.7 PublishedDataSet . . . 246
7.8 DataSetField . . . 248
7.9 WriterGroup . . . 249
7.10 DataSetWriter . . . 251
7.11 SubscribedDataSet . . . 252
7.12 DataSetReader . . . 253
7.13 ReaderGroup . . . 255
7.14 SecurityGroup . . . 257
7.15 Offset Table . . . 260
7.16 PubSub NetworkMessage . . . 261

8 Tutorials **267**
8.1 Working with Data Types . . . 267
8.2 Building a Simple Server . . . 270
8.3 Adding Variables to a Server . . . 272
8.4 Connecting a Variable with a Physical Process . . . 274
8.5 Working with Variable Types . . . 277
8.6 Working with Objects and Object Types . . . 279
8.7 Adding Methods to Objects . . . 285
8.8 Observing Attributes with Local MonitoredItems . . . 289
8.9 Generating events . . . 290

8.10 Using Alarms and Conditions Server . . . 291
8.11 Serving Files and Directories . . . 303
8.12 Building a Simple Client . . . 305
8.13 Working with Publish/Subscribe . . . 306
8.14 Subscribing Fields . . . 311

9 Security **317**
9.1 Supported Security Policies . . . 317
9.2 Crypto Backends . . . 318
9.3 Authentication . . . 319
9.4 Certificates and Trust . . . 320
9.5 ECC Specifics . . . 321

10 Common Definitions **325**
10.1 Attribute Id . . . 325
10.2 Access Level Masks . . . 326
10.3 Write Masks . . . 326
10.4 ValueRank . . . 327
10.5 EventNotifier . . . 327
10.6 Server Capabilities . . . 327
10.7 Rule Handling . . . 328
10.8 Order . . . 328
10.9 Application Notification . . . 329
10.10 Connection State . . . 337
10.11 Statistic Counters . . . 338
10.12 Lifecycle States . . . 338

11 Utility Definitions **341**
11.1 Range Definition . . . 341
11.2 Event-Filter Parsing . . . 341
11.3 Random Number Generator . . . 342
11.4 Translate between DataTypeDescription and UA_DataType . . . 342
11.5 Memory Handling . . . 342
11.6 Key Value Map . . . 342
11.7 Binary Connection Config Parameters . . . 344
11.8 Default Node Attributes . . . 345
11.9 Endpoint URL Parser . . . 345
11.10 Escaping of Strings . . . 346
11.11 RelativePath Expressions . . . 347
11.12 Attribute Path Expression . . . 348
11.13 Convenience macros for complex types . . . 350
11.14 Cryptography Helpers . . . 350
11.15 Trustlist Helpers . . . 350

12 XML Nodeset Compiler **353**
12.1 Manually calling the Nodeset Compiler . . . 358
12.2 Creating object instances . . . 358
12.3 Combination of multiple nodesets . . . 360
12.4 Outstanding Companion Spec Issues . . . 362
12.5 Automatic Nodesetinjection . . . 363

13 StatusCodes **367**

14 Plugin API **389**
14.1 Logging Plugin API 389
14.2 Nodestore Plugin API 392
14.3 AccessControl Plugin API 404
14.4 EventLoop Plugin API 406
14.5 CertificateGroup Plugin API 429
14.6 SecurityPolicy Plugin API 432

15 Drivers **443**
15.1 Multicast DNS Discovery Driver 443
15.2 Alarms & Conditions Driver (Experimental) 444
15.3 GDS Receiver Driver 449
15.4 File Transfer Driver 450

CHAPTER

ONE

INTRODUCTION

open62541 (http://open62541.org) is an open source and free implementation of OPC UA (OPC Unified Architecture) written in the common subset of the C99 and C++98 languages. The library is usable with all major compilers and provides the necessary tools to implement dedicated OPC UA clients and servers, or to integrate OPC UA-based communication into existing applications. open62541 library is platform independent. All platform-specific functionality is implemented via exchangeable plugins. Plugin implementations are provided for the major operating systems.

open62541 is licensed under the Mozilla Public License v2.0 (MPLv2). This allows the open62541 library to be combined and distributed with any proprietary software. Only changes to the open62541 library itself need to be licensed under the MPLv2 when copied and distributed. The plugins, as well as the server and client examples are in the public domain (CC0 license). They can be reused under any license and changes do not have to be published.

An example server based on the 1.4 release family of open62541 was certified by the OPC Foundation for the 'Standard Server 2017' profile. This shows open62541 to be a good basis for products with certification requirements. See https://open62541.org/certification for more details.

1.1 OPC Unified Architecture

OPC UA is a protocol for industrial communication and has been standardized in the IEC 62541 series. At its core, OPC UA defines

- an asynchronous *protocol* (built upon TCP, HTTP or SOAP) that defines the exchange of messages via sessions, (on top of) secure communication channels, (on top of) raw connections,
- a *type system* for protocol messages with a binary and XML-based encoding scheme,
- a meta-model for *information modeling*, that combines object-orientation with semantic triple-relations, and
- a set of 37 standard *services* to interact with server-side information models. The signature of each service is defined as a request and response message in the protocol type system.

The standard itself can be purchased from IEC or downloaded for free on the website of the OPC Foundation at https://opcfoundation.org/ (you need to register with a valid email).

The OPC Foundation drives the continuous improvement of the standard and the development of companion specifications. Companion specifications translate established concepts and reusable components from an application domain into OPC UA. They are created jointly with an established industry council or standardization body from the application domain. Furthermore, the OPC Foundation organizes events for the dissemination of the standard and provides the infrastructure and tools for compliance certification.

1.2 open62541 Features

open62541 implements the OPC UA binary protocol stack as well as a client and server SDK. It currently supports the Micro Embedded Device Server Profile plus some additional features. Server binaries can be well under 100kb in size, depending on the contained information model.

- Communication Stack
  - OPC UA binary protocol
  - Chunking (splitting of large messages)
  - Exchangeable network layer (plugin) for using custom networking APIs (e.g. on embedded targets)
  - Encrypted communication
  - Asynchronous service requests in the client
- Configuration
  - JSON5-based configuration files for both client and server (*Configuration from File*, *Configuration from File*)
  - Flexible configuration API for runtime setup
- Information model
  - Support for all OPC UA node types (including method nodes)
  - Support for adding and removing nodes and references also at runtime.
  - Support for inheritance and instantiation of object- and variable-types (custom constructor/destructor, instantiation of child nodes)
  - Access control for individual nodes
- Subscriptions
  - Support for subscriptions/monitoreditems for data change notifications
  - Very low resource consumption for each monitored value (event-based server architecture)
- Code-Generation
  - Support for generating data types from standard XML definitions
  - Support for generating server-side information models (nodesets) from standard XML definitions

Features on the roadmap for the 0.3 release series but missing in the initial v0.3 release are:

- Encrypted communication in the client
- Events (notifications emitted by objects, data change notifications are implemented)
- Event-loop (background tasks) in the client

1.3 Getting Help

For discussion and help besides this documentation, you can reach the open62541 community via

- the mailing list
- our IRC channel
- the bugtracker

1.4 Contributing

As an open source project, we invite new contributors to help improve open62541. Issue reports, bugfixes and new features are very welcome. The following are good starting points for new contributors:

- Report bugs
- Improve the documentation
- Work on issues marked as good first issue

CHAPTER

TWO

OPC UA BASICS

OPC UA (ISO/IEC 62541) is a framework for *object-oriented information modelling*. The objects typically represent industrial assets for remote monitoring and control. OPC UA further defines a *protocol and a set of services* to efficiently and securely interact with the information models over the network.

OPC UA Companion Specifications define *DataTypes, ObjectTypes* and other reusable definitions. Companion Specifications are the result of industry-specific working groups. Having established a consensus across vendors, Companion Specifications are an enabler for semantic interoperability in their target industry.

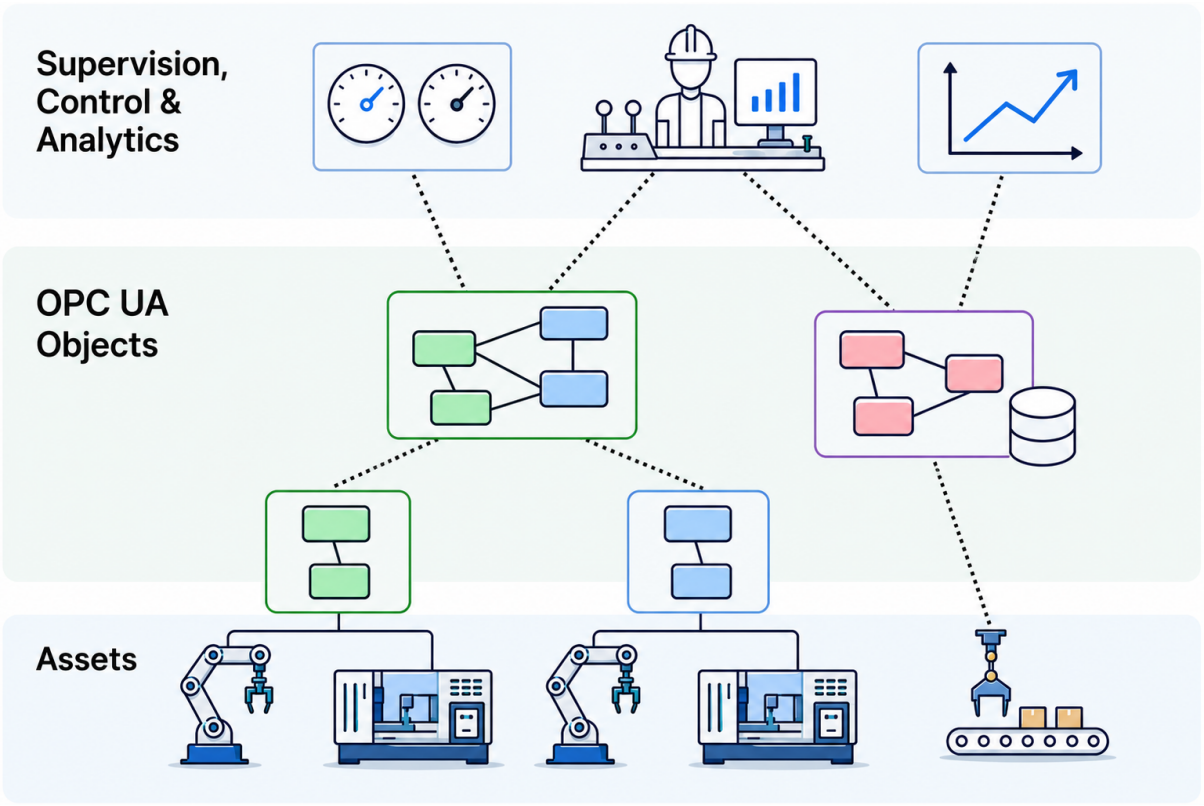


Fig. 2.1: Communication and Object-Oriented Interoperability with OPC UA

The remainder of this section gives an introduction to the core concepts of OPC UA. For the full specification see the OPC UA standard at https://reference.opcfoundation.org/. But beware: The standard is written for SDK implementors and uses precise but cryptic language. Going through this introduction first will help in understanding the definitions from the standard.

2.1 Information Modelling

Information modelling in OPC UA is object-oriented. Object instances are derived from an ObjectTypes with variable and method members. At the same time, the elements of an OPC UA information model are *Nodes* in a graph with *References* between them. This allows for browsing, discovery and introspection of information models at runtime.

Nodes
: There are eight possible NodeClasses: Variable, VariableType, Object, ObjectType, ReferenceType, DataType, Method, and View. The NodeClass defines the *Attributes* a Node can have.

References
: References are links between Nodes. References are typed (with a ReferenceType) and directed.

Each Node is identified by a unique (within the server) *NodeId*. References can be thought of as triples of the form $(\text{source-nodeid}, \text{referencetype-nodeid}, \text{target-nodeid})$. The references of a Node can be retrieved by calling the *Browse Services*. (The target-nodeid is actually an *ExpandedNodeId* which can point to a remote server.) An example reference between nodes is a hasTypeDefinition reference between a Variable and its VariableType. Some ReferenceTypes are *hierarchical* and must not form *directed loops*. See the section on *ReferenceTypes* for more details.

Besides the NodeId (which is mandatory), each NodeClass has additional mandatory (M) and optional (O) attributes. The following table gives an overview of the Attributes defined for each NodeClass. In open62541 the optional attributes are always defined - with sensible defaults if users do not change them.

Table 2.1: Node attributes for the different NodeClasses

Attribute	DataType	Vari-able	Vari-able-Type	Ob-ject	Ob-ject-Type	Refer-ence-Type	DataTy	Methc	View
NodeId	NodeId	M	M	M	M	M	M	M	M
NodeClass	NodeClass	M	M	M	M	M	M	M	M
BrowseName	Qualified-Name	M	M	M	M	M	M	M	M
DisplayName	LocalizedText	M	M	M	M	M	M	M	M
Description	LocalizedText	O	O	O	O	O	O	O	O
WriteMask	UInt32 (*Write Masks*)	O	O	O	O	O	O	O	O
UserWriteMask	UInt32	O	O	O	O	O	O	O	O
IsAbstract	Boolean		M		M	M	M		
Symmetric	Boolean					M			
InverseName	LocalizedText					O			
Contain-sNoLoops	Boolean								M
EventNotifier	Byte (*Event-Notifier*)			M					M
Value	Variant	M	O						
DataType	NodeId	M	M						
ValueRank	Int32 (*ValueR-ank*)	M	M						
ArrayDimen-sions	[UInt32]	O	O						
AccessLevel	Byte (*Access Level Masks*)	M							
UserAc-cessLevel	Byte	M							
MinimumSam-plingInterval	Double	O							
Historizing	Boolean	M							
Executable	Boolean							M	
UserExe-cutable	Boolean							M	
DataTypeDefi-nition	DataTypeDef-inition						O		
RolePermis-sions	[RolePermis-sionType]	O	O	O	O	O	O	O	O
UserRolePer-missions	[RolePermis-sionType]	O	O	O	O	O	O	O	O
AccessRestric-tions	AccessRe-strictionType	O	O	O	O	O	O	O	O
AccessLevelEx	UInt32	O							

Each attribute is referenced by a numerical *Attribute Id*. For example, the *Read Service* takes the NodeId and AttributeId as the argument for every read operation.

Some numerical attributes are used as bitfields or come with special semantics. In particular, see the

sections on *Access Level Masks*, *Write Masks*, *ValueRank* and *EventNotifier*.

The AccessRestrictions Attribute is not yet supported by the open62541 server.

2.1.1 VariableNode

Variables store values in a *DataValue* together with metadata for introspection. Most notably, the attributes data type, value rank and array dimensions constrain the possible values the variable can take on.

Variables come in two flavours: properties and datavariables. Properties are related to a parent with a $\mathrm{hasProperty}$ reference and may not have child nodes themselves. Datavariables may contain properties ($\mathrm{hasProperty}$) and also datavariables ($\mathrm{hasComponents}$).

All variables are instances of some *VariableTypeNode* in return constraining the possible data type, value rank and array dimensions attributes.

Data Type

The (scalar) data type of the variable is constrained to be of a specific type or one of its children in the type hierarchy. The data type is given as a NodeId pointing to a *DataTypeNode* in the type hierarchy. See the Section *DataTypeNode* for more details.

If the data type attribute points to $\mathrm{UInt32}$, then the value attribute must be of that exact type since $\mathrm{UInt32}$ does not have children in the type hierarchy. If the data type attribute points Number, then the type of the value attribute may still be $\mathrm{UInt32}$, but also Float or Byte.

Consistency between the data type attribute in the variable and its *VariableTypeNode* is ensured.

ValueRank

This attribute indicates whether the value attribute of the variable is an array and how many dimensions the array has. It may have the following values:

- $\mathrm{n} >= 1$: the value is an array with the specified number of dimensions
- $\mathrm{n} = 0$: the value is an array with one or more dimensions
- $\mathrm{n} = -1$: the value is a scalar
- $\mathrm{n} = -2$: the value can be a scalar or an array with any number of dimensions
- $\mathrm{n} = -3$: the value can be a scalar or a one dimensional array

Some helper macros for ValueRanks are defined *here*.

The consistency between the value rank attribute of a VariableNode and its *VariableTypeNode* is tested within the server.

Array Dimensions

If the value rank permits the value to be a (multi-dimensional) array, the exact length in each dimensions can be further constrained with this attribute.

- For positive lengths, the variable value must have a dimension length less or equal to the array dimension length defined in the VariableNode.
- The dimension length zero is a wildcard and the actual value may have any length in this dimension. Note that a value (variant) must have array dimensions that are positive (not zero).

Consistency between the array dimensions attribute in the variable and its *VariableTypeNode* is ensured. However, we consider that an array of length zero (can also be a null-array with undefined length) has implicit array dimensions $[0,0,...]$. These always match the required array dimensions.

2.1.2 VariableTypeNode

VariableTypes are used to provide type definitions for variables. VariableTypes constrain the data type, value rank and array dimensions attributes of variable instances. Furthermore, instantiating from a specific variable type may provide semantic information. For example, an instance from $\mathrm{MotorTemperatureVariableType}$ is more meaningful than a float variable instantiated from $\mathrm{BaseDataVariable}$.

2.1.3 ObjectNode

Objects are used to represent systems, system components, real-world objects and software objects. Objects are instances of an *object type* and may contain variables, methods and further objects.

2.1.4 ObjectTypeNode

ObjectTypes provide definitions for Objects. Abstract objects cannot be instantiated. See *Node Lifecycle: Constructors, Destructors and Node Contexts* for the use of constructor and destructor callbacks.

2.1.5 ReferenceTypeNode

Each reference between two nodes is typed with a ReferenceType that gives meaning to the relation. The OPC UA standard defines a set of ReferenceTypes as a mandatory part of OPC UA information models.

- Abstract ReferenceTypes cannot be used in actual references and are only used to structure the ReferenceTypes hierarchy
- Symmetric references have the same meaning from the perspective of the source and target node

The figure below shows the hierarchy of the standard ReferenceTypes. Refer to Part 3 of the OPC UA specification for the full semantics of each ReferenceType.

```
References (Abstract, Symmetric)

    > HierarchicalReferences (Abstract)

        > HasChild (Abstract)

            > Aggregates (Abstract)

                > HasComponent

                    > HasOrderedComponent
```

(continues on next page)

(continued from previous page)

- HasProperty
- HasSubtype
- Organizes
- HasEventSource
- HasNotifier

- NonHierarchicalReferences (Abstract, Symmetric)
  - HasModellingRule
  - HasTypeDefinition
  - HasEncoding
  - HasDescription
  - GeneratesEvent
    - AlwaysGeneratesEvent

The ReferenceType hierarchy can be extended with user-defined ReferenceTypes. Many Companion Specifications for OPC UA define new ReferenceTypes to be used in their domain of interest.

For the following example of custom ReferenceTypes, we attempt to model the structure of a technical system. For this, we introduce two custom ReferenceTypes. First, the hierarchical $\mathrm{contains}$ ReferenceType indicates that a system (represented by an OPC UA object) contains a component (or subsystem). This gives rise to a tree-structure of containment relations. For example, the motor (object) is contained in the car and the crankshaft is contained in the motor. Second, the symmetric $\mathrm{connectedTo}$ ReferenceType indicates that two components are connected. For example, the motor's crankshaft is connected to the gear box. Connections are independent of the containment hierarchy and can induce a general graph-structure. Further subtypes of $\mathrm{connectedTo}$ could be used to differentiate between physical, electrical and information related connections. A client can then learn

the layout of a (physical) system represented in an OPC UA information model based on a common understanding of just two custom reference types.

2.1.6 DataTypeNode

DataTypes represent simple and structured data types. DataTypes may contain arrays. But they always describe the structure of a single instance. In open62541, DataTypeNodes in the information model hierarchy are matched to $\mathrm{UA_DataType}$ type descriptions for *Generic Type Handling* via their NodeId.

Abstract DataTypes (e.g. Number) cannot be the type of actual values. They are used to constrain values to possible child DataTypes (e.g. $\mathrm{UInt32}$).

2.1.7 MethodNode

Methods define callable functions and are invoked using the *Call* service. MethodNodes may have special properties (variable children with a $\mathrm{hasProperty}$ reference) with the *QualifiedName* $(0, \text{"InputArguments"})$ and $(0, \text{"OutputArguments"})$. The input and output arguments are both described via an array of $\mathrm{UA_Argument}$. While the Call service uses a generic array of *Variant* for input and output, the actual argument values are checked to match the signature of the MethodNode.

Note that the same MethodNode may be referenced from several objects (and object types). For this, the NodeId of the method *and of the object providing context* is part of a Call request message.

2.1.8 ViewNode

Each View defines a subset of the Nodes in the AddressSpace. Views can be used when browsing an information model to focus on a subset of nodes and references only. ViewNodes can be created and be interacted with. But their use in the *Browse* service is currently unsupported in open62541.

2.2 Protocol

We focus on the TCP-based binary protocol since it is by far the most common transport layer for OPC UA. The general concepts also translate to HTTP and SOAP-based communication defined in the standard. Communication in OPC UA is best understood by starting with the following key principles:

Request / Response
: All communication is based on the Request/Response pattern. Only clients can send a request to a server. And servers can only send responses to a matching request. Often the server is hosted close to a (physical) device, such as a sensor or a machine tool.

Asynchronous Responses
: A server does not have to immediately respond to requests and responses may be sent in a different order. This keeps the server responsive when it takes time until a specific request has been processed (e.g. a method call or when reading from a sensor with delay). Subscriptions (push-notifications) are implemented via special requests where the response is delayed until a notification is published.

> **Note**
>
> *OPC UA PubSub* (Part 14 of the standard) is an extension for the integration of many-to-many communication with OPC UA. PubSub does not use the client-server protocol. Rather, OPC UA PubSub integrates with either existing broker-based protocols such as MQTT, UDP-multicast or

Ethernet-based communication. Typically an OPC UA server (accessed via the client-server protocol) is used to configured PubSub communication.

Note that the client-server protocol also supports Subscriptions for one-to-one communication and does not depend on PubSub for this feature.

A client-server connection for the OPC UA binary protocol consists of three nested levels: The stateful TCP connection, a SecureChannel and the Session. For full details, see Part 6 of the OPC UA standard.

TCP Connection

The TCP connection is opened to the corresponding hostname and port with an initial handshake of HEL/ACK messages. The handshake establishes the basic settings of the connection, such as the maximum message length. The *Reverse Connect* extension of OPC UA allows the server to initiate the underlying TCP connection.

SecureChannel

SecureChannels are created on top of the raw TCP connection. A SecureChannel is established with an *OpenSecureChannel* request and response message pair. **Attention!** Even though a SecureChannel is mandatory, encryption might still be disabled. The *SecurityMode* of a SecureChannel can be either None, Sign, or $\mathrm{SignAndEncrypt}$. As of version 0.2 of open62541, message signing and encryption is still under ongoing development.

With message signing or encryption enabled, the *OpenSecureChannel* messages are encrypted using an asymmetric encryption algorithm (public-key cryptography)[1]. As part of the *OpenSecureChannel* messages, client and server establish a common secret over an initially unsecure channel. For subsequent messages, the common secret is used for symmetric encryption, which has the advantage of being much faster.

Different *SecurityPolicies* – defined in part 7 of the OPC UA standard – specify the algorithms for asymmetric and symmetric encryption, encryption key lengths, hash functions for message signing, and so on. Example SecurityPolicies are None for transmission of cleartext and $\mathrm{Basic256Sha256}$ which mandates a variant of RSA with SHA256 certificate hashing for asymmetric encryption and AES256 for symmetric encryption.

The possible SecurityPolicies of a server are described with a list of *Endpoints*. An endpoint jointly defines the SecurityMode, SecurityPolicy and means for authenticating a session (discussed in the next section) in order to connect to a certain server. The *GetEndpoints* service returns a list of available endpoints. This service can usually be invoked without a session and from an unencrypted SecureChannel. This allows clients to first discover available endpoints and then use an appropriate SecurityPolicy that might be required to open a session.

Session

Sessions are created on top of a SecureChannel. This ensures that users may authenticate without sending their credentials, such as username and password, in cleartext. Currently defined authentication mechanisms are anonymous login, username/password, Kerberos and x509 certificates. The latter requires that the request message is accompanied by a signature to prove that the sender is in possession of the private key with which the certificate was created.

There are two message exchanges required to establish a session: *CreateSession* and *ActivateSession*. The ActivateSession service can be used to switch an existing session to a different

[1] This entails that the client and server exchange so-called public keys. The public keys might come with a certificate from a key-signing authority or be verified against an external key repository. But we will not discuss certificate management in detail in this section.

SecureChannel. This is important, for example when the connection broke down and the existing session is reused with a new SecureChannel.

2.2.1 Structure of a protocol message

Consider the example OPC UA binary conversation in Figure Fig. 2.2, recorded and displayed with Wireshark.

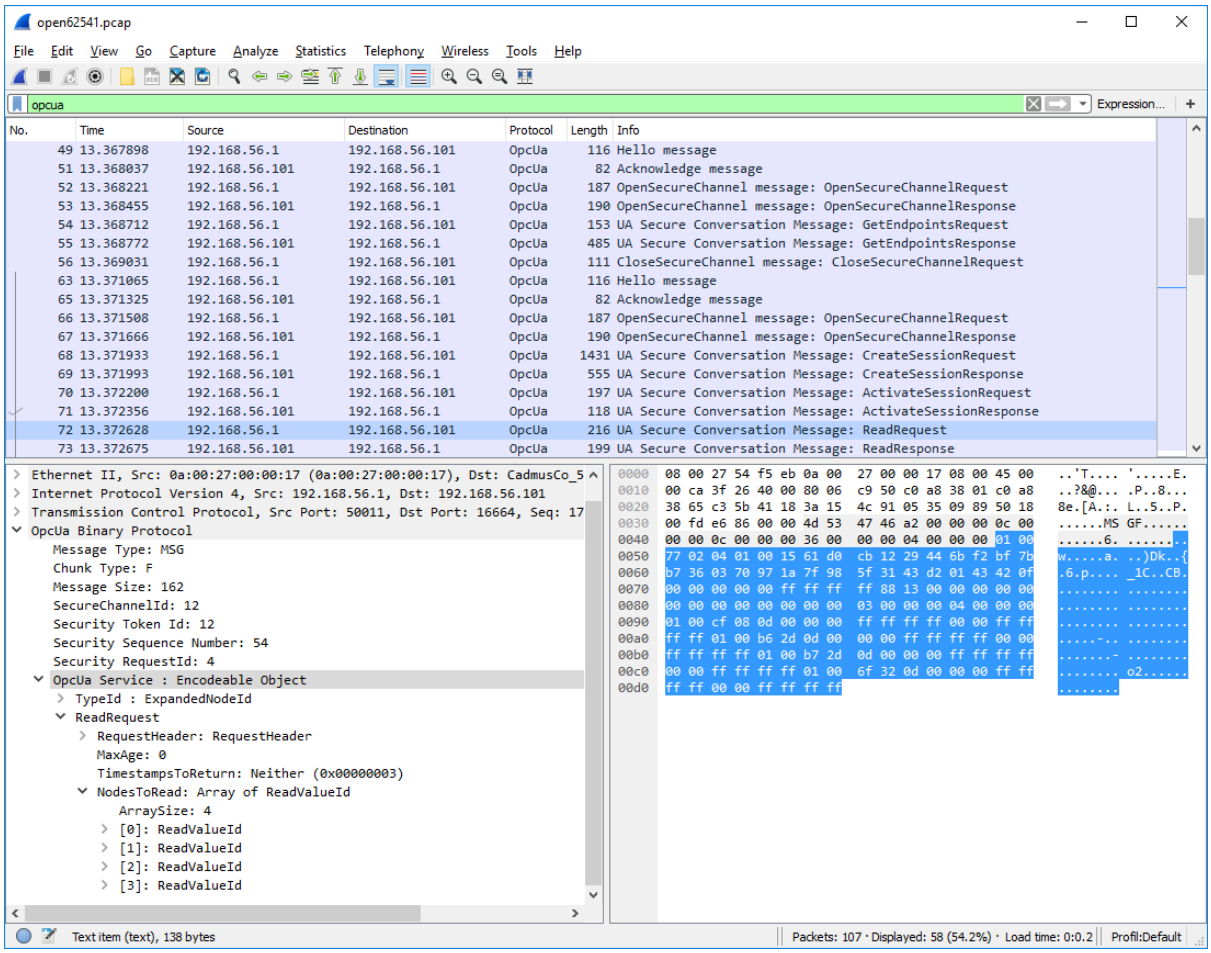


Fig. 2.2: OPC UA conversation displayed in Wireshark

The top part of the Wireshark window shows the messages from the conversation in order. The green line contains the applied filter. Here, we want to see the OPC UA protocol messages only. The first messages (from TCP packets 49 to 56) show the client opening an unencrypted SecureChannel and retrieving the server's endpoints. Then, starting with packet 63, a new connection and SecureChannel are created in conformance with one of the endpoints. On top of this SecureChannel, the client can then create and activate a session. The following *ReadRequest* message is selected and covered in more detail in the bottom windows.

The bottom left window shows the structure of the selected *ReadRequest* message. The purpose of the message is invoking the *Read service*. The message is structured into a header and a message body. Note that we do not consider encryption or signing of messages here.

Message Header

As stated before, OPC UA defines an asynchronous protocol. So responses may be out of order. The message header contains some basic information, such as the length of the message, as well as necessary information to relate messages to a SecureChannel and each request to the

corresponding response. “Chunking” refers to the splitting and reassembling of messages that are longer than the maximum network packet size.

Message Body
: Every OPC UA *service* has a signature in the form of a request and response data structure. These are defined according to the OPC UA protocol *type system*. See especially the *auto-generated type definitions* for the data types corresponding to service requests and responses. The message body begins with the identifier of the following data type. Then, the main payload of the message follows.

The bottom right window shows the binary payload of the selected *ReadRequest* message. The message header is highlighted in light-grey. The message body in blue highlighting shows the encoded *ReadRequest* data structure.

2.3 Services

In OPC UA, all communication is based on service calls, each consisting of a request and a response message. These messages are defined as data structures with a binary encoding and listed in *Generated Data Type Definitions*. Since all Services are pre-defined in the standard, they cannot be modified by the user. But you can use the *Call* service to invoke user-defined methods on the server.

Please refer to the *Client* and *Server* API where the services are exposed to end users. Please see part 4 of the OPC UA standard for the authoritative definition of the services and their behaviour.

2.3.1 Discovery Service Set

This Service Set defines Services used to discover the Endpoints implemented by a Server and to read the security configuration for those Endpoints.

FindServers Service
: Returns the Servers known to a Server or Discovery Server. The Client may reduce the number of results returned by specifying filter criteria

GetEndpoints Service
: Returns the Endpoints supported by a Server and all of the configuration information required to establish a SecureChannel and a Session.

FindServersOnNetwork Service
: Returns the Servers known to a Discovery Server. Unlike FindServer, this Service is only implemented by Discovery Servers. It additionally returns servers which may have been detected through Multicast.

RegisterServer
: Registers a remote server in the local discovery service.

RegisterServer2
: This Service allows a Server to register its DiscoveryUrls and capabilities with a Discovery Server. It extends the registration information from RegisterServer with information necessary for FindServersOnNetwork.

2.3.2 SecureChannel Service Set

This Service Set defines Services used to open a communication channel that ensures the confidentiality and Integrity of all Messages exchanged with the Server.

OpenSecureChannel Service
: Open or renew a SecureChannel that can be used to ensure Confidentiality and Integrity for Message exchange during a Session.

CloseSecureChannel Service
: Used to terminate a SecureChannel.

Session Service Set
: This Service Set defines Services for an application layer connection establishment in the context of a Session.

CreateSession Service
: Used by an OPC UA Client to create a Session and the Server returns two values which uniquely identify the Session. The first value is the sessionId which is used to identify the Session in the audit logs and in the Server's address space. The second is the authenticationToken which is used to associate an incoming request with a Session.

ActivateSession
: Used by the Client to submit its SoftwareCertificates to the Server for validation and to specify the identity of the user associated with the Session. This Service request shall be issued by the Client before it issues any other Service request after CreateSession. Failure to do so shall cause the Server to close the Session.

CloseSession
: Used to terminate a Session.

Cancel Service
: Used to cancel outstanding Service requests. Successfully cancelled service requests shall respond with Bad_RequestCancelledByClient.

2.3.3 NodeManagement Service Set

This Service Set defines Services to add and delete AddressSpace Nodes and References between them. All added Nodes continue to exist in the AddressSpace even if the Client that created them disconnects from the Server.

AddNodes Service
: Used to add one or more Nodes into the AddressSpace hierarchy. If the type or one of the supertypes has any HasInterface references (see OPC 10001-7 - Amendment 7, 4.9.2), the child nodes of the interfaces are added to the new object.

AddReferences Service
: Used to add one or more References to one or more Nodes.

DeleteNodes Service
: Used to delete one or more Nodes from the AddressSpace.

DeleteReferences
: Used to delete one or more References of a Node.

2.3.4 View Service Set

Clients use the browse Services of the View Service Set to navigate through the AddressSpace or through a View which is a subset of the AddressSpace.

Browse Service
: Used to discover the References of a specified Node. The browse can be further limited by the

use of a View. This Browse Service also supports a primitive filtering capability.

BrowseNext Service
Used to request the next set of Browse or BrowseNext response information that is too large to be sent in a single response. “Too large” in this context means that the Server is not able to return a larger response or that the number of results to return exceeds the maximum number of results to return that was specified by the Client in the original Browse request.

TranslateBrowsePathsToNodeIds Service
Used to translate textual node paths to their respective ids.

RegisterNodes Service
Used by Clients to register the Nodes that they know they will access repeatedly (e.g. Write, Call). It allows Servers to set up anything needed so that the access operations will be more efficient.

UnregisterNodes Service
This Service is used to unregister NodeIds that have been obtained via the RegisterNodes service.

2.3.5 Query Service Set

This Service Set is used to issue a Query to a Server. OPC UA Query is generic in that it provides an underlying storage mechanism independent Query capability that can be used to access a wide variety of OPC UA data stores and information management systems. OPC UA Query permits a Client to access data maintained by a Server without any knowledge of the logical schema used for internal storage of the data. Knowledge of the AddressSpace is sufficient.

QueryFirst Service (not implemented)
This Service is used to issue a Query request to the Server.

QueryNext Service (not implemented)
This Service is used to request the next set of QueryFirst or QueryNext response information that is too large to be sent in a single response.

2.3.6 Attribute Service Set

This Service Set provides Services to access Attributes that are part of Nodes.

Read Service
Used to read attributes of nodes. For constructed attribute values whose elements are indexed, such as an array, this Service allows Clients to read the entire set of indexed values as a composite, to read individual elements or to read ranges of elements of the composite.

Write Service
Used to write attributes of nodes. For constructed attribute values whose elements are indexed, such as an array, this Service allows Clients to write the entire set of indexed values as a composite, to write individual elements or to write ranges of elements of the composite.

HistoryRead Service
Used to read historical values or Events of one or more Nodes. Servers may make historical values available to Clients using this Service, although the historical values themselves are not visible in the AddressSpace.

HistoryUpdate Service
Used to update historical values or Events of one or more Nodes. Several request parameters

indicate how the Server is to update the historical value or Event. Valid actions are Insert, Replace or Delete.

2.3.7 Method Service Set

The Method Service Set defines the means to invoke methods. A method shall be a component of an Object. See the section on *MethodNodes* for more information.

Call Service
Used to call (invoke) a methods. Each method call is invoked within the context of an existing Session. If the Session is terminated, the results of the method's execution cannot be returned to the Client and are discarded.

2.3.8 MonitoredItem Service Set

Clients define MonitoredItems to subscribe to data and Events. Each MonitoredItem identifies the item to be monitored and the Subscription to use to send Notifications. The item to be monitored may be any Node Attribute.

CreateMonitoredItems Service
Used to create and add one or more MonitoredItems to a Subscription. A MonitoredItem is deleted automatically by the Server when the Subscription is deleted. Deleting a MonitoredItem causes its entire set of triggered item links to be deleted, but has no effect on the MonitoredItems referenced by the triggered items.

DeleteMonitoredItems Service
Used to remove one or more MonitoredItems of a Subscription. When a MonitoredItem is deleted, its triggered item links are also deleted.

ModifyMonitoredItems Service
Used to modify MonitoredItems of a Subscription. Changes to the MonitoredItem settings shall be applied immediately by the Server. They take effect as soon as practical but not later than twice the new revisedSamplingInterval.

Illegal request values for parameters that can be revised do not generate errors. Instead the server will choose default values and indicate them in the corresponding revised parameter.

SetMonitoringMode Service
Used to set the monitoring mode for one or more MonitoredItems of a Subscription.

SetTriggering Service
Used to create and delete triggering links for a triggering item.

2.3.9 Subscription Service Set

Subscriptions are used to report Notifications to the Client.

CreateSubscription Service
Used to create a Subscription. Subscriptions monitor a set of MonitoredItems for Notifications and return them to the Client in response to Publish requests.

ModifySubscription Service
Used to modify a Subscription.

SetPublishingMode Service
Used to enable sending of Notifications on one or more Subscriptions.

Publish Service
: Used for two purposes. First, it is used to acknowledge the receipt of NotificationMessages for one or more Subscriptions. Second, it is used to request the Server to return a NotificationMessage or a keep-alive Message.

Republish Service
: Requests the Subscription to republish a NotificationMessage from its retransmission queue.

DeleteSubscriptions Service
: Invoked to delete one or more Subscriptions that belong to the Client's Session.

TransferSubscription Service
: Used to transfer a Subscription and its MonitoredItems from one Session to another. For example, a Client may need to reopen a Session and then transfer its Subscriptions to that Session. It may also be used by one Client to take over a Subscription from another Client by transferring the Subscription to its Session.

CHAPTER

THREE

BUILDING OPEN62541

3.1 Building the Library

open62541 uses CMake to build the library and binaries. CMake generates a Makefile or a Visual Studio project. This is then used to perform the actual build.

3.1.1 Building with CMake on Debian/Ubuntu

```
sudo apt-get install git build-essential gcc pkg-config cmake python3

# enable additional features
sudo apt-get install cmake-curses-gui     # for the ccmake graphical interface
sudo apt-get install libmbedtls-dev       # for encryption support
sudo apt-get install check libsubunit-dev # for unit tests
sudo apt-get install libpcap-dev          # for network-replay unit tests
sudo apt-get install python3-sphinx  # for documentation generation
sudo apt-get install python3-sphinx-rtd-theme # documentation style
sudo apt-get install libavahi-client-dev libavahi-common-dev # for LDS-ME (multicast␣
↪discovery)

git clone https://github.com/open62541/open62541.git
cd open62541
mkdir build
cd build
cmake ..
make

# optional: fetch additional dependencies for advanced features
git submodule update --init --recursive

# select additional features
ccmake ..
cmake .. -DUA_ENABLE_DRIVER_MDNS_AVAHI=ON -DUA_ENABLE_DRIVER_
↪MDNS_MDNSD=OFF # use Avahi for multicast discovery
make

# build documentation
make doc # html documentation
make doc_pdf # pdf documentation (requires LaTeX)
```

Note: parallel compilation can be enable by using

```
make -j$(nproc)
```

You can install open62541 using the well known *make install* command. This allows you to use pre-built libraries and headers for your own project. In order to use open62541 as a shared library (.dll or .so) make sure to activate the BUILD_SHARED_LIBS CMake option.

To override the default installation directory use cmake -DCMAKE_INSTALL_PREFIX=/some/path. Based on the SDK Features you selected, as described in *Main Build Options*, these features will also be included in the installation. Thus we recommend to enable as many non-experimental features as possible for the installed binary.

In your own CMake project you can then include the open62541 library. A simple CMake project definition looks as follows:

```
cmake_minimum_required(VERSION 3.5)
project("open62541SampleApplication")
add_executable(main main.c)

# Linux/Unix configuration using pkg-config
find_package(PkgConfig)
pkg_check_modules(open62541 REQUIRED open62541)
target_link_libraries(main open62541)

# Alternative CMake-based library definition.
# This might not be included in some package distributions.
#
#   find_package(open62541 REQUIRED)
#   target_link_libraries(main open62541::open62541)
```

3.1.2 Building with Visual Studio on Windows

Here we explain the build process for Visual Studio 2022 Community.

- **Download and install**
  - Python 3.x: https://python.org/downloads
  - **Visual Studio 2022: https://visualstudio.microsoft.com**
    - When installing Visual Studio select the workload "Desktop development with C++"
- **Open Visual Studio and from the Get started window select "Clone a repository"**
  - Repository location: https://github.com/open62541/open62541.git
  - Select a local path where to download the project and press Clone
- Switch to Folder View from the Solution Explorer
- **Project / CMake Settings for open62541**
  - Customize CMake variables as wanted and save
- Build / Build All
- Build / Install open62541

Note: the solution generated with cmake can also be compiled in parallel with the command

```
msbuild open62541.sln /v:n -t:rebuild -m
```

3.1.3 Building with CMake/MinGW on Windows

To build with MinGW, just replace the compiler selection in the call to CMake.

- **Download and install**
  - MinGW http://sourceforge.net/projects/mingw
  - Python 3.x: https://python.org/downloads
  - CMake: http://www.cmake.org/cmake/resources/software.html
- Download the open62541 sources (using git or as a zipfile from github)
- Open a command shell (cmd) and run

```
cd <path-to>\open62541
mkdir build
cd build
<path-to>\cmake.exe .. -G "MinGW Makefiles"
:: You can use use cmake-gui for a graphical user-interface to select features
make
```

3.1.4 Building on OS X

- Download and install
  - Xcode: https://itunes.apple.com/us/app/xcode/id497799835?ls=1&mt=12
  - Homebrew: http://brew.sh/
  - Pip (a package manager for Python, may be preinstalled): `sudo easy_install pip`
- Run the following in a shell

```
brew install cmake
pip install sphinx # for documentation generation
pip install sphinx_rtd_theme # documentation style
brew install check # for unit tests
```

Follow Ubuntu instructions without the `apt-get` commands as these are taken care of by the above packages.

3.1.5 Building on OpenBSD

The procedure below works on OpenBSD 5.8 with gcc version 4.8.4, cmake version 3.2.3 and Python version 2.7.10.

- Install a recent gcc, python and cmake:

```
pkg_add gcc python cmake
```

- Tell the system to actually use the recent gcc (it gets installed as egcc on OpenBSD):

```
export CC=egcc CXX=eg++
```

- Now proceed as described for Ubuntu/Debian:

```
cd open62541
mkdir build
cd build
cmake ..
make
```

3.1.6 Building Debian Packages inside Docker Container with CMake on Ubuntu or Debian

The python tool with the debian packaging definitions and the debian build files can be found in the external repository: https://salsa.debian.org/jpfr/open62541/-/tree/main/tools?ref_type=heads

This is how to build the Debian packages.

```
# Assume a fresh checkout of open62541 in the ~/open62541 directory
cd ~/open62541

# Create the debian packaging definitions
python3 ./tools/prepare_packaging.py

# Build the package (generated packages and source will be in the ~ folder)
debuild
```

3.2 Build Options

The open62541 project uses CMake to manage the build options, for code generation and to generate build projects for the different systems and IDEs. The tools *ccmake* or *cmake-gui* can be used to graphically set the build options.

Most options can be changed manually in ua_config.h (open62541.h for the single-file release) after the code generation. But usually there is no need to adjust them.

3.2.1 Main Build Options

CMAKE_BUILD_TYPE

- RelWithDebInfo -O2 optimization with debug symbols
- Release -O2 optimization without debug symbols
- Debug -O0 optimization with debug symbols
- MinSizeRel -Os optimization without debug symbols

BUILD_SHARED_LIBS

Build a shared library (.dll/.so) or (an archive of) object files for linking into a static binary. Shared libraries are recommended for a system-wide install. Note that this option modifies the ua_config.h file that is also included in open62541.h for the single-file distribution.

UA_LOGLEVEL

The SDK logs events of the level defined in UA_LOGLEVEL and above only. The logging event levels are as follows:

- 600: Fatal
- 500: Error
- 400: Warning
- 300: Info
- 200: Debug
- 100: Trace

This compilation flag defines which log levels get compiled into the code. In addition, the implementations of *Logging Plugin API* allow to set a filter for the logging level at runtime. So the logging level can be changed in the configuration without recompiling.

UA_MULTITHREADING

Level of multi-threading support. The supported levels are currently as follows:

- 0-99: Multithreading support disabled.
- >=100: API functions marked with the UA_THREADSAFE-macro are protected internally with mutexes. Multiple threads are allowed to call these functions of the SDK at the same time without causing race conditions. Furthermore, this level support the handling of asynchronous method calls from external worker threads.

3.2.2 Select build artefacts

By default only the main library shared object libopen62541.so (open62541.dll) or static linking archive open62541.a (open62541.lib) is built. Additional artifacts can be specified by the following options:

UA_BUILD_EXAMPLES
Compile example servers and clients from $\mathrm{examples}/*.\mathrm{c}$.

UA_BUILD_UNIT_TESTS
Compile unit tests. The tests can be executed with $\mathrm{make\ test}$. An individual test can be executed with $\mathrm{make\ test\ ARGS}=$"-R $<\mathrm{test_name}>$ -V". The list of available tests can be displayed with $\mathrm{make\ test\ ARGS}=$"-N".

3.2.3 Detailed SDK Features

UA_ENABLE_SUBSCRIPTIONS
Enable subscriptions

UA_ENABLE_SUBSCRIPTIONS_EVENTS
Enable the use of events for subscriptions. This is a new feature and currently marked as EXPERIMENTAL.

The Method service set is always compiled.

UA_ENABLE_DRIVER_FILE_TRANSFER
Enable the file transfer driver (default ON on all architectures). The local filesystem backend is available on POSIX and Win32; other architectures can supply a custom backend. The reduced Namespace Zero includes file transfer types when this driver is enabled; the full Namespace Zero always includes them. With $\mathrm{MINIMAL}$ or NONE, load the required types before constructing a file transfer driver.

UA_ENABLE_NODEMANAGEMENT
Enable dynamic addition and removal of nodes at runtime

UA_ENABLE_AMALGAMATION
Compile a single-file release into the files open62541.c and open62541.h. Invoke the CMake target to generate the amalgamation as make open62541-amalgamation.

UA_ENABLE_COVERAGE
Measure the coverage of unit tests

Discovery services are always compiled. Enable registration management with UA_ServerConfig.registeredServersEnabled and network discovery management with UA_ServerConfig.serversOnNetworkEnabled. mDNS announcement and browsing require an attached mDNS driver.

UA_ENABLE_DISCOVERY_SEMAPHORE
Enable Discovery Semaphore support

UA_ENABLE_ENCRYPTION
Enable encryption support and specify the used encryption backend. The possible options are:

- OFF No encryption support. (default)
- MBEDTLS Encryption support using mbed TLS
- OPENSSL Encryption support using OpenSSL
- LIBRESSL EXPERIMENTAL: Encryption support using LibreSSL

ECC security policies (NIST P-256/P-384, Brainpool) are available with OpenSSL or mbedTLS ≥ 3.0. Curve25519 and Curve448 require OpenSSL. See *ECC Specifics* for details.

UA_ENABLE_ENCRYPTION_TPM2

Enable TPM hardware for encryption. The possible options are:

- OFF No TPM encryption support. (default)
- ON TPM encryption support

UA_NAMESPACE_ZERO
Namespace zero contains the standard-defined nodes. The full namespace zero may not be required for all applications. The selectable options are as follows:

- NONE: Do not generate namespace zero from the XML definitions.
- MINIMAL: A barebones namespace zero that is compatible with most clients. But this namespace 0 is so small that it does not pass the CTT (Conformance Testing Tools of the OPC Foundation).
- REDUCED: Small namespace zero that passes the CTT.
- FULL: Full namespace zero generated from the official XML definitions.

UA_ENABLE_DATATYPES_ALL
Generate C definitions, handling functions and UA_TYPES entries for all supported datatypes in the namespace zero schema. Enabled by default. Disabling this option selects the core datatypes and the additional types required by the enabled features, using the tools/schema/datatypes_*.txt lists. This reduces the binary size and the available generated datatype API; it does not disable services or select the namespace zero information model. With

UA_NAMESPACE_ZERO=FULL, all datatypes are generated regardless of this option because the full namespace zero does not use the reduced type lists.

UA_ENABLE_DIAGNOSTICS
Enable diagnostics information exposed by the server. Enabled by default.

UA_ENABLE_JSON_ENCODING
Enable JSON encoding. Enabled by default. The JSON encoding changed with the 1.05 version of the OPC UA specification.

Some options are marked as advanced. The advanced options need to be toggled to be visible in the cmake GUIs.

UA_ENABLE_TYPEDESCRIPTION
Add the type and member names to the UA_DataType structure. Enabled by default.

UA_ENABLE_STATUSCODE_DESCRIPTIONS
Compile the human-readable name of the StatusCodes into the binary. Enabled by default.

UA_ENABLE_LWS
Enable the libwebsockets transport provider. It supplies the generic HTTP client/server ConnectionManager, OPC UA services over HTTPS (opc.https://), and OPC UA Binary over WebSockets (opc.ws:// and opc.wss://). The server transports remain opt-in through httpEnabled and webSocketEnabled. opc.https requires a TLS certificate and private key. Binary and, when UA_ENABLE_JSON_ENCODING is enabled, JSON service bodies can coexist on the same endpoint URL and are exposed as separate Binary and /json endpoint paths. Each logical channel remains pinned to one encoding. Internally, Session routing contexts isolate AuthenticationTokens and nonces between clients without creating a new wire-level SecureChannel for every HTTP request. HTTP content compression is optional in the OPC UA specification.

Session-less HTTPS Authorization and Accept-Language processing are not implemented by this transport. Applications that require those optional profile features must provide them above the generic HTTP ConnectionManager.

UA_ENABLE_HTTP_COMPRESSION
Enable bounded gzip and deflate request/response content coding for HTTP clients and servers. This option requires UA_ENABLE_LWS and zlib. Compressed input is limited independently on the wire and after decompression. Unknown request codings are rejected with HTTP status 415 and unacceptable response coding negotiation with status 406.

UA_ENABLE_LWS_MQTT
Enable the MQTT part of the libwebsockets provider for PubSub MQTT transport protocols (opc.mqtt:// and opc.mqtts://). Requires UA_ENABLE_LWS and libwebsockets compiled with MQTT support (LWS_ROLE_MQTT=ON).

3.2.4 PubSub Build Options

UA_ENABLE_PUBSUB
Enable the experimental OPC UA PubSub support. The option will include the PubSub UDP multicast plugin. Enabled by default.

UA_ENABLE_PUBSUB_FILE_CONFIG
Enable loading OPC UA PubSub configuration from File/ByteString. Enabling PubSub informationmodel methods also will add a method to the Publish/Subscribe object which allows configuring PubSub at runtime. Disabled by default.

UA_ENABLE_PUBSUB_INFORMATIONMODEL
: Enable the information model representation of the PubSub configuration. For more details take a look at the following section *PubSub Information Model Representation*. Enabled by default.

3.2.5 Debug Build Options

This group contains build options mainly useful for development of the library itself.

UA_DEBUG
: Enable assertions and additional definitions not intended for production builds

UA_DEBUG_DUMP_PKGS
: Dump every package received by the server as hexdump format

3.2.6 Minimizing the binary size

The size of the generated binary can be reduced considerably by adjusting the build configuration. With open62541, it is possible to configure minimal servers that require less than 100kB of RAM and ROM.

The following options influence the ROM requirements:

First, in CMake, the build type can be set to CMAKE_BUILD_TYPE=MinSizeRel. This sets the compiler flags to minimize the binary size. The build type also strips out debug information. Second, the binary size can be reduced by removing features via the build-flags described above.

Second, setting UA_NAMESPACE_ZERO to MINIMAL reduces the size of the builtin information model. Setting this option can reduce the binary size by half in some cases.

Third, some features might not be needed and can be disabled to reduce the binary footprint. Examples for this are Subscriptions or encrypted communication.

Last, logging messages take up a lot of space in the binary and might not be needed in embedded scenarios. Setting UA_LOGLEVEL to a value above 600 (FATAL) disables all logging. In addition, the feature-flags UA_ENABLE_TYPEDESCRIPTION and UA_ENABLE_STATUSCODE_DESCRIPTIONS add static information to the binary that is only used for human-readable logging and debugging.

The RAM requirements of a server are mostly due to the following settings:

- The size of the information model
- The number of connected clients
- The configured maximum message size that is preallocated

3.3 Prebuilt packages

3.3.1 Debian/Ubuntu

Official distribution packages are available for Debian and Ubuntu.

Install the development package with:

```
sudo apt-get update
sudo apt-get install libopen62541-1-dev
```

3.3.2 Arch

Arch packages are available in the AUR:

- Stable Builds: https://aur.archlinux.org/packages/open62541/
- Unstable Builds (current master): https://aur.archlinux.org/packages/open62541-git/
- In order to add custom build options (*Main Build Options*), you can set the environment variable OPEN62541_CMAKE_FLAGS

3.3.3 OpenBSD

Starting with OpenBSD 6.7 the ports directory misc/open62541 can build the released version of open62541. Install the binary package from the OpenBSD mirrors:

```
pkg_add open62541
```

3.4 Building the Examples

Make sure that you have installed the shared library as explained in the previous steps. Then the build system should automatically find the includes and the shared library.

```
cp /path-to/examples/tutorial_server_firststeps.c . # copy the example server
gcc -std=c99 -o server tutorial_server_firststeps.c -lopen62541
```

CHAPTER

FOUR

DATA TYPES

The OPC UA protocol defines 25 builtin data types and three ways of combining them into higher-order types: arrays, structures and unions. In open62541, only the builtin data types are defined manually. All other data types are generated from standard XML definitions. Their exact definitions can be looked up at https://opcfoundation.org/UA/schemas/Opc.Ua.Types.bsd.

For users that are new to open62541, take a look at the *tutorial for working with data types* before diving into the implementation details.

4.1 Builtin Types

4.1.1 Boolean

A two-state logical value (true or false).

```
typedef bool UA_Boolean;
#define UA_TRUE true UA_INTERNAL_DEPRECATED
#define UA_FALSE false UA_INTERNAL_DEPRECATED
```

4.1.2 SByte

An integer value between -128 and 127.

```
typedef int8_t UA_SByte;
#define UA_SBYTE_MIN (-128)
#define UA_SBYTE_MAX 127
```

4.1.3 Byte

An integer value between 0 and 255.

```
typedef uint8_t UA_Byte;
#define UA_BYTE_MIN 0
#define UA_BYTE_MAX 255
```

4.1.4 Int16

An integer value between -32 768 and 32 767.

```
typedef int16_t UA_Int16;
#define UA_INT16_MIN (-32768)
#define UA_INT16_MAX 32767
```

4.1.5 UInt16

An integer value between 0 and 65 535.

```
typedef uint16_t UA_UInt16;
#define UA_UINT16_MIN 0
#define UA_UINT16_MAX 65535
```

4.1.6 Int32

An integer value between -2 147 483 648 and 2 147 483 647.

```
typedef int32_t UA_Int32;
#define UA_INT32_MIN ((int32_t)-2147483648LL)
#define UA_INT32_MAX 2147483647L
```

4.1.7 UInt32

An integer value between 0 and 4 294 967 295.

```
typedef uint32_t UA_UInt32;
#define UA_UINT32_MIN 0
#define UA_UINT32_MAX 4294967295UL
```

4.1.8 Int64

An integer value between -9 223 372 036 854 775 808 and 9 223 372 036 854 775 807.

```
typedef int64_t UA_Int64;
#define UA_INT64_MAX (int64_t)9223372036854775807LL
#define UA_INT64_MIN ((int64_t)-UA_INT64_MAX-1LL)
```

4.1.9 UInt64

An integer value between 0 and 18 446 744 073 709 551 615.

```
typedef uint64_t UA_UInt64;
#define UA_UINT64_MIN 0
#define UA_UINT64_MAX (uint64_t)18446744073709551615ULL
```

4.1.10 Float

An IEEE single precision (32 bit) floating point value.

```
typedef float UA_Float;
#define UA_FLOAT_MIN FLT_MIN
#define UA_FLOAT_MAX FLT_MAX
```

4.1.11 Double

An IEEE double precision (64 bit) floating point value.

```
typedef double UA_Double;
#define UA_DOUBLE_MIN DBL_MIN
#define UA_DOUBLE_MAX DBL_MAX
```

4.1.12 StatusCode

A numeric identifier for an error or condition that is associated with a value or an operation. See the section *StatusCodes* for the meaning of a specific code.

Each StatusCode has one of three "severity" bit-flags: Good, Uncertain, Bad. An additional reason is indicated by the SubCode bitfield.

- A StatusCode with severity Good means that the value is of good quality.
- A StatusCode with severity Uncertain means that the quality of the value is uncertain for reasons indicated by the SubCode.
- A StatusCode with severity Bad means that the value is not usable for reasons indicated by the SubCode.

```
typedef uint32_t UA_StatusCode;

/* Returns the human-readable name of the StatusCode. If no matching StatusCode
 * is found, a default string for "Unknown" is returned. This feature might be
 * disabled to create a smaller binary with the
 * UA_ENABLE_STATUSCODE_DESCRIPTIONS build-flag. Then the function returns␣
↪an
 * empty string for every StatusCode. */
const char *
UA_StatusCode_name(UA_StatusCode code);
```

The following methods extract the severity from a StatusCode. See Part 4, Section 7.34 for details.

```
/* (code >> 30) >= 0x02 */
UA_Boolean
UA_StatusCode_isBad(UA_StatusCode code);

/* (code >> 30) == 0x01 */
UA_Boolean
UA_StatusCode_isUncertain(UA_StatusCode code);

/* (code >> 30) == 0x00 */
UA_Boolean
UA_StatusCode_isGood(UA_StatusCode code);

/* Compares the top 16 bits of two StatusCodes for equality. This should only be
 * used when processing user-defined StatusCodes e.g when processing a
 * ReadResponse. As a convention, the lower bits of StatusCodes should not be
 * used internally. */
```

(continues on next page)

(continued from previous page)

```
UA_Boolean UA_StatusCode_equalTop(UA_StatusCode s1, UA_StatusCode s2);
#define UA_StatusCode_isEqualTop(s1, s2) UA_StatusCode_equalTop(s1, s2)
```

4.1.13 String

A sequence of Unicode characters. Strings are just an array of UA_Byte.

```
typedef struct UA_String {
    size_t length; /* The length of the string */
    UA_Byte *data; /* The content (not null-terminated) */
} UA_String;

extern const UA_String UA_STRING_NULL;
UA_Boolean UA_String_isEmpty(const UA_String *s);
#define UA_String_isNull(s) UA_String_isEmpty(s)

/* Returns a string pointing to the original char-array */
UA_String UA_STRING(char *chars);

/* Returns a string-copy of the char-array. Returns a null-string when
 * alloc fails. */
UA_String UA_String_fromChars(const char *src);
#define UA_STRING_ALLOC(CHARS) UA_String_fromChars(CHARS)

/* Define string variable at compile time (in ROM) */
#define UA_STRING_STATIC(CHARS) {sizeof(CHARS)-1, (UA_Byte*)CHARS}

/* Uses realloc to append to the string in the first argument */
UA_StatusCode
UA_String_append(UA_String *s, const UA_String s2);
```

The following methods implement the C standard's printf/vprintf.

In addition to the format specifiers from the C standard, the following can be used also:

- %S - UA_String (not wrapped in quotation marks in the output)
- %N - UA_NodeId (using UA_NodeId_print)
- %Q - UA_QualifiedName (using UA_QualifiedName_print)

Example usage:
UA_NodeId nodeId = UA_NODEID_NUMERIC(1, 4711); UA_String_format(outString, "Test %N", nodeId);

The output is written to the output string in the first argument. Memory of sufficient length is allocated when the output string initially has zero length.

If the string in the first argument initially has non-zero length, then this string is used as buffer for encoding and its length is adjusted accordingly. If the length is too short, then UA_STATUSCODE_BADENCODINGLIMITSEXCEEDED is reported. Also in that case the string is printed as much as possible.

```
UA_StatusCode
UA_String_format(UA_String *str, const char *format, ...);

UA_StatusCode
UA_String_vformat(UA_String *str, const char *format, va_list args);

/* Old API */
#define UA_String_printf(str, format, ...)   \
    UA_String_format(str, format, ___VA_ARGS___)
#define UA_String_vprintf(str, format, args) \
    UA_String_vformat(str, format, args)
```

4.1.14 DateTime

An instance in time. A DateTime value is encoded as a 64-bit signed integer which represents the number of 100 nanosecond intervals since January 1, 1601 (UTC).

The methods providing an interface to the system clock are architecture- specific. Usually, they provide a UTC clock that includes leap seconds. The OPC UA standard allows the use of International Atomic Time (TAI) for the DateTime instead. But this is still unusual and not implemented for most SDKs. Currently (2019), UTC and TAI are 37 seconds apart due to leap seconds.

```
typedef int64_t UA_DateTime;

/* Multiples to convert durations to DateTime */
#define UA_DATETIME_USEC 10LL
#define UA_DATETIME_MSEC (UA_DATETIME_USEC * 1000LL)
#define UA_DATETIME_SEC (UA_DATETIME_MSEC * 1000LL)

/* The current time in UTC time */
UA_DateTime UA_DateTime_now(void);

/* Offset between local time and UTC time. On POSIX, building with
 * UA_ARCH_LOCAL_TIME_IS_UTC enabled returns zero without timezone conversion.
↪ */
UA_Int64 UA_DateTime_localTimeUtcOffset(void);

/* CPU clock invariant to system time changes. Use only to measure durations,
 * not absolute time. */
UA_DateTime UA_DateTime_nowMonotonic(void);

/* Parse the humand-readable DateTime format */
UA_StatusCode
UA_DateTime_parse(UA_DateTime *dst, const UA_String str);

/* Returns zero if parsing fails */
UA_DateTime UA_DATETIME(const char *chars);

/* Represents a Datetime as a structure */
typedef struct UA_DateTimeStruct {
```

(continues on next page)

(continued from previous page)

```
    UA_UInt16 nanoSec;
    UA_UInt16 microSec;
    UA_UInt16 milliSec;
    UA_UInt16 sec;
    UA_UInt16 min;
    UA_UInt16 hour;
    UA_UInt16 day;   /* From 1 to 31 */
    UA_UInt16 month; /* From 1 to 12 */
    UA_Int16 year;   /* Can be negative (BC) */
} UA_DateTimeStruct;

UA_DateTimeStruct UA_DateTime_toStruct(UA_DateTime t);
UA_DateTime UA_DateTime_fromStruct(UA_DateTimeStruct ts);
```

The C99 standard (7.23.1) says: "The range and precision of times representable in clock_t and time_t are implementation-defined." On most systems, time_t is a 4 or 8 byte integer counting seconds since the UTC Unix epoch. The following methods are used for conversion.

```
/* Datetime of 1 Jan 1970 00:00 */
#define UA_DATETIME_UNIX_EPOCH (11644473600LL * UA_DATETIME_SEC)

/* (date - UA_DATETIME_UNIX_EPOCH) / UA_DATETIME_SEC */
UA_Int64
UA_DateTime_toUnixTime(UA_DateTime date);

/* (unixDate * UA_DATETIME_SEC) + UA_DATETIME_UNIX_EPOCH */
UA_DateTime
UA_DateTime_fromUnixTime(UA_Int64 unixDate);
```

4.1.15 Guid

A 16 byte value that can be used as a globally unique identifier.

```
typedef struct UA_Guid {
    UA_UInt32 data1;
    UA_UInt16 data2;
    UA_UInt16 data3;
    UA_Byte   data4[8];
} UA_Guid;

extern const UA_Guid UA_GUID_NULL;

/* Print a Guid in the human-readable format defined in Part 6, 5.1.3
 *
 * Format: C496578A-0DFE-4B8F-870A-745238C6AEAE
 *         |    |  |  |  |         |
 *         0    8  13 18 23        36
 *
 * This allocates memory if the output argument is an empty string. Tries to use
 * the given buffer otherwise. */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode
UA_Guid_print(const UA_Guid *guid, UA_String *output);

/* Parse the humand-readable Guid format */
UA_StatusCode
UA_Guid_parse(UA_Guid *guid, const UA_String str);

/* Shorthand, returns UA_GUID_NULL upon failure to parse */
UA_Guid UA_GUID(const char *chars);
```

4.1.16 ByteString

A sequence of octets.

```
typedef UA_String UA_ByteString;

extern const UA_ByteString UA_BYTESTRING_NULL;

/* Allocates memory of size length for the bytestring.
 * The content is not set to zero. */
UA_StatusCode
UA_ByteString_allocBuffer(UA_ByteString *bs, size_t length);

/* Converts a ByteString to the corresponding
 * base64 representation */
UA_StatusCode
UA_ByteString_toBase64(const UA_ByteString *bs, UA_String *output);

/* Parse a ByteString from a base64 representation */
UA_StatusCode
UA_ByteString_fromBase64(UA_ByteString *bs,
                         const UA_String *input);

#define UA_BYTESTRING(chars) UA_STRING(chars)
#define UA_BYTESTRING_ALLOC(chars) UA_STRING_ALLOC(chars)

/* Returns a non-cryptographic hash of a bytestring */
UA_UInt32
UA_ByteString_hash(UA_UInt32 initialHashValue,
                   const UA_Byte *data, size_t size);
```

4.1.17 XmlElement

An XML element.

```
typedef UA_String UA_XmlElement;
```

4.1.18 NodeId

An identifier for a node in the address space of an OPC UA Server.

```
enum UA_NodeIdType {
    UA_NODEIDTYPE_NUMERIC    = 0, /* In the binary encoding, this can also
                                   * become 1 or 2 (two-byte and four-byte
                                   * encoding of small numeric nodeids) */
    UA_NODEIDTYPE_STRING     = 3,
    UA_NODEIDTYPE_GUID       = 4,
    UA_NODEIDTYPE_BYTESTRING = 5
};

typedef struct UA_NodeId {
    UA_UInt16 namespaceIndex;
    enum UA_NodeIdType identifierType;
    union {
        UA_UInt32     numeric;
        UA_String     string;
        UA_Guid       guid;
        UA_ByteString byteString;
    } identifier;
} UA_NodeId;

extern const UA_NodeId UA_NODEID_NULL;

UA_NodeId
UA_NODEID_NUMERIC(UA_UInt16 nsIndex, UA_UInt32 identifier);

UA_NodeId
UA_NODEID_STRING(UA_UInt16 nsIndex, char *chars);

UA_NodeId
UA_NODEID_STRING_ALLOC(UA_UInt16 nsIndex, const char *chars);

UA_NodeId
UA_NODEID_GUID(UA_UInt16 nsIndex, UA_Guid guid);

UA_NodeId
UA_NODEID_BYTESTRING(UA_UInt16 nsIndex, char *chars);

UA_NodeId
UA_NODEID_BYTESTRING_ALLOC(UA_UInt16 nsIndex,
                           const char *chars);

/* Shorthand for standard-defined NodeIds in Namespace 0.
 * See the generated nodeids.h for the full list. */
#define UA_NS0ID(ID) UA_NODEID_NUMERIC(0, UA_NS0ID_##ID)

UA_Boolean UA_NodeId_isNull(const UA_NodeId *p);
```

(continues on next page)

(continued from previous page)

```
/* Print the NodeId in the human-readable format defined in Part 6.
 *
 * Examples:
 *   UA_NODEID("i=13")
 *   UA_NODEID("ns=10;i=1")
 *   UA_NODEID("ns=10;s=Hello:World")
 *   UA_NODEID("g=09087e75-8e5e-499b-954f-f2a9603db28a")
 *   UA_NODEID("ns=1;b=b3BlbjYyNTQxIQ==") // base64
 *
 * The method can either use a pre-allocated string buffer or allocates memory
 * internally if called with an empty output string. */
UA_StatusCode
UA_NodeId_print(const UA_NodeId *id, UA_String *output);

/* Extended NodeId printing. If nsMapping argument is non-NULL, then the
 * NamespaceIndex is translated to the NamespaceUri. If that is not successful,
 * the numerical NamespaceIndex is used instead. See the section on
 * :ref:`percent-escaping` how NamespaceUris containing semicolons (and
 * whitespace) are encoded.
 *
 * Examples:
 *   nsu=http://widgets.com/schemas/hello;s=Hello World
 */
UA_StatusCode
UA_NodeId_printEx(const UA_NodeId *id, UA_String *output,
                  const UA_NamespaceMapping *nsMapping);

/* Parse the human-readable NodeId format. Attention! String and
 * ByteString NodeIds have their identifier malloc'ed and need to be
 * cleaned up. */
UA_StatusCode
UA_NodeId_parse(UA_NodeId *id, const UA_String str);

/* Extended parsing that uses the provided namespace mapping to find the
 * NamespaceIndex for a provided NamespaceUri.
 *
 * If the NodeId uses an unknown NamespaceUri, then a String-NodeId is returned
 * that uses NamespaceIndex 0 and the full original encoding for the string
 * part.
 *
 * Example:
 *   nsu=my_uri;i=5 => s="nsu=my_uri;i=5" (The quotation marks are for
 *       illustration purposes and not actually included)
 */
UA_StatusCode
UA_NodeId_parseEx(UA_NodeId *id, const UA_String str,
                  const UA_NamespaceMapping *nsMapping);

/* Shorthand, returns UA_NODEID_NULL when parsing fails */
```

(continues on next page)

(continued from previous page)

```
UA_NodeId UA_NODEID(const char *chars);

/* Total ordering of NodeId */
UA_Order
UA_NodeId_order(const UA_NodeId *n1, const UA_NodeId *n2);

/* Returns a non-cryptographic hash for NodeId */
UA_UInt32 UA_NodeId_hash(const UA_NodeId *n);
```

4.1.19 ExpandedNodeId

A NodeId that allows the namespace URI to be specified instead of an index.

```
typedef struct UA_ExpandedNodeId {
    UA_NodeId nodeId;
    UA_String namespaceUri;
    UA_UInt32 serverIndex;
} UA_ExpandedNodeId;

extern const UA_ExpandedNodeId UA_EXPANDEDNODEID_NULL;

UA_ExpandedNodeId
UA_EXPANDEDNODEID_NUMERIC(UA_UInt16 nsIndex, UA_UInt32 identifier);

UA_ExpandedNodeId
UA_EXPANDEDNODEID_STRING(UA_UInt16 nsIndex, char *chars);

UA_ExpandedNodeId
UA_EXPANDEDNODEID_STRING_ALLOC(UA_UInt16 nsIndex, const char *chars);

UA_ExpandedNodeId
UA_EXPANDEDNODEID_STRING_GUID(UA_UInt16 nsIndex, UA_Guid guid);

UA_ExpandedNodeId
UA_EXPANDEDNODEID_BYTESTRING(UA_UInt16 nsIndex, char *chars);

UA_ExpandedNodeId
UA_EXPANDEDNODEID_BYTESTRING_ALLOC(UA_UInt16 nsIndex, const char␣
↪*chars);

UA_ExpandedNodeId UA_EXPANDEDNODEID_NODEID(UA_NodeId nodeId);
#define UA_NODEID2EXPANDEDNODEID(n) UA_EXPANDEDNODEID_NODEID(n)

/* Shorthand for standard-defined NodeIds in Namespace 0.
 * See the generated nodeids.h for the full list. */
#define UA_NS0EXID(ID) UA_EXPANDEDNODEID_NUMERIC(0, UA_NS0ID_##ID)

/* Print the ExpandedNodeId in the humand-readable format defined in Part 6,
 * 5.3.1.11:
```

(continues on next page)

(continued from previous page)

```
 *
 *   svr=<serverindex>;ns=<namespaceindex>;<type>=<value>
 *     or
 *   svr=<serverindex>;nsu=<uri>;<type>=<value>
 *
 * The definitions for svr, ns and nsu is omitted if zero / the empty string.
 *
 * The method can either use a pre-allocated string buffer or allocates memory
 * internally if called with an empty output string. */
UA_StatusCode
UA_ExpandedNodeId_print(const UA_ExpandedNodeId *id, UA_String *output);

/* Extended printing of ExpandedNodeId. It tries to map NamespaceIndex and
 * ServerIndex to a Uri using the provided mapping.
 *
 * Examples:
 *     svu=http://smith.com/west/factory;nsu=tag:acme.com,2023;i=1234
 */
UA_StatusCode
UA_ExpandedNodeId_printEx(const UA_ExpandedNodeId *id, UA_String *output,
                          const UA_NamespaceMapping *nsMapping,
                          size_t serverUrisSize, const UA_String *serverUris);

/* Parse the human-readable NodeId format. Attention! String and
 * ByteString NodeIds have their identifier malloc'ed and need to be
 * cleaned up. */
UA_StatusCode
UA_ExpandedNodeId_parse(UA_ExpandedNodeId *id, const UA_String str);

UA_StatusCode
UA_ExpandedNodeId_parseEx(UA_ExpandedNodeId *id, const UA_String str,
                          const UA_NamespaceMapping *nsMapping,
                          size_t serverUrisSize, const UA_String *serverUris);

/* Shorthand, returns UA_EXPANDEDNODEID_NULL when parsing fails */
UA_ExpandedNodeId
UA_EXPANDEDNODEID(const char *chars);

/* Does the ExpandedNodeId point to a local node? That is, are namespaceUri and
 * serverIndex empty? */
UA_Boolean
UA_ExpandedNodeId_isLocal(const UA_ExpandedNodeId *n);

/* Total ordering of ExpandedNodeId */
UA_Order
UA_ExpandedNodeId_order(const UA_ExpandedNodeId *n1,
                        const UA_ExpandedNodeId *n2);

/* Returns a non-cryptographic hash for ExpandedNodeId. The hash of an
```

(continues on next page)

(continued from previous page)

```
 * ExpandedNodeId is identical to the hash of the embedded (simple) NodeId if
 * the ServerIndex is zero and no NamespaceUri is set. */
UA_UInt32
UA_ExpandedNodeId_hash(const UA_ExpandedNodeId *n);
```

4.1.20 QualifiedName

A name qualified by a namespace.

```
typedef struct UA_QualifiedName {
    UA_UInt16 namespaceIndex;
    UA_String name;
} UA_QualifiedName;

UA_QualifiedName
UA_QUALIFIEDNAME(UA_UInt16 nsIndex, char *chars);

UA_QualifiedName
UA_QUALIFIEDNAME_ALLOC(UA_UInt16 nsIndex, const char *chars);

UA_Boolean
UA_QualifiedName_isNull(const UA_QualifiedName *q);

/* Returns a non-cryptographic hash for QualifiedName */
UA_UInt32
UA_QualifiedName_hash(const UA_QualifiedName *q);

/* Print the human-readable QualifiedName format. QualifiedNames can be printed
 * with either the integer NamespaceIndex or using the NamespaceUri.
 * The Namespace 0 is always omitted.
 *
 * The extended printing tries to translate the NamespaceIndex to the
 * NamespaceUri from the mapping table. When the mapping fails, the integer
 * NamespaceIndex from is used.
 *
 * Examples:
 *    Namespace Zero: HelloWorld
 *    NamespaceIndex Form: 3:HelloWorld
 *    NamespaceUri Form: nsu=http://widgets.com/schemas/hello;HelloWorld
 *
 * The method can either use a pre-allocated string buffer or allocates memory
 * internally if called with an empty output string. */
UA_StatusCode
UA_QualifiedName_print(const UA_QualifiedName *qn, UA_String *output);

UA_StatusCode
UA_QualifiedName_printEx(const UA_QualifiedName *qn, UA_String *output,
                         const UA_NamespaceMapping *nsMapping);
```

(continues on next page)

(continued from previous page)

```
/* Parse the human-readable QualifiedName format.
 *
 * The extended parsing tries to translate the NamespaceIndex to a NamespaceUri
 * from the mapping table. When the mapping fails, the name component gets the
 * entire string. */
UA_StatusCode
UA_QualifiedName_parse(UA_QualifiedName *qn, const UA_String str);

UA_StatusCode
UA_QualifiedName_parseEx(UA_QualifiedName *qn, const UA_String str,
                         const UA_NamespaceMapping *nsMapping);
```

4.1.21 LocalizedText

Human readable text with an optional locale identifier.

```
typedef struct UA_LocalizedText {
    UA_String locale;
    UA_String text;
} UA_LocalizedText;

UA_LocalizedText
UA_LOCALIZEDTEXT(char *locale, char *text);

UA_LocalizedText
UA_LOCALIZEDTEXT_ALLOC(const char *locale, const char *text);
```

4.1.22 NumericRange

NumericRanges are used to indicate subsets of a (multidimensional) array. They no official data type in the OPC UA standard and are transmitted only with a string encoding, such as "1:2,0:3,5". The colon separates min/max index and the comma separates dimensions. A single value indicates a range with a single element (min==max).

```
typedef struct UA_NumericRangeDimension {
    UA_UInt32 min;
    UA_UInt32 max;
} UA_NumericRangeDimension;

typedef struct UA_NumericRange {
    size_t dimensionsSize;
    UA_NumericRangeDimension *dimensions;
} UA_NumericRange;

UA_StatusCode
UA_NumericRange_parse(UA_NumericRange *range, const UA_String str);

/* Returns an empty NumericRange if parsing fails */
UA_NumericRange UA_NUMERICRANGE(const char *s);
```

4.1.23 Variant

Variants may contain values of any type together with a description of the content. See the section on *Generic Type Handling* on how types are described. The standard mandates that variants contain built-in data types only. If the value is not of a builtin type, it is wrapped into an *ExtensionObject*. open62541 hides this wrapping transparently in the encoding layer. If the data type is unknown to the receiver, the variant contains the original ExtensionObject in binary or XML encoding.

Variants may contain a scalar value or an array. For details on the handling of arrays, see the section on *Array handling*. Array variants can have an additional dimensionality (matrix, 3-tensor, ...) defined in an array of dimension lengths. The actual values are kept in an array of dimensions one. For users who work with higher-dimensions arrays directly, keep in mind that dimensions of higher rank are serialized first (the highest rank dimension has stride 1 and elements follow each other directly). Usually it is simplest to interact with higher-dimensional arrays via `UA_NumericRange` descriptions (see *Array handling*).

To differentiate between scalar / array variants, the following definition is used. `UA_Variant_isScalar` provides simplified access to these checks.

- `arrayLength == 0 && data == NULL`: undefined array of length -1
- `arrayLength == 0 && data == UA_EMPTY_ARRAY_SENTINEL`: array of length 0
- `arrayLength == 0 && data > UA_EMPTY_ARRAY_SENTINEL`: scalar value
- `arrayLength > 0`: array of the given length

Variants can also be *empty*. Then, the pointer to the type description is `NULL`.

```
/* Forward declaration. See the section on Generic Type Handling */
struct UA_DataType;
typedef struct UA_DataType UA_DataType;

#define UA_EMPTY_ARRAY_SENTINEL ((void*)0x01)

typedef enum {
    UA_VARIANT_DATA,         /* The data has the same lifecycle as the variant */
    UA_VARIANT_DATA_NODELETE /* The data is "borrowed" by the variant and is
                              * not deleted when the variant is cleared up.
                              * The array dimensions also borrowed. */
} UA_VariantStorageType;

typedef struct UA_Variant {
    const UA_DataType *type;      /* The data type description */
    UA_VariantStorageType storageType;
    size_t arrayLength;           /* The number of elements in the data array */
    void *data;                   /* Points to the scalar or array data */
    size_t arrayDimensionsSize;   /* The number of dimensions */
    UA_UInt32 *arrayDimensions;   /* The length of each dimension */
} UA_Variant;

/* Returns true if the variant has no value defined (contains neither an array
 * nor a scalar value) */
UA_Boolean UA_Variant_isEmpty(const UA_Variant *v);
#define UA_Variant_isNull(v) UA_Variant_isEmpty(v)
```

(continues on next page)

(continued from previous page)

```
/* Returns true if the variant contains a scalar value */
UA_Boolean
UA_Variant_isScalar(const UA_Variant *v);

/* Returns true if the variant contains a scalar value of the given type */
UA_Boolean
UA_Variant_hasScalarType(const UA_Variant *v, const UA_DataType *type);

/* Returns true if the variant contains an array */
UA_Boolean
UA_Variant_isArray(const UA_Variant *v);

/* Returns true if the variant contains an array of the given type */
UA_Boolean
UA_Variant_hasArrayType(const UA_Variant *v, const UA_DataType *type);

/* Set the variant to a scalar value that already resides in memory. The value
 * will be cleared together with the variant. */
void
UA_Variant_setScalar(UA_Variant *v, void *value, const UA_DataType *type);

/* Set the variant to a deep-copy of the provided scalar value */
UA_StatusCode
UA_Variant_setScalarCopy(UA_Variant *v, const void *p, const UA_DataType *type);

/* Set the variant to an existing array value. The array is cleared together
 * with the variant. With arraySize zero, the variant is an empty array (with
 * UA_EMPTY_ARRAY_SENTINEL as data) and the array is not taken over. A NULL
 * array with arraySize zero is a null array. */
void
UA_Variant_setArray(UA_Variant *v, void *array, size_t arraySize,
                    const UA_DataType *type);

/* Set the variant to a deep-copy of the provided array */
UA_StatusCode
UA_Variant_setArrayCopy(UA_Variant *v, const void *array, size_t arraySize,
                        const UA_DataType *type);

/* Copy the variant, but use only a subset of the (multidimensional) array into
 * a variant. Returns an error code if the variant is not an array or if the
 * indicated range does not fit. */
UA_StatusCode
UA_Variant_copyRange(const UA_Variant *src, UA_Variant *dst,
                     const UA_NumericRange range);

/* Insert a range of data into an existing variant. The data array cannot be
 * reused afterwards if it contains types without a fixed size (e.g. strings)
 * since the members are moved into the variant and take on its lifecycle. */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode
UA_Variant_setRange(UA_Variant *v, void *array,
                    size_t arraySize, const UA_NumericRange range);

/* Deep-copy a range of data into the variant */
UA_StatusCode
UA_Variant_setRangeCopy(UA_Variant *v, const void *array,
                        size_t arraySize, const UA_NumericRange range);
```

4.1.24 ExtensionObject

ExtensionObjects may contain scalars of any data type. Even those that are unknown to the receiver. See the section on *Generic Type Handling* on how types are described. If the received data type is unknown, the encoded string and target NodeId is stored instead of the decoded value.

```
typedef enum {
    UA_EXTENSIONOBJECT_ENCODED_NOBODY     = 0,
    UA_EXTENSIONOBJECT_ENCODED_BYTESTRING = 1,
    UA_EXTENSIONOBJECT_ENCODED_XML        = 2,
    UA_EXTENSIONOBJECT_DECODED            = 3,
    UA_EXTENSIONOBJECT_DECODED_NODELETE   = 4, /* Don't delete the␣
↪content
                                                  together with the
                                                  ExtensionObject */
    UA_EXTENSIONOBJECT_ENCODED_JSON       = 5  /* Internal JSON␣
↪representation
                                                 for an unknown datatype */
} UA_ExtensionObjectEncoding;

typedef struct UA_ExtensionObject {
    UA_ExtensionObjectEncoding encoding;
    union {
        struct {
            UA_NodeId typeId;   /* The nodeid of the datatype */
            UA_ByteString body; /* The bytestring of the encoded data */
        } encoded;
        struct {
            const UA_DataType *type;
            void *data;
        } decoded;
    } content;
} UA_ExtensionObject;

/* Initialize the ExtensionObject and set the "decoded" value to the given
 * pointer. The value will be deleted when the ExtensionObject is cleared. */
void
UA_ExtensionObject_setValue(UA_ExtensionObject *eo, void *p,
                            const UA_DataType *type);
```

(continues on next page)

(continued from previous page)

```
/* Initialize the ExtensionObject and set the "decoded" value to the given
 * pointer. The value will *not* be deleted when the ExtensionObject is
 * cleared. */
void
UA_ExtensionObject_setValueNoDelete(UA_ExtensionObject *eo, void *p,
                                    const UA_DataType *type);

/* Initialize the ExtensionObject and set the "decoded" value to a fresh copy of
 * the given value pointer. The value will be deleted when the ExtensionObject
 * is cleared. */
UA_StatusCode
UA_ExtensionObject_setValueCopy(UA_ExtensionObject *eo, void *p,
                                const UA_DataType *type);

/* Returns true if the ExtensionObject contains a decoded value of the type */
UA_Boolean
UA_ExtensionObject_hasDecodedType(const UA_ExtensionObject *eo,
                                  const UA_DataType *type);
```

4.1.25 DataValue

A data value with an associated status code and timestamps.

```
typedef struct UA_DataValue {
    UA_Variant     value;
    UA_DateTime    sourceTimestamp;
    UA_DateTime    serverTimestamp;
    UA_UInt16      sourcePicoseconds;
    UA_UInt16      serverPicoseconds;
    UA_StatusCode status;
    UA_Boolean     hasValue            : 1;
    UA_Boolean     hasStatus           : 1;
    UA_Boolean     hasSourceTimestamp  : 1;
    UA_Boolean     hasServerTimestamp  : 1;
    UA_Boolean     hasSourcePicoseconds : 1;
    UA_Boolean     hasServerPicoseconds : 1;
} UA_DataValue;

/* Copy the DataValue, but use only a subset of the (multidimensional) array of
 * of the variant of the source DataValue. Returns an error code if the variant
 * of the DataValue is not an array or if the indicated range does not fit. */
UA_StatusCode
UA_DataValue_copyRange(const UA_DataValue *src, UA_DataValue *dst,
                       const UA_NumericRange range);
#define UA_DataValue_copyVariantRange(s,d,r) UA_DataValue_copyRange(s,d,r)
```

4.1.26 DiagnosticInfo

A structure that contains detailed error and diagnostic information associated with a StatusCode.

```
typedef struct UA_DiagnosticInfo {
    UA_Boolean    hasSymbolicId          : 1;
    UA_Boolean    hasNamespaceUri        : 1;
    UA_Boolean    hasLocalizedText       : 1;
    UA_Boolean    hasLocale              : 1;
    UA_Boolean    hasAdditionalInfo      : 1;
    UA_Boolean    hasInnerStatusCode     : 1;
    UA_Boolean    hasInnerDiagnosticInfo : 1;
    UA_Int32      symbolicId;
    UA_Int32      namespaceUri;
    UA_Int32      localizedText;
    UA_Int32      locale;
    UA_String     additionalInfo;
    UA_StatusCode innerStatusCode;
    struct UA_DiagnosticInfo *innerDiagnosticInfo;
} UA_DiagnosticInfo;
```

4.2 Generic Type Handling

All information about a (builtin/structured) data type is stored in a `UA_DataType`. The array `UA_TYPES` contains the description of all standard-defined types. This type description is used for the following generic operations that work on all types:

- `void T_init(T *ptr)`: Initialize the data type. This is synonymous with zeroing out the memory, i.e. `memset(ptr, 0, sizeof(T))`.
- `T* T_new()`: Allocate and return the memory for the data type. The value is already initialized.
- `UA_StatusCode T_copy(const T *src, T *dst)`: Copy the content of the data type. Returns `UA_STATUSCODE_GOOD` or `UA_STATUSCODE_BADOUTOFMEMORY`.
- `void T_clear(T *ptr)`: Delete the dynamically allocated content of the data type and perform a `T_init` to reset the type.
- `void T_delete(T *ptr)`: Delete the content of the data type and the memory for the data type itself.
- `void T_equal(T *p1, T *p2)`: Compare whether `p1` and `p2` have identical content. You can use `UA_order` if an absolute ordering is required.

Specializations, such as `UA_Int32_new()` are derived from the generic type operations as static inline functions.

```
/* UA_DataTypeMember describes the fields of structures/unions/etc. In addition
 * it can describe the possible values of enums. Enum members are optional; if
 * present, the memberType value is cast to an integer to represent the enum
 * value. */
typedef struct UA_DataTypeMember {
#ifdef UA_ENABLE_TYPEDESCRIPTION
    const char *memberName;       /* Human-readable member name */
```

(continues on next page)

(continued from previous page)

```
#endif
    const UA_DataType *memberType;/* The member data type description */
    UA_Byte padding    : 6;       /* How much padding is there before this
                                     member element? For arrays this is the
                                     padding before the size_t length member.
                                     (No padding between size_t and the
                                     following ptr.) For unions, the padding
                                     includes the size of the switchfield (the
                                     offset from the start of the union
                                     type). */
    UA_Byte isArray    : 1;       /* The member is an array */
    UA_Byte isOptional : 1;       /* The member is an optional field */
} UA_DataTypeMember;

/* The DataType "kind" is an internal type classification. It is used to
 * dispatch handling to the correct routines. */
#define UA_DATATYPEKINDS 31
typedef enum {
    UA_DATATYPEKIND_BOOLEAN = 0,
    UA_DATATYPEKIND_SBYTE = 1,
    UA_DATATYPEKIND_BYTE = 2,
    UA_DATATYPEKIND_INT16 = 3,
    UA_DATATYPEKIND_UINT16 = 4,
    UA_DATATYPEKIND_INT32 = 5,
    UA_DATATYPEKIND_UINT32 = 6,
    UA_DATATYPEKIND_INT64 = 7,
    UA_DATATYPEKIND_UINT64 = 8,
    UA_DATATYPEKIND_FLOAT = 9,
    UA_DATATYPEKIND_DOUBLE = 10,
    UA_DATATYPEKIND_STRING = 11,
    UA_DATATYPEKIND_DATETIME = 12,
    UA_DATATYPEKIND_GUID = 13,
    UA_DATATYPEKIND_BYTESTRING = 14,
    UA_DATATYPEKIND_XMLELEMENT = 15,
    UA_DATATYPEKIND_NODEID = 16,
    UA_DATATYPEKIND_EXPANDEDNODEID = 17,
    UA_DATATYPEKIND_STATUSCODE = 18,
    UA_DATATYPEKIND_QUALIFIEDNAME = 19,
    UA_DATATYPEKIND_LOCALIZEDTEXT = 20,
    UA_DATATYPEKIND_EXTENSIONOBJECT = 21,
    UA_DATATYPEKIND_DATAVALUE = 22,
    UA_DATATYPEKIND_VARIANT = 23,
    UA_DATATYPEKIND_DIAGNOSTICINFO = 24,
    UA_DATATYPEKIND_DECIMAL = 25,
    UA_DATATYPEKIND_ENUM = 26,
    UA_DATATYPEKIND_STRUCTURE = 27,
    UA_DATATYPEKIND_OPTSTRUCT = 28, /* struct with optional fields */
    UA_DATATYPEKIND_UNION = 29,
    UA_DATATYPEKIND_BITFIELDCLUSTER = 30 /* bitfields + padding */
```

(continues on next page)

(continued from previous page)

```
} UA_DataTypeKind;

#define UA_DATATYPE_MEMBERS_MAX 511

struct UA_DataType {
#ifdef UA_ENABLE_TYPEDESCRIPTION
    const char *typeName;
#endif
    UA_NodeId typeId;           /* The nodeid of the type */
    UA_NodeId binaryEncodingId; /* NodeId of datatype when encoded as binary */
    UA_NodeId xmlEncodingId;    /* NodeId of datatype when encoded as XML */
    UA_UInt32 memSize     : 16; /* Size of the struct in memory */
    UA_UInt32 typeKind    : 5;  /* Dispatch index for the handling routines */
    UA_UInt32 pointerFree : 1;  /* The type (and its members) contains no
                                 * pointers that need to be freed */
    UA_UInt32 overlayable : 1;  /* The type has the identical memory layout
                                 * in memory and on the binary stream. */
    UA_UInt32 membersSize : 9;  /* How many members does the type have? */
    const UA_DataTypeMember *members;
};

/* Clean up type definition with heap-allocated data */
void
UA_DataType_clear(UA_DataType *type);

UA_StatusCode
UA_DataType_copy(const UA_DataType *t1, UA_DataType *t2);

/* Datatype arrays with custom type definitions can be added in a linked list to
 * the client or server configuration. */
typedef struct UA_DataTypeArray {
    struct UA_DataTypeArray *next;
    size_t typesSize;
    const UA_DataType *types;
    UA_Boolean cleanup; /* Free the array structure and its content when the
                         * client or server configuration containing it is
                         * cleaned up */
} UA_DataTypeArray;

/* Returns the offset and type of a structure member. The return value is false
 * if the member was not found.
 *
 * If the member is an array, the offset points to the (size_t) length field.
 * (The array pointer comes after the length field without any padding.) */
#ifdef UA_ENABLE_TYPEDESCRIPTION
UA_Boolean
UA_DataType_getStructMember(const UA_DataType *type,
                            const char *memberName,
                            size_t *outOffset,
```

(continues on next page)

(continued from previous page)

```
                        const UA_DataType **outMemberType,
                        UA_Boolean *outIsArray);
#endif

/* Test if the data type is a numeric builtin data type (via the typeKind field
 * of UA_DataType). This includes integers and floating point numbers. Not
 * included are Boolean, DateTime, StatusCode and Enums. */
UA_Boolean
UA_DataType_isNumeric(const UA_DataType *type);
```

Builtin data types can be accessed as UA_TYPES[UA_TYPES_XXX], where XXX is the name of the data type. If only the NodeId of a type is known, use the following method to retrieve the data type description.

```
/* Returns the data type description for the type's identifier or NULL if no
 * matching data type was found. */
const UA_DataType *
UA_findDataType(const UA_NodeId *typeId);

/* Add custom data types to the search scope of UA_findDataType. */
const UA_DataType *
UA_findDataTypeWithCustom(const UA_NodeId *typeId,
                          const UA_DataTypeArray *customTypes);

/* Datatypes need to have unique name */
#ifdef UA_ENABLE_TYPEDESCRIPTION
const UA_DataType *
UA_findDataTypeByName(const UA_QualifiedName *qn);
#endif
```

The following functions are used for generic handling of data types.

```
/* Allocates and initializes a variable of type dataType */
void * UA_new(const UA_DataType *type);

/* Initializes a variable to default null values */
void UA_init(void *p, const UA_DataType *type);

/* Copies the content of two variables. The pointer is _init'ed internally. If
 * copying fails, then dst is cleared internally to prevent memory leaks. */
UA_StatusCode
UA_copy(const void *src, void *dst, const UA_DataType *type);

/* Deletes the dynamically allocated content of a value (e.g. deallocates all
 * arrays in the variable). At last the entire value is _init'ed. */
void UA_clear(void *p, const UA_DataType *type);

/* Calls UA_clear and then UA_free on the memory */
void UA_delete(void *p, const UA_DataType *type);
```

(continues on next page)

(continued from previous page)

```
/* Pretty-print the value from the datatype. The output is pretty-printed JSON5.
 * Note that this format is non-standard and should not be sent over the
 * network. It can however be read by our own JSON decoding.
 *
 * If the memory for string is already allocated, we try to use the existing
 * string (the length is adjusted down). If the string is empty, memory is
 * allocated for it. */
#ifdef UA_ENABLE_JSON_ENCODING
UA_StatusCode
UA_print(const void *p, const UA_DataType *type, UA_String *output);
#endif

/* Compare two values and return their order.
 *
 * For numerical types (including StatusCodes and Enums), their natural order is
 * used. NaN is the "smallest" value for floating point values. Different bit
 * representations of NaN are considered identical.
 *
 * All other types have *some* absolute ordering so that a < b, b < c -> a < c.
 *
 * The ordering of arrays (also strings) is in "shortlex": A shorter array is
 * always smaller than a longer array. Otherwise the first different element
 * defines the order.
 *
 * When members of different types are permitted (in Variants and
 * ExtensionObjects), the memory address in the "UA_DataType*" pointer
 * determines which variable is smaller. */
UA_Order
UA_order(const void *p1, const void *p2, const UA_DataType *type);

/* Compare if two values are identical */
UA_Boolean
UA_equal(const void *p1, const void *p2, const UA_DataType *type);
```

4.3 Namespace Mapping

Every *NodeId* references a namespace index. Actually the namespace is identified by its URI. The namespace-array of the server maps the URI to the namespace index in the array. Namespace zero always has the URI `http://opcfoundation.org/UA/`. Namespace one has the application URI of the server. All namespaces beyond get a custom assignment.

In order to have predictable NodeIds, a client might predefined its own namespace array that is different from the server's. When a NodeId is decoded from a network message (binary or JSON), a mapping-table can be used to automatically translate between the remote and local namespace index. The mapping is typically done by the client who can generate the mapping table after reading the namespace-array of the server. The reverse mapping is done in the encoding if the mapping table is set in the options.

The mapping table also contains the full URI names. It is also used to translate the NamespaceUri field of an ExpandedNodeId into the namespace index of the NodeId embedded in the Expand-

edNodeId.

```
struct UA_NamespaceMapping {
    /* Namespaces with their local index */
    UA_String *namespaceUris;
    size_t namespaceUrisSize;

    /* Map from local to remote indices */
    UA_UInt16 *local2remote;
    size_t local2remoteSize;

    /* Map from remote to local indices */
    UA_UInt16 *remote2local;
    size_t remote2localSize;
};

/* If the index is unknown, returns (UINT16_MAX - index) */
UA_UInt16
UA_NamespaceMapping_local2Remote(const UA_NamespaceMapping *nm,
                                 UA_UInt16 localIndex);

UA_UInt16
UA_NamespaceMapping_remote2Local(const UA_NamespaceMapping *nm,
                                 UA_UInt16 remoteIndex);

/* Returns an error if the namespace uri was not found.
 * The pointer to the index argument needs to be non-NULL. */
UA_StatusCode
UA_NamespaceMapping_uri2Index(const UA_NamespaceMapping *nm,
                              UA_String uri, UA_UInt16 *index);

/* Upon success, the uri string gets set. The string is not copied and must not
 * outlive the namespace mapping structure. */
UA_StatusCode
UA_NamespaceMapping_index2Uri(const UA_NamespaceMapping *nm,
                              UA_UInt16 index, UA_String *uri);

void
UA_NamespaceMapping_clear(UA_NamespaceMapping *nm);

void
UA_NamespaceMapping_delete(UA_NamespaceMapping *nm);
```

4.4 Binary Encoding/Decoding

Encoding and decoding routines for the binary format. For the binary decoding additional data types can be forwarded.

```
typedef struct UA_EncodeBinaryOptions {
    /* Mapping of namespace indices in NodeIds and of NamespaceUris in
```

(continues on next page)

(continued from previous page)

```
     * ExpandedNodeIds. */
    UA_NamespaceMapping *namespaceMapping;
} UA_EncodeBinaryOptions;

/* Returns the number of bytes the value p takes in binary encoding. Returns
 * zero if an error occurs. */
size_t
UA_calcSizeBinary(const void *p, const UA_DataType *type,
                  UA_EncodeBinaryOptions *options);

/* Encodes a data-structure in the binary format. If outBuf has a length of
 * zero, a buffer of the required size is allocated. Otherwise, encoding into
 * the existing outBuf is attempted (and may fail if the buffer is too
 * small). */
UA_StatusCode
UA_encodeBinary(const void *p, const UA_DataType *type,
                UA_ByteString *outBuf, UA_EncodeBinaryOptions *options);

/* The structure with the decoding options may be extended in the future.
 * Zero-out the entire structure initially to ensure code-compatibility when
 * more fields are added in a later release. */
typedef struct UA_DecodeBinaryOptions {
    /* Begin of a linked list with custom datatype definitions */
    const UA_DataTypeArray *customTypes;

    /* Mapping of namespace indices in NodeIds and of NamespaceUris in
     * ExpandedNodeIds. */
    UA_NamespaceMapping *namespaceMapping;

    /* Override calloc for arena-based memory allocation. Note that allocated
     * memory is not freed if decoding fails afterwards. */
    void *callocContext;
    void * (*calloc)(void *callocContext, size_t nelem, size_t elsize);

    size_t decodedLength; /* After each successful decoding, this contains the
                           * number of decoded bytes. */
} UA_DecodeBinaryOptions;

/* Decodes a data structure from the input buffer in the binary format. It is
 * assumed that `p` points to valid memory (not necessarily zeroed out). The
 * options can be NULL and will be disregarded in that case. */
UA_StatusCode
UA_decodeBinary(const UA_ByteString *inBuf,
                void *p, const UA_DataType *type,
                UA_DecodeBinaryOptions *options);
```

4.5 JSON En/Decoding

The JSON encoding follows OPC UA 1.05 and represents NodeIds and ExpandedNodeIds with their standard string syntax (for example "ns=1;i=42"). The decoder accepts Compact and Verbose values without an a priori mode setting. The encoding is detected independently for nested values, so mixed input is supported. The decoder further allows the following extensions:

- The strict JSON format is relaxed to also allow the JSON5 extensions (https://json5.org/). This allows for more human-readable encoding and adds convenience features such as trailing commas in arrays and comments within JSON documents.
- Int64/UInt64 don't necessarily have to be wrapped into a string.

These extensions are not intended to be used for the OPC UA protocol on the network. They were rather added to allow more convenient configuration file formats that also include data in the OPC UA type system.

```
#ifdef UA_ENABLE_JSON_ENCODING

typedef struct UA_EncodeJsonOptions {
    /* Mapping of namespace indices in NodeIds and of NamespaceUris in
     * ExpandedNodeIds. */
    UA_NamespaceMapping *namespaceMapping;

    const UA_String *serverUris;
    size_t serverUrisSize;

    UA_Boolean prettyPrint;    /* Add newlines and spaces for legibility */

    /* Enabling unquotedKeys leads to non-standard compatible JSON5
     * encoding! Use it for pretty-printing, but not for sending messages over
     * the network. (Our own decoding can still parse it.) */

    UA_Boolean unquotedKeys;  /* Don't print quotes around object element keys */

    /* Use the OPC UA CompactEncoding. The zero-initialized default is the
     * VerboseEncoding. */
    UA_Boolean useCompactEncoding;
} UA_EncodeJsonOptions;

/* Returns the number of bytes the value src takes in JSON encoding. Returns
 * zero if an error occurs. */
size_t
UA_calcSizeJson(const void *src, const UA_DataType *type,
                const UA_EncodeJsonOptions *options);

/* Encodes the scalar value described by type to JSON encoding. If the outBuf
 * already contains memory, this is used (if sufficient) and outBuf->length is
 * adjusted down. Otherwise sufficient memory is allocated. The options can be
 * NULL. */
UA_StatusCode
UA_encodeJson(const void *src, const UA_DataType *type, UA_ByteString *outBuf,
```

(continues on next page)

(continued from previous page)

```
            const UA_EncodeJsonOptions *options);

/* The structure with the decoding options may be extended in the future.
 * Zero-out the entire structure initially to ensure code-compatibility when
 * more fields are added in a later release. */
typedef struct UA_DecodeJsonOptions {
    /* Mapping of namespace indices in NodeIds and of NamespaceUris in
     * ExpandedNodeIds. */
    UA_NamespaceMapping *namespaceMapping;

    const UA_String *serverUris;
    size_t serverUrisSize;

    const UA_DataTypeArray *customTypes; /* Begin of a linked list with custom
                                          * datatype definitions */

    size_t *decodedLength; /* If non-NULL, the length of the decoded input is
                            * stored to the pointer. When this is set, decoding
                            * succeeds also if there is more content after the
                            * first JSON element in the input string. */
} UA_DecodeJsonOptions;

/* Decodes a scalar value described by type from JSON encoding. The dst value is
 * _init'ed initially. It gets cleared internally when an error occurs. The
 * options can be NULL. */
UA_StatusCode
UA_decodeJson(const UA_ByteString *src, void *dst, const UA_DataType *type,
              const UA_DecodeJsonOptions *options);

#endif /* UA_ENABLE_JSON_ENCODING */
```

4.6 XML En/Decoding

The XML decoding can parse the official encoding from the OPC UA specification.

These extensions are not intended to be used for the OPC UA protocol on the network. They were rather added to allow more convenient configuration file formats that also include data in the OPC UA type system.

```
#ifdef UA_ENABLE_XML_ENCODING

/* The structure with the encoding options may be extended in the future.
 * Zero-out the entire structure initially to ensure code-compatibility when
 * more fields are added in a later release. */
typedef struct UA_EncodeXmlOptions {
    UA_NamespaceMapping *namespaceMapping;
    const UA_String *serverUris;
    size_t serverUrisSize;
} UA_EncodeXmlOptions;
```

(continues on next page)

(continued from previous page)

```
/* Returns the number of bytes the value src takes in xml encoding. Returns
 * zero if an error occurs. */
size_t
UA_calcSizeXml(const void *src, const UA_DataType *type,
               const UA_EncodeXmlOptions *options);

/* Encodes the scalar value described by type to XML encoding. If the outBuf
 * already contains memory, this is used (if sufficient) and outBuf->length is
 * adjusted down. Otherwise sufficient memory is allocated. The options can be
 * NULL. */
UA_StatusCode
UA_encodeXml(const void *src, const UA_DataType *type, UA_ByteString *outBuf,
             const UA_EncodeXmlOptions *options);

/* The structure with the decoding options may be extended in the future.
 * Zero-out the entire structure initially to ensure code-compatibility when
 * more fields are added in a later release. */
typedef struct UA_DecodeXmlOptions {
    UA_Boolean unwrapped; /* The value xxx is not wrapped in an XML element - as
                           * in <Type>xxx</Type> */

    UA_NamespaceMapping *namespaceMapping;
    const UA_String *serverUris;
    size_t serverUrisSize;
    const UA_DataTypeArray *customTypes; /* Begin of a linked list with custom
                                          * datatype definitions */
} UA_DecodeXmlOptions;

/* Decodes a scalar value described by type from XML encoding. The dst value is
 * _init'ed initially. It gets cleared internally when an error occurs. The
 * options can be NULL. */
UA_StatusCode
UA_decodeXml(const UA_ByteString *src, void *dst, const UA_DataType *type,
             const UA_DecodeXmlOptions *options);

#endif /* UA_ENABLE_XML_ENCODING */
```

4.7 Array handling

In OPC UA, arrays can have a length of zero or more with the usual meaning. In addition, arrays can be undefined. Then, they don't even have a length. In the binary encoding, this is indicated by an array of length -1.

In open62541 however, we use size_t for array lengths. An undefined array has length 0 and the data pointer is NULL. An array of length 0 also has length 0 but a data pointer UA_EMPTY_ARRAY_SENTINEL.

```
/* Allocates and initializes an array of the given type */
```

(continues on next page)

(continued from previous page)

```
void *
UA_Array_new(size_t size, const UA_DataType *type);

/* Makes a deep-copy of an array. On success, the dst argument is set to point
 * to the allocated memory. */
UA_StatusCode
UA_Array_copy(const void *src, size_t size, void **dst,
              const UA_DataType *type);

/* Resizes (and reallocates) an array. The last entries are initialized to zero
 * if the array length is increased. If the array length is decreased, the last
 * entries are cleared if the size is decreased.
 *
 * The double-pointer to the array and the size-pointer are overwritten upon
 * success. The array remains untouched in case of an internal error. */
UA_StatusCode
UA_Array_resize(void **p, size_t *size, size_t newSize,
                const UA_DataType *type);

/* Append a scalar value at the end of the array. The content is moved (shallow
 * copy) and the original value location is _init'ed if appending is successful.
 * Otherwise similar to UA_Array_resize. */
UA_StatusCode
UA_Array_append(void **p, size_t *size, void *newElem,
                const UA_DataType *type);

/* Append a copy of the given element at the end of the array. The memory of the
 * newValue argument is not written. Otherwise similar to UA_Array_append. */
UA_StatusCode
UA_Array_appendCopy(void **p, size_t *size, const void *newElem,
                    const UA_DataType *type);

/* Deletes an array by calling _clear on the element and freeing the memory */
void
UA_Array_delete(void *p, size_t size, const UA_DataType *type);
```

4.8 Generated Data Type Definitions

The following standard-defined datatypes are auto-generated from XML files that are part of the OPC UA standard. All datatypes are built up from the 25 builtin-in datatypes from the *Data Types* section.

4.8.1 NamingRuleType

```
typedef enum {
    UA_NAMINGRULETYPE_MANDATORY = 1,
    UA_NAMINGRULETYPE_OPTIONAL = 2,
    UA_NAMINGRULETYPE_CONSTRAINT = 3
} UA_NamingRuleType;
```

4.8.2 Enumeration

```
typedef enum {
} UA_Enumeration;
```

4.8.3 ImageBMP

```
typedef UA_ByteString UA_ImageBMP;
```

4.8.4 ImageGIF

```
typedef UA_ByteString UA_ImageGIF;
```

4.8.5 ImageJPG

```
typedef UA_ByteString UA_ImageJPG;
```

4.8.6 ImagePNG

```
typedef UA_ByteString UA_ImagePNG;
```

4.8.7 AudioDataType

```
typedef UA_ByteString UA_AudioDataType;
```

4.8.8 UriString

```
typedef UA_String UA_UriString;
```

4.8.9 BitFieldMaskDataType

```
typedef UA_UInt64 UA_BitFieldMaskDataType;
```

4.8.10 SemanticVersionString

```
typedef UA_String UA_SemanticVersionString;
```

4.8.11 KeyValuePair

```
typedef struct {
    UA_QualifiedName key;
    UA_Variant value;
} UA_KeyValuePair;
```

4.8.12 AdditionalParametersType

```
typedef struct {
    size_t parametersSize;
    UA_KeyValuePair *parameters;
} UA_AdditionalParametersType;
```

4.8.13 EphemeralKeyType

```
typedef struct {
    UA_ByteString publicKey;
    UA_ByteString signature;
} UA_EphemeralKeyType;
```

4.8.14 RationalNumber

```
typedef struct {
    UA_Int32 numerator;
    UA_UInt32 denominator;
} UA_RationalNumber;
```

4.8.15 ThreeDVector

```
typedef struct {
    UA_Double x;
    UA_Double y;
    UA_Double z;
} UA_ThreeDVector;
```

4.8.16 ThreeDCartesianCoordinates

```
typedef struct {
    UA_Double x;
    UA_Double y;
    UA_Double z;
} UA_ThreeDCartesianCoordinates;
```

4.8.17 ThreeDOrientation

```
typedef struct {
    UA_Double a;
    UA_Double b;
    UA_Double c;
} UA_ThreeDOrientation;
```

4.8.18 ThreeDFrame

```
typedef struct {
    UA_ThreeDCartesianCoordinates cartesianCoordinates;
    UA_ThreeDOrientation orientation;
} UA_ThreeDFrame;
```

4.8.19 OpenFileMode

```
typedef enum {
    UA_OPENFILEMODE_READ = 1,
    UA_OPENFILEMODE_WRITE = 2,
    UA_OPENFILEMODE_ERASEEXISTING = 4,
    UA_OPENFILEMODE_APPEND = 8
} UA_OpenFileMode;
```

4.8.20 IdentityCriteriaType

```
typedef enum {
    UA_IDENTITYCRITERIATYPE_USERNAME = 1,
    UA_IDENTITYCRITERIATYPE_THUMBPRINT = 2,
    UA_IDENTITYCRITERIATYPE_ROLE = 3,
    UA_IDENTITYCRITERIATYPE_GROUPID = 4,
    UA_IDENTITYCRITERIATYPE_ANONYMOUS = 5,
    UA_IDENTITYCRITERIATYPE_AUTHENTICATEDUSER = 6,
    UA_IDENTITYCRITERIATYPE_APPLICATION = 7,
    UA_IDENTITYCRITERIATYPE_X509SUBJECT = 8
} UA_IdentityCriteriaType;
```

4.8.21 IdentityMappingRuleType

```
typedef struct {
    UA_IdentityCriteriaType criteriaType;
    UA_String criteria;
} UA_IdentityMappingRuleType;
```

4.8.22 CurrencyUnitType

```
typedef struct {
    UA_Int16 numericCode;
    UA_SByte exponent;
    UA_String alphabeticCode;
    UA_LocalizedText currency;
} UA_CurrencyUnitType;
```

4.8.23 TrustListMasks

```
typedef enum {
    UA_TRUSTLISTMASKS_NONE = 0,
    UA_TRUSTLISTMASKS_TRUSTEDCERTIFICATES = 1,
    UA_TRUSTLISTMASKS_TRUSTEDCRLS = 2,
    UA_TRUSTLISTMASKS_ISSUERCERTIFICATES = 4,
    UA_TRUSTLISTMASKS_ISSUERCRLS = 8,
    UA_TRUSTLISTMASKS_ALL = 15
} UA_TrustListMasks;
```

4.8.24 TrustListDataType

```
typedef struct {
    UA_UInt32 specifiedLists;
    size_t trustedCertificatesSize;
    UA_ByteString *trustedCertificates;
    size_t trustedCrlsSize;
    UA_ByteString *trustedCrls;
    size_t issuerCertificatesSize;
    UA_ByteString *issuerCertificates;
    size_t issuerCrlsSize;
    UA_ByteString *issuerCrls;
} UA_TrustListDataType;
```

4.8.25 DecimalDataType

```
typedef struct {
    UA_Int16 scale;
    UA_ByteString value;
} UA_DecimalDataType;
```

4.8.26 DataTypeDescription

```
typedef struct {
    UA_NodeId dataTypeId;
    UA_QualifiedName name;
} UA_DataTypeDescription;
```

4.8.27 SimpleTypeDescription

```
typedef struct {
    UA_NodeId dataTypeId;
    UA_QualifiedName name;
    UA_NodeId baseDataType;
    UA_Byte builtInType;
} UA_SimpleTypeDescription;
```

4.8.28 PortableQualifiedName

```
typedef struct {
    UA_String namespaceUri;
    UA_String name;
} UA_PortableQualifiedName;
```

4.8.29 PortableNodeId

```
typedef struct {
    UA_String namespaceUri;
    UA_NodeId identifier;
} UA_PortableNodeId;
```

4.8.30 UnsignedRationalNumber

```
typedef struct {
    UA_UInt32 numerator;
    UA_UInt32 denominator;
} UA_UnsignedRationalNumber;
```

4.8.31 PubSubState

```
typedef enum {
    UA_PUBSUBSTATE_DISABLED = 0,
    UA_PUBSUBSTATE_PAUSED = 1,
    UA_PUBSUBSTATE_OPERATIONAL = 2,
    UA_PUBSUBSTATE_ERROR = 3,
    UA_PUBSUBSTATE_PREOPERATIONAL = 4
} UA_PubSubState;
```

4.8.32 DataSetFieldFlags

```
typedef UA_UInt16 UA_DataSetFieldFlags;

#define UA_DATASETFIELDFLAGS_NONE 0
#define UA_DATASETFIELDFLAGS_PROMOTEDFIELD 1
```

4.8.33 ConfigurationVersionDataType

```
typedef struct {
    UA_UInt32 majorVersion;
    UA_UInt32 minorVersion;
} UA_ConfigurationVersionDataType;
```

4.8.34 PublishedVariableDataType

```
typedef struct {
    UA_NodeId publishedVariable;
    UA_UInt32 attributeId;
    UA_Double samplingIntervalHint;
    UA_UInt32 deadbandType;
    UA_Double deadbandValue;
    UA_String indexRange;
    UA_Variant substituteValue;
    size_t metaDataPropertiesSize;
    UA_QualifiedName *metaDataProperties;
} UA_PublishedVariableDataType;
```

4.8.35 PublishedDataItemsDataType

```
typedef struct {
    size_t publishedDataSize;
    UA_PublishedVariableDataType *publishedData;
} UA_PublishedDataItemsDataType;
```

4.8.36 PublishedDataSetCustomSourceDataType

```
typedef struct {
    UA_Boolean cyclicDataSet;
} UA_PublishedDataSetCustomSourceDataType;
```

4.8.37 DataSetFieldContentMask

```
typedef UA_UInt32 UA_DataSetFieldContentMask;

#define UA_DATASETFIELDCONTENTMASK_NONE 0
#define UA_DATASETFIELDCONTENTMASK_STATUSCODE 1
#define UA_DATASETFIELDCONTENTMASK_SOURCETIMESTAMP 2
#define UA_DATASETFIELDCONTENTMASK_SERVERTIMESTAMP 4
#define UA_DATASETFIELDCONTENTMASK_SOURCEPICOSECONDS 8
#define UA_DATASETFIELDCONTENTMASK_SERVERPICOSECONDS 16
#define UA_DATASETFIELDCONTENTMASK_RAWDATA 32
```

4.8.38 DataSetWriterDataType

```
typedef struct {
    UA_String name;
    UA_Boolean enabled;
    UA_UInt16 dataSetWriterId;
    UA_DataSetFieldContentMask dataSetFieldContentMask;
    UA_UInt32 keyFrameCount;
    UA_String dataSetName;
    size_t dataSetWriterPropertiesSize;
```

(continues on next page)

(continued from previous page)

```
    UA_KeyValuePair *dataSetWriterProperties;
    UA_ExtensionObject transportSettings;
    UA_ExtensionObject messageSettings;
} UA_DataSetWriterDataType;
```

4.8.39 NetworkAddressDataType

```
typedef struct {
    UA_String networkInterface;
} UA_NetworkAddressDataType;
```

4.8.40 NetworkAddressUrlDataType

```
typedef struct {
    UA_String networkInterface;
    UA_String url;
} UA_NetworkAddressUrlDataType;
```

4.8.41 OverrideValueHandling

```
typedef enum {
    UA_OVERRIDEVALUEHANDLING_DISABLED = 0,
    UA_OVERRIDEVALUEHANDLING_LASTUSABLEVALUE = 1,
    UA_OVERRIDEVALUEHANDLING_OVERRIDEVALUE = 2
} UA_OverrideValueHandling;
```

4.8.42 StandaloneSubscribedDataSetRefDataType

```
typedef struct {
    UA_String dataSetName;
} UA_StandaloneSubscribedDataSetRefDataType;
```

4.8.43 DataSetOrderingType

```
typedef enum {
    UA_DATASETORDERINGTYPE_UNDEFINED = 0,
    UA_DATASETORDERINGTYPE_ASCENDINGWRITERID = 1,
    UA_DATASETORDERINGTYPE_ASCENDINGWRITERIDSINGLE = 2
} UA_DataSetOrderingType;
```

4.8.44 UadpNetworkMessageContentMask

```
typedef UA_UInt32 UA_UadpNetworkMessageContentMask;

#define UA_UADPNETWORKMESSAGECONTENTMASK_NONE 0
#define UA_UADPNETWORKMESSAGECONTENTMASK_PUBLISHERID 1
#define UA_UADPNETWORKMESSAGECONTENTMASK_GROUPHEADER 2
```

(continues on next page)

(continued from previous page)

```
#define UA_UADPNETWORKMESSAGECONTENTMASK_WRITERGROUPID 4
#define UA_UADPNETWORKMESSAGECONTENTMASK_GROUPVERSION 8
#define UA_UADPNETWORKMESSAGECONTENTMASK_
↪NETWORKMESSAGENUMBER 16
#define UA_UADPNETWORKMESSAGECONTENTMASK_SEQUENCENUMBER 32
#define UA_UADPNETWORKMESSAGECONTENTMASK_PAYLOADHEADER 64
#define UA_UADPNETWORKMESSAGECONTENTMASK_TIMESTAMP 128
#define UA_UADPNETWORKMESSAGECONTENTMASK_PICOSECONDS 256
#define UA_UADPNETWORKMESSAGECONTENTMASK_DATASETCLASSID 512
#define UA_UADPNETWORKMESSAGECONTENTMASK_PROMOTEDFIELDS 1024
```

4.8.45 UadpWriterGroupMessageDataType

```
typedef struct {
    UA_UInt32 groupVersion;
    UA_DataSetOrderingType dataSetOrdering;
    UA_UadpNetworkMessageContentMask networkMessageContentMask;
    UA_Double samplingOffset;
    size_t publishingOffsetSize;
    UA_Double *publishingOffset;
} UA_UadpWriterGroupMessageDataType;
```

4.8.46 UadpDataSetMessageContentMask

```
typedef UA_UInt32 UA_UadpDataSetMessageContentMask;

#define UA_UADPDATASETMESSAGECONTENTMASK_NONE 0
#define UA_UADPDATASETMESSAGECONTENTMASK_TIMESTAMP 1
#define UA_UADPDATASETMESSAGECONTENTMASK_PICOSECONDS 2
#define UA_UADPDATASETMESSAGECONTENTMASK_STATUS 4
#define UA_UADPDATASETMESSAGECONTENTMASK_MAJORVERSION 8
#define UA_UADPDATASETMESSAGECONTENTMASK_MINORVERSION 16
#define UA_UADPDATASETMESSAGECONTENTMASK_SEQUENCENUMBER 32
```

4.8.47 UadpDataSetWriterMessageDataType

```
typedef struct {
    UA_UadpDataSetMessageContentMask dataSetMessageContentMask;
    UA_UInt16 configuredSize;
    UA_UInt16 networkMessageNumber;
    UA_UInt16 dataSetOffset;
} UA_UadpDataSetWriterMessageDataType;
```

4.8.48 UadpDataSetReaderMessageDataType

```
typedef struct {
    UA_UInt32 groupVersion;
    UA_UInt16 networkMessageNumber;
```

(continues on next page)

(continued from previous page)

```
    UA_UInt16 dataSetOffset;
    UA_Guid dataSetClassId;
    UA_UadpNetworkMessageContentMask networkMessageContentMask;
    UA_UadpDataSetMessageContentMask dataSetMessageContentMask;
    UA_Double publishingInterval;
    UA_Double receiveOffset;
    UA_Double processingOffset;
} UA_UadpDataSetReaderMessageDataType;
```

4.8.49 JsonNetworkMessageContentMask

```
typedef UA_UInt32 UA_JsonNetworkMessageContentMask;

#define UA_JSONNETWORKMESSAGECONTENTMASK_NONE 0
#define UA_JSONNETWORKMESSAGECONTENTMASK_
↪NETWORKMESSAGEHEADER 1
#define UA_JSONNETWORKMESSAGECONTENTMASK_
↪DATASETMESSAGEHEADER 2
#define UA_JSONNETWORKMESSAGECONTENTMASK_
↪SINGLEDATASETMESSAGE 4
#define UA_JSONNETWORKMESSAGECONTENTMASK_PUBLISHERID 8
#define UA_JSONNETWORKMESSAGECONTENTMASK_DATASETCLASSID 16
#define UA_JSONNETWORKMESSAGECONTENTMASK_REPLYTO 32
```

4.8.50 JsonWriterGroupMessageDataType

```
typedef struct {
    UA_JsonNetworkMessageContentMask networkMessageContentMask;
} UA_JsonWriterGroupMessageDataType;
```

4.8.51 JsonDataSetMessageContentMask

```
typedef UA_UInt32 UA_JsonDataSetMessageContentMask;

#define UA_JSONDATASETMESSAGECONTENTMASK_NONE 0
#define UA_JSONDATASETMESSAGECONTENTMASK_DATASETWRITERID 1
#define UA_JSONDATASETMESSAGECONTENTMASK_METADATAVERSION 2
#define UA_JSONDATASETMESSAGECONTENTMASK_SEQUENCENUMBER 4
#define UA_JSONDATASETMESSAGECONTENTMASK_TIMESTAMP 8
#define UA_JSONDATASETMESSAGECONTENTMASK_STATUS 16
#define UA_JSONDATASETMESSAGECONTENTMASK_MESSAGETYPE 32
#define UA_JSONDATASETMESSAGECONTENTMASK_DATASETWRITERNAME 64
#define UA_JSONDATASETMESSAGECONTENTMASK_
↪REVERSIBLEFIELDENCODING 128
```

4.8.52 JsonDataSetWriterMessageDataType

```
typedef struct {
    UA_JsonDataSetMessageContentMask dataSetMessageContentMask;
} UA_JsonDataSetWriterMessageDataType;
```

4.8.53 JsonDataSetReaderMessageDataType

```
typedef struct {
    UA_JsonNetworkMessageContentMask networkMessageContentMask;
    UA_JsonDataSetMessageContentMask dataSetMessageContentMask;
} UA_JsonDataSetReaderMessageDataType;
```

4.8.54 TransmitQosPriorityDataType

```
typedef struct {
    UA_String priorityLabel;
} UA_TransmitQosPriorityDataType;
```

4.8.55 ReceiveQosPriorityDataType

```
typedef struct {
    UA_String priorityLabel;
} UA_ReceiveQosPriorityDataType;
```

4.8.56 DatagramConnectionTransportDataType

```
typedef struct {
    UA_ExtensionObject discoveryAddress;
} UA_DatagramConnectionTransportDataType;
```

4.8.57 DatagramConnectionTransport2DataType

```
typedef struct {
    UA_ExtensionObject discoveryAddress;
    UA_UInt32 discoveryAnnounceRate;
    UA_UInt32 discoveryMaxMessageSize;
    UA_String qosCategory;
    size_t datagramQosSize;
    UA_ExtensionObject *datagramQos;
} UA_DatagramConnectionTransport2DataType;
```

4.8.58 DatagramWriterGroupTransportDataType

```
typedef struct {
    UA_Byte messageRepeatCount;
    UA_Double messageRepeatDelay;
} UA_DatagramWriterGroupTransportDataType;
```

4.8.59 DatagramWriterGroupTransport2DataType

```
typedef struct {
    UA_Byte messageRepeatCount;
    UA_Double messageRepeatDelay;
    UA_ExtensionObject address;
    UA_String qosCategory;
    size_t datagramQosSize;
    UA_ExtensionObject *datagramQos;
    UA_UInt32 discoveryAnnounceRate;
    UA_String topic;
} UA_DatagramWriterGroupTransport2DataType;
```

4.8.60 DatagramDataSetReaderTransportDataType

```
typedef struct {
    UA_ExtensionObject address;
    UA_String qosCategory;
    size_t datagramQosSize;
    UA_ExtensionObject *datagramQos;
    UA_String topic;
} UA_DatagramDataSetReaderTransportDataType;
```

4.8.61 BrokerConnectionTransportDataType

```
typedef struct {
    UA_String resourceUri;
    UA_String authenticationProfileUri;
} UA_BrokerConnectionTransportDataType;
```

4.8.62 BrokerTransportQualityOfService

```
typedef enum {
    UA_BROKERTRANSPORTQUALITYOFSERVICE_NOTSPECIFIED = 0,
    UA_BROKERTRANSPORTQUALITYOFSERVICE_BESTEFFORT = 1,
    UA_BROKERTRANSPORTQUALITYOFSERVICE_ATLEASTONCE = 2,
    UA_BROKERTRANSPORTQUALITYOFSERVICE_ATMOSTONCE = 3,
    UA_BROKERTRANSPORTQUALITYOFSERVICE_EXACTLYONCE = 4
} UA_BrokerTransportQualityOfService;
```

4.8.63 BrokerWriterGroupTransportDataType

```
typedef struct {
    UA_String queueName;
    UA_String resourceUri;
    UA_String authenticationProfileUri;
    UA_BrokerTransportQualityOfService requestedDeliveryGuarantee;
} UA_BrokerWriterGroupTransportDataType;
```

4.8.64 BrokerDataSetWriterTransportDataType

```
typedef struct {
    UA_String queueName;
    UA_String resourceUri;
    UA_String authenticationProfileUri;
    UA_BrokerTransportQualityOfService requestedDeliveryGuarantee;
    UA_String metaDataQueueName;
    UA_Double metaDataUpdateTime;
} UA_BrokerDataSetWriterTransportDataType;
```

4.8.65 BrokerDataSetReaderTransportDataType

```
typedef struct {
    UA_String queueName;
    UA_String resourceUri;
    UA_String authenticationProfileUri;
    UA_BrokerTransportQualityOfService requestedDeliveryGuarantee;
    UA_String metaDataQueueName;
} UA_BrokerDataSetReaderTransportDataType;
```

4.8.66 PubSubConfigurationRefMask

```
typedef UA_UInt32 UA_PubSubConfigurationRefMask;

#define UA_PUBSUBCONFIGURATIONREFMASK_NONE 0
#define UA_PUBSUBCONFIGURATIONREFMASK_ELEMENTADD 1
#define UA_PUBSUBCONFIGURATIONREFMASK_ELEMENTMATCH 2
#define UA_PUBSUBCONFIGURATIONREFMASK_ELEMENTMODIFY 4
#define UA_PUBSUBCONFIGURATIONREFMASK_ELEMENTREMOVE 8
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCEWRITER 16
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCEREADER 32
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCEWRITERGROUP 64
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCEREADERGROUP 128
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCECONNECTION 256
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCEPUBDATASET 512
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCESUBDATASET 1024
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCESECURITYGROUP␣
↪2048
#define UA_PUBSUBCONFIGURATIONREFMASK_REFERENCEPUSHTARGET 4096
```

4.8.67 PubSubConfigurationRefDataType

```
typedef struct {
    UA_PubSubConfigurationRefMask configurationMask;
    UA_UInt16 elementIndex;
    UA_UInt16 connectionIndex;
    UA_UInt16 groupIndex;
} UA_PubSubConfigurationRefDataType;
```

4.8.68 PubSubConfigurationValueDataType

```
typedef struct {
    UA_PubSubConfigurationRefDataType configurationElement;
    UA_String name;
    UA_Variant identifier;
} UA_PubSubConfigurationValueDataType;
```

4.8.69 DiagnosticsLevel

```
typedef enum {
    UA_DIAGNOSTICSLEVEL_BASIC = 0,
    UA_DIAGNOSTICSLEVEL_ADVANCED = 1,
    UA_DIAGNOSTICSLEVEL_INFO = 2,
    UA_DIAGNOSTICSLEVEL_LOG = 3,
    UA_DIAGNOSTICSLEVEL_DEBUG = 4
} UA_DiagnosticsLevel;
```

4.8.70 PubSubDiagnosticsCounterClassification

```
typedef enum {
    UA_PUBSUBDIAGNOSTICSCOUNTERCLASSIFICATION_INFORMATION = 0,
    UA_PUBSUBDIAGNOSTICSCOUNTERCLASSIFICATION_ERROR = 1
} UA_PubSubDiagnosticsCounterClassification;
```

4.8.71 AliasNameDataType

```
typedef struct {
    UA_QualifiedName aliasName;
    size_t referencedNodesSize;
    UA_ExpandedNodeId *referencedNodes;
} UA_AliasNameDataType;
```

4.8.72 PasswordOptionsMask

```
typedef UA_UInt32 UA_PasswordOptionsMask;

#define UA_PASSWORDOPTIONSMASK_NONE 0
#define UA_PASSWORDOPTIONSMASK_SUPPORTINITIALPASSWORDCHANGE 1
#define UA_PASSWORDOPTIONSMASK_SUPPORTDISABLEUSER 2
#define UA_PASSWORDOPTIONSMASK_SUPPORTDISABLEDELETEFORUSER 4
#define UA_PASSWORDOPTIONSMASK_SUPPORTNOCHANGEFORUSER 8
#define UA_PASSWORDOPTIONSMASK_SUPPORTDESCRIPTIONFORUSER 16
#define UA_PASSWORDOPTIONSMASK_REQUIRESUPPERCASECHARACTERS 32
#define UA_PASSWORDOPTIONSMASK_REQUIRESLOWERCASECHARACTERS 64
#define UA_PASSWORDOPTIONSMASK_REQUIRESDIGITCHARACTERS 128
#define UA_PASSWORDOPTIONSMASK_REQUIRESSPECIALCHARACTERS 256
```

4.8.73 UserConfigurationMask

```
typedef UA_UInt32 UA_UserConfigurationMask;

#define UA_USERCONFIGURATIONMASK_NONE 0
#define UA_USERCONFIGURATIONMASK_NODELETE 1
#define UA_USERCONFIGURATIONMASK_DISABLED 2
#define UA_USERCONFIGURATIONMASK_NOCHANGEBYUSER 4
#define UA_USERCONFIGURATIONMASK_MUSTCHANGEPASSWORD 8
```

4.8.74 UserManagementDataType

```
typedef struct {
    UA_String userName;
    UA_UserConfigurationMask userConfiguration;
    UA_String description;
} UA_UserManagementDataType;
```

4.8.75 Duplex

```
typedef enum {
    UA_DUPLEX_FULL = 0,
    UA_DUPLEX_HALF = 1,
    UA_DUPLEX_UNKNOWN = 2
} UA_Duplex;
```

4.8.76 InterfaceAdminStatus

```
typedef enum {
    UA_INTERFACEADMINSTATUS_UP = 0,
    UA_INTERFACEADMINSTATUS_DOWN = 1,
    UA_INTERFACEADMINSTATUS_TESTING = 2
} UA_InterfaceAdminStatus;
```

4.8.77 InterfaceOperStatus

```
typedef enum {
    UA_INTERFACEOPERSTATUS_UP = 0,
    UA_INTERFACEOPERSTATUS_DOWN = 1,
    UA_INTERFACEOPERSTATUS_TESTING = 2,
    UA_INTERFACEOPERSTATUS_UNKNOWN = 3,
    UA_INTERFACEOPERSTATUS_DORMANT = 4,
    UA_INTERFACEOPERSTATUS_NOTPRESENT = 5,
    UA_INTERFACEOPERSTATUS_LOWERLAYERDOWN = 6
} UA_InterfaceOperStatus;
```

4.8.78 NegotiationStatus

```
typedef enum {
    UA_NEGOTIATIONSTATUS_INPROGRESS = 0,
    UA_NEGOTIATIONSTATUS_COMPLETE = 1,
    UA_NEGOTIATIONSTATUS_FAILED = 2,
    UA_NEGOTIATIONSTATUS_UNKNOWN = 3,
    UA_NEGOTIATIONSTATUS_NONEGOTIATION = 4
} UA_NegotiationStatus;
```

4.8.79 TsnFailureCode

```
typedef enum {
    UA_TSNFAILURECODE_NOFAILURE = 0,
    UA_TSNFAILURECODE_INSUFFICIENTBANDWIDTH = 1,
    UA_TSNFAILURECODE_INSUFFICIENTRESOURCES = 2,
    UA_TSNFAILURECODE_INSUFFICIENTTRAFFICCLASSBANDWIDTH = 3,
    UA_TSNFAILURECODE_STREAMIDINUSE = 4,
    UA_TSNFAILURECODE_STREAMDESTINATIONADDRESSINUSE = 5,
    UA_TSNFAILURECODE_STREAMPREEMPTEDBYHIGHERRANK = 6,
    UA_TSNFAILURECODE_LATENCYHASCHANGED = 7,
    UA_TSNFAILURECODE_EGRESSPORTNOTAVBCAPABLE = 8,
    UA_TSNFAILURECODE_USEDIFFERENTDESTINATIONADDRESS = 9,
    UA_TSNFAILURECODE_OUTOFMSRPRESOURCES = 10,
    UA_TSNFAILURECODE_OUTOFMMRPRESOURCES = 11,
    UA_TSNFAILURECODE_CANNOTSTOREDESTINATIONADDRESS = 12,
    UA_TSNFAILURECODE_PRIORITYISNOTANSRCCLASS = 13,
    UA_TSNFAILURECODE_MAXFRAMESIZETOOLARGE = 14,
    UA_TSNFAILURECODE_MAXFANINPORTSLIMITREACHED = 15,
    UA_TSNFAILURECODE_FIRSTVALUECHANGEDFORSTREAMID = 16,
    UA_TSNFAILURECODE_VLANBLOCKEDONEGRESS = 17,
    UA_TSNFAILURECODE_VLANTAGGINGDISABLEDONEGRESS = 18,
    UA_TSNFAILURECODE_SRCLASSPRIORITYMISMATCH = 19,
    UA_TSNFAILURECODE_FEATURENOTPROPAGATED = 20,
    UA_TSNFAILURECODE_MAXLATENCYEXCEEDED = 21,
    UA_TSNFAILURECODE_BRIDGEDOESNOTPROVIDENETWORKID = 22,
    UA_TSNFAILURECODE_STREAMTRANSFORMNOTSUPPORTED = 23,
    UA_TSNFAILURECODE_STREAMIDTYPENOTSUPPORTED = 24,
    UA_TSNFAILURECODE_FEATURENOTSUPPORTED = 25
} UA_TsnFailureCode;
```

4.8.80 TsnStreamState

```
typedef enum {
    UA_TSNSTREAMSTATE_DISABLED = 0,
    UA_TSNSTREAMSTATE_CONFIGURING = 1,
    UA_TSNSTREAMSTATE_READY = 2,
    UA_TSNSTREAMSTATE_OPERATIONAL = 3,
    UA_TSNSTREAMSTATE_ERROR = 4
} UA_TsnStreamState;
```

4.8.81 TsnTalkerStatus

```
typedef enum {
    UA_TSNTALKERSTATUS_NONE = 0,
    UA_TSNTALKERSTATUS_READY = 1,
    UA_TSNTALKERSTATUS_FAILED = 2
} UA_TsnTalkerStatus;
```

4.8.82 TsnListenerStatus

```
typedef enum {
    UA_TSNLISTENERSTATUS_NONE = 0,
    UA_TSNLISTENERSTATUS_READY = 1,
    UA_TSNLISTENERSTATUS_PARTIALFAILED = 2,
    UA_TSNLISTENERSTATUS_FAILED = 3
} UA_TsnListenerStatus;
```

4.8.83 PriorityMappingEntryType

```
typedef struct {
    UA_String mappingUri;
    UA_String priorityLabel;
    UA_Byte priorityValue_PCP;
    UA_UInt32 priorityValue_DSCP;
} UA_PriorityMappingEntryType;
```

4.8.84 IdType

```
typedef enum {
    UA_IDTYPE_NUMERIC = 0,
    UA_IDTYPE_STRING = 1,
    UA_IDTYPE_GUID = 2,
    UA_IDTYPE_OPAQUE = 3
} UA_IdType;
```

4.8.85 NodeClass

```
typedef enum {
    UA_NODECLASS_UNSPECIFIED = 0,
    UA_NODECLASS_OBJECT = 1,
    UA_NODECLASS_VARIABLE = 2,
    UA_NODECLASS_METHOD = 4,
    UA_NODECLASS_OBJECTTYPE = 8,
    UA_NODECLASS_VARIABLETYPE = 16,
    UA_NODECLASS_REFERENCETYPE = 32,
    UA_NODECLASS_DATATYPE = 64,
    UA_NODECLASS_VIEW = 128
} UA_NodeClass;
```

4.8.86 PermissionType

```
typedef UA_UInt32 UA_PermissionType;

#define UA_PERMISSIONTYPE_NONE 0
#define UA_PERMISSIONTYPE_BROWSE 1
#define UA_PERMISSIONTYPE_READROLEPERMISSIONS 2
#define UA_PERMISSIONTYPE_WRITEATTRIBUTE 4
#define UA_PERMISSIONTYPE_WRITEROLEPERMISSIONS 8
#define UA_PERMISSIONTYPE_WRITEHISTORIZING 16
#define UA_PERMISSIONTYPE_READ 32
#define UA_PERMISSIONTYPE_WRITE 64
#define UA_PERMISSIONTYPE_READHISTORY 128
#define UA_PERMISSIONTYPE_INSERTHISTORY 256
#define UA_PERMISSIONTYPE_MODIFYHISTORY 512
#define UA_PERMISSIONTYPE_DELETEHISTORY 1024
#define UA_PERMISSIONTYPE_RECEIVEEVENTS 2048
#define UA_PERMISSIONTYPE_CALL 4096
#define UA_PERMISSIONTYPE_ADDREFERENCE 8192
#define UA_PERMISSIONTYPE_REMOVEREFERENCE 16384
#define UA_PERMISSIONTYPE_DELETENODE 32768
#define UA_PERMISSIONTYPE_ADDNODE 65536
```

4.8.87 AccessLevelType

```
typedef UA_Byte UA_AccessLevelType;

#define UA_ACCESSLEVELTYPE_NONE 0
#define UA_ACCESSLEVELTYPE_CURRENTREAD 1
#define UA_ACCESSLEVELTYPE_CURRENTWRITE 2
#define UA_ACCESSLEVELTYPE_HISTORYREAD 4
#define UA_ACCESSLEVELTYPE_HISTORYWRITE 8
#define UA_ACCESSLEVELTYPE_SEMANTICCHANGE 16
#define UA_ACCESSLEVELTYPE_STATUSWRITE 32
#define UA_ACCESSLEVELTYPE_TIMESTAMPWRITE 64
```

4.8.88 AccessLevelExType

```
typedef UA_UInt32 UA_AccessLevelExType;

#define UA_ACCESSLEVELEXTYPE_NONE 0
#define UA_ACCESSLEVELEXTYPE_CURRENTREAD 1
#define UA_ACCESSLEVELEXTYPE_CURRENTWRITE 2
#define UA_ACCESSLEVELEXTYPE_HISTORYREAD 4
#define UA_ACCESSLEVELEXTYPE_HISTORYWRITE 8
#define UA_ACCESSLEVELEXTYPE_SEMANTICCHANGE 16
#define UA_ACCESSLEVELEXTYPE_STATUSWRITE 32
#define UA_ACCESSLEVELEXTYPE_TIMESTAMPWRITE 64
#define UA_ACCESSLEVELEXTYPE_NONATOMICREAD 256
#define UA_ACCESSLEVELEXTYPE_NONATOMICWRITE 512
```

(continues on next page)

(continued from previous page)

```
#define UA_ACCESSLEVELEXTYPE_WRITEFULLARRAYONLY 1024
#define UA_ACCESSLEVELEXTYPE_NOSUBDATATYPES 2048
#define UA_ACCESSLEVELEXTYPE_NONVOLATILE 4096
#define UA_ACCESSLEVELEXTYPE_CONSTANT 8192
```

4.8.89 EventNotifierType

```
typedef UA_Byte UA_EventNotifierType;

#define UA_EVENTNOTIFIERTYPE_NONE 0
#define UA_EVENTNOTIFIERTYPE_SUBSCRIBETOEVENTS 1
#define UA_EVENTNOTIFIERTYPE_HISTORYREAD 4
#define UA_EVENTNOTIFIERTYPE_HISTORYWRITE 8
```

4.8.90 AccessRestrictionType

```
typedef UA_UInt16 UA_AccessRestrictionType;

#define UA_ACCESSRESTRICTIONTYPE_NONE 0
#define UA_ACCESSRESTRICTIONTYPE_SIGNINGREQUIRED 1
#define UA_ACCESSRESTRICTIONTYPE_ENCRYPTIONREQUIRED 2
#define UA_ACCESSRESTRICTIONTYPE_SESSIONREQUIRED 4
#define UA_ACCESSRESTRICTIONTYPE_APPLYRESTRICTIONSTOBROWSE 8
```

4.8.91 RolePermissionType

```
typedef struct {
    UA_NodeId roleId;
    UA_PermissionType permissions;
} UA_RolePermissionType;
```

4.8.92 StructureType

```
typedef enum {
    UA_STRUCTURETYPE_STRUCTURE = 0,
    UA_STRUCTURETYPE_STRUCTUREWITHOPTIONALFIELDS = 1,
    UA_STRUCTURETYPE_UNION = 2,
    UA_STRUCTURETYPE_STRUCTUREWITHSUBTYPEDVALUES = 3,
    UA_STRUCTURETYPE_UNIONWITHSUBTYPEDVALUES = 4
} UA_StructureType;
```

4.8.93 StructureField

```
typedef struct {
    UA_String name;
    UA_LocalizedText description;
    UA_NodeId dataType;
```

(continues on next page)

(continued from previous page)

```
    UA_Int32 valueRank;
    size_t arrayDimensionsSize;
    UA_UInt32 *arrayDimensions;
    UA_UInt32 maxStringLength;
    UA_Boolean isOptional;
} UA_StructureField;
```

4.8.94 StructureDefinition

```
typedef struct {
    UA_NodeId defaultEncodingId;
    UA_NodeId baseDataType;
    UA_StructureType structureType;
    size_t fieldsSize;
    UA_StructureField *fields;
} UA_StructureDefinition;
```

4.8.95 ReferenceNode

```
typedef struct {
    UA_NodeId referenceTypeId;
    UA_Boolean isInverse;
    UA_ExpandedNodeId targetId;
} UA_ReferenceNode;
```

4.8.96 Argument

```
typedef struct {
    UA_String name;
    UA_NodeId dataType;
    UA_Int32 valueRank;
    size_t arrayDimensionsSize;
    UA_UInt32 *arrayDimensions;
    UA_LocalizedText description;
} UA_Argument;
```

4.8.97 EnumValueType

```
typedef struct {
    UA_Int64 value;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
} UA_EnumValueType;
```

4.8.98 EnumField

```
typedef struct {
    UA_Int64 value;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_String name;
} UA_EnumField;
```

4.8.99 OptionSet

```
typedef struct {
    UA_ByteString value;
    UA_ByteString validBits;
} UA_OptionSet;
```

4.8.100 NormalizedString

```
typedef UA_String UA_NormalizedString;
```

4.8.101 DecimalString

```
typedef UA_String UA_DecimalString;
```

4.8.102 DurationString

```
typedef UA_String UA_DurationString;
```

4.8.103 TimeString

```
typedef UA_String UA_TimeString;
```

4.8.104 DateString

```
typedef UA_String UA_DateString;
```

4.8.105 Duration

```
typedef UA_Double UA_Duration;
```

4.8.106 UtcTime

```
typedef UA_DateTime UA_UtcTime;
```

4.8.107 LocaleId

```
typedef UA_String UA_LocaleId;
```

4.8.108 TimeZoneDataType

```
typedef struct {
    UA_Int16 offset;
    UA_Boolean daylightSavingInOffset;
} UA_TimeZoneDataType;
```

4.8.109 Index

```
typedef UA_ByteString UA_Index;
```

4.8.110 IntegerId

```
typedef UA_UInt32 UA_IntegerId;
```

4.8.111 ApplicationType

```
typedef enum {
    UA_APPLICATIONTYPE_SERVER = 0,
    UA_APPLICATIONTYPE_CLIENT = 1,
    UA_APPLICATIONTYPE_CLIENTANDSERVER = 2,
    UA_APPLICATIONTYPE_DISCOVERYSERVER = 3
} UA_ApplicationType;
```

4.8.112 ApplicationDescription

```
typedef struct {
    UA_String applicationUri;
    UA_String productUri;
    UA_LocalizedText applicationName;
    UA_ApplicationType applicationType;
    UA_String gatewayServerUri;
    UA_String discoveryProfileUri;
    size_t discoveryUrlsSize;
    UA_String *discoveryUrls;
} UA_ApplicationDescription;
```

4.8.113 RequestHeader

```
typedef struct {
    UA_NodeId authenticationToken;
    UA_DateTime timestamp;
    UA_UInt32 requestHandle;
    UA_UInt32 returnDiagnostics;
```

(continues on next page)

(continued from previous page)

```
    UA_String auditEntryId;
    UA_UInt32 timeoutHint;
    UA_ExtensionObject additionalHeader;
} UA_RequestHeader;
```

4.8.114 ResponseHeader

```
typedef struct {
    UA_DateTime timestamp;
    UA_UInt32 requestHandle;
    UA_StatusCode serviceResult;
    UA_DiagnosticInfo serviceDiagnostics;
    size_t stringTableSize;
    UA_String *stringTable;
    UA_ExtensionObject additionalHeader;
} UA_ResponseHeader;
```

4.8.115 VersionTime

```
typedef UA_ByteString UA_VersionTime;
```

4.8.116 ServiceFault

```
typedef struct {
    UA_ResponseHeader responseHeader;
} UA_ServiceFault;
```

4.8.117 SessionlessInvokeRequestType

```
typedef struct {
    UA_UInt32 urisVersion;
    size_t namespaceUrisSize;
    UA_String *namespaceUris;
    size_t serverUrisSize;
    UA_String *serverUris;
    size_t localeIdsSize;
    UA_String *localeIds;
    UA_UInt32 serviceId;
} UA_SessionlessInvokeRequestType;
```

4.8.118 SessionlessInvokeResponseType

```
typedef struct {
    size_t namespaceUrisSize;
    UA_String *namespaceUris;
    size_t serverUrisSize;
    UA_String *serverUris;
```

(continues on next page)

(continued from previous page)

```
    UA_UInt32 serviceId;
} UA_SessionlessInvokeResponseType;
```

4.8.119 FindServersRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_String endpointUrl;
    size_t localeIdsSize;
    UA_String *localeIds;
    size_t serverUrisSize;
    UA_String *serverUris;
} UA_FindServersRequest;
```

4.8.120 FindServersResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t serversSize;
    UA_ApplicationDescription *servers;
} UA_FindServersResponse;
```

4.8.121 ServerOnNetwork

```
typedef struct {
    UA_UInt32 recordId;
    UA_String serverName;
    UA_String discoveryUrl;
    size_t serverCapabilitiesSize;
    UA_String *serverCapabilities;
} UA_ServerOnNetwork;
```

4.8.122 FindServersOnNetworkRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 startingRecordId;
    UA_UInt32 maxRecordsToReturn;
    size_t serverCapabilityFilterSize;
    UA_String *serverCapabilityFilter;
} UA_FindServersOnNetworkRequest;
```

4.8.123 FindServersOnNetworkResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_DateTime lastCounterResetTime;
```

(continues on next page)

(continued from previous page)

```
    size_t serversSize;
    UA_ServerOnNetwork *servers;
} UA_FindServersOnNetworkResponse;
```

4.8.124 ApplicationInstanceCertificate

```
typedef UA_ByteString UA_ApplicationInstanceCertificate;
```

4.8.125 MessageSecurityMode

```
typedef enum {
    UA_MESSAGESECURITYMODE_INVALID = 0,
    UA_MESSAGESECURITYMODE_NONE = 1,
    UA_MESSAGESECURITYMODE_SIGN = 2,
    UA_MESSAGESECURITYMODE_SIGNANDENCRYPT = 3
} UA_MessageSecurityMode;
```

4.8.126 UserTokenType

```
typedef enum {
    UA_USERTOKENTYPE_ANONYMOUS = 0,
    UA_USERTOKENTYPE_USERNAME = 1,
    UA_USERTOKENTYPE_CERTIFICATE = 2,
    UA_USERTOKENTYPE_ISSUEDTOKEN = 3
} UA_UserTokenType;
```

4.8.127 UserTokenPolicy

```
typedef struct {
    UA_String policyId;
    UA_UserTokenType tokenType;
    UA_String issuedTokenType;
    UA_String issuerEndpointUrl;
    UA_String securityPolicyUri;
} UA_UserTokenPolicy;
```

4.8.128 EndpointDescription

```
typedef struct {
    UA_String endpointUrl;
    UA_ApplicationDescription server;
    UA_ByteString serverCertificate;
    UA_MessageSecurityMode securityMode;
    UA_String securityPolicyUri;
    size_t userIdentityTokensSize;
    UA_UserTokenPolicy *userIdentityTokens;
    UA_String transportProfileUri;
```

(continues on next page)

(continued from previous page)

```
    UA_Byte securityLevel;
} UA_EndpointDescription;
```

4.8.129 GetEndpointsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_String endpointUrl;
    size_t localeIdsSize;
    UA_String *localeIds;
    size_t profileUrisSize;
    UA_String *profileUris;
} UA_GetEndpointsRequest;
```

4.8.130 GetEndpointsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t endpointsSize;
    UA_EndpointDescription *endpoints;
} UA_GetEndpointsResponse;
```

4.8.131 RegisteredServer

```
typedef struct {
    UA_String serverUri;
    UA_String productUri;
    size_t serverNamesSize;
    UA_LocalizedText *serverNames;
    UA_ApplicationType serverType;
    UA_String gatewayServerUri;
    size_t discoveryUrlsSize;
    UA_String *discoveryUrls;
    UA_String semaphoreFilePath;
    UA_Boolean isOnline;
} UA_RegisteredServer;
```

4.8.132 RegisterServerRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_RegisteredServer server;
} UA_RegisterServerRequest;
```

4.8.133 RegisterServerResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
} UA_RegisterServerResponse;
```

4.8.134 MdnsDiscoveryConfiguration

```
typedef struct {
    UA_String mdnsServerName;
    size_t serverCapabilitiesSize;
    UA_String *serverCapabilities;
} UA_MdnsDiscoveryConfiguration;
```

4.8.135 RegisterServer2Request

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_RegisteredServer server;
    size_t discoveryConfigurationSize;
    UA_ExtensionObject *discoveryConfiguration;
} UA_RegisterServer2Request;
```

4.8.136 RegisterServer2Response

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t configurationResultsSize;
    UA_StatusCode *configurationResults;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_RegisterServer2Response;
```

4.8.137 SecurityTokenRequestType

```
typedef enum {
    UA_SECURITYTOKENREQUESTTYPE_ISSUE = 0,
    UA_SECURITYTOKENREQUESTTYPE_RENEW = 1
} UA_SecurityTokenRequestType;
```

4.8.138 ChannelSecurityToken

```
typedef struct {
    UA_UInt32 channelId;
    UA_UInt32 tokenId;
    UA_DateTime createdAt;
    UA_UInt32 revisedLifetime;
} UA_ChannelSecurityToken;
```

4.8.139 OpenSecureChannelRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 clientProtocolVersion;
    UA_SecurityTokenRequestType requestType;
    UA_MessageSecurityMode securityMode;
    UA_ByteString clientNonce;
    UA_UInt32 requestedLifetime;
} UA_OpenSecureChannelRequest;
```

4.8.140 OpenSecureChannelResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_UInt32 serverProtocolVersion;
    UA_ChannelSecurityToken securityToken;
    UA_ByteString serverNonce;
} UA_OpenSecureChannelResponse;
```

4.8.141 CloseSecureChannelRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
} UA_CloseSecureChannelRequest;
```

4.8.142 CloseSecureChannelResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
} UA_CloseSecureChannelResponse;
```

4.8.143 SignedSoftwareCertificate

```
typedef struct {
    UA_ByteString certificateData;
    UA_ByteString signature;
} UA_SignedSoftwareCertificate;
```

4.8.144 SessionAuthenticationToken

```
typedef UA_NodeId UA_SessionAuthenticationToken;
```

4.8.145 SignatureData

```
typedef struct {
    UA_String algorithm;
    UA_ByteString signature;
} UA_SignatureData;
```

4.8.146 CreateSessionRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_ApplicationDescription clientDescription;
    UA_String serverUri;
    UA_String endpointUrl;
    UA_String sessionName;
    UA_ByteString clientNonce;
    UA_ByteString clientCertificate;
    UA_Double requestedSessionTimeout;
    UA_UInt32 maxResponseMessageSize;
} UA_CreateSessionRequest;
```

4.8.147 CreateSessionResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_NodeId sessionId;
    UA_NodeId authenticationToken;
    UA_Double revisedSessionTimeout;
    UA_ByteString serverNonce;
    UA_ByteString serverCertificate;
    size_t serverEndpointsSize;
    UA_EndpointDescription *serverEndpoints;
    size_t serverSoftwareCertificatesSize;
    UA_SignedSoftwareCertificate *serverSoftwareCertificates;
    UA_SignatureData serverSignature;
    UA_UInt32 maxRequestMessageSize;
} UA_CreateSessionResponse;
```

4.8.148 UserIdentityToken

```
typedef struct {
    UA_String policyId;
} UA_UserIdentityToken;
```

4.8.149 AnonymousIdentityToken

```
typedef struct {
    UA_String policyId;
} UA_AnonymousIdentityToken;
```

4.8.150 UserNameIdentityToken

```
typedef struct {
    UA_String policyId;
    UA_String userName;
    UA_ByteString password;
```

(continues on next page)

(continued from previous page)

```
    UA_String encryptionAlgorithm;
} UA_UserNameIdentityToken;
```

4.8.151 X509IdentityToken

```
typedef struct {
    UA_String policyId;
    UA_ByteString certificateData;
} UA_X509IdentityToken;
```

4.8.152 IssuedIdentityToken

```
typedef struct {
    UA_String policyId;
    UA_ByteString tokenData;
    UA_String encryptionAlgorithm;
} UA_IssuedIdentityToken;
```

4.8.153 RsaEncryptedSecret

```
typedef UA_ByteString UA_RsaEncryptedSecret;
```

4.8.154 EccEncryptedSecret

```
typedef UA_ByteString UA_EccEncryptedSecret;
```

4.8.155 ActivateSessionRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_SignatureData clientSignature;
    size_t clientSoftwareCertificatesSize;
    UA_SignedSoftwareCertificate *clientSoftwareCertificates;
    size_t localeIdsSize;
    UA_String *localeIds;
    UA_ExtensionObject userIdentityToken;
    UA_SignatureData userTokenSignature;
} UA_ActivateSessionRequest;
```

4.8.156 ActivateSessionResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_ByteString serverNonce;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
```

(continues on next page)

(continued from previous page)

```
    UA_DiagnosticInfo *diagnosticInfos;
} UA_ActivateSessionResponse;
```

4.8.157 CloseSessionRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_Boolean deleteSubscriptions;
} UA_CloseSessionRequest;
```

4.8.158 CloseSessionResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
} UA_CloseSessionResponse;
```

4.8.159 CancelRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 requestHandle;
} UA_CancelRequest;
```

4.8.160 CancelResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_UInt32 cancelCount;
} UA_CancelResponse;
```

4.8.161 NodeAttributesMask

```
typedef enum {
    UA_NODEATTRIBUTESMASK_NONE = 0,
    UA_NODEATTRIBUTESMASK_ACCESSLEVEL = 1,
    UA_NODEATTRIBUTESMASK_ARRAYDIMENSIONS = 2,
    UA_NODEATTRIBUTESMASK_BROWSENAME = 4,
    UA_NODEATTRIBUTESMASK_CONTAINSNOLOOPS = 8,
    UA_NODEATTRIBUTESMASK_DATATYPE = 16,
    UA_NODEATTRIBUTESMASK_DESCRIPTION = 32,
    UA_NODEATTRIBUTESMASK_DISPLAYNAME = 64,
    UA_NODEATTRIBUTESMASK_EVENTNOTIFIER = 128,
    UA_NODEATTRIBUTESMASK_EXECUTABLE = 256,
    UA_NODEATTRIBUTESMASK_HISTORIZING = 512,
    UA_NODEATTRIBUTESMASK_INVERSENAME = 1024,
    UA_NODEATTRIBUTESMASK_ISABSTRACT = 2048,
    UA_NODEATTRIBUTESMASK_MINIMUMSAMPLINGINTERVAL = 4096,
```

(continues on next page)

(continued from previous page)

```
    UA_NODEATTRIBUTESMASK_NODECLASS = 8192,
    UA_NODEATTRIBUTESMASK_NODEID = 16384,
    UA_NODEATTRIBUTESMASK_SYMMETRIC = 32768,
    UA_NODEATTRIBUTESMASK_USERACCESSLEVEL = 65536,
    UA_NODEATTRIBUTESMASK_USEREXECUTABLE = 131072,
    UA_NODEATTRIBUTESMASK_USERWRITEMASK = 262144,
    UA_NODEATTRIBUTESMASK_VALUERANK = 524288,
    UA_NODEATTRIBUTESMASK_WRITEMASK = 1048576,
    UA_NODEATTRIBUTESMASK_VALUE = 2097152,
    UA_NODEATTRIBUTESMASK_DATATYPEDEFINITION = 4194304,
    UA_NODEATTRIBUTESMASK_ROLEPERMISSIONS = 8388608,
    UA_NODEATTRIBUTESMASK_ACCESSRESTRICTIONS = 16777216,
    UA_NODEATTRIBUTESMASK_ALL = 33554431,
    UA_NODEATTRIBUTESMASK_BASENODE = 26501220,
    UA_NODEATTRIBUTESMASK_OBJECT = 26501348,
    UA_NODEATTRIBUTESMASK_OBJECTTYPE = 26503268,
    UA_NODEATTRIBUTESMASK_VARIABLE = 26571383,
    UA_NODEATTRIBUTESMASK_VARIABLETYPE = 28600438,
    UA_NODEATTRIBUTESMASK_METHOD = 26632548,
    UA_NODEATTRIBUTESMASK_REFERENCETYPE = 26537060,
    UA_NODEATTRIBUTESMASK_VIEW = 26501356
} UA_NodeAttributesMask;
```

4.8.162 NodeAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
} UA_NodeAttributes;
```

4.8.163 ObjectAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Byte eventNotifier;
} UA_ObjectAttributes;
```

4.8.164 VariableAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Variant value;
    UA_NodeId dataType;
    UA_Int32 valueRank;
    size_t arrayDimensionsSize;
    UA_UInt32 *arrayDimensions;
    UA_Byte accessLevel;
    UA_Byte userAccessLevel;
    UA_Double minimumSamplingInterval;
    UA_Boolean historizing;
} UA_VariableAttributes;
```

4.8.165 MethodAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Boolean executable;
    UA_Boolean userExecutable;
} UA_MethodAttributes;
```

4.8.166 ObjectTypeAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Boolean isAbstract;
} UA_ObjectTypeAttributes;
```

4.8.167 VariableTypeAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
```

(continues on next page)

(continued from previous page)

```
    UA_Variant value;
    UA_NodeId dataType;
    UA_Int32 valueRank;
    size_t arrayDimensionsSize;
    UA_UInt32 *arrayDimensions;
    UA_Boolean isAbstract;
} UA_VariableTypeAttributes;
```

4.8.168 ReferenceTypeAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Boolean isAbstract;
    UA_Boolean symmetric;
    UA_LocalizedText inverseName;
} UA_ReferenceTypeAttributes;
```

4.8.169 DataTypeAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Boolean isAbstract;
} UA_DataTypeAttributes;
```

4.8.170 ViewAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    UA_Boolean containsNoLoops;
    UA_Byte eventNotifier;
} UA_ViewAttributes;
```

4.8.171 GenericAttributeValue

```
typedef struct {
    UA_UInt32 attributeId;
    UA_Variant value;
} UA_GenericAttributeValue;
```

4.8.172 GenericAttributes

```
typedef struct {
    UA_UInt32 specifiedAttributes;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
    UA_UInt32 writeMask;
    UA_UInt32 userWriteMask;
    size_t attributeValuesSize;
    UA_GenericAttributeValue *attributeValues;
} UA_GenericAttributes;
```

4.8.173 AddNodesItem

```
typedef struct {
    UA_ExpandedNodeId parentNodeId;
    UA_NodeId referenceTypeId;
    UA_ExpandedNodeId requestedNewNodeId;
    UA_QualifiedName browseName;
    UA_NodeClass nodeClass;
    UA_ExtensionObject nodeAttributes;
    UA_ExpandedNodeId typeDefinition;
} UA_AddNodesItem;
```

4.8.174 AddNodesResult

```
typedef struct {
    UA_StatusCode statusCode;
    UA_NodeId addedNodeId;
} UA_AddNodesResult;
```

4.8.175 AddNodesRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t nodesToAddSize;
    UA_AddNodesItem *nodesToAdd;
} UA_AddNodesRequest;
```

4.8.176 AddNodesResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_AddNodesResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_AddNodesResponse;
```

4.8.177 AddReferencesItem

```
typedef struct {
    UA_NodeId sourceNodeId;
    UA_NodeId referenceTypeId;
    UA_Boolean isForward;
    UA_String targetServerUri;
    UA_ExpandedNodeId targetNodeId;
    UA_NodeClass targetNodeClass;
} UA_AddReferencesItem;
```

4.8.178 AddReferencesRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t referencesToAddSize;
    UA_AddReferencesItem *referencesToAdd;
} UA_AddReferencesRequest;
```

4.8.179 AddReferencesResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_AddReferencesResponse;
```

4.8.180 DeleteNodesItem

```
typedef struct {
    UA_NodeId nodeId;
    UA_Boolean deleteTargetReferences;
} UA_DeleteNodesItem;
```

4.8.181 DeleteNodesRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t nodesToDeleteSize;
    UA_DeleteNodesItem *nodesToDelete;
} UA_DeleteNodesRequest;
```

4.8.182 DeleteNodesResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_DeleteNodesResponse;
```

4.8.183 DeleteReferencesItem

```
typedef struct {
    UA_NodeId sourceNodeId;
    UA_NodeId referenceTypeId;
    UA_Boolean isForward;
    UA_ExpandedNodeId targetNodeId;
    UA_Boolean deleteBidirectional;
} UA_DeleteReferencesItem;
```

4.8.184 DeleteReferencesRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t referencesToDeleteSize;
    UA_DeleteReferencesItem *referencesToDelete;
} UA_DeleteReferencesRequest;
```

4.8.185 DeleteReferencesResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_DeleteReferencesResponse;
```

4.8.186 AttributeWriteMask

```
typedef UA_UInt32 UA_AttributeWriteMask;

#define UA_ATTRIBUTEWRITEMASK_NONE 0
#define UA_ATTRIBUTEWRITEMASK_ACCESSLEVEL 1
#define UA_ATTRIBUTEWRITEMASK_ARRAYDIMENSIONS 2
#define UA_ATTRIBUTEWRITEMASK_BROWSENAME 4
#define UA_ATTRIBUTEWRITEMASK_CONTAINSNOLOOPS 8
#define UA_ATTRIBUTEWRITEMASK_DATATYPE 16
#define UA_ATTRIBUTEWRITEMASK_DESCRIPTION 32
#define UA_ATTRIBUTEWRITEMASK_DISPLAYNAME 64
#define UA_ATTRIBUTEWRITEMASK_EVENTNOTIFIER 128
#define UA_ATTRIBUTEWRITEMASK_EXECUTABLE 256
#define UA_ATTRIBUTEWRITEMASK_HISTORIZING 512
#define UA_ATTRIBUTEWRITEMASK_INVERSENAME 1024
#define UA_ATTRIBUTEWRITEMASK_ISABSTRACT 2048
#define UA_ATTRIBUTEWRITEMASK_MINIMUMSAMPLINGINTERVAL 4096
#define UA_ATTRIBUTEWRITEMASK_NODECLASS 8192
#define UA_ATTRIBUTEWRITEMASK_NODEID 16384
#define UA_ATTRIBUTEWRITEMASK_SYMMETRIC 32768
#define UA_ATTRIBUTEWRITEMASK_USERACCESSLEVEL 65536
#define UA_ATTRIBUTEWRITEMASK_USEREXECUTABLE 131072
#define UA_ATTRIBUTEWRITEMASK_USERWRITEMASK 262144
#define UA_ATTRIBUTEWRITEMASK_VALUERANK 524288
#define UA_ATTRIBUTEWRITEMASK_WRITEMASK 1048576
#define UA_ATTRIBUTEWRITEMASK_VALUEFORVARIABLETYPE 2097152
#define UA_ATTRIBUTEWRITEMASK_DATATYPEDEFINITION 4194304
#define UA_ATTRIBUTEWRITEMASK_ROLEPERMISSIONS 8388608
#define UA_ATTRIBUTEWRITEMASK_ACCESSRESTRICTIONS 16777216
#define UA_ATTRIBUTEWRITEMASK_ACCESSLEVELEX 33554432
```

4.8.187 BrowseDirection

```
typedef enum {
    UA_BROWSEDIRECTION_FORWARD = 0,
    UA_BROWSEDIRECTION_INVERSE = 1,
    UA_BROWSEDIRECTION_BOTH = 2,
    UA_BROWSEDIRECTION_INVALID = 3
} UA_BrowseDirection;
```

4.8.188 ViewDescription

```
typedef struct {
    UA_NodeId viewId;
    UA_DateTime timestamp;
    UA_UInt32 viewVersion;
} UA_ViewDescription;
```

4.8.189 BrowseDescription

```
typedef struct {
    UA_NodeId nodeId;
    UA_BrowseDirection browseDirection;
    UA_NodeId referenceTypeId;
    UA_Boolean includeSubtypes;
    UA_UInt32 nodeClassMask;
    UA_UInt32 resultMask;
} UA_BrowseDescription;
```

4.8.190 BrowseResultMask

```
typedef enum {
    UA_BROWSERESULTMASK_NONE = 0,
    UA_BROWSERESULTMASK_REFERENCETYPEID = 1,
    UA_BROWSERESULTMASK_ISFORWARD = 2,
    UA_BROWSERESULTMASK_NODECLASS = 4,
    UA_BROWSERESULTMASK_BROWSENAME = 8,
    UA_BROWSERESULTMASK_DISPLAYNAME = 16,
    UA_BROWSERESULTMASK_TYPEDEFINITION = 32,
    UA_BROWSERESULTMASK_ALL = 63,
    UA_BROWSERESULTMASK_REFERENCETYPEINFO = 3,
    UA_BROWSERESULTMASK_TARGETINFO = 60
} UA_BrowseResultMask;
```

4.8.191 ReferenceDescription

```
typedef struct {
    UA_NodeId referenceTypeId;
    UA_Boolean isForward;
    UA_ExpandedNodeId nodeId;
    UA_QualifiedName browseName;
    UA_LocalizedText displayName;
    UA_NodeClass nodeClass;
    UA_ExpandedNodeId typeDefinition;
} UA_ReferenceDescription;
```

4.8.192 ContinuationPoint

```
typedef UA_ByteString UA_ContinuationPoint;
```

4.8.193 BrowseResult

```
typedef struct {
    UA_StatusCode statusCode;
    UA_ByteString continuationPoint;
    size_t referencesSize;
    UA_ReferenceDescription *references;
} UA_BrowseResult;
```

4.8.194 BrowseRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_ViewDescription view;
    UA_UInt32 requestedMaxReferencesPerNode;
    size_t nodesToBrowseSize;
    UA_BrowseDescription *nodesToBrowse;
} UA_BrowseRequest;
```

4.8.195 BrowseResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_BrowseResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_BrowseResponse;
```

4.8.196 BrowseNextRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_Boolean releaseContinuationPoints;
    size_t continuationPointsSize;
    UA_ByteString *continuationPoints;
} UA_BrowseNextRequest;
```

4.8.197 BrowseNextResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_BrowseResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_BrowseNextResponse;
```

4.8.198 RelativePathElement

```
typedef struct {
    UA_NodeId referenceTypeId;
    UA_Boolean isInverse;
    UA_Boolean includeSubtypes;
    UA_QualifiedName targetName;
} UA_RelativePathElement;
```

4.8.199 RelativePath

```
typedef struct {
    size_t elementsSize;
    UA_RelativePathElement *elements;
} UA_RelativePath;
```

4.8.200 BrowsePath

```
typedef struct {
    UA_NodeId startingNode;
    UA_RelativePath relativePath;
} UA_BrowsePath;
```

4.8.201 BrowsePathTarget

```
typedef struct {
    UA_ExpandedNodeId targetId;
    UA_UInt32 remainingPathIndex;
} UA_BrowsePathTarget;
```

4.8.202 BrowsePathResult

```
typedef struct {
    UA_StatusCode statusCode;
    size_t targetsSize;
    UA_BrowsePathTarget *targets;
} UA_BrowsePathResult;
```

4.8.203 TranslateBrowsePathsToNodeIdsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t browsePathsSize;
    UA_BrowsePath *browsePaths;
} UA_TranslateBrowsePathsToNodeIdsRequest;
```

4.8.204 TranslateBrowsePathsToNodeIdsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_BrowsePathResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_TranslateBrowsePathsToNodeIdsResponse;
```

4.8.205 RegisterNodesRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t nodesToRegisterSize;
    UA_NodeId *nodesToRegister;
} UA_RegisterNodesRequest;
```

4.8.206 RegisterNodesResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t registeredNodeIdsSize;
    UA_NodeId *registeredNodeIds;
} UA_RegisterNodesResponse;
```

4.8.207 UnregisterNodesRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t nodesToUnregisterSize;
    UA_NodeId *nodesToUnregister;
} UA_UnregisterNodesRequest;
```

4.8.208 UnregisterNodesResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
} UA_UnregisterNodesResponse;
```

4.8.209 Counter

```
typedef UA_UInt32 UA_Counter;
```

4.8.210 OpaqueNumericRange

```
typedef UA_String UA_OpaqueNumericRange;
```

4.8.211 EndpointConfiguration

```
typedef struct {
    UA_Int32 operationTimeout;
    UA_Boolean useBinaryEncoding;
    UA_Int32 maxStringLength;
    UA_Int32 maxByteStringLength;
    UA_Int32 maxArrayLength;
    UA_Int32 maxMessageSize;
    UA_Int32 maxBufferSize;
```

(continues on next page)

(continued from previous page)

```
    UA_Int32 channelLifetime;
    UA_Int32 securityTokenLifetime;
} UA_EndpointConfiguration;
```

4.8.212 QueryDataDescription

```
typedef struct {
    UA_RelativePath relativePath;
    UA_UInt32 attributeId;
    UA_String indexRange;
} UA_QueryDataDescription;
```

4.8.213 NodeTypeDescription

```
typedef struct {
    UA_ExpandedNodeId typeDefinitionNode;
    UA_Boolean includeSubTypes;
    size_t dataToReturnSize;
    UA_QueryDataDescription *dataToReturn;
} UA_NodeTypeDescription;
```

4.8.214 FilterOperator

```
typedef enum {
    UA_FILTEROPERATOR_EQUALS = 0,
    UA_FILTEROPERATOR_ISNULL = 1,
    UA_FILTEROPERATOR_GREATERTHAN = 2,
    UA_FILTEROPERATOR_LESSTHAN = 3,
    UA_FILTEROPERATOR_GREATERTHANOREQUAL = 4,
    UA_FILTEROPERATOR_LESSTHANOREQUAL = 5,
    UA_FILTEROPERATOR_LIKE = 6,
    UA_FILTEROPERATOR_NOT = 7,
    UA_FILTEROPERATOR_BETWEEN = 8,
    UA_FILTEROPERATOR_INLIST = 9,
    UA_FILTEROPERATOR_AND = 10,
    UA_FILTEROPERATOR_OR = 11,
    UA_FILTEROPERATOR_CAST = 12,
    UA_FILTEROPERATOR_INVIEW = 13,
    UA_FILTEROPERATOR_OFTYPE = 14,
    UA_FILTEROPERATOR_RELATEDTO = 15,
    UA_FILTEROPERATOR_BITWISEAND = 16,
    UA_FILTEROPERATOR_BITWISEOR = 17
} UA_FilterOperator;
```

4.8.215 QueryDataSet

```
typedef struct {
    UA_ExpandedNodeId nodeId;
    UA_ExpandedNodeId typeDefinitionNode;
    size_t valuesSize;
    UA_Variant *values;
} UA_QueryDataSet;
```

4.8.216 NodeReference

```
typedef struct {
    UA_NodeId nodeId;
    UA_NodeId referenceTypeId;
    UA_Boolean isForward;
    size_t referencedNodeIdsSize;
    UA_NodeId *referencedNodeIds;
} UA_NodeReference;
```

4.8.217 ContentFilterElement

```
typedef struct {
    UA_FilterOperator filterOperator;
    size_t filterOperandsSize;
    UA_ExtensionObject *filterOperands;
} UA_ContentFilterElement;
```

4.8.218 ContentFilter

```
typedef struct {
    size_t elementsSize;
    UA_ContentFilterElement *elements;
} UA_ContentFilter;
```

4.8.219 ElementOperand

```
typedef struct {
    UA_UInt32 index;
} UA_ElementOperand;
```

4.8.220 LiteralOperand

```
typedef struct {
    UA_Variant value;
} UA_LiteralOperand;
```

4.8.221 AttributeOperand

```
typedef struct {
    UA_NodeId nodeId;
    UA_String alias;
    UA_RelativePath browsePath;
    UA_UInt32 attributeId;
    UA_String indexRange;
} UA_AttributeOperand;
```

4.8.222 SimpleAttributeOperand

```
typedef struct {
    UA_NodeId typeDefinitionId;
    size_t browsePathSize;
    UA_QualifiedName *browsePath;
    UA_UInt32 attributeId;
    UA_String indexRange;
} UA_SimpleAttributeOperand;
```

4.8.223 ContentFilterElementResult

```
typedef struct {
    UA_StatusCode statusCode;
    size_t operandStatusCodesSize;
    UA_StatusCode *operandStatusCodes;
    size_t operandDiagnosticInfosSize;
    UA_DiagnosticInfo *operandDiagnosticInfos;
} UA_ContentFilterElementResult;
```

4.8.224 ContentFilterResult

```
typedef struct {
    size_t elementResultsSize;
    UA_ContentFilterElementResult *elementResults;
    size_t elementDiagnosticInfosSize;
    UA_DiagnosticInfo *elementDiagnosticInfos;
} UA_ContentFilterResult;
```

4.8.225 ParsingResult

```
typedef struct {
    UA_StatusCode statusCode;
    size_t dataStatusCodesSize;
    UA_StatusCode *dataStatusCodes;
    size_t dataDiagnosticInfosSize;
    UA_DiagnosticInfo *dataDiagnosticInfos;
} UA_ParsingResult;
```

4.8.226 QueryFirstRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_ViewDescription view;
    size_t nodeTypesSize;
    UA_NodeTypeDescription *nodeTypes;
    UA_ContentFilter filter;
    UA_UInt32 maxDataSetsToReturn;
    UA_UInt32 maxReferencesToReturn;
} UA_QueryFirstRequest;
```

4.8.227 QueryFirstResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t queryDataSetsSize;
    UA_QueryDataSet *queryDataSets;
    UA_ByteString continuationPoint;
    size_t parsingResultsSize;
    UA_ParsingResult *parsingResults;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
    UA_ContentFilterResult filterResult;
} UA_QueryFirstResponse;
```

4.8.228 QueryNextRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_Boolean releaseContinuationPoint;
    UA_ByteString continuationPoint;
} UA_QueryNextRequest;
```

4.8.229 QueryNextResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t queryDataSetsSize;
    UA_QueryDataSet *queryDataSets;
    UA_ByteString revisedContinuationPoint;
} UA_QueryNextResponse;
```

4.8.230 TimestampsToReturn

```
typedef enum {
    UA_TIMESTAMPSTORETURN_SOURCE = 0,
    UA_TIMESTAMPSTORETURN_SERVER = 1,
    UA_TIMESTAMPSTORETURN_BOTH = 2,
```

(continues on next page)

(continued from previous page)

```
    UA_TIMESTAMPSTORETURN_NEITHER = 3,
    UA_TIMESTAMPSTORETURN_INVALID = 4
} UA_TimestampsToReturn;
```

4.8.231 ReadValueId

```
typedef struct {
    UA_NodeId nodeId;
    UA_UInt32 attributeId;
    UA_String indexRange;
    UA_QualifiedName dataEncoding;
} UA_ReadValueId;
```

4.8.232 ReadRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_Double maxAge;
    UA_TimestampsToReturn timestampsToReturn;
    size_t nodesToReadSize;
    UA_ReadValueId *nodesToRead;
} UA_ReadRequest;
```

4.8.233 ReadResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_DataValue *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_ReadResponse;
```

4.8.234 HistoryReadValueId

```
typedef struct {
    UA_NodeId nodeId;
    UA_String indexRange;
    UA_QualifiedName dataEncoding;
    UA_ByteString continuationPoint;
} UA_HistoryReadValueId;
```

4.8.235 HistoryReadResult

```
typedef struct {
    UA_StatusCode statusCode;
    UA_ByteString continuationPoint;
```

(continues on next page)

(continued from previous page)

```
    UA_ExtensionObject historyData;
} UA_HistoryReadResult;
```

4.8.236 ReadRawModifiedDetails

```
typedef struct {
    UA_Boolean isReadModified;
    UA_DateTime startTime;
    UA_DateTime endTime;
    UA_UInt32 numValuesPerNode;
    UA_Boolean returnBounds;
} UA_ReadRawModifiedDetails;
```

4.8.237 ReadAtTimeDetails

```
typedef struct {
    size_t reqTimesSize;
    UA_DateTime *reqTimes;
    UA_Boolean useSimpleBounds;
} UA_ReadAtTimeDetails;
```

4.8.238 ReadAnnotationDataDetails

```
typedef struct {
    size_t reqTimesSize;
    UA_DateTime *reqTimes;
} UA_ReadAnnotationDataDetails;
```

4.8.239 HistoryData

```
typedef struct {
    size_t dataValuesSize;
    UA_DataValue *dataValues;
} UA_HistoryData;
```

4.8.240 HistoryReadRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_ExtensionObject historyReadDetails;
    UA_TimestampsToReturn timestampsToReturn;
    UA_Boolean releaseContinuationPoints;
    size_t nodesToReadSize;
    UA_HistoryReadValueId *nodesToRead;
} UA_HistoryReadRequest;
```

4.8.241 HistoryReadResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_HistoryReadResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_HistoryReadResponse;
```

4.8.242 WriteValue

```
typedef struct {
    UA_NodeId nodeId;
    UA_UInt32 attributeId;
    UA_String indexRange;
    UA_DataValue value;
} UA_WriteValue;
```

4.8.243 WriteRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t nodesToWriteSize;
    UA_WriteValue *nodesToWrite;
} UA_WriteRequest;
```

4.8.244 WriteResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_WriteResponse;
```

4.8.245 HistoryUpdateDetails

```
typedef struct {
    UA_NodeId nodeId;
} UA_HistoryUpdateDetails;
```

4.8.246 HistoryUpdateType

```
typedef enum {
    UA_HISTORYUPDATETYPE_INSERT = 1,
    UA_HISTORYUPDATETYPE_REPLACE = 2,
    UA_HISTORYUPDATETYPE_UPDATE = 3,
```

(continues on next page)

(continued from previous page)

```
    UA_HISTORYUPDATETYPE_DELETE = 4
} UA_HistoryUpdateType;
```

4.8.247 PerformUpdateType

```
typedef enum {
    UA_PERFORMUPDATETYPE_INSERT = 1,
    UA_PERFORMUPDATETYPE_REPLACE = 2,
    UA_PERFORMUPDATETYPE_UPDATE = 3,
    UA_PERFORMUPDATETYPE_REMOVE = 4
} UA_PerformUpdateType;
```

4.8.248 UpdateDataDetails

```
typedef struct {
    UA_NodeId nodeId;
    UA_PerformUpdateType performInsertReplace;
    size_t updateValuesSize;
    UA_DataValue *updateValues;
} UA_UpdateDataDetails;
```

4.8.249 UpdateStructureDataDetails

```
typedef struct {
    UA_NodeId nodeId;
    UA_PerformUpdateType performInsertReplace;
    size_t updateValuesSize;
    UA_DataValue *updateValues;
} UA_UpdateStructureDataDetails;
```

4.8.250 DeleteRawModifiedDetails

```
typedef struct {
    UA_NodeId nodeId;
    UA_Boolean isDeleteModified;
    UA_DateTime startTime;
    UA_DateTime endTime;
} UA_DeleteRawModifiedDetails;
```

4.8.251 DeleteAtTimeDetails

```
typedef struct {
    UA_NodeId nodeId;
    size_t reqTimesSize;
    UA_DateTime *reqTimes;
} UA_DeleteAtTimeDetails;
```

4.8.252 DeleteEventDetails

```
typedef struct {
    UA_NodeId nodeId;
    size_t eventIdsSize;
    UA_ByteString *eventIds;
} UA_DeleteEventDetails;
```

4.8.253 HistoryUpdateResult

```
typedef struct {
    UA_StatusCode statusCode;
    size_t operationResultsSize;
    UA_StatusCode *operationResults;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_HistoryUpdateResult;
```

4.8.254 HistoryUpdateRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t historyUpdateDetailsSize;
    UA_ExtensionObject *historyUpdateDetails;
} UA_HistoryUpdateRequest;
```

4.8.255 HistoryUpdateResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_HistoryUpdateResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_HistoryUpdateResponse;
```

4.8.256 CallMethodRequest

```
typedef struct {
    UA_NodeId objectId;
    UA_NodeId methodId;
    size_t inputArgumentsSize;
    UA_Variant *inputArguments;
} UA_CallMethodRequest;
```

4.8.257 CallMethodResult

```
typedef struct {
    UA_StatusCode statusCode;
    size_t inputArgumentResultsSize;
    UA_StatusCode *inputArgumentResults;
    size_t inputArgumentDiagnosticInfosSize;
    UA_DiagnosticInfo *inputArgumentDiagnosticInfos;
    size_t outputArgumentsSize;
    UA_Variant *outputArguments;
} UA_CallMethodResult;
```

4.8.258 CallRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t methodsToCallSize;
    UA_CallMethodRequest *methodsToCall;
} UA_CallRequest;
```

4.8.259 CallResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_CallMethodResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_CallResponse;
```

4.8.260 MonitoringMode

```
typedef enum {
    UA_MONITORINGMODE_DISABLED = 0,
    UA_MONITORINGMODE_SAMPLING = 1,
    UA_MONITORINGMODE_REPORTING = 2
} UA_MonitoringMode;
```

4.8.261 DataChangeTrigger

```
typedef enum {
    UA_DATACHANGETRIGGER_STATUS = 0,
    UA_DATACHANGETRIGGER_STATUSVALUE = 1,
    UA_DATACHANGETRIGGER_STATUSVALUETIMESTAMP = 2
} UA_DataChangeTrigger;
```

4.8.262 DeadbandType

```
typedef enum {
    UA_DEADBANDTYPE_NONE = 0,
    UA_DEADBANDTYPE_ABSOLUTE = 1,
    UA_DEADBANDTYPE_PERCENT = 2
} UA_DeadbandType;
```

4.8.263 DataChangeFilter

```
typedef struct {
    UA_DataChangeTrigger trigger;
    UA_UInt32 deadbandType;
    UA_Double deadbandValue;
} UA_DataChangeFilter;
```

4.8.264 EventFilter

```
typedef struct {
    size_t selectClausesSize;
    UA_SimpleAttributeOperand *selectClauses;
    UA_ContentFilter whereClause;
} UA_EventFilter;
```

4.8.265 AggregateConfiguration

```
typedef struct {
    UA_Boolean useServerCapabilitiesDefaults;
    UA_Boolean treatUncertainAsBad;
    UA_Byte percentDataBad;
    UA_Byte percentDataGood;
    UA_Boolean useSlopedExtrapolation;
} UA_AggregateConfiguration;
```

4.8.266 AggregateFilter

```
typedef struct {
    UA_DateTime startTime;
    UA_NodeId aggregateType;
    UA_Double processingInterval;
    UA_AggregateConfiguration aggregateConfiguration;
} UA_AggregateFilter;
```

4.8.267 EventFilterResult

```
typedef struct {
    size_t selectClauseResultsSize;
    UA_StatusCode *selectClauseResults;
    size_t selectClauseDiagnosticInfosSize;
```

(continues on next page)

(continued from previous page)

```
    UA_DiagnosticInfo *selectClauseDiagnosticInfos;
    UA_ContentFilterResult whereClauseResult;
} UA_EventFilterResult;
```

4.8.268 AggregateFilterResult

```
typedef struct {
    UA_DateTime revisedStartTime;
    UA_Double revisedProcessingInterval;
    UA_AggregateConfiguration revisedAggregateConfiguration;
} UA_AggregateFilterResult;
```

4.8.269 MonitoringParameters

```
typedef struct {
    UA_UInt32 clientHandle;
    UA_Double samplingInterval;
    UA_ExtensionObject filter;
    UA_UInt32 queueSize;
    UA_Boolean discardOldest;
} UA_MonitoringParameters;
```

4.8.270 MonitoredItemCreateRequest

```
typedef struct {
    UA_ReadValueId itemToMonitor;
    UA_MonitoringMode monitoringMode;
    UA_MonitoringParameters requestedParameters;
} UA_MonitoredItemCreateRequest;
```

4.8.271 MonitoredItemCreateResult

```
typedef struct {
    UA_StatusCode statusCode;
    UA_UInt32 monitoredItemId;
    UA_Double revisedSamplingInterval;
    UA_UInt32 revisedQueueSize;
    UA_ExtensionObject filterResult;
} UA_MonitoredItemCreateResult;
```

4.8.272 CreateMonitoredItemsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    UA_TimestampsToReturn timestampsToReturn;
    size_t itemsToCreateSize;
```

(continues on next page)

(continued from previous page)

```
    UA_MonitoredItemCreateRequest *itemsToCreate;
} UA_CreateMonitoredItemsRequest;
```

4.8.273 CreateMonitoredItemsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_MonitoredItemCreateResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_CreateMonitoredItemsResponse;
```

4.8.274 MonitoredItemModifyRequest

```
typedef struct {
    UA_UInt32 monitoredItemId;
    UA_MonitoringParameters requestedParameters;
} UA_MonitoredItemModifyRequest;
```

4.8.275 MonitoredItemModifyResult

```
typedef struct {
    UA_StatusCode statusCode;
    UA_Double revisedSamplingInterval;
    UA_UInt32 revisedQueueSize;
    UA_ExtensionObject filterResult;
} UA_MonitoredItemModifyResult;
```

4.8.276 ModifyMonitoredItemsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    UA_TimestampsToReturn timestampsToReturn;
    size_t itemsToModifySize;
    UA_MonitoredItemModifyRequest *itemsToModify;
} UA_ModifyMonitoredItemsRequest;
```

4.8.277 ModifyMonitoredItemsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_MonitoredItemModifyResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_ModifyMonitoredItemsResponse;
```

4.8.278 SetMonitoringModeRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    UA_MonitoringMode monitoringMode;
    size_t monitoredItemIdsSize;
    UA_UInt32 *monitoredItemIds;
} UA_SetMonitoringModeRequest;
```

4.8.279 SetMonitoringModeResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_SetMonitoringModeResponse;
```

4.8.280 SetTriggeringRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    UA_UInt32 triggeringItemId;
    size_t linksToAddSize;
    UA_UInt32 *linksToAdd;
    size_t linksToRemoveSize;
    UA_UInt32 *linksToRemove;
} UA_SetTriggeringRequest;
```

4.8.281 SetTriggeringResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t addResultsSize;
    UA_StatusCode *addResults;
    size_t addDiagnosticInfosSize;
    UA_DiagnosticInfo *addDiagnosticInfos;
    size_t removeResultsSize;
    UA_StatusCode *removeResults;
    size_t removeDiagnosticInfosSize;
    UA_DiagnosticInfo *removeDiagnosticInfos;
} UA_SetTriggeringResponse;
```

4.8.282 DeleteMonitoredItemsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    size_t monitoredItemIdsSize;
    UA_UInt32 *monitoredItemIds;
} UA_DeleteMonitoredItemsRequest;
```

4.8.283 DeleteMonitoredItemsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_DeleteMonitoredItemsResponse;
```

4.8.284 CreateSubscriptionRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_Double requestedPublishingInterval;
    UA_UInt32 requestedLifetimeCount;
    UA_UInt32 requestedMaxKeepAliveCount;
    UA_UInt32 maxNotificationsPerPublish;
    UA_Boolean publishingEnabled;
    UA_Byte priority;
} UA_CreateSubscriptionRequest;
```

4.8.285 CreateSubscriptionResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_UInt32 subscriptionId;
    UA_Double revisedPublishingInterval;
    UA_UInt32 revisedLifetimeCount;
    UA_UInt32 revisedMaxKeepAliveCount;
} UA_CreateSubscriptionResponse;
```

4.8.286 ModifySubscriptionRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    UA_Double requestedPublishingInterval;
    UA_UInt32 requestedLifetimeCount;
    UA_UInt32 requestedMaxKeepAliveCount;
```

(continues on next page)

(continued from previous page)

```
    UA_UInt32 maxNotificationsPerPublish;
    UA_Byte priority;
} UA_ModifySubscriptionRequest;
```

4.8.287 ModifySubscriptionResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_Double revisedPublishingInterval;
    UA_UInt32 revisedLifetimeCount;
    UA_UInt32 revisedMaxKeepAliveCount;
} UA_ModifySubscriptionResponse;
```

4.8.288 SetPublishingModeRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_Boolean publishingEnabled;
    size_t subscriptionIdsSize;
    UA_UInt32 *subscriptionIds;
} UA_SetPublishingModeRequest;
```

4.8.289 SetPublishingModeResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_SetPublishingModeResponse;
```

4.8.290 NotificationMessage

```
typedef struct {
    UA_UInt32 sequenceNumber;
    UA_DateTime publishTime;
    size_t notificationDataSize;
    UA_ExtensionObject *notificationData;
} UA_NotificationMessage;
```

4.8.291 MonitoredItemNotification

```
typedef struct {
    UA_UInt32 clientHandle;
    UA_DataValue value;
} UA_MonitoredItemNotification;
```

4.8.292 EventFieldList

```
typedef struct {
    UA_UInt32 clientHandle;
    size_t eventFieldsSize;
    UA_Variant *eventFields;
} UA_EventFieldList;
```

4.8.293 HistoryEventFieldList

```
typedef struct {
    size_t eventFieldsSize;
    UA_Variant *eventFields;
} UA_HistoryEventFieldList;
```

4.8.294 StatusChangeNotification

```
typedef struct {
    UA_StatusCode status;
    UA_DiagnosticInfo diagnosticInfo;
} UA_StatusChangeNotification;
```

4.8.295 SubscriptionAcknowledgement

```
typedef struct {
    UA_UInt32 subscriptionId;
    UA_UInt32 sequenceNumber;
} UA_SubscriptionAcknowledgement;
```

4.8.296 PublishRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t subscriptionAcknowledgementsSize;
    UA_SubscriptionAcknowledgement *subscriptionAcknowledgements;
} UA_PublishRequest;
```

4.8.297 PublishResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_UInt32 subscriptionId;
    size_t availableSequenceNumbersSize;
    UA_UInt32 *availableSequenceNumbers;
    UA_Boolean moreNotifications;
    UA_NotificationMessage notificationMessage;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
```

(continues on next page)

(continued from previous page)

```
    UA_DiagnosticInfo *diagnosticInfos;
} UA_PublishResponse;
```

4.8.298 RepublishRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    UA_UInt32 subscriptionId;
    UA_UInt32 retransmitSequenceNumber;
} UA_RepublishRequest;
```

4.8.299 RepublishResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    UA_NotificationMessage notificationMessage;
} UA_RepublishResponse;
```

4.8.300 TransferResult

```
typedef struct {
    UA_StatusCode statusCode;
    size_t availableSequenceNumbersSize;
    UA_UInt32 *availableSequenceNumbers;
} UA_TransferResult;
```

4.8.301 TransferSubscriptionsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t subscriptionIdsSize;
    UA_UInt32 *subscriptionIds;
    UA_Boolean sendInitialValues;
} UA_TransferSubscriptionsRequest;
```

4.8.302 TransferSubscriptionsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_TransferResult *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_TransferSubscriptionsResponse;
```

4.8.303 DeleteSubscriptionsRequest

```
typedef struct {
    UA_RequestHeader requestHeader;
    size_t subscriptionIdsSize;
    UA_UInt32 *subscriptionIds;
} UA_DeleteSubscriptionsRequest;
```

4.8.304 DeleteSubscriptionsResponse

```
typedef struct {
    UA_ResponseHeader responseHeader;
    size_t resultsSize;
    UA_StatusCode *results;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_DeleteSubscriptionsResponse;
```

4.8.305 BuildInfo

```
typedef struct {
    UA_String productUri;
    UA_String manufacturerName;
    UA_String productName;
    UA_String softwareVersion;
    UA_String buildNumber;
    UA_DateTime buildDate;
} UA_BuildInfo;
```

4.8.306 RedundancySupport

```
typedef enum {
    UA_REDUNDANCYSUPPORT_NONE = 0,
    UA_REDUNDANCYSUPPORT_COLD = 1,
    UA_REDUNDANCYSUPPORT_WARM = 2,
    UA_REDUNDANCYSUPPORT_HOT = 3,
    UA_REDUNDANCYSUPPORT_TRANSPARENT = 4,
    UA_REDUNDANCYSUPPORT_HOTANDMIRRORED = 5
} UA_RedundancySupport;
```

4.8.307 ServerState

```
typedef enum {
    UA_SERVERSTATE_RUNNING = 0,
    UA_SERVERSTATE_FAILED = 1,
    UA_SERVERSTATE_NOCONFIGURATION = 2,
    UA_SERVERSTATE_SUSPENDED = 3,
    UA_SERVERSTATE_SHUTDOWN = 4,
    UA_SERVERSTATE_TEST = 5,
```

(continues on next page)

(continued from previous page)

```
    UA_SERVERSTATE_COMMUNICATIONFAULT = 6,
    UA_SERVERSTATE_UNKNOWN = 7
} UA_ServerState;
```

4.8.308 RedundantServerDataType

```
typedef struct {
    UA_String serverId;
    UA_Byte serviceLevel;
    UA_ServerState serverState;
} UA_RedundantServerDataType;
```

4.8.309 EndpointUrlListDataType

```
typedef struct {
    size_t endpointUrlListSize;
    UA_String *endpointUrlList;
} UA_EndpointUrlListDataType;
```

4.8.310 NetworkGroupDataType

```
typedef struct {
    UA_String serverUri;
    size_t networkPathsSize;
    UA_EndpointUrlListDataType *networkPaths;
} UA_NetworkGroupDataType;
```

4.8.311 SamplingIntervalDiagnosticsDataType

```
typedef struct {
    UA_Double samplingInterval;
    UA_UInt32 monitoredItemCount;
    UA_UInt32 maxMonitoredItemCount;
    UA_UInt32 disabledMonitoredItemCount;
} UA_SamplingIntervalDiagnosticsDataType;
```

4.8.312 ServerDiagnosticsSummaryDataType

```
typedef struct {
    UA_UInt32 serverViewCount;
    UA_UInt32 currentSessionCount;
    UA_UInt32 cumulatedSessionCount;
    UA_UInt32 securityRejectedSessionCount;
    UA_UInt32 rejectedSessionCount;
    UA_UInt32 sessionTimeoutCount;
    UA_UInt32 sessionAbortCount;
    UA_UInt32 currentSubscriptionCount;
```

(continues on next page)

(continued from previous page)

```
    UA_UInt32 cumulatedSubscriptionCount;
    UA_UInt32 publishingIntervalCount;
    UA_UInt32 securityRejectedRequestsCount;
    UA_UInt32 rejectedRequestsCount;
} UA_ServerDiagnosticsSummaryDataType;
```

4.8.313 ServerStatusDataType

```
typedef struct {
    UA_DateTime startTime;
    UA_DateTime currentTime;
    UA_ServerState state;
    UA_BuildInfo buildInfo;
    UA_UInt32 secondsTillShutdown;
    UA_LocalizedText shutdownReason;
} UA_ServerStatusDataType;
```

4.8.314 SessionSecurityDiagnosticsDataType

```
typedef struct {
    UA_NodeId sessionId;
    UA_String clientUserIdOfSession;
    size_t clientUserIdHistorySize;
    UA_String *clientUserIdHistory;
    UA_String authenticationMechanism;
    UA_String encoding;
    UA_String transportProtocol;
    UA_MessageSecurityMode securityMode;
    UA_String securityPolicyUri;
    UA_ByteString clientCertificate;
} UA_SessionSecurityDiagnosticsDataType;
```

4.8.315 ServiceCounterDataType

```
typedef struct {
    UA_UInt32 totalCount;
    UA_UInt32 errorCount;
} UA_ServiceCounterDataType;
```

4.8.316 StatusResult

```
typedef struct {
    UA_StatusCode statusCode;
    UA_DiagnosticInfo diagnosticInfo;
} UA_StatusResult;
```

4.8.317 SubscriptionDiagnosticsDataType

```
typedef struct {
    UA_NodeId sessionId;
    UA_UInt32 subscriptionId;
    UA_Byte priority;
    UA_Double publishingInterval;
    UA_UInt32 maxKeepAliveCount;
    UA_UInt32 maxLifetimeCount;
    UA_UInt32 maxNotificationsPerPublish;
    UA_Boolean publishingEnabled;
    UA_UInt32 modifyCount;
    UA_UInt32 enableCount;
    UA_UInt32 disableCount;
    UA_UInt32 republishRequestCount;
    UA_UInt32 republishMessageRequestCount;
    UA_UInt32 republishMessageCount;
    UA_UInt32 transferRequestCount;
    UA_UInt32 transferredToAltClientCount;
    UA_UInt32 transferredToSameClientCount;
    UA_UInt32 publishRequestCount;
    UA_UInt32 dataChangeNotificationsCount;
    UA_UInt32 eventNotificationsCount;
    UA_UInt32 notificationsCount;
    UA_UInt32 latePublishRequestCount;
    UA_UInt32 currentKeepAliveCount;
    UA_UInt32 currentLifetimeCount;
    UA_UInt32 unacknowledgedMessageCount;
    UA_UInt32 discardedMessageCount;
    UA_UInt32 monitoredItemCount;
    UA_UInt32 disabledMonitoredItemCount;
    UA_UInt32 monitoringQueueOverflowCount;
    UA_UInt32 nextSequenceNumber;
    UA_UInt32 eventQueueOverflowCount;
} UA_SubscriptionDiagnosticsDataType;
```

4.8.318 ModelChangeStructureVerbMask

```
typedef enum {
    UA_MODELCHANGESTRUCTUREVERBMASK_NODEADDED = 1,
    UA_MODELCHANGESTRUCTUREVERBMASK_NODEDELETED = 2,
    UA_MODELCHANGESTRUCTUREVERBMASK_REFERENCEADDED = 4,
    UA_MODELCHANGESTRUCTUREVERBMASK_REFERENCEDELETED = 8,
    UA_MODELCHANGESTRUCTUREVERBMASK_DATATYPECHANGED = 16
} UA_ModelChangeStructureVerbMask;
```

4.8.319 ModelChangeStructureDataType

```
typedef struct {
    UA_NodeId affected;
    UA_NodeId affectedType;
    UA_Byte verb;
} UA_ModelChangeStructureDataType;
```

4.8.320 SemanticChangeStructureDataType

```
typedef struct {
    UA_NodeId affected;
    UA_NodeId affectedType;
} UA_SemanticChangeStructureDataType;
```

4.8.321 Range

```
typedef struct {
    UA_Double low;
    UA_Double high;
} UA_Range;
```

4.8.322 EUInformation

```
typedef struct {
    UA_String namespaceUri;
    UA_Int32 unitId;
    UA_LocalizedText displayName;
    UA_LocalizedText description;
} UA_EUInformation;
```

4.8.323 AxisScaleEnumeration

```
typedef enum {
    UA_AXISSCALEENUMERATION_LINEAR = 0,
    UA_AXISSCALEENUMERATION_LOG = 1,
    UA_AXISSCALEENUMERATION_LN = 2
} UA_AxisScaleEnumeration;
```

4.8.324 ComplexNumberType

```
typedef struct {
    UA_Float real;
    UA_Float imaginary;
} UA_ComplexNumberType;
```

4.8.325 DoubleComplexNumberType

```
typedef struct {
    UA_Double real;
    UA_Double imaginary;
} UA_DoubleComplexNumberType;
```

4.8.326 AxisInformation

```
typedef struct {
    UA_EUInformation engineeringUnits;
    UA_Range eURange;
    UA_LocalizedText title;
    UA_AxisScaleEnumeration axisScaleType;
    size_t axisStepsSize;
    UA_Double *axisSteps;
} UA_AxisInformation;
```

4.8.327 XVType

```
typedef struct {
    UA_Double x;
    UA_Float value;
} UA_XVType;
```

4.8.328 ProgramDiagnosticDataType

```
typedef struct {
    UA_NodeId createSessionId;
    UA_String createClientName;
    UA_DateTime invocationCreationTime;
    UA_DateTime lastTransitionTime;
    UA_String lastMethodCall;
    UA_NodeId lastMethodSessionId;
    size_t lastMethodInputArgumentsSize;
    UA_Argument *lastMethodInputArguments;
    size_t lastMethodOutputArgumentsSize;
    UA_Argument *lastMethodOutputArguments;
    UA_DateTime lastMethodCallTime;
    UA_StatusResult lastMethodReturnStatus;
} UA_ProgramDiagnosticDataType;
```

4.8.329 ProgramDiagnostic2DataType

```
typedef struct {
    UA_NodeId createSessionId;
    UA_String createClientName;
    UA_DateTime invocationCreationTime;
    UA_DateTime lastTransitionTime;
```

(continues on next page)

(continued from previous page)

```
    UA_String lastMethodCall;
    UA_NodeId lastMethodSessionId;
    size_t lastMethodInputArgumentsSize;
    UA_Argument *lastMethodInputArguments;
    size_t lastMethodOutputArgumentsSize;
    UA_Argument *lastMethodOutputArguments;
    size_t lastMethodInputValuesSize;
    UA_Variant *lastMethodInputValues;
    size_t lastMethodOutputValuesSize;
    UA_Variant *lastMethodOutputValues;
    UA_DateTime lastMethodCallTime;
    UA_StatusCode lastMethodReturnStatus;
} UA_ProgramDiagnostic2DataType;
```

4.8.330 Annotation

```
typedef struct {
    UA_String message;
    UA_String userName;
    UA_DateTime annotationTime;
} UA_Annotation;
```

4.8.331 ExceptionDeviationFormat

```
typedef enum {
    UA_EXCEPTIONDEVIATIONFORMAT_ABSOLUTEVALUE = 0,
    UA_EXCEPTIONDEVIATIONFORMAT_PERCENTOFVALUE = 1,
    UA_EXCEPTIONDEVIATIONFORMAT_PERCENTOFRANGE = 2,
    UA_EXCEPTIONDEVIATIONFORMAT_PERCENTOFEURANGE = 3,
    UA_EXCEPTIONDEVIATIONFORMAT_UNKNOWN = 4
} UA_ExceptionDeviationFormat;
```

4.8.332 EndpointType

```
typedef struct {
    UA_String endpointUrl;
    UA_MessageSecurityMode securityMode;
    UA_String securityPolicyUri;
    UA_String transportProfileUri;
} UA_EndpointType;
```

4.8.333 StructureDescription

```
typedef struct {
    UA_NodeId dataTypeId;
    UA_QualifiedName name;
    UA_StructureDefinition structureDefinition;
} UA_StructureDescription;
```

4.8.334 FieldMetaData

```
typedef struct {
    UA_String name;
    UA_LocalizedText description;
    UA_DataSetFieldFlags fieldFlags;
    UA_Byte builtInType;
    UA_NodeId dataType;
    UA_Int32 valueRank;
    size_t arrayDimensionsSize;
    UA_UInt32 *arrayDimensions;
    UA_UInt32 maxStringLength;
    UA_Guid dataSetFieldId;
    size_t propertiesSize;
    UA_KeyValuePair *properties;
} UA_FieldMetaData;
```

4.8.335 PublishedEventsDataType

```
typedef struct {
    UA_NodeId eventNotifier;
    size_t selectedFieldsSize;
    UA_SimpleAttributeOperand *selectedFields;
    UA_ContentFilter filter;
} UA_PublishedEventsDataType;
```

4.8.336 PubSubGroupDataType

```
typedef struct {
    UA_String name;
    UA_Boolean enabled;
    UA_MessageSecurityMode securityMode;
    UA_String securityGroupId;
    size_t securityKeyServicesSize;
    UA_EndpointDescription *securityKeyServices;
    UA_UInt32 maxNetworkMessageSize;
    size_t groupPropertiesSize;
    UA_KeyValuePair *groupProperties;
} UA_PubSubGroupDataType;
```

4.8.337 WriterGroupDataType

```
typedef struct {
    UA_String name;
    UA_Boolean enabled;
    UA_MessageSecurityMode securityMode;
    UA_String securityGroupId;
    size_t securityKeyServicesSize;
    UA_EndpointDescription *securityKeyServices;
    UA_UInt32 maxNetworkMessageSize;
```

(continues on next page)

(continued from previous page)

```
    size_t groupPropertiesSize;
    UA_KeyValuePair *groupProperties;
    UA_UInt16 writerGroupId;
    UA_Double publishingInterval;
    UA_Double keepAliveTime;
    UA_Byte priority;
    size_t localeIdsSize;
    UA_String *localeIds;
    UA_String headerLayoutUri;
    UA_ExtensionObject transportSettings;
    UA_ExtensionObject messageSettings;
    size_t dataSetWritersSize;
    UA_DataSetWriterDataType *dataSetWriters;
} UA_WriterGroupDataType;
```

4.8.338 FieldTargetDataType

```
typedef struct {
    UA_Guid dataSetFieldId;
    UA_String receiverIndexRange;
    UA_NodeId targetNodeId;
    UA_UInt32 attributeId;
    UA_String writeIndexRange;
    UA_OverrideValueHandling overrideValueHandling;
    UA_Variant overrideValue;
} UA_FieldTargetDataType;
```

4.8.339 SubscribedDataSetMirrorDataType

```
typedef struct {
    UA_String parentNodeName;
    size_t rolePermissionsSize;
    UA_RolePermissionType *rolePermissions;
} UA_SubscribedDataSetMirrorDataType;
```

4.8.340 SecurityGroupDataType

```
typedef struct {
    UA_String name;
    size_t securityGroupFolderSize;
    UA_String *securityGroupFolder;
    UA_Double keyLifetime;
    UA_String securityPolicyUri;
    UA_UInt32 maxFutureKeyCount;
    UA_UInt32 maxPastKeyCount;
    UA_String securityGroupId;
    size_t rolePermissionsSize;
    UA_RolePermissionType *rolePermissions;
```

(continues on next page)

(continued from previous page)

```
    size_t groupPropertiesSize;
    UA_KeyValuePair *groupProperties;
} UA_SecurityGroupDataType;
```

4.8.341 PubSubKeyPushTargetDataType

```
typedef struct {
    UA_String applicationUri;
    size_t pushTargetFolderSize;
    UA_String *pushTargetFolder;
    UA_String endpointUrl;
    UA_String securityPolicyUri;
    UA_UserTokenPolicy userTokenType;
    UA_UInt16 requestedKeyCount;
    UA_Double retryInterval;
    size_t pushTargetPropertiesSize;
    UA_KeyValuePair *pushTargetProperties;
    size_t securityGroupsSize;
    UA_String *securityGroups;
} UA_PubSubKeyPushTargetDataType;
```

4.8.342 EnumDefinition

```
typedef struct {
    size_t fieldsSize;
    UA_EnumField *fields;
} UA_EnumDefinition;
```

4.8.343 ReadEventDetails

```
typedef struct {
    UA_UInt32 numValuesPerNode;
    UA_DateTime startTime;
    UA_DateTime endTime;
    UA_EventFilter filter;
} UA_ReadEventDetails;
```

4.8.344 ReadProcessedDetails

```
typedef struct {
    UA_DateTime startTime;
    UA_DateTime endTime;
    UA_Double processingInterval;
    size_t aggregateTypeSize;
    UA_NodeId *aggregateType;
    UA_AggregateConfiguration aggregateConfiguration;
} UA_ReadProcessedDetails;
```

4.8.345 ModificationInfo

```
typedef struct {
    UA_DateTime modificationTime;
    UA_HistoryUpdateType updateType;
    UA_String userName;
} UA_ModificationInfo;
```

4.8.346 HistoryModifiedData

```
typedef struct {
    size_t dataValuesSize;
    UA_DataValue *dataValues;
    size_t modificationInfosSize;
    UA_ModificationInfo *modificationInfos;
} UA_HistoryModifiedData;
```

4.8.347 HistoryEvent

```
typedef struct {
    size_t eventsSize;
    UA_HistoryEventFieldList *events;
} UA_HistoryEvent;
```

4.8.348 UpdateEventDetails

```
typedef struct {
    UA_NodeId nodeId;
    UA_PerformUpdateType performInsertReplace;
    UA_EventFilter filter;
    size_t eventDataSize;
    UA_HistoryEventFieldList *eventData;
} UA_UpdateEventDetails;
```

4.8.349 DataChangeNotification

```
typedef struct {
    size_t monitoredItemsSize;
    UA_MonitoredItemNotification *monitoredItems;
    size_t diagnosticInfosSize;
    UA_DiagnosticInfo *diagnosticInfos;
} UA_DataChangeNotification;
```

4.8.350 EventNotificationList

```
typedef struct {
    size_t eventsSize;
    UA_EventFieldList *events;
} UA_EventNotificationList;
```

4.8.351 SessionDiagnosticsDataType

```
typedef struct {
    UA_NodeId sessionId;
    UA_String sessionName;
    UA_ApplicationDescription clientDescription;
    UA_String serverUri;
    UA_String endpointUrl;
    size_t localeIdsSize;
    UA_String *localeIds;
    UA_Double actualSessionTimeout;
    UA_UInt32 maxResponseMessageSize;
    UA_DateTime clientConnectionTime;
    UA_DateTime clientLastContactTime;
    UA_UInt32 currentSubscriptionsCount;
    UA_UInt32 currentMonitoredItemsCount;
    UA_UInt32 currentPublishRequestsInQueue;
    UA_ServiceCounterDataType totalRequestCount;
    UA_UInt32 unauthorizedRequestCount;
    UA_ServiceCounterDataType readCount;
    UA_ServiceCounterDataType historyReadCount;
    UA_ServiceCounterDataType writeCount;
    UA_ServiceCounterDataType historyUpdateCount;
    UA_ServiceCounterDataType callCount;
    UA_ServiceCounterDataType createMonitoredItemsCount;
    UA_ServiceCounterDataType modifyMonitoredItemsCount;
    UA_ServiceCounterDataType setMonitoringModeCount;
    UA_ServiceCounterDataType setTriggeringCount;
    UA_ServiceCounterDataType deleteMonitoredItemsCount;
    UA_ServiceCounterDataType createSubscriptionCount;
    UA_ServiceCounterDataType modifySubscriptionCount;
    UA_ServiceCounterDataType setPublishingModeCount;
    UA_ServiceCounterDataType publishCount;
    UA_ServiceCounterDataType republishCount;
    UA_ServiceCounterDataType transferSubscriptionsCount;
    UA_ServiceCounterDataType deleteSubscriptionsCount;
    UA_ServiceCounterDataType addNodesCount;
    UA_ServiceCounterDataType addReferencesCount;
    UA_ServiceCounterDataType deleteNodesCount;
    UA_ServiceCounterDataType deleteReferencesCount;
    UA_ServiceCounterDataType browseCount;
    UA_ServiceCounterDataType browseNextCount;
    UA_ServiceCounterDataType translateBrowsePathsToNodeIdsCount;
    UA_ServiceCounterDataType queryFirstCount;
    UA_ServiceCounterDataType queryNextCount;
    UA_ServiceCounterDataType registerNodesCount;
    UA_ServiceCounterDataType unregisterNodesCount;
} UA_SessionDiagnosticsDataType;
```

4.8.352 EnumDescription

```
typedef struct {
    UA_NodeId dataTypeId;
    UA_QualifiedName name;
    UA_EnumDefinition enumDefinition;
    UA_Byte builtInType;
} UA_EnumDescription;
```

4.8.353 UABinaryFileDataType

```
typedef struct {
    size_t namespacesSize;
    UA_String *namespaces;
    size_t structureDataTypesSize;
    UA_StructureDescription *structureDataTypes;
    size_t enumDataTypesSize;
    UA_EnumDescription *enumDataTypes;
    size_t simpleDataTypesSize;
    UA_SimpleTypeDescription *simpleDataTypes;
    UA_String schemaLocation;
    size_t fileHeaderSize;
    UA_KeyValuePair *fileHeader;
    UA_Variant body;
} UA_UABinaryFileDataType;
```

4.8.354 DataSetMetaDataType

```
typedef struct {
    size_t namespacesSize;
    UA_String *namespaces;
    size_t structureDataTypesSize;
    UA_StructureDescription *structureDataTypes;
    size_t enumDataTypesSize;
    UA_EnumDescription *enumDataTypes;
    size_t simpleDataTypesSize;
    UA_SimpleTypeDescription *simpleDataTypes;
    UA_String name;
    UA_LocalizedText description;
    size_t fieldsSize;
    UA_FieldMetaData *fields;
    UA_Guid dataSetClassId;
    UA_ConfigurationVersionDataType configurationVersion;
} UA_DataSetMetaDataType;
```

4.8.355 PublishedDataSetDataType

```
typedef struct {
    UA_String name;
    size_t dataSetFolderSize;
```

(continues on next page)

(continued from previous page)

```
    UA_String *dataSetFolder;
    UA_DataSetMetaDataType dataSetMetaData;
    size_t extensionFieldsSize;
    UA_KeyValuePair *extensionFields;
    UA_ExtensionObject dataSetSource;
} UA_PublishedDataSetDataType;
```

4.8.356 DataSetReaderDataType

```
typedef struct {
    UA_String name;
    UA_Boolean enabled;
    UA_Variant publisherId;
    UA_UInt16 writerGroupId;
    UA_UInt16 dataSetWriterId;
    UA_DataSetMetaDataType dataSetMetaData;
    UA_DataSetFieldContentMask dataSetFieldContentMask;
    UA_Double messageReceiveTimeout;
    UA_UInt32 keyFrameCount;
    UA_String headerLayoutUri;
    UA_MessageSecurityMode securityMode;
    UA_String securityGroupId;
    size_t securityKeyServicesSize;
    UA_EndpointDescription *securityKeyServices;
    size_t dataSetReaderPropertiesSize;
    UA_KeyValuePair *dataSetReaderProperties;
    UA_ExtensionObject transportSettings;
    UA_ExtensionObject messageSettings;
    UA_ExtensionObject subscribedDataSet;
} UA_DataSetReaderDataType;
```

4.8.357 TargetVariablesDataType

```
typedef struct {
    size_t targetVariablesSize;
    UA_FieldTargetDataType *targetVariables;
} UA_TargetVariablesDataType;
```

4.8.358 StandaloneSubscribedDataSetDataType

```
typedef struct {
    UA_String name;
    size_t dataSetFolderSize;
    UA_String *dataSetFolder;
    UA_DataSetMetaDataType dataSetMetaData;
    UA_ExtensionObject subscribedDataSet;
} UA_StandaloneSubscribedDataSetDataType;
```

4.8.359 DataTypeSchemaHeader

```
typedef struct {
    size_t namespacesSize;
    UA_String *namespaces;
    size_t structureDataTypesSize;
    UA_StructureDescription *structureDataTypes;
    size_t enumDataTypesSize;
    UA_EnumDescription *enumDataTypes;
    size_t simpleDataTypesSize;
    UA_SimpleTypeDescription *simpleDataTypes;
} UA_DataTypeSchemaHeader;
```

4.8.360 ReaderGroupDataType

```
typedef struct {
    UA_String name;
    UA_Boolean enabled;
    UA_MessageSecurityMode securityMode;
    UA_String securityGroupId;
    size_t securityKeyServicesSize;
    UA_EndpointDescription *securityKeyServices;
    UA_UInt32 maxNetworkMessageSize;
    size_t groupPropertiesSize;
    UA_KeyValuePair *groupProperties;
    UA_ExtensionObject transportSettings;
    UA_ExtensionObject messageSettings;
    size_t dataSetReadersSize;
    UA_DataSetReaderDataType *dataSetReaders;
} UA_ReaderGroupDataType;
```

4.8.361 PubSubConnectionDataType

```
typedef struct {
    UA_String name;
    UA_Boolean enabled;
    UA_Variant publisherId;
    UA_String transportProfileUri;
    UA_ExtensionObject address;
    size_t connectionPropertiesSize;
    UA_KeyValuePair *connectionProperties;
    UA_ExtensionObject transportSettings;
    size_t writerGroupsSize;
    UA_WriterGroupDataType *writerGroups;
    size_t readerGroupsSize;
    UA_ReaderGroupDataType *readerGroups;
} UA_PubSubConnectionDataType;
```

4.8.362 PubSubConfigurationDataType

```
typedef struct {
    size_t publishedDataSetsSize;
    UA_PublishedDataSetDataType *publishedDataSets;
    size_t connectionsSize;
    UA_PubSubConnectionDataType *connections;
    UA_Boolean enabled;
} UA_PubSubConfigurationDataType;
```

4.8.363 PubSubConfiguration2DataType

```
typedef struct {
    size_t publishedDataSetsSize;
    UA_PublishedDataSetDataType *publishedDataSets;
    size_t connectionsSize;
    UA_PubSubConnectionDataType *connections;
    UA_Boolean enabled;
    size_t subscribedDataSetsSize;
    UA_StandaloneSubscribedDataSetDataType *subscribedDataSets;
    size_t dataSetClassesSize;
    UA_DataSetMetaDataType *dataSetClasses;
    size_t defaultSecurityKeyServicesSize;
    UA_EndpointDescription *defaultSecurityKeyServices;
    size_t securityGroupsSize;
    UA_SecurityGroupDataType *securityGroups;
    size_t pubSubKeyPushTargetsSize;
    UA_PubSubKeyPushTargetDataType *pubSubKeyPushTargets;
    UA_UInt32 configurationVersion;
    size_t configurationPropertiesSize;
    UA_KeyValuePair *configurationProperties;
} UA_PubSubConfiguration2DataType;
```

```
/* stop-doc-generation */
```

CHAPTER

FIVE

SERVER

An OPC UA server contains an object-oriented information model and makes it accessible to clients over the network via the OPC UA *Services*. The information model can be used either used to store "passive data" or as an "active database" that integrates with data-sources and devices. For the latter, user-defined callbacks can be attached to VariableNodes and MethodNodes.

5.1 Server Lifecycle

This section describes the API for creating, running and deleting a server. At runtime, the server continuously listens on the network, accepts incoming connections and processes received messages. Furthermore, timed (cyclic) callbacks are executed.

```
/* Create a new server with a default configuration that adds plugins for
 * networking, security, logging and so on. See the "server_config_default.h"
 * for more detailed options.
 *
 * The default configuration can be used as the starting point to adjust the
 * server configuration to individual needs. UA_Server_new is implemented in the
 * /plugins folder under the CC0 license. Furthermore the server configuration
 * only uses the public server API.
 *
 * Returns the configured server or NULL if an error occurs. */
UA_Server *
UA_Server_new(void);

/* Creates a new server. Moves the config into the server with a shallow copy.
 * The config content is cleared together with the server. */
UA_Server *
UA_Server_newWithConfig(UA_ServerConfig *config);

/* Delete the server and its configuration */
UA_StatusCode
UA_Server_delete(UA_Server *server);

/* Get the configuration. Always succeeds as this simply resolves a pointer.
 * Attention! Do not adjust the configuration while the server is running! */
UA_ServerConfig *
UA_Server_getConfig(UA_Server *server);
```

(continues on next page)

(continued from previous page)

```
/* Get the current server lifecycle state */
UA_LifecycleState
UA_Server_getLifecycleState(UA_Server *server);

/* Runs the server until until "running" is set to false. The logical sequence
 * is as follows:
 *
 * - UA_Server_run_startup
 * - Loop UA_Server_run_iterate while "running" is true
 * - UA_Server_run_shutdown */
UA_StatusCode
UA_Server_run(UA_Server *server, const volatile UA_Boolean *running);

/* Runs the server until interrupted. On Unix/Windows this registers an
 * interrupt for SIGINT (ctrl-c). The method only returns after having received
 * the interrupt or upon an error condition. The logical sequence is as follows:
 *
 * - Register the interrupt
 * - UA_Server_run_startup
 * - Loop until interrupt: UA_Server_run_iterate
 * - UA_Server_run_shutdown
 * - Deregister the interrupt
 *
 * Attention! This method is implemented individually for the different
 * platforms (POSIX/Win32/etc.). The default implementation is in
 * /plugins/ua_config_default.c under the CC0 license. Adjust as needed. */
UA_StatusCode
UA_Server_runUntilInterrupt(UA_Server *server);

/* The prologue part of UA_Server_run (no need to use if you call
 * UA_Server_run or UA_Server_runUntilInterrupt). Fails if a transport (TCP,
 * WebSocket, HTTP) cannot open its server socket and the server cannot be
 * reached otherwise (no other transport listens, no reverse connect is
 * registered). Then the startup is undone as in UA_Server_run_shutdown. The
 * lifecycle notifications report STOPPING and STOPPED without STARTED. */
UA_StatusCode
UA_Server_run_startup(UA_Server *server);

/* Executes a single iteration of the server's main loop.
 *
 * @param server The server object.
 * @param waitInternal Should we wait for messages in the networklayer?
 *        Otherwise, the timeouts for the networklayers are set to zero.
 *        The default max wait time is 500ms.
 * @return Returns how long we can wait until the next scheduled
 *         callback (in ms) */
UA_UInt16
UA_Server_run_iterate(UA_Server *server, UA_Boolean waitInternal);
```

(continues on next page)

(continued from previous page)

```
/* The epilogue part of UA_Server_run (no need to use if you call
 * UA_Server_run or UA_Server_runUntilInterrupt).
 *
 * Runs the EventLoop until the server is fully stopped. An external EventLoop
 * is left running for its other users; an owned EventLoop is stopped as well.
 * Calling this on an already stopped server succeeds without further action.
 * Use UA_Server_run_shutdown_async if shutdown requires application work
 * between EventLoop iterations or is initiated from an EventLoop callback.
 *
 * @param server The server object.
 * @return GOOD once the server is STOPPED, or an EventLoop error if shutdown
 *         could not be completed. Returns BADINVALIDARGUMENT for NULL. */
UA_StatusCode
UA_Server_run_shutdown(UA_Server *server);

/* Initiates asynchronous shutdown and advances it on subsequent calls, without
 * running the EventLoop. While the server is STOPPING, the caller must drive
 * the EventLoop (directly or via UA_Server_run_iterate) and call this function
 * again to complete shutdown.
 *
 * Use UA_Server_getLifecycleState to read the current state, or the configured
 * lifecycle notification callbacks to observe state transitions.
 *
 * An external EventLoop is left running; an owned EventLoop is stopped once
 * all drivers have stopped.
 * Repeated calls do not repeat stop requests or lifecycle notifications.
 *
 * @param server The server object. NULL is treated as already stopped.
 * @return The lifecycle state after this shutdown step: STOPPING while shutdown
 *         is in progress, or STOPPED if the server is fully stopped. */
UA_LifecycleState
UA_Server_run_shutdown_async(UA_Server *server);
```

5.2 Timed Callbacks

Timed callback are executed at their defined timestamp. The callback can also be registered with a cyclic repetition interval.

```
typedef void (*UA_ServerCallback)(UA_Server *server, void *data);

/* Add a callback for execution at a specified time. If the indicated time lies
 * in the past, then the callback is executed at the next iteration of the
 * server's main loop.
 *
 * @param server The server object.
 * @param callback The callback that shall be added.
 * @param data Data that is forwarded to the callback.
 * @param date The timestamp for the execution time.
 * @param callbackId Set to the identifier of the repeated callback . This can
```

(continues on next page)

(continued from previous page)

```
 *       be used to cancel the callback later on. If the pointer is null, the
 *       identifier is not set.
 * @return Upon success, ``UA_STATUSCODE_GOOD`` is returned. An error code
 *         otherwise. */
UA_StatusCode UA_THREADSAFE
UA_Server_addTimedCallback(UA_Server *server, UA_ServerCallback callback,
                           void *data, UA_DateTime date, UA_UInt64 *callbackId);

/* Add a callback for cyclic repetition to the server.
 *
 * @param server The server object.
 * @param callback The callback that shall be added.
 * @param data Data that is forwarded to the callback.
 * @param interval_ms The callback shall be repeatedly executed with the given
 *       interval (in ms). The interval must be positive. The first execution
 *       occurs at now() + interval at the latest.
 * @param callbackId Set to the identifier of the repeated callback . This can
 *       be used to cancel the callback later on. If the pointer is null, the
 *       identifier is not set.
 * @return Upon success, ``UA_STATUSCODE_GOOD`` is returned. An error code
 *         otherwise. */
UA_StatusCode UA_THREADSAFE
UA_Server_addRepeatedCallback(UA_Server *server, UA_ServerCallback callback,
                              void *data, UA_Double interval_ms,
                              UA_UInt64 *callbackId);

UA_StatusCode UA_THREADSAFE
UA_Server_changeRepeatedCallbackInterval(UA_Server *server, UA_UInt64 callbackId,
                                         UA_Double interval_ms);

/* Remove a repeated callback. Does nothing if the callback is not found. */
void UA_THREADSAFE
UA_Server_removeCallback(UA_Server *server, UA_UInt64 callbackId);

#define UA_Server_removeRepeatedCallback(server, callbackId) \
    UA_Server_removeCallback(server, callbackId)
```

5.3 Application Notification

The server defines callbacks to notify the application on defined triggering points. These callbacks are executed with the (re-entrant) server-mutex held.

The different types of callback are disambiguated by their type enum. Besides the global notification callback (which is always triggered), the server configuration contains specialized callbacks that trigger only for specific notifications. This can reduce the burden of high-frequency notifications.

If a specialized notification callback is set, it always gets called before the global notification callback for the same triggering point.

See the section on the Application Notification enum for more documentation on the notifications and their defined payload.

```
typedef void (*UA_ServerNotificationCallback)(UA_Server *server,
                                              UA_ApplicationNotificationType type,
                                              const UA_KeyValueMap payload);
```

5.4 SecureChannel Handling

The server opens new SecureChannels internally when a server socket is active. Information about SecureChannels and their state is notified with UA_APPLICATIONNOTIFICATIONTYPE_SECURECHANNEL. SecureChannels can be manually closed. This leaves any attached session alive so that it can potentially reconnect.

```
UA_StatusCode UA_THREADSAFE
UA_Server_closeSecureChannel(UA_Server *server, UA_UInt32 channelId,
                             UA_ShutdownReason reason);
```

A SecureChannel carries application-defined attributes in a key-value map, mirroring the *session attributes* below. This lets the application attach its own per-channel state, for example from the secureChannelNotificationCallback once it has looked at the channel's SecurityMode.

Namespace 0 is reserved for the server itself and restricted to a predefined set of keys, for both read and write – any other 0:... key is rejected with UA_STATUSCODE_BADNOTFOUND (get) or UA_STATUSCODE_BADNOTWRITABLE (set/delete). An application is free to use any key in a non-zero namespace as generic, arbitrary storage.

Most of the predefined ns0 keys are read-only: their value is computed fresh from the live SecureChannel on every access (nothing is cached), mirroring the background information sent with the UA_APPLICATIONNOTIFICATIONTYPE_SECURECHANNEL_OPENED/_CLOSED notification (see common.h): 0:securechannel-id, 0:connection-manager-name, 0:connection-id, 0:remote-address, 0:protocol-version, 0:recv-buffer-size, 0:recv-max-message-size, 0:recv-max-chunk-count, 0:send-buffer-size, 0:send-max-message-size, 0:send-max-chunk-count, 0:endpoint-url, 0:security-mode, 0:security-policy-url, 0:certificate-type-id and 0:remote-certificate. UA_Server_setSecureChannelAttribute and UA_Server_deleteSecureChannelAttribute reject writes to these with UA_STATUSCODE_BADNOTWRITABLE.

One further ns0 key is interpreted by the server itself, and is the one ns0 attribute the application may write (stored in the generic map like any application-defined attribute):

- 0:maxMessageSize (UA_UInt32): If set, tightens (but never loosens) UA_ServerConfig.tcpMaxMsgSize for this specific channel. Useful to cap messages on a channel the application does not yet consider trusted, e.g. because it is unsigned/unencrypted or has no Session with a completed ActivateSession. Delete the attribute (or never set it) to fall back to the server-wide tcpMaxMsgSize.

```
/* Returns a shallow copy of the attribute (don't _clear or _delete manually).
 * While the method is thread-safe, the returned value is not protected. Only
 * use it in a (callback) context where the server is locked for the current
 * thread. */
UA_StatusCode UA_THREADSAFE
UA_Server_getSecureChannelAttribute(UA_Server *server, UA_UInt32 channelId,
                                    const UA_QualifiedName key,
                                    UA_Variant *outValue);
```

(continues on next page)

(continued from previous page)

```
/* Return a deep copy of the attribute */
UA_StatusCode UA_THREADSAFE
UA_Server_getSecureChannelAttributeCopy(UA_Server *server, UA_UInt32 channelId,
                                        const UA_QualifiedName key,
                                        UA_Variant *outValue);

/* Returns UA_STATUSCODE_BADNOTFOUND if the attribute is not defined, not a
 * scalar or not of the right datatype. Otherwise a shallow copy of the
 * scalar value is written to the target location of the void pointer (don't
 * _clear or _delete manually). While the method is thread-safe, the
 * returned value is not protected. Only use it in a (callback) context
 * where the server is locked for the current thread. */
UA_StatusCode UA_THREADSAFE
UA_Server_getSecureChannelAttribute_scalar(UA_Server *server,
                                           UA_UInt32 channelId,
                                           const UA_QualifiedName key,
                                           const UA_DataType *type,
                                           void *outValue);

UA_StatusCode UA_THREADSAFE
UA_Server_setSecureChannelAttribute(UA_Server *server, UA_UInt32 channelId,
                                    const UA_QualifiedName key,
                                    const UA_Variant *value);

UA_StatusCode UA_THREADSAFE
UA_Server_deleteSecureChannelAttribute(UA_Server *server, UA_UInt32 channelId,
                                       const UA_QualifiedName key);
```

5.5 Session Handling

Sessions are managed via the OPC UA Session Service Set (CreateSession, ActivateSession, CloseSession). The identifier of sessions is generated internally in the server and is always a Guid-NodeId.

The creation of sessions is passed to the *AccessControl Plugin API*. There, the authentication information is evaluated and a context-pointer is attached to the new session. The context pointer (and the session identifier) are then forwarded to all user-defined callbacks that can be triggere by a session.

When the operations from the OPC UA Services are invoked locally via the C-API, this implies that the operations are executed with the access rights of the "admin-session" that is always present in a server. Any AccessControl checks are omitted for the admin-session.

The admin-session has the identifier g=00000001-0000-0000-0000-000000000000. Its session context pointer needs to be manually set (NULL by default).

```
void
UA_Server_setAdminSessionContext(UA_Server *server, void *context);

/* Manually close a session */
UA_StatusCode UA_THREADSAFE
UA_Server_closeSession(UA_Server *server, const UA_NodeId *sessionId);
```

Besides the session context pointer from the AccessControl plugin, a session carries attributes in a key-value map. Always defined (and read-only) session attributes are:

- 0:localeIds (UA_String): List of preferred languages
- 0:clientDescription (UA_ApplicationDescription): Client description
- 0:sessionName (String): Client-defined name of the session
- 0:clientUserId (String): User identifier used to activate the session

Additional attributes can be set manually with the API below.

```
/* Returns a shallow copy of the attribute (don't _clear or _delete manually).
 * While the method is thread-safe, the returned value is not protected. Only
 * use it in a (callback) context where the server is locked for the current
 * thread. */
UA_StatusCode UA_THREADSAFE
UA_Server_getSessionAttribute(UA_Server *server, const UA_NodeId *sessionId,
                              const UA_QualifiedName key, UA_Variant *outValue);

/* Return a deep copy of the attribute */
UA_StatusCode UA_THREADSAFE
UA_Server_getSessionAttributeCopy(UA_Server *server, const UA_NodeId *sessionId,
                                  const UA_QualifiedName key, UA_Variant *outValue);

/* Returns NULL if the attribute is not defined or not a scalar or not of the
 * right datatype. Otherwise a shallow copy of the scalar value is created at
 * the target location of the void pointer (don't _clear or _delete manually).
 * While the method is thread-safe, the returned value is not protected. Only
 * use it in a (callback) context where the server is locked for the current
 * thread. */
UA_StatusCode UA_THREADSAFE
UA_Server_getSessionAttribute_scalar(UA_Server *server,
                                     const UA_NodeId *sessionId,
                                     const UA_QualifiedName key,
                                     const UA_DataType *type,
                                     void *outValue);

UA_StatusCode UA_THREADSAFE
UA_Server_setSessionAttribute(UA_Server *server, const UA_NodeId *sessionId,
                              const UA_QualifiedName key,
                              const UA_Variant *value);

UA_StatusCode UA_THREADSAFE
UA_Server_deleteSessionAttribute(UA_Server *server, const UA_NodeId *sessionId,
                                 const UA_QualifiedName key);
```

5.6 Attribute Service Set

The functions for reading and writing node attributes call the regular read and write service in the background that are also used over the network.

The following attributes cannot be read, since the local “admin” user always has full rights.

- UserWriteMask
- UserAccessLevel
- UserExecutable

```
/* Read an attribute of a node. Returns a deep copy. */
UA_DataValue UA_THREADSAFE
UA_Server_read(UA_Server *server, const UA_ReadValueId *item,
               UA_TimestampsToReturn timestamps);
```

The following specialized read methods are a shorthand for the regular read and set a deep copy of the attribute to the `out` pointer (when successful).

```
UA_THREADSAFE UA_StatusCode
UA_Server_readNodeId(UA_Server *server, const UA_NodeId nodeId,
                     UA_NodeId *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readNodeClass(UA_Server *server, const UA_NodeId nodeId,
                        UA_NodeClass *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readBrowseName(UA_Server *server, const UA_NodeId nodeId,
                         UA_QualifiedName *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readDisplayName(UA_Server *server, const UA_NodeId nodeId,
                          UA_LocalizedText *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readDescription(UA_Server *server, const UA_NodeId nodeId,
                          UA_LocalizedText *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readWriteMask(UA_Server *server, const UA_NodeId nodeId,
                        UA_UInt32 *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readIsAbstract(UA_Server *server, const UA_NodeId nodeId,
                         UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readSymmetric(UA_Server *server, const UA_NodeId nodeId,
                        UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readInverseName(UA_Server *server, const UA_NodeId nodeId,
                          UA_LocalizedText *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readContainsNoLoops(UA_Server *server, const UA_NodeId nodeId,
```

(continues on next page)

(continued from previous page)

```
                          UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readEventNotifier(UA_Server *server, const UA_NodeId nodeId,
                            UA_Byte *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readValue(UA_Server *server, const UA_NodeId nodeId,
                    UA_Variant *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readDataType(UA_Server *server, const UA_NodeId nodeId,
                       UA_NodeId *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readValueRank(UA_Server *server, const UA_NodeId nodeId,
                        UA_Int32 *out);

/* Returns a variant with an uint32 array */
UA_THREADSAFE UA_StatusCode
UA_Server_readArrayDimensions(UA_Server *server, const UA_NodeId nodeId,
                              UA_Variant *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readAccessLevel(UA_Server *server, const UA_NodeId nodeId,
                          UA_Byte *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readAccessLevelEx(UA_Server *server, const UA_NodeId nodeId,
                            UA_UInt32 *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readMinimumSamplingInterval(UA_Server *server, const UA_NodeId nodeId,
                                      UA_Double *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readHistorizing(UA_Server *server, const UA_NodeId nodeId,
                          UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readExecutable(UA_Server *server, const UA_NodeId nodeId,
                         UA_Boolean *out);

/* Returns a variant with a UA_RolePermissionType array */
UA_THREADSAFE UA_StatusCode
UA_Server_readRolePermissions(UA_Server *server, const UA_NodeId nodeId,
                              UA_Variant *out);

/* Returns a variant with a UA_RolePermissionType array */
```

(continues on next page)

(continued from previous page)

```
UA_THREADSAFE UA_StatusCode
UA_Server_readUserRolePermissions(UA_Server *server, const UA_NodeId nodeId,
                                  UA_Variant *out);

UA_THREADSAFE UA_StatusCode
UA_Server_readAccessRestrictions(UA_Server *server, const UA_NodeId nodeId,
                                 UA_AccessRestrictionType *out);
```

The following node attributes cannot be written once a node has been created:

- NodeClass
- NodeId
- Symmetric
- ContainsNoLoops

The following attributes cannot be written from C-API, as they are specific to the session (context set by the access control callback):

- UserWriteMask
- UserAccessLevel
- UserExecutable

```
UA_THREADSAFE UA_StatusCode
UA_Server_write(UA_Server *server, const UA_WriteValue *value);

UA_THREADSAFE UA_StatusCode
UA_Server_writeBrowseName(UA_Server *server, const UA_NodeId nodeId,
                          const UA_QualifiedName browseName);

UA_THREADSAFE UA_StatusCode
UA_Server_writeDisplayName(UA_Server *server, const UA_NodeId nodeId,
                           const UA_LocalizedText displayName);

UA_THREADSAFE UA_StatusCode
UA_Server_writeDescription(UA_Server *server, const UA_NodeId nodeId,
                           const UA_LocalizedText description);

UA_THREADSAFE UA_StatusCode
UA_Server_writeWriteMask(UA_Server *server, const UA_NodeId nodeId,
                         const UA_UInt32 writeMask);

UA_THREADSAFE UA_StatusCode
UA_Server_writeIsAbstract(UA_Server *server, const UA_NodeId nodeId,
                          const UA_Boolean isAbstract);

UA_THREADSAFE UA_StatusCode
UA_Server_writeInverseName(UA_Server *server, const UA_NodeId nodeId,
                           const UA_LocalizedText inverseName);
```

(continues on next page)

(continued from previous page)

```
UA_THREADSAFE UA_StatusCode
UA_Server_writeEventNotifier(UA_Server *server, const UA_NodeId nodeId,
                             const UA_Byte eventNotifier);

/* The value attribute is a DataValue. Here only a variant is provided. The
 * StatusCode is set to UA_STATUSCODE_GOOD, sourceTimestamp and serverTimestamp
 * are set to UA_DateTime_now(). See below for setting the full DataValue. */
UA_THREADSAFE UA_StatusCode
UA_Server_writeValue(UA_Server *server, const UA_NodeId nodeId,
                     const UA_Variant value);

UA_THREADSAFE UA_StatusCode
UA_Server_writeDataValue(UA_Server *server, const UA_NodeId nodeId,
                         const UA_DataValue value);

UA_THREADSAFE UA_StatusCode
UA_Server_writeDataType(UA_Server *server, const UA_NodeId nodeId,
                        const UA_NodeId dataType);

UA_THREADSAFE UA_StatusCode
UA_Server_writeValueRank(UA_Server *server, const UA_NodeId nodeId,
                         const UA_Int32 valueRank);

UA_THREADSAFE UA_StatusCode
UA_Server_writeArrayDimensions(UA_Server *server, const UA_NodeId nodeId,
                               const UA_Variant arrayDimensions);

UA_THREADSAFE UA_StatusCode
UA_Server_writeAccessLevel(UA_Server *server, const UA_NodeId nodeId,
                           const UA_Byte accessLevel);

UA_THREADSAFE UA_StatusCode
UA_Server_writeAccessLevelEx(UA_Server *server, const UA_NodeId nodeId,
                             const UA_UInt32 accessLevelEx);

UA_THREADSAFE UA_StatusCode
UA_Server_writeMinimumSamplingInterval(UA_Server *server, const UA_NodeId nodeId,
                                       const UA_Double miniumSamplingInterval);

UA_THREADSAFE UA_StatusCode
UA_Server_writeHistorizing(UA_Server *server, const UA_NodeId nodeId,
                           const UA_Boolean historizing);

UA_THREADSAFE UA_StatusCode
UA_Server_writeExecutable(UA_Server *server, const UA_NodeId nodeId,
                          const UA_Boolean executable);

UA_THREADSAFE UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_Server_writeRolePermissions(UA_Server *server, const UA_NodeId nodeId,
                               const UA_Variant rolePermissions);

UA_THREADSAFE UA_StatusCode
UA_Server_writeAccessRestrictions(UA_Server *server, const UA_NodeId nodeId,
                                  const UA_AccessRestrictionType accessRestrictions);
```

5.7 Method Service Set

The Method Service Set defines the means to invoke methods. A MethodNode is a component of an ObjectNode or of an ObjectTypeNode. The input and output arguments of a method are a list of UA_Variant. The type- and size-requirements of the arguments can be retrieved from the **InputArguments** and **OutputArguments** variable below the MethodNode.

For calling a method, both methodId and objectId need to be defined by their NodeId. This is required because the same MethodNode can be referenced from multiple objects.

In this server implementation, when an object is instantiated from a an ObjectType, all (mandatory) methods are automatically added to the new object instance. This is done by adding an additional reference to the original MethodNode. It is however possible to add a custom MethodNode directly to the object instance. It is also possible to remove a (optional) MethodNode that exists in the ObjectType from an instance.

The methodId can point to a MethodNode that exists in the ObjectType but not in the object instance. It is resolved to the actual MethodNode of the object instance by taking the *BrowseName* attribute of the methodId-MethodNode and looking up the member of the objectId object with the same BrowseName.

The resolved MethodNode then is used to

- Check permissions for the current Session to call the method
- Obtain the UA_MethodCallback to execute
- Forwarded as methodId to said callback

To showcase the resolution of the MethodNode with an example, consider this information model:

```
 ObjectType                 ObjectType                Object
Creature (i=10)  <-isSubTypeOf-  Insect (i=20)  <-hasTypeDef-   Ant (i=30)
     |                           |                         |
 hasComponent                hasComponent              hasComponent
     |                           |                         |
     v                           v                         v
  Methods                     Methods                   Methods
- Walk (i=11)                - Walk (i=21)             - Walk (i=31)
- Fly (i=12)                 - Fly (i=22)              - Amount (i=33)
- Amount (i=13)              - Amount (i=23)
```

The following table shows what methodId - objectId combinations are allowed to be used as parameters for the Call service and the resolved methodId.

objectId	methodId	Corresponds to in OO-languages	Resolved methodId
i=30	i=31	Ant a; a.Walk();	i=31
i=30	i=21	Ant a; Insect i = a; i.Walk();	i=31
i=30	i=11	Ant a; Creature c = a; c.Walk();	i=31
i=20	i=23	Insect::Amount();	i=23
i=10	i=13	Creature::Amount();	i=13

The next table shows methodId - objectId combinations that are not allowed. Note that an ObjecType cannot execute a methodId from a subtype or instance.

objectId	methodId	Reason
i=30	i=22	Object “Ant” does not own a method “Fly”
i=30	i=12	Object “Ant” does not own a method “Fly”
i=10	i=23	The method is not owned by the object type “Creature”
i=20	i=13	The method is not owned by the object type “Insect”

```
UA_CallMethodResult UA_THREADSAFE
UA_Server_call(UA_Server *server, const UA_CallMethodRequest *request);
```

5.8 View Service Set

The View Service Set allows Clients to discover Nodes by browsing the information model.

```
/* Browse the references of a particular node. See the definition of
 * BrowseDescription structure for details. */
UA_BrowseResult UA_THREADSAFE
UA_Server_browse(UA_Server *server, UA_UInt32 maxReferences,
                 const UA_BrowseDescription *bd);

UA_BrowseResult UA_THREADSAFE
UA_Server_browseNext(UA_Server *server, UA_Boolean releaseContinuationPoint,
                     const UA_ByteString *continuationPoint);

/* Non-standard version of the Browse service that recurses into child nodes.
 *
 * Possible loops (that can occur for non-hierarchical references) are handled
 * internally. Every node is added at most once to the results array.
 *
 * Nodes are only added if they match the NodeClassMask in the
 * BrowseDescription. However, child nodes are still recursed into if the
 * NodeClass does not match. So it is possible, for example, to get all
 * VariableNodes below a certain ObjectNode, with additional objects in the
 * hierarchy below. */
UA_StatusCode UA_THREADSAFE
UA_Server_browseRecursive(UA_Server *server, const UA_BrowseDescription *bd,
                          size_t *resultsSize, UA_ExpandedNodeId **results);
```

(continues on next page)

(continued from previous page)

```
/* Translate abrowse path to (potentially several) NodeIds. Each browse path is
 * constructed of a starting Node and a RelativePath. The specified starting
 * Node identifies the Node from which the RelativePath is based. The
 * RelativePath contains a sequence of ReferenceTypes and BrowseNames. */
UA_BrowsePathResult UA_THREADSAFE
UA_Server_translateBrowsePathToNodeIds(UA_Server *server,
                                       const UA_BrowsePath *browsePath);

/* A simplified TranslateBrowsePathsToNodeIds based on the
 * SimpleAttributeOperand type (Part 4, 7.4.4.5).
 *
 * This specifies a relative path using a list of BrowseNames instead of the
 * RelativePath structure. The list of BrowseNames is equivalent to a
 * RelativePath that specifies forward references which are subtypes of the
 * HierarchicalReferences ReferenceType. All Nodes followed by the browsePath
 * shall be of the NodeClass Object or Variable. */
UA_BrowsePathResult UA_THREADSAFE
UA_Server_browseSimplifiedBrowsePath(UA_Server *server, const UA_NodeId origin,
                                     size_t browsePathSize,
                                     const UA_QualifiedName *browsePath);

/* Returns the target of a "HasTypeDefinition" reference (or inverse
 * "HasSubtype" reference for type nodes) */
UA_StatusCode UA_THREADSAFE
UA_Server_getNodeType(UA_Server *server, const UA_NodeId nodeId,
                      UA_NodeId *outTypeId);

/* Iterate over all nodes referenced by parentNodeId by calling the callback
 * function for each child node (in ifdef because GCC/CLANG handle include order
 * differently) */
typedef UA_StatusCode
(*UA_ServerNodeIteratorCallback)(UA_NodeId childId, UA_Boolean isInverse,
                                 UA_NodeId referenceTypeId, void *handle);

UA_StatusCode UA_THREADSAFE
UA_Server_forEachChildNodeCall(UA_Server *server, UA_NodeId parentNodeId,
                               UA_ServerNodeIteratorCallback callback,
                               void *handle);
```

5.9 MonitoredItem Service Set

MonitoredItems are used with the Subscription mechanism of OPC UA to transported notifications for data changes and events. MonitoredItems can also be registered locally. Notifications are then forwarded to a user-defined callback instead of a remote client.

Local MonitoredItems are delivered asynchronously. That is, the notification is inserted as a *Delayed Callback* for the EventLoop. The callback is then triggered when the control flow next returns to the EventLoop.

```
#ifdef UA_ENABLE_SUBSCRIPTIONS

/* Delete a local MonitoredItem. Used for both DataChange- and
 * Event-MonitoredItems. */
UA_StatusCode UA_THREADSAFE
UA_Server_deleteMonitoredItem(UA_Server *server, UA_UInt32 monitoredItemId);

typedef void (*UA_Server_DataChangeNotificationCallback)
    (UA_Server *server, UA_UInt32 monitoredItemId, void *monitoredItemContext,
     const UA_NodeId *nodeId, void *nodeContext, UA_UInt32 attributeId,
     const UA_DataValue *value);
```

DataChange MonitoredItem use a sampling interval and filter criteria to notify the userland about value changes. Note that the sampling interval can also be zero to be notified about changes "right away". For this we hook the MonitoredItem into the observed Node and check the filter after every call of the Write-Service.

```
/* Create a local MonitoredItem to detect data changes.
 *
 * @param server The server executing the MonitoredItem
 * @param timestampsToReturn Shall timestamps be added to the value for the
 *        callback?
 * @param item The parameters of the new MonitoredItem. Note that the attribute
 *        of the ReadValueId (the node that is monitored) can not be
 *        ``UA_ATTRIBUTEID_EVENTNOTIFIER``. See below for event notifications.
 * @param monitoredItemContext A pointer that is forwarded with the callback
 * @param callback The callback that is executed on detected data changes
 * @return Returns a description of the created MonitoredItem. The structure
 *         also contains a StatusCode (in case of an error) and the identifier
 *         of the new MonitoredItem. */
UA_MonitoredItemCreateResult UA_THREADSAFE
UA_Server_createDataChangeMonitoredItem(UA_Server *server,
        UA_TimestampsToReturn timestampsToReturn,
        const UA_MonitoredItemCreateRequest item,
        void *monitoredItemContext,
        UA_Server_DataChangeNotificationCallback callback);
```

See the section on :ref‘events‘ for how to emit events in the server.

Event-MonitoredItems emit notifications with a list of "fields" (variants). The fields are specified as *SimpleAttributeOperands* in the select-clause of the MonitoredItem's event filter. For the local event callback, instead of using a list of variants, we use a key-value map for the event fields. They key names are generated with `UA_SimpleAttributeOperand_print` to get a human-readable representation.

The received event-fields map could look like this:

```
/Severity   => UInt16(1000)
/Message    => LocalizedText("en-US", "My Event Message")
/EventType  => NodeId(i=50831)
/SourceNode => NodeId(i=2253)
```

The order of the keys is identical to the order of SimpleAttributeOperands in the select-clause.

```
#ifdef UA_ENABLE_SUBSCRIPTIONS_EVENTS

typedef void (*UA_Server_EventNotificationCallback)
    (UA_Server *server, UA_UInt32 monitoredItemId, void *monitoredItemContext,
     const UA_KeyValueMap eventFields);

/* Create a local MonitoredItem for Events. The API is simplified compared to a
 * UA_MonitoredItemCreateRequest. The unavailable options are not relevant for
 * local MonitoredItems (e.g. the queue size) or not relevant for Event
 * MonitoredItems (e.g. the sampling interval).
 *
 * @param server The server executing the MonitoredItem
 * @param nodeId The node where events are collected. Note that events "bubble
 *        up" to their parents (via hierarchical references).
 * @param filter The filter defined which event fields are selected (select
 *        clauses) and which events are considered for this particular
 *        MonitoredItem (where clause).
 * @param monitoredItemContext A pointer that is forwarded with the callback
 * @param callback The callback that is executed for each event
 * @return Returns a description of the created MonitoredItem. The structure
 *         also contains a StatusCode (in case of an error) and the identifier
 *         of the new MonitoredItem. */
UA_MonitoredItemCreateResult UA_THREADSAFE
UA_Server_createEventMonitoredItem(UA_Server *server, const UA_NodeId nodeId,
                                   const UA_EventFilter filter,
                                   void *monitoredItemContext,
                                   UA_Server_EventNotificationCallback callback);

/* Extended version UA_Server_createEventMonitoredItem that allows setting of
 * uncommon parameters (for local MonitoredItems) like the MonitoringMode and
 * queue sizes.
 *
 * @param server The server executing the MonitoredItem
 * @param item The description of the MonitoredItem. Must use
 *        UA_ATTRIBUTEID_EVENTNOTIFIER and an EventFilter.
 * @param monitoredItemContext A pointer that is forwarded with the callback
 * @param callback The callback that is executed for each event
 * @return Returns a description of the created MonitoredItem. The structure
 *         also contains a StatusCode (in case of an error) and the identifier
 *         of the new MonitoredItem. */
UA_MonitoredItemCreateResult UA_THREADSAFE
UA_Server_createEventMonitoredItemEx(UA_Server *server,
                                     const UA_MonitoredItemCreateRequest item,
                                     void *monitoredItemContext,
                                     UA_Server_EventNotificationCallback callback);

#endif /* UA_ENABLE_SUBSCRIPTIONS_EVENTS */

#endif /* UA_ENABLE_SUBSCRIPTIONS */
```

5.10 Node Management Service Set

When creating dynamic node instances at runtime, chances are that you will not care about the specific NodeId of the new node, as long as you can reference it later. When passing numeric NodeIds with a numeric identifier 0, the stack evaluates this as "select a random unassigned numeric NodeId in that namespace". To find out which NodeId was actually assigned to the new node, you may pass a pointer *outNewNodeId*, which will (after a successful node insertion) contain the nodeId of the new node. You may also pass a NULL pointer if this result is not needed.

See the Section *Node Lifecycle: Constructors, Destructors and Node Contexts* on constructors and on attaching user-defined data to nodes.

The Section *Default Node Attributes* contains useful starting points for defining node attributes. Forgetting to set the ValueRank or the AccessLevel leads to errors that can be hard to track down for new users. The default attributes have a high likelihood to "do the right thing".

The methods for node addition and deletion take mostly const arguments that are not modified. When creating a node, a deep copy of the node identifier, node attributes, etc. is created. Therefore, it is possible to call for example UA_Server_addVariablenode with a value attribute (a *Variant*) pointing to a memory location on the stack.

5.10.1 VariableNode

Variables store values as well as constraints for possible values. There are three options for storing the value: Internal in the VariableNode data structure itself, external with a double-pointer (to switch to an updated value with an atomic pointer-replacing operation) or with a callback registered by the application.

```
typedef enum {
    UA_VALUESOURCETYPE_INTERNAL = 0,
    UA_VALUESOURCETYPE_EXTERNAL = 1,
    UA_VALUESOURCETYPE_CALLBACK = 2
} UA_ValueSourceType;

typedef struct {
    /* Notify the application before the value attribute is read. Ignored if
     * NULL. It is possible to write into the value attribute during onRead
     * (using the write service). The node is re-retrieved from the Nodestore
     * afterwards so that changes are considered in the following read
     * operation.
     *
     * @param handle Points to user-provided data for the callback.
     * @param nodeid The identifier of the node.
     * @param data Points to the current node value.
     * @param range Points to the numeric range the client wants to read from
     *        (or NULL). */
    void (*onRead)(UA_Server *server, const UA_NodeId *sessionId,
                   void *sessionContext, const UA_NodeId *nodeid,
                   void *nodeContext, const UA_NumericRange *range,
                   const UA_DataValue *value);

    /* Notify the application after writing the value attribute. Ignored if
     * NULL. The node is re-retrieved after writing, so that the new value is
```

(continues on next page)

(continued from previous page)

```
     * visible in the callback.
     *
     * @param server The server executing the callback
     * @sessionId The identifier of the session
     * @sessionContext Additional data attached to the session
     *                 in the access control layer
     * @param nodeid The identifier of the node.
     * @param nodeUserContext Additional data attached to the node by
     *        the user.
     * @param nodeConstructorContext Additional data attached to the node
     *        by the type constructor(s).
     * @param range Points to the numeric range the client wants to write to (or
     *        NULL). */
    void (*onWrite)(UA_Server *server, const UA_NodeId *sessionId,
                    void *sessionContext, const UA_NodeId *nodeId,
                    void *nodeContext, const UA_NumericRange *range,
                    const UA_DataValue *data);
} UA_ValueSourceNotifications;

typedef struct {
    /* Copies the data from the source into the provided value.
     *
     * !! ZERO-COPY OPERATIONS POSSIBLE !!
     * It is not required to return a copy of the actual content data. You can
     * return a pointer to memory owned by the user. Memory can be reused
     * between read callbacks of a DataSource, as the result is already encoded
     * on the network buffer between each read operation.
     *
     * To use zero-copy reads, set the value of the `value->value` Variant
     * without copying, e.g. with `UA_Variant_setScalar`. Then, also set
     * `value->value.storageType` to `UA_VARIANT_DATA_NODELETE` to prevent the
     * memory being cleaned up. Don't forget to also set `value->hasValue` to
     * true to indicate the presence of a value.
     *
     * To make an async read, return UA_STATUSCODE_
↪GOODCOMPLETESASYNCHRONOUSLY.
     * The result can then be set at a later time using
     * UA_Server_setAsyncReadResult. Note that the server might cancel the async
     * read by calling serverConfig->asyncOperationCancelCallback.
     *
     * @param server The server executing the callback
     * @param sessionId The identifier of the session
     * @param sessionContext Additional data attached to the session in the
     *        access control layer
     * @param nodeId The identifier of the node being read from
     * @param nodeContext Additional data attached to the node by the user
     * @param includeSourceTimeStamp If true, then the datasource is expected to
     *        set the source timestamp in the returned value
     * @param range If not null, then the datasource shall return only a
```

(continues on next page)

(continued from previous page)

```
     *        selection of the (nonscalar) data. Set
     *        UA_STATUSCODE_BADINDEXRANGEINVALID in the value if this does not
     *        apply
     * @param value The (non-null) DataValue that is returned to the client. The
     *        data source sets the read data, the result status and optionally a
     *        sourcetimestamp.
     * @return Returns a status code for logging. Error codes intended for the
     *         original caller are set in the value. If an error is returned,
     *         then no releasing of the value is done. */
    UA_StatusCode (*read)(UA_Server *server, const UA_NodeId *sessionId,
                          void *sessionContext, const UA_NodeId *nodeId,
                          void *nodeContext, UA_Boolean includeSourceTimeStamp,
                          const UA_NumericRange *range, UA_DataValue *value);

    /* Write into a data source. This method pointer can be NULL if the
     * operation is unsupported.
     *
     * To make an async write, return UA_STATUSCODE_
↪GOODCOMPLETESASYNCHRONOUSLY.
     * The result can then be set at a later time using
     * UA_Server_setAsyncWriteResult. Note that the server might cancel the
     * async read by calling serverConfig->asyncOperationCancelCallback.
     *
     * @param server The server executing the callback
     * @param sessionId The identifier of the session
     * @param sessionContext Additional data attached to the session in the
     *        access control layer
     * @param nodeId The identifier of the node being written to
     * @param nodeContext Additional data attached to the node by the user
     * @param range If not NULL, then the datasource shall return only a
     *        selection of the (nonscalar) data. Set
     *        UA_STATUSCODE_BADINDEXRANGEINVALID in the value if this does not
     *        apply
     * @param value The (non-NULL) DataValue that has been written by the client.
     *        The data source contains the written data, the result status and
     *        optionally a sourcetimestamp
     * @return Returns a status code for logging. Error codes intended for the
     *         original caller are set in the value. If an error is returned,
     *         then no releasing of the value is done. */
    UA_StatusCode (*write)(UA_Server *server, const UA_NodeId *sessionId,
                           void *sessionContext, const UA_NodeId *nodeId,
                           void *nodeContext, const UA_NumericRange *range,
                           const UA_DataValue *value);
} UA_CallbackValueSource;
```

By default, when adding a VariableNode, the value from the UA_VariableAttributes is used. The methods following afterwards can be used to override the value source.

```
UA_THREADSAFE UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_Server_addVariableNode(UA_Server *server, const UA_NodeId requestedNewNodeId,
                          const UA_NodeId parentNodeId,
                          const UA_NodeId referenceTypeId,
                          const UA_QualifiedName browseName,
                          const UA_NodeId typeDefinition,
                          const UA_VariableAttributes attr,
                          void *nodeContext, UA_NodeId *outNewNodeId);

/* Add a VariableNode with a callback value-source */
UA_StatusCode UA_THREADSAFE
UA_Server_addCallbackValueSourceVariableNode(UA_Server *server,
                                             const UA_NodeId requestedNewNodeId,
                                             const UA_NodeId parentNodeId,
                                             const UA_NodeId referenceTypeId,
                                             const UA_QualifiedName browseName,
                                             const UA_NodeId typeDefinition,
                                             const UA_VariableAttributes attr,
                                             const UA_CallbackValueSource evs,
                                             void *nodeContext, UA_NodeId *outNewNodeId);

/* Legacy API */
#define UA_Server_addDataSourceVariableNode(server, requestedNewNodeId,␣
↪parentNodeId,    \
                                            referenceTypeId, browseName, typeDefinition, \
                                            attr, dataSource, nodeContext, outNewNodeId) \
   UA_Server_addCallbackValueSourceVariableNode(server, requestedNewNodeId,          ␣
↪\
                                                parentNodeId, referenceTypeId,        \
                                                browseName, typeDefinition,           \
                                                attr, dataSource, nodeContext,        \
                                                outNewNodeId)

/* Set an internal value source. Both the value argument and the notifications
 * argument can be NULL. If value is NULL, an existing internal value is kept;
 * switching from another value source creates an empty internal value. If
 * notifications is NULL, then all onRead/onWrite notifications are disabled. */
UA_StatusCode UA_THREADSAFE
UA_Server_setVariableNode_internalValueSource(UA_Server *server,
   const UA_NodeId nodeId, const UA_DataValue *value,
   const UA_ValueSourceNotifications *notifications);

/* For the external value, no initial copy is made. The node "just" points to
 * the provided double-pointer. Otherwise identical to the internal data
 * source. */
UA_StatusCode UA_THREADSAFE
UA_Server_setVariableNode_externalValueSource(UA_Server *server,
   const UA_NodeId nodeId, UA_DataValue** value,
   const UA_ValueSourceNotifications *notifications);
```

(continues on next page)

(continued from previous page)

```
/* It is expected that the read callback is implemented. Whenever the value
 * attribute is read, the function will be called and asked to fill a
 * UA_DataValue structure that contains the value content and additional
 * metadata like timestamps.
 *
 * The write callback can be set to a null-pointer. Then writing into the value
 * is disabled. */
UA_StatusCode UA_THREADSAFE
UA_Server_setVariableNode_callbackValueSource(UA_Server *server,
    const UA_NodeId nodeId, const UA_CallbackValueSource evs);

/* Deprecated API */
typedef UA_CallbackValueSource UA_DataSource;
#define UA_Server_setVariableNode_dataSource(server, nodeId, dataSource) \
    UA_Server_setVariableNode_callbackValueSource(server, nodeId, dataSource)

/* Deprecated API */
typedef UA_ValueSourceNotifications UA_ValueCallback;
#define UA_Server_setVariableNode_valueCallback(server, nodeId, callback) \
    UA_Server_setVariableNode_internalValueSource(server, nodeId, NULL, &callback)

/* VariableNodes that are "dynamic" (default for user-created variables) receive
 * and store a SourceTimestamp. For non-dynamic VariableNodes the current time
 * is used for the SourceTimestamp. */
UA_StatusCode UA_THREADSAFE
UA_Server_setVariableNodeDynamic(UA_Server *server, const UA_NodeId nodeId,
                                 UA_Boolean isDynamic);
```

5.10.2 VariableTypeNode

```
UA_THREADSAFE UA_StatusCode
UA_Server_addVariableTypeNode(UA_Server *server,
                              const UA_NodeId requestedNewNodeId,
                              const UA_NodeId parentNodeId,
                              const UA_NodeId referenceTypeId,
                              const UA_QualifiedName browseName,
                              const UA_NodeId typeDefinition,
                              const UA_VariableTypeAttributes attr,
                              void *nodeContext, UA_NodeId *outNewNodeId);
```

5.10.3 MethodNode

Please refer to the *Method Service Set* to get information about which MethodNodes may get executed and would thus require callbacks to be registered.

```
typedef UA_StatusCode
(*UA_MethodCallback)(UA_Server *server,
                     const UA_NodeId *sessionId, void *sessionContext,
                     const UA_NodeId *methodId, void *methodContext,
```

(continues on next page)

(continued from previous page)

```
                const UA_NodeId *objectId, void *objectContext,
                size_t inputSize, const UA_Variant *input,
                size_t outputSize, UA_Variant *output);


UA_THREADSAFE UA_StatusCode
UA_Server_addMethodNode(UA_Server *server, const UA_NodeId requestedNewNodeId,
                        const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
                        const UA_QualifiedName browseName, const UA_MethodAttributes attr,
                        UA_MethodCallback method,
                        size_t inputArgumentsSize, const UA_Argument *inputArguments,
                        size_t outputArgumentsSize, const UA_Argument *outputArguments,
                        void *nodeContext, UA_NodeId *outNewNodeId);

/* Extended version, allows the additional definition of fixed NodeIds for the
 * InputArgument/OutputArgument child variables */
UA_StatusCode UA_THREADSAFE
UA_Server_addMethodNodeEx(UA_Server *server, const UA_NodeId␣
↪requestedNewNodeId,
                          const UA_NodeId parentNodeId,
                          const UA_NodeId referenceTypeId,
                          const UA_QualifiedName browseName,
                          const UA_MethodAttributes attr, UA_MethodCallback method,
                          size_t inputArgumentsSize, const UA_Argument *inputArguments,
                          const UA_NodeId inputArgumentsRequestedNewNodeId,
                          UA_NodeId *inputArgumentsOutNewNodeId,
                          size_t outputArgumentsSize, const UA_Argument *outputArguments,
                          const UA_NodeId outputArgumentsRequestedNewNodeId,
                          UA_NodeId *outputArgumentsOutNewNodeId,
                          void *nodeContext, UA_NodeId *outNewNodeId);

UA_StatusCode UA_THREADSAFE
UA_Server_setMethodNodeCallback(UA_Server *server,
                                const UA_NodeId methodNodeId,
                                UA_MethodCallback methodCallback);

/* Backwards compatibility definition */
#define UA_Server_setMethodNode_callback(server, methodNodeId, methodCallback) \
    UA_Server_setMethodNodeCallback(server, methodNodeId, methodCallback)

UA_StatusCode UA_THREADSAFE
UA_Server_getMethodNodeCallback(UA_Server *server,
                                const UA_NodeId methodNodeId,
                                UA_MethodCallback *outMethodCallback);
```

5.10.4 ObjectNode

```
UA_THREADSAFE UA_StatusCode
UA_Server_addObjectNode(UA_Server *server, const UA_NodeId requestedNewNodeId,
                        const UA_NodeId parentNodeId,
                        const UA_NodeId referenceTypeId,
                        const UA_QualifiedName browseName,
                        const UA_NodeId typeDefinition,
                        const UA_ObjectAttributes attr,
                        void *nodeContext, UA_NodeId *outNewNodeId);
```

5.10.5 ObjectTypeNode

```
UA_THREADSAFE UA_StatusCode
UA_Server_addObjectTypeNode(UA_Server *server, const UA_NodeId␣
↪requestedNewNodeId,
                            const UA_NodeId parentNodeId,
                            const UA_NodeId referenceTypeId,
                            const UA_QualifiedName browseName,
                            const UA_ObjectTypeAttributes attr,
                            void *nodeContext, UA_NodeId *outNewNodeId);
```

5.10.6 ReferenceTypeNode

```
UA_THREADSAFE UA_StatusCode
UA_Server_addReferenceTypeNode(UA_Server *server,
                               const UA_NodeId requestedNewNodeId,
                               const UA_NodeId parentNodeId,
                               const UA_NodeId referenceTypeId,
                               const UA_QualifiedName browseName,
                               const UA_ReferenceTypeAttributes attr,
                               void *nodeContext, UA_NodeId *outNewNodeId);
```

5.10.7 DataTypeNode

```
UA_THREADSAFE UA_StatusCode
UA_Server_addDataTypeNode(UA_Server *server,
                          const UA_NodeId requestedNewNodeId,
                          const UA_NodeId parentNodeId,
                          const UA_NodeId referenceTypeId,
                          const UA_QualifiedName browseName,
                          const UA_DataTypeAttributes attr,
                          void *nodeContext, UA_NodeId *outNewNodeId);
```

Due to the history of development, the DataTypeAttributes structure used in the AddNodes Service does not describe the layout of the DataType. But the (newer) structures for describing DataTypes do:

- SimpleTypeDescription
- EnumDescription

- StructureDescription

The UA_Server_addDataTypeFromDescription function translates the DataTypeDescription into a UA_DataType structure and adds it to an internal array of the server. Then the DataType is automatically decoded in messages received by the server. Also the DataTypeDefinition attribute of the corresponding DataTypeNode can then be read via the Read service.

The memory layout of the internally generated UA_DataType corresponds to the matching C-structure including padding.

Note that a DataTypeDescription can be added only once during the lifetime of the server. This protects against existing instances of the DataType to having their layout changed.

```
/* Use the DataType description to create an internal UA_DataType entry in the
 * server */
UA_THREADSAFE UA_StatusCode
UA_Server_addDataTypeFromDescription(UA_Server *server,
                                     const UA_ExtensionObject *description);

/* The same as UA_Server_addDataTypeFromDescription, but with the description
 * already converted into a UA_DataType. Makes a copy of the UA_DataType
 * internally. */
UA_THREADSAFE UA_StatusCode
UA_Server_addDataType(UA_Server *server, const UA_NodeId parentNodeId,
                      const UA_DataType *type);

/* Get the entry to the linked list of custom datatypes. This includes both the
 * datatypes from serverConfig->customDataTypes and the internal custom data
 * types from UA_Server_addDataType.
 *
 * Attention! The output pointer is only valid until the next call to
 * UA_Server_addDataType. */
UA_THREADSAFE const UA_DataTypeArray *
UA_Server_getDataTypes(UA_Server *server);
```

5.10.8 ViewNode

```
UA_THREADSAFE UA_StatusCode
UA_Server_addViewNode(UA_Server *server, const UA_NodeId requestedNewNodeId,
                      const UA_NodeId parentNodeId,
                      const UA_NodeId referenceTypeId,
                      const UA_QualifiedName browseName,
                      const UA_ViewAttributes attr,
                      void *nodeContext, UA_NodeId *outNewNodeId);
```

5.10.9 NodeSet Import

Load a UANodeSet XML document into the server.

Load dependencies first by calling this function once for each document. The XML input is borrowed for the duration of the call. Individual nodes that cannot be added are logged and skipped. The return status reports failures that prevent loading the document as a whole. The operation is not transactional; nodes and namespaces added before a failure remain in the server.

```
UA_StatusCode
UA_Server_loadNodeset(UA_Server *server, const UA_XmlElement nodesetXml);
```

5.10.10 Node Lifecycle: Constructors, Destructors and Node Contexts

To finalize the instantiation of a node, a (user-defined) constructor callback is executed. There can be both a global constructor for all nodes and node-type constructor specific to the TypeDefinition of the new node (attached to an ObjectTypeNode or VariableTypeNode).

In the hierarchy of ObjectTypes and VariableTypes, only the constructor of the (lowest) type defined for the new node is executed. Note that every Object and Variable can have only one isTypeOf reference. But type-nodes can technically have several hasSubType references to implement multiple inheritance. Issues of (multiple) inheritance in the constructor need to be solved by the user.

When a node is destroyed, the node-type destructor is called before the global destructor. So the overall node lifecycle is as follows:

1. Global Early Constructor (set in the server config)
2. Node-Type Early Constructor (for Variables or Objects)
3. Recursive instantiation of the node's children
4. Global Constructor (set in the server config)
5. Node-Type Constructor (for Variables or Objects)
6. (Usage-period of the Node)
7. Node-Type Destructor
8. Global Destructor

The constructor and destructor callbacks can be set to NULL and are not used in that case. If a constructor fails, the global destructor will be called before removing the node. The destructors are assumed to never fail.

Every node carries a user-context and a constructor-context pointer. The user-context is used to attach custom data to a node. But the (user-defined) constructors and destructors may replace the user-context pointer if they wish to do so. The initial value for the constructor-context is NULL. When the AddNodes service is used over the network, the user-context pointer of the new node is also initially set to NULL.

```
UA_StatusCode UA_THREADSAFE
UA_Server_getNodeContext(UA_Server *server, UA_NodeId nodeId, void **nodeContext);

/* Careful! The user has to ensure that the destructor callbacks still work. */
UA_StatusCode UA_THREADSAFE
UA_Server_setNodeContext(UA_Server *server, UA_NodeId nodeId, void *nodeContext);
```

Global constructor and destructor callbacks used for every node. It gets set in the server config.

```
typedef struct {
    /* Can be NULL. May replace the nodeContext */
    UA_StatusCode (*constructor)(UA_Server *server,
                                 const UA_NodeId *sessionId, void *sessionContext,
```

(continues on next page)

(continued from previous page)

```
                        const UA_NodeId *nodeId, void **nodeContext);

    /* Can be NULL. The context cannot be replaced since the node is destroyed
     * immediately afterwards anyway. */
    void (*destructor)(UA_Server *server,
                       const UA_NodeId *sessionId, void *sessionContext,
                       const UA_NodeId *nodeId, void *nodeContext);

    /* Can be NULL. Called during recursive node instantiation. While mandatory
     * child nodes are automatically created if not already present, optional child
     * nodes are not. This callback can be used to define whether an optional child
     * node should be created.
     *
     * @param server The server executing the callback
     * @param sessionId The identifier of the session
     * @param sessionContext Additional data attached to the session in the
     *        access control layer
     * @param sourceNodeId Source node from the type definition. If the new node
     *        shall be created, it will be a copy of this node.
     * @param targetParentNodeId Parent of the potential new child node
     * @param referenceTypeId Identifies the reference type which that the parent
     *        node has to the new node.
     * @return Return UA_TRUE if the child node shall be instantiated,
     *         UA_FALSE otherwise. */
    UA_Boolean (*createOptionalChild)(UA_Server *server,
                                      const UA_NodeId *sessionId,
                                      void *sessionContext,
                                      const UA_NodeId *sourceNodeId,
                                      const UA_NodeId *targetParentNodeId,
                                      const UA_NodeId *referenceTypeId);

    /* Can be NULL. Called when a node is to be copied during recursive
     * node instantiation. Allows definition of the NodeId for the new node.
     * If the callback is set to NULL or the resulting NodeId is UA_NODEID_NUMERIC(X,
↪0)
     * an unused nodeid in namespace X will be used. E.g. passing UA_NODEID_NULL will
     * result in a NodeId in namespace 0.
     *
     * @param server The server executing the callback
     * @param sessionId The identifier of the session
     * @param sessionContext Additional data attached to the session in the
     *        access control layer
     * @param sourceNodeId Source node of the copy operation
     * @param targetParentNodeId Parent node of the new node
     * @param referenceTypeId Identifies the reference type which that the parent
     *        node has to the new node. */
    UA_StatusCode (*generateChildNodeId)(UA_Server *server,
                                         const UA_NodeId *sessionId, void *sessionContext,
                                         const UA_NodeId *sourceNodeId,
```

(continues on next page)

(continued from previous page)

```
                                const UA_NodeId *targetParentNodeId,
                                const UA_NodeId *referenceTypeId,
                                UA_NodeId *targetNodeId);

    /* Can be NULL. Called after the node has been inserted into the Nodestore
     * and its parent and TypeDefinition references have been added, but before
     * automatic child instantiation. This allows the callback to add children
     * that shall take the place of children declared by the TypeDefinition.
     * May replace the nodeContext. */
    UA_StatusCode (*earlyConstructor)(UA_Server *server,
                                      const UA_NodeId *sessionId,
                                      void *sessionContext,
                                      const UA_NodeId *nodeId,
                                      void **nodeContext);
} UA_GlobalNodeLifecycle;
```

The following node-type lifecycle can be set for VariableTypeNodes and ObjectTypeNodes. It gets called for instances of this node-type.

```
typedef struct {
    /* Can be NULL. May replace the nodeContext */
    UA_StatusCode (*constructor)(UA_Server *server,
                                 const UA_NodeId *sessionId, void *sessionContext,
                                 const UA_NodeId *typeNodeId, void *typeNodeContext,
                                 const UA_NodeId *nodeId, void **nodeContext);

    /* Can be NULL. May replace the nodeContext. */
    void (*destructor)(UA_Server *server,
                       const UA_NodeId *sessionId, void *sessionContext,
                       const UA_NodeId *typeNodeId, void *typeNodeContext,
                       const UA_NodeId *nodeId, void **nodeContext);

    /* Can be NULL. Called after the global earlyConstructor and before
     * automatic child instantiation. May replace the nodeContext. */
    UA_StatusCode (*earlyConstructor)(UA_Server *server,
                                      const UA_NodeId *sessionId,
                                      void *sessionContext,
                                      const UA_NodeId *typeNodeId,
                                      void *typeNodeContext,
                                      const UA_NodeId *nodeId,
                                      void **nodeContext);
} UA_NodeTypeLifecycle;

UA_StatusCode UA_THREADSAFE
UA_Server_setNodeTypeLifecycle(UA_Server *server, UA_NodeId nodeId,
                               UA_NodeTypeLifecycle lifecycle);
```

5.10.11 Detailed Node Construction

The method pair UA_Server_addNode_begin and _finish splits the AddNodes service in two parts. This is useful if the node shall be modified before finish the instantiation. For example to add children with specific NodeIds. Otherwise, mandatory children (e.g. of an ObjectType) are added with pseudo-random unique NodeIds. Existing children are detected during the _finish part via their matching BrowseName.

The _begin method:

- prepares the node and adds it to the nodestore
- copies some unassigned attributes from the TypeDefinition node internally
- adds the references to the parent (and the TypeDefinition if applicable)
- performs type-checking of variables
- calls the global and node-type earlyConstructors, if configured.

You can add an object node without a parent if you set the parentNodeId and referenceTypeId to UA_NODE_ID_NULL. Then you need to add the parent reference and hasTypeDef reference yourself before calling the _finish method. Not that this is only allowed for object nodes.

The _finish method:

- copies mandatory children
- calls the node constructor(s) at the end
- may remove the node if it encounters an error.

The special UA_Server_addMethodNode_finish method needs to be used for method nodes, since there you need to explicitly specify the input and output arguments which are added in the finish step (if not yet already there)

```
/* The ``attr`` argument must have a type according to the NodeClass.
 * ``VariableAttributes`` for variables, ``ObjectAttributes`` for objects, and
 * so on. Missing attributes are taken from the TypeDefinition node if
 * applicable. */
UA_StatusCode UA_THREADSAFE
UA_Server_addNode_begin(UA_Server *server, const UA_NodeClass nodeClass,
                        const UA_NodeId requestedNewNodeId,
                        const UA_NodeId parentNodeId,
                        const UA_NodeId referenceTypeId,
                        const UA_QualifiedName browseName,
                        const UA_NodeId typeDefinition,
                        const void *attr, const UA_DataType *attributeType,
                        void *nodeContext, UA_NodeId *outNewNodeId);

UA_StatusCode UA_THREADSAFE
UA_Server_addNode_finish(UA_Server *server, const UA_NodeId nodeId);


UA_StatusCode UA_THREADSAFE
UA_Server_addMethodNode_finish(UA_Server *server, const UA_NodeId nodeId,
                        UA_MethodCallback method,
```

(continues on next page)

(continued from previous page)

```
                    size_t inputArgumentsSize, const UA_Argument *inputArguments,
                    size_t outputArgumentsSize, const UA_Argument *outputArguments);


/* Deletes a node and optionally all references leading to the node. */
UA_StatusCode UA_THREADSAFE
UA_Server_deleteNode(UA_Server *server, const UA_NodeId nodeId,
                     UA_Boolean deleteReferences);
```

5.10.12 Reference Management

```
UA_StatusCode UA_THREADSAFE
UA_Server_addReference(UA_Server *server, const UA_NodeId sourceId,
                       const UA_NodeId refTypeId,
                       const UA_ExpandedNodeId targetId, UA_Boolean isForward);

UA_StatusCode UA_THREADSAFE
UA_Server_deleteReference(UA_Server *server, const UA_NodeId sourceNodeId,
                          const UA_NodeId referenceTypeId, UA_Boolean isForward,
                          const UA_ExpandedNodeId targetNodeId,
                          UA_Boolean deleteBidirectional);
```

5.11 Async Operations

Some operations can take time, such as reading a sensor that needs to warm up first. In order not to block the server, a long-running operation can be handled asynchronously and the result returned at a later time. The core idea is that a userland callback can return UA_STATUSCODE_GOODCOMPLETESASYNCHRONOUSLY as the statuscode to signal that it wishes to complete the operation later.

Currently, async operations are supported for the services

- Read
- Write
- Call

with the caveat that read/write need a CallbackValueSource registered for the variable. Values that are stored directly in a VariableNode are written and read immediately.

Note that an async operation can be cancelled (e.g. after a timeout period or if the caller cannot wait for the result). This is signaled in the configured asyncOperationCancelCallback. The provided memory locations to store the operation output are then no longer valid.

```
/* When the UA_MethodCallback returns UA_STATUSCODE_
↪GOODCOMPLETESASYNCHRONOUSLY,
 * then an async operation is created in the server for later completion. The
 * output pointer from the method callback is used to identify the async
 * operation. Do not access the output pointer after the operation has been
 * cancelled or after setting the result. */
UA_THREADSAFE UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_Server_setAsyncCallMethodResult(UA_Server *server, UA_Variant *output,
                                   UA_StatusCode result);

/* See the UA_CallbackValueSource documentation */
UA_THREADSAFE UA_StatusCode
UA_Server_setAsyncReadResult(UA_Server *server, UA_DataValue *result);

/* See the UA_CallbackValueSource documentation. The value needs to be the
 * pointer used in the write callback. The statuscode is the result signal to be
 * returned asynchronously. */
UA_THREADSAFE UA_StatusCode
UA_Server_setAsyncWriteResult(UA_Server *server, const UA_DataValue *value,
                              UA_StatusCode result);
```

The server supports asynchronous "local" read/write/call operations. The user supplies a result-callback that gets called either synchronously (if the operation terminates right away) or asynchronously at a later time. The result-callback is called exactly one time for each operation, also if the operation is cancelled. In this case a StatusCode like UA_STATUSCODE_BADTIMEOUT or UA_STATUSCODE_BADSHUTDOWN is set.

If an operation returns asynchronously, then the result-callback is executed only in the next iteration of the Eventloop. An exception to this is UA_Server_cancelAsync, which can optionally call the result-callback right away (e.g. as part of a cleanup where the context of the result-callback gets removed).

Async operations incur a small overhead since memory is allocated to persist the operation over time.

The operation timeout is defined in milliseconds. A timeout of zero means infinite.

```
typedef void(*UA_ServerAsyncReadResultCallback)
    (UA_Server *server, void *asyncOpContext, const UA_DataValue *result);
typedef void(*UA_ServerAsyncWriteResultCallback)
    (UA_Server *server, void *asyncOpContext, UA_StatusCode result);
typedef void(*UA_ServerAsyncMethodResultCallback)
    (UA_Server *server, void *asyncOpContext, const UA_CallMethodResult *result);

UA_StatusCode UA_THREADSAFE
UA_Server_read_async(UA_Server *server, const UA_ReadValueId *operation,
                     UA_TimestampsToReturn timestamps,
                     UA_ServerAsyncReadResultCallback callback,
                     void *asyncOpContext, UA_UInt32 timeout);

UA_StatusCode UA_THREADSAFE
UA_Server_write_async(UA_Server *server, const UA_WriteValue *operation,
                      UA_ServerAsyncWriteResultCallback callback,
                      void *asyncOpContext, UA_UInt32 timeout);

UA_StatusCode UA_THREADSAFE
UA_Server_call_async(UA_Server *server, const UA_CallMethodRequest *operation,
                     UA_ServerAsyncMethodResultCallback callback,
                     void *asyncOpContext, UA_UInt32 timeout);
```

Local async operations can be manually cancelled (besides an internal cancel due to a timeout or server shutdown). The local async operations to be cancelled are selected by matching their asyncOpContext pointer. This can cancel multiple operations that use the same context pointer.

For operations where the async result was not yet set, the asyncOperationCancelCallback from the server-config gets called and the cancel-status is set in the operation result.

For async operations where the result has already been set, but not yet notified with the result-callback (to be done in the next EventLoop iteration), the asyncOperationCancelCallback is not called and no cancel status is set in the result.

Each operation's result-callback gets called exactly once. When the operation is cancelled, the result-callback can be called synchronously using the synchronousResultCallback flag. Otherwise the result gets returned "normally" in the next EventLoop iteration. The synchronous option ensures that all (matching) async operations are fully cancelled right away. This can be important in a cleanup situation where the asyncOpContext is no longer valid in the future.

```
void UA_THREADSAFE
UA_Server_cancelAsync(UA_Server *server, void *asyncOpContext,
                      UA_StatusCode status,
                      UA_Boolean synchronousResultCallback);
```

5.12 Events

Events are emitted by objects in the OPC UA information model. Starting at the source-node, the events "bubble up" in the hierarchy of objects and are caught by MonitoredItems listening for them.

EventTypes are special ObjectTypeNodes that describe the (data) fields of an event instance. An EventType can simply contain a flat list of VariableNodes. But (deep) nesting of objects and variables is also allowed. The individual MonitoredItems then contain an EventFilter (with a select-clause) that defines the event fields to be transmitted to a particular client.

In open62541, there are three possible sources for the event fields. When the select-clause of an EventFilter is resolved, the sources are evaluated in the following order:

1. An key-value map that defines event fields. The key of its entries is a "path-string", a :ref:human-readable encoding of a SimpleAttributeOperand<parse-sao>`. For example ``/SourceNode or /EventType.
2. A NodeId pointing to an ObjectNode that instantiates an EventType. The SimpleAttributeOperands from the EventFilter are resolved in its context.
3. The event fields defined as mandatory for the *BaseEventType* have a default that gets used if they are not defined otherwise:

/EventId	ByteString to uniquely identify the event instance (default: random 16-byte ByteString)
/EventType	NodeId of the EventType (default: argument of _createEvent)
/SourceNode	NodeId of the emitting node (default: argument of _createEvent)
/SourceName	LocalizedText with the DisplayName of the source node (default: read from the information model)

/Time	DateTime with the timestamp when the event occurred (default: current time)
/ReceiveTime	DateTime when the server received the information about the event from an underlying device (default: current time)
/Message	LocalizedText with a human-readable description of the event (default: argument of _createEvent)
/Severity	UInt16 for the urgency of the event defined to be between 1 (lowest) and 1000 (catastrophic) (default: argument of _createEvent)

The "path-string" (SimpleAttributeOperand expression) can use namespace-indices and point into nested objects and variables. For example /1:Truck/2:Wheel.

The key-value map source for the event-fields uses a QualifiedName for the key. The NamespaceIndex from the key is used as the default NamespaceIndex for the path elements that do not define it explicitly. So the key 2:"/1:Truck/Wheel" becomes /1:Truck/2:Wheel.

An event field that is missing from all sources resolves to an empty variant.

It is typically faster to define event-fields in the key-value map than to look them up from an event instance in the information model. This is particularly important for events emitted at a high frequency.

```
#ifdef UA_ENABLE_SUBSCRIPTIONS_EVENTS

/* Create an event in the server. The eventFields and eventInstance pointer can
 * be NULL and are then not considered as a source of event fields. The
 * outEventId pointer can be NULL. If set, the EventId of a successfully created
 * Event gets copied into the argument. */
UA_StatusCode UA_THREADSAFE
UA_Server_createEvent(UA_Server *server, const UA_NodeId sourceNode,
                      const UA_NodeId eventType, UA_UInt16 severity,
                      const UA_LocalizedText message,
                      const UA_KeyValueMap *eventFields,
                      const UA_NodeId *eventInstance,
                      UA_ByteString *outEventId);

/* Extended version of the _createEvent API. The members of the
 * UA_EventDescription structure have the same meaning as above.
 *
 * In addition, the extended version allows the filtering of Events to be only
 * transmitted to a particular Session/Subscription/MonitoredItem. The filtering
 * criteria can be NULL. But the subscriptionId requires a sessionId and the
 * monitoredItemId requires a subscriptionId as context. */

typedef struct {
    /* Event fields */
    UA_NodeId sourceNode;
    UA_NodeId eventType;
    UA_UInt16 severity;
    UA_LocalizedText message;
    const UA_KeyValueMap *eventFields;
```

(continues on next page)

(continued from previous page)

```
    const UA_NodeId *eventInstance;

    /* Restrict who can receive the event */
    const UA_NodeId *sessionId;
    const UA_UInt32 *subscriptionId;
    const UA_UInt32 *monitoredItemId;
} UA_EventDescription;

UA_StatusCode UA_THREADSAFE
UA_Server_createEventEx(UA_Server *server,
                        const UA_EventDescription *ed,
                        UA_ByteString *outEventId);

#endif /* UA_ENABLE_SUBSCRIPTIONS_EVENTS */
```

5.12.1 Model and Semantic Changes

With UA_ENABLE_SUBSCRIPTIONS_EVENTS, the server automatically emits the standard GeneralModelChangeEventType and SemanticChangeEventType for changes made through OPC UA Services and the corresponding local UA_Server_* APIs. Changes made while the server's namespace is initially populated are suppressed. Both EventTypes are emitted by the Server object.

A structural change is reported only if the affected Node has a scalar NodeVersion Property with the String DataType. The server updates that Property with the decimal representation of a server-wide, increasing Int64. Nodes without a suitable NodeVersion Property are not included in a ModelChangeEvent. The following successful operations are reported:

Operation	ModelChange verb
Add a Node	NodeAdded
Delete a Node	NodeDeleted
Add a Reference	ReferenceAdded
Delete a Reference	ReferenceDeleted
Change the DataType Attribute	DataTypeChanged

Changes to ValueRank and ArrayDimensions are not structural ModelChanges.

A SemanticChange is reported when a successful Value write changes a Variable whose AccessLevel contains UA_ACCESSLEVELMASK_SEMANTICCHANGE. The Variable must be a Property connected to its owner by HasProperty or a subtype. The owner is reported as the affected Node. Same-value writes are suppressed when the previous value is directly available. For callback-based value sources the previous value may not be available for comparison, so a successful write is treated as a SemanticChange.

A SemanticChange also marks Value MonitoredItems on the affected Variable. Their next DataChange notification contains the SemanticsChanged StatusCode bit. MonitoredItems with a zero SamplingInterval are sampled immediately. For cyclic sampling the bit remains pending until the next notification.

Changes are accumulated until the outermost local operation or decoded Service request completes. Entries for the same affected Node are coalesced by combining their ModelChange verbs. Model and

Semantic changes collected together are emitted as separate standard Events. Failed operations are not reported.

See examples/events/server_modelchange.c for a complete local example.

5.13 Remote Discovery Server Registration

The current server can register itself at a discovery. For that it requires to open a client connection.

```
/* Register the given server instance at a discovery server. This should be
 * called periodically, for example every 10 minutes, depending on the
 * configuration of the discovery server.
 *
 * The supplied client configuration is used to create a new client to connect
 * to the discovery server. The client configuration is moved over to the server
 * and eventually cleaned up internally. The structure pointed at by `cc` is
 * zeroed to avoid accessing outdated information.
 *
 * The eventloop and logging plugins in the client configuration are replaced by
 * those configured in the server. */
UA_StatusCode UA_THREADSAFE
UA_Server_registerDiscovery(UA_Server *server, UA_ClientConfig *cc,
                            const UA_String discoveryServerUrl,
                            const UA_String semaphoreFilePath);

/* Deregister the given server instance from the discovery server.
 * This should be called when the server is shutting down. */
UA_StatusCode UA_THREADSAFE
UA_Server_deregisterDiscovery(UA_Server *server, UA_ClientConfig *cc,
                              const UA_String discoveryServerUrl);
```

5.14 Local Discovery Server Records

The Discovery Service-Set allows the registering of local (for FindServers) and also remote servers (for FindServersOnNetwork).

We uniquely identify records by their combination of ServerUri + one matching DiscoveryUrl.

```
UA_StatusCode UA_THREADSAFE
UA_Server_findServers(UA_Server *server, UA_String endpointUrl,
                      size_t localeIdsSize, UA_LocaleId *localeIds,
                      size_t serverUrisSize, UA_String *serverUris,
                      size_t *outServersSize,
                      UA_ApplicationDescription **outServers);

/* Local API for the RegisterServer2 service. If configurationResults is
 * non-Null, then it must point to an array of discoveryConfigurationSize
 * length. */
UA_StatusCode UA_THREADSAFE
```

(continues on next page)

(continued from previous page)

```
UA_Server_registerServer(UA_Server *server,
                         const UA_RegisteredServer *registeredServer,
                         const size_t discoveryConfigurationSize,
                         const UA_ExtensionObject *discoveryConfiguration,
                         UA_StatusCode *configurationResults);

/* Remove the servers matching the ServerUri and at least one of the
 * provided DiscoveryUrls */
UA_StatusCode UA_THREADSAFE
UA_Server_deregisterServer(UA_Server *server, const UA_String serverUri,
                           size_t discoveryUrlsSize,
                           const UA_String *discoveryUrls);
```

The server internally manages the ServersOnNetwork list. Multicast discovery is implemented on top in a driver outside of the core library.

```
UA_StatusCode UA_THREADSAFE
UA_Server_findServersOnNetwork(UA_Server *server, UA_String endpointUrl,
                               UA_UInt32 startingRecordId,
                               UA_UInt32 maxRecordsToReturn,
                               size_t serverCapabilityFilterSize,
                               const UA_String *serverCapabilityFilter,
                               UA_DateTime *outLastCounterResetTime,
                               size_t *outServersSize,
                               UA_ServerOnNetwork **outServers);

/* Register a remote server. If the server name was previously known, the
 * existing entry gets updated. The parameter kv-map can be extended with
 * additional parameters in the future. Currently supported are:
 *
 * 0:remote-address [String]
 *    IP-address or other host identifier from which the information
 *    was received.
 * 0:ttl [UInt32]
 *    Time-to-live of DNS information. Zero means infinite. */
UA_StatusCode UA_THREADSAFE
UA_Server_registerServerOnNetwork(UA_Server *server,
                                  const UA_ServerOnNetwork *son,
                                  const UA_KeyValueMap params);

/* Remove the entry of the remote server with the matching ServerName */
UA_StatusCode UA_THREADSAFE
UA_Server_deregisterServerOnNetwork(UA_Server *server,
                                    UA_String serverName);
```

5.15 Drivers

Drivers are different from other "plugins" in that they have an explicit stateful lifecycle and can be started/stopped at runtime. Their lifecycle is however dependent on the server into which the drivers are embedded. When the server shuts down, the drivers are also stopped.

Drivers can use the server's public API to the full extent. For example add/remove nodes, or register connections and timers in the server's EventLoop.

Some drivers define core functionality and are added internally in the server implementation.

```
/* GENERIC permits multiple instances. Other driver types
 * can be added only once to a server. */
typedef enum {
    UA_DRIVERTYPE_GENERIC = 0,
    UA_DRIVERTYPE_GDS_RECEIVER
} UA_DriverType;

struct UA_Driver;
typedef struct UA_Driver UA_Driver;

/* Callback through which the server notifies the driver
 * about runtime changes and internal events. */
typedef void
(*UA_DriverNotificationCallback)(UA_Driver *drv,
                                 UA_ApplicationNotificationType type,
                                 const UA_KeyValueMap payload);

struct UA_Driver {
    UA_Driver *next; /* linked-list */

    /*
     * Configuration
     */

    UA_DriverType driverType;
    UA_String name;

    /* See the driver-specific documentation for possible parameters.
     * The params need to be cleaned up within the _free method. */
    UA_KeyValueMap params;

    /* Backpointer to the server. Must be set before _start is called. If NULL
     * this is set by the server during registering. Generally the server must
     * not be switched out once the driver has been started. */
    UA_Server *server;

    /* The server forwards its internal notifications to all drivers. In order
     * to avoid overload, the filter must be set. The top 32bit are ANDed with
     * the notification type to see if the driver is interested in the
     * notification. See the common.h for details on the notification types and
     * their payload. */
```

(continues on next page)

(continued from previous page)

```
    UA_DriverNotificationCallback notificationCallback;
    UA_ApplicationNotificationType notificationFilter;

    /*
     * Lifecycle management
     */

    UA_LifecycleState state;

    /* Start the Driver. It will typically register timers/connections in the
     * EventLoop and may add nodes in the server's information model. Starting
     * can fail if the server is not already started also.
     *
     * During startup, the server calls start on all registered drivers. */
    UA_StatusCode (*start)(UA_Driver *sc);

    /* Stopping is asynchronous and might need a few iterations of the eventloop
     * to succeed. All Drivers are stopped during the shutdown of the server.
     * Once fully stopped, the Driver must no longer rely on the server
     * backpointer. So it can be detached and _free'd at runtime of the
     * server. */
    void (*stop)(UA_Driver *sc);

    /* Clean up and delete the Driver. Can fail if it is not fully stopped. When
     * successfully removed, the Driver must no longer be accessed from the
     * server.
     *
     * Drivers are all free'd when the server is deleted. If a Driver is
     * manually removed before, then it needs to be unlinked from the server's
     * internal linked-list before. */
    UA_StatusCode (*free)(UA_Driver *sc);
};

/* Adds the Driver to the server. Starts the driver if the server is already
 * started and returns the start result. A failed start rolls back registration
 * and leaves ownership with the caller. The server backpointer is retained
 * for cleanup; stop the driver if needed before freeing it. */
UA_StatusCode
UA_Server_addDriver(UA_Server *server, UA_Driver *drv);

/* Remove the Driver from the server. This will fail if the driver is not fully
 * stopped. */
UA_StatusCode
UA_Server_removeDriver(UA_Server *server, UA_Driver *drv);

/* Get the first entry of the server's driver linked list. */
UA_Driver *
UA_Server_getDrivers(UA_Server *server);
```

5.16 Statistics

Statistic counters keeping track of the current state of the stack. Counters are structured per OPC UA communication layer.

```
typedef struct {
  UA_SecureChannelStatistics scs;
  UA_SessionStatistics ss;
} UA_ServerStatistics;

UA_ServerStatistics UA_THREADSAFE
UA_Server_getStatistics(UA_Server *server);
```

5.17 Reverse Connect

The reverse connect feature of OPC UA permits the server instead of the client to establish the connection. The client must expose the listening port so the server is able to reach it.

```
/* The reverse connect state change callback is called whenever the state of a
 * reverse connect is changed by a connection attempt, a successful connection
 * or a connection loss.
 *
 * The reverse connect states reflect the state of the secure channel currently
 * associated with a reverse connect. The state will remain
 * UA_SECURECHANNELSTATE_CONNECTING while the server attempts repeatedly to
 * establish a connection. */
typedef void (*UA_Server_ReverseConnectStateCallback)(UA_Server *server,
                                                      UA_UInt64 handle,
                                                      UA_SecureChannelState state,
                                                      void *context);

/* Registers a reverse connect in the server. The server periodically attempts
 * to establish a connection if the initial connect fails or if the connection
 * breaks.
 *
 * @param server The server object
 * @param url The URL of the remote client
 * @param stateCallback The callback which will be called on state changes
 * @param callbackContext The context for the state callback
 * @param handle Is set to the handle of the reverse connect if not NULL
 * @return Returns UA_STATUSCODE_GOOD if the reverse connect has been registered␣
↪*/
UA_StatusCode
UA_Server_addReverseConnect(UA_Server *server, UA_String url,
                            UA_Server_ReverseConnectStateCallback stateCallback,
                            void *callbackContext, UA_UInt64 *handle);

/* Removes a reverse connect from the server and closes the connection if it is
 * currently open.
 *
```

(continues on next page)

(continued from previous page)

```
 * @param server The server object
 * @param handle The handle of the reverse connect to remove
 * @return Returns UA_STATUSCODE_GOOD if the reverse connect has been
 *         successfully removed */
UA_StatusCode
UA_Server_removeReverseConnect(UA_Server *server, UA_UInt64 handle);
```

5.18 Utility Functions

```
/* Lookup a datatype by its NodeId. Takes the custom types in the server
 * configuration into account. Return NULL if none found. */
const UA_DataType *
UA_Server_findDataType(UA_Server *server, const UA_NodeId *typeId);

/* Add a new namespace to the server. Returns the index of the new namespace */
UA_UInt16 UA_THREADSAFE
UA_Server_addNamespace(UA_Server *server, const char* name);

/* Get namespace by name from the server. */
UA_StatusCode UA_THREADSAFE
UA_Server_getNamespaceByName(UA_Server *server, const UA_String namespaceUri,
                             size_t* foundIndex);

/* Get namespace by id from the server. */
UA_StatusCode UA_THREADSAFE
UA_Server_getNamespaceByIndex(UA_Server *server, const size_t namespaceIndex,
                              UA_String *foundUri);
```

Some convenience functions are provided to simplify the interaction with objects.

```
/* Write an object property. The property is represented as a VariableNode with
 * a ``HasProperty`` reference from the ObjectNode. The VariableNode is
 * identified by its BrowseName. Writing the property sets the value attribute
 * of the VariableNode.
 *
 * @param server The server object
 * @param objectId The identifier of the object (node)
 * @param propertyName The name of the property
 * @param value The value to be set for the event attribute
 * @return The StatusCode for setting the event attribute */
UA_StatusCode UA_THREADSAFE
UA_Server_writeObjectProperty(UA_Server *server, const UA_NodeId objectId,
                              const UA_QualifiedName propertyName,
                              const UA_Variant value);

/* Directly point to the scalar value instead of a variant */
UA_StatusCode UA_THREADSAFE
UA_Server_writeObjectProperty_scalar(UA_Server *server, const UA_NodeId objectId,
```

(continues on next page)

(continued from previous page)

```
                                const UA_QualifiedName propertyName,
                                const void *value, const UA_DataType *type);

/* Read an object property.
 *
 * @param server The server object
 * @param objectId The identifier of the object (node)
 * @param propertyName The name of the property
 * @param value Contains the property value after reading. Must not be NULL.
 * @return The StatusCode for setting the event attribute */
UA_StatusCode UA_THREADSAFE
UA_Server_readObjectProperty(UA_Server *server, const UA_NodeId objectId,
                             const UA_QualifiedName propertyName,
                             UA_Variant *value);
```

5.19 Role-Based Access Control (RBAC)

Role-Based Access Control implementation per OPC UA Part 18 v1.05.

WARNING: This feature is EXPERIMENTAL and NOT FOR PRODUCTION USE. The RBAC implementation is under active development and the API may change. Use only for testing and development purposes.

RBAC allows fine-grained access control by assigning roles to sessions and defining permissions per role on individual nodes or entire namespaces.

5.19.1 Type Definitions

```
#ifdef UA_ENABLE_RBAC

/* UA_RolePermission
 * Maps a single role to its permissions bitmask. Used in the server
 * configuration to define presets and in the public API to set or query
 * role permissions on nodes. */
typedef struct {
    UA_NodeId roleId;
    UA_PermissionType permissions;
} UA_RolePermission;

/* UA_RolePermissionSet
 * A set of role-permission mappings. Used in the server configuration
 * to define initial role-permission presets. */
typedef struct {
    size_t rolePermissionsSize;
    UA_RolePermission *rolePermissions;
} UA_RolePermissionSet;

/* UA_RolePermissionSet Type Management */
void
UA_RolePermissionSet_init(UA_RolePermissionSet *rps);
```

(continues on next page)

(continued from previous page)

```
void
UA_RolePermissionSet_clear(UA_RolePermissionSet *rps);

UA_StatusCode
UA_RolePermissionSet_copy(const UA_RolePermissionSet *src,
                          UA_RolePermissionSet *dst);

/* UA_Role
 * Represents an OPC UA role with identity mapping rules and optional
 * application/endpoint restrictions per OPC UA Part 18 v1.05 §4.4. */
typedef struct {
    UA_NodeId roleId;
    UA_QualifiedName roleName;              /* BrowseName of the role */

    /* Identity Mapping Rules - determine which sessions get this role */
    size_t identityMappingRulesSize;
    UA_IdentityMappingRuleType *identityMappingRules;

    /* Application restrictions  (empty list = ignore) */
    UA_Boolean applicationsExclude;
    size_t applicationsSize;
    UA_String *applications;

    /* Endpoint restrictions (empty list = ignore) */
    UA_Boolean endpointsExclude;
    size_t endpointsSize;
    UA_EndpointType *endpoints;
} UA_Role;

/* UA_Role Type Management */
void
UA_Role_init(UA_Role *role);

void
UA_Role_clear(UA_Role *role);

UA_StatusCode
UA_Role_copy(const UA_Role *src, UA_Role *dst);

UA_Boolean
UA_Role_equal(const UA_Role *r1, const UA_Role *r2);

#endif /* UA_ENABLE_RBAC */
```

5.20 Server Configuration

The configuration structure is passed to the server during initialization. The server expects that the configuration is not modified during runtime. Currently, only one server can use a configuration at a time. During shutdown, the server will clean up the parts of the configuration that are modified at runtime through the provided API.

Examples for configurations are provided in the /plugins folder. The usual usage is as follows:

1. Create a server configuration with default settings as a starting point
2. Modify the configuration, e.g. by adding a server certificate
3. Instantiate a server with it
4. After shutdown of the server, clean up the configuration (free memory)

The *Tutorials* provide a good starting point for this.

```
/* Encryption mode requirement for OPC UA Binary over WebSockets */
typedef enum {
    UA_WEBSOCKET_ENCRYPTION_OPTIONAL = 0, /* Allow both opc.ws:// (unencrypted) and opc.wss:// (TLS) */
    UA_WEBSOCKET_ENCRYPTION_REQUIRED = 1, /* Allow only opc.wss:// (TLS); reject opc.ws:// */
    UA_WEBSOCKET_ENCRYPTION_DISABLED = 2  /* Allow only opc.ws:// (unencrypted); reject opc.wss:// */
} UA_WebSocketEncryptionMode;

struct UA_ServerConfig {
    void *context; /* Used to attach custom data to a server config. This can
                    * then be retrieved e.g. in a callback that forwards a
                    * pointer to the server. */
    UA_Logger *logging; /* Plugin for log output */

    /* Server Description
     * ~~~~~~~~~~~~~~~~~~
     * The description must be internally consistent. The ApplicationUri set in
     * the ApplicationDescription must match the URI set in the server
     * certificate.
     * The applicationType is not just descriptive, it changes the actual
     * functionality of the server. The RegisterServer service is available only
     * if the server is a DiscoveryServer and the applicationType is set to the
     * appropriate value.*/
    UA_BuildInfo buildInfo;
    UA_ApplicationDescription applicationDescription;

    /* Server Lifecycle
     * ~~~~~~~~~~~~~~~~
     * Delay in ms from the shutdown signal (ctrl-c) until the actual shutdown.
     * Clients need to be able to get a notification ahead of time. */
    UA_Double shutdownDelay;

    /* If an asynchronous server shutdown is used, this callback notifies about
```

(continues on next page)

(continued from previous page)

```
     * the current lifecycle state (notably the STOPPING -> STOPPED
     * transition). */
    void (*notifyLifecycleState)(UA_Server *server, UA_LifecycleState state);

    /* Rule Handling
     * ~~~~~~~~~~~~~
     * Override the handling of standard-defined behavior. These settings are
     * used to balance the following contradicting requirements:
     *
     * - Strict conformance with the standard (for certification).
     * - Ensure interoperability with old/non-conforming implementations
     *   encountered in the wild.
     *
     * The defaults are set for compatibility with the largest number of OPC UA
     * vendors (with log warnings activated). Cf. Postel's Law "be conservative
     * in what you send, be liberal in what you accept".
     *
     * See the section :ref:`rule-handling` for the possible settings. */

    /* Verify that the server sends a timestamp in the request header */
    UA_RuleHandling verifyRequestTimestamp;

    /* Variables (that don't have a DataType of BaseDataType) must not have an
     * empty variant value. The default behaviour is to auto-create a matching
     * zeroed-out value for empty VariableNodes when they are added. */
    UA_RuleHandling allowEmptyVariables;

    /* Methods shall be the target of at least one HasComponent reference from
     * an Object or ObjectType. Some legacy NodeSets contain detached Methods.
     * DEFAULT and WARN log and accept them, ABORT rejects them and ACCEPT
     * accepts them silently. */
    UA_RuleHandling allowUnattachedMethods;

    UA_RuleHandling allowAllCertificateUris;

    /* Verify that client ApplicationInstanceCertificates and X509IdentityTokens
     * permit clientAuth. For RSA application certificates the EKU extension is
     * mandatory. For ECC application certificates and user certificates it is
     * optional, but must permit clientAuth when present. ABORT rejects invalid
     * usage, DEFAULT and WARN log and continue, and ACCEPT continues silently. */
    UA_RuleHandling certificateEkuRule;

    /* Custom Data Types
     * ~~~~~~~~~~~~~~~~~
     * The following is a linked list of arrays with custom data types. All data
     * types that are accessible from here are automatically considered for the
     * decoding of received messages. Custom data types are not cleaned up
     * together with the configuration. So it is possible to allocate them on
     * ROM.
```

(continues on next page)

(continued from previous page)

```
     *
     * See the section on :ref:`generic-types`. Examples for working with custom
     * data types are provided in ``/examples/custom_datatype/``. */
    UA_DataTypeArray *customDataTypes;

    /* EventLoop
     * ~~~~~~~~~
     * A shared EventLoop must be external to every client and server using it.
     * Otherwise, one of them can stop or delete it while others still use it.
     * The application owns the shared EventLoop's lifetime. */
    UA_EventLoop *eventLoop;
    UA_Boolean externalEventLoop; /* Do not stop or delete the EventLoop */

    /* Application Notification
     * ~~~~~~~~~~~~~~~~~~~~~~~~
     * The notification callbacks can be NULL. The global callback receives all
     * notifications. The specialized callbacks receive only the subset
     * indicated by their name. */
    UA_ServerNotificationCallback globalNotificationCallback;
    UA_ServerNotificationCallback lifecycleNotificationCallback;
    UA_ServerNotificationCallback secureChannelNotificationCallback;
    UA_ServerNotificationCallback sessionNotificationCallback;
    UA_ServerNotificationCallback serviceNotificationCallback;
    UA_ServerNotificationCallback subscriptionNotificationCallback;
    UA_ServerNotificationCallback discoveryNotificationCallback;
#ifdef UA_ENABLE_AUDITING
    UA_ServerNotificationCallback auditNotificationCallback;
#endif

    /* Networking
     * ~~~~~~~~~~
     * The `serverUrls` array contains the server URLs like
     * `opc.tcp://my-server:4840`, `opc.wss://localhost:443` or
     * `opc.https://localhost:443/ua`. The non-standard `opc.ws` and
     * `opc.http` schemes can be enabled explicitly for testing. The URLs are
     * used both for discovery and to set up the server sockets based on the
     * defined hostnames, ports and WebSocket paths.
     *
     * - If the list is empty: Listen on all network interfaces with TCP port 4840.
     * - If the hostname of a URL is empty: Use the define protocol and port and
     *   listen on all interfaces. */
    UA_String *serverUrls;
    size_t serverUrlsSize;

    /* The following settings are specific to OPC UA with TCP transport. */
    UA_Boolean tcpEnabled;    /* Enable the TCP listener (default: true) */
    UA_UInt32 tcpBufSize;    /* Max length of sent and received chunks (packets)
                              * (default: 64kB) */
    UA_UInt32 tcpMaxMsgSize; /* Max length of messages
```

(continues on next page)

(continued from previous page)

```
                         * (default: 0 -> unbounded) */
    UA_UInt32 tcpMaxChunks;  /* Max number of chunks per message
                         * (default: 0 -> unbounded) */
    UA_Boolean tcpReuseAddr;

    /* The following settings are specific to OPC UA Binary over WebSockets.
     * The transport is opt-in and controlled via webSocketEnabled (default: false).
     * TLS credentials protect opc.wss:// endpoints independently of OPC UA␣
↪SecurityPolicies. */
    UA_Boolean webSocketEnabled; /* Enable the WebSocket listener (default: false) */
    UA_Boolean webSocketAllowUnencrypted; /* Allow non-standard unencrypted opc.ws://
↪ endpoints (default: false) */
    UA_WebSocketEncryptionMode webSocketEncryptionMode; /* Encryption requirement␣
↪(default: UA_WEBSOCKET_ENCRYPTION_OPTIONAL) */
    UA_UInt32 webSocketBufSize;    /* Max length of sent and received chunks
                               * (default: 64kB) */
    UA_UInt32 webSocketMaxMsgSize; /* Max length of messages
                               * (default: 0 -> unbounded) */
    UA_UInt32 webSocketMaxChunks;  /* Max number of chunks per message
                               * (default: 0 -> unbounded) */
    UA_UInt32 webSocketMaxQueueSize; /* Max bytes queued for a slow WebSocket
                                 * peer (default: 16 * webSocketBufSize) */
    UA_ByteString webSocketCertificate; /* TLS certificate, DER or PEM */
    UA_ByteString webSocketPrivateKey;  /* TLS private key, DER or PEM */
    UA_String webSocketPrivateKeyPassword;

    UA_Boolean httpEnabled; /* Default: false */
    /* Allow the non-standard, unencrypted opc.http scheme. This removes
     * transport confidentiality and is intended for testing only. Passwords
     * still require an encrypted UserTokenPolicy unless
     * allowNonePolicyPassword is enabled separately. */
    UA_Boolean httpAllowUnencrypted; /* Default: false */

    /* OPC UA services over HTTP(S) (Part 6, 7.4 defines HTTPS). Unlike
     * opc.wss, service structures are encoded directly in the request and
     * response body. The opc.http scheme is an open62541 extension. */
    UA_UInt32 httpMaxMsgSize;       /* Default: 0 -> unbounded */
    UA_UInt32 httpMaxDecompressedMsgSize; /* Expanded request-body limit.
                                     * Default: 0 -> 64 MiB */
    UA_UInt16 httpTimeout;          /* Request timeout in seconds (default: 60) */
    /* Bind address for HTTP(S) listeners. A null String uses the hostname from
     * the corresponding ServerUrl. An explicitly set empty String
     * listens on all interfaces without changing the advertised hostname. */
    UA_String httpListenAddress;
    UA_ByteString httpCertificate;  /* TLS certificate, DER or PEM */
    UA_ByteString httpPrivateKey;   /* TLS private key, DER or PEM */
    UA_String httpPrivateKeyPassword;

    /* Security and Encryption
```

(continues on next page)

(continued from previous page)

```
     * ~~~~~~~~~~~~~~~~~~~~~~~~~ */
    size_t securityPoliciesSize;
    UA_SecurityPolicy* securityPolicies;

    /* Endpoints with combinations of SecurityPolicy and SecurityMode. If the
     * UserIdentityToken array of the Endpoint is not set, then it will be
     * filled by the server for all UserTokenPolicies that are configured in the
     * AccessControl plugin. */
    size_t endpointsSize;
    UA_EndpointDescription *endpoints;

    /* Only allow the following discovery services to be executed on a
     * SecureChannel with SecurityPolicyNone: GetEndpointsRequest,
     * FindServersRequest and FindServersOnNetworkRequest.
     *
     * Only enable this option if there is no endpoint with SecurityPolicy#None
     * in the endpoints list. The SecurityPolicy#None must be present in the
     * securityPolicies list. */
    UA_Boolean securityPolicyNoneDiscoveryOnly;

    /* Allow clients without encryption support to connect with username and password.
     * This requires to transmit the password in plain text over the network which is
     * why this option is disabled by default.
     * Make sure you really need this before enabling plain text passwords. */
    UA_Boolean allowNonePolicyPassword;

    /* Different sets of certificates are trusted for SecureChannel / Session.
     * They correspond to the CertificateGroups "DefaultApplicationGroup" and
     * "DefaultUserTokenGroup" from Part 12.
     *
     * If the client authenticates with an X509IdentityToken (ActivateSession
     * Service), then this certificate is validated with the sessionPKI before
     * forwarding the token to the AccessControl plugin. */
    UA_CertificateGroup secureChannelPKI;
    UA_CertificateGroup sessionPKI;

    /* See the AccessControl Plugin API */
    UA_AccessControl accessControl;

    /* Nodes and Node Lifecycle
     * ~~~~~~~~~~~~~~~~~~~~~~~~
     * See the section for :ref:`node lifecycle handling<node-lifecycle>`. */
    UA_Nodestore *nodestore;
    UA_GlobalNodeLifecycle *nodeLifecycle;

    /* Copy the HasModellingRule reference in instances from the type
     * definition in UA_Server_addObjectNode and UA_Server_addVariableNode.
     *
     * Part 3 - 6.4.4: [...] it is not required that newly created or referenced
```

(continues on next page)

(continued from previous page)

```
    * instances based on InstanceDeclarations have a ModellingRule, however, it
    * is allowed that they have any ModellingRule independent of the
    * ModellingRule of their InstanceDeclaration */
   UA_Boolean modellingRulesOnInstances;

   /* Copy Method instance declarations into each Object instance instead of
    * adding a reference to the Method on the ObjectType. The default is false
    * for backwards compatibility. */
   UA_Boolean copyMethodsOnInstances;

   /* Limits
    * ~~~~~~ */
   /* Limits for SecureChannels */
   UA_UInt16 maxSecureChannels; /* 0 => unlimited */
   UA_UInt32 maxSecurityTokenLifetime; /* in ms */

   /* Limits for Sessions */
   UA_UInt16 maxSessions; /* 0 => unlimited */
   UA_Double maxSessionTimeout; /* in ms */

   /* Operation limits */
   UA_UInt32 maxNodesPerRead;
   UA_UInt32 maxNodesPerWrite;
   UA_UInt32 maxNodesPerMethodCall;
   UA_UInt32 maxNodesPerBrowse;
   UA_UInt32 maxNodesPerRegisterNodes;
   UA_UInt32 maxNodesPerTranslateBrowsePathsToNodeIds;
   UA_UInt32 maxNodesPerNodeManagement;
   UA_UInt32 maxMonitoredItemsPerCall;

   /* Limits for Requests */
   UA_UInt32 maxReferencesPerNode;

   /* Reverse Connect
    * ~~~~~~~~~~~~~~~ */
   UA_UInt32 reverseReconnectInterval; /* Default is 15000 ms */

   /* Async Operations
    * ~~~~~~~~~~~~~~~~
    * See the section for :ref:`async operations<async-operations>`. */
   UA_Double asyncOperationTimeout;   /* in ms, 0 => unlimited */
   size_t maxAsyncOperationQueueSize; /* 0 => unlimited */

   /* Notifies the userland that an async operation has been canceled. The
    * memory for setting the output value is then freed internally and should
    * not be touched afterwards. */
   void (*asyncOperationCancelCallback)(UA_Server *server, const void *out);

#ifdef UA_ENABLE_ENCRYPTION
```

(continues on next page)

(continued from previous page)

```
    /* Limits for TrustList */
    UA_UInt32 maxTrustListSize; /* in bytes, 0 => unlimited */
    UA_UInt32 maxRejectedListSize; /* 0 => unlimited */
#endif

    /* Discovery
     * ~~~~~~~~~ */
    /* Bitmask of the UA_SERVERCAPABILITY_* constants. At startup, the
     * capabilities that are announced are derived from this bitmask and the
     * ApplicationType (Part 12, 7.10.14). */
    UA_UInt64 serverCapabilities;

    /* Enable the internal management of RegisteredServers (besides the local
     * server itself) via the RegisterServer and FindServer services. */
    UA_Boolean registeredServersEnabled;

    /* Timeout in seconds when to automatically remove a registered server from
     * the list, if it doesn't re-register within the given time frame. A value
     * of 0 disables automatic removal. Default is 60 Minutes (60*60). Must be
     * bigger than 10 seconds, because cleanup is only triggered approximately
     * every 10 seconds. The server will still be removed depending on the
     * state of the semaphore file. */
    UA_UInt32 registeredServerCleanupTimeout;

    /* mDNS based Announcement and Discovery is implemented in a driver that
     * attach to a server outside the main configuration. The
     * FindServersOnNetwork Server is however implemented by the server itself.
     * The drivers interact with the server via the Discovery API. For example
     * to add/update/remove a ServerOnNetwork structure. */

    /* Enable the internal management of ServerOnNetwork entries and the
     * FindServersOnNetwork Service. This is used in conjunction with mDNS to
     * implement a Discovery Server that detects other servers on the
     * network. */
    UA_Boolean serversOnNetworkEnabled;

    /* Subscriptions
     * ~~~~~~~~~~~~~ */
    UA_Boolean subscriptionsEnabled;
#ifdef UA_ENABLE_SUBSCRIPTIONS
    /* Limits for Subscriptions */
    UA_UInt32 maxSubscriptions; /* 0 => unlimited */
    UA_UInt32 maxSubscriptionsPerSession; /* 0 => unlimited */
    UA_DurationRange publishingIntervalLimits; /* in ms (must not be less than 5) */
    UA_UInt32Range lifeTimeCountLimits;
    UA_UInt32Range keepAliveCountLimits;
    UA_UInt32 maxNotificationsPerPublish;
    UA_Boolean enableRetransmissionQueue;
    UA_UInt32 maxRetransmissionQueueSize; /* 0 -> unlimited size */
```

(continues on next page)

(continued from previous page)

```
# ifdef UA_ENABLE_SUBSCRIPTIONS_EVENTS
    UA_UInt32 maxEventsPerNode; /* 0 -> unlimited size */
# endif

    /* Limits for MonitoredItems */
    UA_UInt32 maxMonitoredItems; /* 0 => unlimited */
    UA_UInt32 maxMonitoredItemsPerSubscription; /* 0 => unlimited */
    UA_DurationRange samplingIntervalLimits; /* in ms (must not be less than 5) */
    UA_UInt32Range queueSizeLimits; /* Negotiated with the client */

    /* Limits for PublishRequests */
    UA_UInt32 maxPublishReqPerSession;

    /* Register MonitoredItem in Userland.
     * Deprecated now. Use the ServerNotificationCallback mechanism with
     * UA_APPLICATIONNOTIFICATIONTYPE_MONITOREDITEM for the same␣
↪feature.
    void (*monitoredItemRegisterCallback)(UA_Server *server,
                                          const UA_NodeId *sessionId,
                                          void *sessionContext,
                                          const UA_NodeId *nodeId,
                                          void *nodeContext,
                                          UA_UInt32 attibuteId,
                                          UA_Boolean removed);
    */
#endif

    /* PubSub
     * ~~~~~~ */
#ifdef UA_ENABLE_PUBSUB
    UA_Boolean pubsubEnabled;
    UA_PubSubConfiguration pubSubConfig;
#endif

    /* Auditing
     * ~~~~~~~~
     * Drops audit events into the auditNotificationCallback and generates
     * the corresponding Audit Events (if Events are enabled). */
    UA_Boolean auditingEnabled;
#ifdef UA_ENABLE_AUDITING
    UA_Boolean auditWriteUpdateEnabled;  /* Mind the runtime overhead */
    UA_Boolean auditMethodUpdateEnabled; /* Mind the runtime overhead */
#endif

    /* Historical Access
     * ~~~~~~~~~~~~~~~~~ */
    UA_Boolean historizingEnabled;
#ifdef UA_ENABLE_HISTORIZING
    UA_HistoryDatabase historyDatabase;
```

(continues on next page)

(continued from previous page)

```
    UA_Boolean accessHistoryDataCapability;
    UA_UInt32  maxReturnDataValues; /* 0 -> unlimited size */

    UA_Boolean accessHistoryEventsCapability;
    UA_UInt32  maxReturnEventValues; /* 0 -> unlimited size */

    UA_Boolean insertDataCapability;
    UA_Boolean insertEventCapability;
    UA_Boolean insertAnnotationsCapability;

    UA_Boolean replaceDataCapability;
    UA_Boolean replaceEventCapability;

    UA_Boolean updateDataCapability;
    UA_Boolean updateEventCapability;

    UA_Boolean deleteRawCapability;
    UA_Boolean deleteEventCapability;
    UA_Boolean deleteAtTimeDataCapability;
#endif

    /* Certificate Password Callback
     * ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~ */
#ifdef UA_ENABLE_ENCRYPTION
    /* If the private key is in PEM format and password protected, this callback
     * is called during initialization to get the password to decrypt the
     * private key. The memory containing the password is freed by the client
     * after use. The callback should be set early, other parts of the client
     * config setup may depend on it. */
    UA_StatusCode (*privateKeyPasswordCallback)(UA_ServerConfig *sc,
                                                UA_ByteString *password);
#endif

#ifdef UA_ENABLE_RBAC
    /* Initial Role-Permission Presets
     * ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
     * Array of initial role-permission presets. Each entry is a set of
     * UA_RolePermission mappings that define which roles get which
     * permissions on nodes that reference this preset.
     *
     * During server startup, these presets are copied into the server's
     * internal role-permissions array. The preset entries form the initial
     * configuration with the following guarantees:
     *
     * - Preset entries are **never deleted** during server runtime.
     * - Their position (index) in the internal array is **kept stable**,
     *   ensuring that nodes assigned to a preset continue to reference
     *   the same configuration throughout the server's lifetime.
```

(continues on next page)

(continued from previous page)

```
     * - Custom nodestore implementations should be aware of these
     *   index-stability guarantees when managing node permission storage.
     *
     * Additional role-permission sets can be added at runtime through
     * the server API (UA_Server_setNodeRolePermissions). Runtime entries
     * may be garbage-collected when no longer referenced by any node. */
    size_t rolePermissionPresetsSize;
    UA_RolePermissionSet *rolePermissionPresets;

    /* Initial Role Definitions
     * ~~~~~~~~~~~~~~~~~~~~~~~~
     * Array of initial role definitions. During server startup, these roles
     * are copied into the server's internal role registry. Config roles are
     * treated as **protected**: they cannot be removed at runtime via
     * UA_Server_removeRole.
     *
     * Additional roles can be added at runtime through UA_Server_addRole.
     * Runtime-added roles can be removed via UA_Server_removeRole. */
    size_t rolesSize;
    UA_Role *roles;

    /* If true, all permissions are granted regardless of roles.
     * WARNING: Effectively disables authorization. Use for testing only. */
    UA_Boolean allPermissionsForAnonymous;
#endif
};

void
UA_ServerConfig_clear(UA_ServerConfig *config);

UA_DEPRECATED static UA_INLINE void
UA_ServerConfig_clean(UA_ServerConfig *config) {
    UA_ServerConfig_clear(config);
}

/* Adds certificates and Certificate Revocation Lists (CRLs) to a specific
 * certificate group on the server.
 *
 * @param server The server object
 * @param certificateGroupId The NodeId of the certificate group where
 *        certificates will be added
 * @param certificates The certificates to be added
 * @param certificatesSize The number of certificates
 * @param crls The associated CRLs for the certificates, required when adding
 *        issuer certificates
 * @param crlsSize The number of CRLs
 * @param isTrusted Indicates whether the certificates should be added to the
 *        trusted list or the issuer list
 * @param appendCertificates Indicates whether the certificates should be added
```

(continues on next page)

(continued from previous page)

```
 *       to the list or replace the existing list
 * @return ``UA_STATUSCODE_GOOD`` on success */
UA_StatusCode
UA_Server_addCertificates(UA_Server *server,
                          const UA_NodeId certificateGroupId,
                          UA_ByteString *certificates,
                          size_t certificatesSize,
                          UA_ByteString *crls,
                          size_t crlsSize,
                          const UA_Boolean isTrusted,
                          const UA_Boolean appendCertificates);

/* Removes certificates from a specific certificate group on the server. The
 * corresponding CRLs are removed automatically.
 *
 * @param server The server object
 * @param certificateGroupId The NodeId of the certificate group from which
 *       certificates will be removed
 * @param certificates The certificates to be removed
 * @param certificatesSize The number of certificates
 * @param isTrusted Indicates whether the certificates are being removed from
 *       the trusted list or the issuer list
 * @return ``UA_STATUSCODE_GOOD`` on success */
UA_StatusCode
UA_Server_removeCertificates(UA_Server *server,
                             const UA_NodeId certificateGroupId,
                             UA_ByteString *certificates,
                             size_t certificatesSize,
                             const UA_Boolean isTrusted);
```

5.20.1 RBAC API

```
#ifdef UA_ENABLE_RBAC
```

Node Role-Permission Management

Functions for managing role permissions on individual nodes.

```
/* Set role permissions for a node.
 *
 * Assigns the given set of role-permission mappings to the specified node.
 * If an identical permission configuration already exists internally, the
 * node will reference the existing configuration (deduplication). Otherwise,
 * a new internal entry is created.
 *
 * @param server The server instance
 * @param nodeId The NodeId of the node
 * @param rolePermissionsSize Number of role-permission entries
 * @param rolePermissions Array of role-permission mappings
```

(continues on next page)

(continued from previous page)

```
 * @param recursive If true, also set for all hierarchically referenced
 *        child nodes
 * @param options Reserved for future use (e.g. to restrict the reference
 *        type for recursive traversal). Pass NULL for now.
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_setNodeRolePermissions(UA_Server *server,
                                 const UA_NodeId nodeId,
                                 size_t rolePermissionsSize,
                                 const UA_RolePermission *rolePermissions,
                                 UA_Boolean recursive,
                                 const UA_KeyValueMap *options);

/* Get the role permissions of a node.
 *
 * Returns a copy of the role-permission mappings currently assigned to
 * the node. The output array and its entries are allocated and must be
 * freed by the caller.
 *
 * If the node has no specific role permissions assigned, the output size
 * is set to 0 and the output pointer to NULL.
 *
 * @param server The server instance
 * @param nodeId The NodeId of the node
 * @param rolePermissionsSize Output: number of entries
 * @param rolePermissions Output: deep-copy array of role-permission mappings
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_getNodeRolePermissions(UA_Server *server,
                                 const UA_NodeId nodeId,
                                 size_t *rolePermissionsSize,
                                 UA_RolePermission **rolePermissions);

/* Remove role permissions from a node.
 *
 * Resets the node to have no specific role permissions. Default access
 * control behavior then applies. The internal reference count for the
 * previously assigned permission configuration is decremented.
 *
 * @param server The server instance
 * @param nodeId The NodeId of the node
 * @param recursive If true, also remove from all hierarchically referenced
 *        child nodes
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_removeNodeRolePermissions(UA_Server *server,
                                    const UA_NodeId nodeId,
                                    UA_Boolean recursive);
```

Role Management

Functions for managing the server's role registry. Roles define which sessions get which access rights. Config-provided roles are protected and cannot be removed at runtime.

```
/* Add a role to the server's role registry.
 *
 * The role's BrowseName (roleName) is the primary unique identifier,
 * per OPC UA Part 18 v1.05 §4.2.2 (AddRole: "The BrowseName shall be
 * unique within the RoleSet Object"). A role with the same roleName or
 * roleId must not already exist.
 *
 * If role->roleId is null, the server auto-assigns a random numeric
 * NodeId in namespace 0. To control the namespace or identifier, set
 * role->roleId before calling.
 *
 * @param server The server instance
 * @param role The role definition to add (deep-copied)
 * @param outRoleNodeId Output: the assigned NodeId (deep copy). May be NULL.
 * @return UA_STATUSCODE_GOOD on success,
 *         UA_STATUSCODE_BADALREADYEXISTS if a role with the same
 *         roleName or roleId already exists */
UA_StatusCode UA_THREADSAFE
UA_Server_addRole(UA_Server *server, const UA_Role *role,
                  UA_NodeId *outRoleNodeId);

/* Remove a role from the server's role registry.
 *
 * Config-provided (protected) roles cannot be removed.
 *
 * @param server The server instance
 * @param roleName The BrowseName (QualifiedName) of the role to remove
 * @return UA_STATUSCODE_GOOD on success,
 *         UA_STATUSCODE_BADUSERACCESSDENIED if the role is protected,
 *         UA_STATUSCODE_BADNOTFOUND if the role does not exist */
UA_StatusCode UA_THREADSAFE
UA_Server_removeRole(UA_Server *server,
                     const UA_QualifiedName roleName);

/* Get a copy of a role by its BrowseName.
 *
 * @param server The server instance
 * @param roleName The BrowseName (QualifiedName) of the role
 * @param outRole Output: deep copy of the role (caller must clear)
 * @return UA_STATUSCODE_GOOD on success,
 *         UA_STATUSCODE_BADNOTFOUND if the role does not exist */
UA_StatusCode UA_THREADSAFE
UA_Server_getRole(UA_Server *server,
                  const UA_QualifiedName roleName,
                  UA_Role *outRole);
```

(continues on next page)

(continued from previous page)

```
/* Get a copy of a role by its NodeId.
 *
 * @param server The server instance
 * @param roleId The NodeId of the role
 * @param outRole Output: deep copy of the role (caller must clear)
 * @return UA_STATUSCODE_GOOD on success,
 *         UA_STATUSCODE_BADNOTFOUND if the role does not exist */
UA_StatusCode UA_THREADSAFE
UA_Server_getRoleById(UA_Server *server, UA_NodeId roleId,
                      UA_Role *outRole);

/* Get the BrowseNames of all registered roles.
 *
 * @param server The server instance
 * @param rolesSize Output: number of roles
 * @param roleNames Output: array of role BrowseNames (caller must
 *        clear each entry and free the array)
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_getRoles(UA_Server *server, size_t *rolesSize,
                   UA_QualifiedName **roleNames);

/* Update a role in the server's role registry.
 *
 * The existing role is matched by roleId, roleName (QualifiedName) or
 * both. At least one must be set. If both are provided they must
 * identify the same role. Replaces all mutable fields (identityMappingRules,
 * applications, endpoints and their exclude flags) with deep copies
 * from the provided role. The roleId and roleName of the stored role
 * are not changed.
 *
 * Anonymous and AuthenticatedUser are well-known roles defined by the
 * OPC UA specification and cannot be modified.
 *
 * @param server The server instance
 * @param role The role with updated fields
 * @return UA_STATUSCODE_GOOD on success,
 *         UA_STATUSCODE_BADINVALIDARGUMENT if neither roleId nor
 *         roleName is set,
 *         UA_STATUSCODE_BADNOTFOUND if no matching role exists,
 *         UA_STATUSCODE_BADUSERACCESSDENIED if the matched role is
 *         Anonymous or AuthenticatedUser */
UA_StatusCode UA_THREADSAFE
UA_Server_updateRole(UA_Server *server, const UA_Role *role);
```

Session Role Management

Session roles are managed via the generic session attribute API using the key UA_QUALIFIEDNAME(0, "roles"). The value is a UA_NodeId[] array of the roles assigned to the session.

All role NodeIds are validated against the server's role registry on write.

Set roles:

```
UA_NodeId roles[2] = { role1Id, role2Id };
UA_Variant v;
UA_Variant_setArray(&v, roles, 2, &UA_TYPES[UA_TYPES_NODEID]);
UA_Server_setSessionAttribute(server, &sessionId,
                              UA_QUALIFIEDNAME(0, "roles"), &v);
```

Get roles (deep copy):

```
UA_Variant out;
UA_Server_getSessionAttributeCopy(server, &sessionId,
                                  UA_QUALIFIEDNAME(0, "roles"), &out);
UA_NodeId *roles = (UA_NodeId *)out.data;
size_t count = out.arrayLength;
// ... use roles ...
UA_Variant_clear(&out);
```

Clear roles:

```
UA_Server_deleteSessionAttribute(server, &sessionId,
                                 UA_QUALIFIEDNAME(0, "roles"));
```

```
/* Convenience: Get role QualifiedNames assigned to a session.
 * Returns a deep copy of the role names. Free the result with
 * UA_Array_delete(roleNames, count, &UA_TYPES[UA_TYPES_QUALIFIEDNAME]).
 *
 * @param server The server instance
 * @param sessionId The session to query
 * @param outSize Output: number of roles
 * @param outRoleNames Output: deep-copy array of QualifiedNames
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_getSessionRoleNames(UA_Server *server, const UA_NodeId sessionId,
                              size_t *outSize, UA_QualifiedName **outRoleNames);

/* Convenience bitmask: all permission bits set */
#define UA_PERMISSIONTYPE_ALL ((UA_PermissionType)0xFFFFFFFF)
```

Per-Role Node Permission Management

Functions that add or remove permissions for a specific role on a node. Unlike setNodeRolePermissions (which replaces the entire set), these functions modify individual role entries within the node's permission configuration.

```
/* Add role permissions to a node for a specific role.
 *
 * @param server The server instance
 * @param nodeId The node to modify
 * @param roleId The role to add permissions for
 * @param permissions Permission bitmask to set
 * @param overwriteExisting If true, replace the role's existing permission
 *        bitmask entirely; if false, OR (merge) the new bits into the
 *        existing bitmask
 * @param recursive If true, apply recursively to child nodes
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_addRolePermissions(UA_Server *server, const UA_NodeId nodeId,
                             const UA_NodeId roleId,
                             UA_PermissionType permissions,
                             UA_Boolean overwriteExisting,
                             UA_Boolean recursive);

/* Remove role permissions from a node for a specific role.
 *
 * @param server The server instance
 * @param nodeId The node to modify
 * @param roleId The role to remove permissions for
 * @param permissions Permission bitmask to clear
 * @param recursive If true, apply recursively to child nodes
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_removeRolePermissions(UA_Server *server, const UA_NodeId nodeId,
                                const UA_NodeId roleId,
                                UA_PermissionType permissions,
                                UA_Boolean recursive);
```

5.20.2 Namespace Default Role Permissions

Per OPC UA Part 5: if a node has no explicit RolePermissions, the DefaultRolePermissions from the NamespaceMetadata apply.

Permission resolution order:

1. Explicit node RolePermissions (set via addRolePermissions)
2. Namespace default RolePermissions (set via this API)

```
/* Set default role permissions for a namespace.
 * Overwrites any previously set defaults for the given namespace.
 *
 * @param server The server instance
 * @param namespaceIndex The namespace index
 * @param entriesSize Number of role-permission entries
 * @param entries Array of role-permission entries (deep-copied)
 * @return UA_STATUSCODE_GOOD on success */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode UA_THREADSAFE
UA_Server_setNamespaceDefaultRolePermissions(UA_Server *server,
                                             UA_UInt16 namespaceIndex,
                                             size_t entriesSize,
                                             const UA_RolePermission *entries);

/* Get default role permissions for a namespace.
 * Returns a deep copy. The caller must free each entry's roleId
 * with UA_NodeId_clear and the array with UA_free.
 *
 * @param server The server instance
 * @param namespaceIndex The namespace index
 * @param entriesSize Output: number of entries
 * @param entries Output: deep-copied array (caller must free)
 * @return UA_STATUSCODE_GOOD on success */
UA_StatusCode UA_THREADSAFE
UA_Server_getNamespaceDefaultRolePermissions(UA_Server *server,
                                             UA_UInt16 namespaceIndex,
                                             size_t *entriesSize,
                                             UA_RolePermission **entries);

#endif /* UA_ENABLE_RBAC */
```

5.21 Configuration from File

The server can be configured from JSON5-formatted content stored in a UA_ByteString.

The example files examples/json_config/server_json_config.json5 and examples/server_json_config.c document the intended workflow and the currently supported keys. They cover the common runtime limits as well as optional blocks for discovery, subscriptions, historizing, PubSub and security policy configuration. TCP is enabled by default. WebSockets and OPC UA services over HTTPS are disabled by default. When UA_ENABLE_LWS is compiled in, configure them with webSocketEnabled / webSocket and httpEnabled / http respectively.

The following functions require JSON encoding support (UA_ENABLE_JSON_ENCODING).

```
#ifdef UA_ENABLE_JSON_ENCODING

/* Loads the server configuration from a Json5 file into the server.
 *
 * @param jsonConfig The configuration in json5 format.
 */
UA_Server *
UA_Server_newFromFile(const UA_ByteString jsonConfig);

/* Loads a server configuration from a file.  The passed server configuration
 * is cleared.  Memory will be allocated for fields in config.
 *
 * @param config The server configuration.
 * @param jsonConfig The configuration in json5 format.
```

(continues on next page)

(continued from previous page)

```
 */
UA_StatusCode
UA_ServerConfig_loadFromFile(UA_ServerConfig *config, const UA_ByteString␣
↪jsonConfig);

#endif /* UA_ENABLE_JSON_ENCODING */
```

CHAPTER

SIX

CLIENT

The client implementation allows remote access to all OPC UA services. For convenience, some functionality has been wrapped in *high-level abstractions*.

Attention: The client does not start its own thread. The user has to periodically call *UA_Client_run_iterate* to ensure that asynchronous events and housekeeping tasks are handled, including keeping a secure connection established. See more about *asynchronicity* and *subscriptions*.

6.1 Client Lifecycle

```
/* Create a new client with a default configuration that adds plugins for
 * networking, security, logging and so on. See `client_config_default.h` for
 * more detailed options.
 *
 * The default configuration can be used as the starting point to adjust the
 * client configuration to individual needs. UA_Client_new is implemented in the
 * /plugins folder under the CC0 license. Furthermore the client configuration
 * only uses the public server API.
 *
 * @return Returns the configured client or NULL if an error occurs. */
UA_Client * UA_Client_new(void);

/* Creates a new client. Moves the config into the client with a shallow copy.
 * The config content is cleared together with the client. */
UA_Client *
UA_Client_newWithConfig(const UA_ClientConfig *config);

/* Returns the current state. All arguments except ``client`` can be NULL. */
void UA_THREADSAFE
UA_Client_getState(UA_Client *client,
                   UA_SecureChannelState *channelState,
                   UA_SessionState *sessionState,
                   UA_StatusCode *connectStatus);

/* Get the client configuration */
UA_ClientConfig *
UA_Client_getConfig(UA_Client *client);

/* Get the client context (inside the context) */
```

(continues on next page)

(continued from previous page)

```
#define UA_Client_getContext(client) \
    UA_Client_getConfig(client)->clientContext

/* (Disconnect and) delete the client */
void
UA_Client_delete(UA_Client *client);

/* Listen on the network and process arriving asynchronous responses in the
 * background. Internal housekeeping, renewal of SecureChannels and subscription
 * management is done as well. Running _run_iterate is required for asynchronous
 * operations. */
UA_StatusCode UA_THREADSAFE
UA_Client_run_iterate(UA_Client *client, UA_UInt32 timeout);

/* Runs the client until interrupted. On Unix/Windows this registers an
 * interrupt for SIGINT (ctrl-c). The method only returns after having received
 * the interrupt or upon an error condition. The logical sequence is as follows:
 *
 * - Register the interrupt
 * - Loop until interrupt: UA_Client_run_iterate
 * - Deregister the interrupt
 *
 * The inner loop is also aborted when the client is disconnected and fails the
 * automatic reconnect.
 *
 * Attention! This method does not call UA_Client_connect() / _disconnect()
 * internally. This needs to be handled before and after.
 *
 * Attention! This method is implemented individually for the different
 * platforms (POSIX/Win32/etc.). The default implementation is in
 * /plugins/ua_config_default.c under the CC0 license. Adjust as needed. */
UA_StatusCode
UA_Client_runUntilInterrupt(UA_Client *client);
```

6.2 Connect to a Server

A Client connecting to an OPC UA Server first opens a SecureChannel and on top of that opens a Session. The Session can also be moved to another SecureChannel. The Session lifetime is typically longer than the SecureChannel.

The methods below are used to connect a client with a server. The supplied arguments (for example the EndpointUrl or username/password) are first copied into the client config. This is the same as modifying the client config first and then connecting without these extra arguments.

Once a connection is established, the client keeps the connection open and reconnects if necessary. But as the client has no dedicated thread, it might be necessary to call `UA_Client_run_iterate` in an interval for the housekeeping tasks. Note normal Service-calls (like reading a value) also triggers the scheduled house-keeping tasks.

```
/* Copy the given endpointUrl into the configuration and connect. If the
 * endpointURL is NULL, then the client configuration is not modified. */
UA_StatusCode UA_THREADSAFE
UA_Client_connect(UA_Client *client, const char *endpointUrl);

/* Connect to the server and create+activate a Session with the given username
 * and password. This first set the UserIdentityToken in the client config and
 * then calls the regular connect method. */
UA_StatusCode UA_THREADSAFE
UA_Client_connectUsername(UA_Client *client, const char *endpointUrl,
                          const char *username, const char *password);

/* Connect to the server with a SecureChannel, but without creating a Session */
UA_StatusCode UA_THREADSAFE
UA_Client_connectSecureChannel(UA_Client *client, const char *endpointUrl);

/* Connect async (non-blocking) to the server. After initiating the connection,
 * call UA_Client_run_iterate repeatedly until the connection is fully
 * established. You can set a callback to client->config.stateCallback to be
 * notified when the connection status changes. Or use UA_Client_getState to get
 * the state manually. */
UA_StatusCode UA_THREADSAFE
UA_Client_connectAsync(UA_Client *client, const char *endpointUrl);

/* Connect async to the server with a SecureChannel, but without creating a
 * Session */
UA_StatusCode UA_THREADSAFE
UA_Client_connectSecureChannelAsync(UA_Client *client, const char *endpointUrl);

/* Sets up a listening socket for incoming reverse connect requests by OPC UA
 * servers. After the first server has connected, the listening socket is
 * removed. The client state callback is also used for reverse connect. An
 * implementation could for example issue a new call to
 * UA_Client_startListeningForReverseConnect after the server has closed the
 * connection. If the client is connected to any server while
 * UA_Client_startListeningForReverseConnect is called, the connection will be
 * closed.
 *
 * The reverse connect is closed by calling the standard disconnect functions
 * like for a "normal" connection that was initiated by the client. Calling one
 * of the connect methods will also close the listening socket and the
 * connection to the remote server. */
UA_StatusCode
UA_Client_startListeningForReverseConnect(UA_Client *client,
                                          const UA_String *listenHostnames,
                                          size_t listenHostnamesLength,
                                          UA_UInt16 port);

/* Disconnect and close a connection to the selected server. Disconnection is
 * always performed async (without blocking). */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode UA_THREADSAFE
UA_Client_disconnect(UA_Client *client);

/* Disconnect async. Run UA_Client_run_iterate until the callback notifies that
 * all connections are closed. */
UA_StatusCode UA_THREADSAFE
UA_Client_disconnectAsync(UA_Client *client);

/* Disconnect the SecureChannel but keep the Session intact (if it exists). */
UA_StatusCode UA_THREADSAFE
UA_Client_disconnectSecureChannel(UA_Client *client);

/* Disconnect the SecureChannel but keep the Session intact (if it exists). This
 * is an async operation. Iterate the client until the SecureChannel was fully
 * cleaned up. */
UA_StatusCode UA_THREADSAFE
UA_Client_disconnectSecureChannelAsync(UA_Client *client);

/* Get the AuthenticationToken and ServerNonce required to activate the current
 * Session on a different SecureChannel. */
UA_StatusCode UA_THREADSAFE
UA_Client_getSessionAuthenticationToken(UA_Client *client,
                                        UA_NodeId *authenticationToken,
                                        UA_ByteString *serverNonce);

/* Re-activate the current session. A change of preferred locales can be done by
 * updating the client configuration. */
UA_StatusCode UA_THREADSAFE
UA_Client_activateCurrentSession(UA_Client *client);

/* Async version of UA_Client_activateCurrentSession */
UA_StatusCode UA_THREADSAFE
UA_Client_activateCurrentSessionAsync(UA_Client *client);

/* Activate an already created Session. This allows a Session to be transferred
 * from a different client instance. The AuthenticationToken and ServerNonce
 * must be provided for this. Both can be retrieved for an activated Session
 * with UA_Client_getSessionAuthenticationToken.
 *
 * The UserIdentityToken used for authentication must be identical to the
 * original activation of the Session. The UserIdentityToken is set in the
 * client configuration.
 *
 * Note the noNewSession option if there should not be a new Session
 * automatically created when this one closes. */
UA_StatusCode UA_THREADSAFE
UA_Client_activateSession(UA_Client *client,
                          const UA_NodeId authenticationToken,
                          const UA_ByteString serverNonce);
```

(continues on next page)

(continued from previous page)

```
/* Async version of UA_Client_activateSession */
UA_StatusCode UA_THREADSAFE
UA_Client_activateSessionAsync(UA_Client *client,
                               const UA_NodeId authenticationToken,
                               const UA_ByteString serverNonce);
```

6.3 Discovery

```
/* Gets a list of endpoints of a server
 *
 * @param client to use. Must be connected to the same endpoint given in
 *        serverUrl or otherwise in disconnected state.
 * @param serverUrl url to connect (for example "opc.tcp://localhost:4840")
 * @param endpointDescriptionsSize size of the array of endpoint descriptions
 * @param endpointDescriptions array of endpoint descriptions that is allocated
 *        by the function (you need to free manually)
 * @return Indicates whether the operation succeeded or returns an error code */
UA_StatusCode UA_THREADSAFE
UA_Client_getEndpoints(UA_Client *client, const char *serverUrl,
                       size_t* endpointDescriptionsSize,
                       UA_EndpointDescription** endpointDescriptions);

/* Gets a list of all registered servers at the given server.
 *
 * You can pass an optional filter for serverUris. If the given server is not
 * registered, an empty array will be returned. If the server is registered,
 * only that application description will be returned.
 *
 * Additionally you can optionally indicate which locale you want for the server
 * name in the returned application description. The array indicates the order
 * of preference. A server may have localized names.
 *
 * @param client to use. Must be connected to the same endpoint given in
 *        serverUrl or otherwise in disconnected state.
 * @param serverUrl url to connect (for example "opc.tcp://localhost:4840")
 * @param serverUrisSize Optional filter for specific server uris
 * @param serverUris Optional filter for specific server uris
 * @param localeIdsSize Optional indication which locale you prefer
 * @param localeIds Optional indication which locale you prefer
 * @param registeredServersSize size of returned array, i.e., number of
 *        found/registered servers
 * @param registeredServers array containing found/registered servers
 * @return Indicates whether the operation succeeded or returns an error code */
UA_StatusCode UA_THREADSAFE
UA_Client_findServers(UA_Client *client, const char *serverUrl,
                      size_t serverUrisSize, UA_String *serverUris,
                      size_t localeIdsSize, UA_String *localeIds,
```

(continues on next page)

(continued from previous page)

```
                    size_t *registeredServersSize,
                    UA_ApplicationDescription **registeredServers);

/* Get a list of all known server in the network. Only supported by LDS servers.
 *
 * @param client to use. Must be connected to the same endpoint given in
 * serverUrl or otherwise in disconnected state.
 * @param serverUrl url to connect (for example "opc.tcp://localhost:4840")
 * @param startingRecordId optional. Only return the records with an ID higher
 *        or equal the given. Can be used for pagination to only get a subset of
 *        the full list
 * @param maxRecordsToReturn optional. Only return this number of records

 * @param serverCapabilityFilterSize optional. Filter the returned list to only
 *        get servers with given capabilities, e.g. "LDS"
 * @param serverCapabilityFilter optional. Filter the returned list to only get
 *        servers with given capabilities, e.g. "LDS"
 * @param serverOnNetworkSize size of returned array, i.e., number of
 *        known/registered servers
 * @param serverOnNetwork array containing known/registered servers
 * @return Indicates whether the operation succeeded or returns an error code */
UA_StatusCode UA_THREADSAFE
UA_Client_findServersOnNetwork(UA_Client *client, const char *serverUrl,
                               UA_UInt32 startingRecordId,
                               UA_UInt32 maxRecordsToReturn,
                               size_t serverCapabilityFilterSize,
                               UA_String *serverCapabilityFilter,
                               size_t *serverOnNetworkSize,
                               UA_ServerOnNetwork **serverOnNetwork);
```

6.4 Services

The raw OPC UA services are exposed to the client. A Request-message typically contains an array of operations to be performed by the server. The convenience-methods from ua_client_highlevel.h are used to call individual operations and are often easier to use. For performance reasons it might however be better to request many operations in a single message.

The raw Service-calls return the entire Response-message. Check the ResponseHeader for the overall StatusCode. The operations within the Service-call may return individual StatusCodes in addition.

Note that some Services are not exposed here. For example, there is a dedicated client API for Subscriptions and for Session management.

6.4.1 Attribute Service Set

This Service Set provides Services to access Attributes that are part of Nodes.

```
UA_ReadResponse UA_THREADSAFE
UA_Client_Service_read(UA_Client *client, const UA_ReadRequest req);
```

(continues on next page)

(continued from previous page)

```
UA_WriteResponse UA_THREADSAFE
UA_Client_Service_write(UA_Client *client, const UA_WriteRequest req);

/*
 * Historical Access Service Set
 * ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~ */
#ifdef UA_ENABLE_HISTORIZING

UA_HistoryReadResponse UA_THREADSAFE
UA_Client_Service_historyRead(UA_Client *client,
                              const UA_HistoryReadRequest req);

UA_HistoryUpdateResponse UA_THREADSAFE
UA_Client_Service_historyUpdate(UA_Client *client,
                                const UA_HistoryUpdateRequest req);

#endif
```

6.4.2 Method Service Set

Methods represent the function calls of Objects. The Method Service Set defines the means to invoke Methods.

```
UA_CallResponse UA_THREADSAFE
UA_Client_Service_call(UA_Client *client, const UA_CallRequest req);
```

6.4.3 NodeManagement Service Set

This Service Set defines Services to add and delete AddressSpace Nodes and References between them.

```
UA_AddNodesResponse UA_THREADSAFE
UA_Client_Service_addNodes(UA_Client *client, const UA_AddNodesRequest req);

UA_AddReferencesResponse UA_THREADSAFE
UA_Client_Service_addReferences(UA_Client *client,
                                const UA_AddReferencesRequest req);

UA_DeleteNodesResponse UA_THREADSAFE
UA_Client_Service_deleteNodes(UA_Client *client,
                              const UA_DeleteNodesRequest req);

UA_DeleteReferencesResponse UA_THREADSAFE
UA_Client_Service_deleteReferences(UA_Client *client,
                                   const UA_DeleteReferencesRequest req);
```

6.4.4 View Service Set

Clients use the browse Services of the View Service Set to navigate through the AddressSpace or through a View which is a subset of the AddressSpace.

```
UA_BrowseResponse UA_THREADSAFE
UA_Client_Service_browse(UA_Client *client, const UA_BrowseRequest req);

UA_BrowseNextResponse UA_THREADSAFE
UA_Client_Service_browseNext(UA_Client *client,
                             const UA_BrowseNextRequest req);

UA_TranslateBrowsePathsToNodeIdsResponse UA_THREADSAFE
UA_Client_Service_translateBrowsePathsToNodeIds(UA_Client *client,
    const UA_TranslateBrowsePathsToNodeIdsRequest req);

UA_RegisterNodesResponse UA_THREADSAFE
UA_Client_Service_registerNodes(UA_Client *client,
                                const UA_RegisterNodesRequest req);

UA_UnregisterNodesResponse UA_THREADSAFE
UA_Client_Service_unregisterNodes(UA_Client *client,
                                  const UA_UnregisterNodesRequest req);
```

6.4.5 Query Service Set

This Service Set is used to issue a Query to a Server.

```
#ifdef UA_ENABLE_QUERY

UA_QueryFirstResponse UA_THREADSAFE
UA_Client_Service_queryFirst(UA_Client *client,
                             const UA_QueryFirstRequest req);

UA_QueryNextResponse UA_THREADSAFE
UA_Client_Service_queryNext(UA_Client *client,
                            const UA_QueryNextRequest req);

#endif
```

6.5 Client Utility Functions

```
/* Lookup a datatype by its NodeId. Takes the custom types in the client
 * configuration into account. Return NULL if none found. */
const UA_DataType *
UA_Client_findDataType(UA_Client *client, const UA_NodeId *typeId);

/* The string is allocated and needs to be cleared */
UA_StatusCode UA_THREADSAFE
UA_Client_getNamespaceUri(UA_Client *client, UA_UInt16 index,
```

(continues on next page)

(continued from previous page)

```
                        UA_String *nsUri);

UA_StatusCode UA_THREADSAFE
UA_Client_getNamespaceIndex(UA_Client *client, const UA_String nsUri,
                            UA_UInt16 *outIndex);

/* Returns the old index of the namespace already exists */
UA_StatusCode UA_THREADSAFE
UA_Client_addNamespace(UA_Client *client, const UA_String nsUri,
                       UA_UInt16 *outIndex);

/* Retrieve the DataTypes defined on the server. The customTypes
 * UA_DataTypeArray gets allocated and populated internally, and its pointer is
 * set for the output. But the UA_DataTypeArray does not get added to the client
 * configuration automatically.
 *
 * - Only DataTypes that are previously unknown in the client config get added
 *   to the UA_DataTypeArray array.
 * - If the dataTypesNodeSize is zero, then the type hierarchy in the server is
 *   browsed to find any unknown DataTypes.
 * - The "cleanup"-flag in the UA_DataTypeArray is set to true, so it is cleaned
 *   up together with the client configuration -- if it is added there. */
UA_StatusCode UA_THREADSAFE
UA_Client_getRemoteDataTypes(UA_Client *client,
                             size_t dataTypesNodesSize,
                             const UA_NodeId *dataTypesNodes,
                             UA_DataTypeArray **customTypes);
```

6.5.1 Connection Attributes

Besides the client configuration, some attributes of the connection are defined only at runtime. For example the choice of SecurityPolicy or the ApplicationDescripton from the server. This API allows to access such connection attributes.

The currently defined connection attributes are:

- 0:serverDescription (UA_ApplicationDescription): Server description
- 0:securityPolicyUri (UA_String): Uri of the SecurityPolicy used
- 0:securityMode (UA_MessageSecurityMode): SecurityMode of the SecureChannel

```
/* Returns a shallow copy of the attribute. Don't _clear or _delete the value
 * variant. Don't use the value after returning the control flow to the client.
 * Also don't use this in a multi-threaded application. */
UA_StatusCode
UA_Client_getConnectionAttribute(UA_Client *client, const UA_QualifiedName key,
                                 UA_Variant *outValue);

/* Return a deep copy of the attribute */
UA_StatusCode UA_THREADSAFE
```

(continues on next page)

(continued from previous page)

```
UA_Client_getConnectionAttributeCopy(UA_Client *client, const UA_QualifiedName key,
                                     UA_Variant *outValue);

/* Returns NULL if the attribute is not defined or not a scalar or not of the
 * right datatype. Otherwise a shallow copy of the scalar value is created at
 * the target location of the void pointer. Hence don't use this in a
 * multi-threaded application. */
UA_StatusCode
UA_Client_getConnectionAttribute_scalar(UA_Client *client,
                                        const UA_QualifiedName key,
                                        const UA_DataType *type,
                                        void *outValue);
```

6.5.2 Timed Callbacks

Repeated callbacks can be attached to a client and will be executed in the defined interval.

```
typedef void (*UA_ClientCallback)(UA_Client *client, void *data);

/* Add a callback for execution at a specified time. If the indicated time lies
 * in the past, then the callback is executed at the next iteration of the
 * server's main loop.
 *
 * @param client The client object.
 * @param callback The callback that shall be added.
 * @param data Data that is forwarded to the callback.
 * @param date The timestamp for the execution time.
 * @param callbackId Set to the identifier of the repeated callback. This can
 *        be used to cancel the callback later on. If the pointer is null, the
 *        identifier is not set.
 * @return Upon success, UA_STATUSCODE_GOOD is returned. An error code
 *         otherwise. */
UA_StatusCode UA_THREADSAFE
UA_Client_addTimedCallback(UA_Client *client, UA_ClientCallback callback,
                           void *data, UA_DateTime date, UA_UInt64 *callbackId);

/* Add a callback for cyclic repetition to the client.
 *
 * @param client The client object.
 * @param callback The callback that shall be added.
 * @param data Data that is forwarded to the callback.
 * @param interval_ms The callback shall be repeatedly executed with the given
 *        interval (in ms). The interval must be positive. The first execution
 *        occurs at now() + interval at the latest.
 * @param callbackId Set to the identifier of the repeated callback. This can
 *        be used to cancel the callback later on. If the pointer is null, the
 *        identifier is not set.
 * @return Upon success, UA_STATUSCODE_GOOD is returned. An error code
 *         otherwise. */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode UA_THREADSAFE
UA_Client_addRepeatedCallback(UA_Client *client, UA_ClientCallback callback,
                              void *data, UA_Double interval_ms,
                              UA_UInt64 *callbackId);

UA_StatusCode UA_THREADSAFE
UA_Client_changeRepeatedCallbackInterval(UA_Client *client,
                                         UA_UInt64 callbackId,
                                         UA_Double interval_ms);

void UA_THREADSAFE
UA_Client_removeCallback(UA_Client *client, UA_UInt64 callbackId);

#define UA_Client_removeRepeatedCallback(server, callbackId)     \
    UA_Client_removeCallback(server, callbackId);
```

6.6 Application Notification

The client defines callbacks to notify the application on defined triggering points. These callbacks are executed with the (re-entrant) client-mutex held.

The different types of callback are disambiguated by their type enum. Besides the global notification callback (which is always triggered), the client configuration contains specialized callbacks that trigger only for specific notifications. This can reduce the burden of high-frequency notifications.

See the section on the Application Notification enum for more documentation on the notifications and their defined payload.

```
typedef void (*UA_ClientNotificationCallback)(UA_Client *client,
                                              UA_ApplicationNotificationType type,
                                              const UA_KeyValueMap payload);
```

6.7 Client Configuration

The configuration used for setting connection parameters and additional client settings. The *Tutorials* provide examples for many of the client settings.

```
/* Previous policy arrays retained by the default configuration plugin. Open
 * SecureChannels and Sessions may still reference them. Freed with the config. */
typedef struct UA_ClientConfig_PolicyHistory {
    struct UA_ClientConfig_PolicyHistory *next;
    UA_SecurityPolicy *policies;
    size_t policiesSize;
} UA_ClientConfig_PolicyHistory;

struct UA_ClientConfig {
    void *clientContext; /* User-defined pointer attached to the client */
    UA_Logger *logging;  /* Plugin for log output */
```

(continues on next page)

(continued from previous page)

```
    /* Response timeout in ms (0 -> no timeout). If the server does not answer a
     * request within this time a StatusCode UA_STATUSCODE_BADTIMEOUT is
     * returned. This timeout can be overridden for individual requests by
     * setting a non-null "timeoutHint" in the request header. */
    UA_UInt32 timeout;

    /* Self-description of the client.
     * The ApplicationUri set in the ApplicationDescription must match the URI
     * set in the certificate */
    UA_ApplicationDescription clientDescription;

    /* Connection configuration
     * ~~~~~~~~~~~~~~~~~~~~~~~~
     * The following configuration elements reduce the "degrees of freedom" the
     * client has when connecting to a server. If no connection can be made
     * under these restrictions, then the connection will abort with an error
     * message. */

    UA_String endpointUrl; /* The URI for the client to connect to, e.g.
                            * "opc.tcp://host:port" or
                            * "opc.wss://host:port/path" */

    /* Session config */
    UA_ExtensionObject userIdentityToken; /* UserIdentityToken containing e.g.
                                           * username+pwd for authentication.
                                           * Set by the _connect API. */
    UA_String sessionName;         /* Client-defined session name */
    UA_LocaleId *sessionLocaleIds; /* Preferred locales of the client */
    size_t sessionLocaleIdsSize;

    /* Basic connection behavior
     *
     * While the connectStatus is Good, the client re-dials a lost connection on
     * its own. Any other status is final: the connection is closed and nothing
     * is retried until UA_Client_connect* or a synchronous service call
     * connects again. The status becomes final when the handshake, discovery
     * or Session setup fails, the transport connection is refused on connect,
     * a synchronous connect times out (BadTimeout), or the connection is lost
     * with noReconnect set.
     *
     * The first re-dial starts right away. If it fails, the next one waits
     * reconnectInterval, then twice as long, up to reconnectIntervalMax. If the
     * re-dial after the longest wait fails too, the client gives up with
     * BadConnectionRejected or BadConnectionClosed. With the defaults, this
     * takes about 26 seconds. Synchronous calls don't wait for a pending
     * re-dial, they connect right away.
     *
     * UA_Client_connect* re-activates the previous Session if the server still
     * has it. Call UA_Client_disconnect first to get a new Session. Until the
```

(continues on next page)

(continued from previous page)

```
     * Session is activated, asynchronous requests that need a Session fail
     * with BadSessionNotActivated. */
    UA_Boolean noSession;    /* Only open a SecureChannel, but no Session */
    UA_Boolean noReconnect;  /* Don't reconnect SecureChannel when the connection
                              * is lost without explicitly closing. */
    UA_UInt32 reconnectInterval;    /* Wait in ms after the first failed
                                     * re-dial, doubled for every further one
                                     * (default: 250) */
    UA_UInt32 reconnectIntervalMax; /* Longest wait between re-dials in ms
                                     * (default: 10000) */
    UA_Boolean noNewSession; /* Don't automatically create a new Session when
                              * the initial one is lost. Instead abort the
                              * connection when the Session is lost. */

    /* Advanced Connection Settings */
    UA_UInt32 secureChannelLifeTime; /* Lifetime in ms (then the channel needs
                                      * to be renewed) */
    UA_UInt32 requestedSessionTimeout; /* Session timeout in ms */
    UA_ConnectionConfig localConnectionConfig;
    UA_UInt32 connectivityCheckInterval; /* Connectivity check interval in ms.
                                          * 0 -> background task disabled */
    UA_Boolean tcpReuseAddr; /* Specific to OPC UA with TCP transport. */
#ifdef UA_ENABLE_LWS
    UA_ByteString webSocketCaCertificate; /* DER or PEM encoded CA certificate
                                           * for WebSocket TLS. The system trust
                                           * store is used when empty. */
    UA_UInt32 webSocketMaxQueueSize; /* Max bytes queued for a slow WebSocket
                                      * peer (default: 16 * sendBufferSize) */
#endif
    UA_Boolean httpAllowUnencrypted; /* Allow the non-standard opc.http scheme */
    UA_ByteString httpCaCertificate; /* DER or PEM CA; empty uses system trust */
    UA_ByteString httpClientCertificate; /* DER or PEM TLS client certificate */
    UA_ByteString httpClientPrivateKey;  /* DER or PEM TLS client private key */
    UA_String httpClientPrivateKeyPassword; /* Password for encrypted TLS key */
    UA_UInt16 httpTimeout;           /* HTTP binding/request timeout in seconds */
    UA_UInt32 httpMaxMsgSize;        /* Maximum encoded HTTP message size */
    UA_UInt32 httpMaxDecompressedMsgSize; /* Maximum expanded HTTP response */

    /* Endpoint Selection
     * ~~~~~~~~~~~~~~~~~~
     * The Endpoint and UserTokenPolicy define the encryption for the
     * SecureChannel and for the UserIdentityToken (e.g. to encrypt
     * username/password). They can be set directly or selected by the client
     * from the GetEndpoints response via filter criteria. */

    /* Direct definition of Endpoint and UserTokenPolicy. Note that an Endpoint
     * can define multiple UserTokenPolicies to select. */
    UA_EndpointDescription endpoint;
    UA_UserTokenPolicy userTokenPolicy;
```

(continues on next page)

(continued from previous page)

```
    /* Endpoint filtering */
    UA_String applicationUri; /* Filters the servers considered in the
                               * FindServers service and the Endpoints
                               * considered in the GetEndpoints service. */
    UA_MessageSecurityMode securityMode; /* None, Sign, SignAndEncrypt. The
                                          * default is "invalid". This lets the
                                          * client select any matching
                                          * endpoint. */
    UA_String securityPolicyUri; /* SecurityPolicy for the SecureChannel. An
                                  * empty string indicates the client to select
                                  * any available SecurityPolicy. */
    UA_String authSecurityPolicyUri; /* For authentication (UserIdentityToken).
                                      * The empty string lets the client select
                                      * any matching policy. */

    /* Security Configuration
     * ~~~~~~~~~~~~~~~~~~~~~~
     * Additional settings and the plugins providing encryption support. */

    /* Available SecurityPolicies (Plugins) */
    size_t securityPoliciesSize;
    UA_SecurityPolicy *securityPolicies;

    /* Certificate Verification Plugin */
    UA_CertificateGroup certificateVerification;

    /* Verify that the server ApplicationInstanceCertificate permits serverAuth.
     * For RSA SecurityPolicies the EKU extension is mandatory. For ECC
     * SecurityPolicies it is optional, but must permit serverAuth when present.
     * ABORT rejects invalid usage, DEFAULT and WARN log and continue, and ACCEPT
     * continues silently. */
    UA_RuleHandling certificateEkuRule;

    /* Compare the selected EndpointDescription from discovery with the
     * authenticated EndpointDescriptions returned by CreateSession. Only
     * server.applicationUri, endpointUrl, securityMode, securityPolicyUri,
     * userIdentityTokens, transportProfileUri and securityLevel are compared,
     * as required by Part 4. ABORT and DEFAULT reject a mismatch, WARN logs and
     * continues, and ACCEPT continues silently. Certificate verification is
     * separate and unaffected by this rule. The comparison is skipped when
     * an EndpointDescription was configured directly. */
    UA_RuleHandling endpointDescriptionRule;

    /* SecurityPolicies for authentication with an x509 certificate. The
     * UserIdentityToken contains only the certificate. The certificate gets
     * matched against the SecurityPolicy instance from this array. Only the
     * SecurityPolicy instance carries the private key. See
     * UA_ClientConfig_setAuthenticationCert. */
```

(continues on next page)

(continued from previous page)

```
    size_t authSecurityPoliciesSize;
    UA_SecurityPolicy *authSecurityPolicies;
    UA_ClientConfig_PolicyHistory *securityPolicyHistory; /* Internal ownership */

    /* Allow clients without encryption support to connect with username and
     * password. This requires to transmit the password in plain text over the
     * network which is why this option is disabled by default. Make sure you
     * really need this before enabling plain text passwords. */
    UA_Boolean allowNonePolicyPassword;

#ifdef UA_ENABLE_ENCRYPTION
    /* Limits for TrustList */
    UA_UInt32 maxTrustListSize; /* in bytes, 0 => unlimited */
    UA_UInt32 maxRejectedListSize; /* 0 => unlimited */

    /* If the private key is in PEM format and password protected, this callback
     * is called during initialization to get the password to decrypt the
     * private key. The memory containing the password is freed by the client
     * after use. The callback should be set early, other parts of the client
     * config setup may depend on it. */
    UA_StatusCode (*privateKeyPasswordCallback)(UA_ClientConfig *cc,
                                                UA_ByteString *password);
#endif

    /* Custom Data Types
     * ~~~~~~~~~~~~~~~~~
     * The following is a linked list of arrays with custom data types. All data
     * types that are accessible from here are automatically considered for the
     * decoding of received messages. Custom data types are not cleaned up
     * together with the configuration. So it is possible to allocate them on
     * ROM.
     *
     * See the section on :ref:`generic-types`. Examples for working with custom
     * data types are provided in ``/examples/custom_datatype/``. */

    UA_DataTypeArray *customDataTypes;

    /* Namespace Mapping
     * ~~~~~~~~~~~~~~~~~
     * The namespaces index is "just" a mapping to the Uris in the namespace
     * array of the server. In order to have stable NodeIds across servers, the
     * client keeps a list of predefined namespaces. Use
     * ``UA_Client_addNamespaceUri``, ``UA_Client_getNamespaceUri`` and
     * ``UA_Client_getNamespaceIndex`` to interact with the local namespace
     * mapping.
     *
     * The namespace indices are assigned internally in the client as follows:
     *
     * - Ns0 and Ns1 are pre-defined by the standard. Ns0 is always
```

(continues on next page)

(continued from previous page)

```
     *   ```http://opcfoundation.org/UA/``` and used for standard-defined
     *   NodeIds. Ns1 corresponds to the application uri of the individual
     *   server.
     * - The next namespaces are added in-order from the list below at startup
     *   (starting at index 2).
     * - The local API ``UA_Client_addNamespaceUri`` can be used to add more
     *   namespaces.
     * - When the client connects, the namespace array of the server is read.
     *   All previously unknown namespaces are added from this to the internal
     *   array of the client. */

    UA_String *namespaces;
    size_t namespacesSize;

    /* Application Notification
     * ~~~~~~~~~~~~~~~~~~~~~~~~
     * The notification callbacks can be NULL. The global callback receives all
     * notifications. The specialized callbacks receive only the subset
     * indicated by their name. */

    UA_ClientNotificationCallback globalNotificationCallback;
    UA_ClientNotificationCallback lifecycleNotificationCallback;
    UA_ClientNotificationCallback serviceNotificationCallback;

    /* Below are the old state-change callbacks */

    /* Callback for state changes. The client state is differentated into the
     * SecureChannel state and the Session state. The connectStatus is set if
     * the client connection (including reconnects) has failed and the client
     * has to ”give up”. If the connectStatus is not set, the client still has
     * hope to connect or recover. */
    void (*stateCallback)(UA_Client *client,
                          UA_SecureChannelState channelState,
                          UA_SessionState sessionState,
                          UA_StatusCode connectStatus);

    /* When connectivityCheckInterval is greater than 0, every
     * connectivityCheckInterval (in ms), an async read request is performed on
     * the server. inactivityCallback is called when the client receives no
     * response for this read request. The connection can be closed, this in an
     * attempt to recreate a healthy connection. */
    void (*inactivityCallback)(UA_Client *client);

    /* If the client does not receive a PublishResponse after the defined delay
     * of ``(sub->publishingInterval * sub->maxKeepAliveCount) +
     * client->config.timeout)``, then subscriptionInactivityCallback is called
     * for the subscription.. */
    void (*subscriptionInactivityCallback)(UA_Client *client,
                                           UA_UInt32 subscriptionId,
```

(continues on next page)

(continued from previous page)

```
                                    void *subContext);

    /* Advanced Settings
     * ~~~~~~~~~~~~~~~~~ */

    /* Maximum number of application-level asynchronous service calls that may
     * be outstanding. A value of 0 disables the limit. The default is 32.
     * Internal requests for connection maintenance and Publish are not counted.
     *
     * When the limit is reached, UA_RULEHANDLING_ABORT (and DEFAULT) rejects a
     * new call with BadTooManyOperations. UA_RULEHANDLING_WARN logs a warning
     * and waits for capacity. UA_RULEHANDLING_ACCEPT waits silently. Waiting
     * runs the EventLoop and can therefore execute user callbacks. */
    UA_UInt32 maxAsyncServiceCalls;
    UA_RuleHandling asyncServiceCallRule;

    /* Number of PublishResponse queued up in the server */
    UA_UInt16 outStandingPublishRequests;

    /* EventLoop
     * ~~~~~~~~~
     * A shared EventLoop must be external to every client and server using it.
     * Otherwise, one of them can stop or delete it while others still use it.
     * The application owns the shared EventLoop's lifetime. */
    UA_EventLoop *eventLoop;
    UA_Boolean externalEventLoop; /* Do not stop or delete the EventLoop */
};

/* Makes a deep copy of the config. Only The copies of the plugins (logger,
 * eventloop, securitypolicy) are shallow. Therefore calling _clear on dst will
 * also delete the plugins of src. */
UA_StatusCode
UA_ClientConfig_copy(UA_ClientConfig const *src, UA_ClientConfig *dst);

/* Cleans the client config and frees the pointer */
void
UA_ClientConfig_delete(UA_ClientConfig *config);

/* Cleans the client config */
void
UA_ClientConfig_clear(UA_ClientConfig *config);

/* Configure Username/Password for the Session authentication. Also see
 * UA_ClientConfig_setAuthenticationCert for x509-based authentication, which is
 * implemented as a plugin (as it can be based on different crypto
 * libraries). */
UA_StatusCode
UA_ClientConfig_setAuthenticationUsername(UA_ClientConfig *config,
                                          const char *username,
```

(continues on next page)

(continued from previous page)

```
                                const char *password);

/* This method operates on the client config in two ways:
 *
 * - Sets the certificate as a X509IdentityToken for authentication.
 * - Replaces the authSecurityPolicies array with new instances of the available
 *   SecurityPolicies using the certificate and associated private-key.
 *
 * During session activation, the certificate from the X509IdentityToken gets
 * matched to a SecurityPolicy for the UserTokenPolicy supported by the server.
 * The SecurityPolicy (with the private key) is then used for a signature that
 * proves ownership of the certificate.
 *
 * This method is implemented in the /plugins directory as it depends on the
 * platform-specific crypto implementations. */
#if defined(UA_ENABLE_ENCRYPTION_OPENSSL) || defined(UA_ENABLE_
↪ENCRYPTION_MBEDTLS)
UA_StatusCode
UA_ClientConfig_setAuthenticationCert(UA_ClientConfig *config,
                                      UA_ByteString certificateAuth,
                                      UA_ByteString privateKeyAuth);
#endif
```

6.8 Configuration from File

The client can be configured from JSON5-formatted content stored in a UA_ByteString.

The example files examples/json_config/client_json_config.json5 and examples/client_json_config.c document the intended workflow and the currently supported keys.

The following functions require JSON encoding support (UA_ENABLE_JSON_ENCODING).

```
#ifdef UA_ENABLE_JSON_ENCODING

/* Create a new client from a file.  The client configuration is loaded from a
 * Json5 file.
 *
 * @param jsonConfig The configuration in json5 format.
 */
UA_Client *
UA_Client_newFromFile(const UA_ByteString jsonConfig);

/* Loads a client configuration from a file.  The passed client configuration is
 * cleared.  Memory will be allocated for fields in config.
 *
 * @param config The client configuration.
 * @param jsonConfig The configuration in json5 format.
 */
UA_StatusCode
UA_ClientConfig_loadFromFile(UA_ClientConfig *config, const UA_ByteString␣
```

(continues on next page)

(continued from previous page)

```
↪jsonConfig);

#endif /* UA_ENABLE_JSON_ENCODING */
```

6.9 Subscriptions

Subscriptions in OPC UA are asynchronous. That is, the client sends several PublishRequests to the server. The server returns PublishResponses with notifications. But only when a notification has been generated. The client does not wait for the responses and continues normal operations.

Note the difference between Subscriptions and MonitoredItems. Subscriptions are used to report back notifications. MonitoredItems are used to generate notifications. Every MonitoredItem is attached to exactly one Subscription. And a Subscription can contain many MonitoredItems.

The client automatically processes PublishResponses (with a callback) in the background and keeps enough PublishRequests in transit. The PublishResponses may be received during a synchronous service call or in UA_Client_run_iterate. See more about *asynchronicity*.

```
/* Callbacks defined for Subscriptions */
typedef void (*UA_Client_DeleteSubscriptionCallback)
    (UA_Client *client, UA_UInt32 subId, void *subContext);

typedef void (*UA_Client_StatusChangeNotificationCallback)
    (UA_Client *client, UA_UInt32 subId, void *subContext,
     UA_StatusChangeNotification *notification);

/* Provides default values for a new subscription.
 *
 * RequestedPublishingInterval:  500.0 [ms]
 * RequestedLifetimeCount: 10000
 * RequestedMaxKeepAliveCount: 10
 * MaxNotificationsPerPublish: 0 (unlimited)
 * PublishingEnabled: true
 * Priority: 0 */
static UA_INLINE UA_CreateSubscriptionRequest
UA_CreateSubscriptionRequest_default(void) {
    UA_CreateSubscriptionRequest request;
    UA_CreateSubscriptionRequest_init(&request);

    request.requestedPublishingInterval = 500.0;
    request.requestedLifetimeCount = 10000;
    request.requestedMaxKeepAliveCount = 10;
    request.maxNotificationsPerPublish = 0;
    request.publishingEnabled = true;
    request.priority = 0;
    return request;
}

UA_CreateSubscriptionResponse UA_THREADSAFE
UA_Client_Subscriptions_create(UA_Client *client,
```

(continues on next page)

(continued from previous page)

```
    const UA_CreateSubscriptionRequest request,
    void *subscriptionContext,
    UA_Client_StatusChangeNotificationCallback statusChangeCallback,
    UA_Client_DeleteSubscriptionCallback deleteCallback);

typedef void
(*UA_ClientAsyncCreateSubscriptionCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_CreateSubscriptionResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_Subscriptions_create_async(UA_Client *client,
    const UA_CreateSubscriptionRequest request,
    void *subscriptionContext,
    UA_Client_StatusChangeNotificationCallback statusChangeCallback,
    UA_Client_DeleteSubscriptionCallback deleteCallback,
    UA_ClientAsyncCreateSubscriptionCallback callback,
    void *userdata, UA_UInt32 *requestId);

UA_ModifySubscriptionResponse UA_THREADSAFE
UA_Client_Subscriptions_modify(UA_Client *client,
    const UA_ModifySubscriptionRequest request);

typedef void
(*UA_ClientAsyncModifySubscriptionCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_ModifySubscriptionResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_Subscriptions_modify_async(UA_Client *client,
    const UA_ModifySubscriptionRequest request,
    UA_ClientAsyncModifySubscriptionCallback callback,
    void *userdata, UA_UInt32 *requestId);

UA_DeleteSubscriptionsResponse UA_THREADSAFE
UA_Client_Subscriptions_delete(UA_Client *client,
    const UA_DeleteSubscriptionsRequest request);

typedef void
(*UA_ClientAsyncDeleteSubscriptionsCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_DeleteSubscriptionsResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_Subscriptions_delete_async(UA_Client *client,
    const UA_DeleteSubscriptionsRequest request,
    UA_ClientAsyncDeleteSubscriptionsCallback callback,
    void *userdata, UA_UInt32 *requestId);
```

(continues on next page)

(continued from previous page)

```
/* Delete a single subscription */
UA_StatusCode UA_THREADSAFE
UA_Client_Subscriptions_deleteSingle(UA_Client *client,
                                     UA_UInt32 subscriptionId);

/* Retrieve or change the user supplied subscription contexts */
UA_StatusCode UA_THREADSAFE
UA_Client_Subscriptions_getContext(UA_Client *client,
                                   UA_UInt32 subscriptionId,
                                   void **subContext);

UA_StatusCode UA_THREADSAFE
UA_Client_Subscriptions_setContext(UA_Client *client,
                                   UA_UInt32 subscriptionId,
                                   void *subContext);

UA_SetPublishingModeResponse UA_THREADSAFE
UA_Client_Subscriptions_setPublishingMode(UA_Client *client,
    const UA_SetPublishingModeRequest request);
```

6.9.1 MonitoredItems

MonitoredItems for Events indicate the EventNotifier attribute. This indicates to the server not to monitor changes of the attribute, but to forward Event notifications from that node.

During the creation of a MonitoredItem, the server may return changed adjusted parameters. Check the returned UA_CreateMonitoredItemsResponse to get the current parameters.

Be aware that the client may process incoming notifications before receiving the CreateMonitoredItemsResponse. This is due to the behavior of some server SDKs. Without this "early processing" we would miss the initial value, which can be an issue if the value changes at a slow rate.

```
/* Provides default values for a new monitored item. */
static UA_INLINE UA_MonitoredItemCreateRequest
UA_MonitoredItemCreateRequest_default(UA_NodeId nodeId) {
    UA_MonitoredItemCreateRequest request;
    UA_MonitoredItemCreateRequest_init(&request);
    request.itemToMonitor.nodeId = nodeId;
    request.itemToMonitor.attributeId = UA_ATTRIBUTEID_VALUE;
    request.monitoringMode = UA_MONITORINGMODE_REPORTING;
    request.requestedParameters.samplingInterval = 250;
    request.requestedParameters.discardOldest = true;
    request.requestedParameters.queueSize = 1;
    return request;
}
```

The clientHandle parameter cannot be set by the user, any value will be replaced by the client with a unique internal ClientHandle value before sending the request to the server.

```
/* Callback for the deletion of a MonitoredItem */
typedef void (*UA_Client_DeleteMonitoredItemCallback)
```

(continues on next page)

(continued from previous page)

```
    (UA_Client *client, UA_UInt32 subId, void *subContext,
     UA_UInt32 monId, void *monContext);

/* Callback for DataChange notifications */
typedef void (*UA_Client_DataChangeNotificationCallback)
    (UA_Client *client, UA_UInt32 subId, void *subContext,
     UA_UInt32 monId, void *monContext,
     UA_DataValue *value);

/* Callback for Event notifications */
typedef void (*UA_Client_EventNotificationCallback)
    (UA_Client *client, UA_UInt32 subId, void *subContext,
     UA_UInt32 monId, void *monContext,
     const UA_KeyValueMap eventFields);

/* Don't use to monitor the EventNotifier attribute */
UA_CreateMonitoredItemsResponse UA_THREADSAFE
UA_Client_MonitoredItems_createDataChanges(UA_Client *client,
    const UA_CreateMonitoredItemsRequest request, void **contexts,
    UA_Client_DataChangeNotificationCallback *callbacks,
    UA_Client_DeleteMonitoredItemCallback *deleteCallbacks);

typedef void
(*UA_ClientAsyncCreateMonitoredItemsCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_CreateMonitoredItemsResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_createDataChanges_async(UA_Client *client,
    const UA_CreateMonitoredItemsRequest request, void **contexts,
    UA_Client_DataChangeNotificationCallback *callbacks,
    UA_Client_DeleteMonitoredItemCallback *deleteCallbacks,
    UA_ClientAsyncCreateMonitoredItemsCallback createCallback,
    void *userdata, UA_UInt32 *requestId);

UA_MonitoredItemCreateResult UA_THREADSAFE
UA_Client_MonitoredItems_createDataChange(UA_Client *client,
    UA_UInt32 subscriptionId,
    UA_TimestampsToReturn timestampsToReturn,
    const UA_MonitoredItemCreateRequest item,
    void *context, UA_Client_DataChangeNotificationCallback callback,
    UA_Client_DeleteMonitoredItemCallback deleteCallback);

/* Monitor the EventNotifier attribute only */
UA_CreateMonitoredItemsResponse UA_THREADSAFE
UA_Client_MonitoredItems_createEvents(UA_Client *client,
    const UA_CreateMonitoredItemsRequest request, void **contexts,
    UA_Client_EventNotificationCallback *callback,
    UA_Client_DeleteMonitoredItemCallback *deleteCallback);
```

(continues on next page)

(continued from previous page)

```
/* Monitor the EventNotifier attribute only */
UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_createEvents_async(UA_Client *client,
    const UA_CreateMonitoredItemsRequest request, void **contexts,
    UA_Client_EventNotificationCallback *callbacks,
    UA_Client_DeleteMonitoredItemCallback *deleteCallbacks,
    UA_ClientAsyncCreateMonitoredItemsCallback createCallback,
    void *userdata, UA_UInt32 *requestId);

UA_MonitoredItemCreateResult UA_THREADSAFE
UA_Client_MonitoredItems_createEvent(UA_Client *client,
    UA_UInt32 subscriptionId,
    UA_TimestampsToReturn timestampsToReturn,
    const UA_MonitoredItemCreateRequest item,
    void *context, UA_Client_EventNotificationCallback callback,
    UA_Client_DeleteMonitoredItemCallback deleteCallback);

UA_DeleteMonitoredItemsResponse UA_THREADSAFE
UA_Client_MonitoredItems_delete(UA_Client *client,
    const UA_DeleteMonitoredItemsRequest);

typedef void
(*UA_ClientAsyncDeleteMonitoredItemsCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_DeleteMonitoredItemsResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_delete_async(UA_Client *client,
    const UA_DeleteMonitoredItemsRequest request,
    UA_ClientAsyncDeleteMonitoredItemsCallback callback,
    void *userdata, UA_UInt32 *requestId);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_deleteSingle(UA_Client *client,
    UA_UInt32 subscriptionId, UA_UInt32 monitoredItemId);
```

The “ClientHandle” is part of the MonitoredItem configuration. The handle is set internally and not exposed to the user. A modification uses a new ClientHandle. Until the first notification with that handle arrives, queued notifications with the old handle are processed with the old settings.

If the same MonitoredItem is modified again before a notification with the pending handle arrives, the newer modification supersedes the pending one. Notifications with the superseded handle are ignored.

```
UA_ModifyMonitoredItemsResponse UA_THREADSAFE
UA_Client_MonitoredItems_modify(UA_Client *client,
    const UA_ModifyMonitoredItemsRequest request);

typedef void
```

(continues on next page)

(continued from previous page)

```
(*UA_ClientAsyncModifyMonitoredItemsCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_ModifyMonitoredItemsResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_modify_async(UA_Client *client,
    const UA_ModifyMonitoredItemsRequest request,
    UA_ClientAsyncModifyMonitoredItemsCallback callback,
    void *userdata, UA_UInt32 *requestId);

UA_SetMonitoringModeResponse UA_THREADSAFE
UA_Client_MonitoredItems_setMonitoringMode(
    UA_Client *client, const UA_SetMonitoringModeRequest request);

typedef void
(*UA_ClientAsyncSetMonitoringModeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_SetMonitoringModeResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_setMonitoringMode_async(
    UA_Client *client, const UA_SetMonitoringModeRequest request,
    UA_ClientAsyncSetMonitoringModeCallback callback,
    void *userdata, UA_UInt32 *requestId);

UA_SetTriggeringResponse UA_THREADSAFE
UA_Client_MonitoredItems_setTriggering(
    UA_Client *client, const UA_SetTriggeringRequest request);

typedef void
(*UA_ClientAsyncSetTriggeringCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_SetTriggeringResponse *response);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItems_setTriggering_async(
    UA_Client *client, const UA_SetTriggeringRequest request,
    UA_ClientAsyncSetTriggeringCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Retrieve or change the user supplied MonitoredItem context */
UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItem_getContext(UA_Client *client,
    UA_UInt32 subscriptionId, UA_UInt32 monitoredItemId,
    void **monContext);

UA_StatusCode UA_THREADSAFE
UA_Client_MonitoredItem_setContext(UA_Client *client,
    UA_UInt32 subscriptionId, UA_UInt32 monitoredItemId,
```

(continues on next page)

(continued from previous page)

```
    void *monContext);
```

6.10 Highlevel Client Functionality

The following definitions are convenience functions making use of the standard OPC UA services in the background. This is a less flexible way of handling the stack, because at many places sensible defaults are presumed; at the same time using these functions is the easiest way of implementing an OPC UA application, as you will not have to consider all the details that go into the OPC UA services. If more flexibility is needed, you can always achieve the same functionality using the raw *OPC UA services*.

6.10.1 Read Attributes

The following functions can be used to retrieve a single node attribute. Use the regular service to read several attributes at once.

```
UA_DataValue UA_THREADSAFE
UA_Client_read(UA_Client *client, const UA_ReadValueId *rvi);

UA_THREADSAFE UA_StatusCode
UA_Client_readNodeIdAttribute(UA_Client *client, const UA_NodeId nodeId,
                              UA_NodeId *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readNodeClassAttribute(UA_Client *client, const UA_NodeId nodeId,
                                 UA_NodeClass *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readBrowseNameAttribute(UA_Client *client, const UA_NodeId nodeId,
                                  UA_QualifiedName *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readDisplayNameAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   UA_LocalizedText *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readDescriptionAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   UA_LocalizedText *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readWriteMaskAttribute(UA_Client *client, const UA_NodeId nodeId,
                                 UA_UInt32 *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readUserWriteMaskAttribute(UA_Client *client, const UA_NodeId nodeId,
                                     UA_UInt32 *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readIsAbstractAttribute(UA_Client *client, const UA_NodeId nodeId,
```

(continues on next page)

(continued from previous page)

```
                                UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readSymmetricAttribute(UA_Client *client, const UA_NodeId nodeId,
                                 UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readInverseNameAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   UA_LocalizedText *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readContainsNoLoopsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                       UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readEventNotifierAttribute(UA_Client *client, const UA_NodeId nodeId,
                                     UA_Byte *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readValueAttribute(UA_Client *client, const UA_NodeId nodeId,
                             UA_Variant *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readDataTypeAttribute(UA_Client *client, const UA_NodeId nodeId,
                                UA_NodeId *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readValueRankAttribute(UA_Client *client, const UA_NodeId nodeId,
                                 UA_Int32 *out);

UA_StatusCode UA_THREADSAFE
UA_Client_readArrayDimensionsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                       size_t *outArrayDimensionsSize,
                                       UA_UInt32 **outArrayDimensions);

UA_THREADSAFE UA_StatusCode
UA_Client_readAccessLevelAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   UA_Byte *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readAccessLevelExAttribute(UA_Client *client, const UA_NodeId nodeId,
                                     UA_UInt32 *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readUserAccessLevelAttribute(UA_Client *client, const UA_NodeId nodeId,
                                       UA_Byte *out);

/* Returns a variant with a UA_RolePermissionType array */
UA_THREADSAFE UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_Client_readRolePermissionsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                       UA_Variant *out);

/* Returns a variant with a UA_RolePermissionType array */
UA_THREADSAFE UA_StatusCode
UA_Client_readUserRolePermissionsAttribute(UA_Client *client, const UA_NodeId␣
↪nodeId,
                                           UA_Variant *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readAccessRestrictionsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                          UA_AccessRestrictionType *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readMinimumSamplingIntervalAttribute(UA_Client *client,
                                               const UA_NodeId nodeId,
                                               UA_Double *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readHistorizingAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readExecutableAttribute(UA_Client *client, const UA_NodeId nodeId,
                                  UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readUserExecutableAttribute(UA_Client *client,
                                      const UA_NodeId nodeId,
                                      UA_Boolean *out);

UA_THREADSAFE UA_StatusCode
UA_Client_readDatatypeDefinitionAttribute(UA_Client* client, const UA_NodeId nodeId,
                                          UA_StructureDefinition *out);
```

6.10.2 Historical Access

The following functions can be used to read a single node historically. Use the regular service to read several nodes at once.

```
typedef UA_Boolean
(*UA_HistoricalIteratorCallback)(
    UA_Client *client, const UA_NodeId *nodeId, UA_Boolean moreDataAvailable,
    const UA_ExtensionObject *data, void *callbackContext);

UA_StatusCode
UA_Client_HistoryRead_events(
    UA_Client *client, const UA_NodeId *nodeId,
    const UA_HistoricalIteratorCallback callback, UA_DateTime startTime,
```

(continues on next page)

(continued from previous page)

```
    UA_DateTime endTime, UA_String indexRange, const UA_EventFilter filter,
    UA_UInt32 numValuesPerNode, UA_TimestampsToReturn timestampsToReturn,
    void *callbackContext);

UA_StatusCode
UA_Client_HistoryRead_raw(
    UA_Client *client, const UA_NodeId *nodeId,
    const UA_HistoricalIteratorCallback callback, UA_DateTime startTime,
    UA_DateTime endTime, UA_String indexRange, UA_Boolean returnBounds,
    UA_UInt32 numValuesPerNode, UA_TimestampsToReturn timestampsToReturn,
    void *callbackContext);

UA_StatusCode
UA_Client_HistoryRead_modified(
    UA_Client *client, const UA_NodeId *nodeId,
    const UA_HistoricalIteratorCallback callback, UA_DateTime startTime,
    UA_DateTime endTime, UA_String indexRange, UA_Boolean returnBounds,
    UA_UInt32 numValuesPerNode, UA_TimestampsToReturn timestampsToReturn,
    void *callbackContext);

UA_StatusCode
UA_Client_HistoryUpdate_insert(
    UA_Client *client, const UA_NodeId *nodeId, UA_DataValue *value);

UA_StatusCode
UA_Client_HistoryUpdate_replace(
    UA_Client *client, const UA_NodeId *nodeId, UA_DataValue *value);

UA_StatusCode
UA_Client_HistoryUpdate_update(
    UA_Client *client, const UA_NodeId *nodeId, UA_DataValue *value);

UA_StatusCode
UA_Client_HistoryUpdate_deleteRaw(
    UA_Client *client, const UA_NodeId *nodeId,
    UA_DateTime startTimestamp, UA_DateTime endTimestamp);
```

6.10.3 Write Attributes

The following functions can be use to write a single node attribute at a time. Use the regular write service to write several attributes at once.

```
UA_StatusCode UA_THREADSAFE
UA_Client_write(UA_Client *client, const UA_WriteValue *wv);

UA_THREADSAFE UA_StatusCode
UA_Client_writeNodeIdAttribute(UA_Client *client, const UA_NodeId nodeId,
                               const UA_NodeId *newNodeId);
```

(continues on next page)

(continued from previous page)

```
UA_THREADSAFE UA_StatusCode
UA_Client_writeNodeClassAttribute(UA_Client *client, const UA_NodeId nodeId,
                                  const UA_NodeClass *newNodeClass);

UA_THREADSAFE UA_StatusCode
UA_Client_writeBrowseNameAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   const UA_QualifiedName *newBrowseName);

UA_THREADSAFE UA_StatusCode
UA_Client_writeDisplayNameAttribute(UA_Client *client, const UA_NodeId nodeId,
                                    const UA_LocalizedText *newDisplayName);

UA_THREADSAFE UA_StatusCode
UA_Client_writeDescriptionAttribute(UA_Client *client, const UA_NodeId nodeId,
                                    const UA_LocalizedText *newDescription);

UA_THREADSAFE UA_StatusCode
UA_Client_writeWriteMaskAttribute(UA_Client *client, const UA_NodeId nodeId,
                                  const UA_UInt32 *newWriteMask);

UA_THREADSAFE UA_StatusCode
UA_Client_writeUserWriteMaskAttribute(UA_Client *client, const UA_NodeId nodeId,
                                      const UA_UInt32 *newUserWriteMask);

UA_THREADSAFE UA_StatusCode
UA_Client_writeIsAbstractAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   const UA_Boolean *newIsAbstract);

UA_THREADSAFE UA_StatusCode
UA_Client_writeSymmetricAttribute(UA_Client *client, const UA_NodeId nodeId,
                                  const UA_Boolean *newSymmetric);

UA_THREADSAFE UA_StatusCode
UA_Client_writeInverseNameAttribute(UA_Client *client, const UA_NodeId nodeId,
                                    const UA_LocalizedText *newInverseName);

UA_THREADSAFE UA_StatusCode
UA_Client_writeContainsNoLoopsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                        const UA_Boolean *newContainsNoLoops);

UA_THREADSAFE UA_StatusCode
UA_Client_writeEventNotifierAttribute(UA_Client *client, const UA_NodeId nodeId,
                                      const UA_Byte *newEventNotifier);

UA_THREADSAFE UA_StatusCode
UA_Client_writeValueAttribute(UA_Client *client, const UA_NodeId nodeId,
                              const UA_Variant *newValue);

UA_THREADSAFE UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_Client_writeValueAttribute_scalar(UA_Client *client, const UA_NodeId nodeId,
                                     const void *newValue,
                                     const UA_DataType *valueType);

/* Write a DataValue that can include timestamps and status codes */
UA_THREADSAFE UA_StatusCode
UA_Client_writeValueAttributeEx(UA_Client *client, const UA_NodeId nodeId,
                                const UA_DataValue *newValue);

UA_THREADSAFE UA_StatusCode
UA_Client_writeDataTypeAttribute(UA_Client *client, const UA_NodeId nodeId,
                                 const UA_NodeId *newDataType);

UA_THREADSAFE UA_StatusCode
UA_Client_writeValueRankAttribute(UA_Client *client, const UA_NodeId nodeId,
                                  const UA_Int32 *newValueRank);

UA_StatusCode UA_THREADSAFE
UA_Client_writeArrayDimensionsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                        size_t newArrayDimensionsSize,
                                        const UA_UInt32 *newArrayDimensions);

UA_THREADSAFE UA_StatusCode
UA_Client_writeAccessLevelAttribute(UA_Client *client, const UA_NodeId nodeId,
                                    const UA_Byte *newAccessLevel);

UA_THREADSAFE UA_StatusCode
UA_Client_writeAccessLevelExAttribute(UA_Client *client, const UA_NodeId nodeId,
                                      UA_UInt32 *newAccessLevelEx);

UA_THREADSAFE UA_StatusCode
UA_Client_writeUserAccessLevelAttribute(UA_Client *client, const UA_NodeId nodeId,
                                        const UA_Byte *newUserAccessLevel);

UA_THREADSAFE UA_StatusCode
UA_Client_writeRolePermissionsAttribute(UA_Client *client, const UA_NodeId nodeId,
                                        const UA_Variant *newRolePermissions);

UA_THREADSAFE UA_StatusCode
UA_Client_writeAccessRestrictionsAttribute(
    UA_Client *client, const UA_NodeId nodeId,
    const UA_AccessRestrictionType *newAccessRestrictions);

UA_THREADSAFE UA_StatusCode
UA_Client_writeMinimumSamplingIntervalAttribute(UA_Client *client,
                                                const UA_NodeId nodeId,
                                                const UA_Double *newMinInterval);

UA_THREADSAFE UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_Client_writeHistorizingAttribute(UA_Client *client, const UA_NodeId nodeId,
                                    const UA_Boolean *newHistorizing);

UA_THREADSAFE UA_StatusCode
UA_Client_writeExecutableAttribute(UA_Client *client, const UA_NodeId nodeId,
                                   const UA_Boolean *newExecutable);

UA_THREADSAFE UA_StatusCode
UA_Client_writeUserExecutableAttribute(UA_Client *client, const UA_NodeId nodeId,
                                       const UA_Boolean *newUserExecutable);
```

6.10.4 Method Calling

```
UA_StatusCode UA_THREADSAFE
UA_Client_call(UA_Client *client,
               const UA_NodeId objectId, const UA_NodeId methodId,
               size_t inputSize, const UA_Variant *input,
               size_t *outputSize, UA_Variant **output);
```

6.10.5 Browsing

```
UA_THREADSAFE UA_BrowseResult
UA_Client_browse(UA_Client *client,
                 const UA_ViewDescription *view,
                 UA_UInt32 requestedMaxReferencesPerNode,
                 const UA_BrowseDescription *nodesToBrowse);

UA_THREADSAFE UA_BrowseResult
UA_Client_browseNext(UA_Client *client,
                     UA_Boolean releaseContinuationPoint,
                     UA_ByteString continuationPoint);

UA_THREADSAFE UA_BrowsePathResult
UA_Client_translateBrowsePathToNodeIds(UA_Client *client,
                                       const UA_BrowsePath *browsePath);
```

6.10.6 Node Management

See the section on *server-side node management*.

```
UA_StatusCode
UA_Client_addReference(UA_Client *client, const UA_NodeId sourceNodeId,
                       const UA_NodeId referenceTypeId, UA_Boolean isForward,
                       const UA_String targetServerUri,
                       const UA_ExpandedNodeId targetNodeId,
                       UA_NodeClass targetNodeClass);

UA_StatusCode
UA_Client_deleteReference(UA_Client *client, const UA_NodeId sourceNodeId,
```

(continues on next page)

(continued from previous page)

```
                    const UA_NodeId referenceTypeId, UA_Boolean isForward,
                    const UA_ExpandedNodeId targetNodeId,
                    UA_Boolean deleteBidirectional);

UA_StatusCode
UA_Client_deleteNode(UA_Client *client, const UA_NodeId nodeId,
                     UA_Boolean deleteTargetReferences);

UA_THREADSAFE UA_StatusCode
UA_Client_addVariableNode(UA_Client *client, const UA_NodeId requestedNewNodeId,
                    const UA_NodeId parentNodeId,
                    const UA_NodeId referenceTypeId,
                    const UA_QualifiedName browseName,
                    const UA_NodeId typeDefinition,
                    const UA_VariableAttributes attr,
                    UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addVariableTypeNode(UA_Client *client,
                        const UA_NodeId requestedNewNodeId,
                        const UA_NodeId parentNodeId,
                        const UA_NodeId referenceTypeId,
                        const UA_QualifiedName browseName,
                        const UA_VariableTypeAttributes attr,
                        UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addObjectNode(UA_Client *client, const UA_NodeId requestedNewNodeId,
                    const UA_NodeId parentNodeId,
                    const UA_NodeId referenceTypeId,
                    const UA_QualifiedName browseName,
                    const UA_NodeId typeDefinition,
                    const UA_ObjectAttributes attr, UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addObjectTypeNode(UA_Client *client, const UA_NodeId␣
↪requestedNewNodeId,
                        const UA_NodeId parentNodeId,
                        const UA_NodeId referenceTypeId,
                        const UA_QualifiedName browseName,
                        const UA_ObjectTypeAttributes attr,
                        UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addViewNode(UA_Client *client, const UA_NodeId requestedNewNodeId,
                    const UA_NodeId parentNodeId,
                    const UA_NodeId referenceTypeId,
                    const UA_QualifiedName browseName,
                    const UA_ViewAttributes attr,
```

(continues on next page)

(continued from previous page)

```
                    UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addReferenceTypeNode(UA_Client *client,
                               const UA_NodeId requestedNewNodeId,
                               const UA_NodeId parentNodeId,
                               const UA_NodeId referenceTypeId,
                               const UA_QualifiedName browseName,
                               const UA_ReferenceTypeAttributes attr,
                               UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addDataTypeNode(UA_Client *client, const UA_NodeId requestedNewNodeId,
                          const UA_NodeId parentNodeId,
                          const UA_NodeId referenceTypeId,
                          const UA_QualifiedName browseName,
                          const UA_DataTypeAttributes attr,
                          UA_NodeId *outNewNodeId);

UA_THREADSAFE UA_StatusCode
UA_Client_addMethodNode(UA_Client *client, const UA_NodeId requestedNewNodeId,
                        const UA_NodeId parentNodeId,
                        const UA_NodeId referenceTypeId,
                        const UA_QualifiedName browseName,
                        const UA_MethodAttributes attr,
                        UA_NodeId *outNewNodeId);
```

6.10.7 Misc Highlevel Functionality

```
/* Get the namespace-index of a namespace-URI
 *
 * @param client The UA_Client struct for this connection
 * @param namespaceUri The interested namespace URI
 * @param namespaceIndex The namespace index of the URI. The value is unchanged
 *        in case of an error
 * @return Indicates whether the operation succeeded or returns an error code */
UA_StatusCode UA_THREADSAFE
UA_Client_NamespaceGetIndex(UA_Client *client, UA_String *namespaceUri,
                            UA_UInt16 *namespaceIndex);

/* Iterate over all nodes referenced by parentNodeId by calling the callback
 * function for each child node */
typedef UA_StatusCode
(*UA_ClientNodeIteratorCallback)(UA_NodeId childId, UA_Boolean isInverse,
                                 UA_NodeId referenceTypeId, void *handle);

UA_StatusCode
UA_Client_forEachChildNodeCall(
    UA_Client *client, UA_NodeId parentNodeId,
```

(continues on next page)

(continued from previous page)

```
    UA_ClientNodeIteratorCallback callback, void *handle);
```

6.11 Asynchronous Services

All OPC UA services are asynchronous in nature. So several service calls can be made without waiting for the individual responses. Depending on the server's priorities responses may come in a different ordering than sent.

Connection and session management are performed in *UA_Client_run_iterate*, so to keep a connection healthy any client needs to consider how and when it is appropriate to do the call. This is especially true for the periodic renewal of a SecureChannel's SecurityToken which is designed to have a limited lifetime and will invalidate the connection if not renewed.

If there is an error after an async service has been dispatched, the callback is called with an "empty" response where the StatusCode has been set accordingly. This is also done if the client is shutting down and the list of dispatched async services is emptied. The StatusCode received when the client is shutting down is UA_STATUSCODE_BADSHUTDOWN. The StatusCode received when the client doesn't receive response after the specified in config->timeout (can be overridden via the "timeoutHint" in the request header) is UA_STATUSCODE_BADTIMEOUT.

The number of outstanding application-level async calls can be limited with config->maxAsyncServiceCalls. Depending on config->asyncServiceCallRule, a call at the limit either returns UA_STATUSCODE_BADTOOMANYOPERATIONS or waits for capacity. Waiting runs the EventLoop and can execute user callbacks before the async call returns.

The userdata and requestId arguments can be NULL. The (optional) requestId output can be used to cancel the service while it is still pending. The requestId is unique for each service request. Alternatively the requestHandle can be manually set (non necessarily unique) in the request header for full service call. This can be used to cancel all outstanding requests using that handle together. Note that the client will auto-generate a requestHandle >100,000 if none is defined. Avoid these when manually setting a requetHandle in the requestHeader to avoid clashes.

```
/* Generalized asynchronous service call. This can be used for any
 * request/response datatype pair whenever no type-stable specialization is
 * defined below. */
typedef void
(*UA_ClientAsyncServiceCallback)(UA_Client *client, void *userdata,
                                 UA_UInt32 requestId, void *response);

UA_StatusCode UA_THREADSAFE
__UA_Client_AsyncService(UA_Client *client, const void *request,
                         const UA_DataType *requestType,
                         UA_ClientAsyncServiceCallback callback,
                         const UA_DataType *responseType,
                         void *userdata, UA_UInt32 *requestId);

/* Cancel all dispatched requests with the given requestHandle.
 * The number if cancelled requests is returned by the server.
 * The output argument cancelCount is not set if NULL. */
UA_THREADSAFE UA_StatusCode
UA_Client_cancelByRequestHandle(UA_Client *client, UA_UInt32 requestHandle,
```

(continues on next page)

(continued from previous page)

```
                                UA_UInt32 *cancelCount);

/* Map the requestId to the requestHandle used for that request and call the
 * Cancel service for that requestHandle. */
UA_THREADSAFE UA_StatusCode
UA_Client_cancelByRequestId(UA_Client *client, UA_UInt32 requestId,
                            UA_UInt32 *cancelCount);

/* Force the manual renewal of the SecureChannel. This is useful to renew the
 * SecureChannel during a downtime when no time-critical operations are
 * performed. This method is asynchronous. The renewal is triggered (the OPN
 * message is sent) but not completed. The OPN response is handled with
 * ``UA_Client_run_iterate`` or a synchronous service-call operation.
 *
 * @return The return value is UA_STATUSCODE_GOODCALLAGAIN if the␣
↪SecureChannel
 *         has not elapsed at least 75% of its lifetime. Otherwise the
 *         ``connectStatus`` is returned. */
UA_StatusCode UA_THREADSAFE
UA_Client_renewSecureChannel(UA_Client *client);
```

6.11.1 Asynchronous Service Calls

Call OPC UA Services asynchronously with a callback. The (optional) requestId output can be used to cancel the service while it is still pending.

```
typedef void
(*UA_ClientAsyncReadCallback)(UA_Client *client, void *userdata,
                              UA_UInt32 requestId, UA_ReadResponse *rr);

UA_StatusCode UA_THREADSAFE
UA_Client_sendAsyncReadRequest(UA_Client *client, UA_ReadRequest *request,
                               UA_ClientAsyncReadCallback readCallback,
                               void *userdata, UA_UInt32 *reqId);

typedef void
(*UA_ClientAsyncWriteCallback)(UA_Client *client, void *userdata,
                               UA_UInt32 requestId, UA_WriteResponse *wr);

UA_StatusCode UA_THREADSAFE
UA_Client_sendAsyncWriteRequest(UA_Client *client, UA_WriteRequest *request,
                                UA_ClientAsyncWriteCallback writeCallback,
                                void *userdata, UA_UInt32 *reqId);

typedef void
(*UA_ClientAsyncBrowseCallback)(UA_Client *client, void *userdata,
                                UA_UInt32 requestId, UA_BrowseResponse *wr);

UA_StatusCode UA_THREADSAFE
```

(continues on next page)

(continued from previous page)

```
UA_Client_sendAsyncBrowseRequest(UA_Client *client, UA_BrowseRequest *request,
                                 UA_ClientAsyncBrowseCallback browseCallback,
                                 void *userdata, UA_UInt32 *reqId);

typedef void
(*UA_ClientAsyncBrowseNextCallback)(UA_Client *client, void *userdata,
                                    UA_UInt32 requestId,
                                    UA_BrowseNextResponse *wr);

UA_StatusCode UA_THREADSAFE
UA_Client_sendAsyncBrowseNextRequest(
    UA_Client *client, UA_BrowseNextRequest *request,
    UA_ClientAsyncBrowseNextCallback browseNextCallback,
    void *userdata, UA_UInt32 *reqId);
```

6.11.2 Asynchronous Operations

Many Services can be called with an array of operations. For example, a request to the Read Service contains an array of ReadValueId, each corresponding to a single read operation. For convenience, wrappers are provided to call single operations for the most common Services.

All async operations have a callback of the following structure: The returned StatusCode is split in two parts. The status indicates the overall success of the request and the operation. The result argument is non-NULL only if the status is no good.

```
typedef void
(*UA_ClientAsyncOperationCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, void *result);
```

Read Attribute

Asynchronously read a single attribute. The attribute is unpacked from the response as the datatype of the attribute is known ahead of time. Value attributes are variants.

Note that the last argument (value pointer) of the callbacks can be NULL if the status of the operation is not good.

```
/* Reading a single attribute */
typedef void
(*UA_ClientAsyncReadAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_DataValue *attribute);

UA_StatusCode UA_THREADSAFE
UA_Client_readAttribute_async(
    UA_Client *client, const UA_ReadValueId *rvi,
    UA_TimestampsToReturn timestampsToReturn,
    UA_ClientAsyncReadAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);
```

(continues on next page)

(continued from previous page)

```
/* Read a single Value attribute */
typedef void
(*UA_ClientAsyncReadValueAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_DataValue *value);

UA_StatusCode UA_THREADSAFE
UA_Client_readValueAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadValueAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single DataType attribute */
typedef void
(*UA_ClientAsyncReadDataTypeAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_NodeId *dataType);

UA_StatusCode UA_THREADSAFE
UA_Client_readDataTypeAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadDataTypeAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single ArrayDimensions attribute. If the status is good, the variant
 * carries an UInt32 array. */
typedef void
(*UA_ClientReadArrayDimensionsAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Variant *arrayDimensions);

UA_StatusCode UA_THREADSAFE
UA_Client_readArrayDimensionsAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientReadArrayDimensionsAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single NodeClass attribute */
typedef void
(*UA_ClientAsyncReadNodeClassAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_NodeClass *nodeClass);

UA_StatusCode UA_THREADSAFE
UA_Client_readNodeClassAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadNodeClassAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);
```

(continues on next page)

(continued from previous page)

```
/* Read a single BrowseName attribute */
typedef void
(*UA_ClientAsyncReadBrowseNameAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_QualifiedName *browseName);

UA_StatusCode UA_THREADSAFE
UA_Client_readBrowseNameAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadBrowseNameAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single DisplayName attribute */
typedef void
(*UA_ClientAsyncReadDisplayNameAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_LocalizedText *displayName);

UA_StatusCode UA_THREADSAFE
UA_Client_readDisplayNameAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadDisplayNameAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single Description attribute */
typedef void
(*UA_ClientAsyncReadDescriptionAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_LocalizedText *description);

UA_StatusCode UA_THREADSAFE
UA_Client_readDescriptionAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadDescriptionAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single WriteMask attribute */
typedef void
(*UA_ClientAsyncReadWriteMaskAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_UInt32 *writeMask);

UA_StatusCode UA_THREADSAFE
UA_Client_readWriteMaskAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadWriteMaskAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);
```

(continues on next page)

(continued from previous page)

```
/* Read a single UserWriteMask attribute */
typedef void
(*UA_ClientAsyncReadUserWriteMaskAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_UInt32 *writeMask);

UA_StatusCode UA_THREADSAFE
UA_Client_readUserWriteMaskAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadUserWriteMaskAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single IsAbstract attribute */
typedef void
(*UA_ClientAsyncReadIsAbstractAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Boolean *isAbstract);

UA_StatusCode UA_THREADSAFE
UA_Client_readIsAbstractAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadIsAbstractAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single Symmetric attribute */
typedef void
(*UA_ClientAsyncReadSymmetricAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Boolean *symmetric);

UA_StatusCode UA_THREADSAFE
UA_Client_readSymmetricAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadSymmetricAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single InverseName attribute */
typedef void
(*UA_ClientAsyncReadInverseNameAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_LocalizedText *inverseName);

UA_StatusCode UA_THREADSAFE
UA_Client_readInverseNameAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadInverseNameAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single ContainsNoLoops attribute */
```

(continues on next page)

(continued from previous page)

```
typedef void
(*UA_ClientAsyncReadContainsNoLoopsAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Boolean *containsNoLoops);

UA_StatusCode UA_THREADSAFE
UA_Client_readContainsNoLoopsAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadContainsNoLoopsAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single EventNotifier attribute */
typedef void
(*UA_ClientAsyncReadEventNotifierAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Byte *eventNotifier);

UA_StatusCode UA_THREADSAFE
UA_Client_readEventNotifierAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadEventNotifierAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single ValueRank attribute */
typedef void
(*UA_ClientAsyncReadValueRankAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Int32 *valueRank);

UA_StatusCode UA_THREADSAFE
UA_Client_readValueRankAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadValueRankAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single AccessLevel attribute */
typedef void
(*UA_ClientAsyncReadAccessLevelAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Byte *accessLevel);

UA_StatusCode UA_THREADSAFE
UA_Client_readAccessLevelAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadAccessLevelAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single AccessLevelEx attribute */
typedef void
```

(continues on next page)

(continued from previous page)

```
(*UA_ClientAsyncReadAccessLevelExAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_UInt32 *accessLevelEx);

UA_StatusCode UA_THREADSAFE
UA_Client_readAccessLevelExAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadAccessLevelExAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single UserAccessLevel attribute */
typedef void
(*UA_ClientAsyncReadUserAccessLevelAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Byte *userAccessLevel);

UA_StatusCode UA_THREADSAFE
UA_Client_readUserAccessLevelAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadUserAccessLevelAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single RolePermissions attribute. If the status is good, the variant
 * carries a UA_RolePermissionType array. */
typedef void
(*UA_ClientAsyncReadRolePermissionsAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Variant *rolePermissions);

UA_StatusCode UA_THREADSAFE
UA_Client_readRolePermissionsAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadRolePermissionsAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single UserRolePermissions attribute. If the status is good, the
 * variant carries a UA_RolePermissionType array. */
typedef void
(*UA_ClientAsyncReadUserRolePermissionsAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Variant *userRolePermissions);

UA_StatusCode UA_THREADSAFE
UA_Client_readUserRolePermissionsAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadUserRolePermissionsAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single AccessRestrictions attribute */
```

(continues on next page)

(continued from previous page)

```
typedef void
(*UA_ClientAsyncReadAccessRestrictionsAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_AccessRestrictionType *accessRestrictions);

UA_StatusCode UA_THREADSAFE
UA_Client_readAccessRestrictionsAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadAccessRestrictionsAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single MinimumSamplingInterval attribute */
typedef void
(*UA_ClientAsyncReadMinimumSamplingIntervalAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Double *minimumSamplingInterval);

UA_StatusCode UA_THREADSAFE
UA_Client_readMinimumSamplingIntervalAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadMinimumSamplingIntervalAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single Historizing attribute */
typedef void
(*UA_ClientAsyncReadHistorizingAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Boolean *historizing);

UA_StatusCode UA_THREADSAFE
UA_Client_readHistorizingAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadHistorizingAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single Executable attribute */
typedef void
(*UA_ClientAsyncReadExecutableAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Boolean *executable);

UA_StatusCode UA_THREADSAFE
UA_Client_readExecutableAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadExecutableAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);

/* Read a single UserExecutable attribute */
typedef void
```

(continues on next page)

(continued from previous page)

```
(*UA_ClientAsyncReadUserExecutableAttributeCallback)(
    UA_Client *client, void *userdata, UA_UInt32 requestId,
    UA_StatusCode status, UA_Boolean *userExecutable);

UA_StatusCode UA_THREADSAFE
UA_Client_readUserExecutableAttribute_async(
    UA_Client *client, const UA_NodeId nodeId,
    UA_ClientAsyncReadUserExecutableAttributeCallback callback,
    void *userdata, UA_UInt32 *requestId);
```

Write Attribute

The methods for async writing of attributes all have a similar API. We generate the method signatures with a macro.

```
#define UA_CLIENT_ASYNCWRITE(NAME, ATTR_TYPE)                           \
    UA_StatusCode UA_THREADSAFE                                         \
    NAME(UA_Client *client, const UA_NodeId nodeId,                     \
         const ATTR_TYPE *attr, UA_ClientAsyncWriteCallback callback,   \
         void *userdata, UA_UInt32 *reqId);

UA_CLIENT_ASYNCWRITE(UA_Client_writeNodeIdAttribute_async, UA_NodeId)
UA_CLIENT_ASYNCWRITE(UA_Client_writeNodeClassAttribute_async, UA_
↪NodeClass)
UA_CLIENT_ASYNCWRITE(UA_Client_writeBrowseNameAttribute_async, UA_
↪QualifiedName)
UA_CLIENT_ASYNCWRITE(UA_Client_writeDisplayNameAttribute_async, UA_
↪LocalizedText)
UA_CLIENT_ASYNCWRITE(UA_Client_writeDescriptionAttribute_async, UA_
↪LocalizedText)
UA_CLIENT_ASYNCWRITE(UA_Client_writeWriteMaskAttribute_async, UA_UInt32)
UA_CLIENT_ASYNCWRITE(UA_Client_writeIsAbstractAttribute_async, UA_Boolean)
UA_CLIENT_ASYNCWRITE(UA_Client_writeSymmetricAttribute_async, UA_Boolean)
UA_CLIENT_ASYNCWRITE(UA_Client_writeInverseNameAttribute_async, UA_
↪LocalizedText)
UA_CLIENT_ASYNCWRITE(UA_Client_writeContainsNoLoopsAttribute_async, UA_
↪Boolean)
UA_CLIENT_ASYNCWRITE(UA_Client_writeEventNotifierAttribute_async, UA_Byte)
UA_CLIENT_ASYNCWRITE(UA_Client_writeValueAttribute_async, UA_Variant)
UA_CLIENT_ASYNCWRITE(UA_Client_writeDataTypeAttribute_async, UA_NodeId)
UA_CLIENT_ASYNCWRITE(UA_Client_writeValueRankAttribute_async, UA_Int32)
UA_CLIENT_ASYNCWRITE(UA_Client_writeAccessLevelAttribute_async, UA_Byte)
UA_CLIENT_ASYNCWRITE(UA_Client_writeMinimumSamplingIntervalAttribute_
↪async, UA_Double)
UA_CLIENT_ASYNCWRITE(UA_Client_writeHistorizingAttribute_async, UA_Boolean)
UA_CLIENT_ASYNCWRITE(UA_Client_writeExecutableAttribute_async, UA_Boolean)
UA_CLIENT_ASYNCWRITE(UA_Client_writeAccessLevelExAttribute_async, UA_
↪UInt32)
UA_CLIENT_ASYNCWRITE(UA_Client_writeRolePermissionsAttribute_async, UA_
```

(continues on next page)

(continued from previous page)

```
↪Variant)
UA_CLIENT_ASYNCWRITE(UA_Client_writeAccessRestrictionsAttribute_async,
                     UA_AccessRestrictionType)
```

Method Calling

```
typedef void
(*UA_ClientAsyncCallCallback)(
    UA_Client *client, void *userdata,
    UA_UInt32 requestId, UA_CallResponse *cr);

UA_StatusCode UA_THREADSAFE
UA_Client_call_async(UA_Client *client, const UA_NodeId objectId,
                     const UA_NodeId methodId, size_t inputSize,
                     const UA_Variant *input,
                     UA_ClientAsyncCallCallback callback,
                     void *userdata, UA_UInt32 *reqId);
```

Node Management

```
typedef void
(*UA_ClientAsyncAddNodesCallback)(
    UA_Client *client, void *userdata,
    UA_UInt32 requestId, UA_AddNodesResponse *ar);

UA_StatusCode UA_THREADSAFE
UA_Client_addVariableNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_NodeId typeDefinition,
    const UA_VariableAttributes attr, UA_NodeId *outNewNodeId,
    UA_ClientAsyncAddNodesCallback callback, void *userdata,
    UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addVariableTypeNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_VariableTypeAttributes attr,
    UA_NodeId *outNewNodeId, UA_ClientAsyncAddNodesCallback callback,
    void *userdata, UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addObjectNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_NodeId typeDefinition,
    const UA_ObjectAttributes attr, UA_NodeId *outNewNodeId,
    UA_ClientAsyncAddNodesCallback callback, void *userdata,
```

(continues on next page)

(continued from previous page)

```
    UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addObjectTypeNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_ObjectTypeAttributes attr,
    UA_NodeId *outNewNodeId, UA_ClientAsyncAddNodesCallback callback,
    void *userdata, UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addViewNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName,
    const UA_ViewAttributes attr, UA_NodeId *outNewNodeId,
    UA_ClientAsyncAddNodesCallback callback, void *userdata,
    UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addReferenceTypeNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_ReferenceTypeAttributes attr,
    UA_NodeId *outNewNodeId, UA_ClientAsyncAddNodesCallback callback,
    void *userdata, UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addDataTypeNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_DataTypeAttributes attr,
    UA_NodeId *outNewNodeId, UA_ClientAsyncAddNodesCallback callback,
    void *userdata, UA_UInt32 *reqId);

UA_StatusCode UA_THREADSAFE
UA_Client_addMethodNode_async(
    UA_Client *client, const UA_NodeId requestedNewNodeId,
    const UA_NodeId parentNodeId, const UA_NodeId referenceTypeId,
    const UA_QualifiedName browseName, const UA_MethodAttributes attr,
    UA_NodeId *outNewNodeId, UA_ClientAsyncAddNodesCallback callback,
    void *userdata, UA_UInt32 *reqId);
```

CHAPTER

SEVEN

PUBSUB

In PubSub the participating OPC UA Applications take their roles as Publishers and Subscribers. Publishers are the sources of data, while Subscribers consume that data. Communication in PubSub is message-based. Publishers send messages to a Message Oriented Middleware, without knowledge of what, if any, Subscribers there may be. Similarly, Subscribers express interest in specific types of data, and process messages that contain this data, without knowledge of what Publishers there are.

Message Oriented Middleware is software or hardware infrastructure that supports sending and receiving messages between distributed systems. OPC UA PubSub supports two different Message Oriented Middleware variants, namely the broker-less form and broker-based form. A broker-less form is where the Message Oriented Middleware is the network infrastructure that is able to route datagram-based messages. Subscribers and Publishers use datagram protocols like UDP. In a broker-based form, the core component of the Message Oriented Middleware is a message Broker. Subscribers and Publishers use standard messaging protocols like AMQP or MQTT to communicate with the Broker.

This makes PubSub suitable for applications where location independence and/or scalability are required.

The Publish/Subscribe (PubSub) extension for OPC UA enables fast and efficient 1:m communication. The PubSub extension is protocol agnostic and can be used with broker based protocols like MQTT and AMQP or brokerless implementations like UDP-Multicasting.

The figure below shows how the PubSub components are related. The PubSub Tutorials have more examples about the API usage:

```
Server

    > PubSubPublishedDataSet <

        > DataSetField

    > PubSubConnection
```

(continues on next page)

(continued from previous page)

```
        > WriterGroup

            > DataSetWriter <

    > ReaderGroup

        > DataSetReader

            > SubscribedDataSet

                > TargetVariablesDataType

                > SubscribedDataSetMirrorDataType
```

7.1 PubSub Information Model Representation

The complete PubSub configuration is available inside the information model. The entry point is the node 'PublishSubscribe', located under the Server node. The standard defines for PubSub no new Service set. The configuration can optionally be done over methods inside the information model. The information model representation of the current PubSub configuration is generated automatically. This feature can be enabled/disabled by changing the UA_ENABLE_PUBSUB_INFORMATIONMODEL option.

7.2 PublisherId

Valid PublisherId types are defined in Part 14, 7.2.2.2.2 NetworkMessage. The PublisherId is sometimes encoded as a variant in the standard (e.g. in some configuration structures). We do however use our own tagged-union structure where we can.

```
typedef enum {
    UA_PUBLISHERIDTYPE_BYTE   = 0, /* 000 Byte (default) */
    UA_PUBLISHERIDTYPE_UINT16 = 1, /* 001 UInt16 */
    UA_PUBLISHERIDTYPE_UINT32 = 2, /* 010 UInt32 */
    UA_PUBLISHERIDTYPE_UINT64 = 3, /* 011 UInt64 */
```

(continues on next page)

(continued from previous page)

```
    UA_PUBLISHERIDTYPE_STRING = 4  /* 100 String */
} UA_PublisherIdType;

typedef struct {
    UA_PublisherIdType idType;
    union {
        UA_Byte byte;
        UA_UInt16 uint16;
        UA_UInt32 uint32;
        UA_UInt64 uint64;
        UA_String string;
    } id;
} UA_PublisherId;

UA_StatusCode
UA_PublisherId_copy(const UA_PublisherId *src, UA_PublisherId *dst);

void
UA_PublisherId_clear(UA_PublisherId *p);

/* The variant must contain a scalar of the five possible identifier types */
UA_StatusCode
UA_PublisherId_fromVariant(UA_PublisherId *p, const UA_Variant *src);

/* Makes a shallow copy (no malloc) in the variant */
void
UA_PublisherId_toVariant(const UA_PublisherId *p, UA_Variant *dst);
```

7.3 PubSub Components

A PubSubComponent is either a PubSubConnection, DataSetReader, ReaderGroup, DataSetWriter, WriterGroup, PublishedDataSet or SubscribedDataSet. The PubSubComponents are represented in the information model (if that is enabled for the server). In the C-API they are identified by a unique NodeId that is assigned during creation.

```
typedef enum  {
    UA_PUBSUBCOMPONENT_CONNECTION  = 0,
    UA_PUBSUBCOMPONENT_WRITERGROUP  = 1,
    UA_PUBSUBCOMPONENT_DATASETWRITER  = 2,
    UA_PUBSUBCOMPONENT_READERGROUP  = 3,
    UA_PUBSUBCOMPONENT_DATASETREADER  = 4,
    UA_PUBSUBCOMPONENT_PUBLISHEDDATASET  = 5,
    UA_PUBSUBCOMPONENT_SUBSCRIBEDDDATASET = 6
} UA_PubSubComponentType;
```

The datasets are static “configuration containers”. The other PubSubComponents are active and have a state machine governing their runtime behavior and state transitions. The state machine API (in C and in the information model) exposes only _enable and _disable methods for the different PubSubComponents. Their actual state is more detailed and emerges from the internal behavior. This

UA_PubSubState defines five possible states, part 14 contains a diagram for the possible state transitions:

- DISABLED
- PAUSED
- OPERATIONAL
- ERROR
- PREOPERATIONAL

In open62541 we classify all PubSubStates as either *enabled* or *disabled*. The disabled states are DISABLED and ERROR. These need to be manually enabled to trigger a state change. All other states are enabled and "want to become OPERATIONAL". The state machine triggers internally to automatically reach the OPERATIONAL state when the external conditions allow it. The PREOPERATIONAL state indicates that necessary measures to become OPERATIONAL have been taken, but the OPERATIONAL state has not yet been achieved. For example, a ReaderGroup only becomes OPERATIONAL, once the first message for it has been received.

Notably, the state machines of the PubSubComponents are cascading. That is, the state depends on the state of the parent component. For example, an OPERATIONAL WriterGroup becomes PAUSED when its parent PubSubConnection goes to DISABLED. The PAUSED WriterGroup automatically returns to OPERATIONAL once the parent PubSubConnection becomes OPERATIONAL again.

```
/* Enable all PubSub components. They are triggered in the following order:
 * DataSetWriter, WriterGroups, DataSetReader, ReaderGroups, PubSubConnections.
 * Returns the ORed statuscodes from enabling the individual components. */
UA_StatusCode
UA_Server_enableAllPubSubComponents(UA_Server *server);

/* Disable all PubSubComponents.
 * They are triggered in the following order:
 * DataSetReader, ReaderGroups, DataSetWriter, WriterGroups, PubSubConnections.
 *
 * A timeout might occur if the writers are disabled before the readers
 * in case of a loopback configuration on the same server.
 * So disable the reader side before the writer side.
 */
void
UA_Server_disableAllPubSubComponents(UA_Server *server);
```

The default implementation of the state machines manages resources via the configured EventLoop. Most notably these are connections (sockets) and timed callbacks. A *custom state machine* can be configured when these resources needto be managed outside of the EventLoop. An example are realtime-capable connections that should not end up in the same select syscall as the server TCP connections. Custom state machines are set in the configuration structure of the individual PubSubComponents. The custom state machine only needs to handle the state of its "local" PubSubComponent. The open62541 implementation automatically triggers the child PubSubComponents after a state change.

The following definitions are part of the configuration for all active PubSubComponents (not the dataset PubSubComponents).

```
#define UA_PUBSUBCOMPONENT_COMMON                                       \
    UA_String name; /* For logging and in the information model */     \
    void *context;  /* Custom pointer forwarded to callbacks */        \
    UA_Boolean enabled; /* Component is auto-enabled at creation */    \
                                                                        \
    /* The custom state machine callback is optional (can be */        \
    /* NULL). It gets called with a request to change the state to */  \
    /* targetState. The state pointer has the old (and afterwards */   \
    /* the new) state. When the state machine returns a bad */         \
    /* statuscode, the state must be set to ERROR before. */           \
    UA_StatusCode (*customStateMachine)(UA_Server *server,              \
                                        const UA_NodeId componentId,   \
                                        void *componentContext,        \
                                        UA_PubSubState *state,         \
                                        UA_PubSubState targetState);   \
```

7.4 Global PubSub Configuration

The following PubSub configuration structure is part of the server-config. It configures behavior that is valid for all PubSubComponents.

```
typedef struct {
    /* Notify the application when a new PubSubComponent is added or removed.
     * That way it is possible to keep track of the changes from the methods in
     * the information model.
     *
     * When the return StatusCode is not good, then adding/removing the
     * component is aborted. When a component is added, it is possible to call
     * the public API _getConfig and _updateConfig methods on it from within the
     * lifecycle callback. */
    UA_StatusCode
    (*componentLifecycleCallback)(UA_Server *server, const UA_NodeId id,
                                  const UA_PubSubComponentType componentType,
                                  UA_Boolean remove);

    /* The callback is executed first thing in the state machine. The component
     * config can be modified from within the beforeStateChangeCallback if the
     * component is not enabled. Also the TargetState can be changed. For
     * example to prevent a component that is not ready from getting enabled. */
    void (*beforeStateChangeCallback)(UA_Server *server, const UA_NodeId id,
                                      UA_PubSubState *targetState);

    /* Callback to notify the application about PubSub component state changes.
     * The status code provides additional information. */
    void (*stateChangeCallback)(UA_Server *server, const UA_NodeId id,
                                UA_PubSubState state, UA_StatusCode status);

    UA_Boolean enableDeltaFrames;
```

(continues on next page)

(continued from previous page)

```
#ifdef UA_ENABLE_PUBSUB_INFORMATIONMODEL
    UA_Boolean enableInformationModelMethods;
#endif

    /* PubSub security policies */
    size_t securityPoliciesSize;
    UA_PubSubSecurityPolicy *securityPolicies;
} UA_PubSubConfiguration;
```

7.5 PubSub Custom State Machine

All PubSubComponents (Connection, Reader, ReaderGroup, ...) have a two configuration items in common: A void context-pointer and a callback to override the default state machine with a custom implementation.

When a custom state machine is set, then internally no sockets are opened and no periodic callbacks are registered. All "active behavior" has to be managed/configured entirely in the custom state machine.

```
/* The custom state machine callback is optional (can be NULL). It gets called
 * with a request to change the state targetState. The state pointer contains
 * the old (and afterwards the new) state. The notification stateChangeCallback
 * is called afterwards. When a bad statuscode is returned, the component must
 * be set to an ERROR state. */
#define UA_PUBSUB_COMPONENT_CONTEXT                                     \
    void *context;                                                      \
    UA_StatusCode (*customStateMachine)(UA_Server *server,              \
                                        const UA_NodeId componentId,    \
                                        void *componentContext,         \
                                        UA_PubSubState *state,          \
                                        UA_PubSubState targetState);    \
```

The following methods are used to retrieve the metadata of PubSubComponents. So gar they are implemented to operate only on the components with a state machine (connection, ReaderGroup, Reder, WriterGroup, Writer).

```
/* Get the component-type enum from the identifier */
UA_StatusCode
UA_Server_getPubSubComponentType(UA_Server *server, UA_NodeId componentId,
                                 UA_PubSubComponentType *outType);

/* Get the parent of a PubSubComponent (PubSubConnections have no parent).
 * Returns a deep copy of the parent's NodeId. */
UA_StatusCode
UA_Server_getPubSubComponentParent(UA_Server *server, UA_NodeId componentId,
                                   UA_NodeId *outParent);

/* Get the list of child-components. Allocates the output array. For
 * PubSubConnections, both the ReaderGroups and WriterGroups attached to it are
```

(continues on next page)

(continued from previous page)

```
 * returned. */
UA_StatusCode
UA_Server_getPubSubComponentChildren(UA_Server *server, UA_NodeId componentId,
                                     size_t *outChildrenSize,
                                     UA_NodeId **outChildren);
```

7.6 PubSubConnection

PubSubConnections are the abstraction between the concrete transport protocol and the PubSub functionality. It is possible to create multiple PubSubConnections with (possibly) different transport protocols at runtime.

```
typedef struct {
    UA_PUBSUBCOMPONENT_COMMON
    /* Configuration parameters from PubSubConnectionDataType */
    UA_PublisherId publisherId;
    UA_String transportProfileUri;
    UA_Variant address;
    UA_KeyValueMap connectionProperties;
    UA_Variant connectionTransportSettings;
} UA_PubSubConnectionConfig;

UA_StatusCode
UA_PubSubConnectionConfig_copy(const UA_PubSubConnectionConfig *src,
                               UA_PubSubConnectionConfig *dst);

void
UA_PubSubConnectionConfig_clear(UA_PubSubConnectionConfig *cfg);

/* Add a PubSub connection to the server. The connectionId can be NULL. If
 * defined, it is set to the NodeId of the created PubSubConnection. */
UA_StatusCode UA_THREADSAFE
UA_Server_addPubSubConnection(UA_Server *server,
                              const UA_PubSubConnectionConfig *connectionConfig,
                              UA_NodeId *connectionId);

UA_StatusCode UA_THREADSAFE
UA_Server_enablePubSubConnection(UA_Server *server,
                                 const UA_NodeId connectionId);

UA_StatusCode UA_THREADSAFE
UA_Server_disablePubSubConnection(UA_Server *server,
                                  const UA_NodeId connectionId);

/* Manually "inject" a packet as if it had been received by the
 * PubSubConnection. This is intended to be used in combination with a custom
 * state machine where sockets (connections) are handled by user code. */
UA_StatusCode UA_THREADSAFE
UA_Server_processPubSubConnectionReceive(UA_Server *server,
```

(continues on next page)

(continued from previous page)

```
                                    const UA_NodeId connectionId,
                                    const UA_ByteString packet);

/* Returns a deep copy of the config */
UA_StatusCode UA_THREADSAFE
UA_Server_getPubSubConnectionConfig(UA_Server *server,
                                    const UA_NodeId connectionId,
                                    UA_PubSubConnectionConfig *config);

/* The PubSubConnection must be disabled to update the config */
UA_StatusCode UA_THREADSAFE
UA_Server_updatePubSubConnectionConfig(UA_Server *server,
                                       const UA_NodeId connectionId,
                                       const UA_PubSubConnectionConfig *config);

/* Deletion of a PubSubConnection removes all "below" WriterGroups and
 * ReaderGroups. This can fail if the PubSubConnection is enabled. */
UA_StatusCode UA_THREADSAFE
UA_Server_removePubSubConnection(UA_Server *server,
                                 const UA_NodeId connectionId);
```

7.7 PublishedDataSet

The PublishedDataSets (PDS) are containers for the published information. The PDS contain the published variables and meta information. The metadata is commonly autogenerated or given as constant argument as part of the template functions. The template functions are standard defined and intended for configuration tools. You should normally create an empty PDS and call the functions to add new fields.

```
typedef enum {
    UA_PUBSUB_DATASET_PUBLISHEDITEMS,
    UA_PUBSUB_DATASET_PUBLISHEDEVENTS,
    UA_PUBSUB_DATASET_PUBLISHEDITEMS_TEMPLATE,
    UA_PUBSUB_DATASET_PUBLISHEDEVENTS_TEMPLATE,
} UA_PublishedDataSetType;

typedef struct {
    UA_DataSetMetaDataType metaData;
    size_t variablesToAddSize;
    UA_PublishedVariableDataType *variablesToAdd;
} UA_PublishedDataItemsTemplateConfig;

typedef struct {
    UA_NodeId eventNotfier;
    UA_ContentFilter filter;
} UA_PublishedEventConfig;

typedef struct {
    UA_DataSetMetaDataType metaData;
```

(continues on next page)

(continued from previous page)

```
    UA_NodeId eventNotifier;
    size_t selectedFieldsSize;
    UA_SimpleAttributeOperand *selectedFields;
    UA_ContentFilter filter;
} UA_PublishedEventTemplateConfig;

/* Configuration structure for PublishedDataSet */
typedef struct {
    UA_String name;
    UA_PublishedDataSetType publishedDataSetType;
    union {
        /* The UA_PUBSUB_DATASET_PUBLISHEDITEMS has currently no additional
         * members and thus no dedicated config structure.*/
        UA_PublishedDataItemsTemplateConfig itemsTemplate;
        UA_PublishedEventConfig event;
        UA_PublishedEventTemplateConfig eventTemplate;
    } config;

    void *context; /* Context Configuration (PublishedDataSet has no state
                    * machine) */
} UA_PublishedDataSetConfig;

void
UA_PublishedDataSetConfig_clear(UA_PublishedDataSetConfig *pdsConfig);

typedef struct {
    UA_StatusCode addResult;
    size_t fieldAddResultsSize;
    UA_StatusCode *fieldAddResults;
    UA_ConfigurationVersionDataType configurationVersion;
} UA_AddPublishedDataSetResult;

UA_AddPublishedDataSetResult UA_THREADSAFE
UA_Server_addPublishedDataSet(UA_Server *server,
                              const UA_PublishedDataSetConfig *pdsConfig,
                              UA_NodeId *pdsId);

/* Return an owned copy; release it with UA_PublishedDataSetConfig_clear. */
UA_StatusCode UA_THREADSAFE
UA_Server_getPublishedDataSetConfig(UA_Server *server, const UA_NodeId pdsId,
                                    UA_PublishedDataSetConfig *config);

/* Returns a deep copy of the DataSetMetaData for an specific PDS */
UA_StatusCode UA_THREADSAFE
UA_Server_getPublishedDataSetMetaData(UA_Server *server, const UA_NodeId pdsId,
                                      UA_DataSetMetaDataType *metaData);

/* Remove PublishedDataSet, identified by the NodeId. Deletion of PDS removes
 * all contained and linked PDS Fields and the connected DataSetWriters. Returns
```

(continues on next page)

(continued from previous page)

```
 * Bad_ConfigurationError while a connected DataSetWriter is enabled and
 * Bad_InvalidState (without changes) while the WriterGroup of a connected
 * DataSetWriter is enabled. */
UA_StatusCode UA_THREADSAFE
UA_Server_removePublishedDataSet(UA_Server *server, const UA_NodeId pdsId);
```

7.8 DataSetField

The description of published variables is named DataSetField. Each DataSetField contains the selection of one information model node. The DataSetField has additional parameters for the publishing, sampling and error handling process.

```
typedef struct {
    UA_ConfigurationVersionDataType configurationVersion;
    UA_String fieldNameAlias;
    UA_Boolean promotedField;
    UA_PublishedVariableDataType publishParameters;

    UA_UInt32 maxStringLength;
    UA_LocalizedText description;
    /* If dataSetFieldId is not set, the GUID will be generated on adding the
     * field */
    UA_Guid dataSetFieldId;
} UA_DataSetVariableConfig;

typedef enum {
    UA_PUBSUB_DATASETFIELD_VARIABLE,
    UA_PUBSUB_DATASETFIELD_EVENT
} UA_DataSetFieldType;

typedef struct {
    UA_DataSetFieldType dataSetFieldType;
    union {
        /* events need other config later */
        UA_DataSetVariableConfig variable;
    } field;
} UA_DataSetFieldConfig;

void
UA_DataSetFieldConfig_clear(UA_DataSetFieldConfig *dataSetFieldConfig);

typedef struct {
    UA_StatusCode result;
    UA_ConfigurationVersionDataType configurationVersion;
} UA_DataSetFieldResult;

UA_DataSetFieldResult UA_THREADSAFE
UA_Server_addDataSetField(UA_Server *server,
                          const UA_NodeId publishedDataSet,
```

(continues on next page)

(continued from previous page)

```
                        const UA_DataSetFieldConfig *fieldConfig,
                        UA_NodeId *fieldId);

/* Returns a deep copy of the config */
UA_StatusCode UA_THREADSAFE
UA_Server_getDataSetFieldConfig(UA_Server *server, const UA_NodeId dsfId,
                                UA_DataSetFieldConfig *config);

UA_DataSetFieldResult UA_THREADSAFE
UA_Server_removeDataSetField(UA_Server *server, const UA_NodeId dsfId);
```

7.9 WriterGroup

All WriterGroups are created within a PubSubConnection and automatically deleted if the connection is removed. The WriterGroup is primary used as container for *DataSetWriter* and network message settings. The WriterGroup can be imagined as producer of the network messages. The creation of network messages is controlled by parameters like the publish interval, which is e.g. contained in the WriterGroup.

```
typedef enum {
    UA_PUBSUB_ENCODING_UADP = 0,
    UA_PUBSUB_ENCODING_JSON
} UA_PubSubEncodingType;

typedef struct {
    UA_PUBSUBCOMPONENT_COMMON
    UA_UInt16 writerGroupId;
    UA_Duration publishingInterval;
    UA_Double keepAliveTime;
    UA_Byte priority;
    UA_ExtensionObject transportSettings;
    UA_ExtensionObject messageSettings;
    UA_KeyValueMap groupProperties;
    UA_PubSubEncodingType encodingMimeType;

    /* non std. config parameter. maximum count of embedded DataSetMessage in
     * one NetworkMessage */
    UA_UInt16 maxEncapsulatedDataSetMessageCount;

    /* Security Configuration
     * Message are encrypted if a SecurityPolicy is configured and the
     * securityMode set accordingly. The symmetric key is a runtime information
     * and has to be set via UA_Server_setWriterGroupEncryptionKey. */
    UA_MessageSecurityMode securityMode; /* via the UA_WriterGroupDataType */
    UA_PubSubSecurityPolicy *securityPolicy;
    UA_String securityGroupId;

    /* Fields defined by PubSubGroupDataType and preserved by file-config
     * save/load. securityPolicy remains the runtime policy implementation. */
```

(continues on next page)

(continued from previous page)

```
    size_t securityKeyServicesSize;
    UA_EndpointDescription *securityKeyServices;
    UA_UInt32 maxNetworkMessageSize;
    size_t localeIdsSize;
    UA_String *localeIds;
    UA_String headerLayoutUri;
} UA_WriterGroupConfig;

void
UA_WriterGroupConfig_clear(UA_WriterGroupConfig *writerGroupConfig);

UA_StatusCode UA_THREADSAFE
UA_Server_addWriterGroup(UA_Server *server, const UA_NodeId connection,
                         const UA_WriterGroupConfig *writerGroupConfig,
                         UA_NodeId *wgId);

/* Returns a deep copy of the config */
UA_StatusCode UA_THREADSAFE
UA_Server_getWriterGroupConfig(UA_Server *server, const UA_NodeId wgId,
                               UA_WriterGroupConfig *config);

/* The WriterGroup must be disabled to update the config */
UA_StatusCode UA_THREADSAFE
UA_Server_updateWriterGroupConfig(UA_Server *server, const UA_NodeId wgId,
                                  const UA_WriterGroupConfig *config);

UA_StatusCode UA_THREADSAFE
UA_Server_getWriterGroupState(UA_Server *server, const UA_NodeId wgId,
                              UA_PubSubState *state);

UA_StatusCode UA_THREADSAFE
UA_Server_triggerWriterGroupPublish(UA_Server *server,
                                    const UA_NodeId wgId);

UA_StatusCode UA_THREADSAFE
UA_Server_getWriterGroupLastPublishTimestamp(UA_Server *server,
                                             const UA_NodeId wgId,
                                             UA_DateTime *timestamp);

/* Remove the group and its DataSetWriters. A failed child removal is returned
 * and leaves the group pending deletion; retry after resolving the failure.
 * New writers cannot be added once group deletion has started. */
UA_StatusCode UA_THREADSAFE
UA_Server_removeWriterGroup(UA_Server *server, const UA_NodeId wgId);

UA_StatusCode UA_THREADSAFE
UA_Server_enableWriterGroup(UA_Server *server, const UA_NodeId wgId);

UA_StatusCode UA_THREADSAFE
```

(continues on next page)

(continued from previous page)

```
UA_Server_disableWriterGroup(UA_Server *server, const UA_NodeId wgId);

/* Set the group key for the message encryption */
UA_StatusCode UA_THREADSAFE
UA_Server_setWriterGroupEncryptionKeys(UA_Server *server, const UA_NodeId wgId,
                                       UA_UInt32 securityTokenId,
                                       const UA_ByteString signingKey,
                                       const UA_ByteString encryptingKey,
                                       const UA_ByteString keyNonce);

/* Legacy API */
#define UA_Server_setWriterGroupOperational(server, wgId) \
    UA_Server_enableWriterGroup(server, wgId)
#define UA_Server_setWriterGroupDisabled(server, wgId) \
    UA_Server_disableWriterGroup(server, wgId)
#define UA_Server_WriterGroup_getState(server, wgId, state) \
    UA_Server_getWriterGroupState(server, wgId, state)
#define UA_WriterGroup_lastPublishTimestamp(server, wgId, timestamp) \
    UA_Server_getWriterGroupLastPublishTimestamp(server, wgId, timestamp)
#define UA_Server_WriterGroup_publish(server, wgId) \
    UA_Server_triggerWriterGroupPublish(server, wgId)
```

7.10 DataSetWriter

The DataSetWriters are the glue between the WriterGroups and the PublishedDataSets. The DataSetWriter contain configuration parameters and flags which influence the creation of DataSet messages. These messages are encapsulated inside the network message. The DataSetWriter must be linked with an existing PublishedDataSet and be contained within a WriterGroup.

```
typedef struct {
    UA_PUBSUBCOMPONENT_COMMON
    UA_UInt16 dataSetWriterId;
    UA_DataSetFieldContentMask dataSetFieldContentMask;
    UA_UInt32 keyFrameCount;
    UA_ExtensionObject messageSettings;
    UA_ExtensionObject transportSettings;
    UA_String dataSetName;
    UA_KeyValueMap dataSetWriterProperties;
} UA_DataSetWriterConfig;

void
UA_DataSetWriterConfig_clear(UA_DataSetWriterConfig *pdsConfig);

/* Add a new DataSetWriter to an existing WriterGroup. The DataSetWriter must be
 * coupled with a PublishedDataSet on creation.
 *
 * Part 14, 7.1.5.2.1 defines: The link between the PublishedDataSet and
 * DataSetWriter shall be created when an instance of the DataSetWriterType is
 * created. */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode UA_THREADSAFE
UA_Server_addDataSetWriter(UA_Server *server,
                           const UA_NodeId writerGroup, const UA_NodeId dataSet,
                           const UA_DataSetWriterConfig *dataSetWriterConfig,
                           UA_NodeId *dswId);

/* Returns a deep copy of the config */
UA_StatusCode UA_THREADSAFE
UA_Server_getDataSetWriterConfig(UA_Server *server, const UA_NodeId dswId,
                                 UA_DataSetWriterConfig *config);

/* The DataSetWriter must be disabled to update the config */
UA_StatusCode UA_THREADSAFE
UA_Server_updateDataSetWriterConfig(UA_Server *server, const UA_NodeId dswId,
                                    const UA_DataSetWriterConfig *config);

UA_StatusCode UA_THREADSAFE
UA_Server_enableDataSetWriter(UA_Server *server, const UA_NodeId dswId);

UA_StatusCode UA_THREADSAFE
UA_Server_disableDataSetWriter(UA_Server *server, const UA_NodeId dswId);

UA_StatusCode UA_THREADSAFE
UA_Server_getDataSetWriterState(UA_Server *server, const UA_NodeId dswId,
                                UA_PubSubState *state);

UA_StatusCode UA_THREADSAFE
UA_Server_removeDataSetWriter(UA_Server *server, const UA_NodeId dswId);

/* Legacy API */
#define UA_Server_DataSetWriter_getState(server, dswId, state) \
    UA_Server_getDataSetWriterState(server, dswId, state)
```

7.11 SubscribedDataSet

With the OPC UA Part 14 1.0.5, the concept of StandaloneSubscribedDataSet (SSDS) was introduced. The SSDS is the counterpart to the PublishedDataSet and has its own lifecycle. The SSDS can be connected to exactly one DataSetReader. In general, the SSDS is optional and a DataSetReader can still be defined without referencing a SSDS.

The SubscribedDataSet has two sub-types called the TargetVariablesType and SubscribedDataSetMirrorType. SubscribedDataSetMirrorType is currently not supported. SubscribedDataSet is set to TargetVariablesType and then the list of target Variables are created in the Subscriber AddressSpace. TargetVariables are a list of variables that are to be added in the Subscriber AddressSpace. It defines a list of Variable mappings between received DataSet fields and added Variables in the Subscriber AddressSpace.

```
typedef enum {
    UA_PUBSUB_SDS_TARGET,
```

(continues on next page)

(continued from previous page)

```
    UA_PUBSUB_SDS_MIRROR
} UA_SubscribedDataSetType;

typedef struct {
    UA_String name;
    UA_SubscribedDataSetType subscribedDataSetType;
    union {
        /* DataSetMirror is currently not implemented */
        UA_TargetVariablesDataType target;
    } subscribedDataSet;
    UA_DataSetMetaDataType dataSetMetaData;

    void *context; /* Context Configuration (SubscribedDataSet has no state
                    * machine) */
} UA_SubscribedDataSetConfig;

void
UA_SubscribedDataSetConfig_clear(UA_SubscribedDataSetConfig *sdsConfig);

UA_StatusCode UA_THREADSAFE
UA_Server_addSubscribedDataSet(UA_Server *server,
                               const UA_SubscribedDataSetConfig *sdsConfig,
                               UA_NodeId *sdsId);

UA_StatusCode UA_THREADSAFE
UA_Server_removeSubscribedDataSet(UA_Server *server, const UA_NodeId sdsId);

/* TODO: Implementation of SubscribedDataSetMirrorType */
```

7.12 DataSetReader

DataSetReader can receive NetworkMessages with the DataSetMessage of interest sent by the Publisher. DataSetReaders represent the configuration necessary to receive and process DataSetMessages on the Subscriber side. DataSetReader must be linked with a SubscribedDataSet and be contained within a ReaderGroup.

```
typedef struct {
    UA_PUBSUBCOMPONENT_COMMON
    UA_PublisherId publisherId;
    /* A zero-initialized config keeps the legacy wildcard behavior. Set this
     * to true to filter explicitly for the legitimate Byte PublisherId 0. */
    UA_Boolean publisherIdFilterEnabled;
    UA_UInt16 writerGroupId;
    UA_UInt16 dataSetWriterId;
    UA_DataSetMetaDataType dataSetMetaData;
    UA_DataSetFieldContentMask dataSetFieldContentMask;
    UA_Double messageReceiveTimeout; /* The maximum interval (in milliseconds)
                                      * after which we want to receive a
                                      * message. Gets reset after every received
```

(continues on next page)

(continued from previous page)

```
                                * message. If <= 0.0, then no timeout is
                                * configured. */
    UA_UInt32 keyFrameCount; /* Maximum key-frame period. A value <= 1
                              * accepts key frames only. */
    UA_String headerLayoutUri;
    UA_KeyValueMap dataSetReaderProperties;
    UA_ExtensionObject messageSettings;
    UA_ExtensionObject transportSettings;
    UA_SubscribedDataSetType subscribedDataSetType;
    union {
        /* TODO: UA_SubscribedDataSetMirrorDataType subscribedDataSetMirror */
        UA_TargetVariablesDataType target;
    } subscribedDataSet;
    /* non std. fields */
    UA_String linkedStandaloneSubscribedDataSetName;
} UA_DataSetReaderConfig;

UA_StatusCode
UA_DataSetReaderConfig_copy(const UA_DataSetReaderConfig *src,
                            UA_DataSetReaderConfig *dst);

void
UA_DataSetReaderConfig_clear(UA_DataSetReaderConfig *cfg);

/* Get the configuration (deep copy) of the DataSetReader */
UA_StatusCode UA_THREADSAFE
UA_Server_getDataSetReaderConfig(UA_Server *server, const UA_NodeId dsrId,
                                 UA_DataSetReaderConfig *config);

UA_StatusCode UA_THREADSAFE
UA_Server_getDataSetReaderState(UA_Server *server, const UA_NodeId dsrId,
                                UA_PubSubState *state);

UA_StatusCode UA_THREADSAFE
UA_Server_addDataSetReader(UA_Server *server, UA_NodeId readerGroupId,
                           const UA_DataSetReaderConfig *config,
                           UA_NodeId *dsrId);

/* The DataSetReader must be disabled to update the config */
UA_StatusCode UA_THREADSAFE
UA_Server_updateDataSetReaderConfig(UA_Server *server,
                                    const UA_NodeId dsrId,
                                    const UA_DataSetReaderConfig *config);

UA_StatusCode UA_THREADSAFE
UA_Server_removeDataSetReader(UA_Server *server, const UA_NodeId dsrId);

UA_StatusCode UA_THREADSAFE
UA_Server_enableDataSetReader(UA_Server *server, const UA_NodeId dsrId);
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode UA_THREADSAFE
UA_Server_disableDataSetReader(UA_Server *server, const UA_NodeId dsrId);

UA_StatusCode UA_THREADSAFE
UA_Server_setDataSetReaderTargetVariables(
    UA_Server *server, const UA_NodeId dsrId,
    size_t targetVariablesSize,
    const UA_FieldTargetDataType *targetVariables);

/* Legacy API */
#define UA_Server_DataSetReader_getConfig(server, dsrId, config) \
    UA_Server_getDataSetReaderConfig(server, dsrId, config)
#define UA_Server_DataSetReader_getState(server, dsrId, state) \
    UA_Server_getDataSetReaderState(server, dsrId, state)
#define UA_Server_DataSetReader_createTargetVariables(server, dsrId, \
                                                      tvsSize, tvs)  \
    UA_Server_setDataSetReaderTargetVariables(server, dsrId, tvsSize, tvs)
```

7.13 ReaderGroup

ReaderGroups contain a list of DataSetReaders. All ReaderGroups are created within a PubSubConnection and automatically deleted if the connection is removed. All network message related filters are only available in the DataSetReader.

The RT-levels go along with different requirements. The below listed levels can be configured for a ReaderGroup.

```
typedef struct {
    UA_PUBSUBCOMPONENT_COMMON

    /* non std. fields */
    UA_KeyValueMap groupProperties;
    UA_PubSubEncodingType encodingMimeType;
    UA_ExtensionObject transportSettings;
    UA_ExtensionObject messageSettings;

    /* Messages are decrypted if a SecurityPolicy is configured and the
     * securityMode set accordingly. The symmetric key is a runtime information
     * and has to be set via UA_Server_setReaderGroupEncryptionKey. */
    UA_MessageSecurityMode securityMode;
    UA_PubSubSecurityPolicy *securityPolicy;
    UA_String securityGroupId;
    size_t securityKeyServicesSize;
    UA_EndpointDescription *securityKeyServices;
    UA_UInt32 maxNetworkMessageSize;
} UA_ReaderGroupConfig;

void
```

(continues on next page)

(continued from previous page)

```
UA_ReaderGroupConfig_clear(UA_ReaderGroupConfig *readerGroupConfig);

/* Get configuration of ReaderGroup (deep copy) */
UA_StatusCode UA_THREADSAFE
UA_Server_getReaderGroupConfig(UA_Server *server, const UA_NodeId rgId,
                               UA_ReaderGroupConfig *config);

UA_StatusCode UA_THREADSAFE
UA_Server_getReaderGroupState(UA_Server *server, const UA_NodeId rgId,
                              UA_PubSubState *state);

UA_StatusCode UA_THREADSAFE
UA_Server_addReaderGroup(UA_Server *server, const UA_NodeId connectionId,
                         const UA_ReaderGroupConfig *config,
                         UA_NodeId *rgId);

/* The ReaderGroup must be disabled to update the config */
UA_StatusCode UA_THREADSAFE
UA_Server_updateReaderGroupConfig(UA_Server *server,
                                  const UA_NodeId rgId,
                                  const UA_ReaderGroupConfig *config);

UA_StatusCode UA_THREADSAFE
UA_Server_removeReaderGroup(UA_Server *server, const UA_NodeId rgId);

UA_StatusCode UA_THREADSAFE
UA_Server_enableReaderGroup(UA_Server *server, const UA_NodeId rgId);

UA_StatusCode UA_THREADSAFE
UA_Server_disableReaderGroup(UA_Server *server, const UA_NodeId rgId);

/* Set the group key for the message encryption */
UA_StatusCode UA_THREADSAFE
UA_Server_setReaderGroupEncryptionKeys(UA_Server *server,
                                       const UA_NodeId rgId,
                                       UA_UInt32 securityTokenId,
                                       const UA_ByteString signingKey,
                                       const UA_ByteString encryptingKey,
                                       const UA_ByteString keyNonce);

/* Return an owned copy; release it with UA_SubscribedDataSetConfig_clear. */
UA_StatusCode UA_THREADSAFE
UA_Server_getSubscribedDataSetConfig(UA_Server *server, const UA_NodeId id,
                                     UA_SubscribedDataSetConfig *config);

/* Update the settings of a disabled dataset in place. Name and NodeId are
 * immutable. Disable attached writers/readers before updating and restore
 * their states afterwards. Plain PublishedItems fields are retained. */
UA_StatusCode UA_THREADSAFE
```

(continues on next page)

(continued from previous page)

```
UA_Server_updatePublishedDataSetConfig(UA_Server *server, const UA_NodeId id,
                                       const UA_PublishedDataSetConfig *config);
UA_StatusCode UA_THREADSAFE
UA_Server_updateSubscribedDataSetConfig(UA_Server *server, const UA_NodeId id,
                                        const UA_SubscribedDataSetConfig *config);

#ifdef UA_ENABLE_PUBSUB_FILE_CONFIG

/* Decodes the information from the ByteString. If the decoded content is a
 * PubSubConfiguration in a UABinaryFileDataType-object. It will overwrite the
 * current PubSub configuration from the server. The added components are
 * enabled automatically if their enabled-flag is set in the config.
 * Child-components are enabled first.
 *
 * Note that you need to disable all components with
 * UA_Server_disableAllPubSubComponents before loading the config. */
UA_StatusCode
UA_Server_loadPubSubConfigFromByteString(UA_Server *server,
                                         const UA_ByteString buffer);

/* Saves the current PubSub configuration of a server in a byteString. */
UA_StatusCode
UA_Server_writePubSubConfigurationToByteString(UA_Server *server,
                                               UA_ByteString *buffer);
#endif

/* Legacy API */
#define UA_Server_ReaderGroup_getConfig(server, rgId, config) \
    UA_Server_getReaderGroupConfig(server, rgId, config)
#define UA_Server_ReaderGroup_getState(server, rgId, state) \
    UA_Server_getReaderGroupState(server, rgId, state)
#define UA_Server_setReaderGroupOperational(server, rgId) \
    UA_Server_enableReaderGroup(server, rgId)
#define UA_Server_setReaderGroupDisabled(server, rgId) \
    UA_Server_disableReaderGroup(server, rgId)

#ifdef UA_ENABLE_PUBSUB_SKS
```

7.14 SecurityGroup

A SecurityGroup is an abstraction that represents the message security settings and security keys for a subset of NetworkMessages exchanged between Publishers and Subscribers. The SecurityGroup objects are created on a Security Key Service (SKS). The SKS manages the access to the keys based on the role permission for a user assigned to a SecurityGroup Object. A SecurityGroup is identified with a unique identifier called the SecurityGroupId. It is unique within the SKS.

Note

The access to the SecurityGroup and therefore the securitykeys managed by SKS requires management of Roles and Permissions in the SKS. The Role Permission model is not supported at the time of writing. However, the access control plugin can be used to create and manage role permission on SecurityGroup object.

```
typedef struct {
    UA_String securityGroupName;
    UA_Duration keyLifeTime;
    UA_String securityPolicyUri;
    UA_UInt32 maxFutureKeyCount;
    UA_UInt32 maxPastKeyCount;
} UA_SecurityGroupConfig;

/* Creates a SecurityGroup object and add it to the list in PubSub Manager. If
 * the information model is enabled then the SecurityGroup object Node is also
 * created in the server. A keyStorage with initial list of keys is created with
 * a SecurityGroup. A callback is added to new SecurityGroup which updates the
 * keys periodically at each KeyLifeTime expire.
 *
 * @param server The server instance
 * @param securityGroupFolderNodeId The parent node of the SecurityGroup. It
 *        must be of SecurityGroupFolderType
 * @param securityGroupConfig The security settings of a SecurityGroup
 * @param securityGroupNodeId The output nodeId of the new SecurityGroup
 * @return UA_StatusCode The return status code */
UA_StatusCode UA_THREADSAFE
UA_Server_addSecurityGroup(UA_Server *server,
                           UA_NodeId securityGroupFolderNodeId,
                           const UA_SecurityGroupConfig *securityGroupConfig,
                           UA_NodeId *securityGroupNodeId);

/* Removes the SecurityGroup from PubSub Manager. It removes the KeyStorage
 * associated with the SecurityGroup from the server.
 *
 * @param server The server instance
 * @param securityGroup The nodeId of the securityGroup to be removed
 * @return UA_StatusCode The returned status code. */
UA_StatusCode UA_THREADSAFE
UA_Server_removeSecurityGroup(UA_Server *server,
                              const UA_NodeId securityGroup);

/* This is a repeated callback which is triggered on each iteration of SKS Pull
 * request. The server uses this callback to notify user about the status of
 * current Pull request iteration. The period is calculated based on the
 * KeylifeTime of specified in the SecurityGroup object node on the SKS server.
 *
 * @param server The server instance managing the publisher/subscriber.
```

(continues on next page)

(continued from previous page)

```
 * @param sksPullRequestStatus The current status of sks pull request.
 * @param context The pointer to user defined data passed to this callback. */
typedef void
(*UA_Server_sksPullRequestCallback)(UA_Server *server,
                                    UA_StatusCode sksPullRequestStatus,
                                    void* context);

/* Sets the SKS client config used to call the GetSecurityKeys Method on SKS and
 * get the initial set of keys for a SecurityGroupId and adds timedCallback for
 * the next GetSecurityKeys method Call. This uses async Client API for SKS Pull
 * request. The SKS Client instance is created and destroyed at runtime on each
 * iteration of SKS Pull request by the server. The key Rollover mechanism will
 * check if the new keys are needed then it will call the getSecurityKeys Method
 * on SKS Server. At the end of SKS Pull request iteration, the sks client will
 * be deleted by a delayed callback (in next server iteration).
 *
 * Note: It is be called before setting Reader/Writer Group into Operational
 * because this also allocates a channel context for the pubsub security policy.
 *
 * Note: The stateCallback of sksClientConfig will be overwritten by an internal
 * callback.
 *
 * @param server the server instance
 * @param clientConfig holds the required configuration to make encrypted
 *        connection with SKS Server. The input client config takes the
 *        lifecycle as long as SKS request are made. It is deleted with its
 *        plugins when the server is deleted or the last Reader/Writer Group of
 *        the securityGroupId is deleted. The input config is copied to an
 *        internal config object and the content of input config object will be
 *        reset to zero.
 * @param endpointUrl holds the endpointUrl of the SKS server. It is copied and
 *        does not need to outlive this call.
 * @param securityGroupId the SecurityGroupId of the securityGroup on SKS and
 *        reader/writergroups
 * @param callback the user defined callback to notify the user about the status
 *        of SKS Pull request.
 * @param context passed to the callback function
 * @return UA_StatusCode the returned status */
UA_StatusCode
UA_Server_setSksClient(UA_Server *server, UA_String securityGroupId,
                       UA_ClientConfig *clientConfig, const char *endpointUrl,
                       UA_Server_sksPullRequestCallback callback, void *context);

UA_StatusCode UA_THREADSAFE
UA_Server_setReaderGroupActivateKey(UA_Server *server,
                                    const UA_NodeId readerGroupId);

UA_StatusCode UA_THREADSAFE
UA_Server_setWriterGroupActivateKey(UA_Server *server,
```

(continues on next page)

(continued from previous page)

```
                                const UA_NodeId writerGroup);

#endif /* UA_ENABLE_PUBSUB_SKS */
```

7.15 Offset Table

When the content of a PubSub Networkmessage has a fixed length, then only a few “content bytes” at known locations within the NetworkMessage change between publish cycles. The so-called offset table exposes this to enable fast-path implementations for realtime applications.

String and ByteString fields with RawData encoding have a fixed length when MaxStringLength is configured in the FieldMetaData. This is supported for direct fields, but not for String and ByteString members nested inside structures.

```
typedef enum {
    UA_PUBSUBOFFSETTYPE_NETWORKMESSAGE_GROUPVERSION,   /* UInt32␣
↪*/
    UA_PUBSUBOFFSETTYPE_NETWORKMESSAGE_SEQUENCENUMBER, /*␣
↪UInt16 */
    UA_PUBSUBOFFSETTYPE_NETWORKMESSAGE_TIMESTAMP,      /* DateTime␣
↪*/
    UA_PUBSUBOFFSETTYPE_NETWORKMESSAGE_PICOSECONDS,    /* UInt16 */
    UA_PUBSUBOFFSETTYPE_DATASETMESSAGE, /* no content, marks the DSM␣
↪beginning */
    UA_PUBSUBOFFSETTYPE_DATASETMESSAGE_SEQUENCENUMBER, /*␣
↪UInt16 */
    UA_PUBSUBOFFSETTYPE_DATASETMESSAGE_STATUS,         /* UInt16 */
    UA_PUBSUBOFFSETTYPE_DATASETMESSAGE_TIMESTAMP,      /* DateTime */
    UA_PUBSUBOFFSETTYPE_DATASETMESSAGE_PICOSECONDS,    /* UInt16 */
    UA_PUBSUBOFFSETTYPE_DATASETFIELD_DATAVALUE,
    UA_PUBSUBOFFSETTYPE_DATASETFIELD_VARIANT,
    UA_PUBSUBOFFSETTYPE_DATASETFIELD_RAW
} UA_PubSubOffsetType;

typedef struct {
    UA_PubSubOffsetType offsetType; /* Content type at the offset */
    size_t offset;                  /* Offset in the NetworkMessage */

    /* The PubSub component that originates / receives the offset content.
     * - For NetworkMessage-offsets this is the ReaderGroup / WriterGroup.
     * - For DataSetMessage-offsets this is DataSetReader / DataSetWriter.
     * - For DataSetFields this is the NodeId associated with the field:
     *   - For Writers the NodeId of the DataSetField (in a PublishedDataSet).
     *   - For Readers the TargetNodeId of the FieldTargetDataType (this can
     *     come from a SubscribedDataSet or a StandaloneSubscribedDataSets).
     *     Access more metadata from the FieldTargetVariable by counting the
     *     index of the current DataSetField-offset within the DataSetMessage
     *     and use that index for the lookup in the DataSetReader configuration. */
    UA_NodeId component;
```

(continues on next page)

(continued from previous page)

```
} UA_PubSubOffset;

typedef struct {
    UA_PubSubOffset *offsets;       /* Array of offset entries */
    size_t offsetsSize;             /* Number of entries */
    UA_ByteString networkMessage;  /* Current NetworkMessage in binary encoding */
} UA_PubSubOffsetTable;

void
UA_PubSubOffsetTable_clear(UA_PubSubOffsetTable *ot);

/* Compute the offset table for a WriterGroup */
UA_StatusCode UA_THREADSAFE
UA_Server_computeWriterGroupOffsetTable(UA_Server *server,
                                        const UA_NodeId writerGroupId,
                                        UA_PubSubOffsetTable *ot);
```

For ReaderGroups we cannot compute the offset table up front, because it is not ensured that all Readers end up with their DataSetMessage in the same NetworkMessage. Furthermore the ReaderGroup might receive messages from multiple different publishers.

Instead the offset tables are computed beforehand for each DataSetReader. At runtime, use UA_NetworkMessage_decodeBinaryHeaders to decode the NetworkMessage headers. The information therein (e.g. the MessageCount and and the DataSetWriterIds) can then be used to iterate over the DataSetMessages in the payload with their respective offset tables.

```
/* The offsets begin at zero for the DataSetMessage */
UA_StatusCode UA_THREADSAFE
UA_Server_computeDataSetReaderOffsetTable(UA_Server *server,
                                          const UA_NodeId dataSetReaderId,
                                          UA_PubSubOffsetTable *ot);

#endif /* UA_ENABLE_PUBSUB */
```

7.16 PubSub NetworkMessage

The following definitions enable to work directly with PubSub messages. This is not required when *PubSub is integrated with a server*.

7.16.1 DataSet Message

```
typedef enum {
    UA_FIELDENCODING_VARIANT   = 0,
    UA_FIELDENCODING_RAWDATA   = 1,
    UA_FIELDENCODING_DATAVALUE = 2,
    UA_FIELDENCODING_UNKNOWN   = 3
} UA_FieldEncoding;

typedef enum {
```

(continues on next page)

(continued from previous page)

```
    UA_DATASETMESSAGETYPE_DATAKEYFRAME   = 0,
    UA_DATASETMESSAGE_DATAKEYFRAME       = 0,
    UA_DATASETMESSAGETYPE_DATADELTAFRAME = 1,
    UA_DATASETMESSAGE_DATADELTAFRAME     = 1,
    UA_DATASETMESSAGETYPE_EVENT          = 2,
    UA_DATASETMESSAGE_EVENT              = 2,
    UA_DATASETMESSAGETYPE_KEEPALIVE      = 3,
    UA_DATASETMESSAGE_KEEPALIVE          = 3
} UA_DataSetMessageType;

typedef struct {
    /* Settings and message fields enabled with the DataSetFlags1 */
    UA_Boolean dataSetMessageValid;

    UA_FieldEncoding fieldEncoding;

    UA_Boolean dataSetMessageSequenceNrEnabled;
    /* UADP encodes this as UInt16; JSON uses UInt32. */
    UA_UInt32 dataSetMessageSequenceNr;

    UA_Boolean statusEnabled;
    /* UADP encodes the high 16 bits; JSON uses the full StatusCode. */
    UA_StatusCode status;

    UA_Boolean configVersionMajorVersionEnabled;
    UA_UInt32 configVersionMajorVersion;

    UA_Boolean configVersionMinorVersionEnabled;
    UA_UInt32 configVersionMinorVersion;

    /* Settings and message fields enabled with the DataSetFlags2 */
    UA_DataSetMessageType dataSetMessageType;

    UA_Boolean timestampEnabled;
    UA_UtcTime timestamp;

    UA_Boolean picoSecondsIncluded;
    UA_UInt16 picoSeconds;
} UA_DataSetMessageHeader;

typedef struct {
    UA_UInt16 index;
    UA_DataValue value;
} UA_DataSetMessage_DeltaFrameField;

typedef struct {
    UA_DataSetMessageHeader header;
    UA_UInt16 fieldCount;
    union { /* Array of fields (cf. header->dataSetMessageType) */
```

(continues on next page)

(continued from previous page)

```
        UA_DataValue *keyFrameFields;
        UA_DataSetMessage_DeltaFrameField *deltaFrameFields;
    } data;
} UA_DataSetMessage;

void UA_DataSetMessage_clear(UA_DataSetMessage *p);
```

7.16.2 Network Message

```
typedef enum {
    UA_NETWORKMESSAGE_DATASET = 0,
    UA_NETWORKMESSAGE_DISCOVERY_REQUEST = 1,
    UA_NETWORKMESSAGE_DISCOVERY_RESPONSE = 2
} UA_NetworkMessageType;

typedef struct {
    UA_Boolean writerGroupIdEnabled;
    UA_UInt16 writerGroupId;

    UA_Boolean groupVersionEnabled;
    UA_UInt32 groupVersion;

    UA_Boolean networkMessageNumberEnabled;
    UA_UInt16 networkMessageNumber;

    UA_Boolean sequenceNumberEnabled;
    UA_UInt16 sequenceNumber;
} UA_NetworkMessageGroupHeader;

#define UA_NETWORKMESSAGE_MAX_NONCE_LENGTH 16

typedef struct {
    UA_Boolean networkMessageSigned;

    UA_Boolean networkMessageEncrypted;

    UA_Boolean securityFooterEnabled;
    UA_UInt16 securityFooterSize;

    UA_Boolean forceKeyReset;

    UA_UInt32 securityTokenId;

    UA_UInt16 messageNonceSize;
    UA_Byte messageNonce[UA_NETWORKMESSAGE_MAX_NONCE_LENGTH];
} UA_NetworkMessageSecurityHeader;

#define UA_NETWORKMESSAGE_MAXMESSAGECOUNT 32
```

(continues on next page)

(continued from previous page)

```
typedef struct {
    UA_Byte version;

    /* Fields defined via the UADPFlags */

    UA_Boolean publisherIdEnabled;
    UA_PublisherId publisherId;

    UA_Boolean groupHeaderEnabled;
    UA_NetworkMessageGroupHeader groupHeader;

    /* Fields defined via the Extended1Flags */

    UA_Boolean dataSetClassIdEnabled;
    UA_Guid dataSetClassId;

    UA_Boolean securityEnabled;
    UA_NetworkMessageSecurityHeader securityHeader;

    UA_Boolean timestampEnabled;
    UA_DateTime timestamp;

    UA_Boolean picosecondsEnabled;
    UA_UInt16 picoseconds;

    /* Fields defined via the Extended2Flags */

    UA_Boolean chunkMessage;

    UA_Boolean promotedFieldsEnabled;
    UA_UInt16 promotedFieldsSize;
    UA_Variant *promotedFields; /* BaseDataType */

    /* For Json NetworkMessage */
    UA_Boolean messageIdEnabled;
    UA_String messageId;
    UA_Boolean jsonSingleDataSetMessage; /* Messages contains an object instead of an␣
↪array */

    /* The PayloadHeader contains the number of DataSetMessages and the
     * DataSetWriterId for each of them. If the PayloadHeader is disabled, then
     * the number of DataSetMessages is determined as follows:
     *
     * - If the UA_NetworkMessage_EncodingOptions contain metadata, they define
     *   the number and the order of the DataSetMessages.
     * - Otherwise we assume exactly one DataSetMessage which takes up all of the
     *   remaining NetworkMessage length.
     *
     * There is an upper bound for the number of DataSetMessages, so that the
```

(continues on next page)

(continued from previous page)

```
     * DataSetWriterIds can be parsed as part of the headers without allocating
     * memory. */
    UA_Boolean payloadHeaderEnabled;
    UA_Byte messageCount;
    UA_UInt16 dataSetWriterIds[UA_NETWORKMESSAGE_MAXMESSAGECOUNT];

    /* TODO: Add support for Discovery Messages */
    UA_NetworkMessageType networkMessageType;
    union {
        /* The DataSetMessages are an array of messageCount length.
         * Can be NULL if only the headers have been decoded. */
        UA_DataSetMessage *dataSetMessages;
    } payload;

    UA_ByteString securityFooter;
} UA_NetworkMessage;

void
UA_NetworkMessage_clear(UA_NetworkMessage* p);
```

7.16.3 NetworkMessage Encoding

The en/decoding translates the NetworkMessage structure to/from a binary or JSON encoding. The en/decoding of PubSub NetworkMessages can require additional metadata. For example, during decoding, the DataType of raw encoded fields must be already known. As an example for encoding, the configuredSize may define zero-padding after a DataSetMessage.

In the below methods, the different encoding options can be a NULL pointer and will then be ignored.

```
typedef struct {
    /* The WriterId is used to find the matching encoding metadata. If the
     * NetworkMessage/DataSetMessage does not transmit the identifier, then the
     * encoding metadata is used in-order for the received fields. */
    UA_UInt16 dataSetWriterId;

    /* FieldMetaData for JSON and RAW encoding */
    size_t fieldsSize;
    UA_FieldMetaData *fields;

    /* Zero-padding if the DataSetMessage is shorter (UADP) */
    UA_UInt16 configuredSize;
} UA_DataSetMessage_EncodingMetaData;

typedef struct {
    size_t metaDataSize;
    UA_DataSetMessage_EncodingMetaData *metaData;
} UA_NetworkMessage_EncodingOptions;

/* The output buffer is allocated to the required size if initially empty.
 * Otherwise, upon success, the length is adjusted. */
```

(continues on next page)

(continued from previous page)

```
UA_StatusCode
UA_NetworkMessage_encodeBinary(const UA_NetworkMessage* src,
                               UA_ByteString *outBuf,
                               const UA_NetworkMessage_EncodingOptions *eo);

size_t
UA_NetworkMessage_calcSizeBinary(const UA_NetworkMessage *p,
                                 const UA_NetworkMessage_EncodingOptions *eo);

UA_StatusCode
UA_NetworkMessage_decodeBinary(const UA_ByteString *src,
                               UA_NetworkMessage *dst,
                               const UA_NetworkMessage_EncodingOptions *eo,
                               const UA_DecodeBinaryOptions *bo);

/* Decode only the headers before the payload */
UA_StatusCode
UA_NetworkMessage_decodeBinaryHeaders(const UA_ByteString *src,
                                      UA_NetworkMessage *dst,
                                      const UA_NetworkMessage_EncodingOptions *eo,
                                      const UA_DecodeBinaryOptions *bo,
                                      size_t *payloadOffset);

#ifdef UA_ENABLE_JSON_ENCODING

/* The output buffer is allocated to the required size if initially empty.
 * Otherwise, upon success, the length is adjusted. */
UA_StatusCode
UA_NetworkMessage_encodeJson(const UA_NetworkMessage *src,
                             UA_ByteString *outBuf,
                             const UA_NetworkMessage_EncodingOptions *eo,
                             const UA_EncodeJsonOptions *jo);

size_t
UA_NetworkMessage_calcSizeJson(const UA_NetworkMessage *src,
                               const UA_NetworkMessage_EncodingOptions *eo,
                               const UA_EncodeJsonOptions *jo);

UA_StatusCode
UA_NetworkMessage_decodeJson(const UA_ByteString *src,
                             UA_NetworkMessage *dst,
                             const UA_NetworkMessage_EncodingOptions *eo,
                             const UA_DecodeJsonOptions *jo);

#endif /* UA_ENABLE_JSON_ENCODING */
#endif /* UA_ENABLE_PUBSUB */
```

CHAPTER

EIGHT

TUTORIALS

8.1 Working with Data Types

OPC UA defines a type system for values that can be encoded in the protocol messages. This tutorial shows some examples for available data types and their use. See the section on *Data Types* for the full definitions.

8.1.1 Basic Data Handling

This section shows the basic interaction patterns for data types. Make sure to compare with the type definitions in types.h.

```
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>
#include <open62541/server_config_default.h>

#include <stdlib.h>
#include <stdio.h>

static void
variables_basic(void) {
    /* Int32 */
    UA_Int32 i = 5;
    UA_Int32 j;
    UA_Int32_copy(&i, &j);

    UA_Int32 *ip = UA_Int32_new();
    UA_Int32_copy(&i, ip);
    UA_Int32_delete(ip);

    /* String */
    UA_String s;
    UA_String_init(&s); /* _init zeroes out the entire memory of the datatype */
    char *test = "test";
    s.length = strlen(test);
    s.data = (UA_Byte*)test;

    UA_String s2;
    UA_String_copy(&s, &s2);
    UA_String_clear(&s2); /* Copying heap-allocated the dynamic content */
```

(continues on next page)

(continued from previous page)

```
    UA_String s3 = UA_STRING("test2");
    UA_String s4 = UA_STRING_ALLOC("test2"); /* Copies the content to the heap */
    UA_Boolean eq = UA_String_equal(&s3, &s4);
    UA_String_clear(&s4);
    if(!eq)
        return;

    /* Structured Type */
    UA_ReadRequest rr;
    UA_init(&rr, &UA_TYPES[UA_TYPES_READREQUEST]); /* Generic method */
    UA_ReadRequest_init(&rr); /* Shorthand for the previous line */

    rr.requestHeader.timestamp = UA_DateTime_now(); /* Members of a structure */

    rr.nodesToRead = (UA_ReadValueId *)UA_Array_new(5, &UA_TYPES[UA_TYPES_
↪READVALUEID]);
    rr.nodesToReadSize = 5; /* Array size needs to be made known */

    UA_ReadRequest *rr2 = UA_ReadRequest_new();
    UA_copy(&rr, rr2, &UA_TYPES[UA_TYPES_READREQUEST]);
    UA_ReadRequest_clear(&rr);
    UA_ReadRequest_delete(rr2);
}
```

8.1.2 NodeIds

An OPC UA information model is made up of nodes and references between nodes. Every node has a unique *NodeId*. NodeIds refer to a namespace with an additional identifier value that can be an integer, a string, a guid or a bytestring.

```
static void
variables_nodeids(void) {
    UA_NodeId id1 = UA_NODEID_NUMERIC(1, 1234);
    id1.namespaceIndex = 3;

    UA_NodeId id2 = UA_NODEID_STRING(1, "testid"); /* the string is static */
    UA_Boolean eq = UA_NodeId_equal(&id1, &id2);
    if(eq)
        return;

    UA_NodeId id3;
    UA_NodeId_copy(&id2, &id3);
    UA_NodeId_clear(&id3);

    UA_NodeId id4 = UA_NODEID_STRING_ALLOC(1, "testid"); /* the string is copied
                                                            to the heap */
    UA_NodeId_clear(&id4);
}
```

8.1.3 Variants

The datatype *Variant* belongs to the built-in datatypes of OPC UA and is used as a container type. A variant can hold any other datatype as a scalar (except variant) or as an array. Array variants can additionally denote the dimensionality of the data (e.g. a 2x3 matrix) in an additional integer array.

```
static void
variables_variants(void) {
    /* Set a scalar value */
    UA_Variant v;
    UA_Int32 i = 42;
    UA_Variant_setScalar(&v, &i, &UA_TYPES[UA_TYPES_INT32]);

    /* Make a copy */
    UA_Variant v2;
    UA_Variant_copy(&v, &v2);
    UA_Variant_clear(&v2);

    /* Set an array value */
    UA_Variant v3;
    UA_Double d[9] = {1.0, 2.0, 3.0,
                      4.0, 5.0, 6.0,
                      7.0, 8.0, 9.0};
    UA_Variant_setArrayCopy(&v3, d, 9, &UA_TYPES[UA_TYPES_DOUBLE]);

    /* Set array dimensions */
    v3.arrayDimensions = (UA_UInt32 *)UA_Array_new(2, &UA_TYPES[UA_TYPES_
↪UINT32]);
    v3.arrayDimensionsSize = 2;
    v3.arrayDimensions[0] = 3;
    v3.arrayDimensions[1] = 3;
    UA_Variant_clear(&v3);
}

#ifdef UA_ENABLE_JSON_ENCODING
static void
prettyprint(void) {
    UA_ReadRequest rr;
    UA_ReadRequest_init(&rr);
    UA_ReadValueId rvi[2];
    UA_ReadValueId_init(rvi);
    UA_ReadValueId_init(&rvi[1]);
    rr.nodesToRead = rvi;
    rr.nodesToReadSize = 2;
    UA_String out = UA_STRING_NULL;
    UA_print(&rr, &UA_TYPES[UA_TYPES_READREQUEST], &out);

    printf("%.*s\n", (int)out.length, out.data);
    UA_String_clear(&out);

    UA_ReadResponse resp;
```

(continues on next page)

(continued from previous page)

```
    UA_ReadResponse_init(&resp);
    UA_print(&resp, &UA_TYPES[UA_TYPES_READRESPONSE], &out);

    printf("%.*s\n", (int)out.length, out.data);
    UA_String_clear(&out);

    UA_ReferenceDescription br;
    UA_ReferenceDescription_init(&br);
    br.nodeClass = (UA_NodeClass)5;
    UA_print(&br, &UA_TYPES[UA_TYPES_REFERENCEDESCRIPTION], &out);
    printf("%.*s\n", (int)out.length, out.data);
    UA_String_clear(&out);

    UA_Float matrix[4] = {1.0, 2.0, 3.0, 4.0};
    UA_UInt32 matrix_dims[2] = {2, 2};
    UA_DataValue dv;
    UA_DataValue_init(&dv);
    UA_Variant_setArray(&dv.value, &matrix, 4, &UA_TYPES[UA_TYPES_FLOAT]);
    dv.value.arrayDimensions = matrix_dims;
    dv.value.arrayDimensionsSize = 2;
    dv.hasValue = true;
    dv.hasStatus = true;
    dv.hasServerTimestamp = true;
    dv.hasSourcePicoseconds = true;
    UA_print(&dv, &UA_TYPES[UA_TYPES_DATAVALUE], &out);
    printf("%.*s\n", (int)out.length, out.data);
    UA_String_clear(&out);
}
#endif
```

It follows the main function, making use of the above definitions.

```
int main(void) {
    variables_basic();
    variables_nodeids();
    variables_variants();
```

8.2 Building a Simple Server

This series of tutorial guide you through your first steps with open62541. For compiling the examples, you need a compiler (MS Visual Studio 2015 or newer, GCC, Clang and MinGW32 are all known to be working). The compilation instructions are given for GCC but should be straightforward to adapt.

It will also be very helpful to install an OPC UA Client with a graphical frontend, such as UAExpert by Unified Automation. That will enable you to examine the information model of any OPC UA server.

To get started, download the open62541 single-file release from http://open62541.org or generate it according to the *build instructions* with the "amalgamation" option enabled. From now on, we assume you have the open62541.c/.h files in the current folder. Now create a new C source-file called myServer.c with the following content:

```c
#include <open62541/server.h>

int main(void) {
    UA_Server *server = UA_Server_new();
    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

This is all that is needed for a simple OPC UA server. With the GCC compiler, the following command produces an executable:

```
$ gcc -std=c99 myServer.c -lopen62541 -o myServer
```

In a MinGW environment, the Winsock library must be added.

```
$ gcc -std=c99 myServer.c -lopen62541 -lws2_32 -o myServer.exe
```

Now start the server (stop with ctrl-c):

```
$ ./myServer
```

You have now compiled and run your first OPC UA server. You can go ahead and browse the information model with client. The server is listening on opc.tcp://localhost:4840.

8.2.1 Server Configuration and Plugins

open62541 provides a flexible framework for building OPC UA servers and clients. The goals is to have a core library that accommodates for all use cases and runs on all platforms. Users can then adjust the library to fit their use case via configuration and by developing (platform-specific) plugins. The core library is based on C99 only and does not even require basic POSIX support. For example, the lowlevel networking code is implemented as an exchangeable plugin. But don't worry. *open62541* provides plugin implementations for most platforms and sensible default configurations out-of-the-box.

In the above server code, we simply take the default server configuration and add a single TCP network layer that is listerning on port 4840.

8.2.2 Server Lifecycle

The code in this example shows the three parts for server lifecycle management: Creating a server, running the server, and deleting the server. Creating and deleting a server is trivial once the configuration is set up. The server is started with UA_Server_run. Internally, the server schedules regular tasks. Between the timeouts, the server listens on the network layer for incoming messages.

In order to integrate OPC UA in a single-threaded application with its own mainloop (for example provided by a GUI toolkit), one can alternatively drive the server manually. See the section of the server documentation on *Server Lifecycle* for details.

8.3 Adding Variables to a Server

This tutorial shows how to work with data types and how to add variable nodes to a server. First, we add a new variable to the server. Take a look at the definition of the UA_VariableAttributes structure to see the list of all attributes defined for VariableNodes.

Note that the default settings have the AccessLevel of the variable value as read only. See below for making the variable writable.

```
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>
#include <stdio.h>

static void
addVariable(UA_Server *server) {
    /* Define the attribute of the myInteger variable node */
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    UA_Int32 myInteger = 42;
    UA_Variant_setScalar(&attr.value, &myInteger, &UA_TYPES[UA_TYPES_INT32]);
    attr.description = UA_LOCALIZEDTEXT("en-US","the answer");
    attr.displayName = UA_LOCALIZEDTEXT("en-US","the answer");
    attr.dataType = UA_TYPES[UA_TYPES_INT32].typeId;
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;

    /* Add the variable node to the information model */
    UA_NodeId myIntegerNodeId = UA_NODEID_STRING(1, "the.answer");
    UA_QualifiedName myIntegerName = UA_QUALIFIEDNAME(1, "the answer");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_Server_addVariableNode(server, myIntegerNodeId, parentNodeId,
                              parentReferenceNodeId, myIntegerName,
                              UA_NS0ID(BASEDATAVARIABLETYPE), attr, NULL, NULL);
}

static void
addMatrixVariable(UA_Server *server) {
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en-US", "Double Matrix");
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;

    /* Set the variable value constraints */
    attr.dataType = UA_TYPES[UA_TYPES_DOUBLE].typeId;
    attr.valueRank = UA_VALUERANK_TWO_DIMENSIONS;
    UA_UInt32 arrayDims[2] = {2,2};
    attr.arrayDimensions = arrayDims;
    attr.arrayDimensionsSize = 2;

    /* Set the value. The array dimensions need to be the same for the value. */
    UA_Double zero[4] = {0.0, 0.0, 0.0, 0.0};
```

(continues on next page)

(continued from previous page)

```
    UA_Variant_setArray(&attr.value, zero, 4, &UA_TYPES[UA_TYPES_DOUBLE]);
    attr.value.arrayDimensions = arrayDims;
    attr.value.arrayDimensionsSize = 2;

    UA_NodeId myIntegerNodeId = UA_NODEID_STRING(1, "double.matrix");
    UA_QualifiedName myIntegerName = UA_QUALIFIEDNAME(1, "double matrix");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_Server_addVariableNode(server, myIntegerNodeId, parentNodeId,
                              parentReferenceNodeId, myIntegerName,
                              UA_NS0ID(BASEDATAVARIABLETYPE), attr, NULL, NULL);
}
```

Now we change the value with the write service. This uses the same service implementation that can also be reached over the network by an OPC UA client.

```
static void
writeVariable(UA_Server *server) {
    UA_NodeId myIntegerNodeId = UA_NODEID_STRING(1, "the.answer");

    /* Write a different integer value */
    UA_Int32 myInteger = 43;
    UA_Variant myVar;
    UA_Variant_init(&myVar);
    UA_Variant_setScalar(&myVar, &myInteger, &UA_TYPES[UA_TYPES_INT32]);
    UA_Server_writeValue(server, myIntegerNodeId, myVar);

    /* Set the status code of the value to an error code. The function
     * UA_Server_write provides access to the raw service. The above
     * UA_Server_writeValue is syntactic sugar for writing a specific node
     * attribute with the write service. */
    UA_WriteValue wv;
    UA_WriteValue_init(&wv);
    wv.nodeId = myIntegerNodeId;
    wv.attributeId = UA_ATTRIBUTEID_VALUE;
    wv.value.status = UA_STATUSCODE_BADNOTCONNECTED;
    wv.value.hasStatus = true;
    UA_Server_write(server, &wv);

    /* Reset the variable to a good statuscode with a value */
    wv.value.hasStatus = false;
    wv.value.value = myVar;
    wv.value.hasValue = true;
    UA_Server_write(server, &wv);
}
```

Note how we initially set the DataType attribute of the variable node to the NodeId of the Int32 data type. This forbids writing values that are not an Int32. The following code shows how this consistency check is performed for every write.

```
static void
writeWrongVariable(UA_Server *server) {
    UA_NodeId myIntegerNodeId = UA_NODEID_STRING(1, "the.answer");

    /* Write a string */
    UA_String myString = UA_STRING("test");
    UA_Variant myVar;
    UA_Variant_init(&myVar);
    UA_Variant_setScalar(&myVar, &myString, &UA_TYPES[UA_TYPES_STRING]);
    UA_StatusCode retval = UA_Server_writeValue(server, myIntegerNodeId, myVar);
    printf("Writing a string returned statuscode %s\n", UA_StatusCode_name(retval));
}
```

It follows the main server code, making use of the above definitions.

```
int main(void) {
    UA_Server *server = UA_Server_new();

    addVariable(server);
    addMatrixVariable(server);
    writeVariable(server);
    writeWrongVariable(server);

    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

8.4 Connecting a Variable with a Physical Process

In OPC UA-based architectures, servers are typically situated near the source of information. In an industrial context, this translates into servers being near the physical process and clients consuming the data at runtime. In the previous tutorial, we saw how to add variables to an OPC UA information model. This tutorial shows how to connect a variable to runtime information, for example from measurements of a physical process. For simplicity, we take the system clock as the underlying "process".

The following code snippets are each concerned with a different way of updating variable values at runtime. Taken together, the code snippets define a compilable source file.

8.4.1 Updating variables manually

As a starting point, assume that a variable for a value of type *DateTime* has been created in the server with the identifier "ns=1,s=current-time-value-callback". Assuming that our application gets triggered when a new value arrives from the underlying process, we can just write into the variable.

```
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>

static void
updateCurrentTime(UA_Server *server) {
```

(continues on next page)

(continued from previous page)

```
    UA_DateTime now = UA_DateTime_now();
    UA_Variant value;
    UA_Variant_setScalar(&value, &now, &UA_TYPES[UA_TYPES_DATETIME]);
    UA_NodeId currentNodeId = UA_NODEID_STRING(1, "current-time-value-callback");
    UA_Server_writeValue(server, currentNodeId, value);
}

static void
addCurrentTimeVariable(UA_Server *server) {
    UA_DateTime now = 0;
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en-US", "Current time - value callback");
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;
    UA_Variant_setScalar(&attr.value, &now, &UA_TYPES[UA_TYPES_DATETIME]);

    UA_NodeId currentNodeId = UA_NODEID_STRING(1, "current-time-value-callback");
    UA_QualifiedName currentName = UA_QUALIFIEDNAME(1, "current-time-value-
↪callback");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_NodeId variableTypeNodeId = UA_NS0ID(BASEDATAVARIABLETYPE);
    UA_Server_addVariableNode(server, currentNodeId, parentNodeId,
                              parentReferenceNodeId, currentName,
                              variableTypeNodeId, attr, NULL, NULL);

    updateCurrentTime(server);
}
```

8.4.2 Variable Value Callback

When a value changes continuously, such as the system time, updating the value in a tight loop would take up a lot of resources. Value callbacks allow to synchronize a variable value with an external representation. They attach callbacks to the variable that are executed before every read and after every write operation.

```
static void
beforeReadTime(UA_Server *server,
               const UA_NodeId *sessionId, void *sessionContext,
               const UA_NodeId *nodeid, void *nodeContext,
               const UA_NumericRange *range, const UA_DataValue *data) {
    updateCurrentTime(server);
}

static void
afterWriteTime(UA_Server *server,
               const UA_NodeId *sessionId, void *sessionContext,
               const UA_NodeId *nodeId, void *nodeContext,
               const UA_NumericRange *range, const UA_DataValue *data) {
```

(continues on next page)

(continued from previous page)

```
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                "The variable was updated");
}

static void
addValueCallbackToCurrentTimeVariable(UA_Server *server) {
    UA_NodeId currentNodeId = UA_NODEID_STRING(1, "current-time-value-callback");
    UA_ValueCallback callback ;
    callback.onRead = beforeReadTime;
    callback.onWrite = afterWriteTime;
    UA_Server_setVariableNode_valueCallback(server, currentNodeId, callback);
}
```

8.4.3 Variable Data Sources

With value callbacks, the value is still stored in the variable node. So-called data sources go one step further. The server redirects every read and write request to a callback function. Upon reading, the callback provides a copy of the current value. Internally, the data source needs to implement its own memory management.

```
static UA_StatusCode
readCurrentTime(UA_Server *server,
                const UA_NodeId *sessionId, void *sessionContext,
                const UA_NodeId *nodeId, void *nodeContext,
                UA_Boolean sourceTimeStamp, const UA_NumericRange *range,
                UA_DataValue *dataValue) {
    UA_DateTime now = UA_DateTime_now();
    UA_Variant_setScalarCopy(&dataValue->value, &now,
                             &UA_TYPES[UA_TYPES_DATETIME]);
    dataValue->hasValue = true;
    return UA_STATUSCODE_GOOD;
}

static UA_StatusCode
writeCurrentTime(UA_Server *server,
                 const UA_NodeId *sessionId, void *sessionContext,
                 const UA_NodeId *nodeId, void *nodeContext,
                 const UA_NumericRange *range, const UA_DataValue *data) {
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                "Changing the system time is not implemented");
    return UA_STATUSCODE_BADINTERNALERROR;
}

static void
addCurrentTimeDataSourceVariable(UA_Server *server) {
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en-US", "Current time - data source");
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;
```

(continues on next page)

(continued from previous page)

```
    UA_NodeId currentNodeId = UA_NODEID_STRING(1, "current-time-datasource");
    UA_QualifiedName currentName = UA_QUALIFIEDNAME(1, "current-time-datasource
↪");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_NodeId variableTypeNodeId = UA_NS0ID(BASEDATAVARIABLETYPE);

    UA_DataSource timeDataSource;
    timeDataSource.read = readCurrentTime;
    timeDataSource.write = writeCurrentTime;
    UA_Server_addDataSourceVariableNode(server, currentNodeId, parentNodeId,
                                        parentReferenceNodeId, currentName,
                                        variableTypeNodeId, attr,
                                        timeDataSource, NULL, NULL);
}
```

It follows the main server code, making use of the above definitions.

```
int main(void) {
    UA_Server *server = UA_Server_new();

    addCurrentTimeVariable(server);
    addValueCallbackToCurrentTimeVariable(server);
    addCurrentTimeDataSourceVariable(server);

    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

8.5 Working with Variable Types

Variable types have three functions:

- Constrain the possible data type, value rank and array dimensions of the variables of that type. This allows interface code to be written against the generic type definition, so it is applicable for all instances.
- Provide a sensible default value
- Enable a semantic interpretation of the variable based on its type

In the example of this tutorial, we represent a point in 2D space by an array of double values. The following function adds the corresponding VariableTypeNode to the hierarchy of variable types.

```
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>

static UA_NodeId pointTypeId;
```

(continues on next page)

(continued from previous page)

```
static void
addVariableType2DPoint(UA_Server *server) {
    UA_VariableTypeAttributes vtAttr = UA_VariableTypeAttributes_default;
    vtAttr.dataType = UA_TYPES[UA_TYPES_DOUBLE].typeId;
    vtAttr.valueRank = UA_VALUERANK_ONE_DIMENSION;
    UA_UInt32 arrayDims[1] = {2};
    vtAttr.arrayDimensions = arrayDims;
    vtAttr.arrayDimensionsSize = 1;
    vtAttr.displayName = UA_LOCALIZEDTEXT("en-US", "2DPoint Type");

    /* a matching default value is required */
    UA_Double zero[2] = {0.0, 0.0};
    UA_Variant_setArray(&vtAttr.value, zero, 2, &UA_TYPES[UA_TYPES_DOUBLE]);

    UA_Server_addVariableTypeNode(server, UA_NODEID_NULL,
                                  UA_NS0ID(BASEDATAVARIABLETYPE), UA_
↪NS0ID(HASSUBTYPE),
                                  UA_QUALIFIEDNAME(1, "2DPoint Type"), UA_NODEID_
↪NULL,
                                  vtAttr, NULL, &pointTypeId);
}
```

Now the new variable type for *2DPoint* can be referenced during the creation of a new variable. If no value is given, the default from the variable type is copied during instantiation.

```
static UA_NodeId pointVariableId;

static void
addVariable(UA_Server *server) {
    /* Prepare the node attributes */
    UA_VariableAttributes vAttr = UA_VariableAttributes_default;
    vAttr.dataType = UA_TYPES[UA_TYPES_DOUBLE].typeId;
    vAttr.valueRank = UA_VALUERANK_ONE_DIMENSION;
    UA_UInt32 arrayDims[1] = {2};
    vAttr.arrayDimensions = arrayDims;
    vAttr.arrayDimensionsSize = 1;
    vAttr.displayName = UA_LOCALIZEDTEXT("en-US", "2DPoint Variable");
    vAttr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;
    /* vAttr.value is left empty, the server instantiates with the default value */

    /* Add the node */
    UA_Server_addVariableNode(server, UA_NODEID_NULL,
                              UA_NS0ID(OBJECTSFOLDER), UA_NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "2DPoint Type"), pointTypeId,
                              vAttr, NULL, &pointVariableId);
}
```

The constraints of the variable type are enforced when creating new variable instances of the type. In the following function, adding a variable of *2DPoint* type with a string value fails because the value

does not match the variable type constraints.

```
static void
addVariableFail(UA_Server *server) {
    /* Prepare the node attributes */
    UA_VariableAttributes vAttr = UA_VariableAttributes_default;
    vAttr.dataType = UA_TYPES[UA_TYPES_DOUBLE].typeId;
    vAttr.valueRank = UA_VALUERANK_SCALAR; /* a scalar. this is not allowed per
                                            * the variable type */
    vAttr.displayName = UA_LOCALIZEDTEXT("en-US", "2DPoint Variable (fail)");
    UA_String s = UA_STRING("2dpoint?");
    UA_Variant_setScalar(&vAttr.value, &s, &UA_TYPES[UA_TYPES_STRING]);

    /* Add the node will return UA_STATUSCODE_BADTYPEMISMATCH*/
    UA_Server_addVariableNode(server, UA_NODEID_NULL,
                              UA_NS0ID(OBJECTSFOLDER), UA_NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "2DPoint Type (fail)"), pointTypeId,
                              vAttr, NULL, NULL);
}
```

The constraints of the variable type are enforced when writing the datatype, valuerank and arraydimensions attributes of the variable. This, in turn, constrains the value attribute of the variable.

```
static void
writeVariable(UA_Server *server) {
    UA_StatusCode retval =
        UA_Server_writeValueRank(server, pointVariableId,
                                 UA_VALUERANK_ONE_OR_MORE_DIMENSIONS);
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                "Setting the Value Rank failed with Status Code %s",
                UA_StatusCode_name(retval));
}
```

It follows the main server code, making use of the above definitions.

```
int main(void) {
    UA_Server *server = UA_Server_new();
    addVariableType2DPoint(server);
    addVariable(server);
    addVariableFail(server);
    writeVariable(server);
    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

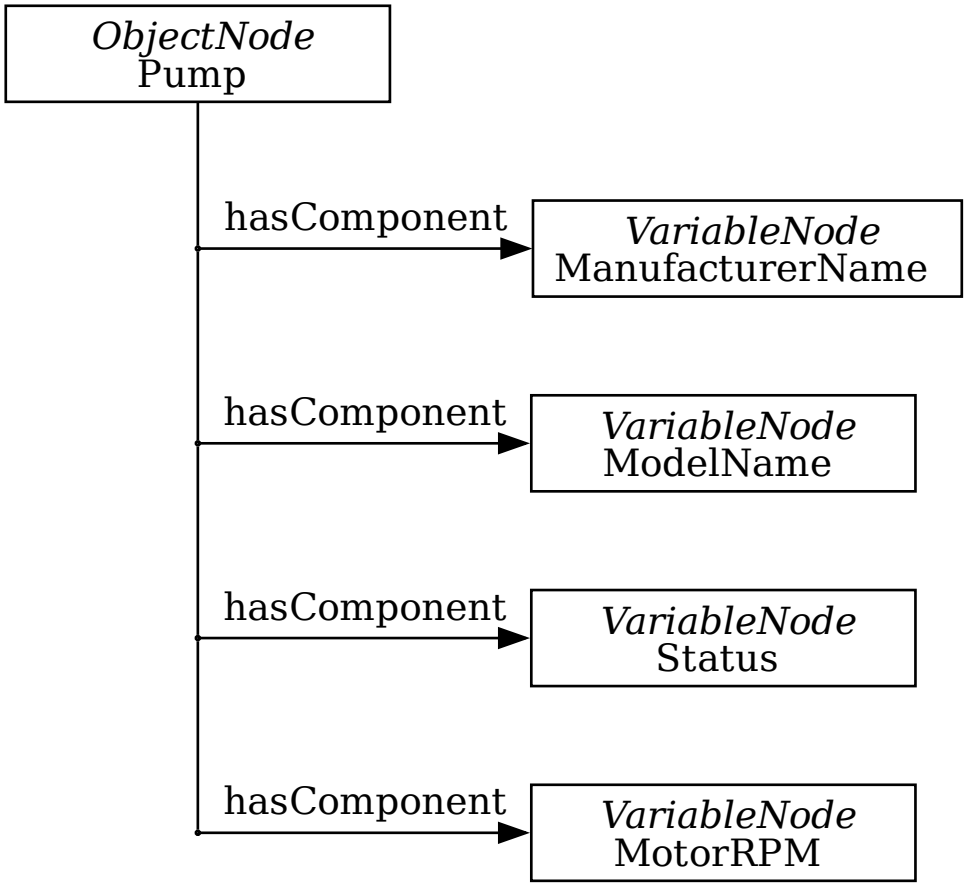
8.6 Working with Objects and Object Types

8.6.1 Using objects to structure information models

Assume a situation where we want to model a set of pumps and their runtime state in an OPC UA information model. Of course, all pump representations should follow the same basic structure, For

example, we might have graphical representation of pumps in a SCADA visualisation that shall be reusable for all pumps.

Following the object-oriented programming paradigm, every pump is represented by an object with the following layout:



The following code manually defines a pump and its member variables. We omit setting constraints on the variable values as this is not the focus of this tutorial and was already covered.

```
static void
manuallyDefinePump(UA_Server *server) {
    UA_NodeId pumpId; /* get the nodeid assigned by the server */
    UA_ObjectAttributes oAttr = UA_ObjectAttributes_default;
    oAttr.displayName = UA_LOCALIZEDTEXT("en-US", "Pump (Manual)");
    UA_Server_addObjectNode(server, UA_NODEID_NULL, UA_
↪NS0ID(OBJECTSFOLDER),
                            UA_NS0ID(ORGANIZES), UA_QUALIFIEDNAME(1, "Pump␣
↪(Manual)"),
                            UA_NS0ID(BASEOBJECTTYPE),
                            oAttr, NULL, &pumpId);
```

(continues on next page)

(continued from previous page)

```
    UA_VariableAttributes mnAttr = UA_VariableAttributes_default;
    UA_String manufacturerName = UA_STRING("Pump King Ltd.");
    UA_Variant_setScalar(&mnAttr.value, &manufacturerName, &UA_TYPES[UA_
↪TYPES_STRING]);
    mnAttr.displayName = UA_LOCALIZEDTEXT("en-US", "ManufacturerName");
    UA_Server_addVariableNode(server, UA_NODEID_NULL, pumpId, UA_
↪NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "ManufacturerName"),
                              UA_NS0ID(BASEDATAVARIABLETYPE), mnAttr, NULL, NULL);

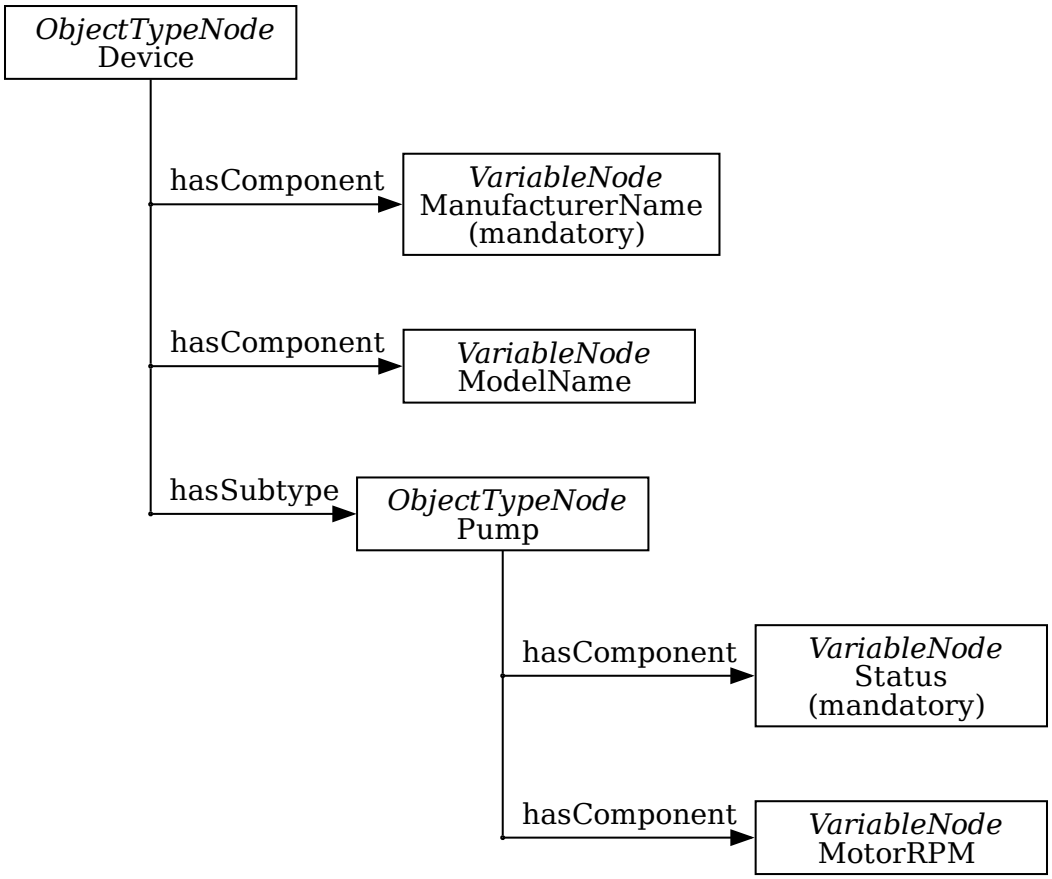
    UA_VariableAttributes modelAttr = UA_VariableAttributes_default;
    UA_String modelName = UA_STRING("Mega Pump 3000");
    UA_Variant_setScalar(&modelAttr.value, &modelName, &UA_TYPES[UA_TYPES_
↪STRING]);
    modelAttr.displayName = UA_LOCALIZEDTEXT("en-US", "ModelName");
    UA_Server_addVariableNode(server, UA_NODEID_NULL, pumpId, UA_
↪NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "ModelName"),
                              UA_NS0ID(BASEDATAVARIABLETYPE), modelAttr, NULL,␣
↪NULL);

    UA_VariableAttributes statusAttr = UA_VariableAttributes_default;
    UA_Boolean status = true;
    UA_Variant_setScalar(&statusAttr.value, &status, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);
    statusAttr.displayName = UA_LOCALIZEDTEXT("en-US", "Status");
    UA_Server_addVariableNode(server, UA_NODEID_NULL, pumpId, UA_
↪NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "Status"), UA_
↪NS0ID(BASEDATAVARIABLETYPE),
                              statusAttr, NULL, NULL);

    UA_VariableAttributes rpmAttr = UA_VariableAttributes_default;
    UA_Double rpm = 50.0;
    UA_Variant_setScalar(&rpmAttr.value, &rpm, &UA_TYPES[UA_TYPES_DOUBLE]);
    rpmAttr.displayName = UA_LOCALIZEDTEXT("en-US", "MotorRPM");
    UA_Server_addVariableNode(server, UA_NODEID_NULL, pumpId, UA_
↪NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "MotorRPMs"),
                              UA_NS0ID(BASEDATAVARIABLETYPE),
                              rpmAttr, NULL, NULL);
}
```

8.6.2 Object types, type hierarchies and instantiation

Building up each object manually requires us to write a lot of code. Furthermore, there is no way for clients to detect that an object represents a pump. (We might use naming conventions or similar to detect pumps. But that's not exactly a clean solution.) Furthermore, we might have more devices than just pumps. And we require all devices to share some common structure. The solution is to define ObjectTypes in a hierarchy with inheritance relations.



Children that are marked mandatory are automatically instantiated together with the parent object. This is indicated by a *hasModellingRule* reference to an object that represents the *mandatory* modelling rule.

```
/* predefined identifier for later use */
static UA_NodeId pumpTypeId = {1, UA_NODEIDTYPE_NUMERIC, {1001}};

static void
defineObjectTypes(UA_Server *server) {
    /* Define the object type for "Device" */
    UA_NodeId deviceTypeId; /* get the nodeid assigned by the server */
    UA_ObjectTypeAttributes dtAttr = UA_ObjectTypeAttributes_default;
    dtAttr.displayName = UA_LOCALIZEDTEXT("en-US", "DeviceType");
    UA_Server_addObjectTypeNode(server, UA_NODEID_NULL,
                                UA_NS0ID(BASEOBJECTTYPE), UA_NS0ID(HASSUBTYPE),
                                UA_QUALIFIEDNAME(1, "DeviceType"), dtAttr,
                                NULL, &deviceTypeId);

    UA_VariableAttributes mnAttr = UA_VariableAttributes_default;
    mnAttr.displayName = UA_LOCALIZEDTEXT("en-US", "ManufacturerName");
    UA_NodeId manufacturerNameId;
```

(continues on next page)

(continued from previous page)

```
    UA_Server_addVariableNode(server, UA_NODEID_NULL, deviceTypeId,
                              UA_NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "ManufacturerName"),
                              UA_NS0ID(BASEDATAVARIABLETYPE),
                              mnAttr, NULL, &manufacturerNameId);
    /* Make the manufacturer name mandatory */
    UA_Server_addReference(server, manufacturerNameId, UA_
↪NS0ID(HASMODELLINGRULE),
                           UA_NS0EXID(MODELLINGRULE_MANDATORY), true);


    UA_VariableAttributes modelAttr = UA_VariableAttributes_default;
    modelAttr.displayName = UA_LOCALIZEDTEXT("en-US", "ModelName");
    UA_Server_addVariableNode(server, UA_NODEID_NULL, deviceTypeId,
                              UA_NS0ID(HASCOMPONENT),
                              UA_QUALIFIEDNAME(1, "ModelName"),
                              UA_NS0ID(BASEDATAVARIABLETYPE),
                              modelAttr, NULL, NULL);

    /* Define the object type for "Pump" */
    UA_ObjectTypeAttributes ptAttr = UA_ObjectTypeAttributes_default;
    ptAttr.displayName = UA_LOCALIZEDTEXT("en-US", "PumpType");
    UA_Server_addObjectTypeNode(server, pumpTypeId, deviceTypeId, UA_
↪NS0ID(HASSUBTYPE),
                                UA_QUALIFIEDNAME(1, "PumpType"), ptAttr,
                                NULL, NULL);

    UA_VariableAttributes statusAttr = UA_VariableAttributes_default;
    statusAttr.displayName = UA_LOCALIZEDTEXT("en-US", "Status");
    statusAttr.valueRank = UA_VALUERANK_SCALAR;
    UA_NodeId statusId;
    UA_Server_addVariableNode(server, UA_NODEID_NULL, pumpTypeId,
                              UA_NS0ID(HASCOMPONENT), UA_QUALIFIEDNAME(1, "Status
↪"),
                              UA_NS0ID(BASEDATAVARIABLETYPE), statusAttr, NULL, &
↪statusId);
    /* Make the status variable mandatory */
    UA_Server_addReference(server, statusId, UA_NS0ID(HASMODELLINGRULE),
                           UA_NS0EXID(MODELLINGRULE_MANDATORY), true);

    UA_VariableAttributes rpmAttr = UA_VariableAttributes_default;
    rpmAttr.displayName = UA_LOCALIZEDTEXT("en-US", "MotorRPM");
    rpmAttr.valueRank = UA_VALUERANK_SCALAR;
    UA_Server_addVariableNode(server, UA_NODEID_NULL, pumpTypeId,
                              UA_NS0ID(HASCOMPONENT), UA_QUALIFIEDNAME(1,
↪"MotorRPMs"),
                              UA_NS0ID(BASEDATAVARIABLETYPE), rpmAttr, NULL, NULL);
}
```

Now we add the derived ObjectType for the pump that inherits from the device object type. The

resulting object contains all mandatory child variables. These are simply copied over from the object type. The object has a reference of type hasTypeDefinition to the object type, so that clients can detect the type-instance relation at runtime.

```
static void
addPumpObjectInstance(UA_Server *server, char *name) {
    UA_ObjectAttributes oAttr = UA_ObjectAttributes_default;
    oAttr.displayName = UA_LOCALIZEDTEXT("en-US", name);
    UA_Server_addObjectNode(server, UA_NODEID_NULL,
                            UA_NS0ID(OBJECTSFOLDER), UA_NS0ID(ORGANIZES),
                            UA_QUALIFIEDNAME(1, name),
                            pumpTypeId, /* this refers to the object type
                                           identifier */
                            oAttr, NULL, NULL);
}
```

Often we want to run a constructor function on a new object. This is especially useful when an object is instantiated at runtime (with the AddNodes service) and the integration with an underlying process cannot be manually defined. In the following constructor example, we simply set the pump status to on.

```
static UA_StatusCode
pumpTypeConstructor(UA_Server *server,
                    const UA_NodeId *sessionId, void *sessionContext,
                    const UA_NodeId *typeId, void *typeContext,
                    const UA_NodeId *nodeId, void **nodeContext) {
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION, "New␣
↪pump created");

    /* Find the NodeId of the status child variable */
    UA_RelativePathElement rpe;
    UA_RelativePathElement_init(&rpe);
    rpe.referenceTypeId = UA_NS0ID(HASCOMPONENT);
    rpe.isInverse = false;
    rpe.includeSubtypes = false;
    rpe.targetName = UA_QUALIFIEDNAME(1, "Status");

    UA_BrowsePath bp;
    UA_BrowsePath_init(&bp);
    bp.startingNode = *nodeId;
    bp.relativePath.elementsSize = 1;
    bp.relativePath.elements = &rpe;

    UA_BrowsePathResult bpr =
        UA_Server_translateBrowsePathToNodeIds(server, &bp);
    if(bpr.statusCode != UA_STATUSCODE_GOOD ||
       bpr.targetsSize < 1)
        return bpr.statusCode;

    /* Set the status value */
    UA_Boolean status = true;
```

(continues on next page)

(continued from previous page)

```
    UA_Variant value;
    UA_Variant_setScalar(&value, &status, &UA_TYPES[UA_TYPES_BOOLEAN]);
    UA_Server_writeValue(server, bpr.targets[0].targetId.nodeId, value);
    UA_BrowsePathResult_clear(&bpr);

    /* At this point we could replace the node context .. */

    return UA_STATUSCODE_GOOD;
}

static void
addPumpTypeConstructor(UA_Server *server) {
    UA_NodeTypeLifecycle lifecycle = {0};
    lifecycle.constructor = pumpTypeConstructor;
    lifecycle.destructor = NULL;
    UA_Server_setNodeTypeLifecycle(server, pumpTypeId, lifecycle);
}
```

It follows the main server code, making use of the above definitions.

```
int main(void) {
    UA_Server *server = UA_Server_new();

    manuallyDefinePump(server);
    defineObjectTypes(server);
    addPumpObjectInstance(server, "pump2");
    addPumpObjectInstance(server, "pump3");
    addPumpTypeConstructor(server);
    addPumpObjectInstance(server, "pump4");
    addPumpObjectInstance(server, "pump5");

    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

8.7 Adding Methods to Objects

An object in an OPC UA information model may contain methods similar to objects in a programming language. Methods are represented by a MethodNode. Note that several objects may reference the same MethodNode. When an object type is instantiated, a reference to the method is added instead of copying the MethodNode. Therefore, the identifier of the context object is always explicitly stated when a method is called.

The method callback takes as input a custom data pointer attached to the method node, the identifier of the object from which the method is called, and two arrays for the input and output arguments. The input and output arguments are all of type *Variant*. Each variant may in turn contain a (multi-dimensional) array or scalar of any data type.

Constraints for the method arguments are defined in terms of data type, value rank and array dimen-

sion (similar to variable definitions). The argument definitions are stored in child VariableNodes of the MethodNode with the respective BrowseNames $(0, \text{"InputArguments"})$ and $(0, \text{"OutputArguments"})$.

8.7.1 Example: Hello World Method

The method takes a string scalar and returns a string scalar with "Hello " prepended. The type and length of the input arguments is checked internally by the SDK, so that we don't have to verify the arguments in the callback.

```
#include <open62541/client_config_default.h>
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>

static UA_StatusCode
helloWorldMethodCallback(UA_Server *server,
                         const UA_NodeId *sessionId, void *sessionHandle,
                         const UA_NodeId *methodId, void *methodContext,
                         const UA_NodeId *objectId, void *objectContext,
                         size_t inputSize, const UA_Variant *input,
                         size_t outputSize, UA_Variant *output) {
    UA_String *inputStr = (UA_String*)input->data;
    UA_String tmp = UA_STRING_ALLOC("Hello ");
    if(inputStr->length > 0) {
        tmp.data = (UA_Byte *)UA_realloc(tmp.data, tmp.length + inputStr->length);
        memcpy(&tmp.data[tmp.length], inputStr->data, inputStr->length);
        tmp.length += inputStr->length;
    }
    UA_Variant_setScalarCopy(output, &tmp, &UA_TYPES[UA_TYPES_STRING]);
    UA_String_clear(&tmp);
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_SERVER, "Hello World was␣
↪called");
    return UA_STATUSCODE_GOOD;
}

static void
addHelloWorldMethod(UA_Server *server) {
    UA_Argument inputArgument;
    UA_Argument_init(&inputArgument);
    inputArgument.description = UA_LOCALIZEDTEXT("en-US", "A String");
    inputArgument.name = UA_STRING("MyInput");
    inputArgument.dataType = UA_TYPES[UA_TYPES_STRING].typeId;
    inputArgument.valueRank = UA_VALUERANK_SCALAR;

    UA_Argument outputArgument;
    UA_Argument_init(&outputArgument);
    outputArgument.description = UA_LOCALIZEDTEXT("en-US", "A String");
    outputArgument.name = UA_STRING("MyOutput");
    outputArgument.dataType = UA_TYPES[UA_TYPES_STRING].typeId;
    outputArgument.valueRank = UA_VALUERANK_SCALAR;
```

(continues on next page)

(continued from previous page)

```
    UA_MethodAttributes helloAttr = UA_MethodAttributes_default;
    helloAttr.description = UA_LOCALIZEDTEXT("en-US","Say `Hello World`");
    helloAttr.displayName = UA_LOCALIZEDTEXT("en-US","Hello World");
    helloAttr.executable = true;
    helloAttr.userExecutable = true;
    UA_Server_addMethodNode(server, UA_NODEID_NUMERIC(1,62541),
                            UA_NS0ID(OBJECTSFOLDER),
                            UA_NS0ID(HASCOMPONENT),
                            UA_QUALIFIEDNAME(1, "hello world"),
                            helloAttr, &helloWorldMethodCallback,
                            1, &inputArgument, 1, &outputArgument, NULL, NULL);
}
```

8.7.2 Increase Array Values Method

The method takes an array of 5 integers and a scalar as input. It returns a copy of the array with every entry increased by the scalar.

```
static UA_StatusCode
IncInt32ArrayMethodCallback(UA_Server *server,
                            const UA_NodeId *sessionId, void *sessionContext,
                            const UA_NodeId *methodId, void *methodContext,
                            const UA_NodeId *objectId, void *objectContext,
                            size_t inputSize, const UA_Variant *input,
                            size_t outputSize, UA_Variant *output) {
    UA_Int32 *inputArray = (UA_Int32*)input[0].data;
    UA_Int32 delta = *(UA_Int32*)input[1].data;

    /* Copy the input array */
    UA_StatusCode retval = UA_Variant_setArrayCopy(output, inputArray, 5,
                                                   &UA_TYPES[UA_TYPES_INT32]);
    if(retval != UA_STATUSCODE_GOOD)
        return retval;

    /* Increase the elements */
    UA_Int32 *outputArray = (UA_Int32*)output->data;
    for(size_t i = 0; i < input->arrayLength; i++)
        outputArray[i] = inputArray[i] + delta;

    return UA_STATUSCODE_GOOD;
}

static void
addIncInt32ArrayMethod(UA_Server *server) {
    /* Two input arguments */
    UA_Argument inputArguments[2];
    UA_Argument_init(&inputArguments[0]);
    inputArguments[0].description = UA_LOCALIZEDTEXT("en-US", "int32[5] array");
    inputArguments[0].name = UA_STRING("int32 array");
```

(continues on next page)

(continued from previous page)

```
    inputArguments[0].dataType = UA_TYPES[UA_TYPES_INT32].typeId;
    inputArguments[0].valueRank = UA_VALUERANK_ONE_DIMENSION;
    UA_UInt32 pInputDimension = 5;
    inputArguments[0].arrayDimensionsSize = 1;
    inputArguments[0].arrayDimensions = &pInputDimension;

    UA_Argument_init(&inputArguments[1]);
    inputArguments[1].description = UA_LOCALIZEDTEXT("en-US", "int32 delta");
    inputArguments[1].name = UA_STRING("int32 delta");
    inputArguments[1].dataType = UA_TYPES[UA_TYPES_INT32].typeId;
    inputArguments[1].valueRank = UA_VALUERANK_SCALAR;

    /* One output argument */
    UA_Argument outputArgument;
    UA_Argument_init(&outputArgument);
    outputArgument.description = UA_LOCALIZEDTEXT("en-US", "int32[5] array");
    outputArgument.name = UA_STRING("each entry is incremented by the delta");
    outputArgument.dataType = UA_TYPES[UA_TYPES_INT32].typeId;
    outputArgument.valueRank = UA_VALUERANK_ONE_DIMENSION;
    UA_UInt32 pOutputDimension = 5;
    outputArgument.arrayDimensionsSize = 1;
    outputArgument.arrayDimensions = &pOutputDimension;

    /* Add the method node */
    UA_MethodAttributes incAttr = UA_MethodAttributes_default;
    incAttr.description = UA_LOCALIZEDTEXT("en-US", "IncInt32ArrayValues");
    incAttr.displayName = UA_LOCALIZEDTEXT("en-US", "IncInt32ArrayValues");
    incAttr.executable = true;
    incAttr.userExecutable = true;
    UA_Server_addMethodNode(server, UA_NODEID_STRING(1, "IncInt32ArrayValues"),
                            UA_NS0ID(OBJECTSFOLDER),
                            UA_NS0ID(HASCOMPONENT),
                            UA_QUALIFIEDNAME(1, "IncInt32ArrayValues"),
                            incAttr, &IncInt32ArrayMethodCallback,
                            2, inputArguments, 1, &outputArgument,
                            NULL, NULL);
}
```

It follows the main server code, making use of the above definitions.

```
int main(void) {
    UA_Server *server = UA_Server_new();

    addHelloWorldMethod(server);
    addIncInt32ArrayMethod(server);

    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

8.8 Observing Attributes with Local MonitoredItems

A client that is interested in the current value of a variable does not need to regularly poll the variable. Instead, the client can use the Subscription mechanism to be notified about changes.

So-called MonitoredItems define which values (node attributes) and events the client wants to monitor. Under the right conditions, a notification is created and added to the Subscription. The notifications currently in the queue are regularly sent to the client.

The local user can add MonitoredItems as well. Locally, the MonitoredItems do not go via a Subscription and each have an individual callback method and a context pointer.

```
#include <open62541/client_subscriptions.h>
#include <open62541/server.h>
#include <open62541/plugin/log_stdout.h>

static void
dataChangeNotificationCallback(UA_Server *server, UA_UInt32 monitoredItemId,
                               void *monitoredItemContext, const UA_NodeId *nodeId,
                               void *nodeContext, UA_UInt32 attributeId,
                               const UA_DataValue *value) {
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION, "Received␣
↪Notification");
}

static void
addMonitoredItemToCurrentTimeVariable(UA_Server *server) {
    UA_NodeId currentTimeNodeId = UA_NS0ID(SERVER_SERVERSTATUS_
↪CURRENTTIME);
    UA_MonitoredItemCreateRequest monRequest =
        UA_MonitoredItemCreateRequest_default(currentTimeNodeId);
    monRequest.requestedParameters.samplingInterval = 100.0; /* 100 ms interval */
    UA_Server_createDataChangeMonitoredItem(server, UA_TIMESTAMPSTORETURN_
↪SOURCE,
                                            monRequest, NULL,
                                            dataChangeNotificationCallback);
}
```

It follows the main server code, making use of the above definitions.

```
int main(void) {
    UA_Server *server = UA_Server_new();

    addMonitoredItemToCurrentTimeVariable(server);

    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

8.9 Generating events

To make sense of the many things going on in a server, monitoring items can be useful. Though in many cases, data change does not convey enough information to be the optimal solution. Events can be generated at any time, hold a lot of information and can be filtered so the client only receives the specific attributes of interest.

8.9.1 Emitting events by calling methods

The following example will be based on the server method tutorial. We will be creating a method node which generates an event from the server node.

The event we want to generate should be very simple. Since the *BaseEventType* is abstract, we will have to create our own event type. *EventTypes* are saved internally as *ObjectTypes*, so add the type as you would a new *ObjectType*.

```
static UA_NodeId eventType;

static UA_StatusCode
addNewEventType(UA_Server *server) {
    UA_ObjectTypeAttributes attr = UA_ObjectTypeAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en-US", "SimpleEventType");
    attr.description = UA_LOCALIZEDTEXT("en-US", "The simple event type we created
↪");
    return UA_Server_addObjectTypeNode(server, UA_NODEID_NULL,
                                       UA_NS0ID(BASEEVENTTYPE), UA_
↪NS0ID(HASSUBTYPE),
                                       UA_QUALIFIEDNAME(0, "SimpleEventType"),
                                       attr, NULL, &eventType);
}
```

8.9.2 Creating an event

In order to set up the event, we can first use UA_Server_createEvent to give us a node representation of the event. All we need for this is our *EventType*. Once we have our event node, which is saved internally as an *ObjectNode*, we can define the attributes the event has the same way we would define the attributes of an object node. It is not necessary to define the attributes *EventId*, *ReceiveTime*, *SourceNode* or *EventType* since these are set automatically by the server. In this example, we will be setting the fields 'Message' and 'Severity' in addition to *Time* which is needed to make the example UaExpert compliant.

```
static UA_StatusCode
createEventMethodCallback(UA_Server *server,
                          const UA_NodeId *sessionId, void *sessionHandle,
                          const UA_NodeId *methodId, void *methodContext,
                          const UA_NodeId *objectId, void *objectContext,
                          size_t inputSize, const UA_Variant *input,
                          size_t outputSize, UA_Variant *output) {
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION, "Creating␣
↪event");

    UA_UInt16 severity = 100;
```

(continues on next page)

(continued from previous page)

```
    UA_LocalizedText message = UA_LOCALIZEDTEXT("en-US", "An event has been␣
↪generated.");
    UA_StatusCode res = UA_Server_createEvent(server, UA_NS0ID(SERVER),␣
↪eventType,
                                              severity, message, NULL, NULL, NULL);
    if(res != UA_STATUSCODE_GOOD) {
        UA_LOG_WARNING(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                       "Creating event failed. StatusCode %s",
                       UA_StatusCode_name(res));
    }
    return res;
}
```

Now, all that is left to do is to create a method node which triggers the event creation for the example.

```
int main(void) {
    UA_Server *server = UA_Server_new();

    addNewEventType(server);

    UA_MethodAttributes generateAttr = UA_MethodAttributes_default;
    generateAttr.description = UA_LOCALIZEDTEXT("en-US","Generate an event.");
    generateAttr.displayName = UA_LOCALIZEDTEXT("en-US","Generate Event");
    generateAttr.executable = true;
    generateAttr.userExecutable = true;
    UA_Server_addMethodNode(server, UA_NODEID_NUMERIC(1, 62541),
                            UA_NS0ID(OBJECTSFOLDER), UA_NS0ID(HASCOMPONENT),
                            UA_QUALIFIEDNAME(1, "Generate Event"),
                            generateAttr, &createEventMethodCallback,
                            0, NULL, 0, NULL, NULL, NULL);

    UA_Server_runUntilInterrupt(server);
    UA_Server_delete(server);
    return 0;
}
```

8.10 Using Alarms and Conditions Server

Besides the usage of monitored items and events to observe the changes in the server, it is also important to make use of the Alarms and Conditions Server Model. Alarms are events which are triggered automatically by the server dependent on internal server logic or user specific logic when the states of server components change. The state of a component is represented through a condition. So the values of all the condition children (Fields) are the actual state of the component.

8.10.1 Trigger Alarm events by changing States

The following example will be based on the server events tutorial. Please make sure to understand the principle of normal events before proceeding with this example!

```
static UA_NodeId conditionSource;
static UA_NodeId conditionInstance_1;
static UA_NodeId conditionInstance_2;
static UA_AlarmConditionsDriver *acDriver;
static UA_Boolean running = true;

static void
stopHandler(int sign) {
    (void)sign;
    running = false;
}

static UA_StatusCode
addConditionSourceObject(UA_Server *server) {
    UA_ObjectAttributes object_attr = UA_ObjectAttributes_default;
    object_attr.eventNotifier = 1;

    object_attr.displayName = UA_LOCALIZEDTEXT("en", "ConditionSourceObject");
    UA_StatusCode retval =  UA_Server_addObjectNode(server, UA_NODEID_NULL,
                                UA_NS0ID(OBJECTSFOLDER), UA_NS0ID(ORGANIZES),
                                UA_QUALIFIEDNAME(0, "ConditionSourceObject"),
                                UA_NS0ID(BASEOBJECTTYPE),
                                object_attr, NULL, &conditionSource);

    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Creating Condition Source failed. StatusCode %s",
                     UA_StatusCode_name(retval));
    }

    /* ConditionSource should be EventNotifier of another Object (usually the
     * Server Object). If this Reference is not created by user then the A&C
     * Server will create "HasEventSource" reference to the Server Object
     * automatically when the condition is created*/
    retval = UA_Server_addReference(server, UA_NS0ID(SERVER), UA_
↪NS0ID(HASNOTIFIER),
                                UA_EXPANDEDNODEID_NUMERIC(conditionSource.
↪namespaceIndex,
                                                          conditionSource.identifier.numeric),
                                UA_TRUE);

    return retval;
}
```

Create a condition instance from OffNormalAlarmType. The condition source is the Object created in addConditionSourceObject(). The condition will be exposed in Address Space through the HasComponent reference to the condition source.

```
static UA_StatusCode
addCondition_1(UA_Server *server) {
```

(continues on next page)

(continued from previous page)

```
    UA_StatusCode retval = addConditionSourceObject(server);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "creating Condition Source failed. StatusCode %s",
                     UA_StatusCode_name(retval));
    }

    retval = acDriver->createCondition(acDriver, UA_NODEID_NULL, UA_
↪NS0ID(OFFNORMALALARMTYPE),
                                       UA_QUALIFIEDNAME(0, "Condition 1"), conditionSource,
                                       UA_NS0ID(HASCOMPONENT), &conditionInstance_1);

    return retval;
}
```

Create a condition instance from OffNormalAlarmType. The condition source is the server Object. The condition won't be exposed in Address Space.

```
static UA_StatusCode
addCondition_2(UA_Server *server) {
    UA_StatusCode retval =
        acDriver->createCondition(acDriver, UA_NODEID_NULL, UA_
↪NS0ID(OFFNORMALALARMTYPE),
                                  UA_QUALIFIEDNAME(0, "Condition 2"), UA_NS0ID(SERVER),
                                  UA_NODEID_NULL, &conditionInstance_2);

    return retval;
}

static void
addVariable_1_triggerAlarmOfCondition_1(UA_Server *server, UA_NodeId* outNodeId) {
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en", "Activate Condition 1");
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;
    UA_Boolean tboolValue = UA_FALSE;
    UA_Variant_setScalar(&attr.value, &tboolValue, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);

    UA_QualifiedName CallbackTestVariableName = UA_QUALIFIEDNAME(0, "Activate␣
↪Condition 1");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_NodeId variableTypeNodeId = UA_NS0ID(BASEDATAVARIABLETYPE);
    UA_Server_addVariableNode(server, UA_NODEID_NULL, parentNodeId,
                              parentReferenceNodeId, CallbackTestVariableName,
                              variableTypeNodeId, attr, NULL, outNodeId);
}
```

(continues on next page)

(continued from previous page)

```
static void
addVariable_2_changeSeverityOfCondition_2(UA_Server *server,
                                          UA_NodeId* outNodeId) {
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en", "Change Severity Condition 2");
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;
    UA_UInt16 severityValue = 0;
    UA_Variant_setScalar(&attr.value, &severityValue, &UA_TYPES[UA_TYPES_
↪UINT16]);

    UA_QualifiedName CallbackTestVariableName =
        UA_QUALIFIEDNAME(0, "Change Severity Condition 2");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_NodeId variableTypeNodeId = UA_NS0ID(BASEDATAVARIABLETYPE);
    UA_Server_addVariableNode(server, UA_NODEID_NULL, parentNodeId,
                              parentReferenceNodeId, CallbackTestVariableName,
                              variableTypeNodeId, attr, NULL, outNodeId);
}

static void
addVariable_3_returnCondition_1_toNormalState(UA_Server *server,
                                              UA_NodeId* outNodeId) {
    UA_VariableAttributes attr = UA_VariableAttributes_default;
    attr.displayName = UA_LOCALIZEDTEXT("en", "Return to Normal Condition 1");
    attr.accessLevel = UA_ACCESSLEVELMASK_READ | UA_ACCESSLEVELMASK_
↪WRITE;
    UA_Boolean rtn = 0;
    UA_Variant_setScalar(&attr.value, &rtn, &UA_TYPES[UA_TYPES_BOOLEAN]);

    UA_QualifiedName CallbackTestVariableName =
        UA_QUALIFIEDNAME(0, "Return to Normal Condition 1");
    UA_NodeId parentNodeId = UA_NS0ID(OBJECTSFOLDER);
    UA_NodeId parentReferenceNodeId = UA_NS0ID(ORGANIZES);
    UA_NodeId variableTypeNodeId = UA_NS0ID(BASEDATAVARIABLETYPE);
    UA_Server_addVariableNode(server, UA_NODEID_NULL, parentNodeId,
                              parentReferenceNodeId, CallbackTestVariableName,
                              variableTypeNodeId, attr, NULL, outNodeId);
}

static void
afterWriteCallbackVariable_1(UA_Server *server, const UA_NodeId *sessionId,
                             void *sessionContext, const UA_NodeId *nodeId,
                             void *nodeContext, const UA_NumericRange *range,
                             const UA_DataValue *data) {
    UA_QualifiedName activeStateField = UA_QUALIFIEDNAME(0,"ActiveState");
    UA_QualifiedName activeStateIdField = UA_QUALIFIEDNAME(0,"Id");
    UA_Variant value;
```

(continues on next page)

(continued from previous page)

```
    UA_StatusCode retval =
        UA_Server_writeObjectProperty_scalar(server, conditionInstance_1,
                                             UA_QUALIFIEDNAME(0, "Time"),
                                             &data->sourceTimestamp,
                                             &UA_TYPES[UA_TYPES_DATETIME]);

    if(*(UA_Boolean *)(data->value.data) == true) {
        /* By writing "true" in ActiveState/Id, the A&C server will set the
         * related fields automatically and then will trigger event
         * notification. */
        UA_Boolean activeStateId = true;
        UA_Variant_setScalar(&value, &activeStateId, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);
        retval |= acDriver->setConditionVariableFieldProperty(acDriver, conditionInstance_1,
                                                              &value, activeStateField,
                                                              activeStateIdField);
        if(retval != UA_STATUSCODE_GOOD) {
            UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                         "Setting ActiveState/Id Field failed. StatusCode %s",
                         UA_StatusCode_name(retval));
            return;
        }
    } else {
        /* By writing "false" in ActiveState/Id, the A&C server will set only
         * the ActiveState field automatically to the value "Inactive". The user
         * should trigger the event manually by calling
         * acDriver->triggerConditionEvent inside the application or call
         * ConditionRefresh method with client to update the event notification. */
        UA_Boolean activeStateId = false;
        UA_Variant_setScalar(&value, &activeStateId, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);
        retval = acDriver->setConditionVariableFieldProperty(acDriver, conditionInstance_1,
                                                             &value, activeStateField,
                                                             activeStateIdField);
        if(retval != UA_STATUSCODE_GOOD) {
            UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                         "Setting ActiveState/Id Field failed. StatusCode %s",
                         UA_StatusCode_name(retval));
            return;
        }

        retval = acDriver->triggerConditionEvent(acDriver, conditionInstance_1,
                                                 conditionSource, NULL);
        if(retval != UA_STATUSCODE_GOOD) {
            UA_LOG_WARNING(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                           "Triggering condition event failed. StatusCode %s",
                           UA_StatusCode_name(retval));
            return;
```

(continues on next page)

(continued from previous page)

```
        }
    }
}
```

The callback only changes the severity field of the condition 2. The severity field is of ConditionVariableType, so changes in it triggers an event notification automatically by the server.

```
static void
afterWriteCallbackVariable_2(UA_Server *server, const UA_NodeId *sessionId,
                             void *sessionContext, const UA_NodeId *nodeId,
                             void *nodeContext, const UA_NumericRange *range,
                             const UA_DataValue *data) {
  /* Another way to set fields of conditions */
  UA_Server_writeObjectProperty_scalar(server, conditionInstance_2,
                                       UA_QUALIFIEDNAME(0, "Severity"),
                                       (UA_UInt16 *)data->value.data,
                                       &UA_TYPES[UA_TYPES_UINT16]);
}
```

RTN = return to normal.

Retain will be set to false, thus no events will be generated for condition 1 (although EnabledState/=true). To set Retain to true again, the disable and enable methods should be called respectively.

```
static void
afterWriteCallbackVariable_3(UA_Server *server,
                const UA_NodeId *sessionId, void *sessionContext,
                const UA_NodeId *nodeId, void *nodeContext,
                const UA_NumericRange *range, const UA_DataValue *data) {

    //UA_QualifiedName enabledStateField = UA_QUALIFIEDNAME(0,"EnabledState");
    UA_QualifiedName ackedStateField = UA_QUALIFIEDNAME(0,"AckedState");
    UA_QualifiedName confirmedStateField = UA_QUALIFIEDNAME(0,"ConfirmedState");
    UA_QualifiedName activeStateField = UA_QUALIFIEDNAME(0,"ActiveState");
    UA_QualifiedName severityField = UA_QUALIFIEDNAME(0,"Severity");
    UA_QualifiedName messageField = UA_QUALIFIEDNAME(0,"Message");
    UA_QualifiedName commentField = UA_QUALIFIEDNAME(0,"Comment");
    UA_QualifiedName retainField = UA_QUALIFIEDNAME(0,"Retain");
    UA_QualifiedName idField = UA_QUALIFIEDNAME(0,"Id");

    UA_StatusCode retval =
        UA_Server_writeObjectProperty_scalar(server, conditionInstance_1,
                                       UA_QUALIFIEDNAME(0, "Time"),
                                       &data->serverTimestamp,
                                       &UA_TYPES[UA_TYPES_DATETIME]);
    UA_Variant value;
    UA_Boolean idValue = false;
    UA_Variant_setScalar(&value, &idValue, &UA_TYPES[UA_TYPES_BOOLEAN]);
    retval |= acDriver->setConditionVariableFieldProperty(acDriver, conditionInstance_1,
```

(continues on next page)

(continued from previous page)

```
                                        &value, activeStateField,
                                        idField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting ActiveState/Id Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return;
    }

    retval = acDriver->setConditionVariableFieldProperty(acDriver, conditionInstance_1,
                                        &value, ackedStateField,
                                        idField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting AckedState/Id Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return;
    }

    retval = acDriver->setConditionVariableFieldProperty(acDriver, conditionInstance_1,
                                        &value, confirmedStateField,
                                        idField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting ConfirmedState/Id Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return;
    }

    UA_UInt16 severityValue = 100;
    UA_Variant_setScalar(&value, &severityValue, &UA_TYPES[UA_TYPES_UINT16]);
    retval = acDriver->setConditionField(acDriver, conditionInstance_1,
                                &value, severityField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting Severity Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return;
    }

    UA_LocalizedText messageValue =
        UA_LOCALIZEDTEXT("en", "Condition returned to normal state");
    UA_Variant_setScalar(&value, &messageValue, &UA_TYPES[UA_TYPES_
↪LOCALIZEDTEXT]);
    retval = acDriver->setConditionField(acDriver, conditionInstance_1,
                                &value, messageField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting Message Field failed. StatusCode %s",
```

(continues on next page)

(continued from previous page)

```
                    UA_StatusCode_name(retval));
        return;
    }

    UA_LocalizedText commentValue = UA_LOCALIZEDTEXT("en", "Normal State");
    UA_Variant_setScalar(&value, &commentValue, &UA_TYPES[UA_TYPES_
↪LOCALIZEDTEXT]);
    retval = acDriver->setConditionField(acDriver, conditionInstance_1,
                                         &value, commentField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting Comment Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return;
    }

    UA_Boolean retainValue = false;
    UA_Variant_setScalar(&value, &retainValue, &UA_TYPES[UA_TYPES_BOOLEAN]);
    retval = acDriver->setConditionField(acDriver, conditionInstance_1,
                                         &value, retainField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting Retain Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return;
    }

    retval = acDriver->triggerConditionEvent(acDriver, conditionInstance_1,
                                             conditionSource, NULL);
    if (retval != UA_STATUSCODE_GOOD) {
     UA_LOG_WARNING(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                    "Triggering condition event failed. StatusCode %s",
                    UA_StatusCode_name(retval));
     return;
    }
}

static UA_StatusCode
enteringEnabledStateCallback(UA_Server *server, const UA_NodeId *condition) {
    UA_Boolean retain = true;
    return UA_Server_writeObjectProperty_scalar(server, *condition,
                                                UA_QUALIFIEDNAME(0, "Retain"),
                                                &retain,
                                                &UA_TYPES[UA_TYPES_BOOLEAN]);
}
```

This is user specific function which will be called upon acknowledging an alarm notification. In this example we will set the Alarm to Inactive state. The server is responsible of setting standard fields related to Acknowledge Method and triggering the alarm notification.

```
static UA_StatusCode
enteringAckedStateCallback(UA_Server *server, const UA_NodeId *condition) {
    /* deactivate Alarm when acknowledging*/
    UA_Boolean activeStateId = false;
    UA_Variant value;
    UA_QualifiedName activeStateField = UA_QUALIFIEDNAME(0,"ActiveState");
    UA_QualifiedName activeStateIdField = UA_QUALIFIEDNAME(0,"Id");

    UA_Variant_setScalar(&value, &activeStateId, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);
    UA_StatusCode retval =
        acDriver->setConditionVariableFieldProperty(acDriver, *condition,
                                            &value, activeStateField,
                                            activeStateIdField);

    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting ActiveState/Id Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
    }

    return retval;
}

static UA_StatusCode
enteringConfirmedStateCallback(UA_Server *server, const UA_NodeId *condition) {
    /* Deactivate Alarm and put it out of the interesting state (by writing
     * false to Retain field) when confirming*/
    UA_Boolean activeStateId = false;
    UA_Boolean retain = false;
    UA_Variant value;
    UA_QualifiedName activeStateField = UA_QUALIFIEDNAME(0,"ActiveState");
    UA_QualifiedName activeStateIdField = UA_QUALIFIEDNAME(0,"Id");
    UA_QualifiedName retainField = UA_QUALIFIEDNAME(0,"Retain");

    UA_Variant_setScalar(&value, &activeStateId, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);
    UA_StatusCode retval =
        acDriver->setConditionVariableFieldProperty(acDriver, *condition,
                                            &value, activeStateField,
                                            activeStateIdField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting ActiveState/Id Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return retval;
    }

    UA_Variant_setScalar(&value, &retain, &UA_TYPES[UA_TYPES_BOOLEAN]);
    retval = acDriver->setConditionField(acDriver, *condition,
```

(continues on next page)

(continued from previous page)

```
                                    &value, retainField);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting ActiveState/Id Field failed. StatusCode %s",
                     UA_StatusCode_name(retval));
    }

    return retval;
}

static UA_StatusCode
setUpEnvironment(UA_Server *server) {
    UA_NodeId variable_1;
    UA_NodeId variable_2;
    UA_NodeId variable_3;
    UA_ValueCallback callback;
    callback.onRead = NULL;

    /* Exposed condition 1. We will add to it user specific callbacks when
     * entering enabled state, when acknowledging and when confirming. */
    UA_StatusCode retval = addCondition_1(server);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "adding condition 1 failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return retval;
    }

    UA_TwoStateVariableChangeCallback userSpecificCallback =␣
↪enteringEnabledStateCallback;
    retval = acDriver->setConditionTwoStateVariableCallback(acDriver, conditionInstance_1,
                                                            conditionSource, false,
                                                            userSpecificCallback,
                                                            UA_ENTERING_ENABLEDSTATE);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "adding entering enabled state callback failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return retval;
    }

    userSpecificCallback = enteringAckedStateCallback;
    retval = acDriver->setConditionTwoStateVariableCallback(acDriver, conditionInstance_1,
                                                            conditionSource, false,
                                                            userSpecificCallback,
                                                            UA_ENTERING_ACKEDSTATE);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "adding entering acked state callback failed. StatusCode %s",
```

(continues on next page)

(continued from previous page)

```
                UA_StatusCode_name(retval));
    return retval;
}

userSpecificCallback = enteringConfirmedStateCallback;
retval = acDriver->setConditionTwoStateVariableCallback(acDriver, conditionInstance_1,
                                                        conditionSource, false,
                                                        userSpecificCallback,
                                                        UA_ENTERING_CONFIRMEDSTATE);
if(retval != UA_STATUSCODE_GOOD) {
    UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                 "adding entering confirmed state callback failed. StatusCode %s",
                 UA_StatusCode_name(retval));
    return retval;
}

/* Unexposed condition 2. No user specific callbacks, so the server will
 * behave in a standard manner upon entering enabled state, acknowledging
 * and confirming. We will set Retain field to true and enable the condition
 * so we can receive event notifications (we cannot call enable method on
 * unexposed condition using a client like UaExpert or Softing). */
retval = addCondition_2(server);
if(retval != UA_STATUSCODE_GOOD) {
    UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                 "adding condition 2 failed. StatusCode %s",
                 UA_StatusCode_name(retval));
    return retval;
}

UA_Boolean retain = UA_TRUE;
UA_Server_writeObjectProperty_scalar(server, conditionInstance_2,
                                     UA_QUALIFIEDNAME(0, "Retain"),
                                     &retain, &UA_TYPES[UA_TYPES_BOOLEAN]);

UA_Variant value;
UA_Boolean enabledStateId = true;
UA_QualifiedName enabledStateField = UA_QUALIFIEDNAME(0,"EnabledState");
UA_QualifiedName enabledStateIdField = UA_QUALIFIEDNAME(0,"Id");
UA_Variant_setScalar(&value, &enabledStateId, &UA_TYPES[UA_TYPES_
↪BOOLEAN]);
retval = acDriver->setConditionVariableFieldProperty(acDriver, conditionInstance_2,
                                                     &value, enabledStateField,
                                                     enabledStateIdField);

if(retval != UA_STATUSCODE_GOOD) {
    UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                 "Setting EnabledState/Id Field failed. StatusCode %s",
                 UA_StatusCode_name(retval));
    return retval;
```

(continues on next page)

(continued from previous page)

```
    }


    /* Add 3 variables to trigger condition events */
    addVariable_1_triggerAlarmOfCondition_1(server, &variable_1);

    callback.onWrite = afterWriteCallbackVariable_1;
    retval = UA_Server_setVariableNode_valueCallback(server, variable_1, callback);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting variable 1 Callback failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return retval;
    }

    /* Severity can change internally also when the condition disabled and
     * retain is false. However, in this case no events will be generated. */
    addVariable_2_changeSeverityOfCondition_2(server, &variable_2);

    callback.onWrite = afterWriteCallbackVariable_2;
    retval = UA_Server_setVariableNode_valueCallback(server, variable_2, callback);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting variable 2 Callback failed. StatusCode %s",
                     UA_StatusCode_name(retval));
        return retval;
    }

    addVariable_3_returnCondition_1_toNormalState(server, &variable_3);

    callback.onWrite = afterWriteCallbackVariable_3;
    retval = UA_Server_setVariableNode_valueCallback(server, variable_3, callback);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                     "Setting variable 3 Callback failed. StatusCode %s",
                     UA_StatusCode_name(retval));
    }

    return retval;
}
```

It follows the main server code, making use of the above definitions.

```
int main (void) {
    signal(SIGINT, stopHandler);
    signal(SIGTERM, stopHandler);

    UA_Server *server = UA_Server_new();
```

(continues on next page)

(continued from previous page)

```
    acDriver = UA_AlarmsConditionsDriver(UA_KEYVALUEMAP_NULL);
    if(!acDriver ||
       UA_Server_addDriver(server, &acDriver->drv) != UA_STATUSCODE_GOOD) {
        if(acDriver)
            acDriver->drv.free(&acDriver->drv);
        UA_Server_delete(server);
        return EXIT_FAILURE;
    }

    UA_StatusCode retval = UA_Server_run_startup(server);
    if(retval != UA_STATUSCODE_GOOD) {
        UA_Server_delete(server);
        return EXIT_FAILURE;
    }

    setUpEnvironment(server);

    while(running)
        UA_Server_run_iterate(server, true);

    UA_Server_run_shutdown(server);
    UA_Server_delete(server);
    return 0;
}
```

8.11 Serving Files and Directories

Build with UA_BUILD_EXAMPLES, Method calls and a reduced or full Namespace Zero. This example serves a local directory and a separate read-only file through the OPC UA Part 20 Methods.

Create a backend, construct its driver, then register it with the server. Registration starts the driver if the server is already running; otherwise it starts with the server. Shutdown stops drivers; deleting the server frees them. For parameters and backend callbacks, see *File Transfer Driver*. A custom in-memory backend is shown in examples/filetransfer/server_filetransfer_custom_backend.c.

8.11.1 Serving a local directory

The default root Object is named “FileSystem”. Its entries are mirrored on start and refreshed periodically. Client Methods operate on storage directly.

```
static UA_StatusCode
mountLocalDirectory(UA_Server *server, const char *rootPath) {
    UA_FileTransferBackend backend;
    UA_StatusCode res =
        UA_FileTransferBackend_localDirectory(UA_STRING((char*)(uintptr_t)rootPath),
                                              &backend);
    if(res != UA_STATUSCODE_GOOD)
        return res;

    UA_Driver *ftd = NULL;
```

(continues on next page)

(continued from previous page)

```
    res = UA_FileTransferDriver_newDirectory(server, &backend, NULL, NULL, &ftd);
    if(res != UA_STATUSCODE_GOOD) {
        backend.file.clear(&backend.file);
        return res;
    }
    res = UA_Server_addDriver(server, ftd);
    if(res != UA_STATUSCODE_GOOD)
        ftd->free(ftd);
    return res;
}
```

8.11.2 Exposing a single file

Individual files (a configuration file, a manual, a log) can be exposed as standalone FileType Objects without mirroring a whole directory. Here the file is served read-only: the Writable/UserWritable Properties are false and opening the file for writing fails.

```
static UA_StatusCode
addReadOnlyFile(UA_Server *server, const char *rootPath) {
    char filePath[512];
    int length = snprintf(filePath, sizeof(filePath), "%s/readme.txt", rootPath);
    if(length < 0 || (size_t)length >= sizeof(filePath))
        return UA_STATUSCODE_BADINVALIDARGUMENT;
    UA_FileTransferFileBackend backend;
    UA_StatusCode res =
        UA_FileTransferFileBackend_localFile(UA_STRING(filePath), &backend);
    if(res != UA_STATUSCODE_GOOD)
        return res;

    UA_Driver *ftd = NULL;
    UA_FileTransferNodeDescription description;
    memset(&description, 0, sizeof(description));
    description.browseName = UA_QUALIFIEDNAME(1, "ReadMe");
    res = UA_FileTransferDriver_newFile(server, &backend, UA_STRING_NULL,
                                        &description, NULL, &ftd);
    if(res != UA_STATUSCODE_GOOD) {
        backend.clear(&backend);
        return res;
    }
    /* Configure the driver before adding it: registration starts it immediately
     * when the server is already running. */
    UA_Boolean readOnly = true;
    res = UA_KeyValueMap_setScalar(&ftd->params, UA_QUALIFIEDNAME(0, "read-
↪only"),
                                   &readOnly, &UA_TYPES[UA_TYPES_BOOLEAN]);
    if(res == UA_STATUSCODE_GOOD)
        res = UA_Server_addDriver(server, ftd);
    if(res != UA_STATUSCODE_GOOD)
        ftd->free(ftd);
```

(continues on next page)

(continued from previous page)

```
    return res;
}

/* Prepare the served directory with demo content */
static void
prepareDemoDirectory(const char *rootPath) {
```

8.12 Building a Simple Client

You should already have a basic server from the previous tutorials. open62541 provides both a server- and clientside API, so creating a client is as easy as creating a server. Copy the following into a file *myClient.c*:

```
#include <open62541/client_config_default.h>
#include <open62541/client_highlevel.h>
#include <open62541/plugin/log_stdout.h>

int main(void) {
    UA_Client *client = UA_Client_new();
    UA_ClientConfig_setDefault(UA_Client_getConfig(client));
    UA_StatusCode retval = UA_Client_connect(client, "opc.tcp://localhost:4840");
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                    "The connection failed with status code %s",
                    UA_StatusCode_name(retval));
        UA_Client_delete(client);
        return 0;
    }

    /* Read the value attribute of the node. UA_Client_readValueAttribute is a
     * wrapper for the raw read service available as UA_Client_Service_read. */
    UA_Variant value; /* Variants can hold scalar values and arrays of any type */
    UA_Variant_init(&value);

    /* NodeId of the variable holding the current time */
    const UA_NodeId nodeId = UA_NS0ID(SERVER_SERVERSTATUS_
↪CURRENTTIME);
    retval = UA_Client_readValueAttribute(client, nodeId, &value);

    if(retval == UA_STATUSCODE_GOOD &&
       UA_Variant_hasScalarType(&value, &UA_TYPES[UA_TYPES_DATETIME])) {
        UA_DateTime raw_date = *(UA_DateTime *) value.data;
        UA_DateTimeStruct dts = UA_DateTime_toStruct(raw_date);
        UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
                    "date is: %u-%u-%u %u:%u:%u.%03u",
                    dts.day, dts.month, dts.year, dts.hour,
                    dts.min, dts.sec, dts.milliSec);
    } else {
        UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_APPLICATION,
```

(continues on next page)

(continued from previous page)

```
                "Reading the value failed with status code %s",
                UA_StatusCode_name(retval));
    }

    /* Clean up */
    UA_Variant_clear(&value);
    UA_Client_delete(client); /* Disconnects the client internally */
    return 0;
}
```

Compilation is similar to the server example.

```
$ gcc -std=c99 myClient.c -lopen62541 -o myClient
```

In a MinGW environment, the Winsock library must be added.

```
$ gcc -std=c99 myClient.c -lopen62541 -lws2_32 -o myClient.exe
```

8.12.1 Further tasks

- Try to connect to some other OPC UA server by changing opc.tcp://localhost:4840 to an appropriate address (remember that the queried node is contained in any OPC UA server).
- Try to set the value of the variable node (ns=1,i="the.answer") containing an Int32 from the example server (which is built in *Building a Simple Server*) using "UA_Client_write" function. The example server needs some more modifications, i.e., changing request types. The answer can be found in examples/client.c.

8.13 Working with Publish/Subscribe

Work in progress: This Tutorial will be continuously extended during the next PubSub batches. More details about the PubSub extension and corresponding open62541 API are located here: *PubSub*.

8.13.1 Publishing Fields

The PubSub publish example demonstrates the simplest way to publish information from the information model over UDP multicast using the UADP encoding.

Connection handling

PubSubConnections can be created and deleted on runtime. More details about the system preconfiguration and connection can be found in tutorial_pubsub_connection.c.

```
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>
#include <open62541/server_pubsub.h>
#include <open62541/types.h>

#include <stdio.h>
#include <stdlib.h>
```

(continues on next page)

(continued from previous page)

```
static UA_NodeId connectionIdent, publishedDataSetIdent, writerGroupIdent,
    dataSetWriterIdent;

static void
addPubSubConnection(UA_Server *server, UA_String *transportProfile,
                    UA_NetworkAddressUrlDataType *networkAddressUrl){
    /* Details about the connection configuration and handling are located
     * in the pubsub connection tutorial */
    UA_PubSubConnectionConfig connectionConfig;
    memset(&connectionConfig, 0, sizeof(connectionConfig));
    connectionConfig.name = UA_STRING("UADP Connection 1");
    connectionConfig.transportProfileUri = *transportProfile;
    UA_Variant_setScalar(&connectionConfig.address, networkAddressUrl,
                         &UA_TYPES[UA_TYPES_NETWORKADDRESSURLDATATYPE]);
    /* Changed to static publisherId from random generation to identify
     * the publisher on Subscriber side */
    connectionConfig.publisherId.idType = UA_PUBLISHERIDTYPE_UINT16;
    connectionConfig.publisherId.id.uint16 = 2234;
    UA_Server_addPubSubConnection(server, &connectionConfig, &connectionIdent);
}
```

PublishedDataSet handling

The PublishedDataSet (PDS) and PubSubConnection are the toplevel entities and can exist alone. The PDS contains the collection of the published fields. All other PubSub elements are directly or indirectly linked with the PDS or connection.

```
static void
addPublishedDataSet(UA_Server *server) {
    /* The PublishedDataSetConfig contains all necessary public
    * information for the creation of a new PublishedDataSet */
    UA_PublishedDataSetConfig publishedDataSetConfig;
    memset(&publishedDataSetConfig, 0, sizeof(UA_PublishedDataSetConfig));
    publishedDataSetConfig.publishedDataSetType = UA_PUBSUB_DATASET_
↪PUBLISHEDITEMS;
    publishedDataSetConfig.name = UA_STRING("Demo PDS");
    UA_Server_addPublishedDataSet(server, &publishedDataSetConfig, &
↪publishedDataSetIdent);
}
```

DataSetField handling

The DataSetField (DSF) is part of the PDS and describes exactly one published field.

```
static void
addDataSetField(UA_Server *server) {
    /* Add a field to the previous created PublishedDataSet */
    UA_NodeId dataSetFieldIdent;
    UA_DataSetFieldConfig dataSetFieldConfig;
    memset(&dataSetFieldConfig, 0, sizeof(UA_DataSetFieldConfig));
    dataSetFieldConfig.dataSetFieldType = UA_PUBSUB_DATASETFIELD_VARIABLE;
```

(continues on next page)

(continued from previous page)

```
    dataSetFieldConfig.field.variable.fieldNameAlias = UA_STRING("Server localtime");
    dataSetFieldConfig.field.variable.promotedField = false;
    dataSetFieldConfig.field.variable.publishParameters.publishedVariable =
        UA_NS0ID(SERVER_SERVERSTATUS_CURRENTTIME);
    dataSetFieldConfig.field.variable.publishParameters.attributeId = UA_ATTRIBUTEID_
↪VALUE;
    UA_Server_addDataSetField(server, publishedDataSetIdent,
                              &dataSetFieldConfig, &dataSetFieldIdent);
}
```

WriterGroup handling

The WriterGroup (WG) is part of the connection and contains the primary configuration parameters for the message creation.

```
static void
addWriterGroup(UA_Server *server) {
    /* Now we create a new WriterGroupConfig and add the group to the existing
     * PubSubConnection. */
    UA_WriterGroupConfig writerGroupConfig;
    memset(&writerGroupConfig, 0, sizeof(UA_WriterGroupConfig));
    writerGroupConfig.name = UA_STRING("Demo WriterGroup");
    writerGroupConfig.publishingInterval = 100;
    writerGroupConfig.writerGroupId = 100;
    writerGroupConfig.encodingMimeType = UA_PUBSUB_ENCODING_UADP;

    /* Change message settings of writerGroup to send PublisherId,
     * WriterGroupId in GroupHeader and DataSetWriterId in PayloadHeader
     * of NetworkMessage */
    UA_UadpWriterGroupMessageDataType writerGroupMessage;
    UA_UadpWriterGroupMessageDataType_init(&writerGroupMessage);
    writerGroupMessage.networkMessageContentMask =
        (UA_UadpNetworkMessageContentMask)(UA_
↪UADPNETWORKMESSAGECONTENTMASK_PUBLISHERID |
                                           UA_UADPNETWORKMESSAGECONTENTMASK_
↪GROUPHEADER |
                                           UA_UADPNETWORKMESSAGECONTENTMASK_
↪WRITERGROUPID |
                                           UA_UADPNETWORKMESSAGECONTENTMASK_
↪PAYLOADHEADER);

    /* The configuration flags for the messages are encapsulated inside the
     * message- and transport settings extension objects. These extension
     * objects are defined by the standard. e.g.
     * UadpWriterGroupMessageDataType */
    UA_ExtensionObject_setValue(&writerGroupConfig.messageSettings, &
↪writerGroupMessage,
                                &UA_TYPES[UA_TYPES_
↪UADPWRITERGROUPMESSAGEDATATYPE]);
```

(continues on next page)

(continued from previous page)

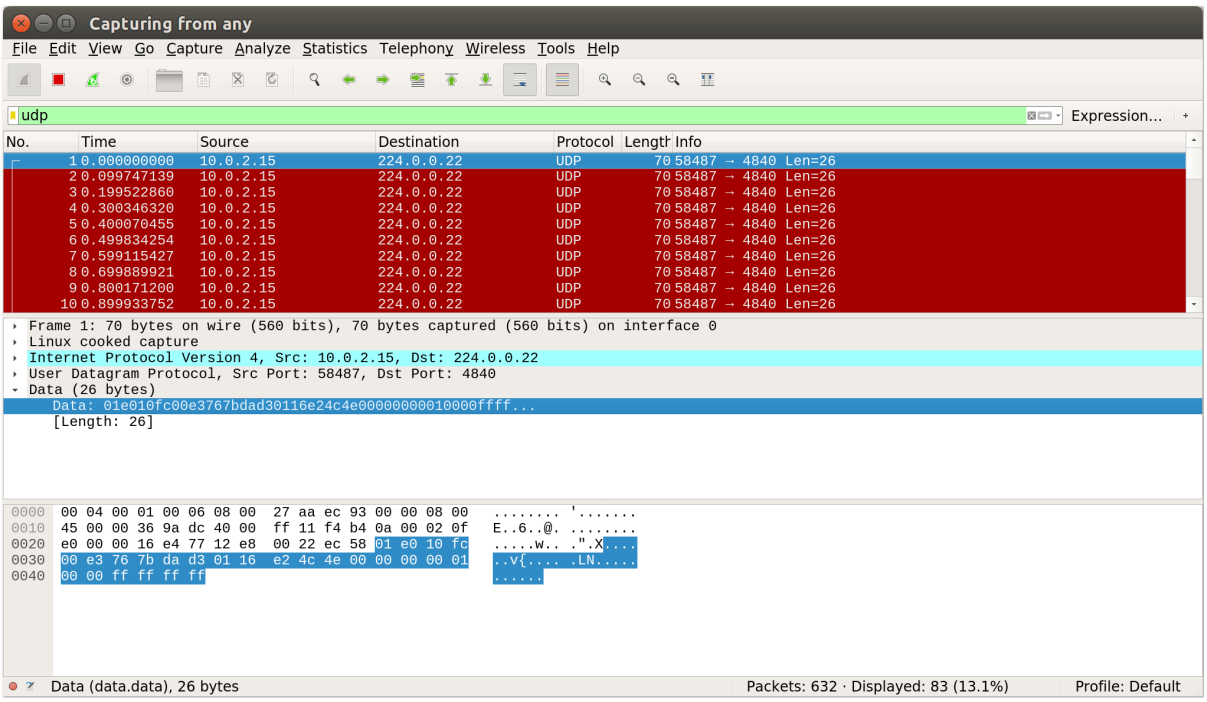
```
    UA_Server_addWriterGroup(server, connectionIdent, &writerGroupConfig, &
↪writerGroupIdent);
}
```

DataSetWriter handling

A DataSetWriter (DSW) is the glue between the WG and the PDS. The DSW is linked to exactly one PDS and contains additional information for the message generation.

```
static void
addDataSetWriter(UA_Server *server) {
    /* We need now a DataSetWriter within the WriterGroup. This means we must
     * create a new DataSetWriterConfig and add call the addWriterGroup function. */
    UA_DataSetWriterConfig dataSetWriterConfig;
    memset(&dataSetWriterConfig, 0, sizeof(UA_DataSetWriterConfig));
    dataSetWriterConfig.name = UA_STRING("Demo DataSetWriter");
    dataSetWriterConfig.dataSetWriterId = 62541;
    dataSetWriterConfig.keyFrameCount = 10;
    UA_Server_addDataSetWriter(server, writerGroupIdent, publishedDataSetIdent,
                               &dataSetWriterConfig, &dataSetWriterIdent);
}
```

That's it! You're now publishing the selected fields. Open a packet inspection tool of trust e.g. wireshark and take a look on the outgoing packages. The following graphic figures out the packages created by this tutorial.



The open62541 subscriber API will be released later. If you want to process the datagrams, take a look on the ua_pubsub_networkmessage_binary.c which already contains the decoding code for UADP messages.

It follows the main server code, making use of the above definitions.

```
static int
run(UA_String *transportProfile, UA_NetworkAddressUrlDataType *networkAddressUrl) {
    /* Create a server */
    UA_Server *server = UA_Server_new();

    /* Add the PubSub components. They are initially disabled */
    addPubSubConnection(server, transportProfile, networkAddressUrl);
    addPublishedDataSet(server);
    addDataSetField(server);
    addWriterGroup(server);
    addDataSetWriter(server);

    /* Enable the PubSubComponents */
    UA_Server_enableAllPubSubComponents(server);

    /* Run the server */
    UA_StatusCode retval = UA_Server_runUntilInterrupt(server);

    /* Delete the server */
    UA_Server_delete(server);
    return retval == UA_STATUSCODE_GOOD ? EXIT_SUCCESS : EXIT_FAILURE;
}

static void
usage(char *progname) {
    printf("usage: %s <uri> [device]\n", progname);
}

int main(int argc, char **argv) {
    UA_String transportProfile =
        UA_STRING("http://opcfoundation.org/UA-Profile/Transport/pubsub-udp-uadp");
    UA_NetworkAddressUrlDataType networkAddressUrl =
        {UA_STRING_NULL , UA_STRING("opc.udp://224.0.0.22:4840/")};

    if (argc > 1) {
        if (strcmp(argv[1], "-h") == 0) {
            usage(argv[0]);
            return EXIT_SUCCESS;
        } else if (strncmp(argv[1], "opc.udp://", 10) == 0) {
            networkAddressUrl.url = UA_STRING(argv[1]);
        } else if (strncmp(argv[1], "opc.eth://", 10) == 0) {
            transportProfile =
                UA_STRING("http://opcfoundation.org/UA-Profile/Transport/pubsub-eth-uadp
↪");
            if (argc < 3) {
                printf("Error: UADP/ETH needs an interface name\n");
                return EXIT_FAILURE;
            }
            networkAddressUrl.url = UA_STRING(argv[1]);
        } else {
```

(continues on next page)

(continued from previous page)

```
            printf("Error: unknown URI\n");
            return EXIT_FAILURE;
        }
    }
    if (argc > 2) {
        networkAddressUrl.networkInterface = UA_STRING(argv[2]);
    }

    return run(&transportProfile, &networkAddressUrl);
}
```

8.14 Subscribing Fields

The PubSub subscribe example demonstrates the simplest way to receive information over two transport layers such as UDP and Ethernet, that are published by tutorial_pubsub_publish example and update values in the TargetVariables of Subscriber Information Model.

```
#include <open62541/plugin/log_stdout.h>
#include <open62541/server.h>
#include <open62541/server_pubsub.h>

#include <stdio.h>
#include <stdlib.h>

static UA_NodeId connectionIdentifier;
static UA_NodeId readerGroupIdentifier;
static UA_NodeId readerIdentifier;

static UA_DataSetReaderConfig readerConfig;

static void fillTestDataSetMetaData(UA_DataSetMetaDataType *pMetaData);

/* Add new connection to the server */
static void
addPubSubConnection(UA_Server *server, UA_String *transportProfile,
                    UA_NetworkAddressUrlDataType *networkAddressUrl) {
    /* Configuration creation for the connection */
    UA_PubSubConnectionConfig connectionConfig;
    memset (&connectionConfig, 0, sizeof(UA_PubSubConnectionConfig));
    connectionConfig.name = UA_STRING("UDPMC Connection 1");
    connectionConfig.transportProfileUri = *transportProfile;
    UA_Variant_setScalar(&connectionConfig.address, networkAddressUrl,
                         &UA_TYPES[UA_TYPES_NETWORKADDRESSURLDATATYPE]);
    connectionConfig.publisherId.idType = UA_PUBLISHERIDTYPE_UINT32;
    connectionConfig.publisherId.id.uint32 = UA_UInt32_random();
    UA_Server_addPubSubConnection (server, &connectionConfig, &connectionIdentifier);
}
```

ReaderGroup

ReaderGroup is used to group a list of DataSetReaders. All ReaderGroups are created within a PubSubConnection and automatically deleted if the connection is removed. All network message related filters are only available in the DataSetReader.

```
static void
addReaderGroup(UA_Server *server) {
    UA_ReaderGroupConfig readerGroupConfig;
    memset(&readerGroupConfig, 0, sizeof(UA_ReaderGroupConfig));
    readerGroupConfig.name = UA_STRING("ReaderGroup1");
    UA_Server_addReaderGroup(server, connectionIdentifier, &readerGroupConfig,
                             &readerGroupIdentifier);
}
```

DataSetReader

DataSetReader can receive NetworkMessages with the DataSetMessage of interest sent by the Publisher. DataSetReader provides the configuration necessary to receive and process DataSetMessages on the Subscriber side. DataSetReader must be linked with a SubscribedDataSet and be contained within a ReaderGroup.

```
static void
addDataSetReader(UA_Server *server) {
    memset (&readerConfig, 0, sizeof(UA_DataSetReaderConfig));
    readerConfig.name = UA_STRING("DataSet Reader 1");
    /* Parameters to filter which DataSetMessage has to be processed
     * by the DataSetReader */
    /* The following parameters are used to show that the data published by
     * tutorial_pubsub_publish.c is being subscribed and is being updated in
     * the information model */
    UA_UInt16 publisherIdentifier = 2234;
    readerConfig.publisherId.idType = UA_PUBLISHERIDTYPE_UINT16;
    readerConfig.publisherId.id.uint16 = publisherIdentifier;
    readerConfig.writerGroupId    = 100;
    readerConfig.dataSetWriterId  = 62541;

    /* Setting up Meta data configuration in DataSetReader */
    fillTestDataSetMetaData(&readerConfig.dataSetMetaData);

    UA_Server_addDataSetReader(server, readerGroupIdentifier, &readerConfig,
                               &readerIdentifier);
}
```

SubscribedDataSet

Set SubscribedDataSet type to TargetVariables data type. Add subscribedvariables to the DataSetReader

```
static void
addSubscribedVariables (UA_Server *server, UA_NodeId dataSetReaderId) {
    UA_NodeId folderId;
    UA_String folderName = readerConfig.dataSetMetaData.name;
    UA_ObjectAttributes oAttr = UA_ObjectAttributes_default;
```

(continues on next page)

(continued from previous page)

```
    UA_QualifiedName folderBrowseName;
    if(folderName.length > 0) {
        oAttr.displayName.locale = UA_STRING ("en-US");
        oAttr.displayName.text = folderName;
        folderBrowseName.namespaceIndex = 1;
        folderBrowseName.name = folderName;
    }
    else {
        oAttr.displayName = UA_LOCALIZEDTEXT ("en-US", "Subscribed Variables");
        folderBrowseName = UA_QUALIFIEDNAME (1, "Subscribed Variables");
    }

    UA_Server_addObjectNode(server, UA_NODEID_NULL,
                            UA_NS0ID(OBJECTSFOLDER), UA_NS0ID(ORGANIZES),
                            folderBrowseName, UA_NS0ID(BASEOBJECTTYPE), oAttr, NULL,␣
↪&folderId);
```

TargetVariables

The SubscribedDataSet option TargetVariables defines a list of Variable mappings between received DataSet fields and target Variables in the Subscriber AddressSpace. The values subscribed from the Publisher are updated in the value field of these variables

```
    /* Create the TargetVariables with respect to DataSetMetaData fields */
    UA_FieldTargetDataType *targetVars = (UA_FieldTargetDataType *)
            UA_calloc(readerConfig.dataSetMetaData.fieldsSize, sizeof(UA_
↪FieldTargetDataType));
    for(size_t i = 0; i < readerConfig.dataSetMetaData.fieldsSize; i++) {
        /* Variable to subscribe data */
        UA_VariableAttributes vAttr = UA_VariableAttributes_default;
        UA_LocalizedText_copy(&readerConfig.dataSetMetaData.fields[i].description,
                              &vAttr.description);
        vAttr.displayName.locale = UA_STRING("en-US");
        vAttr.displayName.text = readerConfig.dataSetMetaData.fields[i].name;
        vAttr.dataType = readerConfig.dataSetMetaData.fields[i].dataType;

        UA_NodeId newNode;
        UA_Server_addVariableNode(server, UA_NODEID_NUMERIC(1, (UA_UInt32)i +␣
↪50000),
                                  folderId, UA_NS0ID(HASCOMPONENT),
                                  UA_QUALIFIEDNAME(1, (char *)readerConfig.dataSetMetaData.
↪fields[i].name.data),
                                  UA_NS0ID(BASEDATAVARIABLETYPE),
                                  vAttr, NULL, &newNode);

        /* For creating Targetvariables */
        targetVars[i].attributeId  = UA_ATTRIBUTEID_VALUE;
        targetVars[i].targetNodeId = newNode;
    }
```

(continues on next page)

(continued from previous page)

```
    UA_Server_DataSetReader_createTargetVariables(server, dataSetReaderId,
                                                  readerConfig.dataSetMetaData.fieldsSize,
                                                  targetVars);

    UA_free(targetVars);
    UA_free(readerConfig.dataSetMetaData.fields);
}
```

DataSetMetaData

The DataSetMetaData describes the content of a DataSet. It provides the information necessary to decode DataSetMessages on the Subscriber side. DataSetMessages received from the Publisher are decoded into DataSet and each field is updated in the Subscriber based on datatype match of TargetVariable fields of Subscriber and PublishedDataSetFields of Publisher

```
static void
fillTestDataSetMetaData(UA_DataSetMetaDataType *pMetaData) {
    UA_DataSetMetaDataType_init (pMetaData);
    pMetaData->name = UA_STRING ("DataSet 1");

    /* Static definition of number of fields size to 4 to create four different
     * targetVariables of distinct datatype
     * Currently the publisher sends only DateTime data type */
    pMetaData->fieldsSize = 4;
    pMetaData->fields = (UA_FieldMetaData*)UA_Array_new (pMetaData->fieldsSize,
                        &UA_TYPES[UA_TYPES_FIELDMETADATA]);

    /* DateTime DataType */
    UA_FieldMetaData_init (&pMetaData->fields[0]);
    UA_NodeId_copy (&UA_TYPES[UA_TYPES_DATETIME].typeId,
                    &pMetaData->fields[0].dataType);
    pMetaData->fields[0].builtInType = UA_NS0ID_DATETIME;
    pMetaData->fields[0].name =  UA_STRING ("DateTime");
    pMetaData->fields[0].valueRank = -1; /* scalar */

    /* Int32 DataType */
    UA_FieldMetaData_init (&pMetaData->fields[1]);
    UA_NodeId_copy(&UA_TYPES[UA_TYPES_INT32].typeId,
                   &pMetaData->fields[1].dataType);
    pMetaData->fields[1].builtInType = UA_NS0ID_INT32;
    pMetaData->fields[1].name =  UA_STRING ("Int32");
    pMetaData->fields[1].valueRank = -1; /* scalar */

    /* Int64 DataType */
    UA_FieldMetaData_init (&pMetaData->fields[2]);
    UA_NodeId_copy(&UA_TYPES[UA_TYPES_INT64].typeId,
                   &pMetaData->fields[2].dataType);
    pMetaData->fields[2].builtInType = UA_NS0ID_INT64;
    pMetaData->fields[2].name =  UA_STRING ("Int64");
    pMetaData->fields[2].valueRank = -1; /* scalar */
```

(continues on next page)

(continued from previous page)

```
    /* Boolean DataType */
    UA_FieldMetaData_init (&pMetaData->fields[3]);
    UA_NodeId_copy (&UA_TYPES[UA_TYPES_BOOLEAN].typeId,
                    &pMetaData->fields[3].dataType);
    pMetaData->fields[3].builtInType = UA_NS0ID_BOOLEAN;
    pMetaData->fields[3].name =  UA_STRING ("BoolToggle");
    pMetaData->fields[3].valueRank = -1; /* scalar */
}
```

Followed by the main server code, making use of the above definitions

```
static int
run(UA_String *transportProfile, UA_NetworkAddressUrlDataType *networkAddressUrl) {
    UA_Server *server = UA_Server_new();

    addPubSubConnection(server, transportProfile, networkAddressUrl);
    addReaderGroup(server);
    addDataSetReader(server);
    addSubscribedVariables(server, readerIdentifier);

    UA_Server_enableAllPubSubComponents(server);
    UA_Server_runUntilInterrupt(server);

    UA_Server_delete(server);
    return 0;
}

static void
usage(char *progname) {
    printf("usage: %s <uri> [device]\n", progname);
}

int main(int argc, char **argv) {
    UA_String transportProfile =
        UA_STRING("http://opcfoundation.org/UA-Profile/Transport/pubsub-udp-uadp");
    UA_NetworkAddressUrlDataType networkAddressUrl =
        {UA_STRING_NULL , UA_STRING("opc.udp://224.0.0.22:4840/")};
    if(argc > 1) {
        if(strcmp(argv[1], "-h") == 0) {
            usage(argv[0]);
            return EXIT_SUCCESS;
        } else if(strncmp(argv[1], "opc.udp://", 10) == 0) {
            networkAddressUrl.url = UA_STRING(argv[1]);
        } else if(strncmp(argv[1], "opc.eth://", 10) == 0) {
            transportProfile =
                UA_STRING("http://opcfoundation.org/UA-Profile/Transport/pubsub-eth-uadp
↪");
            if(argc < 3) {
```

(continues on next page)

(continued from previous page)

```
            printf("Error: UADP/ETH needs an interface name\n");
            return EXIT_FAILURE;
        }

        networkAddressUrl.url = UA_STRING(argv[1]);
    } else {
        printf ("Error: unknown URI\n");
        return EXIT_FAILURE;
    }
}
if (argc > 2) {
    networkAddressUrl.networkInterface = UA_STRING(argv[2]);
}

return run(&transportProfile, &networkAddressUrl);
}
```

CHAPTER

NINE

SECURITY

OPC UA security rests on two layers: the *SecureChannel* protects the wire between client and server, and the *Session* authenticates the user on top of an open channel. open62541 ships with a complete, pluggable security stack covering both.

This section documents the parts you actually configure and operate:

9.1 Supported Security Policies

A *SecurityPolicy* is a named bundle of algorithms for asymmetric signing, asymmetric encryption, symmetric encryption, key derivation, and message signing – defined in Part 7 of the OPC UA standard. A server advertises its supported policies through its *Endpoints* (returned by GetEndpoints). The client picks one when opening a SecureChannel.

The table below lists every standard profile implemented in open62541. Adding a policy to a server is the same call regardless of backend – UA_ServerConfig_addSecurityPolicy*() (see *ECC Specifics* for the full call list).

Policy	Type	Notes
None	Universal	No signing, no encryption. Required for unencrypted discovery.
Basic128Rsa15	RSA	Legacy. RSA-1024 + AES-128. Deprecated, do not use for new systems.
Basic256	RSA	RSA-2048 + AES-256. Replaced by Basic256Sha256.
Basic256Sha256	RSA	RSA-2048 + SHA-256 + AES-256. The current RSA baseline.
Aes128_Sha256_RsaOaep	RSA	RSA-OAEP + AES-128 + SHA-256. Modern RSA profile.
Aes256_Sha256_RsaPss	RSA	RSA-PSS + AES-256 + SHA-256. Strongest RSA profile.
ECC_nistP256	ECC	NIST P-256 (prime256v1). See *ECC Specifics*.
ECC_nistP384	ECC	NIST P-384 (secp384r1). See *ECC Specifics*.
ECC_brainpoolP256r1	ECC	Brainpool P-256r1. See *ECC Specifics*.
ECC_brainpoolP384r1	ECC	Brainpool P-384r1. See *ECC Specifics*.
ECC_curve25519	ECC (EdDSA)	Curve25519 / Ed25519. OpenSSL only. See *ECC Specifics*.
ECC_curve448	ECC (EdDSA)	Curve448 / Ed448. OpenSSL only. See *ECC Specifics*.
PubSub-Aes128-CTR	PubSub	Symmetric-only, AES-128-CTR. Used with PubSub message security.
PubSub-Aes256-CTR	PubSub	Symmetric-only, AES-256-CTR. Used with PubSub message security.

Which of these you can actually enable depends on the *crypto backend* and the certificate you load. ECC-specifics – the curve support matrix, certificate requirements, and the UA_ServerConfig_addSecurityPolicy*Ecc*() helpers – are documented in *ECC Specifics*.

9.2 Crypto Backends

open62541 does not implement cryptography itself. It delegates to an external library through a thin plugin layer. The build option UA_ENABLE_ENCRYPTION selects which library is compiled in. No security policy from *Supported Security Policies* is available when this option is OFF.

9.2.1 Backend selection

CMake option	Effect
UA_ENABLE_ENCRYPTION=OFF	No encryption support. Default. Only the None policy is usable.
UA_ENABLE_ENCRYPTION=MBEDTLS	mbedTLS. All RSA profiles + the four NIST/Brainpool ECC curves.
UA_ENABLE_ENCRYPTION=OPENSSL	OpenSSL. All RSA profiles + every ECC curve, including Curve25519 and Curve448 (EdDSA).
UA_ENABLE_ENCRYPTION=LIBRESSL	LibreSSL. Experimental. Coverage is similar to OpenSSL.

The two boolean aliases UA_ENABLE_ENCRYPTION_OPENSSL and UA_ENABLE_ENCRYPTION_MBEDTLS are still accepted for backward compatibility. Prefer the unified UA_ENABLE_ENCRYPTION form in new projects.

9.2.2 Hardware-backed key storage (TPM2)

UA_ENABLE_ENCRYPTION_TPM2=ON adds TPM2-backed key storage on top of an already-selected UA_ENABLE_ENCRYPTION backend. It is **not** a standalone crypto library, so it must be combined with MBEDTLS or OPENSSL. When enabled, the private key used by the server lives on a TPM2 device instead of in a DER file. See the server_encryption_tpm_keystore and client_encryption_tpm_keystore examples, and the README_client_server_tpm_keystore.txt next to them for the required software stack.

9.2.3 Library version requirements

- **mbedTLS ≥ 3.0** is required for the four NIST/Brainpool ECC curves. mbedTLS does not support Curve25519 / Curve448.
- **OpenSSL ≥ 3.0** is required for full ECC support including EdDSA curves.
- See *Building open62541* for the exact flag set and the UA_ENABLE_ENCRYPTION_TPM2 option.

9.2.4 Registering policies

The backend is selected at build time, not at runtime. Once built with UA_ENABLE_ENCRYPTION=OPENSSL (or MBEDTLS), adding a policy to a server is one call per policy and the same call works with either backend:

```
UA_ServerConfig_addSecurityPolicyNone(config, &cert, &key);
UA_ServerConfig_addSecurityPolicyBasic256Sha256(config, &cert, &key);
UA_ServerConfig_addSecurityPolicyEccNistP256(config, &cert, &key);
/* ... or, in one shot: */
UA_ServerConfig_addAllSecurityPolicies(config, &cert, &key);
```

UA_ServerConfig_addAllSecurityPolicies() auto-detects the key type (RSA vs EC) and registers all policies matching the key. The ECC-specific helpers and the full set of addSecurityPolicy*() calls are listed in *ECC Specifics*.

9.3 Authentication

Authentication in OPC UA happens at the *Session* layer, on top of an already-open SecureChannel. The client presents an *IdentityToken* during ActivateSession. The server validates it through its *AccessControl* plugin and either accepts or rejects the session.

9.3.1 Identity token types

The standard defines four token types. All four are accepted by open62541's default access-control plugin and may be enabled in any combination.

Token	Type constant	Typical use
Anonymous	UA_AnonymousIdentityToken	Public, read-only data, discovery. Opt out for protected servers.
Username / Password	UA_UserNameIdentityToken	Simple human-user login. Credentials checked against an in-server list or a custom callback.
X.509 Certificate	UA_X509IdentityToken	Mutual-TLS-style login. The client proves possession of the private key matching a certificate the server trusts.
Issued Token	UA_IssuedIdentityToken	External STS / OAuth-style flows. The server forwards the token to an authorization service.

9.3.2 Default server configuration

UA_AccessControl_default() (in plugins/ua_accesscontrol_default.c) wires up the standard validator. It checks anonymous logins, looks up username/password pairs in a static list, and (when configured) verifies X.509 chains through a UA_CertificateGroup. The actual session-validation routine is activateSession_default() in the same file.

Useful entry points in the public plugin header plugins/include/open62541/plugin/accesscontrol_default.h:

- UA_AccessControl_default() – construct a default access-control plugin with a username/password allow-list and an optional UA_UsernamePasswordLoginCallback for custom validation.
- verifyX509 – a UA_CertificateGroup used to validate X.509 identity tokens. See *Certificates and Trust*.

For the protocol-level description of how authentication fits into CreateSession / ActivateSession, see *OPC UA Basics* ("Session" section).

9.4 Certificates and Trust

Every SecureChannel protected by anything other than the None policy requires an X.509 certificate and matching private key on both sides. Trust is managed by *CertificateGroup* plugins, which store the trusted issuer list, the rejected-certificate list, and the CRLs used during validation.

9.4.1 CertificateGroup backends

open62541 ships three CertificateGroup backends. Pick one when you configure your server.

Backend	Notes
UA_CertificateGroup_AcceptAll	Trust every certificate. Useful for tests and demos, **not for production**.
UA_CertificateGroup_Memorystore	Trust list kept in RAM. Configurable size limits (0:max-trust-listsize defaults to 64 KiB; 0:max-rejected-listsize defaults to 100).
UA_CertificateGroup_Filestore	Trust list on disk under a PKI directory. Default location is the current working directory, override with UA_CertificateGroup_Filestore's path argument.

9.4.2 Filestore PKI layout

The Filestore backend reads and writes a fixed directory layout rooted at the configured path:

```
pki/
  ApplCerts/
    issuer/{certs,crl}/
    own/{certs,private}/
    rejected/certs/
    trusted/{certs,crl}/
  UserTokenCerts/
    (same layout as ApplCerts)
```

9.4.3 Generate a certificate

For development and self-signed certificates, UA_CreateCertificate (in plugins/include/open62541/plugin/create_certificate.h) generates a key pair and a self-signed certificate in one call. RSA key sizes 2048 / 4096 are supported (4096 default). ECC usage is documented in *ECC Specifics*.

9.4.4 Certificate utility functions

The UA_CertificateUtils_* helpers in include/open62541/plugin/certificategroup.h extract common information from a DER certificate without exposing the underlying crypto library:

- UA_CertificateUtils_verifyApplicationUri() – check that the cert's SubjectAltName URI matches the application's ApplicationUri.
- UA_CertificateUtils_getExpirationDate() – X.509 notAfter.

- UA_CertificateUtils_getSubjectName() – X.509 Subject DN.
- UA_CertificateUtils_getThumbprint() – SHA-1 thumbprint.
- UA_CertificateUtils_getKeySize() – asymmetric key length in bits.
- UA_CertificateUtils_comparePublicKeys() – compare public keys from a certificate and a CSR.

9.4.5 Revocation

When a certificate fails trust validation, the standard OPC UA status codes used by open62541 are:

- UA_STATUSCODE_BADCERTIFICATEREVOCATIONUNKNOWN
- UA_STATUSCODE_BADCERTIFICATEISSUERREVOCATIONUNKNOWN
- UA_STATUSCODE_BADCERTIFICATEISSUERREVOKED
- UA_STATUSCODE_BADCERTIFICATEISSUERUSENOTALLOWED

The Filestore backend's getCertificateCrls() operation returns the CRLs relevant to a certificate, used by the verification step.

For the full plugin API, see *CertificateGroup Plugin API*.

9.5 ECC Specifics

ECC policies give equivalent or stronger security than the RSA-based profiles with much shorter key lengths, which makes them well suited for resource-constrained and modern deployments. This page documents only the ECC-specific configuration. For the general picture read *Security* first.

9.5.1 Supported curves

Each row corresponds to one SecurityPolicy. The presence of a check mark means that backend can register the policy given a matching certificate and private key.

OPC UA Policy	Curve	OpenSSL	mbedTLS ≥ 3.0
EccNistP256	NIST P-256 (prime256v1)	✓	✓
EccNistP384	NIST P-384 (secp384r1)	✓	✓
EccBrainpoolP256r1	Brainpool P-256r1	✓	✓
EccBrainpoolP384r1	Brainpool P-384r1	✓	✓
EccCurve25519	Curve25519 (Ed25519)	✓	☒
EccCurve448	Curve448 (Ed448)	✓	☒

Curve25519 and Curve448 require OpenSSL ≥ 3.0. The NIST and Brainpool curves require mbedTLS ≥ 3.0 or OpenSSL ≥ 3.0. See *Crypto Backends* for the build flags.

9.5.2 Certificate requirements

Each ECC policy requires its own certificate and private key, and the key must match the curve. A single certificate **cannot** be shared across different curves. RSA and ECC policies can coexist on the same server. Each carries its own key pair.

X.509 extensions to set when creating an ECC certificate:

- **Key Usage** – keyAgreement is required (for ECDH key exchange). The recommended set for ECDSA curves is:

```
digitalSignature, nonRepudiation, keyAgreement, keyCertSign
```

  RSA certificates use keyEncipherment instead.

- **SubjectAltName** – must include at least URI:<ApplicationUri> and DNS:<hostname> (or IP:...).

9.5.3 Generating ECC certificates

There are three ways to obtain an ECC certificate for use with open62541.

1. "UA_CreateCertificate" (built-in, recommended for development)

Pass key-type = "EC" and the desired ecc-curve value in the key-value map. Supported curves: prime256v1, secp384r1, brainpoolP256r1, brainpoolP384r1, ed25519, ed448 (last two require OpenSSL).

```
UA_KeyValueMap *kvm = UA_KeyValueMap_new();
UA_UInt16 expiresIn = 365;
UA_KeyValueMap_setScalar(kvm, UA_QUALIFIEDNAME(0, "expires-in-days"),
                         &expiresIn, &UA_TYPES[UA_TYPES_UINT16]);
UA_String keyType = UA_STRING_STATIC("EC");
UA_KeyValueMap_setScalar(kvm, UA_QUALIFIEDNAME(0, "key-type"),
                         &keyType, &UA_TYPES[UA_TYPES_STRING]);
UA_String curve = UA_STRING_STATIC("prime256v1");
UA_KeyValueMap_setScalar(kvm, UA_QUALIFIEDNAME(0, "ecc-curve"),
                         &curve, &UA_TYPES[UA_TYPES_STRING]);

UA_ByteString privateKey = UA_BYTESTRING_NULL;
UA_ByteString certificate = UA_BYTESTRING_NULL;
UA_CreateCertificate(UA_Log_Stdout, subject, lenSubject,
                     subjectAltName, lenSubjectAltName,
                     UA_CERTIFICATEFORMAT_DER, kvm,
                     &privateKey, &certificate);
UA_KeyValueMap_delete(kvm);
```

2. OpenSSL CLI (production, full control)

ECDSA curves (P-256, P-384, Brainpool):

```
openssl ecparam -name prime256v1 -genkey -noout -out server_nistP256.key.pem
openssl req -new -x509 -key server_nistP256.key.pem \
    -out server_nistP256.cert.pem -days 3650 -sha256 \
    -subj "/C=DE/O=MyOrg/CN=OPC-UA-Server" \
    -addext "subjectAltName=URI:urn:my.app:server,DNS:localhost,IP:127.0.0.1" \
    -addext "basicConstraints=critical,CA:FALSE" \
    -addext "keyUsage=critical,digitalSignature,nonRepudiation,keyAgreement" \
    -addext "extendedKeyUsage=serverAuth,clientAuth"
openssl x509  -in  server_nistP256.cert.pem -outform DER -out server_nistP256.cert.der
openssl pkey  -in  server_nistP256.key.pem  -outform DER -out server_nistP256.key.der
```

Use -sha384 and prime256v1 replaced with secp384r1 or brainpoolP256r1 / brainpoolP384r1 accordingly.

EdDSA curves (Curve25519, Curve448):

```
openssl genpkey -algorithm ed25519 -out server_curve25519.key.pem
openssl req -new -x509 -key server_curve25519.key.pem \
    -out server_curve25519.cert.pem -days 3650 \
    -subj "/C=DE/O=MyOrg/CN=OPC-UA-Server" \
    -addext "subjectAltName=URI:urn:my.app:server,DNS:localhost,IP:127.0.0.1" \
    -addext "basicConstraints=critical,CA:FALSE" \
    -addext "keyUsage=critical,digitalSignature,nonRepudiation,keyAgreement" \
    -addext "extendedKeyUsage=serverAuth,clientAuth"
openssl x509 -in  server_curve25519.cert.pem -outform DER -out server_curve25519.cert.der
openssl pkey -in  server_curve25519.key.pem  -outform DER -out server_curve25519.key.der
```

Replace ed25519 with ed448 for Curve448.

3. "tools/certs/create_self-signed.py" (development helper)

```
# Single ECC certificate
python3 tools/certs/create_self-signed.py /tmp/certs --ecc prime256v1

# All supported ECC curves at once
python3 tools/certs/create_self-signed.py /tmp/certs --ecc-all
```

The --ecc-all mode produces DER files named server_c_<curve>.cert.der / server_c_<curve>.key.der ready for --ecc-cert-dir.

- *Supported Security Policies* – the standard OPC UA SecurityPolicies supported by open62541 and how to register them on a server.
- *Crypto Backends* – the underlying crypto library (mbedTLS, OpenSSL, or LibreSSL) and the build flags that select it.
- *Authentication* – the identity-token types used to authenticate Sessions (Anonymous, Username/Password, X.509, Issued Token).
- *Certificates and Trust* – X.509 certificate management: trust lists, the three CertificateGroup backends (AcceptAll, Memorystore, Filestore), and the UA_CreateCertificate helper.
- *ECC Specifics* – ECC specifics: supported curves, certificate requirements, generation, and the related examples.

For the protocol-level overview (SecureChannel, SecurityMode, Endpoint discovery) – see *OPC UA Basics*. This section focuses on the configuration and operational side.

For the auto-generated plugin API, see *SecurityPolicy Plugin API* and *CertificateGroup Plugin API*.

CHAPTER

TEN

COMMON DEFINITIONS

Common definitions for Client, Server and PubSub.

10.1 Attribute Id

Every node in an OPC UA information model contains attributes depending on the node type. Possible attributes are as follows:

```
typedef enum {
    UA_ATTRIBUTEID_INVALID           = 0,
    UA_ATTRIBUTEID_NODEID            = 1,
    UA_ATTRIBUTEID_NODECLASS         = 2,
    UA_ATTRIBUTEID_BROWSENAME        = 3,
    UA_ATTRIBUTEID_DISPLAYNAME       = 4,
    UA_ATTRIBUTEID_DESCRIPTION       = 5,
    UA_ATTRIBUTEID_WRITEMASK         = 6,
    UA_ATTRIBUTEID_USERWRITEMASK     = 7,
    UA_ATTRIBUTEID_ISABSTRACT        = 8,
    UA_ATTRIBUTEID_SYMMETRIC         = 9,
    UA_ATTRIBUTEID_INVERSENAME       = 10,
    UA_ATTRIBUTEID_CONTAINSNOLOOPS   = 11,
    UA_ATTRIBUTEID_EVENTNOTIFIER     = 12,
    UA_ATTRIBUTEID_VALUE             = 13,
    UA_ATTRIBUTEID_DATATYPE          = 14,
    UA_ATTRIBUTEID_VALUERANK         = 15,
    UA_ATTRIBUTEID_ARRAYDIMENSIONS   = 16,
    UA_ATTRIBUTEID_ACCESSLEVEL       = 17,
    UA_ATTRIBUTEID_USERACCESSLEVEL   = 18,
    UA_ATTRIBUTEID_MINIMUMSAMPLINGINTERVAL = 19,
    UA_ATTRIBUTEID_HISTORIZING       = 20,
    UA_ATTRIBUTEID_EXECUTABLE        = 21,
    UA_ATTRIBUTEID_USEREXECUTABLE    = 22,
    UA_ATTRIBUTEID_DATATYPEDEFINITION = 23,
    UA_ATTRIBUTEID_ROLEPERMISSIONS   = 24,
    UA_ATTRIBUTEID_USERROLEPERMISSIONS = 25,
    UA_ATTRIBUTEID_ACCESSRESTRICTIONS = 26,
    UA_ATTRIBUTEID_ACCESSLEVELEX     = 27
} UA_AttributeId;
```

Returns a readable attribute name like “NodeId” or “Invalid” if the attribute does not exist.

```
const char *
UA_AttributeId_name(UA_AttributeId attrId);
```

10.2 Access Level Masks

The access level to a node is given by the following constants that are ANDed with the overall access level.

```
#define UA_ACCESSLEVELMASK_READ           (0x01u << 0u)
#define UA_ACCESSLEVELMASK_CURRENTREAD    (0x01u << 0u)
#define UA_ACCESSLEVELMASK_WRITE          (0x01u << 1u)
#define UA_ACCESSLEVELMASK_CURRENTWRITE   (0x01u << 1u)
#define UA_ACCESSLEVELMASK_HISTORYREAD    (0x01u << 2u)
#define UA_ACCESSLEVELMASK_HISTORYWRITE   (0x01u << 3u)
#define UA_ACCESSLEVELMASK_SEMANTICCHANGE (0x01u << 4u)
#define UA_ACCESSLEVELMASK_STATUSWRITE    (0x01u << 5u)
#define UA_ACCESSLEVELMASK_TIMESTAMPWRITE (0x01u << 6u)
```

10.3 Write Masks

The write mask and user write mask is given by the following constants that are ANDed for the overall write mask. Part 3: 5.2.7 Table 2

```
#define UA_WRITEMASK_ACCESSLEVEL             (0x01u << 0u)
#define UA_WRITEMASK_ARRAYDIMENSIONS         (0x01u << 1u)
#define UA_WRITEMASK_ARRRAYDIMENSIONS        UA_WRITEMASK_
↪ARRAYDIMENSIONS /* legacy typo alias */
#define UA_WRITEMASK_BROWSENAME              (0x01u << 2u)
#define UA_WRITEMASK_CONTAINSNOLOOPS         (0x01u << 3u)
#define UA_WRITEMASK_DATATYPE                (0x01u << 4u)
#define UA_WRITEMASK_DESCRIPTION             (0x01u << 5u)
#define UA_WRITEMASK_DISPLAYNAME             (0x01u << 6u)
#define UA_WRITEMASK_EVENTNOTIFIER           (0x01u << 7u)
#define UA_WRITEMASK_EXECUTABLE              (0x01u << 8u)
#define UA_WRITEMASK_HISTORIZING             (0x01u << 9u)
#define UA_WRITEMASK_INVERSENAME             (0x01u << 10u)
#define UA_WRITEMASK_ISABSTRACT              (0x01u << 11u)
#define UA_WRITEMASK_MINIMUMSAMPLINGINTERVAL (0x01u << 12u)
#define UA_WRITEMASK_NODECLASS               (0x01u << 13u)
#define UA_WRITEMASK_NODEID                  (0x01u << 14u)
#define UA_WRITEMASK_SYMMETRIC               (0x01u << 15u)
#define UA_WRITEMASK_USERACCESSLEVEL         (0x01u << 16u)
#define UA_WRITEMASK_USEREXECUTABLE          (0x01u << 17u)
#define UA_WRITEMASK_USERWRITEMASK           (0x01u << 18u)
#define UA_WRITEMASK_VALUERANK               (0x01u << 19u)
#define UA_WRITEMASK_WRITEMASK               (0x01u << 20u)
#define UA_WRITEMASK_VALUEFORVARIABLETYPE    (0x01u << 21u)
#define UA_WRITEMASK_DATATYPEDEFINITION      (0x01u << 22u)
#define UA_WRITEMASK_ROLEPERMISSIONS         (0x01u << 23u)
```

(continues on next page)

(continued from previous page)

```
#define UA_WRITEMASK_ACCESSRESTRICTIONS      (0x01u << 24u)
#define UA_WRITEMASK_ACCESSLEVELEX           (0x01u << 25u)
```

10.4 ValueRank

The following are the most common ValueRanks used for Variables, VariableTypes and method arguments. ValueRanks higher than 3 are valid as well (but less common).

```
#define UA_VALUERANK_SCALAR_OR_ONE_DIMENSION  -3
#define UA_VALUERANK_ANY                      -2
#define UA_VALUERANK_SCALAR                   -1
#define UA_VALUERANK_ONE_OR_MORE_DIMENSIONS    0
#define UA_VALUERANK_ONE_DIMENSION             1
#define UA_VALUERANK_TWO_DIMENSIONS            2
#define UA_VALUERANK_THREE_DIMENSIONS          3
```

10.5 EventNotifier

The following are the available EventNotifier used for Nodes. The EventNotifier Attribute is used to indicate if the Node can be used to subscribe to Events or to read / write historic Events. Part 3: 5.4 Table 10

```
#define UA_EVENTNOTIFIER_SUBSCRIBE_TO_EVENT (0x01u << 0u)
#define UA_EVENTNOTIFIER_HISTORY_READ       (0x01u << 2u)
#define UA_EVENTNOTIFIER_HISTORY_WRITE      (0x01u << 3u)
```

10.6 Server Capabilities

The ServerCapabilities announce the features of a server during discovery. The identifiers are defined in Part 12, Annex D. The normative list is maintained by the OPC Foundation in http://www.opcfoundation.org/UA/schemas/ServerCapabilities.csv.

```
#define UA_SERVERCAPABILITY_NA     (0x01ULL << 0u)  /* Not available */
#define UA_SERVERCAPABILITY_DA     (0x01ULL << 1u)  /* Current data */
#define UA_SERVERCAPABILITY_HD     (0x01ULL << 2u)  /* Historical data */
#define UA_SERVERCAPABILITY_AC     (0x01ULL << 3u)  /* Alarms and Conditions */
#define UA_SERVERCAPABILITY_HE     (0x01ULL << 4u)  /* Historical events */
#define UA_SERVERCAPABILITY_GDS    (0x01ULL << 5u)  /* Global Discovery */
#define UA_SERVERCAPABILITY_LDS    (0x01ULL << 6u)  /* Discovery only */
#define UA_SERVERCAPABILITY_DI     (0x01ULL << 7u)  /* Device Integration */
#define UA_SERVERCAPABILITY_ADI    (0x01ULL << 8u)  /* Analyser Devices */
#define UA_SERVERCAPABILITY_FDI    (0x01ULL << 9u)  /* Field Devices */
#define UA_SERVERCAPABILITY_FDIC   (0x01ULL << 10u) /* FDI Communication */
#define UA_SERVERCAPABILITY_PLC    (0x01ULL << 11u) /* PLCopen */
#define UA_SERVERCAPABILITY_S95    (0x01ULL << 12u) /* ISA-95 */
#define UA_SERVERCAPABILITY_RCP    (0x01ULL << 13u) /* Reverse Connect */
#define UA_SERVERCAPABILITY_PUB    (0x01ULL << 14u) /* PubSub Publisher */
#define UA_SERVERCAPABILITY_NTRS   (0x01ULL << 15u) /* Redundant Server Set */
```

(continues on next page)

(continued from previous page)

```
#define UA_SERVERCAPABILITY_AUTOID (0x01ULL << 16u) /* AutoID */
#define UA_SERVERCAPABILITY_MDIS   (0x01ULL << 17u) /* MDIS */
#define UA_SERVERCAPABILITY_CNC    (0x01ULL << 18u) /* CNC Systems */
#define UA_SERVERCAPABILITY_PLK    (0x01ULL << 19u) /* POWERLINK */
#define UA_SERVERCAPABILITY_FDT    (0x01ULL << 20u) /* FDT */
#define UA_SERVERCAPABILITY_TMC    (0x01ULL << 21u) /* Tobacco Machines */
#define UA_SERVERCAPABILITY_CSPP   (0x01ULL << 22u) /* CSP+ for Machine */
#define UA_SERVERCAPABILITY_61850  (0x01ULL << 23u) /* IEC 61850 */
#define UA_SERVERCAPABILITY_PACKML (0x01ULL << 24u) /* PackML */
#define UA_SERVERCAPABILITY_MTC    (0x01ULL << 25u) /* MTConnect */
#define UA_SERVERCAPABILITY_AUTOML (0x01ULL << 26u) /* AutomationML */
#define UA_SERVERCAPABILITY_SERCOS (0x01ULL << 27u) /* Sercos */
#define UA_SERVERCAPABILITY_MIMOSA (0x01ULL << 28u) /* MIMOSA */
#define UA_SERVERCAPABILITY_WITSML (0x01ULL << 29u) /* WITSML */
#define UA_SERVERCAPABILITY_DEXPI  (0x01ULL << 30u) /* DEXPI */
#define UA_SERVERCAPABILITY_IOLINK (0x01ULL << 31u) /* IO-Link */
#define UA_SERVERCAPABILITY_VROBOT (0x01ULL << 32u) /* VDMA Robotics */
#define UA_SERVERCAPABILITY_PNO    (0x01ULL << 33u) /* PROFINET */
#define UA_SERVERCAPABILITY_PADIM  (0x01ULL << 34u) /* Process Automation */
#define UA_SERVERCAPABILITY_ALIAS  (0x01ULL << 35u) /* AliasNames */
#define UA_SERVERCAPABILITY_SKS    (0x01ULL << 36u) /* Security Key Server */
#define UA_SERVERCAPABILITY_FXAC   (0x01ULL << 37u) /* FX␣
↪AutomationComponent */
#define UA_SERVERCAPABILITY_FXCM   (0x01ULL << 38u) /* FX ConnectionManager␣
↪*/
```

10.7 Rule Handling

The RuleHanding settings define how error cases that result from rules in the OPC UA specification shall be handled. The rule handling can be softened, e.g. to workaround misbehaving implementations or to mitigate the impact of additional rules that are introduced in later versions of the OPC UA specification.

```
typedef enum {
    UA_RULEHANDLING_DEFAULT = 0,
    UA_RULEHANDLING_ABORT,  /* Abort the operation and return an error code */
    UA_RULEHANDLING_WARN,   /* Print a message in the logs and continue */
    UA_RULEHANDLING_ACCEPT, /* Continue and disregard the broken rule */
} UA_RuleHandling;
```

10.8 Order

The Order enum is used to establish an absolute ordering between elements.

```
typedef enum {
    UA_ORDER_LESS = -1,
    UA_ORDER_EQ = 0,
    UA_ORDER_MORE = 1
```

(continues on next page)

(continued from previous page)

```
} UA_Order;
```

10.9 Application Notification

The ApplicationNotification mechanism is for runtime notifications where the server/client wants to make the local application aware of state changes and internal events.

The identifier for the different notifications is a UA_UInt64 integer. The high 32bits contain a bitfield for the notification type. The low 32bits are for detailed differentiation within each type. This allows for easy filtering with bitwise operations.

The notifications comes with a key-value map for the payload. Future additional payload members are added to the end of the payload. So that the names, type and also index of the payload members is stable.

```
typedef uint64_t UA_ApplicationNotificationType;

/* Lifecycle notifications, no payload */
#define UA_APPLICATIONNOTIFICATIONTYPE_LIFECYCLE          \
    (0x01ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_LIFECYCLE_STARTED  \
    ((0x01ULL << 32) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_LIFECYCLE_SHUTDOWN \
    ((0x01ULL << 32) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_LIFECYCLE_STOPPING \
    ((0x01ULL << 32) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_LIFECYCLE_STOPPED  \
    ((0x01ULL << 32) | 0x04)
```

(Server only) Give background information after a SecureChannel is opened or closed.

0:securechannel-id [UInt32]
Identifier of the SecureChannel to which the Session is connected.

0:connection-manager-name [String]
Name of the ConnectionManager (configured in the EventLoop) from which the connection was opened.

0:connection-id [UInt64]
Identifier of the connection in the context of the EventLoop. This is often the socket identifier, but that is not necessarily the case.

0:remote-address [String]
Address (hostname or IP that opened the SecureChannel.

0:protocol-version [UInt32]
The version of the UACP protocol requested by the Client.

0:recv-buffer-size [UInt32]
The largest buffer (chunk size) we can receive over the channel.

0:recv-max-message-size [UInt32]
The maximum size of received messages.

0:recv-max-chunk-count [UInt32]
The maximum number of chunks for received messages.

0:send-buffer-size [UInt32]
The largest buffer (chunk size) we can send over the channel.

0:send-max-message-size [UInt32]
The maximum size of sent messages.

0:send-max-chunk-count [UInt32]
The maximum number of chunks for sent messages.

0:endpoint-url
The target EndpointUri (for the server) indicated by the client.

0:security-mode [MessageSecurityMode]
The SecurityChannel can be unsigned, signed or signed+encrypted.

0:security-policy-uri [String]
Uri of the SecurityPolicy for this SecyrityChannel.

0:certificate-type-id [NodeId]
Certificate type used by the SecurityPolicy for this SecureChannel.

0:remote-certificate [ByteString]
Certificate used by the remote side during OpenSecureChannel.

```
#define UA_APPLICATIONNOTIFICATIONTYPE_SECURECHANNEL       \
    (0x02ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_SECURECHANNEL_OPENED \
    ((0x02ULL << 32) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_SECURECHANNEL_CLOSED \
    ((0x02ULL << 32) | 0x02)
```

(Server only) Give background information for Sessions. The _DEACTIVATE notification occurs when a Session is unbound from its original SecureChannel. Either because the SecureChannel is closed or because the session is activated on another SecureChannel.

0:session-id [NodeId]
Identifier of the Session.

0:securechannel-id [UInt32]
Identifier of the SecureChannel on which the Session is activated. Zero if the Session is not bound to any SecureChannel.

0:session-name [String]
Name of the Session as defined by the client.

0:client-description [ApplicationDescription]
Name of the Session as defined by the client.

0:client-user-id [String]
User identifier used to activate the session. This is extracted from the UserIdentityToken (e.g. username but not the password).

0:locale-ids [Array of String]
List of preferred languages.

Any additional attributes set via UA_Server_setSessionAttribute are appended to the above list.

```
#define UA_APPLICATIONNOTIFICATIONTYPE_SESSION            \
    (0x04ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_SESSION_CREATED    \
    ((0x04ULL << 32) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_SESSION_ACTIVATED  \
    ((0x04ULL << 32) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_SESSION_DEACTIVATED \
    ((0x04ULL << 32) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_SESSION_CLOSED     \
    ((0x04ULL << 32) | 0x04)
```

Processing of a service request or response. The server-side processing of a request can be asynchronous. The existence of a yet-unfinished async operation from the request is signaled with the _SERVICE_ASYNC enum. The _SERVICE_END enum is signalled eventually, once all async operations from the service request are completed.

0:securechannel-id [UInt32]
Identifier of the SecureChannel to which the Session is connected.

0:session-id [NodeId]
Identifier of the Session for/from which the Service is requested. This is the ns=0;i=0 NodeId if no Session is bound to the receiving SecureChannel.

0:request-id [UInt32]
Request/Response correlation identifier. On the server this is zero for transports that do not carry a RequestId.

0:service-type [NodeId]
DataType identifier for the Request (server) or Response (client).

```
#define UA_APPLICATIONNOTIFICATIONTYPE_SERVICE      \
    (0x08ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_SERVICE_BEGIN \
    ((0x08ULL << 32) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_SERVICE_ASYNC \
    ((0x08ULL << 32) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_SERVICE_END  \
    ((0x08ULL << 32) | 0x03)
```

(Server only) Signals the creation or modification of a Subscription.

0:session-id [NodeId]
Identifier of the Session for which the Subscription is created. If the subscription is not bound to any Session, then the NodeId ns=0;i=0 is returned.

0:subscription-id [UInt32]
Identifier of the Subscription (unique for the Session).

0:publishing-interval [Double]
Frequency at which accumulated notifications are sent out.

0:lifetime-count [UInt32]
Number of consecutive publishing interval with a missing PublishRequest before the Subscription is starved (deleted).

0:max-keepalive-count [UInt32]
Number of consecutive publishing intervals without a PublishResponse before a keepalive is sent.

0:max-notifications-per-publish [UInt32]
Number of notifications that can be in a PublishResponse.

0:priority [Byte]
Higher-priority subscriptions send out PublishResponses first.

0:publishing-enabled [Boolean]
What the name says.

```
#define UA_APPLICATIONNOTIFICATIONTYPE_SUBSCRIPTION                \
    (0x10ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_SUBSCRIPTION_CREATED        \
    ((0x10ULL << 32) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_SUBSCRIPTION_MODIFIED       \
    ((0x10ULL << 32) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_SUBSCRIPTION_
↪PUBLISHINGMODE \
    ((0x10ULL << 32) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_SUBSCRIPTION_TRANSFERRED␣
↪  \
    ((0x10ULL << 32) | 0x04)
#define UA_APPLICATIONNOTIFICATIONTYPE_SUBSCRIPTION_DELETED        \
    ((0x10ULL << 32) | 0x05)
```

(Server only) Signals the creation or modification of a MonitoredItem.

0:session-id [NodeId]
Identifier of the Session for which the Subscription is created. If the subscription is not bound to any Session, then the NodeId ns=0;i=0 is returned.

0:subscription-id [UInt32]
Identifier of the Subscription (unique for the Session).

0:monitoreditem-id [UInt32]
Identifier of the MonitoredItem (unique for the Subscription).

0:target-node [NodeId]
Identifier of the Node that is monitored.

0:attribute-id [UInt32]
Node-attribute that is being monitored. Conforms to the values from the UA_AttributeId enum.

0:index-range [String]
Defines if only part of an array value is monitored.

0:timestamps-to-return [TimestampsToReturn]
Enum with the options SOURCE | SERVER | BOTH | NEITHER.

0:monitorimg-mode [MonitoringMode]
Enum with the options DISABLED | SAMPLING | REPORTING.

0:client-handle [UInt32]
Client-supplied identifier of the MonitoredItem.

0:samping-interval [Double]
Interval to evaluate the MonitoredItem in milliseconds.

0:filter [Empty | DataChangeFilter | EventFilter | AggregateFilter]
The filter used to emit notifications.

0:queue-size [UInt32]
Maximum number of notifications waiting to be published.

0:discard-oldest [Boolean]
When the queue overflows, delete the newest or the oldest notification.

```
#define UA_APPLICATIONNOTIFICATIONTYPE_MONITOREDITEM                \
    (0x20ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_MONITOREDITEM_CREATED
↪ \
    ((0x20ULL << 32) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_MONITOREDITEM_MODIFIED
↪ \
    ((0x20ULL << 32) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_MONITOREDITEM_
↪MONITORINGMODE \
    ((0x20ULL << 32) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_MONITOREDITEM_DELETED
↪ \
    ((0x20ULL << 32) | 0x04)
```

(Server only) Signals the creation of an audit event.

The key-value map for audit application notifications follows the properties defined for the AuditEventType and its subtypes. The key-string is the human-readable encoding for the SimpleAttributeOperand of the event properties (value attribute only). The properties should be looked up from their string-key and not via their order-index in the key-value map.

Examples properties are (datatype in brackets):

- /ActionTimeStamp [DateTime]
- **/Status** [Boolean]
- **/ServerId** [String]

See the definition of the AuditEventType and its subtypes in part 5 of the OPC UA specification. Below follows the hierarchy of the event types. Properties from the super-type are inherited.

- AuditEventType
  - AuditSecurityEventType
    - AuditChannelEventType
      - AuditOpenSecureChannelEventType
    - AuditSessionEventType
      - AuditCreateSessionEventType
      - AuditActivateSessionEventType
      - AuditCancelEventType

        * AuditCertificateEventType
            - AuditCertificateDataMismatchEventType
            - AuditCertificateExpiredEventType
            - AuditCertificateInvalidEventType
            - AuditCertificateUntrustedEventType
            - AuditCertificateRevokedEventType
            - AuditCertificateMismatchEventType
    - AuditNodeManagementEventType
        * AuditAddNodesEventType
        * AuditDeleteNodesEventType
        * AuditAddReferencesEventType
        * AuditDeleteReferencesEventType
    - AuditUpdateEventType
        * AuditWriteUpdateEventType
        * AuditHistoryUpdateEventType
    - AuditUpdateMethodEventType
    - AuditClientEventType
        * AuditClientUpdateMethodResultEventType

```
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT                                  \
    (0x40ULL << 32)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY                         ␣
↪    \
    ((0x40ULL << 32) | (0x01 << 16))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_CHANNEL                 ␣
↪           \
    ((0x40ULL << 32) | (0x01 << 16) | (0x01 << 8))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_CHANNEL_
↪OPEN            \
    ((0x40ULL << 32) | (0x01 << 16) | (0x01 << 8) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_SESSION                 ␣
↪         \
    ((0x40ULL << 32) | (0x01 << 16) | (0x02 << 8))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_SESSION_
↪CREATE          \
    ((0x40ULL << 32) | (0x01 << 16) | (0x02 << 8) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_SESSION_
↪ACTIVATE        \
    ((0x40ULL << 32) | (0x01 << 16) | (0x02 << 8) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_SESSION_
↪CANCEL          \
    ((0x40ULL << 32) | (0x01 << 16) | (0x02 << 8) | 0x03)
```

(continues on next page)

(continued from previous page)

```
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE              \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE_DATAMISMATCH \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE_EXPIRED      \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE_INVALID      \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE_UNTRUSTED    \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8) | 0x04)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE_REVOKED      \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8) | 0x05)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_SECURITY_
↪CERTIFICATE_MISMATCH     \
   ((0x40ULL << 32) | (0x01 << 16) | (0x04 << 8) | 0x06)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_NODE                ␣
↪ \
   ((0x40ULL << 32) | (0x02 << 16))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_NODE_ADD            ␣
↪     \
   ((0x40ULL << 32) | (0x02 << 16) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_NODE_DELETE         ␣
↪       \
   ((0x40ULL << 32) | (0x02 << 16) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_NODE_
↪ADDREFERENCES            \
   ((0x40ULL << 32) | (0x02 << 16) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_NODE_
↪DELETEREFERENCES         \
   ((0x40ULL << 32) | (0x02 << 16) | 0x04)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_UPDATE              ␣
↪   \
   ((0x40ULL << 32) | (0x04 << 16))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_UPDATE_WRITE        ␣
↪       \
   ((0x40ULL << 32) | (0x04 << 16) | 0x01)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_UPDATE_HISTORY      ␣
↪         \
   ((0x40ULL << 32) | (0x04 << 16) | 0x02)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_UPDATE_METHOD       ␣
↪          \
   ((0x40ULL << 32) | (0x04 << 16) | 0x03)
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_CLIENT              ␣
```

(continues on next page)

(continued from previous page)

```
↪ \
   ((0x40ULL << 32) | (0x08 << 16))
#define UA_APPLICATIONNOTIFICATIONTYPE_AUDIT_CLIENT_
↪UPDATEMETHOD                \
   ((0x40ULL << 32) | (0x08 << 16) | 0x01)
```

(Server only) Signals for the Discovery service set.

We uniquely identify servers by the combination of the following information. This is used to decide whether to update an existing record or whether to create a new one.

- For FindServers: ServerUri + one matching DiscoveryUrl
- For FindServerOnNetwork: ServerName + DiscoveryUrl

```
#define UA_APPLICATIONNOTIFICATIONTYPE_DISCOVERY (0x80ULL << 32)

/* A server was added via the RegisterServer service or the local API.
 * Updates and removals are also notified.
 *
 * 0:registered-server [RegisteredServer]
 *    Received server information.
 * 0:discovery-configuration [Array of ExtensionObject]
 *    Additional data, typically a UA_MdnsDiscoveryConfiguration.
 * 0:server-added [Boolean]
 *    They entry was newly added.
 * 0:server-updated [Boolean]
 *    An existing entry was updated.
 * 0:server-removed [Boolean]
 *    The entry was removed.
 * 0:securechannel-id [UInt32]
 *    Identifier of the SecureChannel from which the information was
 *    received (0 for locally triggered operations).
 * 0:session-id [NodeId]
 *    Identifier of the Session that called the RegisterServer service.
 *    The Null-NodeId if a SecureChannel without Session made the call. */
#define UA_APPLICATIONNOTIFICATIONTYPE_DISCOVERY_REGISTERSERVER \
   ((0x80ULL << 32) | 0x01)

/* Information about a server over multicast DNS or the local API.
 * Updates over time and removal is also notified (e.g after the DNS
 * TTL (time-to-live) runs out).
 *
 * 0:server-on-network [ServerOnNetwork] Server information received.
 *    (The RecordId datatype member is defined internally in the
 *    server. It does not have semantic meaning here.)
 * 0:remote-address [String]
 *    IP-address or other host identifier from which the information
 *    was received.
 * 0:ttl [UInt32]
 *    Time-to-live of DNS information. Zero means infinite
```

(continues on next page)

(continued from previous page)

```
 *    (if the server is not currently getting removed).
 * 0:server-added [Boolean]
 *    They entry was added.
 * 0:server-updated [Boolean]
 *    An existing entry was updated.
 * 0:server-removed [Boolean]
 *    The entry was removed. */
#define UA_APPLICATIONNOTIFICATIONTYPE_DISCOVERY_
↪SERVERONNETWORK \
   ((0x80ULL << 32) | 0x02)
```

10.10 Connection State

```
typedef enum {
   UA_CONNECTIONSTATE_CLOSED,     /* The socket has been closed and the␣
↪connection
                                   * will be deleted */
   UA_CONNECTIONSTATE_OPENING,    /* The socket is open, but the connection␣
↪not yet
                                     fully established */
   UA_CONNECTIONSTATE_ESTABLISHED,/* The socket is open and the connection
                                   * configured */
   UA_CONNECTIONSTATE_CLOSING,    /* The socket is closing down */
   UA_CONNECTIONSTATE_BLOCKING,   /* Listening disabled (e.g. max connections␣
↪reached) */
   UA_CONNECTIONSTATE_REOPENING   /* Listening resumed after being blocked */
} UA_ConnectionState;

typedef enum {
   UA_SECURECHANNELSTATE_CLOSED = 0,
   UA_SECURECHANNELSTATE_REVERSE_LISTENING,
   UA_SECURECHANNELSTATE_CONNECTING,
   UA_SECURECHANNELSTATE_CONNECTED,
   UA_SECURECHANNELSTATE_REVERSE_CONNECTED,
   UA_SECURECHANNELSTATE_RHE_SENT,
   UA_SECURECHANNELSTATE_HEL_SENT,
   UA_SECURECHANNELSTATE_HEL_RECEIVED,
   UA_SECURECHANNELSTATE_ACK_SENT,
   UA_SECURECHANNELSTATE_ACK_RECEIVED,
   UA_SECURECHANNELSTATE_OPN_SENT,
   UA_SECURECHANNELSTATE_OPEN,
   UA_SECURECHANNELSTATE_CLOSING,
} UA_SecureChannelState;

typedef enum {
   UA_SESSIONSTATE_CLOSED = 0,
   UA_SESSIONSTATE_CREATE_REQUESTED,
   UA_SESSIONSTATE_CREATED,
```

(continues on next page)

(continued from previous page)

```
    UA_SESSIONSTATE_ACTIVATE_REQUESTED,
    UA_SESSIONSTATE_ACTIVATED,
    UA_SESSIONSTATE_CLOSING
} UA_SessionState;
```

10.11 Statistic Counters

The stack manages statistic counters for SecureChannels and Sessions.

The Session layer counters are matching the counters of the ServerDiagnosticsSummaryDataType that are defined in the OPC UA Part 5 specification. The SecureChannel counters are not defined in the OPC UA spec, but are harmonized with the Session layer counters if possible.

```
typedef enum {
    UA_SHUTDOWNREASON_CLOSE = 0,
    UA_SHUTDOWNREASON_REJECT,
    UA_SHUTDOWNREASON_SECURITYREJECT,
    UA_SHUTDOWNREASON_TIMEOUT,
    UA_SHUTDOWNREASON_ABORT,
    UA_SHUTDOWNREASON_PURGE
} UA_ShutdownReason;

typedef struct {
    size_t currentChannelCount;
    size_t cumulatedChannelCount;
    size_t rejectedChannelCount;
    size_t channelTimeoutCount; /* only used by servers */
    size_t channelAbortCount;
    size_t channelPurgeCount;   /* only used by servers */
} UA_SecureChannelStatistics;

typedef struct {
    size_t currentSessionCount;
    size_t cumulatedSessionCount;
    size_t securityRejectedSessionCount; /* only used by servers */
    size_t rejectedSessionCount;
    size_t sessionTimeoutCount;          /* only used by servers */
    size_t sessionAbortCount;            /* only used by servers */
} UA_SessionStatistics;
```

10.12 Lifecycle States

Generic lifecycle states. The STOPPING state indicates that the lifecycle is being terminated. But it might take time to (asynchronously) perform a graceful shutdown.

```
typedef enum {
    UA_LIFECYCLESTATE_STOPPED = 0,
    UA_LIFECYCLESTATE_STARTED,
```

(continues on next page)

(continued from previous page)

```
    UA_LIFECYCLESTATE_STOPPING
} UA_LifecycleState;
```

CHAPTER

ELEVEN

UTILITY DEFINITIONS

Utility functions are used by both client and server. Different from the *Common Definitions*, the utility functions can use the *Data Types*.

11.1 Range Definition

```
typedef struct {
    UA_UInt32 min;
    UA_UInt32 max;
} UA_UInt32Range;

typedef struct {
    UA_Duration min;
    UA_Duration max;
} UA_DurationRange;
```

11.2 Event-Filter Parsing

```
#ifdef UA_ENABLE_SUBSCRIPTIONS_EVENTS
#ifdef UA_ENABLE_JSON_ENCODING

typedef struct {
    const UA_Logger *logger;
} UA_EventFilterParserOptions;

UA_StatusCode
UA_EventFilter_parse(UA_EventFilter *filter, UA_ByteString content,
                     UA_EventFilterParserOptions *options);

#endif
#endif
```

11.3 Random Number Generator

If UA_MULTITHREADING is defined, then the seed is stored in thread local storage. The seed is initialized for every thread in the server/client.

```
/* Initialize the RNG with the seed number and UA_DateTime_now() */
void
UA_random_seed(UA_UInt64 seed);

/* Initialize the RNG with the seed number (only) */
void
UA_random_seed_deterministic(UA_UInt64 seed);

UA_UInt32
UA_UInt32_random(void); /* no cryptographic entropy */

UA_Guid
UA_Guid_random(void);   /* no cryptographic entropy */
```

11.4 Translate between DataTypeDescription and UA_DataType

The ExtensionObject for the description is either of

- SimpleTypeDescription
- EnumDescription
- StructureDescription

The generated UA_DataType faithfully generates the padding of the corresponding C-structures.

```
#ifdef UA_TYPES_STRUCTUREDESCRIPTION
UA_StatusCode
UA_DataType_fromDescription(UA_DataType *type, const UA_ExtensionObject *descr,
                            const UA_DataTypeArray *customTypes);

UA_StatusCode
UA_DataType_toDescription(const UA_DataType *type, UA_ExtensionObject *descr);
#endif
```

11.5 Memory Handling

```
/* Force casting a pointer. Attention, unsafe! */
#define UA_FORCE_CAST_PTR(t, p) ((t)(uintptr_t)p)
```

11.6 Key Value Map

Helper functions to work with configuration parameters in an array of UA_KeyValuePair. Lookup is linear. So this is for small numbers of keys. The methods below that accept a *const UA_KeyValueMap* as an argument also accept NULL for that argument and treat it as an empty map.

```
/* The layout is identical to UA_AdditionalParametersType (casting possible) */
typedef struct {
    size_t mapSize;
    UA_KeyValuePair *map;
} UA_KeyValueMap;

extern const UA_KeyValueMap UA_KEYVALUEMAP_NULL;

UA_KeyValueMap *
UA_KeyValueMap_new(void);

void
UA_KeyValueMap_clear(UA_KeyValueMap *map);

void
UA_KeyValueMap_delete(UA_KeyValueMap *map);

/* Is the map empty (or NULL)? */
UA_Boolean
UA_KeyValueMap_isEmpty(const UA_KeyValueMap *map);

/* Does the map contain an entry for the key? */
UA_Boolean
UA_KeyValueMap_contains(const UA_KeyValueMap *map, const UA_QualifiedName key);

/* Insert a copy of the value. Can reallocate the underlying array. This
 * invalidates pointers into the previous array. If the key exists already, the
 * value is overwritten (upsert semantics). */
UA_StatusCode
UA_KeyValueMap_set(UA_KeyValueMap *map,
                   const UA_QualifiedName key,
                   const UA_Variant *value);

/* The same as _set, but inserts a shallow copy of the value. Set
 * UA_VARIANT_DATA_NODELETE if the value should not be _cleared together with
 * the map. */
UA_StatusCode
UA_KeyValueMap_setShallow(UA_KeyValueMap *map,
                          const UA_QualifiedName key,
                          UA_Variant *value);

/* Helper function for scalar insertion that internally calls
 * `UA_KeyValueMap_set` */
UA_StatusCode
UA_KeyValueMap_setScalar(UA_KeyValueMap *map,
                         const UA_QualifiedName key,
                         const void *p,
                         const UA_DataType *type);

UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_KeyValueMap_setScalarShallow(UA_KeyValueMap *map,
                                const UA_QualifiedName key,
                                void *p,
                                const UA_DataType *type);

/* Returns a pointer to the value or NULL if the key is not found */
const UA_Variant *
UA_KeyValueMap_get(const UA_KeyValueMap *map,
                   const UA_QualifiedName key);

/* Returns NULL if the value for the key is not defined, not of the right
 * datatype or not a scalar */
const void *
UA_KeyValueMap_getScalar(const UA_KeyValueMap *map,
                         const UA_QualifiedName key,
                         const UA_DataType *type);

/* Remove a single entry. To delete the entire map, use `UA_KeyValueMap_clear`. */
UA_StatusCode
UA_KeyValueMap_remove(UA_KeyValueMap *map,
                      const UA_QualifiedName key);

/* Create a deep copy of the given KeyValueMap */
UA_StatusCode
UA_KeyValueMap_copy(const UA_KeyValueMap *src, UA_KeyValueMap *dst);

/* Copy entries from the right-hand-side into the left-hand-size. Reallocates
 * previous memory in the left-hand-side. If the operation fails, both maps are
 * left untouched. */
UA_StatusCode
UA_KeyValueMap_merge(UA_KeyValueMap *lhs, const UA_KeyValueMap *rhs);
```

11.7 Binary Connection Config Parameters

```
typedef struct {
    UA_UInt32 protocolVersion;
    UA_UInt32 recvBufferSize;
    UA_UInt32 sendBufferSize;
    UA_UInt32 localMaxMessageSize;  /* (0 = unbounded) */
    UA_UInt32 remoteMaxMessageSize; /* (0 = unbounded) */
    UA_UInt32 localMaxChunkCount;   /* (0 = unbounded) */
    UA_UInt32 remoteMaxChunkCount;  /* (0 = unbounded) */
} UA_ConnectionConfig;
```

11.8 Default Node Attributes

Default node attributes to simplify the use of the AddNodes services. For example, Setting the ValueRank and AccessLevel to zero is often an unintended setting and leads to errors that are hard to track down.

```
/* The default for variables is "BaseDataType" for the datatype, -2 for the
 * valuerank and a read-accesslevel. */
extern const UA_VariableAttributes UA_VariableAttributes_default;
extern const UA_VariableTypeAttributes UA_VariableTypeAttributes_default;

/* Methods are executable by default */
extern const UA_MethodAttributes UA_MethodAttributes_default;

/* The remaining attribute definitions are currently all zeroed out */
extern const UA_ObjectAttributes UA_ObjectAttributes_default;
extern const UA_ObjectTypeAttributes UA_ObjectTypeAttributes_default;
extern const UA_ReferenceTypeAttributes UA_ReferenceTypeAttributes_default;
extern const UA_DataTypeAttributes UA_DataTypeAttributes_default;
extern const UA_ViewAttributes UA_ViewAttributes_default;
```

11.9 Endpoint URL Parser

The endpoint URL parser is generally useful for the implementation of network layer plugins.

```
/* Split the given endpoint url into hostname, port and path. All arguments must
 * be non-NULL. EndpointUrls have the form "opc.tcp://hostname:port/path", port
 * and path may be omitted (together with the prefix colon and slash).
 *
 * @param endpointUrl The endpoint URL.
 * @param outHostname Set to the parsed hostname. The string points into the
 *        original endpointUrl, so no memory is allocated. If an IPv6 address is
 *        given, hostname contains e.g. '[2001:0db8:85a3::8a2e:0370:7334]'
 * @param outPort Set to the port of the url or left unchanged.
 * @param outPath Set to the path if one is present in the endpointUrl. Can be
 *        NULL. Then not path is returned. Starting or trailing '/' are NOT
 *        included in the path. The string points into the original endpointUrl,
 *        so no memory is allocated.
 * @return Returns UA_STATUSCODE_BADTCPENDPOINTURLINVALID if parsing␣
↪failed. */
UA_StatusCode
UA_parseEndpointUrl(const UA_String *endpointUrl, UA_String *outHostname,
                    UA_UInt16 *outPort, UA_String *outPath);

/* Split the given endpoint url into hostname, vid and pcp. All arguments must
 * be non-NULL. EndpointUrls have the form "opc.eth://<host>[:<VID>[.PCP]]".
 * The host is a MAC address, an IP address or a registered name like a
 * hostname. The format of a MAC address is six groups of hexadecimal digits,
 * separated by hyphens (e.g. 01-23-45-67-89-ab). A system may also accept
 * hostnames and/or IP addresses if it provides means to resolve it to a MAC
```

(continues on next page)

(continued from previous page)

```
 * address (e.g. DNS and Reverse-ARP).
 *
 * Note: currently only parsing MAC address is supported.
 *
 * @param endpointUrl The endpoint URL.
 * @param vid Set to VLAN ID.
 * @param pcp Set to Priority Code Point.
 * @return Returns UA_STATUSCODE_BADINTERNALERROR if parsing failed. */
UA_StatusCode
UA_parseEndpointUrlEthernet(const UA_String *endpointUrl, UA_String *target,
                            UA_UInt16 *vid, UA_Byte *pcp);

/* Convert given byte string to a positive number. Returns the number of valid
 * digits. Stops if a non-digit char is found and returns the number of digits
 * up to that point. */
size_t
UA_readNumber(const UA_Byte *buf, size_t buflen, UA_UInt32 *number);

/* Same as UA_ReadNumber but with a base parameter */
size_t
UA_readNumberWithBase(const UA_Byte *buf, size_t buflen,
                      UA_UInt32 *number, UA_Byte base);

#ifndef UA_MIN
#define UA_MIN(A, B) ((A) > (B) ? (B) : (A))
#endif

#ifndef UA_MAX
#define UA_MAX(A, B) ((A) > (B) ? (A) : (B))
#endif
```

11.10 Escaping of Strings

11.10.1 And-Escaping

The “and-escaping” of strings is described in Part 4, A2. The & character is used to escape the reserved characters /.<>:#!&. So the string My.String becomes My&.String.

In addition to the standard we define “extended-and-escaping” where additionally commas, semicolons, brackets and whitespace characters are escaped. This improves the parsing in a larger context, as a lexer can find the end of the escaped string. The additionally reserved characters for the extended escaping are ,()[] \t\n\v\f\r.

This documentation always states whether “and-escaping” or the “extended-and-escaping is used.

11.10.2 Percent-Escaping

Percent-escaped characters get encoded as %xx where xx is the hex-string for an 8-bit octet. Its use is stated in Part 6 of the OPC UA specification for human-readable QualifiedNames, NodeIds and ExpandedNodeIds. The specification (only) requires that the ; characters gets escaped, and that decoders allow percent-escaping of any character.

Percent-escaping was originally defined for URIs. RFC 3986 defines a set of reserved characters (delimiters) that require escaping when they are used within URI components. But the reserved characters can be used directly in an URI to "delimit" between its components.

For the **percent-escaping** (e.g. used for the NamespaceUri part of the NodeId in UA_NodeId_printEx), the reserved characters are the semicolon, the percent character, whitespaces and unprintable ASCII control codes. Besides those, any UTF8 encoding is allowed.

In some contexts the semicolon alone is not sufficient as the delimiter. For our non-standard **extended-percent-escaping** the reserved characters are the following (in addition to whitespaces and unprintable ASCII control codes).:

```
' ' - %20    '(' - %28    '<' - %3C
'"' - %22    ')' - %29    '>' - %3E
'#' - %23    ',' - %2C    '[' - %5B
'%' - %25    '/' - %2F    '\' - %5C
'&' - %26    ':' - %3A    ']' - %5D
''' - %27    ';' - %3B    '`' - %60
```

11.11 RelativePath Expressions

Parse a RelativePath according to the format defined in Part 4, A2. This is used e.g. for the BrowsePath structure.

RelativePath := (ReferenceType BrowseName)+

The ReferenceType has one of the following formats:

- /: *HierarchicalReferences* and subtypes
- .: *Aggregates* ReferenceTypes and subtypes
- < [!#]* BrowseName >: The ReferenceType is indicated by its BrowseName. Reserved characters in the BrowseName are and-escaped. The following prefix-modifiers are defined for the ReferenceType.
  - ! switches to inverse References
  - # excludes subtypes of the ReferenceType.
  - As a non-standard extension we allow the ReferenceType in angle-brackets as a NodeId. For example <ns=1;i=345>. If a string NodeId is used, the string identifier is and-escaped.

The BrowseName is a QualifiedName. It consist of an optional NamespaceIndex and the name itself. The NamespaceIndex can be left out for the default Namespace zero. The name component is and-escaped (see above).

BrowseName := ([0-9]+ ":")? Name

The last BrowseName in a RelativePath can be omitted. This acts as a wildcard that matches any BrowseName.

11.11.1 Example RelativePaths

- /2:Block&.Output
- /3:Truck.0:NodeVersion
- <0:HasProperty>1:Boiler/1:HeatSensor
- <0:HasChild>2:Wheel
- <#Aggregates>1:Boiler/
- <!HasChild>Truck
- <HasChild>

```
UA_StatusCode
UA_RelativePath_parse(UA_RelativePath *rp, const UA_String str);

/* Supports the lookup of non-standard ReferenceTypes by their browse name in
 * the information model of a server. The first matching result in the
 * ReferenceType hierarchy is used. */
UA_StatusCode
UA_RelativePath_parseWithServer(UA_Server *server, UA_RelativePath *rp,
                                const UA_String str);

/* The out-string can be pre-allocated. Then the size is adjusted or an error
 * returned. If the out-string is NULL, then memory is allocated for it. */
UA_StatusCode
UA_RelativePath_print(const UA_RelativePath *rp, UA_String *out);
```

11.12 Attribute Path Expression

The data types ReadValueId, AttributeOperand and SimpleAttributeOperand define the location of a value. That is, they define an attribute (with an optional index-range) of a node in the information model. The AttributeOperand data type has the most features. The other two are similar but simplified:

ReadValueId :=
NodeId? (“#” Attribute)? (“[” IndexRange “]”)?

AttributeOperand :=
NodeId? RelativePath? (“#” Attribute)? (“[” IndexRange “]”)?

SimpleAttributeOperand :=
TypeDefId? (“/” BrowseName)* (“#” Attribute)? (“[” IndexRange “]”)?

The ReadValueId is used for the Read Service. The default NodeId is i=0. The default attribute is #Value.

The AttributeOperand is used for the Query Service. It extends the ReadValueId with a RelativePath to be followed from the initial node. The default NodeId is the ObjectsFolder i=85. The default attribute is #Value.

The SimpleAttributeOperand is used for the Query Service and for *Event-Filter Parsing*. It is a simplified AttributeOperand where all references in the RelativePath are (subtypes of) HierarchicalReferences. So the / separator is mandatory. The initial NodeId is called *TypeDefinitionId* because Sim-

pleAttributeOperands are typically used in EventFilters to reference the child nodes of an EventType to select the values reported for each event instance.

The different parts of the expression are parsed as follows:

- The *NodeId* uses the human-readable format of Part 6, 5.3.1.10. String NodeIds use the extended-and-escaping from above.
- The *RelativePath Expressions* use the human-readable format with extended-and-escaping for strings.
- The *Attribute Id* is the human-readable name of the selected node attribute.
- For the index range, see the section on *NumericRange*.

11.12.1 Example SimpleAttributeOperands

- ns=2;s=TruckEventType/3:Truck/5:Wheel#Value[1:3]
- /3:Truck/5:Wheel
- #BrowseName

```
#ifdef UA_TYPES_ATTRIBUTEOPERAND
UA_StatusCode
UA_ReadValueId_parse(UA_ReadValueId *rvi,
                     const UA_String str);
#endif

#ifdef UA_TYPES_ATTRIBUTEOPERAND
UA_StatusCode
UA_AttributeOperand_parse(UA_AttributeOperand *ao,
                          const UA_String str);
#endif

#ifdef UA_TYPES_SIMPLEATTRIBUTEOPERAND
UA_StatusCode
UA_SimpleAttributeOperand_parse(UA_SimpleAttributeOperand *sao,
                                const UA_String str);
#endif

/* The out-string can be pre-allocated. Then the size is adjusted or an error
 * returned. If the out-string is NULL, then memory is allocated for it. */
UA_StatusCode
UA_ReadValueId_print(const UA_ReadValueId *rvi,
                     UA_String *out);

#ifdef UA_TYPES_ATTRIBUTEOPERAND
UA_StatusCode
UA_AttributeOperand_print(const UA_AttributeOperand *ao,
                          UA_String *out);
#endif

#ifdef UA_TYPES_SIMPLEATTRIBUTEOPERAND
UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_SimpleAttributeOperand_print(const UA_SimpleAttributeOperand *sao,
                                UA_String *out);
#endif
```

11.13 Convenience macros for complex types

```
#define UA_PRINTF_GUID_FORMAT                                              \
    "%08" PRIx32 "-%04" PRIx16 "-%04" PRIx16 "-%02" PRIx8 "%02" PRIx8    \
    "-%02" PRIx8 "%02" PRIx8 "%02" PRIx8 "%02" PRIx8  "%02" PRIx8 "%02" PRIx8 "%02" PRIx8
#define UA_PRINTF_GUID_DATA(GUID) (GUID).data1, (GUID).data2, (GUID).data3,␣
↪\
        (GUID).data4[0], (GUID).data4[1], (GUID).data4[2], (GUID).data4[3], \
        (GUID).data4[4], (GUID).data4[5], (GUID).data4[6], (GUID).data4[7]

#define UA_PRINTF_STRING_FORMAT "\"%.*s\""
#define UA_PRINTF_STRING_DATA(STRING) (int)(STRING).length, (STRING).data
```

11.14 Cryptography Helpers

```
/* Compare memory in constant time to mitigate timing attacks.
 * Returns true if ptr1 and ptr2 are equal for length bytes. */
UA_Boolean
UA_constantTimeEqual(const void *ptr1, const void *ptr2, size_t length);

/* Zero-out memory in a way that is not removed by compiler-optimizations. Use
 * this to ensure cryptographic secrets don't leave traces after the memory was
 * freed. */
void
UA_ByteString_memZero(UA_ByteString *bs);
```

11.15 Trustlist Helpers

```
/* Adds all of the certificates from the src trusted list to the dst trusted list. */
UA_StatusCode
UA_TrustListDataType_add(const UA_TrustListDataType *src, UA_TrustListDataType␣
↪*dst);

/* Replaces the contents of the destination trusted list with the certificates from the source␣
↪trusted list. */
UA_StatusCode
UA_TrustListDataType_set(const UA_TrustListDataType *src, UA_TrustListDataType␣
↪*dst);

/* Removes all of the certificates from the dst trust list that are specified
 * in the src trust list. */
UA_StatusCode
```

(continues on next page)

(continued from previous page)

```
UA_TrustListDataType_remove(const UA_TrustListDataType *src, UA_
↪TrustListDataType *dst);

/* Checks if the certificate is present in the trust list.
 * The mask parameter can be used to specify the part of the trust list to check. */
UA_Boolean
UA_TrustListDataType_contains(const UA_TrustListDataType *trustList,
                              const UA_ByteString *certificate,
                              UA_TrustListMasks mask);

/* Returns the size of the TrustList in bytes. */
UA_UInt32
UA_TrustListDataType_getSize(const UA_TrustListDataType *trustList);
```

CHAPTER

TWELVE

XML NODESET COMPILER

When writing an application, it is more comfortable to create information models using GUI tools. Most tools can export data according the OPC UA Nodeset XML schema. open62541 contains a Python based nodeset compiler that transforms Nodeset XML files into code for the open62541 SDK.

We take the following information model snippet as the starting point of the following tutorial. A more detailed tutorial on how to create your own information model and NodeSet2.xml can be found in this blog post: https://profanter.medium.com/how-to-create-custom-opc-ua-information-models-1e9a461f5b58

```
<UANodeSet xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xmlns:uax="http://opcfoundation.org/UA/2008/02/Types.xsd"
        xmlns="http://opcfoundation.org/UA/2011/03/UANodeSet.xsd"
        xmlns:s1="http://yourorganisation.org/example_nodeset/"
        xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <NamespaceUris>
    <Uri>http://yourorganisation.org/example_nodeset/</Uri>
  </NamespaceUris>
  <Aliases>
    <Alias Alias="Boolean">i=1</Alias>
    <Alias Alias="UInt32">i=7</Alias>
    <Alias Alias="String">i=12</Alias>
    <Alias Alias="HasModellingRule">i=37</Alias>
    <Alias Alias="HasTypeDefinition">i=40</Alias>
    <Alias Alias="HasSubtype">i=45</Alias>
    <Alias Alias="HasProperty">i=46</Alias>
    <Alias Alias="HasComponent">i=47</Alias>
    <Alias Alias="Argument">i=296</Alias>
  </Aliases>
  <Extensions>
    <Extension>
      <ModelInfo Tool="UaModeler" Hash="Zs8w1AQI71W8P/GOk3k/xQ=="
          Version="1.3.4"/>
    </Extension>
  </Extensions>
  <UAReferenceType NodeId="ns=1;i=4001" BrowseName="1:providesInputTo">
    <DisplayName>providesInputTo</DisplayName>
    <References>
      <Reference ReferenceType="HasSubtype" IsForward="false">
        i=33
```

(continues on next page)

(continued from previous page)

```
        </Reference>
    </References>
    <InverseName Locale="en-US">inputProcidedBy</InverseName>
</UAReferenceType>
<UAObjectType IsAbstract="true" NodeId="ns=1;i=1001"
            BrowseName="1:FieldDevice">
    <DisplayName>FieldDevice</DisplayName>
    <References>
        <Reference ReferenceType="HasSubtype" IsForward="false">
            i=58
        </Reference>
        <Reference ReferenceType="HasComponent">ns=1;i=6001</Reference>
        <Reference ReferenceType="HasComponent">ns=1;i=6002</Reference>
    </References>
</UAObjectType>
<UAVariable DataType="String" ParentNodeId="ns=1;i=1001"
          NodeId="ns=1;i=6001" BrowseName="1:ManufacturerName"
          UserAccessLevel="3" AccessLevel="3">
    <DisplayName>ManufacturerName</DisplayName>
    <References>
        <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
        <Reference ReferenceType="HasModellingRule">i=78</Reference>
        <Reference ReferenceType="HasComponent" IsForward="false">
            ns=1;i=1001
        </Reference>
    </References>
</UAVariable>
<UAVariable DataType="String" ParentNodeId="ns=1;i=1001"
          NodeId="ns=1;i=6002" BrowseName="1:ModelName"
          UserAccessLevel="3" AccessLevel="3">
    <DisplayName>ModelName</DisplayName>
    <References>
        <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
        <Reference ReferenceType="HasModellingRule">i=78</Reference>
        <Reference ReferenceType="HasComponent" IsForward="false">
            ns=1;i=1001
        </Reference>
    </References>
</UAVariable>
<UAObjectType NodeId="ns=1;i=1002" BrowseName="1:Pump">
    <DisplayName>Pump</DisplayName>
    <References>
        <Reference ReferenceType="HasComponent">ns=1;i=6003</Reference>
        <Reference ReferenceType="HasComponent">ns=1;i=6004</Reference>
        <Reference ReferenceType="HasSubtype" IsForward="false">
            ns=1;i=1001
        </Reference>
        <Reference ReferenceType="HasComponent">ns=1;i=7001</Reference>
        <Reference ReferenceType="HasComponent">ns=1;i=7002</Reference>
```

(continues on next page)

(continued from previous page)

```
    </References>
  </UAObjectType>
  <UAVariable DataType="Boolean" ParentNodeId="ns=1;i=1002"
          NodeId="ns=1;i=6003" BrowseName="1:isOn" UserAccessLevel="3"
          AccessLevel="3">
    <DisplayName>isOn</DisplayName>
    <References>
      <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
      <Reference ReferenceType="HasModellingRule">i=78</Reference>
      <Reference ReferenceType="HasComponent" IsForward="false">
        ns=1;i=1002
      </Reference>
    </References>
  </UAVariable>
  <UAVariable DataType="UInt32" ParentNodeId="ns=1;i=1002"
          NodeId="ns=1;i=6004" BrowseName="1:MotorRPM"
          UserAccessLevel="3" AccessLevel="3">
    <DisplayName>MotorRPM</DisplayName>
    <References>
      <Reference ReferenceType="HasTypeDefinition">i=63</Reference>
      <Reference ReferenceType="HasModellingRule">i=78</Reference>
      <Reference ReferenceType="HasComponent" IsForward="false">
        ns=1;i=1002
      </Reference>
    </References>
  </UAVariable>
  <UAMethod ParentNodeId="ns=1;i=1002" NodeId="ns=1;i=7001"
        BrowseName="1:startPump">
    <DisplayName>startPump</DisplayName>
    <References>
      <Reference ReferenceType="HasModellingRule">i=78</Reference>
      <Reference ReferenceType="HasProperty">ns=1;i=6005</Reference>
      <Reference ReferenceType="HasComponent" IsForward="false">
        ns=1;i=1002
      </Reference>
    </References>
  </UAMethod>
  <UAVariable DataType="Argument" ParentNodeId="ns=1;i=7001" ValueRank="1"
          NodeId="ns=1;i=6005" ArrayDimensions="1"
          BrowseName="OutputArguments">
    <DisplayName>OutputArguments</DisplayName>
    <References>
      <Reference ReferenceType="HasModellingRule">i=78</Reference>
      <Reference ReferenceType="HasProperty"
              IsForward="false">ns=1;i=7001</Reference>
      <Reference ReferenceType="HasTypeDefinition">i=68</Reference>
    </References>
    <Value>
      <ListOfExtensionObject>
```

(continues on next page)

(continued from previous page)

```
            <ExtensionObject>
               <TypeId>
                  <Identifier>i=297</Identifier>
               </TypeId>
               <Body>
                  <Argument>
                     <Name>started</Name>
                     <DataType>
                        <Identifier>i=1</Identifier>
                     </DataType>
                     <ValueRank>-1</ValueRank>
                     <ArrayDimensions></ArrayDimensions>
                     <Description/>
                  </Argument>
               </Body>
            </ExtensionObject>
         </ListOfExtensionObject>
      </Value>
   </UAVariable>
   <UAMethod ParentNodeId="ns=1;i=1002" NodeId="ns=1;i=7002"
         BrowseName="1:stopPump">
      <DisplayName>stopPump</DisplayName>
      <References>
         <Reference ReferenceType="HasModellingRule">i=78</Reference>
         <Reference ReferenceType="HasProperty">ns=1;i=6006</Reference>
         <Reference ReferenceType="HasComponent"
               IsForward="false">ns=1;i=1002</Reference>
      </References>
   </UAMethod>
   <UAVariable DataType="Argument" ParentNodeId="ns=1;i=7002" ValueRank="1"
         NodeId="ns=1;i=6006" ArrayDimensions="1"
         BrowseName="OutputArguments">
      <DisplayName>OutputArguments</DisplayName>
      <References>
         <Reference ReferenceType="HasModellingRule">i=78</Reference>
         <Reference ReferenceType="HasProperty" IsForward="false">
            ns=1;i=7002
         </Reference>
         <Reference ReferenceType="HasTypeDefinition">i=68</Reference>
      </References>
      <Value>
         <ListOfExtensionObject>
            <ExtensionObject>
               <TypeId>
                  <Identifier>i=297</Identifier>
               </TypeId>
               <Body>
                  <Argument>
                     <Name>stopped</Name>
```

(continues on next page)

(continued from previous page)

```
                    <DataType>
                        <Identifier>i=1</Identifier>
                    </DataType>
                    <ValueRank>-1</ValueRank>
                    <ArrayDimensions></ArrayDimensions>
                    <Description/>
                </Argument>
            </Body>
        </ExtensionObject>
    </ListOfExtensionObject>
  </Value>
 </UAVariable>
</UANodeSet>
```

Take the previous snippet and save it to a file myNS.xml. The file is compiled using the following function call in the CMake build system:

```
ua_generate_nodeset(
    NAME "myNs"
    FILE "${PROJECT_SOURCE_DIR}/examples/nodeset/myNS.xml"
    DEPENDS_TYPES "UA_TYPES"
    DEPENDS_NS "${PROJECT_SOURCE_DIR}/deps/ua-nodeset/Schema/Opc.Ua.
↪NodeSet2.xml
)
```

If you look into the files generated by the nodeset compiler, you will see that it generated a method called extern UA_StatusCode myNS(UA_Server *server);. You need to include the header and source file and then call myNS(server) right after creating the server instance with UA_Server_new. This will automatically add all the nodes to the server and return UA_STATUSCODE_GOOD if there weren't any errors. Additionally you need to compile the open62541 stack with the full NS0 by setting UA_NAMESPACE_ZERO=FULL in CMake. Otherwise the stack uses a subset where many nodes are not included and thus adding a custom nodeset may fail.

If you also want to generate custom DataTypes for the nodeset, use the CMake function ua_generate_nodeset_and_datatypes. It uses some best practice settings and you only need to pass a name and the nodeset files. Passing the .csv and .bsd files with the datatype information is optional. If not given, generating datatypes for that nodeset will be skipped. You can also define dependencies between nodesets using the DEPENDS argument. Here are some examples for the DI and PLCOpen nodesets:

```
# Generate types and namespace for DI
ua_generate_nodeset_and_datatypes(
    NAME "di"
    FILE_CSV "${UA_NODESET_DIR}/DI/Opc.Ua.Di.NodeIds.csv"
    FILE_BSD "${UA_NODESET_DIR}/DI/Opc.Ua.Di.Types.bsd"
    FILE_NS "${UA_NODESET_DIR}/DI/Opc.Ua.Di.NodeSet2.xml"
)

# generate PLCopen namespace which is using DI
ua_generate_nodeset_and_datatypes(
```

(continues on next page)

(continued from previous page)

```
    NAME "plc"
    # PLCopen does not define custom types. Only generate the nodeset
    FILE_NS "${UA_NODESET_DIR}/PLCopen/Opc.Ua.PLCopen.NodeSet2_V1.02.xml"
    # PLCopen depends on the di nodeset, which must be generated before
    DEPENDS "di"
)
```

12.1 Manually calling the Nodeset Compiler

Besides the CMake macros, the underlying Python code of the Nodeset compiler can be called directly. The call looks like this:

```
$ python ./nodeset_compiler.py --types-array=UA_TYPES --existing ../../deps/ua-nodeset/
↪Schema/Opc.Ua.NodeSet2.xml --xml myNS.xml myNS
```

The output of the command is:

```
INFO:___main___:Preprocessing (existing) ../../deps/ua-nodeset/Schema/Opc.Ua.NodeSet2.
↪xml
INFO:___main___:Preprocessing myNS.xml
INFO:___main___:Generating Code
INFO:___main___:NodeSet generation code successfully printed
```

The first argument --types-array=UA_TYPES defines the name of the global array in open62541 which contains the corresponding types used within the nodeset in NodeSet2.xml. If you do not define your own datatypes, you can always use the UA_TYPES value. More on that later in this tutorial. The next argument --existing ../../deps/ua-nodeset/Schema/Opc.Ua.NodeSet2.xml points to the XML definition of the standard-defined namespace 0 (NS0). Namespace 0 is assumed to be loaded beforehand and provides definitions for data type, reference types, and so. Since we reference nodes from NS0 in our myNS.xml we need to tell the nodeset compiler that it should also load that nodeset, but not compile it into the output. Note that you may need to initialize the git submodule to get the deps/ua-nodeset folder (git submodule update --init) or download the full NodeSet2.xml manually. The argument --xml myNS.xml points to the user-defined information model, whose nodes will be added to the abstract syntax tree. The script will then create the files myNS.c and myNS.h (indicated by the last argument myNS) containing the C code necessary to instantiate those namespaces.

The help command show which additional options are available:

```
$ python ./nodeset_compiler.py -h
```

12.2 Creating object instances

One of the key benefits of defining object types is being able to create object instances fairly easily. Object instantiation is handled automatically when the TypeDefinition NodeId points to a valid ObjectType node. All Attributes and Methods contained in the objectType definition will be instantiated along with the object node.

Let's look at an example that will create a pump instance given the newly defined objectType from myNS.xml:

```c
/* This work is licensed under a Creative Commons CCZero 1.0 Universal License.
 * See http://creativecommons.org/publicdomain/zero/1.0/ for more information. */

#include <signal.h>
#include <stdio.h>
#include "open62541.h"

/* Files myNS.h and myNS.c are created from myNS.xml */
#include "myNS.h"

UA_Boolean running = true;

static void stopHandler(int sign) {
    UA_LOG_INFO(UA_Log_Stdout, UA_LOGCATEGORY_SERVER, "received ctrl-c");
    running = false;
}

int main(int argc, char **argv) {
    signal(SIGINT, stopHandler);
    signal(SIGTERM, stopHandler);

    UA_Server *server = UA_Server_new();
    UA_ServerConfig_setDefault(UA_Server_getConfig(server));

    UA_StatusCode retval = myNS(server);
    /* Create nodes from nodeset */
    if(retval != UA_STATUSCODE_GOOD) {
        UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_SERVER, "Could not␣
↪add the example nodeset. "
            "Check previous output for any error.");
        retval = UA_STATUSCODE_BADUNEXPECTEDERROR;
    } else {
        UA_NodeId createdNodeId;
        UA_ObjectAttributes object_attr = UA_ObjectAttributes_default;

        object_attr.description = UA_LOCALIZEDTEXT("en-US", "A pump!");
        object_attr.displayName = UA_LOCALIZEDTEXT("en-US", "Pump1");

        // we assume that the myNS nodeset was added in namespace 2.
        // You should always use UA_Server_addNamespace to check what the
        // namespace index is for a given namespace URI. UA_Server_addNamespace
        // will just return the index if it is already added.
        UA_Server_addObjectNode(server, UA_NODEID_NUMERIC(1, 0),
                                UA_NODEID_NUMERIC(0, UA_NS0ID_OBJECTSFOLDER),
                                UA_NODEID_NUMERIC(0, UA_NS0ID_ORGANIZES),
                                UA_QUALIFIEDNAME(1, "Pump1"),
                                UA_NODEID_NUMERIC(2, 1002),
                                object_attr, NULL, &createdNodeId);
```

(continues on next page)

(continued from previous page)

```
    retval = UA_Server_run(server, &running);
  }

  UA_Server_delete(server);
  return (int) retval;
}
```

If you start the server and inspect the nodes, you will find the pump in the objects folder, which look like this Fig. 12.1.

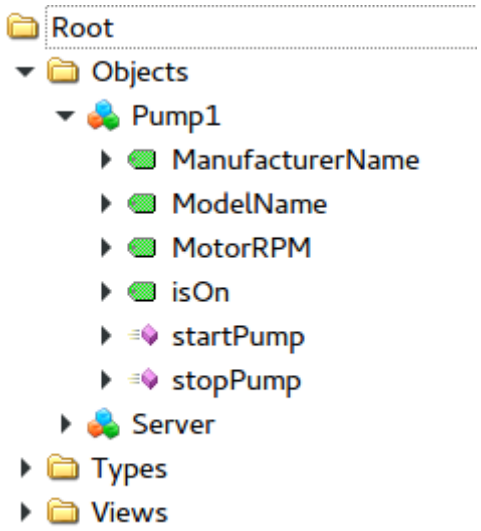


Fig. 12.1: Instantiated Pump Object with inherited children

As you can see the pump has inherited its parents attributes (ManufacturerName and ModelName). Methods, in contrast to objects and variables, are never cloned but instead only linked. The reason is that you will quite probably attach a method callback to a central method, not each object. Objects are instantiated if they are *below* the object you are creating, so any object (like an object called associatedServer of ServerType) that is part of pump will be instantiated as well. Objects *above* you object are never instantiated, so the same ServerType object in Fielddevices would have been omitted (the reason is that the recursive instantiation function protects itself from infinite recursions, which are hard to track when first ascending, then redescending into a tree).

12.3 Combination of multiple nodesets

In the previous section you have seen how you can use the nodeset compiler with one single nodeset which depends on the default nodeset (NS0) `Opc.Ua.NodeSet2.xml`. The nodeset compiler also supports nodesets which depend on more than one nodeset. We will show this use-case with the PLCopen nodeset. The PLCopen nodeset `Opc.Ua.PLCopen.NodeSet2_V1.02.xml` depends on the DI nodeset `Opc.Ua.Di.NodeSet2.xml` which then depends on NS0. This example is also shown in `examples/nodeset/CMakeLists.txt`.

This DI nodeset makes use of some additional data types in `deps/ua-nodeset/DI/Opc.Ua.Di.Types.bsd`. Since we also need these types within the generated code, we first need to compile the types into C code. The generated code is mainly a definition of the binary representation of the types required for encoding and decoding. The generation can be done using the `ua_generate_datatypes` CMake

function, which uses the tools/generate_datatypes.py script:

```
ua_generate_datatypes(
    NAME "ua_types_di"
    TARGET_SUFFIX "types-di"
    FILE_CSV "${UA_NODESET_DIR}/DI/Opc.Ua.Di.NodeIds.csv"
    FILES_BSD "${UA_NODESET_DIR}/DI/Opc.Ua.Di.Types.bsd"
)
```

TARGET_SUFFIX is used to create a new target with the name open62541-generator-TARGET_SUFFIX.

Now you can compile the DI nodeset XML using the following command:

```
ua_generate_nodeset(
    NAME "di"
    FILE "${UA_NODESET_DIR}/DI/Opc.Ua.Di.NodeSet2.xml"
    TYPES_ARRAY "UA_TYPES_DI"
    INTERNAL
    DEPENDS_TYPES "UA_TYPES"
    DEPENDS_NS    "${UA_NODESET_DIR}/Schema/Opc.Ua.NodeSet2.xml"
    DEPENDS_TARGET "open62541-generator-types-di"
)
```

There are now two new arguments: INTERNAL indicates that internal headers (and non public API) should be included within the generated source code. This is currently required for nodesets which use structures as data values, and will probably be fixed in the future. The DEPENDS_TYPES types array argument is matched with the nodesets in the same order as they appear on the DEPENDS_TARGET parameter. It tells the nodeset compiler which types array it should use: UA_TYPES for Opc.Ua.NodeSet2.xml and UA_TYPES_DI for Opc.Ua.Di.NodeSet2.xml. This is the type array generated by the generate_datatypes.py script. The rest is similar to the example in previous section: Opc.Ua.NodeSet2.xml is assumed to exist already and only needs to be loaded for consistency checks, Opc.Ua.Di.NodeSet2.xml will be generated in the output file ua_namespace_di.c/.h

Next we can generate the PLCopen nodeset. Since it doesn't require any additional datatype definitions, we can immediately start with the nodeset compiler command:

```
ua_generate_nodeset(
    NAME "plc"
    FILE "${UA_NODESET_DIR}/PLCopen/Opc.Ua.PLCopen.NodeSet2_V1.02.xml"
    INTERNAL
    DEPENDS_TYPES
        "UA_TYPES" "UA_TYPES_DI"
    DEPENDS_NS
        "${UA_NODESET_DIR}/Schema/Opc.Ua.NodeSet2.xml"
        "${UA_NODESET_DIR}/DI/Opc.Ua.Di.NodeSet2.xml"
    DEPENDS_TARGET "open62541-generator-ns-di"
)
```

This call is quite similar to the compilation of the DI nodeset. As you can see, we do not define any specific types array for the PLCopen nodeset. Since the PLCopen nodeset depends on the NS0 and DI nodeset, we need to tell the nodeset compiler that these two nodesets should be seen as already

existing. Make sure that the order is the same as in your XML file, e.g., in this case the order indicated in Opc.Ua.PLCopen.NodeSet2_V1.02.xml -> UANodeSet -> Models -> Model.

As a result of the previous scripts you will have multiple source files:

- ua_types_di_generated.c
- ua_types_di_generated.h
- ua_types_di_generated_encoding_binary.h
- ua_types_di_generated_handling.h
- ua_namespace_di.c
- ua_namespace_di.h
- ua_namespace_plc.c
- ua_namespace_plc.h

Finally you need to include all these files in your build process and call the corresponding initialization methods for the nodesets. An example application could look like this:

```
UA_Server *server = UA_Server_new();
UA_ServerConfig_setDefault(UA_Server_getConfig(server));

/* Create nodes from nodeset */
UA_StatusCode retval = ua_namespace_di(server);
if(retval != UA_STATUSCODE_GOOD) {
    UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_SERVER,
                 "Adding the DI namespace failed. Please check previous error output.");
    UA_Server_delete(server);
    return (int)UA_STATUSCODE_BADUNEXPECTEDERROR;
}

retval |= ua_namespace_plc(server);
if(retval != UA_STATUSCODE_GOOD) {
    UA_LOG_ERROR(UA_Log_Stdout, UA_LOGCATEGORY_SERVER,
                 "Adding the PLCopen namespace failed. Please check previous error output.");
    UA_Server_delete(server);
    return (int)UA_STATUSCODE_BADUNEXPECTEDERROR;
}

retval = UA_Server_run(server, &running);
```

12.4 Outstanding Companion Spec Issues

There are some Companion Specifications that currently cannot be compiled with the Nodeset compiler. Which Companion Specifications are affected and what causes this is described below.

Safety, Glass, DEXPI
: Do not specify a BSD file or BSD blob in the XML file. The BSD file is considered deprecated. However, it is currently still required by the Nodeser compiler.

I4AAS, RSL, FDI
: Attempting to load will result in a runtime error (“Type-checking failed with error code BadTypeMismatch” or “Parent node not found”).

BACnet
: Defines data types whose fields have the names signed or unsigned. This leads to errors when creating C structures, because signed and unsigned are keywords in C.

12.5 Automatic Nodesetinjection

The nodesetinjector is a mechanism for automatically loading nodeset/companion specifications during server initialization. It provides a fast and easy way to load nodesets in all applications, focusing on the official OPCFoundation/UANodeset Repository (https://github.com/OPCFoundation/UA-Nodeset). Specify the required information models using CMake.

Which nodesets are to be loaded is determined by the Cmake flag UA_INFORMATION_MODEL_AUTOLOAD. All nodesets that are to be loaded automatically are listed here. The naming is based on the folder name of the Companion Specification in the ua-nodeset folder.

A CMake call could look like this.

```
-DCMAKE_BUILD_TYPE=Debug
-DUA_BUILD_EXAMPLES=ON
-DUA_INFORMATION_MODEL_AUTOLOAD=DI;POWERLINK;PROFINET;
↪MachineVision
-DUA_NAMESPACE_ZERO=FULL
```

The order of nodesets is important! Nodesets that build on other nodesets must be placed after them in the list. The following nodesets are currently supported, grouped by dependency depth:

Level 0 — Standalone (no companion-spec dependency, only ns0):

AMB, BACnet, CNC, DEXPI, DI, GDS, I4AAS, IOLinkIODD, IREDES, ISA95-JOBCONTROL, MachineVision, Machinery-Result, MDIS, MTConnect, OpenSCS, PackML, PROFINET, Safety, Scheduler, UAFX-Data, WireHarness-VEC, WoT

Level 1 — Depends on Level-0 specs (typically DI):

ADI (DI), AutoID (DI), CommercialKitchenEquipment (DI), CSPPlusForMachine (DI), FDI5 (DI), FDI7 (DI), FDT (DI), IA (DI), IOLink (DI), Machinery-Jobs (ISA95-JOBCONTROL), Onboarding (GDS), PADIM (DI), PlasticsRubber-GeneralTypes (DI), PLCopen (DI), PNEM (DI), PNENC (DI), PNGSDGM (DI), PNRIO (DI), POWERLINK (DI), Robotics (DI), Sercos (DI), UAFX-AC (DI, UAFX-Data), UAFX-CM (DI, UAFX-Data)

Level 2 — Depends on Level-1 specs (two or more dependency hops):

Each entry lists its required dependencies in parentheses.

CAS (DI, Machinery, IA), ECM (DI, IA), Eumabois (DI, Machinery, Woodworking), Glass (DI, Machinery), LADS (DI, AMB, Machinery), Machinery (DI, IA), Machinery-Energy (DI, IA, ECM), Machinery-ProcessValues (DI, PADIM), MachineTool (DI, Machinery, IA, ISA95-JOBCONTROL, Machinery-Jobs), Mining-General (DI, IA, Machinery), PNDRV (DI, PNENC), PlasticsRubber-TCD (DI, PlasticsRubber-GeneralTypes), PlasticsRubber-IMM2MES (DI, PlasticsRubber-GeneralTypes), PlasticsRubber-LDS (DI, PlasticsRubber-GeneralTypes), PlasticsRubber-HotRunner (DI, PlasticsRubber-GeneralTypes), Pumps (DI, Machinery), Scales (DI, Machinery, PackML), Shotblasting (DI, Machinery, ISA95-JOBCONTROL, Machinery-Jobs), STGeneralTypes (DI, IA, Machinery,

ISA95-JOBCONTROL, Machinery-Jobs), SurfaceTechnology (DI, IA, Machinery, ISA95-JOBCONTROL, Machinery-Jobs), TMC (DI, PackML), Weihenstephan (DI, Machinery, PackML), Woodworking (DI, Machinery, ISA95-JOBCONTROL, Machinery-Jobs)

Level 3 — Deep stacks (depends on Level-2 specs):

Each entry lists its full dependency chain in parentheses, i.e. every spec that must be loaded before it.

AdditiveManufacturing (DI, IA, Machinery, PADIM, Machinery-ProcessValues, ISA95-JOBCONTROL, Machinery-Jobs, MachineTool), CuttingTool (DI, IA, Machinery, ISA95-JOBCONTROL, Machinery-Jobs, Machinery-Result, MachineTool, GMS), GMS (DI, IA, Machinery, ISA95-JOBCONTROL, Machinery-Jobs, Machinery-Result, MachineTool), IJT (DI, AMB, IA, Machinery, Machinery-Result — loads both IJT/Base and IJT/Tightening), LaserSystems (DI, IA, Machinery, ISA95-JOBCONTROL, Machinery-Jobs, MachineTool), MetalForming (DI, IA, Machinery, PADIM, Machinery-ProcessValues, ISA95-JOBCONTROL, Machinery-Jobs, MachineTool), Mining-Extraction-General (DI, IA, Machinery, Mining-General), Mining-Extraction-ShearerLoader (DI, IA, Machinery, Mining-General, Mining-Extraction-General), Mining-Loading-General (DI, IA, Machinery, Mining-General), Mining-Loading-HydraulicExcavator (DI, IA, Machinery, Mining-General, Mining-Loading-General), Mining-DevelopmentSupport-General (DI, IA, Machinery, Mining-General), Mining-DevelopmentSupport-RoofSupportSystem (DI, IA, Machinery, Mining-General, Mining-DevelopmentSupport-General), Mining-DevelopmentSupport-Dozer (DI, IA, Machinery, Mining-General, Mining-DevelopmentSupport-General), Mining-TransportDumping-General (DI, IA, Machinery, Mining-General), Mining-TransportDumping-RearDumpTruck (DI, IA, Machinery, Mining-General, Mining-TransportDumping-General), Mining-TransportDumping-ArmouredFaceConveyor (DI, IA, Machinery, Mining-General, Mining-TransportDumping-General), Mining-MineralProcessing-General (DI, IA, Machinery, Mining-General), Mining-MineralProcessing-RockCrusher (DI, IA, Machinery, Mining-General, Mining-MineralProcessing-General), Mining-PELOServices-General (DI, IA, Machinery, Mining-General), Mining-PELOServices-FaceAlignmentSystem (DI, IA, Machinery, Mining-General, Mining-PELOServices-General), Mining-MonitoringSupervisionServices-General (DI, IA, Machinery, Mining-General), PAEFS (DI, Machinery, PADIM, Machinery-ProcessValues), Powertrain (DI, Machinery, UAFX-Data, UAFX-AC), WMTP (DI, Machinery, PADIM, Machinery-ProcessValues), PlasticsRubber-Extrusion-GeneralTypes (DI, IA, Machinery, PlasticsRubber-GeneralTypes), PlasticsRubber-Extrusion-{ExtrusionLine,Extruder,Die,Filter,MeltPump,HaulOff, Pelletizer,Calender,Calibrator,Corrugator,Cutter} (DI, IA, Machinery, PlasticsRubber-GeneralTypes, PlasticsRubber-Extrusion-GeneralTypes), PlasticsRubber-Extrusion_v2-GeneralTypes (DI, IA, Machinery, PlasticsRubber-GeneralTypes), PlasticsRubber-Extrusion_v2-{ExtrusionLine,Extruder,Die,Filter,MeltPump,HaulOff, Pelletizer,Calender,Calibrator,Corrugator,Cutter} (DI, IA, Machinery, PlasticsRubber-GeneralTypes, PlasticsRubber-Extrusion_v2-GeneralTypes), TTD (DI, IA, Machinery, ISA95-JOBCONTROL, Machinery-Jobs, Machinery-Result, MachineTool), WireHarness (DI, Machinery, ISA95-JOBCONTROL, Machinery-Jobs, Machinery-Result, WireHarness-VEC)

> **Note**
>
> **PADIM** and **Machinery-ProcessValues** are generated as internal helper nodesets (including the IRDI dictionary) when a model that requires them is selected.

Warning

Some companion specifications are not yet supported:

- **GPOS** and **RSL** depend on OPC UA Part 8 spatial types (3DFrame, 3DCartesianCoordinates) and use abstract VariableTypes as TypeDefinitions, which are not yet fully supported by the open62541 type system.
- **Glass** has no embedded .bsd blob, so the compile-time nodeset compiler cannot generate UA_DataType definitions for its custom structs. Loading Glass via the **NodesetLoader** at runtime works (the loader registers types from <Definition> elements dynamically).

Some companion specifications define types with the same C name in different OPC UA namespaces. These cannot be loaded simultaneously:

- **Pumps** and **PAEFS** both define UA_ControlModeEnum.
- **TMC** and **PlasticsRubber-GeneralTypes** both define UA_ControlModeEnumeration / UA_ProductionStatusEnumeration.
- **CommercialKitchenEquipment** and **PlasticsRubber-TCD** both define UA_OperatingModeEnumeration.
- **PlasticsRubber-LDS** and **PlasticsRubber-Extrusion-GeneralTypes** both define UA_ComponentStatusEnumeration.
- **PlasticsRubber-Extrusion** (v1) and **PlasticsRubber-Extrusion_v2** share multiple type names and cannot be loaded together.
- **MachineVision** and **Machinery-Result** both define ProcessingTimesDataType / ResultDataType. Choose Machinery-Result when IJT, GMS, or TTD are needed.
- **PNEM** and **ECM** both define AcPeDataType / AcPpDataType / EnergyStateInformationDataType. Choose ECM when Machinery-Energy is needed.

When the open62541 library is installed on the system, the automatically generated autoinject library is installed alongside it. Additionally, the header files for the specified nodesets will be accessible on the system.

Integrating the autoinject library into an existing CMake project is straightforward. To achieve this, the CMake configuration should include the following:

```
set(open62541_LIBRARIES "")
find_package(open62541 REQUIRED)
list(APPEND open62541_LIBRARIES open62541::open62541 open62541::autoinject)
```

The generated namespace header files can also be included and used with the following code:

```
#include <autoinject/namespace_di_generated.h>
#include <autoinject/namespace_amb_generated.h>
#include <autoinject/namespace_machinery_generated.h>
```

CHAPTER

THIRTEEN

STATUSCODES

StatusCodes are extensively used in the OPC UA protocol and in the open62541 API. They are represented by the *StatusCode* data type. The following definitions are autogenerated from the Opc.Ua.StatusCodes.csv file provided with the OPC UA standard.

```
/* These StatusCodes are manually generated. */
#define UA_STATUSCODE_INFOTYPE_DATAVALUE ((UA_StatusCode) 0x00000400)
#define UA_STATUSCODE_INFOBITS_OVERFLOW ((UA_StatusCode) 0x00000080)
#define UA_STATUSCODE_SEMANTICSCHANGED ((UA_StatusCode) 0x00004000)

/* The operation succeeded. */
#define UA_STATUSCODE_GOOD ((UA_StatusCode) 0x00000000)

/* The operation was uncertain. */
#define UA_STATUSCODE_UNCERTAIN ((UA_StatusCode) 0x40000000)

/* The operation failed. */
#define UA_STATUSCODE_BAD ((UA_StatusCode) 0x80000000)

/* An unexpected error occurred. */
#define UA_STATUSCODE_BADUNEXPECTEDERROR ((UA_StatusCode) 0x80010000)

/* An internal error occurred as a result of a programming or configuration error. */
#define UA_STATUSCODE_BADINTERNALERROR ((UA_StatusCode) 0x80020000)

/* Not enough memory to complete the operation. */
#define UA_STATUSCODE_BADOUTOFMEMORY ((UA_StatusCode) 0x80030000)

/* An operating system resource is not available. */
#define UA_STATUSCODE_BADRESOURCEUNAVAILABLE ((UA_StatusCode)␣
↪0x80040000)

/* A low level communication error occurred. */
#define UA_STATUSCODE_BADCOMMUNICATIONERROR ((UA_StatusCode)␣
↪0x80050000)

/* Encoding halted because of invalid data in the objects being serialized. */
#define UA_STATUSCODE_BADENCODINGERROR ((UA_StatusCode) 0x80060000)

/* Decoding halted because of invalid data in the stream. */
```

(continues on next page)

(continued from previous page)

```
#define UA_STATUSCODE_BADDECODINGERROR ((UA_StatusCode) 0x80070000)

/* The message encoding/decoding limits imposed by the stack have been exceeded. */
#define UA_STATUSCODE_BADENCODINGLIMITSEXCEEDED ((UA_StatusCode)␣
↪0x80080000)

/* The request message size exceeds limits set by the server. */
#define UA_STATUSCODE_BADREQUESTTOOLARGE ((UA_StatusCode) 0x80B80000)

/* The response message size exceeds limits set by the client. */
#define UA_STATUSCODE_BADRESPONSETOOLARGE ((UA_StatusCode)␣
↪0x80B90000)

/* An unrecognized response was received from the server. */
#define UA_STATUSCODE_BADUNKNOWNRESPONSE ((UA_StatusCode) 0x80090000)

/* The operation timed out. */
#define UA_STATUSCODE_BADTIMEOUT ((UA_StatusCode) 0x800A0000)

/* The server does not support the requested service. */
#define UA_STATUSCODE_BADSERVICEUNSUPPORTED ((UA_StatusCode)␣
↪0x800B0000)

/* The operation was cancelled because the application is shutting down. */
#define UA_STATUSCODE_BADSHUTDOWN ((UA_StatusCode) 0x800C0000)

/* The operation could not complete because the client is not connected to the server. */
#define UA_STATUSCODE_BADSERVERNOTCONNECTED ((UA_StatusCode)␣
↪0x800D0000)

/* The server has stopped and cannot process any requests. */
#define UA_STATUSCODE_BADSERVERHALTED ((UA_StatusCode) 0x800E0000)

/* There was nothing to do because the client passed a list of operations with no elements. */
#define UA_STATUSCODE_BADNOTHINGTODO ((UA_StatusCode) 0x800F0000)

/* The request could not be processed because it specified too many operations. */
#define UA_STATUSCODE_BADTOOMANYOPERATIONS ((UA_StatusCode)␣
↪0x80100000)

/* The request could not be processed because there are too many monitored items in the␣
↪subscription. */
#define UA_STATUSCODE_BADTOOMANYMONITOREDITEMS ((UA_StatusCode)␣
↪0x80DB0000)

/* The extension object cannot be (de)serialized because the data type id is not recognized.␣
↪*/
#define UA_STATUSCODE_BADDATATYPEIDUNKNOWN ((UA_StatusCode)␣
↪0x80110000)
```

(continues on next page)

(continued from previous page)

```
/* The certificate provided as a parameter is not valid. */
#define UA_STATUSCODE_BADCERTIFICATEINVALID ((UA_StatusCode)␣
↪0x80120000)

/* An error occurred verifying security. */
#define UA_STATUSCODE_BADSECURITYCHECKSFAILED ((UA_StatusCode)␣
↪0x80130000)

/* The certificate does not meet the requirements of the security policy. */
#define UA_STATUSCODE_BADCERTIFICATEPOLICYCHECKFAILED ((UA_
↪StatusCode) 0x81140000)

/* The certificate has expired or is not yet valid. */
#define UA_STATUSCODE_BADCERTIFICATETIMEINVALID ((UA_StatusCode)␣
↪0x80140000)

/* An issuer certificate has expired or is not yet valid. */
#define UA_STATUSCODE_BADCERTIFICATEISSUERTIMEINVALID ((UA_
↪StatusCode) 0x80150000)

/* The HostName used to connect to a server does not match a HostName in the certificate.␣
↪*/
#define UA_STATUSCODE_BADCERTIFICATEHOSTNAMEINVALID ((UA_
↪StatusCode) 0x80160000)

/* The URI specified in the ApplicationDescription does not match the URI in the certificate.
↪ */
#define UA_STATUSCODE_BADCERTIFICATEURIINVALID ((UA_StatusCode)␣
↪0x80170000)

/* The certificate may not be used for the requested operation. */
#define UA_STATUSCODE_BADCERTIFICATEUSENOTALLOWED ((UA_
↪StatusCode) 0x80180000)

/* The issuer certificate may not be used for the requested operation. */
#define UA_STATUSCODE_BADCERTIFICATEISSUERUSENOTALLOWED ((UA_
↪StatusCode) 0x80190000)

/* The certificate is not trusted. */
#define UA_STATUSCODE_BADCERTIFICATEUNTRUSTED ((UA_StatusCode)␣
↪0x801A0000)

/* It was not possible to determine if the certificate has been revoked. */
#define UA_STATUSCODE_BADCERTIFICATEREVOCATIONUNKNOWN ((UA_
↪StatusCode) 0x801B0000)

/* It was not possible to determine if the issuer certificate has been revoked. */
#define UA_STATUSCODE_BADCERTIFICATEISSUERREVOCATIONUNKNOWN␣
```

(continues on next page)

(continued from previous page)

```
↪((UA_StatusCode) 0x801C0000)

/* The certificate has been revoked. */
#define UA_STATUSCODE_BADCERTIFICATEREVOKED ((UA_StatusCode)␣
↪0x801D0000)

/* The issuer certificate has been revoked. */
#define UA_STATUSCODE_BADCERTIFICATEISSUERREVOKED ((UA_StatusCode)␣
↪0x801E0000)

/* The certificate chain is incomplete. */
#define UA_STATUSCODE_BADCERTIFICATECHAININCOMPLETE ((UA_
↪StatusCode) 0x810D0000)

/* User does not have permission to perform the requested operation. */
#define UA_STATUSCODE_BADUSERACCESSDENIED ((UA_StatusCode) 0x801F0000)

/* The user identity token is not valid. */
#define UA_STATUSCODE_BADIDENTITYTOKENINVALID ((UA_StatusCode)␣
↪0x80200000)

/* The user identity token is valid but the server has rejected it. */
#define UA_STATUSCODE_BADIDENTITYTOKENREJECTED ((UA_StatusCode)␣
↪0x80210000)

/* The specified secure channel is no longer valid. */
#define UA_STATUSCODE_BADSECURECHANNELIDINVALID ((UA_StatusCode)␣
↪0x80220000)

/* The timestamp is outside the range allowed by the server. */
#define UA_STATUSCODE_BADINVALIDTIMESTAMP ((UA_StatusCode) 0x80230000)

/* The nonce does appear to be not a random value or it is not the correct length. */
#define UA_STATUSCODE_BADNONCEINVALID ((UA_StatusCode) 0x80240000)

/* The session id is not valid. */
#define UA_STATUSCODE_BADSESSIONIDINVALID ((UA_StatusCode) 0x80250000)

/* The session was closed by the client. */
#define UA_STATUSCODE_BADSESSIONCLOSED ((UA_StatusCode) 0x80260000)

/* The session cannot be used because ActivateSession has not been called. */
#define UA_STATUSCODE_BADSESSIONNOTACTIVATED ((UA_StatusCode)␣
↪0x80270000)

/* The subscription id is not valid. */
#define UA_STATUSCODE_BADSUBSCRIPTIONIDINVALID ((UA_StatusCode)␣
↪0x80280000)
```

(continues on next page)

(continued from previous page)

```
/* The header for the request is missing or invalid. */
#define UA_STATUSCODE_BADREQUESTHEADERINVALID ((UA_StatusCode)␣
↪0x802A0000)

/* The timestamps to return parameter is invalid. */
#define UA_STATUSCODE_BADTIMESTAMPSTORETURNINVALID ((UA_
↪StatusCode) 0x802B0000)

/* The request was cancelled by the client. */
#define UA_STATUSCODE_BADREQUESTCANCELLEDBYCLIENT ((UA_
↪StatusCode) 0x802C0000)

/* Too many arguments were provided. */
#define UA_STATUSCODE_BADTOOMANYARGUMENTS ((UA_StatusCode)␣
↪0x80E50000)

/* The server requires a license to operate in general or to perform a service or operatio */
#define UA_STATUSCODE_BADLICENSEEXPIRED ((UA_StatusCode) 0x810E0000)

/* The server has limits on number of allowed operations / object */
#define UA_STATUSCODE_BADLICENSELIMITSEXCEEDED ((UA_StatusCode)␣
↪0x810F0000)

/* The server does not have a license which is required to operate in general or to perform a␣
↪service or operation. */
#define UA_STATUSCODE_BADLICENSENOTAVAILABLE ((UA_StatusCode)␣
↪0x81100000)

/* The subscription was transferred to another session. */
#define UA_STATUSCODE_GOODSUBSCRIPTIONTRANSFERRED ((UA_
↪StatusCode) 0x002D0000)

/* The processing will complete asynchronously. */
#define UA_STATUSCODE_GOODCOMPLETESASYNCHRONOUSLY ((UA_
↪StatusCode) 0x002E0000)

/* Sampling has slowed down due to resource limitations. */
#define UA_STATUSCODE_GOODOVERLOAD ((UA_StatusCode) 0x002F0000)

/* The value written was accepted but was clamped. */
#define UA_STATUSCODE_GOODCLAMPED ((UA_StatusCode) 0x00300000)

/* Communication with the data source is define */
#define UA_STATUSCODE_BADNOCOMMUNICATION ((UA_StatusCode) 0x80310000)

/* Waiting for the server to obtain values from the underlying data source. */
#define UA_STATUSCODE_BADWAITINGFORINITIALDATA ((UA_StatusCode)␣
↪0x80320000)
```

(continues on next page)

(continued from previous page)

```
/* The syntax of the node id is not valid. */
#define UA_STATUSCODE_BADNODEIDINVALID ((UA_StatusCode) 0x80330000)

/* The node id refers to a node that does not exist in the server address space. */
#define UA_STATUSCODE_BADNODEIDUNKNOWN ((UA_StatusCode) 0x80340000)

/* The attribute is not supported for the specified Node. */
#define UA_STATUSCODE_BADATTRIBUTEIDINVALID ((UA_StatusCode)␣
↪0x80350000)

/* The syntax of the index range parameter is invalid. */
#define UA_STATUSCODE_BADINDEXRANGEINVALID ((UA_StatusCode)␣
↪0x80360000)

/* No data exists within the range of indexes specified. */
#define UA_STATUSCODE_BADINDEXRANGENODATA ((UA_StatusCode)␣
↪0x80370000)

/* The data encoding is invalid. */
#define UA_STATUSCODE_BADDATAENCODINGINVALID ((UA_StatusCode)␣
↪0x80380000)

/* The server does not support the requested data encoding for the node. */
#define UA_STATUSCODE_BADDATAENCODINGUNSUPPORTED ((UA_StatusCode)␣
↪0x80390000)

/* The access level does not allow reading or subscribing to the Node. */
#define UA_STATUSCODE_BADNOTREADABLE ((UA_StatusCode) 0x803A0000)

/* The access level does not allow writing to the Node. */
#define UA_STATUSCODE_BADNOTWRITABLE ((UA_StatusCode) 0x803B0000)

/* The value was out of range. */
#define UA_STATUSCODE_BADOUTOFRANGE ((UA_StatusCode) 0x803C0000)

/* The requested operation is not supported. */
#define UA_STATUSCODE_BADNOTSUPPORTED ((UA_StatusCode) 0x803D0000)

/* A requested item was not found or a search operation ended without success. */
#define UA_STATUSCODE_BADNOTFOUND ((UA_StatusCode) 0x803E0000)

/* The object cannot be used because it has been deleted. */
#define UA_STATUSCODE_BADOBJECTDELETED ((UA_StatusCode) 0x803F0000)

/* Requested operation is not implemented. */
#define UA_STATUSCODE_BADNOTIMPLEMENTED ((UA_StatusCode) 0x80400000)

/* The monitoring mode is invalid. */
#define UA_STATUSCODE_BADMONITORINGMODEINVALID ((UA_StatusCode)␣
```

(continues on next page)

(continued from previous page)

```
↪0x80410000)

/* The monitoring item id does not refer to a valid monitored item. */
#define UA_STATUSCODE_BADMONITOREDITEMIDINVALID ((UA_StatusCode)␣
↪0x80420000)

/* The monitored item filter parameter is not valid. */
#define UA_STATUSCODE_BADMONITOREDITEMFILTERINVALID ((UA_
↪StatusCode) 0x80430000)

/* The server does not support the requested monitored item filter. */
#define UA_STATUSCODE_BADMONITOREDITEMFILTERUNSUPPORTED ((UA_
↪StatusCode) 0x80440000)

/* A monitoring filter cannot be used in combination with the attribute specified. */
#define UA_STATUSCODE_BADFILTERNOTALLOWED ((UA_StatusCode)␣
↪0x80450000)

/* A mandatory structured parameter was missing or null. */
#define UA_STATUSCODE_BADSTRUCTUREMISSING ((UA_StatusCode) 0x80460000)

/* The event filter is not valid. */
#define UA_STATUSCODE_BADEVENTFILTERINVALID ((UA_StatusCode)␣
↪0x80470000)

/* The content filter is not valid. */
#define UA_STATUSCODE_BADCONTENTFILTERINVALID ((UA_StatusCode)␣
↪0x80480000)

/* An unrecognized operator was provided in a filter. */
#define UA_STATUSCODE_BADFILTEROPERATORINVALID ((UA_StatusCode)␣
↪0x80C10000)

/* A valid operator was provide */
#define UA_STATUSCODE_BADFILTEROPERATORUNSUPPORTED ((UA_
↪StatusCode) 0x80C20000)

/* The number of operands provided for the filter operator was less then expected for the␣
↪operand provided. */
#define UA_STATUSCODE_BADFILTEROPERANDCOUNTMISMATCH ((UA_
↪StatusCode) 0x80C30000)

/* The operand used in a content filter is not valid. */
#define UA_STATUSCODE_BADFILTEROPERANDINVALID ((UA_StatusCode)␣
↪0x80490000)

/* The referenced element is not a valid element in the content filter. */
#define UA_STATUSCODE_BADFILTERELEMENTINVALID ((UA_StatusCode)␣
↪0x80C40000)
```

(continues on next page)

(continued from previous page)

```
/* The referenced literal is not a valid value. */
#define UA_STATUSCODE_BADFILTERLITERALINVALID ((UA_StatusCode)␣
↪0x80C50000)

/* The continuation point provide is longer valid. */
#define UA_STATUSCODE_BADCONTINUATIONPOINTINVALID ((UA_StatusCode)␣
↪0x804A0000)

/* The operation could not be processed because all continuation points have been allocated.
↪ */
#define UA_STATUSCODE_BADNOCONTINUATIONPOINTS ((UA_StatusCode)␣
↪0x804B0000)

/* The reference type id does not refer to a valid reference type node. */
#define UA_STATUSCODE_BADREFERENCETYPEIDINVALID ((UA_StatusCode)␣
↪0x804C0000)

/* The browse direction is not valid. */
#define UA_STATUSCODE_BADBROWSEDIRECTIONINVALID ((UA_StatusCode)␣
↪0x804D0000)

/* The node is not part of the view. */
#define UA_STATUSCODE_BADNODENOTINVIEW ((UA_StatusCode) 0x804E0000)

/* The number was not accepted because of a numeric overflow. */
#define UA_STATUSCODE_BADNUMERICOVERFLOW ((UA_StatusCode) 0x81120000)

/* The ServerUri is not a valid URI. */
#define UA_STATUSCODE_BADSERVERURIINVALID ((UA_StatusCode) 0x804F0000)

/* No ServerName was specified. */
#define UA_STATUSCODE_BADSERVERNAMEMISSING ((UA_StatusCode)␣
↪0x80500000)

/* No DiscoveryUrl was specified. */
#define UA_STATUSCODE_BADDISCOVERYURLMISSING ((UA_StatusCode)␣
↪0x80510000)

/* The semaphore file specified by the client is not valid. */
#define UA_STATUSCODE_BADSEMAPHOREFILEMISSING ((UA_StatusCode)␣
↪0x80520000)

/* The security token request type is not valid. */
#define UA_STATUSCODE_BADREQUESTTYPEINVALID ((UA_StatusCode)␣
↪0x80530000)

/* The security mode does not meet the requirements set by the server. */
#define UA_STATUSCODE_BADSECURITYMODEREJECTED ((UA_StatusCode)␣
```

(continues on next page)

(continued from previous page)

```
↪0x80540000)

/* The security policy does not meet the requirements set by the server. */
#define UA_STATUSCODE_BADSECURITYPOLICYREJECTED ((UA_StatusCode)␣
↪0x80550000)

/* The server has reached its maximum number of sessions. */
#define UA_STATUSCODE_BADTOOMANYSESSIONS ((UA_StatusCode) 0x80560000)

/* The user token signature is missing or invalid. */
#define UA_STATUSCODE_BADUSERSIGNATUREINVALID ((UA_StatusCode)␣
↪0x80570000)

/* The signature generated with the client certificate is missing or invalid. */
#define UA_STATUSCODE_BADAPPLICATIONSIGNATUREINVALID ((UA_
↪StatusCode) 0x80580000)

/* The client did not provide at least one software certificate that is valid and meets the␣
↪profile requirements for the server. */
#define UA_STATUSCODE_BADNOVALIDCERTIFICATES ((UA_StatusCode)␣
↪0x80590000)

/* The server does not support changing the user identity assigned to the session. */
#define UA_STATUSCODE_BADIDENTITYCHANGENOTSUPPORTED ((UA_
↪StatusCode) 0x80C60000)

/* The request was cancelled by the client with the Cancel service. */
#define UA_STATUSCODE_BADREQUESTCANCELLEDBYREQUEST ((UA_
↪StatusCode) 0x805A0000)

/* The parent node id does not to refer to a valid node. */
#define UA_STATUSCODE_BADPARENTNODEIDINVALID ((UA_StatusCode)␣
↪0x805B0000)

/* The reference could not be created because it violates constraints imposed by the data␣
↪model. */
#define UA_STATUSCODE_BADREFERENCENOTALLOWED ((UA_StatusCode)␣
↪0x805C0000)

/* The requested node id was reject because it was either invalid or server does not allow␣
↪node ids to be specified by the client. */
#define UA_STATUSCODE_BADNODEIDREJECTED ((UA_StatusCode) 0x805D0000)

/* The requested node id is already used by another node. */
#define UA_STATUSCODE_BADNODEIDEXISTS ((UA_StatusCode) 0x805E0000)

/* The node class is not valid. */
#define UA_STATUSCODE_BADNODECLASSINVALID ((UA_StatusCode) 0x805F0000)
```

(continues on next page)

(continued from previous page)

```
/* The browse name is invalid. */
#define UA_STATUSCODE_BADBROWSENAMEINVALID ((UA_StatusCode)␣
↪0x80600000)

/* The browse name is not unique among nodes that share the same relationship with the␣
↪parent. */
#define UA_STATUSCODE_BADBROWSENAMEDUPLICATED ((UA_StatusCode)␣
↪0x80610000)

/* The node attributes are not valid for the node class. */
#define UA_STATUSCODE_BADNODEATTRIBUTESINVALID ((UA_StatusCode)␣
↪0x80620000)

/* The type definition node id does not reference an appropriate type node. */
#define UA_STATUSCODE_BADTYPEDEFINITIONINVALID ((UA_StatusCode)␣
↪0x80630000)

/* The source node id does not reference a valid node. */
#define UA_STATUSCODE_BADSOURCENODEIDINVALID ((UA_StatusCode)␣
↪0x80640000)

/* The target node id does not reference a valid node. */
#define UA_STATUSCODE_BADTARGETNODEIDINVALID ((UA_StatusCode)␣
↪0x80650000)

/* The reference type between the nodes is already defined. */
#define UA_STATUSCODE_BADDUPLICATEREFERENCENOTALLOWED ((UA_
↪StatusCode) 0x80660000)

/* The server does not allow this type of self reference on this node. */
#define UA_STATUSCODE_BADINVALIDSELFREFERENCE ((UA_StatusCode)␣
↪0x80670000)

/* The reference type is not valid for a reference to a remote server. */
#define UA_STATUSCODE_BADREFERENCELOCALONLY ((UA_StatusCode)␣
↪0x80680000)

/* The server will not allow the node to be deleted. */
#define UA_STATUSCODE_BADNODELETERIGHTS ((UA_StatusCode) 0x80690000)

/* The server was not able to delete all target references. */
#define UA_STATUSCODE_UNCERTAINREFERENCENOTDELETED ((UA_
↪StatusCode) 0x40BC0000)

/* The server index is not valid. */
#define UA_STATUSCODE_BADSERVERINDEXINVALID ((UA_StatusCode)␣
↪0x806A0000)

/* The view id does not refer to a valid view node. */
```

(continues on next page)

(continued from previous page)

```
#define UA_STATUSCODE_BADVIEWIDUNKNOWN ((UA_StatusCode) 0x806B0000)

/* The view timestamp is not available or not supported. */
#define UA_STATUSCODE_BADVIEWTIMESTAMPINVALID ((UA_StatusCode)␣
↪0x80C90000)

/* The view parameters are not consistent with each other. */
#define UA_STATUSCODE_BADVIEWPARAMETERMISMATCH ((UA_StatusCode)␣
↪0x80CA0000)

/* The view version is not available or not supported. */
#define UA_STATUSCODE_BADVIEWVERSIONINVALID ((UA_StatusCode)␣
↪0x80CB0000)

/* The list of references may not be complete because the underlying system is not available.
↪ */
#define UA_STATUSCODE_UNCERTAINNOTALLNODESAVAILABLE ((UA_
↪StatusCode) 0x40C00000)

/* The server should have followed a reference to a node in a remote server but did not. The␣
↪result set may be incomplete. */
#define UA_STATUSCODE_GOODRESULTSMAYBEINCOMPLETE ((UA_StatusCode)␣
↪0x00BA0000)

/* The provided Nodeid was not a type definition nodeid. */
#define UA_STATUSCODE_BADNOTTYPEDEFINITION ((UA_StatusCode)␣
↪0x80C80000)

/* One of the references to follow in the relative path references to a node in the address␣
↪space in another server. */
#define UA_STATUSCODE_UNCERTAINREFERENCEOUTOFSERVER ((UA_
↪StatusCode) 0x406C0000)

/* The requested operation has too many matches to return. */
#define UA_STATUSCODE_BADTOOMANYMATCHES ((UA_StatusCode) 0x806D0000)

/* The requested operation requires too many resources in the server. */
#define UA_STATUSCODE_BADQUERYTOOCOMPLEX ((UA_StatusCode)␣
↪0x806E0000)

/* The requested operation has no match to return. */
#define UA_STATUSCODE_BADNOMATCH ((UA_StatusCode) 0x806F0000)

/* The max age parameter is invalid. */
#define UA_STATUSCODE_BADMAXAGEINVALID ((UA_StatusCode) 0x80700000)

/* The operation is not permitted over the current secure channel. */
#define UA_STATUSCODE_BADSECURITYMODEINSUFFICIENT ((UA_StatusCode)␣
↪0x80E60000)
```

(continues on next page)

(continued from previous page)

```
/* The history details parameter is not valid. */
#define UA_STATUSCODE_BADHISTORYOPERATIONINVALID ((UA_StatusCode)␣
↪0x80710000)

/* The server does not support the requested operation. */
#define UA_STATUSCODE_BADHISTORYOPERATIONUNSUPPORTED ((UA_
↪StatusCode) 0x80720000)

/* The defined timestamp to return was invalid. */
#define UA_STATUSCODE_BADINVALIDTIMESTAMPARGUMENT ((UA_
↪StatusCode) 0x80BD0000)

/* The server does not support writing the combination of valu */
#define UA_STATUSCODE_BADWRITENOTSUPPORTED ((UA_StatusCode)␣
↪0x80730000)

/* The value supplied for the attribute is not of the same type as the attribute's value. */
#define UA_STATUSCODE_BADTYPEMISMATCH ((UA_StatusCode) 0x80740000)

/* The method id does not refer to a method for the specified object. */
#define UA_STATUSCODE_BADMETHODINVALID ((UA_StatusCode) 0x80750000)

/* The client did not specify all of the input arguments for the method. */
#define UA_STATUSCODE_BADARGUMENTSMISSING ((UA_StatusCode) 0x80760000)

/* The executable attribute does not allow the execution of the method. */
#define UA_STATUSCODE_BADNOTEXECUTABLE ((UA_StatusCode) 0x81110000)

/* The server has reached its maximum number of subscriptions. */
#define UA_STATUSCODE_BADTOOMANYSUBSCRIPTIONS ((UA_StatusCode)␣
↪0x80770000)

/* The server has reached the maximum number of queued publish requests. */
#define UA_STATUSCODE_BADTOOMANYPUBLISHREQUESTS ((UA_StatusCode)␣
↪0x80780000)

/* There is no subscription available for this session. */
#define UA_STATUSCODE_BADNOSUBSCRIPTION ((UA_StatusCode) 0x80790000)

/* The sequence number is unknown to the server. */
#define UA_STATUSCODE_BADSEQUENCENUMBERUNKNOWN ((UA_StatusCode)␣
↪0x807A0000)

/* The Server does not support retransmission queue and acknowledgement of sequence␣
↪numbers is not available. */
#define UA_STATUSCODE_GOODRETRANSMISSIONQUEUENOTSUPPORTED␣
↪((UA_StatusCode) 0x00DF0000)
```

(continues on next page)

(continued from previous page)

```
/* The requested notification message is no longer available. */
#define UA_STATUSCODE_BADMESSAGENOTAVAILABLE ((UA_StatusCode)␣
↪0x807B0000)

/* The client of the current session does not support one or more Profiles that are necessary␣
↪for the subscription. */
#define UA_STATUSCODE_BADINSUFFICIENTCLIENTPROFILE ((UA_StatusCode)␣
↪0x807C0000)

/* The sub-state machine is not currently active. */
#define UA_STATUSCODE_BADSTATENOTACTIVE ((UA_StatusCode) 0x80BF0000)

/* An equivalent rule already exists. */
#define UA_STATUSCODE_BADALREADYEXISTS ((UA_StatusCode) 0x81150000)

/* The server cannot process the request because it is too busy. */
#define UA_STATUSCODE_BADTCPSERVERTOOBUSY ((UA_StatusCode)␣
↪0x807D0000)

/* The type of the message specified in the header invalid. */
#define UA_STATUSCODE_BADTCPMESSAGETYPEINVALID ((UA_StatusCode)␣
↪0x807E0000)

/* The SecureChannelId and/or TokenId are not currently in use. */
#define UA_STATUSCODE_BADTCPSECURECHANNELUNKNOWN ((UA_
↪StatusCode) 0x807F0000)

/* The size of the message chunk specified in the header is too large. */
#define UA_STATUSCODE_BADTCPMESSAGETOOLARGE ((UA_StatusCode)␣
↪0x80800000)

/* There are not enough resources to process the request. */
#define UA_STATUSCODE_BADTCPNOTENOUGHRESOURCES ((UA_StatusCode)␣
↪0x80810000)

/* An internal error occurred. */
#define UA_STATUSCODE_BADTCPINTERNALERROR ((UA_StatusCode)␣
↪0x80820000)

/* The server does not recognize the QueryString specified. */
#define UA_STATUSCODE_BADTCPENDPOINTURLINVALID ((UA_StatusCode)␣
↪0x80830000)

/* The request could not be sent because of a network interruption. */
#define UA_STATUSCODE_BADREQUESTINTERRUPTED ((UA_StatusCode)␣
↪0x80840000)

/* Timeout occurred while processing the request. */
#define UA_STATUSCODE_BADREQUESTTIMEOUT ((UA_StatusCode) 0x80850000)
```

(continues on next page)

```
/* The secure channel has been closed. */
#define UA_STATUSCODE_BADSECURECHANNELCLOSED ((UA_StatusCode)␣
↪0x80860000)

/* The token has expired or is not recognized. */
#define UA_STATUSCODE_BADSECURECHANNELTOKENUNKNOWN ((UA_
↪StatusCode) 0x80870000)

/* The sequence number is not valid. */
#define UA_STATUSCODE_BADSEQUENCENUMBERINVALID ((UA_StatusCode)␣
↪0x80880000)

/* The applications do not have compatible protocol versions. */
#define UA_STATUSCODE_BADPROTOCOLVERSIONUNSUPPORTED ((UA_
↪StatusCode) 0x80BE0000)

/* There is a problem with the configuration that affects the usefulness of the value. */
#define UA_STATUSCODE_BADCONFIGURATIONERROR ((UA_StatusCode)␣
↪0x80890000)

/* The variable should receive its value from another variabl */
#define UA_STATUSCODE_BADNOTCONNECTED ((UA_StatusCode) 0x808A0000)

/* There has been a failure in the device/data source that generates the value that has␣
↪affected the value. */
#define UA_STATUSCODE_BADDEVICEFAILURE ((UA_StatusCode) 0x808B0000)

/* There has been a failure in the sensor from which the value is derived by the device/data␣
↪source. */
#define UA_STATUSCODE_BADSENSORFAILURE ((UA_StatusCode) 0x808C0000)

/* The source of the data is not operational. */
#define UA_STATUSCODE_BADOUTOFSERVICE ((UA_StatusCode) 0x808D0000)

/* The deadband filter is not valid. */
#define UA_STATUSCODE_BADDEADBANDFILTERINVALID ((UA_StatusCode)␣
↪0x808E0000)

/* Communication to the data source has failed. The variable value is the last value that␣
↪had a good quality. */
#define UA_STATUSCODE_UNCERTAINNOCOMMUNICATIONLASTUSABLEVALUE␣
↪((UA_StatusCode) 0x408F0000)

/* Whatever was updating this value has stopped doing so. */
#define UA_STATUSCODE_UNCERTAINLASTUSABLEVALUE ((UA_StatusCode)␣
↪0x40900000)

/* The value is an operational value that was manually overwritten. */
#define UA_STATUSCODE_UNCERTAINSUBSTITUTEVALUE ((UA_StatusCode)␣
```

(continues on next page)

(continued from previous page)

```
↪0x40910000)

/* The value is an initial value for a variable that normally receives its value from another␣
↪variable. */
#define UA_STATUSCODE_UNCERTAININITIALVALUE ((UA_StatusCode)␣
↪0x40920000)

/* The value is at one of the sensor limits. */
#define UA_STATUSCODE_UNCERTAINSENSORNOTACCURATE ((UA_StatusCode)␣
↪0x40930000)

/* The value is outside of the range of values defined for this parameter. */
#define UA_STATUSCODE_UNCERTAINENGINEERINGUNITSEXCEEDED ((UA_
↪StatusCode) 0x40940000)

/* The value is derived from multiple sources and has less than the required number of␣
↪Good sources. */
#define UA_STATUSCODE_UNCERTAINSUBNORMAL ((UA_StatusCode) 0x40950000)

/* The value has been overridden. */
#define UA_STATUSCODE_GOODLOCALOVERRIDE ((UA_StatusCode) 0x00960000)

/* This Condition refresh faile */
#define UA_STATUSCODE_BADREFRESHINPROGRESS ((UA_StatusCode)␣
↪0x80970000)

/* This condition has already been disabled. */
#define UA_STATUSCODE_BADCONDITIONALREADYDISABLED ((UA_StatusCode)␣
↪0x80980000)

/* This condition has already been enabled. */
#define UA_STATUSCODE_BADCONDITIONALREADYENABLED ((UA_StatusCode)␣
↪0x80CC0000)

/* Property not availabl */
#define UA_STATUSCODE_BADCONDITIONDISABLED ((UA_StatusCode)␣
↪0x80990000)

/* The specified event id is not recognized. */
#define UA_STATUSCODE_BADEVENTIDUNKNOWN ((UA_StatusCode) 0x809A0000)

/* The event cannot be acknowledged. */
#define UA_STATUSCODE_BADEVENTNOTACKNOWLEDGEABLE ((UA_
↪StatusCode) 0x80BB0000)

/* The dialog condition is not active. */
#define UA_STATUSCODE_BADDIALOGNOTACTIVE ((UA_StatusCode) 0x80CD0000)

/* The response is not valid for the dialog. */
```

(continues on next page)

(continued from previous page)

```
#define UA_STATUSCODE_BADDIALOGRESPONSEINVALID ((UA_StatusCode)␣
↪0x80CE0000)

/* The condition branch has already been acknowledged. */
#define UA_STATUSCODE_BADCONDITIONBRANCHALREADYACKED ((UA_
↪StatusCode) 0x80CF0000)

/* The condition branch has already been confirmed. */
#define UA_STATUSCODE_BADCONDITIONBRANCHALREADYCONFIRMED ((UA_
↪StatusCode) 0x80D00000)

/* The condition has already been shelved. */
#define UA_STATUSCODE_BADCONDITIONALREADYSHELVED ((UA_StatusCode)␣
↪0x80D10000)

/* The condition is not currently shelved. */
#define UA_STATUSCODE_BADCONDITIONNOTSHELVED ((UA_StatusCode)␣
↪0x80D20000)

/* The shelving time not within an acceptable range. */
#define UA_STATUSCODE_BADSHELVINGTIMEOUTOFRANGE ((UA_StatusCode)␣
↪0x80D30000)

/* No data exists for the requested time range or event filter. */
#define UA_STATUSCODE_BADNODATA ((UA_StatusCode) 0x809B0000)

/* No data found to provide upper or lower bound value. */
#define UA_STATUSCODE_BADBOUNDNOTFOUND ((UA_StatusCode) 0x80D70000)

/* The server cannot retrieve a bound for the variable. */
#define UA_STATUSCODE_BADBOUNDNOTSUPPORTED ((UA_StatusCode)␣
↪0x80D80000)

/* Data is missing due to collection started/stopped/lost. */
#define UA_STATUSCODE_BADDATALOST ((UA_StatusCode) 0x809D0000)

/* Expected data is unavailable for the requested time range due to an un-mounted volum */
#define UA_STATUSCODE_BADDATAUNAVAILABLE ((UA_StatusCode) 0x809E0000)

/* The data or event was not successfully inserted because a matching entry exists. */
#define UA_STATUSCODE_BADENTRYEXISTS ((UA_StatusCode) 0x809F0000)

/* The data or event was not successfully updated because no matching entry exists. */
#define UA_STATUSCODE_BADNOENTRYEXISTS ((UA_StatusCode) 0x80A00000)

/* The client requested history using a timestamp format the server does not support (i.e␣
↪requested ServerTimestamp when server only supports SourceTimestamp). */
#define UA_STATUSCODE_BADTIMESTAMPNOTSUPPORTED ((UA_StatusCode)␣
↪0x80A10000)
```

(continues on next page)

(continued from previous page)

```
/* The data or event was successfully inserted into the historical database. */
#define UA_STATUSCODE_GOODENTRYINSERTED ((UA_StatusCode) 0x00A20000)

/* The data or event field was successfully replaced in the historical database. */
#define UA_STATUSCODE_GOODENTRYREPLACED ((UA_StatusCode) 0x00A30000)

/* The value is derived from multiple values and has less than the required number of Good␣
↪values. */
#define UA_STATUSCODE_UNCERTAINDATASUBNORMAL ((UA_StatusCode)␣
↪0x40A40000)

/* No data exists for the requested time range or event filter. */
#define UA_STATUSCODE_GOODNODATA ((UA_StatusCode) 0x00A50000)

/* The data or event field was successfully replaced in the historical database. */
#define UA_STATUSCODE_GOODMOREDATA ((UA_StatusCode) 0x00A60000)

/* The requested number of Aggregates does not match the requested number of NodeIds. */
#define UA_STATUSCODE_BADAGGREGATELISTMISMATCH ((UA_StatusCode)␣
↪0x80D40000)

/* The requested Aggregate is not support by the server. */
#define UA_STATUSCODE_BADAGGREGATENOTSUPPORTED ((UA_StatusCode)␣
↪0x80D50000)

/* The aggregate value could not be derived due to invalid data inputs. */
#define UA_STATUSCODE_BADAGGREGATEINVALIDINPUTS ((UA_StatusCode)␣
↪0x80D60000)

/* The aggregate configuration is not valid for specified node. */
#define UA_STATUSCODE_BADAGGREGATECONFIGURATIONREJECTED ((UA_
↪StatusCode) 0x80DA0000)

/* The request specifies fields which are not valid for the EventType or cannot be saved by␣
↪the historian. */
#define UA_STATUSCODE_GOODDATAIGNORED ((UA_StatusCode) 0x00D90000)

/* The request was rejected by the server because it did not meet the criteria set by the␣
↪server. */
#define UA_STATUSCODE_BADREQUESTNOTALLOWED ((UA_StatusCode)␣
↪0x80E40000)

/* The request has not been processed by the server yet. */
#define UA_STATUSCODE_BADREQUESTNOTCOMPLETE ((UA_StatusCode)␣
↪0x81130000)

/* The operation is not allowed because a transaction is in progress. */
#define UA_STATUSCODE_BADTRANSACTIONPENDING ((UA_StatusCode)␣
```

(continues on next page)

(continued from previous page)

```
↪0x80E80000)

/* The device identity needs a ticket before it can be accepted. */
#define UA_STATUSCODE_BADTICKETREQUIRED ((UA_StatusCode) 0x811F0000)

/* The device identity needs a ticket before it can be accepted. */
#define UA_STATUSCODE_BADTICKETINVALID ((UA_StatusCode) 0x81200000)

/* The value does not come from the real source and has been edited by the server. */
#define UA_STATUSCODE_GOODEDITED ((UA_StatusCode) 0x00DC0000)

/* There was an error in execution of these post-actions. */
#define UA_STATUSCODE_GOODPOSTACTIONFAILED ((UA_StatusCode)␣
↪0x00DD0000)

/* The related EngineeringUnit has been changed but the Variable Value is still provided␣
↪based on the previous unit. */
#define UA_STATUSCODE_UNCERTAINDOMINANTVALUECHANGED ((UA_
↪StatusCode) 0x40DE0000)

/* A dependent value has been changed but the change has not been applied to the device.␣
↪*/
#define UA_STATUSCODE_GOODDEPENDENTVALUECHANGED ((UA_StatusCode)␣
↪0x00E00000)

/* The related EngineeringUnit has been changed but this change has not been applied to␣
↪the device. The Variable Value is still dependent on the previous unit but its status is␣
↪currently Bad. */
#define UA_STATUSCODE_BADDOMINANTVALUECHANGED ((UA_StatusCode)␣
↪0x80E10000)

/* A dependent value has been changed but the change has not been applied to the device.␣
↪The quality of the dominant variable is uncertain. */
#define UA_STATUSCODE_UNCERTAINDEPENDENTVALUECHANGED ((UA_
↪StatusCode) 0x40E20000)

/* A dependent value has been changed but the change has not been applied to the device.␣
↪The quality of the dominant variable is Bad. */
#define UA_STATUSCODE_BADDEPENDENTVALUECHANGED ((UA_StatusCode)␣
↪0x80E30000)

/* It is delivered with a dominant Variable value when a dependent Variable has changed␣
↪but the change has not been applied. */
#define UA_STATUSCODE_GOODEDITED_DEPENDENTVALUECHANGED ((UA_
↪StatusCode) 0x01160000)

/* It is delivered with a dependent Variable value when a dominant Variable has changed␣
↪but the change has not been applied. */
#define UA_STATUSCODE_GOODEDITED_DOMINANTVALUECHANGED ((UA_
```

(continues on next page)

(continued from previous page)

```
↪StatusCode) 0x01170000)

/* It is delivered with a dependent Variable value when a dominant or dependent Variable␣
↪has changed but change has not been applied. */
#define UA_STATUSCODE_GOODEDITED_DOMINANTVALUECHANGED_
↪DEPENDENTVALUECHANGED ((UA_StatusCode) 0x01180000)

/* It is delivered with a Variable value when Variable has changed but the value is not legal.␣
↪*/
#define UA_STATUSCODE_BADEDITED_OUTOFRANGE ((UA_StatusCode)␣
↪0x81190000)

/* It is delivered with a Variable value when a source Variable has changed but the value is␣
↪not legal. */
#define UA_STATUSCODE_BADINITIALVALUE_OUTOFRANGE ((UA_StatusCode)␣
↪0x811A0000)

/* It is delivered with a dependent Variable value when a dominant Variable has changed␣
↪and the value is not legal. */
#define UA_STATUSCODE_BADOUTOFRANGE_DOMINANTVALUECHANGED␣
↪((UA_StatusCode) 0x811B0000)

/* It is delivered with a dependent Variable value when a dominant Variable has change */
#define UA_STATUSCODE_BADEDITED_OUTOFRANGE_
↪DOMINANTVALUECHANGED ((UA_StatusCode) 0x811C0000)

/* It is delivered with a dependent Variable value when a dominant or dependent Variable␣
↪has changed and the value is not legal. */
#define UA_STATUSCODE_BADOUTOFRANGE_DOMINANTVALUECHANGED_
↪DEPENDENTVALUECHANGED ((UA_StatusCode) 0x811D0000)

/* It is delivered with a dependent Variable value when a dominant or dependent Variable␣
↪has change */
#define UA_STATUSCODE_BADEDITED_OUTOFRANGE_
↪DOMINANTVALUECHANGED_DEPENDENTVALUECHANGED ((UA_StatusCode)␣
↪0x811E0000)

/* The communication layer has raised an event. */
#define UA_STATUSCODE_GOODCOMMUNICATIONEVENT ((UA_StatusCode)␣
↪0x00A70000)

/* The system is shutting down. */
#define UA_STATUSCODE_GOODSHUTDOWNEVENT ((UA_StatusCode) 0x00A80000)

/* The operation is not finished and needs to be called again. */
#define UA_STATUSCODE_GOODCALLAGAIN ((UA_StatusCode) 0x00A90000)

/* A non-critical timeout occurred. */
#define UA_STATUSCODE_GOODNONCRITICALTIMEOUT ((UA_StatusCode)␣
```

(continues on next page)

(continued from previous page)

```
↪0x00AA0000)

/* One or more arguments are invalid. */
#define UA_STATUSCODE_BADINVALIDARGUMENT ((UA_StatusCode) 0x80AB0000)

/* Could not establish a network connection to remote server. */
#define UA_STATUSCODE_BADCONNECTIONREJECTED ((UA_StatusCode)␣
↪0x80AC0000)

/* The server has disconnected from the client. */
#define UA_STATUSCODE_BADDISCONNECT ((UA_StatusCode) 0x80AD0000)

/* The network connection has been closed. */
#define UA_STATUSCODE_BADCONNECTIONCLOSED ((UA_StatusCode)␣
↪0x80AE0000)

/* The operation cannot be completed because the object is close */
#define UA_STATUSCODE_BADINVALIDSTATE ((UA_StatusCode) 0x80AF0000)

/* Cannot move beyond end of the stream. */
#define UA_STATUSCODE_BADENDOFSTREAM ((UA_StatusCode) 0x80B00000)

/* No data is currently available for reading from a non-blocking stream. */
#define UA_STATUSCODE_BADNODATAAVAILABLE ((UA_StatusCode) 0x80B10000)

/* The asynchronous operation is waiting for a response. */
#define UA_STATUSCODE_BADWAITINGFORRESPONSE ((UA_StatusCode)␣
↪0x80B20000)

/* The asynchronous operation was abandoned by the caller. */
#define UA_STATUSCODE_BADOPERATIONABANDONED ((UA_StatusCode)␣
↪0x80B30000)

/* The stream did not return all data requested (possibly because it is a non-blocking␣
↪stream). */
#define UA_STATUSCODE_BADEXPECTEDSTREAMTOBLOCK ((UA_StatusCode)␣
↪0x80B40000)

/* Non blocking behaviour is required and the operation would block. */
#define UA_STATUSCODE_BADWOULDBLOCK ((UA_StatusCode) 0x80B50000)

/* A value had an invalid syntax. */
#define UA_STATUSCODE_BADSYNTAXERROR ((UA_StatusCode) 0x80B60000)

/* The operation could not be finished because all available connections are in use. */
#define UA_STATUSCODE_BADMAXCONNECTIONSREACHED ((UA_StatusCode)␣
↪0x80B70000)

/* Depending on the version of the schema, the following might be already defined: */
```

(continues on next page)

(continued from previous page)

```
#ifndef UA_STATUSCODE_GOOD
# define UA_STATUSCODE_GOOD ((UA_StatusCode) 0x00000000)
#endif
#ifndef UA_STATUSCODE_UNCERTAIN
# define UA_STATUSCODE_UNCERTAIN ((UA_StatusCode) 0x40000000)
#endif
#ifndef UA_STATUSCODE_BAD
# define UA_STATUSCODE_BAD ((UA_StatusCode) 0x80000000)
#endif
```

CHAPTER

FOURTEEN

PLUGIN API

14.1 Logging Plugin API

Servers and clients define a logger in their configuration. The logger is a plugin. A default plugin that logs to stdout is provided as an example. The logger plugin is stateful and can point to custom data. So it is possible to keep open file handlers in the logger context.

Every log message consists of a log level, a log category and a string message content. The timestamp of the log message is created within the logger.

```
typedef enum {
    UA_LOGLEVEL_TRACE   = 100,
    UA_LOGLEVEL_DEBUG   = 200,
    UA_LOGLEVEL_INFO    = 300,
    UA_LOGLEVEL_WARNING = 400,
    UA_LOGLEVEL_ERROR   = 500,
    UA_LOGLEVEL_FATAL   = 600
} UA_LogLevel;

#define UA_LOGCATEGORIES 10

typedef enum {
    UA_LOGCATEGORY_NETWORK = 0,
    UA_LOGCATEGORY_SECURECHANNEL = 1,
    UA_LOGCATEGORY_SESSION = 2,
    UA_LOGCATEGORY_SERVER = 3,
    UA_LOGCATEGORY_CLIENT = 4,
    UA_LOGCATEGORY_APPLICATION= 5,
    UA_LOGCATEGORY_USERLAND = 5, /* == APPLICATION */
    UA_LOGCATEGORY_SECURITY = 6,
    UA_LOGCATEGORY_SECURITYPOLICY = 6, /* == SECURITY */
    UA_LOGCATEGORY_EVENTLOOP = 7,
    UA_LOGCATEGORY_PUBSUB = 8,
    UA_LOGCATEGORY_DISCOVERY = 9
} UA_LogCategory;

typedef struct UA_Logger {
    /* Log a message. The message string and following varargs matching the
     * format specifiers of UA_String_format.
     *
```

(continues on next page)

(continued from previous page)

```
     * !!Attention!! UA_String_format takes additional format specifiers that
     * are not known for standard printf. These special specifiers are getting
     * used throughout the library. */
    void (*log)(void *logContext, UA_LogLevel level, UA_LogCategory category,
                const char *msg, va_list args);

    void *context; /* Logger state */

    void (*clear)(struct UA_Logger *logger); /* Clean up the logger plugin */
} UA_Logger;

static UA_INLINE void
UA_LOG_TRACE(const UA_Logger *logger, UA_LogCategory category, const char *msg,␣
↪...) {
#if UA_LOGLEVEL <= 100
    if(!logger || !logger->log)
        return;
    va_list args; va_start(args, msg);
    logger->log(logger->context, UA_LOGLEVEL_TRACE, category, msg, args);
    va_end(args);
#else
    (void) logger;
    (void) category;
    (void) msg;
#endif
}

static UA_INLINE void
UA_LOG_DEBUG(const UA_Logger *logger, UA_LogCategory category, const char *msg,
↪ ...) {
#if UA_LOGLEVEL <= 200
    if(!logger || !logger->log)
        return;
    va_list args; va_start(args, msg);
    logger->log(logger->context, UA_LOGLEVEL_DEBUG, category, msg, args);
    va_end(args);
#else
    (void) logger;
    (void) category;
    (void) msg;
#endif
}

static UA_INLINE void
UA_LOG_INFO(const UA_Logger *logger, UA_LogCategory category, const char *msg, ...
↪) {
#if UA_LOGLEVEL <= 300
    if(!logger || !logger->log)
        return;
```

(continues on next page)

(continued from previous page)

```
    va_list args; va_start(args, msg);
    logger->log(logger->context, UA_LOGLEVEL_INFO, category, msg, args);
    va_end(args);
#else
    (void) logger;
    (void) category;
    (void) msg;
#endif
}

static UA_INLINE void
UA_LOG_WARNING(const UA_Logger *logger, UA_LogCategory category, const char␣
↪*msg, ...) {
#if UA_LOGLEVEL <= 400
    if(!logger || !logger->log)
        return;
    va_list args; va_start(args, msg);
    logger->log(logger->context, UA_LOGLEVEL_WARNING, category, msg, args);
    va_end(args);
#else
    (void) logger;
    (void) category;
    (void) msg;
#endif
}

static UA_INLINE void
UA_LOG_ERROR(const UA_Logger *logger, UA_LogCategory category, const char *msg,␣
↪...) {
#if UA_LOGLEVEL <= 500
    if(!logger || !logger->log)
        return;
    va_list args; va_start(args, msg);
    logger->log(logger->context, UA_LOGLEVEL_ERROR, category, msg, args);
    va_end(args);
#else
    (void) logger;
    (void) category;
    (void) msg;
#endif
}

static UA_INLINE void
UA_LOG_FATAL(const UA_Logger *logger, UA_LogCategory category, const char *msg, ..
↪.) {
#if UA_LOGLEVEL <= 600
    if(!logger || !logger->log)
        return;
    va_list args; va_start(args, msg);
```

(continues on next page)

(continued from previous page)

```
    logger->log(logger->context, UA_LOGLEVEL_FATAL, category, msg, args);
    va_end(args);
#else
    (void) logger;
    (void) category;
    (void) msg;
#endif
}
```

14.2 Nodestore Plugin API

Warning!! The structures defined in this section are only relevant for the developers of custom Nodestores. The interaction with the information model is possible only via the OPC UA *Services*. So the following sections are mainly for users that seek to understand the underlying representation – which is not directly accessible.

14.2.1 ReferenceType Bitfield Representation

ReferenceTypes have an alternative representation as an index into a bitfield for fast comparison. The index is generated when the corresponding ReferenceTypeNode is added. By bounding the number of ReferenceTypes that can exist in the server, the bitfield can represent a set of an combination of ReferenceTypes.

Every ReferenceTypeNode contains a bitfield with the set of all its subtypes. This speeds up the Browse services substantially.

The following ReferenceTypes have a fixed index. The NS0 bootstrapping creates these ReferenceTypes in-order.

```
#define UA_REFERENCETYPEINDEX_REFERENCES 0
#define UA_REFERENCETYPEINDEX_HASSUBTYPE 1
#define UA_REFERENCETYPEINDEX_AGGREGATES 2
#define UA_REFERENCETYPEINDEX_HIERARCHICALREFERENCES 3
#define UA_REFERENCETYPEINDEX_NONHIERARCHICALREFERENCES 4
#define UA_REFERENCETYPEINDEX_HASCHILD 5
#define UA_REFERENCETYPEINDEX_ORGANIZES 6
#define UA_REFERENCETYPEINDEX_HASEVENTSOURCE 7
#define UA_REFERENCETYPEINDEX_HASMODELLINGRULE 8
#define UA_REFERENCETYPEINDEX_HASENCODING 9
#define UA_REFERENCETYPEINDEX_HASDESCRIPTION 10
#define UA_REFERENCETYPEINDEX_HASTYPEDEFINITION 11
#define UA_REFERENCETYPEINDEX_GENERATESEVENT 12
#define UA_REFERENCETYPEINDEX_HASPROPERTY 13
#define UA_REFERENCETYPEINDEX_HASCOMPONENT 14
#define UA_REFERENCETYPEINDEX_HASNOTIFIER 15
#define UA_REFERENCETYPEINDEX_HASORDEREDCOMPONENT 16
#define UA_REFERENCETYPEINDEX_HASINTERFACE 17

/* The maximum number of ReferenceTypes. Must be a multiple of 32. */
#define UA_REFERENCETYPESET_MAX 256
```

(continues on next page)

(continued from previous page)

```
typedef struct {
    UA_UInt32 bits[UA_REFERENCETYPESET_MAX / 32];
} UA_ReferenceTypeSet;

extern const UA_ReferenceTypeSet UA_REFERENCETYPESET_NONE;
extern const UA_ReferenceTypeSet UA_REFERENCETYPESET_ALL;

static UA_INLINE void
UA_ReferenceTypeSet_init(UA_ReferenceTypeSet *set) {
    memset(set, 0, sizeof(UA_ReferenceTypeSet));
}

static UA_INLINE UA_ReferenceTypeSet
UA_REFTYPESET(UA_Byte index) {
    UA_Byte i = index / 32, j = index % 32;
    UA_ReferenceTypeSet set;
    UA_ReferenceTypeSet_init(&set);
    set.bits[i] |= ((UA_UInt32)1) << j;
    return set;
}

static UA_INLINE void
UA_ReferenceTypeSet_add(UA_ReferenceTypeSet *set,
                        UA_Byte newIndex) {
    UA_Byte i = newIndex / 32, j = newIndex % 32;
    set->bits[i] |= ((UA_UInt32)1) << j;
}

static UA_INLINE UA_ReferenceTypeSet
UA_ReferenceTypeSet_union(const UA_ReferenceTypeSet setA,
                          const UA_ReferenceTypeSet setB) {
    UA_ReferenceTypeSet set;
    for(size_t i = 0; i < UA_REFERENCETYPESET_MAX / 32; i++)
        set.bits[i] = setA.bits[i] | setB.bits[i];
    return set;
}

static UA_INLINE UA_Boolean
UA_ReferenceTypeSet_contains(const UA_ReferenceTypeSet *set, UA_Byte index) {
    UA_Byte i = index / 32, j = index % 32;
    return !!(set->bits[i] & (((UA_UInt32)1) << j));
}
```

14.2.2 Node Pointer

The "native" format for reference between nodes is the ExpandedNodeId. That is, references can also point to external servers. In practice, most references point to local nodes using numerical NodeIds from the standard-defined namespace zero. In order to save space (and time), pointer-tagging is used for compressed "NodePointer" representations. Numerical NodeIds are immediately contained in the pointer. Full NodeIds and ExpandedNodeIds are behind a pointer indirection. If the Nodestore

supports it, a NodePointer can also be an actual pointer to the target node.

Depending on the processor architecture, some numerical NodeIds don't fit into an immediate encoding and are kept as pointers. ExpandedNodeIds may be internally translated to "normal" NodeIds. Use the provided functions to generate NodePointers that fit the assumptions for the local architecture.

```
/* Forward declaration. All node structures begin with the NodeHead. */
struct UA_NodeHead;
typedef struct UA_NodeHead UA_NodeHead;

/* Tagged Pointer structure. */
typedef union {
    uintptr_t immediate;                /* 00: Small numerical NodeId */
    const UA_NodeId *id;                /* 01: Pointer to NodeId */
    const UA_ExpandedNodeId *expandedId; /* 10: Pointer to ExternalNodeId */
    const UA_NodeHead *node;            /* 11: Pointer to a node */
} UA_NodePointer;

/* Sets the pointer to an immediate NodeId "ns=0;i=0" similar to a freshly
 * initialized UA_NodeId */
static UA_INLINE void
UA_NodePointer_init(UA_NodePointer *np) { np->immediate = 0; }

/* NodeId and ExpandedNodeId targets are freed */
void
UA_NodePointer_clear(UA_NodePointer *np);

/* Makes a deep copy */
UA_StatusCode
UA_NodePointer_copy(UA_NodePointer in, UA_NodePointer *out);

/* Test if an ExpandedNodeId or a local NodeId */
UA_Boolean
UA_NodePointer_isLocal(UA_NodePointer np);

UA_Order
UA_NodePointer_order(UA_NodePointer p1, UA_NodePointer p2);

static UA_INLINE UA_Boolean
UA_NodePointer_equal(UA_NodePointer p1, UA_NodePointer p2) {
    return (UA_NodePointer_order(p1, p2) == UA_ORDER_EQ);
}

/* Cannot fail. The resulting NodePointer can point to the memory from the
 * NodeId. Make a deep copy if required. */
UA_NodePointer
UA_NodePointer_fromNodeId(const UA_NodeId *id);

/* Cannot fail. The resulting NodePointer can point to the memory from the
 * ExpandedNodeId. Make a deep copy if required. */
```

(continues on next page)

(continued from previous page)

```
UA_NodePointer
UA_NodePointer_fromExpandedNodeId(const UA_ExpandedNodeId *id);

/* Can point to the memory from the NodePointer */
UA_ExpandedNodeId
UA_NodePointer_toExpandedNodeId(UA_NodePointer np);

/* Can point to the memory from the NodePointer. Discards the ServerIndex and
 * NamespaceUri of a potential ExpandedNodeId inside the NodePointer. Test
 * before if the NodePointer is local. */
UA_NodeId
UA_NodePointer_toNodeId(UA_NodePointer np);
```

14.2.3 Base Node Attributes

Nodes contain attributes according to their node type. The base node attributes are common to all node types. In the OPC UA *Services*, attributes are referred to via the *NodeId* of the containing node and an integer *Attribute Id*.

Internally, open62541 uses UA_Node in places where the exact node type is not known or not important. The nodeClass attribute is used to ensure the correctness of casting from UA_Node to a specific node type.

```
typedef struct {
    UA_NodePointer targetId;  /* Has to be the first entry */
    UA_UInt32 targetNameHash; /* Hash of the target's BrowseName. Set to zero
                               * if the target is remote. */
} UA_ReferenceTarget;

typedef struct UA_ReferenceTargetTreeElem {
    UA_ReferenceTarget target;   /* Has to be the first entry */
    UA_UInt32 targetIdHash;      /* Hash of the targetId */
    struct {
        struct UA_ReferenceTargetTreeElem *left;
        struct UA_ReferenceTargetTreeElem *right;
    } idTreeEntry;
    struct {
        struct UA_ReferenceTargetTreeElem *left;
        struct UA_ReferenceTargetTreeElem *right;
    } nameTreeEntry;
} UA_ReferenceTargetTreeElem;

/* List of reference targets with the same reference type and direction. Uses
 * either an array or a tree structure. The SDK will not change the type of
 * reference target structure internally. The nodestore implementations may
 * switch internally when a node is updated.
 *
 * The recommendation is to switch to a tree once the number of refs > 8. */
typedef struct {
    union {
```

(continues on next page)

(continued from previous page)

```
        /* Organize the references in an array. Uses less memory, but incurs
         * lookups in linear time. Recommended if the number of references is
         * known to be small. */
        UA_ReferenceTarget *array;

        /* Organize the references in a tree for fast lookup. Use
         * UA_Node_addReference and UA_Node_deleteReference to modify the
         * tree-structure. The binary tree implementation (and absolute ordering
         * / duplicate browseNames are allowed) are not exposed otherwise in the
         * public API. */
        struct {
            UA_ReferenceTargetTreeElem *idRoot;   /* Lookup based on target id */
            UA_ReferenceTargetTreeElem *nameRoot; /* Lookup based on browseName*/
        } tree;
    } targets;
    size_t targetsSize;
    UA_Boolean hasRefTree; /* RefTree or RefArray? */
    UA_Byte referenceTypeIndex;
    UA_Boolean isInverse;
} UA_NodeReferenceKind;

/* Iterate over the references. Aborts when the first callback return a non-NULL
 * pointer and returns that pointer. Do not modify the reference targets during
 * the iteration. */
typedef void *
(*UA_NodeReferenceKind_iterateCallback)(void *context, UA_ReferenceTarget *target);

void *
UA_NodeReferenceKind_iterate(UA_NodeReferenceKind *rk,
                             UA_NodeReferenceKind_iterateCallback callback,
                             void *context);

/* Returns the entry for the targetId or NULL if not found */
const UA_ReferenceTarget *
UA_NodeReferenceKind_findTarget(const UA_NodeReferenceKind *rk,
                                const UA_ExpandedNodeId *targetId);

/* Switch between array and tree representation. Does nothing upon error (e.g.
 * out-of-memory). */
UA_StatusCode
UA_NodeReferenceKind_switch(UA_NodeReferenceKind *rk);

#ifdef UA_ENABLE_RBAC

/* Compact index stored in each node to reference a position in the server's
 * role-permissions array. Size configured via UA_ROLEPERMISSIONS_NODE_SIZE_
 ↪BYTE. */
#if UA_ROLEPERMISSIONS_NODE_SIZE_BYTE == 2
typedef UA_UInt16 UA_PermissionIndex;
```

(continues on next page)

(continued from previous page)

```
#define UA_PERMISSION_INDEX_INVALID 0xFFFF
#elif UA_ROLEPERMISSIONS_NODE_SIZE_BYTE == 4
typedef UA_UInt32 UA_PermissionIndex;
#define UA_PERMISSION_INDEX_INVALID 0xFFFFFFFF
#elif UA_ROLEPERMISSIONS_NODE_SIZE_BYTE == 8
typedef UA_UInt64 UA_PermissionIndex;
#define UA_PERMISSION_INDEX_INVALID 0xFFFFFFFFFFFFFFFF
#else
#error "UA_ROLEPERMISSIONS_NODE_SIZE_BYTE must be 2, 4, or 8"
#endif

/* Sentinel value for the refCount of a role-permission entry:
 * The entry originates from initial server configuration presets and
 * must not be deleted during server runtime. A nodestore implementation
 * may need this when managing role-permission storage. */
#define UA_ROLEPERMISSIONS_REFCOUNT_PROTECTED SIZE_MAX

#endif /* UA_ENABLE_RBAC */

/* Singly-linked LocalizedText list */
typedef struct UA_LocalizedTextListEntry {
    struct UA_LocalizedTextListEntry *next;
    UA_LocalizedText localizedText;
} UA_LocalizedTextListEntry;

/* Every Node starts with these attributes */
struct UA_NodeHead {
    UA_NodeId nodeId;
    UA_NodeClass nodeClass;
    UA_QualifiedName browseName;

    /* A node can have different localizations for displayName and description.
     * The server selects a suitable localization depending on the locale ids
     * that are set for the current session.
     *
     * Locales are added simply by writing a LocalizedText value with a new
     * locale. A locale can be removed by writing a LocalizedText value of the
     * corresponding locale with an empty text field. */
    UA_LocalizedTextListEntry *displayName;
    UA_LocalizedTextListEntry *description;

    UA_UInt32 writeMask;
    size_t referencesSize;
    UA_NodeReferenceKind *references;

    /* Members specific to open62541 */
    void *context;
    UA_Boolean constructed; /* Constructors were called */
#ifdef UA_ENABLE_RBAC
```

(continues on next page)

(continued from previous page)

```
    UA_PermissionIndex permissionIndex; /* Index into server's rolePermissions array.
                                         * UA_PERMISSION_INDEX_INVALID means no specific
                                         * permissions configured (use defaults). */
#endif
#ifdef UA_ENABLE_SUBSCRIPTIONS
    UA_MonitoredItem *monitoredItems; /* MonitoredItems for Events and immediate
                                       * DataChanges (no sampling interval). */
#endif
};
```

14.2.4 VariableNode

```
#define UA_NODE_VARIABLEATTRIBUTES                                          \
    /* Constraints on possible values */                                    \
    UA_NodeId dataType;                                                     \
    UA_Int32 valueRank;                                                     \
    size_t arrayDimensionsSize;                                             \
    UA_UInt32 *arrayDimensions;                                             \
                                                                            \
    /* The current value */                                                 \
    UA_ValueSourceType valueSourceType;                                     \
    union {                                                                 \
        struct {                                                            \
            UA_DataValue value;                                             \
            UA_ValueSourceNotifications notifications;                      \
        } internal;                                                         \
        struct {                                                            \
            UA_atomic(UA_DataValue *)* value; /* double-pointer */          \
            UA_ValueSourceNotifications notifications;                      \
        } external;                                                         \
        UA_CallbackValueSource callback;                                    \
    } valueSource;

typedef struct {
    UA_NodeHead head;
    UA_NODE_VARIABLEATTRIBUTES
    UA_Byte accessLevel;
    UA_Double minimumSamplingInterval;
    UA_Boolean historizing;

    /* Members specific to open62541 */
    UA_Boolean isDynamic; /* Some variables are ”static” in the sense that they
                           * are not attached to a dynamic process in the
                           * background. Only dynamic variables conserve source
                           * and server timestamp for the value attribute.
                           * Static variables have timestamps of ”now”. */
} UA_VariableNode;
```

14.2.5 VariableTypeNode

```
typedef struct {
    UA_NodeHead head;
    UA_NODE_VARIABLEATTRIBUTES
    UA_Boolean isAbstract;

    /* Members specific to open62541 */
    UA_NodeTypeLifecycle lifecycle;
} UA_VariableTypeNode;
```

14.2.6 MethodNode

```
typedef struct {
    UA_NodeHead head;
    UA_Boolean executable;

    /* Members specific to open62541 */
    UA_MethodCallback method;
} UA_MethodNode;
```

14.2.7 ObjectNode

```
typedef struct {
    UA_NodeHead head;
    UA_Byte eventNotifier;
} UA_ObjectNode;
```

14.2.8 ObjectTypeNode

```
typedef struct {
    UA_NodeHead head;
    UA_Boolean isAbstract;

    /* Members specific to open62541 */
    UA_NodeTypeLifecycle lifecycle;
} UA_ObjectTypeNode;
```

14.2.9 ReferenceTypeNode

```
typedef struct {
    UA_NodeHead head;
    UA_Boolean isAbstract;
    UA_Boolean symmetric;
    UA_LocalizedText inverseName;

    /* Members specific to open62541 */
    UA_Byte referenceTypeIndex;
    UA_ReferenceTypeSet subTypes; /* contains the type itself as well */
} UA_ReferenceTypeNode;
```

14.2.10 DataTypeNode

```
typedef struct {
    UA_NodeHead head;
    UA_Boolean isAbstract;
} UA_DataTypeNode;
```

14.2.11 ViewNode

```
typedef struct {
    UA_NodeHead head;
    UA_Byte eventNotifier;
    UA_Boolean containsNoLoops;
} UA_ViewNode;
```

14.2.12 Node Union

A union that represents any kind of node. The node head can always be used. Check the NodeClass before accessing specific content.

```
typedef union {
    UA_NodeHead head;
    UA_VariableNode variableNode;
    UA_VariableTypeNode variableTypeNode;
    UA_MethodNode methodNode;
    UA_ObjectNode objectNode;
    UA_ObjectTypeNode objectTypeNode;
    UA_ReferenceTypeNode referenceTypeNode;
    UA_DataTypeNode dataTypeNode;
    UA_ViewNode viewNode;
} UA_Node;
```

14.2.13 Nodestore

The following structurere defines the interaction between the server and Nodestore backends.

```
typedef void (*UA_NodestoreVisitor)(void *visitorCtx, const UA_Node *node);

struct UA_Nodestore {
    /* Nodestore context and lifecycle */
    void (*free)(UA_Nodestore *ns);

    /* The following definitions are used to create empty nodes of the different
     * node types. The memory is managed by the nodestore. Therefore, the node
     * has to be removed via a special deleteNode function. (If the new node is
     * not added to the nodestore.) */
    UA_Node * (*newNode)(UA_Nodestore *ns, UA_NodeClass nodeClass);

    void (*deleteNode)(UA_Nodestore *ns, UA_Node *node);
```

(continues on next page)

(continued from previous page)

```
    /* _getNode returns a pointer to an immutable node. Call _releaseNode to
     * indicate when the pointer is no longer accessed.
     *
     * It can be indicated if only a subset of the attributes and references need
     * to be accessed. That is relevant when the nodestore accesses a slow
     * storage backend for the attributes. The attribute mask is a bitfield with
     * ORed entries from UA_NodeAttributesMask. If the attributes mask is empty,
     * then only the non-standard entries (context-pointer, callbacks, etc.) are
     * returned.
     *
     * The returned node always contains the context-pointer and other fields
     * specific to open626541 (not official attributes).
     *
     * The NodeStore does not complain if attributes and references that don't
     * exist (for that node) are requested. Attributes and references in
     * addition to those specified can be returned. For example, if the full
     * node already is kept in memory by the Nodestore. */
    const UA_Node * (*getNode)(UA_Nodestore *ns, const UA_NodeId *nodeId,
                               UA_UInt32 attributeMask,
                               UA_ReferenceTypeSet references,
                               UA_BrowseDirection referenceDirections);

    /* Similar to the normal _getNode. But it can take advantage of the
     * NodePointer structure, e.g. if it contains a direct pointer. */
    const UA_Node * (*getNodeFromPtr)(UA_Nodestore *ns, UA_NodePointer ptr,
                                      UA_UInt32 attributeMask,
                                      UA_ReferenceTypeSet references,
                                      UA_BrowseDirection referenceDirections);

    /* _getEditNode returns a pointer to a mutable version of the node. A
     * plugin implementation that keeps all nodes in RAM can return the same
     * pointer from _getNode and _getEditNode. The differences are more
     * important if, for example, nodes are stored in a backend database. Then
     * the _getEditNode version is used to indicate that modifications are
     * being made.
     *
     * Call _releaseNode to indicate when editing is done and the pointer is
     * no longer used. Note that changes are not (necessarily) visible in other
     * (const) node-pointers that were previously retrieved. Changes are however
     * visible in all newly retrieved node-pointers for the given NodeId after
     * calling _releaseNode.
     *
     * The attribute-mask and reference-description indicate if only a subset of
     * the attributes and references are to be modified. Other attributes and
     * references shall not be changed. */
    UA_Node * (*getEditNode)(UA_Nodestore *ns, const UA_NodeId *nodeId,
                             UA_UInt32 attributeMask,
                             UA_ReferenceTypeSet references,
                             UA_BrowseDirection referenceDirections);
```

(continues on next page)

(continued from previous page)

```
    /* Similar to _getEditNode. But it can take advantage of the NodePointer
     * structure, e.g. if it contains a direct pointer. */
    UA_Node * (*getEditNodeFromPtr)(UA_Nodestore *ns, UA_NodePointer ptr,
                                    UA_UInt32 attributeMask,
                                    UA_ReferenceTypeSet references,
                                    UA_BrowseDirection referenceDirections);

    /* Release a node that has been retrieved with ``getNode`` or
     * ``getNodeFromPtr``. */
    void (*releaseNode)(UA_Nodestore *ns, const UA_Node *node);

    /* Returns an editable copy of the same logical node (needs to be deleted
     * with the deleteNode function or inserted / replaced into the nodestore).
     * Runtime state such as MonitoredItem associations is not copied. A
     * successful replacement must transfer that state from the old node to
     * the replacement. */
    UA_StatusCode (*getNodeCopy)(UA_Nodestore *ns, const UA_NodeId *nodeId,
                                 UA_Node **outNode);

    /* Inserts a new node into the nodestore. If the NodeId is zero, then a
     * fresh numeric NodeId is assigned. If insertion fails, the node is
     * deleted. */
    UA_StatusCode (*insertNode)(UA_Nodestore *ns, UA_Node *node,
                                UA_NodeId *addedNodeId);

    /* To replace a node, get an editable copy of the node, edit and replace
     * with this function. If the node was already replaced since the copy was
     * made, UA_STATUSCODE_BADINTERNALERROR is returned. If the NodeId is not
     * found, UA_STATUSCODE_BADNODEIDUNKNOWN is returned. In both error␣
↪cases,
     * the editable node is deleted. */
    UA_StatusCode (*replaceNode)(UA_Nodestore *ns, UA_Node *node);

    /* Removes a node from the nodestore. */
    UA_StatusCode (*removeNode)(UA_Nodestore *ns, const UA_NodeId *nodeId);

    /* Maps the ReferenceTypeIndex used for the references to the NodeId of the
     * ReferenceType. The returned pointer is stable until the Nodestore is
     * deleted. */
    const UA_NodeId * (*getReferenceTypeId)(UA_Nodestore *ns,
                                            UA_Byte refTypeIndex);

    /* Execute a callback for every node in the nodestore. */
    void (*iterate)(UA_Nodestore *ns, UA_NodestoreVisitor visitor,
                    void *visitorCtx);
};

/* Attributes must be of a matching type (VariableAttributes, ObjectAttributes,
```

(continues on next page)

(continued from previous page)

```
 * and so on). The attributes are copied. Note that the attributes structs do
 * not contain NodeId, NodeClass and BrowseName. The NodeClass of the node needs
 * to be correctly set before calling this method. This method does not call
 * UA_Node_clear internally when an error is encountered. */
UA_StatusCode
UA_Node_setAttributes(UA_Node *node, const void *attributes,
                      const UA_DataType *attributeType);

/* Nodes can carry multiples locales for DisplayName and Description. The
 * following inserts a new locales or updates an existing one with new text. */
UA_StatusCode
UA_Node_insertOrUpdateDisplayName(UA_Node *node,
                                  const UA_LocalizedText *displayName);

UA_StatusCode
UA_Node_insertOrUpdateDescription(UA_Node *node,
                                  const UA_LocalizedText *description);

/* Reset the destination node and copy the content of the source. Runtime
 * MonitoredItem associations are not copied. */
UA_StatusCode
UA_Node_copy(const UA_Node *src, UA_Node *dst);

/* Allocate new node and copy the values from src */
UA_Node *
UA_Node_copy_alloc(const UA_Node *src);

/* Add a single reference to the node */
UA_StatusCode
UA_Node_addReference(UA_Node *node, UA_Byte refTypeIndex, UA_Boolean isForward,
                     const UA_ExpandedNodeId *targetNodeId,
                     UA_UInt32 targetBrowseNameHash);

/* Delete a single reference from the node */
UA_StatusCode
UA_Node_deleteReference(UA_Node *node, UA_Byte refTypeIndex, UA_Boolean␣
↪isForward,
                        const UA_ExpandedNodeId *targetNodeId);

/* Deletes references from the node which are not matching any type in the given
 * array. Could be used to e.g. delete all the references, except
 * 'HASMODELINGRULE' */
void
UA_Node_deleteReferencesSubset(UA_Node *node, const UA_ReferenceTypeSet *keepSet);

/* Delete all references of the node */
void
UA_Node_deleteReferences(UA_Node *node);
```

(continues on next page)

(continued from previous page)

```
/* Remove all malloc'ed members of the node and reset */
void
UA_Node_clear(UA_Node *node);
```

14.3 AccessControl Plugin API

The access control callback is used to authenticate sessions and grant access rights accordingly.

The sessionId and sessionContext can be both NULL. This is the case when, for example, a MonitoredItem (the underlying Subscription) is detached from its Session but continues to run.

The callbacks activateSession, getUserRightsMask, getUserAccessLevel, getUserExecutable, getUserExecutableOnObject and allowBrowseNode are mandatory. The server does not start without them. The other callbacks can be NULL.

```
struct UA_AccessControl {
    void *context;
    void (*clear)(UA_AccessControl *ac);

    /* Supported login mechanisms. The server endpoints are created from here. */
    size_t userTokenPoliciesSize;
    UA_UserTokenPolicy *userTokenPolicies;

    /* Authenticate a session. The session context is attached to the session
     * and later passed into the node-based access control callbacks. The new
     * session is rejected if a StatusCode other than UA_STATUSCODE_GOOD is
     * returned.
     *
     * Note that this callback can be called several times for a Session. For
     * example when a Session is recovered (activated) on a new
     * SecureChannel.
     *
     * The following checks are performed in the server before calling
     * activateSession:
     *
     * - Select matching Endpoint/UserTokenPolicy (compare token type,
     *   SecureChannel and PolicyId from the UserIdentityToken)
     * - Cryptographic checks:
     *   - Check the encryption algorithm from the UserIdentityToken
     *   - UsernamePassword/IssuedToken: Decrypt the secret
     *   - Check the x509 auth certificate signature and validate the
     *     certificate against the server's sessionPKI */
    UA_StatusCode (*activateSession)(UA_Server *server, UA_AccessControl *ac,
                                     const UA_EndpointDescription *endpointDescription,
                                     const UA_ByteString *secureChannelRemoteCertificate,
                                     const UA_NodeId *sessionId,
                                     const UA_ExtensionObject *userIdentityToken,
                                     void **sessionContext);

    /* Deauthenticate a session and cleanup */
```

(continues on next page)

(continued from previous page)

```
    void (*closeSession)(UA_Server *server, UA_AccessControl *ac,
                         const UA_NodeId *sessionId, void *sessionContext);

    /* Access control for all nodes*/
    UA_UInt32 (*getUserRightsMask)(UA_Server *server, UA_AccessControl *ac,
                                   const UA_NodeId *sessionId, void *sessionContext,
                                   const UA_NodeId *nodeId, void *nodeContext);

    /* Additional access control for variable nodes */
    UA_Byte (*getUserAccessLevel)(UA_Server *server, UA_AccessControl *ac,
                                  const UA_NodeId *sessionId, void *sessionContext,
                                  const UA_NodeId *nodeId, void *nodeContext);

    /* Additional access control for method nodes */
    UA_Boolean (*getUserExecutable)(UA_Server *server, UA_AccessControl *ac,
                                    const UA_NodeId *sessionId, void *sessionContext,
                                    const UA_NodeId *methodId, void *methodContext);

    /* Additional access control for calling a method node in the context of a
     * specific object */
    UA_Boolean (*getUserExecutableOnObject)(UA_Server *server, UA_AccessControl *ac,
                                            const UA_NodeId *sessionId, void *sessionContext,
                                            const UA_NodeId *methodId, void *methodContext,
                                            const UA_NodeId *objectId, void *objectContext);

    /* Allow adding a node */
    UA_Boolean (*allowAddNode)(UA_Server *server, UA_AccessControl *ac,
                               const UA_NodeId *sessionId, void *sessionContext,
                               const UA_AddNodesItem *item);

    /* Allow adding a reference */
    UA_Boolean (*allowAddReference)(UA_Server *server, UA_AccessControl *ac,
                                    const UA_NodeId *sessionId, void *sessionContext,
                                    const UA_AddReferencesItem *item);

    /* Allow deleting a node */
    UA_Boolean (*allowDeleteNode)(UA_Server *server, UA_AccessControl *ac,
                                  const UA_NodeId *sessionId, void *sessionContext,
                                  const UA_DeleteNodesItem *item);

    /* Allow deleting a reference */
    UA_Boolean (*allowDeleteReference)(UA_Server *server, UA_AccessControl *ac,
                                       const UA_NodeId *sessionId, void *sessionContext,
                                       const UA_DeleteReferencesItem *item);

    /* Allow browsing a node */
    UA_Boolean (*allowBrowseNode)(UA_Server *server, UA_AccessControl *ac,
                                  const UA_NodeId *sessionId, void *sessionContext,
                                  const UA_NodeId *nodeId, void *nodeContext);
```

(continues on next page)

(continued from previous page)

```
#ifdef UA_ENABLE_SUBSCRIPTIONS
    /* Allow creating a subscription */
    UA_Boolean (*allowCreateSubscription)(UA_Server *server, UA_AccessControl *ac,
                                          const UA_NodeId *sessionId, void *sessionContext);

    /* Allow transfer of a subscription to another session. The Server shall
     * validate that the Client of that Session is operating on behalf of the
     * same user. The oldSessionId is NULL for a Subscription that is detached
     * from its Session. The server only calls this after it has verified that
     * the new Session has the identity (user, and for anonymous users the
     * client ApplicationUri over a secure SecureChannel) of the Session the
     * Subscription was detached from. */
    UA_Boolean (*allowTransferSubscription)(UA_Server *server, UA_AccessControl *ac,
                                            const UA_NodeId *oldSessionId, void *oldSessionContext,
                                            const UA_NodeId *newSessionId, void␣
↪*newSessionContext);
#endif

#ifdef UA_ENABLE_HISTORIZING
    /* Allow insert,replace,update of historical data */
    UA_Boolean (*allowHistoryUpdateUpdateData)(UA_Server *server, UA_AccessControl␣
↪*ac,
                                               const UA_NodeId *sessionId, void *sessionContext,
                                               const UA_NodeId *nodeId,
                                               UA_PerformUpdateType performInsertReplace,
                                               const UA_DataValue *value);

    /* Allow delete of historical data */
    UA_Boolean (*allowHistoryUpdateDeleteRawModified)(UA_Server *server, UA_
↪AccessControl *ac,
                                                      const UA_NodeId *sessionId, void *sessionContext,
                                                      const UA_NodeId *nodeId,
                                                      UA_DateTime startTimestamp,
                                                      UA_DateTime endTimestamp,
                                                      bool isDeleteModified);
#endif
};
```

14.4 EventLoop Plugin API

An OPC UA-enabled application can have several clients and servers. And server can serve different transport-level protocols for OPC UA. The EventLoop is a central module that provides a unified control-flow for all of these. Hence, several clients and servers can share an EventLoop. When they do, every client and server using that EventLoop must set externalEventLoop in its configuration. Otherwise, one of them may stop and delete the EventLoop during its own shutdown or configuration cleanup, while the others still use it. The application must keep the shared EventLoop alive until all its users have finished, then stop and delete it.

The EventLoop and the ConnectionManager implementation is architecture-specific. The goal is to

have a single call to "poll" (epoll, kqueue, ...) in the EventLoop that covers all ConnectionManagers. Hence the EventLoop plugin implementation must know implementation details of the ConnectionManager implementations. So the EventLoop can extract socket information, etc. from the ConnectionManagers.

14.4.1 Timer Policies

A timer comes with a periodic interval in which a callback is executed. If an application is congested the interval can be missed. Two different policies can be used when this happens. Either schedule the next execution after the interval has elapsed again from the current time onwards or stay within the regular interval with respect to the original basetime.

```
typedef enum {
    UA_TIMERPOLICY_ONCE = 0,        /* Execute the timer once and remove */
    UA_TIMERPOLICY_CURRENTTIME = 1, /* Repeated timer. Upon cycle miss,␣
↪execute
                                     * "now" and wait exactly for the interval
                                     * until the next execution (new basetime). */
    UA_TIMERPOLICY_BASETIME = 2,    /* Repeated timer. Upon cycle miss, execute
                                     * "now" and fall back into the regular
                                     * cycle from the original basetime (the
                                     * next execution might come after less
                                     * delay than the interval defines). */
} UA_TimerPolicy;
```

14.4.2 Event Loop

The EventLoop implementation is part of the selected architecture. For example, "Win32/POSIX" stands for a Windows environment with an EventLoop that uses the POSIX API. Several EventLoops can be instantiated in parallel. But the globally defined functions are the same everywhere.

```
typedef void (*UA_Callback)(void *application, void *context);

/* Delayed callbacks are executed not when they are registered, but in the
 * following EventLoop cycle */
typedef struct UA_DelayedCallback {
    UA_atomic(struct UA_DelayedCallback *)next;
    UA_Callback callback;
    void *application;
    void *context;
} UA_DelayedCallback;

typedef enum {
    UA_EVENTLOOPSTATE_FRESH = 0,
    UA_EVENTLOOPSTATE_STOPPED,
    UA_EVENTLOOPSTATE_STARTED,
    UA_EVENTLOOPSTATE_STOPPING /* Stopping in progress, needs EventLoop
                                * cycles to finish */
} UA_EventLoopState;

struct UA_EventLoop {
```

(continues on next page)

(continued from previous page)

```
    /* Configuration
     * ~~~~~~~~~~~~~~~~
     * The configuration should be set before the EventLoop is started */

    const UA_Logger *logger;

    /* See the implementation-specific documentation for possible parameters.
     * The params map is cleaned up when the EventLoop is _free'd. */
    UA_KeyValueMap params;

    /* EventLoop Lifecycle
     * ~~~~~~~~~~~~~~~~~~~
     * The EventLoop state also controls the state of the configured
     * EventSources. Stopping the EventLoop gracefully closes e.g. the open
     * network connections. The only way to process incoming events is to call
     * the 'run' method. Events are then triggering their respective callbacks
     * from within that method.*/

    const volatile UA_EventLoopState state; /* Only read the state from outside */

    /* Start the EventLoop and start all already registered EventSources */
    UA_StatusCode (*start)(UA_EventLoop *el);

    /* Stop all EventSources. This is asynchronous and might need a few
     * iterations of the main-loop to succeed. */
    void (*stop)(UA_EventLoop *el);

    /* Clean up the EventLoop and free allocated memory. Can fail if the
     * EventLoop is not stopped. */
    UA_StatusCode (*free)(UA_EventLoop *el);

    /* Wait for events and process them for at most "timeout" ms or until an
     * unrecoverable error occurs. If timeout==0, then only already received
     * events are processed. Returns immediately after processing the first
     * (batch of) event(s). */
    UA_StatusCode (*run)(UA_EventLoop *el, UA_UInt32 timeout);

    /* The "run" method is blocking and waits for events during a timeout
     * period. This cancels the "run" method to return immediately. */
    void (*cancel)(UA_EventLoop *el);

    /* EventLoop Time Domain
     * ~~~~~~~~~~~~~~~~~~~~~
     * Each EventLoop instance can manage its own time domain. This affects the
     * execution of timed callbacks and time-based sending of network packets.
     * Managing independent time domains is important when different parts of
     * the same system are synchronized to different external master clocks.
     *
     * Each EventLoop uses a "normal" and a "monotonic" clock. The monotonic
```

(continues on next page)

(continued from previous page)

```
     * clock does not (necessarily) conform to the current wallclock date. But
     * its time intervals are more precise. So it is used for all internally
     * scheduled events of the EventLoop (e.g. timed callbacks and time-based
     * sending of network packets). The normal and monotonic clock sources can
     * be configured via parameters before starting the EventLoop. See the
     * architecture-specific documentation for that.
     *
     * Note that the logger configured in the EventLoop generates timestamps
     * independently. If the logger uses a different time domain than the
     * EventLoop, discrepancies may appear in the logs.
     *
     * The EventLoop clocks can be read via the following functions. See
     * `open62541/types.h` for the documentation of their equivalent globally
     * defined functions. */

    UA_DateTime (*dateTime_now)(UA_EventLoop *el);
    UA_DateTime (*dateTime_nowMonotonic)(UA_EventLoop *el);
    UA_Int64    (*dateTime_localTimeUtcOffset)(UA_EventLoop *el);

    /* Timer Callbacks
     * ~~~~~~~~~~~~~~~
     * Timer callbacks are executed at a defined time or regularly with a
     * periodic interval. The timer subsystem always uses the
     * monotonic clock. */

    /* Time of the next timer. Returns the UA_DATETIME_MAX if no timer is
     * registered. Returns the current monotonic time if a delayed
     * callback is registered for immediate execution. */
    UA_DateTime (*nextTimer)(UA_EventLoop *el);

    /* The execution interval is in ms. The first execution time is baseTime +
     * interval. If baseTime is NULL, then the current time is used for the base
     * time. The timerId is written if the pointer is non-NULL. */
    UA_StatusCode
    (*addTimer)(UA_EventLoop *el, UA_Callback cb, void *application,
                void *data, UA_Double interval_ms, UA_DateTime *baseTime,
                UA_TimerPolicy timerPolicy, UA_UInt64 *timerId);

    /* If baseTime is NULL, use the current time as the base. */
    UA_StatusCode
    (*modifyTimer)(UA_EventLoop *el, UA_UInt64 timerId,
                   UA_Double interval_ms, UA_DateTime *baseTime,
                   UA_TimerPolicy timerPolicy);

    void (*removeTimer)(UA_EventLoop *el, UA_UInt64 timerId);

    /* Delayed Callbacks
     * ~~~~~~~~~~~~~~~~~
     * Delayed callbacks are executed once in the next iteration of the
```

(continues on next page)

(continued from previous page)

```
     * EventLoop and then deregistered automatically. A typical use case is to
     * delay a resource cleanup to a point where it is known that the resource
     * has no remaining users.
     *
     * The delayed callbacks are processed in each cycle of the EventLoop
     * between the handling of periodic callbacks and polling for (network)
     * events. The memory for the delayed callback is *NOT* automatically freed
     * after the execution. But this can be done from within the callback.
     *
     * Delayed callbacks are processed in the order in which they are added.
     *
     * The delayed callback API is thread-safe. addDelayedCallback is
     * non-blocking and can be called from an interrupt context.
     * removeDelayedCallback can take a mutex and is blocking. */

    void (*addDelayedCallback)(UA_EventLoop *el, UA_DelayedCallback *dc);
    void (*removeDelayedCallback)(UA_EventLoop *el, UA_DelayedCallback *dc);

    /* EventSources
     * ~~~~~~~~~~~~
     * EventSources are stored in a singly-linked list for direct access. But
     * only the below methods shall be used for adding and removing - this
     * impacts the lifecycle of the EventSource. For example it may be
     * auto-started if the EventLoop is already running. */

    /* Linked list of EventSources */
    UA_EventSource *eventSources;

    /* Register the ES. Immediately starts the ES if the EventLoop is already
     * started. Otherwise the ES is started together with the EventLoop. */
    UA_StatusCode
    (*registerEventSource)(UA_EventLoop *el, UA_EventSource *es);

    /* Stops the EventSource before deregistrering it */
    UA_StatusCode
    (*deregisterEventSource)(UA_EventLoop *el, UA_EventSource *es);

    /* Locking
     * ~~~~~~~
     *
     * For multi-threading the EventLoop is protected by a mutex. The mutex is
     * expected to be recursive (can be taken more than once from the same
     * thread). A common approach to avoid deadlocks is to establish an absolute
     * ordering between the locks. Where the "lower" locks needs to be taken
     * before the "upper" lock. The EventLoop-mutex is exposed here to allow it
     * to be taken from the outside. */
    void (*lock)(UA_EventLoop *el);
    void (*unlock)(UA_EventLoop *el);
};
```

14.4.3 Event Source

Event Sources are attached to an EventLoop. Typically the event source and the EventLoop are developed together and share a private API in the background.

```
typedef enum {
    UA_EVENTSOURCESTATE_FRESH = 0,
    UA_EVENTSOURCESTATE_STOPPED,      /* Registered but stopped */
    UA_EVENTSOURCESTATE_STARTING,
    UA_EVENTSOURCESTATE_STARTED,
    UA_EVENTSOURCESTATE_STOPPING      /* Stopping in progress, needs
                                       * EventLoop cycles to finish */
} UA_EventSourceState;

/* Type-tag for proper casting of the difference EventSource (e.g. when they are
 * looked up via UA_EventLoop_findEventSource). */
typedef enum {
    UA_EVENTSOURCETYPE_CONNECTIONMANAGER,
    UA_EVENTSOURCETYPE_INTERRUPTMANAGER
} UA_EventSourceType;

struct UA_EventSource {
    struct UA_EventSource *next; /* Singly-linked list for use by the
                                  * application that registered the ES */

    UA_EventSourceType eventSourceType;

    /* Configuration
     * ~~~~~~~~~~~~~ */
    UA_String name;                /* Unique name of the ES */
    UA_EventLoop *eventLoop;       /* EventLoop where the ES is registered */
    UA_KeyValueMap params;

    /* Lifecycle
     * ~~~~~~~~~ */
    UA_EventSourceState state;
    UA_StatusCode (*start)(UA_EventSource *es);
    void (*stop)(UA_EventSource *es); /* Asynchronous. Iterate theven EventLoop
                                       * until the EventSource is stopped. */
    UA_StatusCode (*free)(UA_EventSource *es);
};
```

14.4.4 Connection Manager

Every Connection is created by a ConnectionManager. Every ConnectionManager belongs to just one application. A ConnectionManager can act purely as a passive “Factory” for Connections. But it can also be stateful. For example, it can keep a session to an MQTT broker open which is used by individual connections that are each bound to an MQTT topic.

```
/* The ConnectionCallback is the only interface from the connection back to
 * the application.
```

(continues on next page)

(continued from previous page)

```
 *
 * - The connectionId is initially unknown to the target application and
 *   "announced" to the application when first used first in this callback.
 *
 * - The context is attached to the connection. Initially a default context
 *   is set. The context can be replaced within the callback (via the
 *   double-pointer).
 *
 * - The state argument indicates the lifecycle of the connection. Every
 *   connection calls the callback a last time with UA_CONNECTIONSTATE_CLOSING.
 *   Protocols individually can forward diagnostic information relevant to the
 *   state as part of the key-value parameters.
 *
 * - The parameters are a key-value list with additional information. The
 *   possible keys and their meaning are documented for the individual
 *   ConnectionManager implementations.
 *
 * - The msg ByteString is the message (or packet) received on the
 *   connection. Can be empty. */
typedef void
(*UA_ConnectionManager_connectionCallback)
    (UA_ConnectionManager *cm, uintptr_t connectionId,
     void *application, void **connectionContext, UA_ConnectionState state,
     const UA_KeyValueMap *params, UA_ByteString msg);

struct UA_ConnectionManager {
    /* Every ConnectionManager is treated like an EventSource from the
     * perspective of the EventLoop. */
    UA_EventSource eventSource;

    /* Name of the protocol supported by the ConnectionManager. For example
     * "mqtt", "udp", "mqtt". */
    UA_String protocol;

    /* Open a Connection
     * ~~~~~~~~~~~~~~~~~
     * Connecting is asynchronous. The connection-callback is called when the
     * connection is open (status=GOOD) or aborted (status!=GOOD) when
     * connecting failed.
     *
     * Some ConnectionManagers can also passively listen for new connections.
     * Configuration parameters for this are passed via the key-value list. The
     * `context` pointer of the listening connection is also set as the initial
     * context of newly opened connections.
     *
     * The parameters describe the connection. For example hostname and port
     * (for TCP). Other protocols (e.g. MQTT, AMQP, etc.) may required
     * additional arguments to open a connection in the key-value list.
     *
```

(continues on next page)

(continued from previous page)

```
     * The provided context is set as the initial context attached to this
     * connection. It is already set before the first call to
     * connectionCallback.
     *
     * The connection can be opened synchronously or asynchronously.
     *
     * - For synchronous connection, the connectionCallback is called with the
     *   status UA_CONNECTIONSTATE_ESTABLISHED immediately from within the
     *   openConnection operation.
     *
     * - In the asynchronous case the connectionCallback is called immediately
     *   from within the openConnection operation with the status
     *   UA_CONNECTIONSTATE_OPENING. The connectionCallback is called with the
     *   status UA_CONNECTIONSTATE_ESTABLISHED once the connection has fully
     *   opened.
     *
     * Note that a single call to openConnection might open multiple
     * connections. For example listening on IPv4 and IPv6 for a single
     * hostname. Each protocol implementation documents whether multiple
     * connections might be opened at once. */
    UA_StatusCode
    (*openConnection)(UA_ConnectionManager *cm, const UA_KeyValueMap *params,
                      void *application, void *context,
                      UA_ConnectionManager_connectionCallback connectionCallback);

    /* Send a message over a Connection
     * ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
     * Sending is asynchronous. That is, the function returns before the message
     * is ACKed from remote. The memory for the buffer is expected to be
     * allocated with allocNetworkBuffer and is released internally (also if
     * sending fails).
     *
     * Some ConnectionManagers can accept additional parameters for sending. For
     * example a tx-time for sending in time-synchronized TSN settings. */
    UA_StatusCode
    (*sendWithConnection)(UA_ConnectionManager *cm, uintptr_t connectionId,
                          const UA_KeyValueMap *params, UA_ByteString *buf);

    /* Close a Connection
     * ~~~~~~~~~~~~~~~~~~
     * When a connection is closed its `connectionCallback` is called with
     * (status=BadConnectionClosed, msg=empty). Then the connection is cleared
     * up inside the ConnectionManager. This is the case both for connections
     * that are actively closed and those that are closed remotely. The return
     * code is non-good only if the connection is already closed. */
    UA_StatusCode
    (*closeConnection)(UA_ConnectionManager *cm, uintptr_t connectionId);

    /* Buffer Management
```

(continues on next page)

(continued from previous page)

```
     * ~~~~~~~~~~~~~~~~~~
     * Each ConnectionManager allocates and frees his own memory for the network
     * buffers. This enables, for example, zero-copy neworking mechanisms. The
     * connectionId is part of the API to enable cases where memory is
     * statically allocated for every connection */
    UA_StatusCode
    (*allocNetworkBuffer)(UA_ConnectionManager *cm, uintptr_t connectionId,
                          UA_ByteString *buf, size_t bufSize);
    void
    (*freeNetworkBuffer)(UA_ConnectionManager *cm, uintptr_t connectionId,
                         UA_ByteString *buf);
};
```

14.4.5 Interrupt Manager

The Interrupt Manager allows to register to listen for system interrupts. Triggering the interrupt calls the callback associated with it.

The implementations of the interrupt manager for the different platforms shall be designed such that:

- Registered interrupts are only intercepted from within the running EventLoop
- Processing an interrupt in the EventLoop is handled similarly to handling a network event: all methods and also memory allocation are available from within the interrupt callback.

```
/* Interrupts can have additional key-value 'instanceInfos' for each individual
 * triggering. See the architecture-specific documentation. */
typedef void
(*UA_InterruptCallback)(UA_InterruptManager *im,
                        uintptr_t interruptHandle, void *interruptContext,
                        const UA_KeyValueMap *instanceInfos);

struct UA_InterruptManager {
    /* Every InterruptManager is treated like an EventSource from the
     * perspective of the EventLoop. */
    UA_EventSource eventSource;

    /* Register an interrupt. The handle and context information is passed
     * through to the callback.
     *
     * The interruptHandle is a numerical identifier of the interrupt. In some
     * cases, such as POSIX signals, this is enough information to register
     * callback. For other interrupt systems (architectures) additional
     * parameters may be required and can be passed in via the parameters
     * key-value list. See the implementation-specific documentation.
     *
     * The interruptContext is opaque user-defined information and passed
     * through to the callback without modification. */
    UA_StatusCode
    (*registerInterrupt)(UA_InterruptManager *im, uintptr_t interruptHandle,
                         const UA_KeyValueMap *params,
```

(continues on next page)

(continued from previous page)

```
                    UA_InterruptCallback callback, void *interruptContext);

    /* Remove a registered interrupt. Returns no error code if the interrupt is
     * already deregistered. */
    void
    (*deregisterInterrupt)(UA_InterruptManager *im, uintptr_t interruptHandle);
};

#if (defined(UA_ARCHITECTURE_POSIX) && !defined(UA_ARCHITECTURE_
↪LWIP)) || defined(UA_ARCHITECTURE_WIN32)
```

14.4.6 Native EventLoop Implementations

POSIX uses readiness polling. Win32 uses I/O completion ports. Configuration parameters must be set before starting the EventLoop.

Clock configuration (Linux and BSDs only)

0:clock-source [int32]
: Clock source (default: CLOCK_REALTIME).

0:clock-source-monotonic [int32]:
: Clock source used for time intervals. A non-monotonic source can be used as well. But expect accordingly longer sleep-times for timed events when the clock is set to the past. See the man-page of "clock_gettime" on how to get a clock source id for a character-device such as /dev/ptp0. (default: CLOCK_MONOTONIC_RAW)

```
#if defined(UA_ARCHITECTURE_POSIX) && !defined(UA_ARCHITECTURE_LWIP)
UA_EventLoop *
UA_EventLoop_new_POSIX(const UA_Logger *logger);
#endif

#ifdef UA_ARCHITECTURE_WIN32
UA_EventLoop *
UA_EventLoop_new_WIN32(const UA_Logger *logger);
#endif

#ifdef UA_ENABLE_EVENTLOOP_GLIB
```

14.4.7 GLib EventLoop Implementation

A drop-in alternative to UA_EventLoop_new_POSIX that is driven by a GLib GMainContext instead of directly calling select()/epoll_wait(). All ConnectionManagers documented below (TCP, UDP, Ethernet, …) work unchanged with this EventLoop.

Once started, the EventLoop attaches a GSource to the given GMainContext. From that point on, sockets and timers are serviced automatically whenever that context is iterated – for example by g_main_loop_run(), gtk_main(), g_application_run(), or any other GLib-based application main loop. It is therefore not necessary to call the EventLoop's own run method at all; the open62541 stack can be fully driven by GLib. run is still provided (performing a single bounded iteration of the GMainContext) for backwards compatibility with code that pumps the EventLoop itself (e.g. UA_Server_run).

@param logger The logger for the EventLoop. @param glibMainContext The GMainContext* to attach to. If NULL,

the process-wide default context (g_main_context_default()) is used – the same context iterated by a plain g_main_loop_new(NULL, ...) or by GTK/GNOME applications.

```
UA_EventLoop *
UA_EventLoop_new_GLib(const UA_Logger *logger, void *glibMainContext);

#endif /* UA_ENABLE_EVENTLOOP_GLIB */
```

TCP Connection Manager

Listens on the network and manages TCP connections. This should be available for all architectures.

The *openConnection* callback is used to create both client and server sockets. A server socket listens and accepts incoming connections (creates an active connection). This is distinguished by the key-value parameters passed to *openConnection*. Note that a single call to *openConnection* for a server connection may actually create multiple connections (one per hostname / device).

The *connectionCallback* of the server socket and *context* of the server socket is reused for each new connection. But the key-value parameters for the first callback are different between server and client connections.

Configuration parameters for the ConnectionManager (set before start)

0:max-connections [uint32]
max connections (default: 0 -> unbounded). The server sockets get deactivated if the limit is reached.

0:recv-bufsize [uint32]
Size of the buffer that is statically allocated for receiving messages (default 64kB).

0:send-bufsize [uint32]
Size of the statically allocated buffer for sending messages. The buffer is reused for every *allocNetworkBuffer* to avoid a heap allocation per message; a message larger than the buffer falls back to a fresh allocation. A dedicated send buffer is used (never the recv buffer), so sending while received data is still being processed is safe. If undefined, the send buffer defaults to the recv-bufsize.

Open Connection Parameters:

0:address [string | array of string]
Hostname or IPv4/v6 address for the connection (scalar parameter required for active connections). For listen-connections the address contains the local hostnames or IP addresses for listening. If undefined, listen on all interfaces INADDR_ANY. (default: undefined)

0:port [uint16]
Port of the target host (required).

0:listen [boolean]
Listen-connection or active-connection (default: false)

0:validate [boolean]
If true, the connection setup will act as a dry-run without actually creating any connection but solely validating the provided parameters (default: false)

Active Connection Connection Callback Parameters (first callback only):

0:remote-address [string]
Address of the remote side (hostname or IP address).

Listen Connection Connection Callback Parameters (at establishment):

Listener setup can be asynchronous: openConnection may return before binding and listening have completed. In this case, the initial UA_CONNECTIONSTATE_OPENING callback may omit the bound address and port. These are provided in the UA_CONNECTIONSTATE_ESTABLISHED callback once the listener is ready to accept connections.

0:listen-address [string]
Local address (IP or hostname) for the new listen-connection.

0:listen-port [uint16]
Port on which the new connection listens.

Send Parameters:

No additional parameters for sending over an established TCP socket defined.

```
#if defined(UA_ARCHITECTURE_POSIX) && !defined(UA_ARCHITECTURE_LWIP)
UA_ConnectionManager *
UA_ConnectionManager_new_POSIX_TCP(const UA_String eventSourceName);
#endif

#ifdef UA_ARCHITECTURE_WIN32
UA_ConnectionManager *
UA_ConnectionManager_new_WIN32_TCP(const UA_String eventSourceName);
#endif
```

UDP Connection Manager

Manages UDP connections. This should be available for all architectures. The configuration parameters have to set before calling _start to take effect.

Configuration parameters for the ConnectionManager (set before start)

0:recv-bufsize [uint32]
Size of the buffer that is statically allocated for receiving messages (default 64kB).

0:send-bufsize [uint32]
Size of the statically allocated buffer for sending messages. The buffer is reused for every *allocNetworkBuffer* to avoid a heap allocation per message; a message larger than the buffer falls back to a fresh allocation. A dedicated send buffer is used (never the recv buffer), so sending while received data is still being processed is safe. If undefined, the send buffer defaults to the recv-bufsize.

Open Connection Parameters:

0:listen [boolean]
Use the connection for listening or for sending (default: false)

0:address [string | string array]
Hostname (or IPv4/v6 address) for sending or receiving. A scalar is required for sending. For listening a string array for the list-hostnames is possible as well (default: list on all hostnames).

0:port [uint16]
Port for sending or listening (required).

0:interface [string]
Network interface for listening or sending (e.g. when using multicast addresses). Can be either the IP address of the network interface or the interface name (e.g. 'eth0').

0:ttl [uint32]
Multicast time to live, (optional, default: 1 - meaning multicast is available only to the local subnet).

0:loopback [boolean]
Whether or not to use multicast loopback, enabling local interfaces belonging to the multicast group to receive packages. (default: enabled).

0:reuse [boolean]
Enables sharing of the same listening address on different sockets (default: disabled).

0:sockpriority [uint32]
The socket priority (optional) - only available on linux. packets with a higher priority may be processed first depending on the selected device queueing discipline. Setting a priority outside the range 0 to 6 requires the CAP_NET_ADMIN capability (on Linux).

0:validate [boolean]
If true, the connection setup will act as a dry-run without actually creating any connection but solely validating the provided parameters (default: false)

Connection Callback Parameters:

0:remote-address [string]
Contains the remote IP address.

0:remote-port [uint16]
Contains the remote port.

Send Parameters:

No additional parameters for sending over an UDP connection defined.

```
#if defined(UA_ARCHITECTURE_POSIX) && !defined(UA_ARCHITECTURE_LWIP)
UA_ConnectionManager *
UA_ConnectionManager_new_POSIX_UDP(const UA_String eventSourceName);
#endif

#ifdef UA_ARCHITECTURE_WIN32
UA_ConnectionManager *
UA_ConnectionManager_new_WIN32_UDP(const UA_String eventSourceName);
#endif

#if defined(__linux__) /* Linux only so far */
```

Ethernet Connection Manager

Listens on the network and manages UDP connections. This should be available for all architectures. The configuration parameters have to set before calling _start to take effect.

Configuration parameters for the ConnectionManager (set before start)

0:recv-bufsize [uint32]
Size of the buffer that is statically allocated for receiving messages (default 64kB).

0:send-bufsize [uint32]
: Size of the statically allocated buffer for sending messages. The buffer is reused for every *allocNetworkBuffer* to avoid a heap allocation per message; a message larger than the buffer falls back to a fresh allocation. A dedicated send buffer is used (never the recv buffer), so sending while received data is still being processed is safe. If undefined, the send buffer defaults to the recv-bufsize.

Open Connection Parameters:

0:listen [bool]
: The connection is either for sending or for listening (default: false).

0:interface [string]
: The name of the Ethernet interface to use (required).

0:address [string]
: MAC target address consisting of six groups of hexadecimal digits separated by hyphens such as 01-23-45-67-89-ab. For sending this is a required parameter. For listening this is a multicast address that the connections tries to register for.

0:priority [int32]
: Set the socket priority for sending (cf. SO_PRIORITY)

0:ethertype [uint16]
: EtherType for sending and receiving frames (optional). For listening connections, this filters out all frames with different EtherTypes.

0:promiscuous [bool]
: Receive frames also for different target addresses. Defined only for listening connections (default: false).

0:vid [uint16]
: 12-bit VLAN identifier (optional for send connections).

0:pcp [byte]
: 3-bit priority code point (optional for send connections).

0:dei [bool]
: 1-bit drop eligible indicator (optional for send connections).

0:validate [boolean]
: If true, the connection setup will act as a dry-run without actually creating any connection but solely validating the provided parameters (default: false)

Sending with a txtime (for Time-Sensitive Networking) is possible on recent Linux kernels, If enabled for the socket, then a txtime parameters can be passed to *sendWithConnection*. Note that the clock source for txtime sending is the monotonic clock source set for the entire EventLoop. Check the EventLoop parameters for how to set that e.g. to a PTP clock source. The txtime parameters uses Linux conventions.

0:txtime-enable [bool]
: Enable sending with a txtime for the connection (default: false).

0:txtime-flags [uint32]
: txtime flags set for the socket (default: SOF_TXTIME_REPORT_ERRORS).

Send Parameters (only with txtime enabled for the connection)

0:txtime [datetime]
Time when the message is sent out (Datetime has 100ns precision) for the "monotonic" clock source of the EventLoop.

0:txtime-pico [uint16]
Picoseconds added to the txtime timestamp (default: 0).

0:txtime-drop-late [bool]
Drop message if it cannot be sent in time (default: true).

```
UA_ConnectionManager *
UA_ConnectionManager_new_POSIX_Ethernet(const UA_String eventSourceName);
#endif
```

HTTP Connection Manager

The HTTP ConnectionManager uses libwebsockets for HTTP client and server connections. A server listener announces each accepted peer connection with its own connection id and context. Request bodies are reassembled and delivered in a callback on that accepted connection. Send the response with sendWithConnection and the accepted connection id. Sequential HTTP/1.1 keep-alive requests reuse the same id and context.

Open Connection Parameters:

0:address [string]
Remote hostname or IPv4/IPv6 address for clients (required), or local interface for listeners. Listeners default to all interfaces.

0:port [uint16]
Remote or listening port (required; zero selects a dynamic server port).

0:content-coding-policy [string]
Server responses only. identity forces an uncompressed response; gzip restricts negotiation to gzip or identity. This lets an application protocol narrow the generic HTTP coding set.

0:timeout [uint16]
Connection or request timeout in seconds (default: 30).

0:listen [boolean]
Create a listening connection (default: false).

0:useSSL [bool]
Encrypt the connection with TLS (default: false).

0:certificate [bytestring]
DER or PEM encoded local certificate. For listeners this is the server certificate. For clients it enables mutual TLS.

0:private-key [bytestring]
DER or PEM encoded private key for certificate.

0:private-key-password [string]
Password for an encrypted private key.

0:ca-certificate [bytestring]
DER or PEM encoded CA certificate used to validate the TLS peer. For a listener, setting this requires clients to present a trusted certificate. A client uses the system trust store when this parameter is omitted.

0:recv-max-message-size [uint32]
Maximum size of a reassembled response body for clients or request body on the wire for listeners. Zero or omission means unlimited.

0:recv-max-decompressed-message-size [uint32]
Maximum request body size after HTTP Content-Encoding decompression for listeners, or response body size after decompression for clients. It defaults to recv-max-message-size when that limit is set, otherwise to 64 MiB for listeners. A client only advertises supported compression when this limit is non-zero.

0:send-max-message-size [uint32]
Maximum request body size for clients or response body size for listeners. Zero or omission means unlimited.

0:validate [boolean]
Validate parameters without opening a connection.

Send Parameters:

0:path [string]
Request-target path (default: /).

0:method [string]
HTTP request method (default: GET).

0:status-code [uint16]
HTTP response status for server requests (default: 200). Not valid for client requests.

0:headers [KeyValuePair array]
Additional request or response headers. Every pair uses the header name as its QualifiedName with namespace index zero and a scalar String value. Header names are case-insensitive. Content-Length, Transfer-Encoding and Connection are managed internally and cannot be supplied.

0:request-handle [Variant]
Opaque caller-defined metadata for an HTTP client request. Scalar, array and custom DataTypes are accepted. The value is copied when sending, returned unchanged in every response callback and never transmitted as an HTTP header. Concurrent client requests require handles; callers are responsible for choosing values that distinguish their requests. Without a handle only one request may be outstanding.

0:timeout [uint16]
Timeout in seconds for this request. Zero or omission uses the timeout configured when opening the client binding.

The buf argument passed to sendWithConnection is used as the request body. It may be NULL for an empty body and, like all ConnectionManager send buffers, is released internally even if sending fails.

Listener callbacks provide listen-address and listen-port. An accepted connection is first announced with remote-address and an empty body. Its subsequent request callbacks provide method, path, headers, remote-address and request-random. The latter is a 32-byte ByteString filled by the libwebsockets platform random source. Client response callbacks provide status-code and headers when the response is established. Identity response-body fragments also provide content-length. Decompressed fragments omit it because the wire length no longer describes the delivered body. A final callback contains response-complete set to true and an empty body. It also contains request-status

with the terminal transport StatusCode. All callbacks contain request-handle when it was supplied by the caller. A failed client request does not close sibling requests or the client binding.

Every accepted peer connection has an independent connection context and produces exactly one final UA_CONNECTIONSTATE_CLOSING callback. Its connection id remains valid across sequential requests until it is explicitly closed, the peer disconnects or the configured inactivity timeout expires. Only one request is active at a time on an HTTP/1.1 connection. Closing a listener also closes all of its accepted connections before the ConnectionManager reaches the stopped state. Incoming chunked request bodies are decoded and delivered as one reassembled message.

```
UA_ConnectionManager *
UA_ConnectionManager_new_HTTP(const UA_String eventSourceName);
```

libwebsockets WebSocket Connection Manager

Provides WebSocket client and server connections. Each buffer passed to sendWithConnection is sent as one binary WebSocket message. Incoming fragments belonging to one WebSocket message are reassembled and delivered together in one connection callback.

Open Connection Parameters:

0:address [string]
Remote hostname for clients or local interface for listeners. Listeners default to all interfaces.

0:port [uint16]
Remote or listening port (required; zero selects a dynamic server port).

0:listen [boolean]
Create a listening connection (default: false).

0:path [string]
WebSocket request path (default: /). Clients request this path and listeners reject upgrade requests for any other path.

0:subprotocol [string]
WebSocket subprotocol to request or accept. By default no subprotocol is negotiated. When configured, listeners reject clients that do not offer this exact subprotocol. A client may offer additional protocols.

0:binary-only [boolean]
Reject incoming text messages and accept only binary WebSocket messages (default: true).

0:useSSL [bool]
Encrypt the connection with TLS (default: false). TLS listeners require certificate and private-key.

0:certificate [bytestring]
DER or PEM encoded local certificate. For listeners this is the server certificate. For clients it enables mutual TLS.

0:private-key [bytestring]
DER or PEM encoded private key for certificate.

0:private-key-password [string]
Password for an encrypted private key.

0:ca-certificate [bytestring]
DER or PEM encoded CA certificate used to validate the TLS peer for client connections. If omitted, the system trust store is used.

0:recv-max-message-size [uint32]
Maximum size of one reconstructed incoming WebSocket message. Zero or omission means unlimited. For opcua+uacp this is the negotiated receive buffer size because each WebSocket message carries one UACP MessageChunk.

0:send-max-message-size [uint32]
Maximum size of one outgoing WebSocket message. Zero or omission means unlimited. For opcua+uacp this is the negotiated send buffer size.

0:send-max-queue-size [uint32]
Maximum total payload size queued for sending on one connection. Zero or omission means unlimited.

0:validate [boolean]
Validate parameters without opening a connection.

Listener callbacks provide listen-address and listen-port. Active and accepted connections provide remote-address. Secure client connections validate the server certificate and hostname using either ca-certificate or the system trust store.

```
UA_ConnectionManager *
UA_ConnectionManager_new_LWS_WebSocket(const UA_String eventSourceName);
```

MQTT Connection Manager

The MQTT ConnectionManager reuses the TCP ConnectionManager that is configured in the EventLoop. Hence the MQTT ConnectionManager is platform agnostic and does not require porting. An MQTT connection is for a combination of broker and topic. The MQTT ConnectionManager can group connections to the same broker in the background. Hence adding multiple connections for the same broker is "cheap". To have individual control, separate connections are created for each topic and for each direction (publishing / subscribing).

Open Connection Parameters:

0:address [string]
Hostname or IPv4/v6 address of the MQTT broker (required).

0:port [uint16]
Port of the MQTT broker (default: 1883).

0:username [string]
Username to use (default: none)

0:password [string]
Password to use (default: none)

0:useSSL [bool]
Encrypt the broker connection with TLS (default: false).

0:certificate [bytestring]
DER or PEM encoded client certificate for mutual TLS.

0:private-key [bytestring]
DER or PEM encoded private key for certificate.

0:private-key-password [string]
Password for an encrypted private key.

0:ca-certificate [bytestring]
DER or PEM encoded CA certificate used to validate the broker certificate. If omitted, the system trust store is used. The broker certificate and hostname are always validated for TLS connections.

0:keep-alive [uint16]
Number of seconds for the keep-alive (ping) (default: 400).

0:validate [boolean]
If true, the connection setup will act as a dry-run without actually creating any connection but solely validating the provided parameters (default: false)

0:topic [string]
Topic to which the connection is associated (required).

0:subscribe [bool]
Subscribe to the topic (default: false). Otherwise it is only possible to publish on the topic. Subscribed topics can also be published to.

Connection Callback Parameters:

0:topic [string]
The value set during connect.

0:subscribe [bool]
The value set during connect.

Send Parameters:

No additional parameters for sending over an Ethernet connection defined.

```
UA_ConnectionManager *
UA_ConnectionManager_new_MQTT(const UA_String eventSourceName);
```

Libwebsockets MQTT Connection Manager

Implements the same MQTT ConnectionManager contract and parameters as UA_ConnectionManager_new_MQTT, using the MQTT client role provided by libwebsockets. Enable this implementation with UA_ENABLE_LWS_MQTT. The libwebsockets library must be built with LWS_ROLE_MQTT=ON.

```
UA_ConnectionManager *
UA_ConnectionManager_new_LWS_MQTT(const UA_String eventSourceName);
```

Signal Interrupt Manager

Create an instance of the interrupt manager that handles POSX signals. This interrupt manager takes the numerical interrupt identifiers from <signal.h> for the interruptHandle.

```
#if defined(UA_ARCHITECTURE_POSIX) && !defined(UA_ARCHITECTURE_LWIP)
UA_InterruptManager *
UA_InterruptManager_new_POSIX(const UA_String eventSourceName);
#endif
```

(continues on next page)

(continued from previous page)

```
#ifdef UA_ARCHITECTURE_WIN32
UA_InterruptManager *
UA_InterruptManager_new_WIN32(const UA_String eventSourceName);
#endif

#ifdef UA_ENABLE_EVENTLOOP_GLIB
```

GLib Signal Interrupt Manager

Same contract as UA_InterruptManager_new_POSIX (interruptHandle is a <signal.h> signal number), but each interrupt is registered as a native GLib GSource (g_unix_signal_source_new) attached to the GMainContext of the owning EventLoop (see UA_EventLoop_new_GLib), instead of installing a sigaction handler writing to a self-pipe. GLib handles the OS-level signal reception itself.

Two behavioral differences from UA_InterruptManager_new_POSIX, both inherent to GLib:

- Delivery is asynchronous: GLib forwards a received signal to the EventLoop via an internal worker thread. A signal may therefore not yet be visible to a single non-blocking EventLoop iteration performed immediately after it was raised; callers polling for a specific interrupt should use one or more blocking iterations (a nonzero *run* timeout), not assume synchronous delivery.
- If the same signal number is registered on two independent GLib-backed InterruptManagers (or GSources) in the same process, GLib delivers each occurrence to only one of them, not both – unlike UA_InterruptManager_new_POSIX, which fans a signal out to every one of its own registered listeners. Use different signal numbers if multiple independent GLib InterruptManagers need to run side by side.

Do not register the same signal on both a UA_InterruptManager_new_POSIX and a UA_InterruptManager_new_GLib instance in the same process – they install the OS-level handler through different, mutually-unaware mechanisms and will race over it. Only available on Unix (Linux/Darwin), not Win32.

```
UA_InterruptManager *
UA_InterruptManager_new_GLib(const UA_String eventSourceName);

#endif /* UA_ENABLE_EVENTLOOP_GLIB */

#elif defined(UA_ARCHITECTURE_ZEPHYR)

UA_EventLoop *
UA_EventLoop_new_Zephyr(const UA_Logger *logger);

UA_ConnectionManager *
UA_ConnectionManager_new_Zephyr_TCP(const UA_String eventSourceName);

#elif defined(UA_ARCHITECTURE_LWIP)

struct UA_EventLoopConfiguration;
typedef struct UA_EventLoopConfiguration UA_EventLoopConfiguration;
```

14.4.8 Event Loop Configuration

Defines the configuration parameters and optional callback functions for managing the network interface within the EventLoop.

The interface callbacks are optional. On POSIX, omitted callbacks use the default TAP interface. It and the lwIP stack are shared by all default EventLoops and remain alive until process exit, independently of EventLoop, server, and client lifetimes. The first EventLoop start initializes them. Subsequent loops reuse that configuration; explicitly conflicting IP settings return BadConfigurationError. Omitted IP settings reuse the existing values.

On other platforms the application must provide callbacks or manage the network interface externally. Custom callbacks retain application-defined ownership: netifInit is called during start and retried until it succeeds. The interface then remains initialized through loop stop/start. netifShutdown is called during deletion, after all loop activity has ended, only if netifInit succeeded. Shared custom interfaces must outlive every EventLoop using them.

** Configuration Parameters for the EventLoop**

0:ipaddr [string]
: IPv4 address of the network interface (optional).

0:netmask [string]
: Netmask of the network interface (optional).

0:gateway [string]
: Gateway of the network interface (optional).

```
struct UA_EventLoopConfiguration {
 UA_KeyValueMap params;

 UA_StatusCode (*netifInit)(UA_EventLoop *el, const UA_String *ipaddr,
                            const UA_String *netmask, const UA_String *gw);
 UA_StatusCode (*netifPoll)(UA_EventLoop *el);
 void (*netifShutdown)(UA_EventLoop *el);
};
```

14.4.9 LWIP EventLoop Implementation

This EventLoop is built on LWIP's socket-like API. The configuration parameters must be set before starting the EventLoop.

Clock configuration (Linux and BSDs only)

0:clock-source [int32]
: Clock source (default: CLOCK_REALTIME).

0:clock-source-monotonic [int32]:
: Clock source used for time intervals. A non-monotonic source can be used as well. But expect accordingly longer sleep-times for timed events when the clock is set to the past. See the man-page of "clock_gettime" on how to get a clock source id for a character-device such as /dev/ptp0. (default: CLOCK_MONOTONIC_RAW)

```
UA_EventLoop *
UA_EventLoop_new_LWIP(const UA_Logger *logger, UA_EventLoopConfiguration␣
↪*config);
```

TCP Connection Manager

Listens on the network and manages TCP connections. This should be available for all architectures.

The *openConnection* callback is used to create both client and server sockets. A server socket listens and accepts incoming connections (creates an active connection). This is distinguished by the key-value parameters passed to *openConnection*. Note that a single call to *openConnection* for a server connection may actually create multiple connections (one per hostname / device).

The *connectionCallback* of the server socket and *context* of the server socket is reused for each new connection. But the key-value parameters for the first callback are different between server and client connections.

Configuration parameters for the ConnectionManager (set before start)

0:recv-bufsize [uint32]
Size of the buffer that is statically allocated for receiving messages (default 64kB).

0:send-bufsize [uint32]
Size of the statically allocated buffer for sending messages. The buffer is reused for every *allocNetworkBuffer* to avoid a heap allocation per message; a message larger than the buffer falls back to a fresh allocation. A dedicated send buffer is used (never the recv buffer), so sending while received data is still being processed is safe. If undefined, the send buffer defaults to the recv-bufsize.

Open Connection Parameters:

0:address [string | array of string]
Hostname or IPv4/v6 address for the connection (scalar parameter required for active connections). For listen-connections the address contains the local hostnames or IP addresses for listening. If undefined, listen on all interfaces INADDR_ANY. (default: undefined)

0:port [uint16]
Port of the target host (required).

0:listen [boolean]
Listen-connection or active-connection (default: false)

0:validate [boolean]
If true, the connection setup will act as a dry-run without actually creating any connection but solely validating the provided parameters (default: false)

Active Connection Connection Callback Parameters (first callback only):

0:remote-address [string]
Address of the remote side (hostname or IP address).

Listen Connection Connection Callback Parameters (at establishment):

Listener setup can be asynchronous: openConnection may return before binding and listening have completed. In this case, the initial UA_CONNECTIONSTATE_OPENING callback may omit the bound address and port. These are provided in the UA_CONNECTIONSTATE_ESTABLISHED callback once the listener is ready to accept connections.

0:listen-address [string]
Local address (IP or hostname) for the new listen-connection.

0:listen-port [uint16]
Port on which the new connection listens.

Send Parameters:

No additional parameters for sending over an established TCP socket defined.

```
UA_ConnectionManager *
UA_ConnectionManager_new_LWIP_TCP(const UA_String eventSourceName);
```

UDP Connection Manager

Manages UDP connections. This should be available for all architectures. The configuration parameters have to set before calling _start to take effect.

Configuration parameters for the ConnectionManager (set before start)

0:recv-bufsize [uint32]
Size of the buffer that is statically allocated for receiving messages (default 64kB).

0:send-bufsize [uint32]
Size of the statically allocated buffer for sending messages. The buffer is reused for every *allocNetworkBuffer* to avoid a heap allocation per message; a message larger than the buffer falls back to a fresh allocation. A dedicated send buffer is used (never the recv buffer), so sending while received data is still being processed is safe. If undefined, the send buffer defaults to the recv-bufsize.

Open Connection Parameters:

0:listen [boolean]
Use the connection for listening or for sending (default: false)

0:address [string | string array]
Hostname (or IPv4/v6 address) for sending or receiving. A scalar is required for sending. For listening a string array for the list-hostnames is possible as well (default: list on all hostnames).

0:port [uint16]
Port for sending or listening (required).

0:interface [string]
Network interface for listening or sending (e.g. when using multicast addresses). Can be either the IP address of the network interface or the interface name (e.g. 'eth0').

0:ttl [uint32]
Multicast time to live, (optional, default: 1 - meaning multicast is available only to the local subnet).

0:loopback [boolean]
Whether or not to use multicast loopback, enabling local interfaces belonging to the multicast group to receive packages. (default: enabled).

0:reuse [boolean]
Enables sharing of the same listening address on different sockets (default: disabled).

0:sockpriority [uint32]
The socket priority (optional) - only available on linux. packets with a higher priority may be processed first depending on the selected device queueing discipline. Setting a priority outside the range 0 to 6 requires the CAP_NET_ADMIN capability (on Linux).

0:validate [boolean]
If true, the connection setup will act as a dry-run without actually creating any connection but

solely validating the provided parameters (default: false)

Connection Callback Parameters:

0:remote-address [string]
Contains the remote IP address.

0:remote-port [uint16]
Contains the remote port.

Send Parameters:

No additional parameters for sending over an UDP connection defined.

```
UA_ConnectionManager *
UA_ConnectionManager_new_LWIP_UDP(const UA_String eventSourceName);

#endif
```

14.5 CertificateGroup Plugin API

This plugin verifies that the origin of the certificate is trusted. It does not assign any access rights/roles to the holder of the certificate.

Usually, implementations of the CertificateGroup plugin provide an initialization method that takes a trust-list and a revocation-list as input. This initialization method takes a pointer to the plugin location and therein calls the clear method if valid, before attempting to initialize it anew. The lifecycle of the plugin is attached to a server or client config. The clear method is called automatically when the config is destroyed.

```
struct UA_CertificateGroup {
    /* The NodeId of the certificate group this pki store is associated with */
    UA_NodeId certificateGroupId;

    /* Context-pointer to be set by the CertificateGroup plugin implementation */
    void *context;

    /* Pointer to logging pointer in the server/client configuration. If the
     * logging pointer is changed outside of the plugin, the new logger is used
     * automatically. */
    const UA_Logger *logging;

    UA_StatusCode (*getTrustList)(UA_CertificateGroup *certGroup,
                                  UA_TrustListDataType *trustList);

    UA_StatusCode (*setTrustList)(UA_CertificateGroup *certGroup,
                                  const UA_TrustListDataType *trustList);

    UA_StatusCode (*addToTrustList)(UA_CertificateGroup *certGroup,
                                    const UA_TrustListDataType *trustList);

    UA_StatusCode (*removeFromTrustList)(UA_CertificateGroup *certGroup,
                                         const UA_TrustListDataType *trustList);
```

(continues on next page)

(continued from previous page)

```
    UA_StatusCode (*getRejectedList)(UA_CertificateGroup *certGroup,
                                     UA_ByteString **rejectedList,
                                     size_t *rejectedListSize);

    /* Provides all associated CRLs for a CA certificate. */
    UA_StatusCode (*getCertificateCrls)(UA_CertificateGroup *certGroup,
                                        const UA_ByteString *certificate,
                                        const UA_Boolean isTrusted,
                                        UA_ByteString **crls,
                                        size_t *crlsSize);

    UA_StatusCode (*verifyCertificate)(UA_CertificateGroup *certGroup,
                                       const UA_ByteString *certificate);

    void (*clear)(UA_CertificateGroup *certGroup);
};

/* Verify that the certificate has the applicationURI in the subject name. */
UA_StatusCode
UA_CertificateUtils_verifyApplicationUri(const UA_ByteString *certificate,
                                         const UA_String *applicationUri);

/* Get the expire date from certificate */
UA_StatusCode
UA_CertificateUtils_getExpirationDate(UA_ByteString *certificate,
                                      UA_DateTime *expiryDateTime);

UA_StatusCode
UA_CertificateUtils_getSubjectName(UA_ByteString *certificate,
                                   UA_String *subjectName);

UA_StatusCode
UA_CertificateUtils_getThumbprint(UA_ByteString *certificate,
                                  UA_String *thumbprint);

UA_StatusCode
UA_CertificateUtils_getKeySize(UA_ByteString *certificate,
                               size_t *keySize);

/* Extended key usages are a set. A value of UA_CERTIFICATEEKU_NONE means that
 * the certificate does not contain an Extended Key Usage extension. The
 * UA_CERTIFICATEEKU_OTHER flag is set if the extension contains at least one
 * purpose that is not represented by the other flags. */
typedef enum {
    UA_CERTIFICATEEKU_NONE       = 0,
    UA_CERTIFICATEEKU_SERVERAUTH = 1 << 0,
    UA_CERTIFICATEEKU_CLIENTAUTH = 1 << 1,
    UA_CERTIFICATEEKU_ANY        = 1 << 2,
```

(continues on next page)

(continued from previous page)

```
    UA_CERTIFICATEEKU_OTHER      = 1 << 3
} UA_CertificateEku;

UA_StatusCode
UA_CertificateUtils_getExtendedKeyUsage(const UA_ByteString *certificate,
                                        UA_CertificateEku *extendedKeyUsage);

/* Compares the public keys from two byte strings, which can represent either
 * certificates or Certificate Signing Requests (CSR). This function extracts
 * the public keys from the provided byte strings and compares them to determine
 * if they are identical.
 *
 * @param certificate1 Containing either a certificate or a CSR.
 * @param certificate2 Containing either a certificate or a CSR.
 * @return UA_STATUSCODE_GOOD if the public keys are identical,
 *         UA_STATUSCODE_BADNOMATCH if the public keys do not match,
 *         UA_STATUSCODE_BADINTERNALERROR if an error occurs. */
UA_StatusCode
UA_CertificateUtils_comparePublicKeys(const UA_ByteString *certificate1,
                                      const UA_ByteString *certificate2);

UA_StatusCode
UA_CertificateUtils_checkKeyPair(const UA_ByteString *certificate,
                                 const UA_ByteString *privateKey);

UA_StatusCode
UA_CertificateUtils_checkCA(const UA_ByteString *certificate);

/* Decrypt a private key in PEM format using a password. The output is the key
 * in the binary DER format. Also succeeds if the PEM private key does not
 * require a password or is already in the DER format. The outDerKey memory is
 * allocated internally.
 *
 * Returns UA_STATUSCODE_BADSECURITYCHECKSFAILED if the password is␣
↪wrong. */
UA_StatusCode
UA_CertificateUtils_decryptPrivateKey(const UA_ByteString privateKey,
                                      const UA_ByteString password,
                                      UA_ByteString *outDerKey);

/* OPC UA Part 6 defines requirements for Application Instance Certificates
 * and User Certificates:
 * https://reference.opcfoundation.org/Core/Part6/v105/docs/6.2.2
 * https://reference.opcfoundation.org/Core/Part6/v105/docs/6.2.3
 *
 * Application Instance certificate and User certificate types contain a subject
 * field, which is a distinguished name defined by RFC 5280:
 * https://www.ietf.org/rfc/rfc5280.txt
 *
```

(continues on next page)

(continued from previous page)

```
 * According to the OPC UA specification, the Common Name (CN) attribute shall
 * be specified in the subject field for both Application Instance Certificates
 * and User Certificates.
 *
 * Other subject attributes may also be specified, so the Common Name must be
 * extracted as a single subject attribute instead of parsing all characters
 * after "CN=" from the subject string.
 */
UA_StatusCode
UA_CertificateUtils_getCertCommonName(const UA_ByteString *certificate,
                                      UA_String *commonName);
```

14.6 SecurityPolicy Plugin API

```
typedef struct {
    UA_String uri;

    /* Verifies the signature of the message using the provided keys in the
     * context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The channelContext that contains the key to verify
     *        the supplied message with.
     * @param message The message to which the signature is supposed to belong.
     * @param signature The signature of the message, that should be verified. */
    UA_StatusCode (*verify)(const UA_SecurityPolicy *policy,
                            void *channelContext, const UA_ByteString *message,
                            const UA_ByteString *signature);

    /* Signs the given message using this policy's signing algorithm and the
     * provided keys in the context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The channelContext that contains the key to sign
     *        the supplied message with.
     * @param message The message to sign.
     * @param signature An output buffer to which the signature is written. The
     *        buffer needs to be allocated by the caller. The necessary size can
     *        be acquired with the signatureSize attribute of this module. */
    UA_StatusCode (*sign)(const UA_SecurityPolicy *policy,
                          void *channelContext, const UA_ByteString *message,
                          UA_ByteString *signature);

    /* Gets the signature size that depends on the local (private) key.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The channelContext that contains the
     *        certificate/key.
```

(continues on next page)

(continued from previous page)

```
     * @return The size of the local signature. Returns 0 if no local
     *         certificate was set. */
    size_t (*getLocalSignatureSize)(const UA_SecurityPolicy *policy,
                                    const void *channelContext);

    /* Gets the signature size that depends on the remote (public) key.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The size of the remote signature. Returns 0 if no
     *         remote certificate was set previously. */
    size_t (*getRemoteSignatureSize)(const UA_SecurityPolicy *policy,
                                     const void *channelContext);

    /* Gets the local signing key length.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The length of the signing key in bytes. Returns 0 if no length
     *         can be found. */
    size_t (*getLocalKeyLength)(const UA_SecurityPolicy *policy,
                                const void *channelContext);

    /* Gets the local signing key length.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The length of the signing key in bytes. Returns 0 if no length
     *         can be found. */
    size_t (*getRemoteKeyLength)(const UA_SecurityPolicy *policy,
                                 const void *channelContext);
} UA_SecurityPolicySignatureAlgorithm;

typedef struct {
    UA_String uri;

    /* Encrypt the given data in place. For asymmetric encryption, the block
     * size for plaintext and cypher depend on the remote key (certificate).
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The channelContext which contains information about
     *        the keys to encrypt data.
     * @param data The data that is encrypted. The encrypted data will overwrite
     *        the data that was supplied. */
    UA_StatusCode (*encrypt)(const UA_SecurityPolicy *policy,
                             void *channelContext, UA_ByteString *data);

    /* Decrypts the given ciphertext in place. For asymmetric encryption, the
     * block size for plaintext and cypher depend on the local private key.
```

(continues on next page)

(continued from previous page)

```
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The channelContext which contains information about
     *        the keys needed to decrypt the message.
     * @param data The data to decrypt. The decryption is done in place. */
    UA_StatusCode (*decrypt)(const UA_SecurityPolicy *policy,
                             void *channelContext, UA_ByteString *data);

    /* Returns the length of the key used to encrypt messages in bits. For
     * asymmetric encryption the key length is for the local private key.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The length of the local key. Returns 0 if no
     *         key length is known. */
    size_t (*getLocalKeyLength)(const UA_SecurityPolicy *policy,
                                const void *channelContext);

    /* Returns the length of the key to encrypt messages in bits. Depends on the
     * key (certificate) from the remote side.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The length of the remote key. Returns 0 if no
     *         key length is known. */
    size_t (*getRemoteKeyLength)(const UA_SecurityPolicy *policy,
                                 const void *channelContext);

    /* Returns the size of encrypted blocks for sending. For asymmetric
     * encryption this depends on the remote key (certificate). For symmetric
     * encryption the local and remote encrypted block size are identical.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The size of encrypted blocks in bytes. Returns 0 if no key length
     *         is known. */
    size_t (*getRemoteBlockSize)(const UA_SecurityPolicy *policy,
                                 const void *channelContext);

    /* Returns the size of plaintext blocks for sending. For asymmetric
     * encryption this depends on the remote key (certificate). For symmetric
     * encryption the local and remote plaintext block size are identical.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The size of plaintext blocks in bytes. Returns 0 if no key length
     *         is known. */
    size_t (*getRemotePlainTextBlockSize)(const UA_SecurityPolicy *policy,
                                          const void *channelContext);
```

(continues on next page)

(continued from previous page)

```
    /* Returns the IV (initialization vector) length for symmetric encryption.
     * For AEAD ciphers (e.g. ChaCha20-Poly1305) the IV length differs from the
     * block size. If NULL, the block size is used as the IV length (which is
     * correct for traditional block ciphers like AES-CBC).
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to retrieve data from.
     * @return The IV length in bytes. */
    size_t (*getLocalIvLength)(const UA_SecurityPolicy *policy,
                               const void *channelContext);
} UA_SecurityPolicyEncryptionAlgorithm;

typedef enum {
    UA_SECURITYPOLICYTYPE_NONE = 0,
    UA_SECURITYPOLICYTYPE_RSA = 1,
    UA_SECURITYPOLICYTYPE_ECC = 2,
    UA_SECURITYPOLICYTYPE_ECC_AEAD = 3
} UA_SecurityPolicyType;

struct UA_SecurityPolicy {
    void *policyContext;      /* Context data */
    const UA_Logger *logger; /* Logger to be used by the plugin */
    UA_String policyUri;      /* SecurityPolicyUri */

    /* Value indicating the crypto strength of the policy, with zero for
     * deprecated or none */
    UA_Byte securityLevel;

    /* For special handling of SecurityPolicies using ECC (Elliptic Curve
     * Cryptography) */
    UA_SecurityPolicyType policyType;

    /* The local certificate is specific for each SecurityPolicy since it
     * depends on the used key length. */
    UA_ByteString localCertificate;

    UA_NodeId certificateGroupId;
    UA_NodeId certificateTypeId;

    /* Function pointers grouped into modules */
    UA_SecurityPolicySignatureAlgorithm asymSignatureAlgorithm;
    UA_SecurityPolicyEncryptionAlgorithm asymEncryptionAlgorithm;
    UA_SecurityPolicySignatureAlgorithm symSignatureAlgorithm;
    UA_SecurityPolicyEncryptionAlgorithm symEncryptionAlgorithm;
    UA_SecurityPolicySignatureAlgorithm certSignatureAlgorithm;

    /* Create a new channel context. The caller needs to call delete on the
     * received object to free allocated memory. Memory is only allocated if the
```

(continues on next page)

(continued from previous page)

```
     * function succeeds so there is no need to manually free the memory pointed
     * to by *channelContext or to call delete in case of failure.
     *
     * @param policy The policy for which the channelContext is created.
     * @param remoteCertificate The remote certificate contains the remote
     *        asymmetric key. The certificate will be verified and then stored
     *        in the context so that its details may be accessed.
     * @param channelContext The initialized channelContext that is passed to
     *        functions that work on a context. */
    UA_StatusCode (*newChannelContext)(const UA_SecurityPolicy *policy,
                                       const UA_ByteString *remoteCertificate,
                                       void **channelContext);

    /* Deletes the the channel context. */
    void (*deleteChannelContext)(const UA_SecurityPolicy *policy,
                                 void *channelContext);

    /* Sets the local encrypting key in the supplied context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param key The local encrypting key to store in the context. */
    UA_StatusCode (*setLocalSymEncryptingKey)(const UA_SecurityPolicy *policy,
                                              void *channelContext,
                                              const UA_ByteString *key);

    /* Sets the local signing key in the supplied context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param key The local signing key to store in the context. */
    UA_StatusCode (*setLocalSymSigningKey)(const UA_SecurityPolicy *policy,
                                           void *channelContext,
                                           const UA_ByteString *key);

    /* Sets the local initialization vector in the supplied context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param iv The local initialization vector to store in the context. */
    UA_StatusCode (*setLocalSymIv)(const UA_SecurityPolicy *policy,
                                   void *channelContext,
                                   const UA_ByteString *iv);

    /* Sets the remote encrypting key in the supplied context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param key The remote encrypting key to store in the context. */
```

(continues on next page)

(continued from previous page)

```
    UA_StatusCode (*setRemoteSymEncryptingKey)(const UA_SecurityPolicy *policy,
                                               void *channelContext,
                                               const UA_ByteString *key);

    /* Sets the remote signing key in the supplied context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param key The remote signing key to store in the context. */
    UA_StatusCode (*setRemoteSymSigningKey)(const UA_SecurityPolicy *policy,
                                            void *channelContext,
                                            const UA_ByteString *key);

    /* Sets the remote initialization vector in the supplied context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param iv The remote initialization vector to store in the context. */
    UA_StatusCode (*setRemoteSymIv)(const UA_SecurityPolicy *policy,
                                    void *channelContext,
                                    const UA_ByteString *iv);

    /* Sets the message security parameters for AEAD nonce derivation.
     * Called before each symmetric encrypt/decrypt for ECC_AEAD policies.
     * NULL for non-AEAD policies.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context to work on.
     * @param tokenId The SecurityToken ID for nonce masking.
     * @param previousSequenceNumber The previous sequence number for nonce masking.
     * @param additionalAuthData The header bytes used as AAD for AEAD. */
    UA_StatusCode (*setMessageSecurityParameters)(
        const UA_SecurityPolicy *policy,
        void *channelContext,
        UA_UInt32 tokenId,
        UA_UInt32 previousSequenceNumber,
        const UA_ByteString *additionalAuthData);

    /* Compares the supplied certificate with the remote certificate stored in
     * the channel context.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The channel context data that contains the
     *        certificate to compare to.
     * @param certificate The certificate to compare to the one stored in the context.
     * @return If the certificates match UA_STATUSCODE_GOOD is returned. If they
     *         don't match or an error occurred an error code is returned. */
    UA_StatusCode (*compareCertificate)(const UA_SecurityPolicy *policy,
                                        const void *channelContext,
```

(continues on next page)

(continued from previous page)

```
                                const UA_ByteString *certificate);

    /* Generate the keys for symmetric encryption. For information on what
     * parameters this function receives in what situation, refer to the OPC UA
     * specification Part 6, "Deriving keys".
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context of the SecureChannel
     * @param secret Usually from the nonce. See part 6.
     * @param seed Usually from the nonce. See part 6.
     * @param out An output to write the data to. The length defines the maximum
     *        number of output bytes that are produced. */
    UA_StatusCode (*generateKey)(const UA_SecurityPolicy *policy,
                                 void *channelContext, const UA_ByteString *secret,
                                 const UA_ByteString *seed, UA_ByteString *out);

    /* Random generator for generating nonces.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param channelContext The context of the SecureChannel
     * @param out Pointer to a buffer to store the nonce in. Needs to be
     *        allocated by the caller. The buffer is filled with random data. */
    UA_StatusCode (*generateNonce)(const UA_SecurityPolicy *policy,
                                   void *channelContext, UA_ByteString *out);

    /* The length of the nonce used in the SecureChannel */
    size_t nonceLength;

    /* Generates a thumbprint for the specified certificate.
     *
     * @param policy The policy that generates the thumbprint.
     * @param certificate The certificate to make a thumbprint of.
     * @param thumbprint an output buffer for the resulting thumbprint. */
    UA_StatusCode (*makeCertThumbprint)(const UA_SecurityPolicy *policy,
                                        const UA_ByteString *certificate,
                                        UA_ByteString *thumbprint);

    /* Compares the supplied certificate with the certificate in the
     * SecurityPolicy context.
     *
     * @param policy The policy data that contains the certificate
     *        to compare to.
     * @param thumbprint The certificate thumbprint to compare to the one stored
     *        in the context.
     * @return If the thumbprints match UA_STATUSCODE_GOOD is returned. If they
     *         don't match or an error occurred an error code is returned. */
    UA_StatusCode (*compareCertThumbprint)(const UA_SecurityPolicy *policy,
                                           const UA_ByteString *thumbprint);
```

(continues on next page)

(continued from previous page)

```
    /* Updates the ApplicationInstanceCertificate and the corresponding private
     * key at runtime. This overwrites the localCertificate member of the
     * SecurityPolicy structure and updates the context internally. */
    UA_StatusCode (*updateCertificate)(UA_SecurityPolicy *policy,
                                       const UA_ByteString certificate,
                                       const UA_ByteString privateKey);

    /* Creates a PKCS #10 DER encoded certificate request signed with the
     * server's private key.
     *
     * @param policy The SecurityPolicy to work on.
     * @param subjectName The subject name to use in the Certificate Request. If
     *        not specified the SubjectName from the current Certificate is
     *        used.
     * @param nonce Additional entropy that the caller can provide. It shall be
     *        at least 32 bytes long.
     * @param params A KeyVaue list that can be used for additional parameters
     *        later.
     * @param csr Returns the created CSR. If the passed byte string is not
     *        empty, nothing is created.
     * @param newPrivateKey Returns the private key if a new one needs to be
     *        generated. Alternatively, an existing key can be provided, which
     *        will be used as the CSR key in the security policy. This is
     *        necessary if the CSR was created under a different security policy
     *        and the current one only requires an update. */
    UA_StatusCode (*createSigningRequest)(UA_SecurityPolicy *policy,
                                          const UA_String *subjectName,
                                          const UA_ByteString *nonce,
                                          const UA_KeyValueMap *params,
                                          UA_ByteString *csr,
                                          UA_ByteString *newPrivateKey);

    /* Deletes the dynamic content of the policy */
    void (*clear)(UA_SecurityPolicy *policy);
};

/* True if the SecurityPolicy requires the OPC UA Part 6 v1.05.07
 * SecureChannelEnhancements behavior (see the conformance unit "Security
 * SecureChannelEnhancements True"). The changed behavior is implemented in the
 * core library, not in the SecurityPolicy itself. Detected from the policy URI
 * so it does not depend on a struct member - this keeps the SecurityPolicy
 * layout stable across the 1.5 release family. */
UA_Boolean
UA_SecurityPolicy_isEnhancedSecurity(const UA_SecurityPolicy *policy);

/* True if the SecurityPolicy uses the legacy SequenceNumber handling (the OPC
 * UA reference stack's LegacySequenceNumbers = true): the channel SequenceNumber
 * starts at 1 and rolls over with the "< 1024" rule. This is the case for None
 * and the RSA policies (Basic*, Aes*_RsaOaep/RsaPss). The ECC policies use the
```

(continues on next page)

(continued from previous page)

```
 * non-legacy scheme (start at 0, wrap UA_UINT32_MAX -> 0). Detected from the
 * policy URI so it does not depend on a struct member - this keeps the
 * SecurityPolicy layout stable across the 1.5 release family. A NULL policy is
 * treated as legacy. */
UA_Boolean
UA_SecurityPolicy_useLegacySequenceNumbers(const UA_SecurityPolicy *policy);

/* OPC UA Part 6 v1.05.07 (SecureChannelEnhancements): hash a certificate (the
 * leaf, DER) with the hash of the policy's elliptic curve - SHA-256 for the
 * nistP256 curve, SHA-384 for nistP384. Used to build the channel-bound
 * CreateSession / ActivateSession SignatureData.
 *
 * This is NOT the OPN-header Certificate thumbprint: that thumbprint uses the
 * SecurityPolicy's CertificateThumbprintAlgorithm (SHA-1 by default) and is
 * produced by makeCertThumbprint. The digest here is selected from the policy
 * URI's curve; the implementation is provided by the crypto backend.
 *
 * @param policy The policy whose curve selects the hash algorithm.
 * @param certificate The certificate to hash (DER; leaf is used).
 * @param hash Output buffer; allocated by the callee. */
UA_StatusCode
UA_SecurityPolicy_hashCertificate(const UA_SecurityPolicy *policy,
                                  const UA_ByteString *certificate,
                                  UA_ByteString *hash);
```

14.6.1 PubSub SecurityPolicy

For PubSub encryption, the message nonce is part of the (unencrypted) SecurityHeader. The nonce is required for the de- and encryption and has to be set in the group context before de/encrypting.

```
struct UA_PubSubSecurityPolicy;
typedef struct UA_PubSubSecurityPolicy UA_PubSubSecurityPolicy;

struct UA_PubSubSecurityPolicy {
    void *policyContext;
    const UA_Logger *logger;

    UA_String policyUri; /* The policy uri that identifies the implemented
                          * algorithms */

    /* Create the context for the WriterGroup / ReaderGroup. The keys and nonce
     * can be NULL here. Then they have to be set before the first encryption or
     * signing operation. */
    UA_StatusCode (*newGroupContext)(UA_PubSubSecurityPolicy *policy,
                                     const UA_ByteString *signingKey,
                                     const UA_ByteString *encryptingKey,
                                     const UA_ByteString *keyNonce,
                                     void **gContext);
```

(continues on next page)

(continued from previous page)

```
    /* Delete the WriterGroup / ReaderGroup SecurityPolicy context. */
    void (*deleteGroupContext)(UA_PubSubSecurityPolicy *policy, void *gContext);

    /* Verify a message signature using the keys in the group context. */
    UA_StatusCode (*verify)(const UA_PubSubSecurityPolicy *policy,
                            void *gContext, const UA_ByteString *message,
                            const UA_ByteString *signature);

    /* Sign a message using the keys in the group context. The caller allocates
     * the signature buffer with the size returned by getSignatureSize. */
    UA_StatusCode (*sign)(const UA_PubSubSecurityPolicy *policy,
                          void *gContext, const UA_ByteString *message,
                          UA_ByteString *signature);

    /* Return the signature size in bytes. */
    size_t (*getSignatureSize)(const UA_PubSubSecurityPolicy *policy,
                               const void *gContext);

    /* Return the signing-key length in bytes. The group context may be NULL
     * when querying the policy for SKS key-material processing. */
    size_t (*getSignatureKeyLength)(const UA_PubSubSecurityPolicy *policy,
                                    const void *gContext);

    /* Return the encryption-key length in bytes. The group context may be NULL
     * when querying the policy for SKS key-material processing. */
    size_t (*getEncryptionKeyLength)(const UA_PubSubSecurityPolicy *policy,
                                     const void *gContext);

    /* Encrypt the supplied buffer in place using the group keys and the
     * message nonce installed with setMessageNonce. */
    UA_StatusCode (*encrypt)(const UA_PubSubSecurityPolicy *policy,
                             void *gContext, UA_ByteString *data);

    /* Decrypt the supplied buffer in place using the group keys and the
     * message nonce installed with setMessageNonce. */
    UA_StatusCode (*decrypt)(const UA_PubSubSecurityPolicy *policy,
                             void *gContext, UA_ByteString *data);

    /* Set the keys and key nonce for a WriterGroup / ReaderGroup. These are
     * returned by GetSecurityKeys on a Security Key Service (SKS), or supplied
     * manually via out-of-band transmission. */
    UA_StatusCode (*setSecurityKeys)(UA_PubSubSecurityPolicy *policy,
                                     void *gContext,
                                     const UA_ByteString *signingKey,
                                     const UA_ByteString *encryptingKey,
                                     const UA_ByteString *keyNonce);

    /* Generate the keys for symmetric encryption.
     *
```

(continues on next page)

(continued from previous page)

```
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param gContext The group context.
     * @param secret Secret input for the policy-specific key derivation.
     * @param seed Seed input for the policy-specific key derivation.
     * @param out An output to write the data to. The length defines the maximum
     *        number of output bytes that are produced. */
    UA_StatusCode (*generateKey)(UA_PubSubSecurityPolicy *policy,
                                 void *gContext,
                                 const UA_ByteString *secret,
                                 const UA_ByteString *seed,
                                 UA_ByteString *out);

    /* Random generator for generating nonces.
     *
     * @param policy The SecurityPolicy to which the callback belongs.
     * @param gContext The group context, or NULL for policy-level generation.
     * @param out Pointer to a buffer to store the nonce in. Needs to be
     *        allocated by the caller. The buffer is filled with random data. */
    UA_StatusCode (*generateNonce)(UA_PubSubSecurityPolicy *policy,
                                   void *gContext, UA_ByteString *out);

    /* Total length in bytes of one SKS key-material item. For the CTR policies
     * this is signing key + encryption key + key nonce. */
    size_t keyMaterialLength;

    /* Length of the per-NetworkMessage nonce passed to setMessageNonce. */
    size_t messageNonceLength;

    /* The nonce is contained in the NetworkMessage SecurityHeader. Set before
     * each en-/decryption step. */
    UA_StatusCode (*setMessageNonce)(UA_PubSubSecurityPolicy *policy,
                                     void *gContext,
                                     const UA_ByteString *nonce);

    /* Deletes the dynamic content of the policy */
    void (*clear)(UA_PubSubSecurityPolicy *policy);
};
```

CHAPTER

FIFTEEN

DRIVERS

Drivers integrate external functionality with a server instance. Unlike the core SDK, driver implementations can be architecture- and operating system-specific. They may use platform facilities internally, but their interaction with the server must go through the public server API.

Drivers are created separately and added to a server after the server has been initialized. See the *Drivers* for the generic UA_Driver interface and lifecycle hooks.

15.1 Multicast DNS Discovery Driver

An OPC UA application can announce / retract a server via multicast DNS. Also it receives multicast-based information about other servers. This functionality is implemented as a driver that integrates with open62541 servers. Internally, the drivers use the EventLoop of the server to create and manage UDP sockets. They further use the UA_APPLICATIONNOTIFICATIONTYPE_DISCOVERY category for server notifications to receive relevant updates from the server at runtime.

The following configuration parameters are forwarded in a key-value map during driver instantiation and/or set in the key-value map inside the generic driver structure before starting the driver.

0:listen [**bool**]
: Receive information about OPC UA servers on the network via mDNS. Calls UA_Server_registerServerOnNetwork and UA_Server_deregisterServerOnNetwork to store the server information locally.

0:announce [**bool**]
: Announce registered servers over mDNS. This receives register/deregister notifications from UA_Server_registerServerOnNetwork and forwards the information.

0:announce-ttl [**uint32**]
: Default TTL in seconds for announced mDNS records.

0:query-presence [**bool**]
: Send an mDNS PTR query for _opcua-tcp._tcp.local. on startup. Defaults to false.

0:query-details [**bool**]
: Send targeted SRV/TXT follow-up queries for partially received OPC UA service instances. Defaults to false.

0:query-interval [**uint32**]
: Interval in seconds for repeated presence queries. A value of zero disables repeated queries. Defaults to zero.

0:interface [**string**]
: Network interface for listening or sending. Can be either the IP address of the network interface or the interface name (e.g. 'eth0'). Cf. the UDP ConnectionManager parameters.

```
typedef struct UA_MdnsDriver UA_MdnsDriver;

struct UA_MdnsDriver {
    UA_Driver drv; /* Must be the first member */

    /* Manually publish/remove address records. This is separate from the
     * ServerOnNetwork announcement path as PTR/SRV/TXT records are always
     * derived from the ServerOnNetwork data. The address strings are numeric
     * IPv4/IPv6 literals. */
    UA_StatusCode (*addARecord)(UA_MdnsDriver *mdns, UA_String hostname,
                                UA_String address, UA_UInt32 ttl);
    UA_StatusCode (*removeARecord)(UA_MdnsDriver *mdns, UA_String hostname,
                                   UA_String address);
    UA_StatusCode (*addAAAARecord)(UA_MdnsDriver *mdns, UA_String hostname,
                                   UA_String address, UA_UInt32 ttl);
    UA_StatusCode (*removeAAAARecord)(UA_MdnsDriver *mdns, UA_String hostname,
                                      UA_String address);
};

/* Based on the libmdnsd dependency */
UA_MdnsDriver *
UA_MdnsDriver_Mdnsd(const UA_KeyValueMap params);

/* Based on the libavahi dependency */
UA_MdnsDriver *
UA_MdnsDriver_Avahi(const UA_KeyValueMap params);
```

15.2 Alarms & Conditions Driver (Experimental)

OPC UA Alarms & Conditions model process states that need operator attention: limit violations, abnormal device states, certificate expiry, and similar situations. Unlike plain events, conditions are stateful. They can remain active, be acknowledged or confirmed, carry comments, and be refreshed for clients that reconnect or subscribe later.

The central concept is the Condition instance. A Condition is an Object of a ConditionType, such as OffNormalAlarmType or ExclusiveLimitAlarmType. It is associated with a Condition Source, the node whose state is represented by the Condition. The source is connected to the event hierarchy with HasEventSource so clients can discover and subscribe to relevant alarms.

Conditions expose standardized fields such as EnabledState, ActiveState, AckedState, ConfirmedState, Severity, Message, Retain, and SourceNode. These fields describe the current alarm state and are also copied into emitted condition events. Retain marks conditions that are still relevant to a client. Acknowledgement and confirmation model operator interaction.

This driver provides the server-side bookkeeping for Conditions, retained EventIds, refresh handling, state callbacks, and helper functions to create, update, trigger, and delete Conditions. Create the driver with UA_AlarmsConditionsDriver() and attach it to a server with UA_Server_addDriver(). Only one Alarms & Conditions driver instance can be attached to a server.

```
#if defined(UA_ENABLE_SUBSCRIPTIONS_EVENTS) && defined(UA_GENERATED_
↪NAMESPACE_ZERO_FULL)
```

(continues on next page)

(continued from previous page)

```
_UA_BEGIN_DECLS

typedef struct UA_AlarmConditionsDriver UA_AlarmConditionsDriver;

typedef enum UA_TwoStateVariableCallbackType {
    UA_ENTERING_ENABLEDSTATE,
    UA_ENTERING_ACKEDSTATE,
    UA_ENTERING_CONFIRMEDSTATE,
    UA_ENTERING_ACTIVESTATE
} UA_TwoStateVariableCallbackType;

/* User-defined callback for changes of TwoStateVariable fields. */
typedef UA_StatusCode
(*UA_TwoStateVariableChangeCallback)(UA_Server *server,
                                     const UA_NodeId *condition);

struct UA_AlarmConditionsDriver {
    UA_Driver drv; /* Must be the first member */

    /* Create a condition instance.
     *
     * The conditionType must be a subtype of ConditionType. If the condition
     * source has no inverse HasEventSource reference, a HasEventSource
     * reference from the Server object to the source is created. Pass a
     * hierarchical reference type to expose the condition below its source in
     * the address space. Pass UA_NODEID_NULL to keep the condition unexposed.
     *
     * @param driver The Alarms & Conditions driver
     * @param conditionId The requested Condition Object NodeId. Passing
     *        UA_NODEID_NUMERIC(X,0) selects an unused NodeId in namespace X.
     *        Passing UA_NODEID_NULL selects an unused NodeId in namespace 0.
     * @param conditionType The ConditionType node
     * @param conditionName The condition name
     * @param conditionSource The Condition Source node
     * @param hierarchialReferenceType The hierarchical ReferenceType between
     *        the condition source and the condition
     * @param outConditionId The created Condition node
     * @return The StatusCode of the operation */
    UA_StatusCode (*createCondition)(UA_AlarmConditionsDriver *driver,
                                     const UA_NodeId conditionId,
                                     const UA_NodeId conditionType,
                                     const UA_QualifiedName conditionName,
                                     const UA_NodeId conditionSource,
                                     const UA_NodeId hierarchialReferenceType,
                                     UA_NodeId *outConditionId);

    /* Begin creating a condition instance.
     *
```

(continues on next page)

(continued from previous page)

```
     * This is the first half of createCondition(), mirroring the
     * UA_Server_addNode_begin() / UA_Server_addNode_finish() pattern. Use it
     * when the condition node shall be modified before instantiation is
     * finished, for example to add children with specific NodeIds.
     *
     * @param driver The Alarms & Conditions driver
     * @param conditionId The requested Condition Object NodeId. Passing
     *        UA_NODEID_NUMERIC(X,0) selects an unused NodeId in namespace X.
     *        Passing UA_NODEID_NULL selects an unused NodeId in namespace 0.
     * @param conditionType The ConditionType node. It must be a subtype of the
     *        OPC UA ConditionType.
     * @param conditionName The condition name
     * @param outConditionId The added Condition node
     * @return The StatusCode of the operation */
    UA_StatusCode (*addConditionBegin)(UA_AlarmConditionsDriver *driver,
                                       const UA_NodeId conditionId,
                                       const UA_NodeId conditionType,
                                       const UA_QualifiedName conditionName,
                                       UA_NodeId *outConditionId);

    /* Finish creating a condition instance.
     *
     * This is the second half of the addConditionBegin() /
     * addConditionFinish() pair. It finishes the node, creates missing
     * HasEventSource references, optionally exposes the condition below its
     * source, and initializes the standard condition fields and callbacks.
     *
     * @param driver The Alarms & Conditions driver
     * @param conditionId The unfinished Condition Object node
     * @param conditionSource The Condition Source node
     * @param hierarchialReferenceType The hierarchical ReferenceType between
     *        the condition source and the condition
     * @return The StatusCode of the operation */
    UA_StatusCode (*addConditionFinish)(UA_AlarmConditionsDriver *driver,
                                        const UA_NodeId conditionId,
                                        const UA_NodeId conditionSource,
                                        const UA_NodeId hierarchialReferenceType);

    /* Set the value of a condition field.
     *
     * @param driver The Alarms & Conditions driver
     * @param condition The Condition instance node
     * @param value The value to write
     * @param fieldName The target field name
     * @return The StatusCode of the operation */
    UA_StatusCode (*setConditionField)(UA_AlarmConditionsDriver *driver,
                                       const UA_NodeId condition,
                                       const UA_Variant *value,
                                       const UA_QualifiedName fieldName);
```

(continues on next page)

(continued from previous page)

```
    /* Set a property value of a condition field.
     *
     * @param driver The Alarms & Conditions driver
     * @param condition The Condition instance node
     * @param value The value to write
     * @param variableFieldName The field containing the property
     * @param variablePropertyName The target field property name
     * @return The StatusCode of the operation */
    UA_StatusCode (*setConditionVariableFieldProperty)(
        UA_AlarmConditionsDriver *driver, const UA_NodeId condition,
        const UA_Variant *value,
        const UA_QualifiedName variableFieldName,
        const UA_QualifiedName variablePropertyName);

    /* Trigger an event for an enabled condition.
     *
     * The internal condition state is updated with the generated EventId.
     *
     * @param driver The Alarms & Conditions driver
     * @param condition The Condition instance node
     * @param conditionSource The Condition Source node
     * @param outEventId The generated EventId
     * @return The StatusCode of the operation */
    UA_StatusCode (*triggerConditionEvent)(UA_AlarmConditionsDriver *driver,
                                           const UA_NodeId condition,
                                           const UA_NodeId conditionSource,
                                           UA_ByteString *outEventId);

    /* Add an optional condition field by name.
     *
     * Optional methods are not implemented yet.
     *
     * @param driver The Alarms & Conditions driver
     * @param condition The Condition instance node
     * @param conditionType The ConditionType containing the optional field
     * @param fieldName The optional field name
     * @param outOptionalVariable The created field Variable node
     * @return The StatusCode of the operation */
    UA_StatusCode (*addConditionOptionalField)(UA_AlarmConditionsDriver *driver,
                                               const UA_NodeId condition,
                                               const UA_NodeId conditionType,
                                               const UA_QualifiedName fieldName,
                                               UA_NodeId *outOptionalVariable);

    /* Set a callback for a condition TwoStateVariable field.
     *
     * The callback is called before triggering events when EnabledState/Id,
     * AckedState/Id, ConfirmedState/Id, or ActiveState/Id transitions to true.
```

(continues on next page)

(continued from previous page)

```
     *
     * @param driver The Alarms & Conditions driver
     * @param condition The Condition instance node
     * @param conditionSource The Condition Source node
     * @param removeBranch Not implemented yet
     * @param callback The user callback
     * @param callbackType The state transition for which the callback is used
     * @return The StatusCode of the operation */
    UA_StatusCode (*setConditionTwoStateVariableCallback)(
        UA_AlarmConditionsDriver *driver, const UA_NodeId condition,
        const UA_NodeId conditionSource, UA_Boolean removeBranch,
        UA_TwoStateVariableChangeCallback callback,
        UA_TwoStateVariableCallbackType callbackType);

    /* Delete a condition from the address space and internal state.
     *
     * @param driver The Alarms & Conditions driver
     * @param condition The Condition instance node
     * @param conditionSource The Condition Source node
     * @return The StatusCode of the operation */
    UA_StatusCode (*deleteCondition)(UA_AlarmConditionsDriver *driver,
                                     const UA_NodeId condition,
                                     const UA_NodeId conditionSource);

    /* Set the LimitState of a LimitAlarmType condition.
     *
     * @param driver The Alarms & Conditions driver
     * @param conditionId The Condition instance node
     * @param limitValue The value from the trigger node
     * @return The StatusCode of the operation */
    UA_StatusCode (*setLimitState)(UA_AlarmConditionsDriver *driver,
                                   const UA_NodeId conditionId,
                                   UA_Double limitValue);

    /* Parse a certificate and set the ExpirationDate.
     *
     * @param driver The Alarms & Conditions driver
     * @param conditionId The Condition instance node
     * @param cert The certificate to parse
     * @return The StatusCode of the operation */
    UA_StatusCode (*setExpirationDate)(UA_AlarmConditionsDriver *driver,
                                       const UA_NodeId conditionId,
                                       UA_ByteString cert);
};

UA_AlarmConditionsDriver *
UA_AlarmsConditionsDriver(const UA_KeyValueMap params);
```

15.3 GDS Receiver Driver

The GDS Receiver driver implements the server side of OPC UA PushManagement. It exposes the ServerConfiguration methods for updating the application certificate and the CertificateGroup trust lists. Changes are collected in a transaction and applied together with the ApplyChanges method. The driver also tracks active sessions and SecureChannels so that abandoned transactions can be discarded and affected channels can be revalidated or closed after an update.

Build open62541 with UA_ENABLE_DRIVER_GDS_RECEIVER enabled. PushManagement additionally requires encryption and the full Namespace Zero.

Create the driver with UA_GDSReceiver_new() and attach its drv member to the server with UA_Server_addDriver() before calling UA_Server_run_startup(). A server can have at most one GDS Receiver driver. The server takes ownership of the receiver after it has been added successfully.

```
#ifdef UA_ENABLE_DRIVER_GDS_RECEIVER

_UA_BEGIN_DECLS

typedef struct UA_GDSReceiver {
    UA_Driver drv; /* Must be the first member */
} UA_GDSReceiver;

/* Create a GDS Receiver driver. The returned driver is heap-allocated and
 * must either be passed to UA_Server_addDriver() or released with its ``free``
 * callback. Returns NULL if allocation fails. */
UA_GDSReceiver *
UA_GDSReceiver_new(void);

/* Update the application certificate used by the server endpoints. The GDS
 * Receiver driver must be attached and started. If certificateGroupId is
 * null, the DefaultApplicationGroup is used. Returns BadInvalidArgument for a
 * null receiver or empty certificate and BadInvalidState if the receiver is
 * not attached and started. This function is thread-safe. */
UA_StatusCode
UA_GDSReceiver_updateCertificate(UA_GDSReceiver *receiver,
                                 const UA_NodeId certificateGroupId,
                                 const UA_NodeId certificateTypeId,
                                 const UA_ByteString certificate,
                                 const UA_ByteString *privateKey);

/* Create a PKCS #10 DER-encoded certificate signing request. The GDS Receiver
 * driver must be attached and started. If certificateGroupId is null,
 * the DefaultApplicationGroup is used. Returns BadInvalidArgument for a null
 * receiver or output pointer and BadInvalidState if the receiver is not
 * attached and started. This function is thread-safe. */
UA_StatusCode
UA_GDSReceiver_createSigningRequest(UA_GDSReceiver *receiver,
                                    const UA_NodeId certificateGroupId,
                                    const UA_NodeId certificateTypeId,
                                    const UA_String *subjectName,
                                    const UA_Boolean *regenerateKey,
```

(continues on next page)

(continued from previous page)

```
                                const UA_ByteString *nonce,
                                UA_ByteString *csr);
```

15.4 File Transfer Driver

Expose a file or directory tree through the OPC UA Part 20 FileType and FileDirectoryType Methods. Each driver owns one storage backend; built-in backends serve local files and directories.

Linking the driver requires UA_ENABLE_DRIVER_FILE_TRANSFER. The reduced and full Namespace Zero include the required types; with MINIMAL or NONE, load them before constructing a driver. See *Serving Files and Directories* for setup.

15.4.1 Storage Backend

Use UA_FileTransferFileBackend for one file or UA_FileTransferBackend for a directory tree. Each driver needs an independently owned context. Callbacks return OPC UA StatusCodes. Paths are UTF-8, relative to the backend root, with '/' separators; the empty path denotes the root.

open, close, read, getPosition, setPosition and getInfo are required. Directory backends also require listDirectory. Unless read-only is set, write is required, plus create, remove and rename for directories. Missing callbacks cause Bad_InvalidArgument during construction or start.

Backend handles must be unique within the instance and have independent positions. Client handles belong to their Session and close with it.

Callbacks run synchronously with the server lock held and must not block for extended periods. Scans and recursive copy/delete run in a single call; their duration grows with the tree size.

```
/* Permission bit mask for one Object. For files, read/write access content.
 * For directories, read lists entries, write changes entries, and traverse
 * accesses children. Managed ancestors require traverse; directory mutations
 * require write/traverse, recursive copy requires read/traverse on directories
 * and read on files. Traverse has no meaning for files.
 *
 * Storage rights, read-only configuration and the optional access callbacks
 * are intersected. They govern file operations and Writable/UserWritable;
 * OPC UA Browse remains controlled by accessControl.allowBrowseNode.
 * Owner/group/other selection and ACL evaluation belong to the backend. */
typedef UA_Byte UA_FileAccessRights;

#define UA_FILEACCESS_TRAVERSE ((UA_FileAccessRights)0x01)
#define UA_FILEACCESS_WRITE    ((UA_FileAccessRights)0x02)
#define UA_FILEACCESS_READ     ((UA_FileAccessRights)0x04)

/* Maximum UTF-8 byte length of one filename, excluding the terminator.
 * This is an API limit, not a filesystem limit. Never truncate a name. */
#define UA_FILETRANSFER_FILENAME_MAX 1023

typedef struct {
    /* NUL-terminated UTF-8 basename; empty for a directory root */
    char name[UA_FILETRANSFER_FILENAME_MAX + 1];
```

(continues on next page)

(continued from previous page)

```
    UA_UInt64 size;             /* Size in bytes (files only) */
    UA_Boolean sizeUnknown;     /* The Size Property then reports
                                 * Bad_NotSupported (Part 20, 4.2.1) */
    UA_DateTime lastModified;   /* Last modification time */
    UA_Boolean isDirectory;
    UA_FileAccessRights accessRights; /* Storage-level permissions. Backend
                                       * callbacks can further restrict them. */
    char mimeType[255];         /* Media type, zero-padded; all bytes may be used.
                                 * Empty omits a new MimeType Property and
                                 * leaves an existing one unchanged. */
} UA_FileTransferFileInfo;

/* The caller of an open. The driver passes the Session of the Method call for
 * Open, CreateFile, the copies of MoveOrCopy and
 * UA_FileTransferDriver_openHandle(). Direct calls of the application pass
 * NULL. */
typedef struct {
    UA_Server *server;
    const UA_NodeId *sessionId;
    void *sessionContext;
    void *openContext; /* Of UA_FileTransferDriver_openHandle(), else NULL */
} UA_FileTransferOpenContext;

typedef struct UA_FileTransferFileBackend UA_FileTransferFileBackend;
struct UA_FileTransferFileBackend {
    /* Backend-owned storage and handle state */
    void *context;

    /* Open a file. mode is a UA_Byte bit mask of UA_OPENFILEMODE_* flags,
     * validated by the driver. Append starts at EOF; otherwise start at zero.
     * EraseExisting truncates the file. Return an independent handle.
     * context is the caller or NULL. For a Method call, the driver has checked
     * the mode, the locking (Part 20, 4.2.2), the access rights and the handle
     * limits before, and the result is returned to the client, e.g.
     * Bad_NotSupported for a mode the file does not offer. */
    UA_StatusCode (*open)(UA_FileTransferFileBackend *b, const UA_String path,
                          UA_Byte mode, const UA_FileTransferOpenContext *context,
                          UA_UInt32 *handle);

    /* Close the handle and flush buffered writes. Report flush errors here;
     * a destination close error also fails a driver-managed copy. */
    UA_StatusCode (*close)(UA_FileTransferFileBackend *b, UA_UInt32 handle);

    /* Read up to length bytes and advance the position. Allocate out for the
     * caller to free with UA_ByteString_clear(). Empty out means EOF. */
    UA_StatusCode (*read)(UA_FileTransferFileBackend *b, UA_UInt32 handle,
                          UA_Int32 length, UA_ByteString *out);

    /* Write all data and advance the position. The caller retains the buffer. */
```

(continues on next page)

(continued from previous page)

```
    UA_StatusCode (*write)(UA_FileTransferFileBackend *b, UA_UInt32 handle,
                           const UA_ByteString data);

    /* Get the current position of the handle */
    UA_StatusCode (*getPosition)(UA_FileTransferFileBackend *b, UA_UInt32 handle,
                                 UA_UInt64 *outPosition);

    /* Set the current position of the handle. Positions beyond the end of the
     * file are clamped to the file size (Part 20, 4.2.7). */
    UA_StatusCode (*setPosition)(UA_FileTransferFileBackend *b, UA_UInt32 handle,
                                 UA_UInt64 position);

    /* Get metadata, including the basename (empty for a directory root).
     * outInfo is zero-initialized. Set accessRights explicitly: zero denies
     * access. An overlong name returns Bad_EncodingLimitsExceeded. */
    UA_StatusCode (*getInfo)(UA_FileTransferFileBackend *b, const UA_String path,
                             UA_FileTransferFileInfo *outInfo);

    /* Optional restrictions independent of client identity. outRights starts
     * at zero; errors deny access and propagate to the caller. NULL adds no
     * restriction. Policy state is available in b->context. */
    UA_StatusCode (*getAccessRights)(UA_FileTransferFileBackend *b,
                                     UA_Server *server, const UA_NodeId *nodeId,
                                     UA_FileAccessRights *outRights);

    /* Optional Session-specific restrictions, with the Session identity and
     * context from the Method or Property callback. Same output/error rules
     * as getAccessRights. Open and every Read/Write recheck permissions;
     * Close remains possible after revocation. With either access callback,
     * recursive operations reject unmirrored entries needing an Object check. */
    UA_StatusCode (*getUserAccessRights)(UA_FileTransferFileBackend *b,
                                         UA_Server *server,
                                         const UA_NodeId *sessionId,
                                         void *sessionContext,
                                         const UA_NodeId *nodeId,
                                         UA_FileAccessRights *outRights);

    /* Release context when the driver is freed. Optional; called once if set. */
    void (*clear)(UA_FileTransferFileBackend *b);
};

/* Report one entry with fully initialized metadata and a non-empty,
 * NUL-terminated basename. info is borrowed for the duration of the callback;
 * copy the struct to retain it. */
typedef void
(*UA_FileTransferListCallback)(void *listContext,
                               const UA_FileTransferFileInfo *info);

typedef struct UA_FileTransferBackend UA_FileTransferBackend;
```

(continues on next page)

(continued from previous page)

```
struct UA_FileTransferBackend {
    /* File operations; first member to allow casting b back to this struct */
    UA_FileTransferFileBackend file;

    /* Enumerate entries at path. Return an error if enumeration or reading
     * entry metadata fails, including when an entry disappears during the
     * listing. An error discards the entire listing; a successful incomplete
     * listing could lose data during a move. Never truncate overlong names;
     * return Bad_EncodingLimitsExceeded instead. */
    UA_StatusCode (*listDirectory)(UA_FileTransferBackend *b, const UA_String path,
                                   UA_FileTransferListCallback cb, void *listContext);

    /* Create an empty entry at path; fail if it exists. info is borrowed;
     * name matches the path basename, isDirectory selects the type. Other
     * fields are zeroed and ignored; use backend defaults for metadata. */
    UA_StatusCode (*create)(UA_FileTransferBackend *b, const UA_String path,
                            const UA_FileTransferFileInfo *info);

    /* Remove a file or empty directory. The driver removes trees bottom-up. */
    UA_StatusCode (*remove)(UA_FileTransferBackend *b, const UA_String path);

    /* Move within this backend. Never overwrite or merge an existing target,
     * even if it is not mirrored: return Bad_BrowseNameDuplicated. The driver
     * rejects moves of entries with open handles. */
    UA_StatusCode (*rename)(UA_FileTransferBackend *b, const UA_String fromPath,
                            const UA_String toPath);

    /* Optional file copy within this backend; otherwise the driver uses
     * open/read/write/close. Trees are copied entry by entry. As for rename,
     * do not overwrite targets or copy entries with open handles. Remove any
     * partial destination on failure. */
    UA_StatusCode (*copy)(UA_FileTransferBackend *b, const UA_String fromPath,
                          const UA_String toPath);
};
```

15.4.2 Construction and Parameters

Set driver->params before UA_Server_addDriver(). Parameters are read on each start; all keys use namespace 0. A known key with another type than listed fails the start with Bad_InvalidArgument.

0:read-only [Boolean]
: Disable writes and directory mutations (default: false).

0:max-scan-depth [UInt32]
: Maximum depth of mirrored entries; the root's entries have depth 1 and directories at the limit appear empty. 0 means unlimited (default). CreateFile, CreateDirectory and MoveOrCopy return Bad_ResourceUnavailable for targets beyond the limit. Lowering the limit keeps deeper Objects, but they are no longer refreshed.

0:max-nodes [UInt32]
: Maximum mirrored Objects, including the root; 0 means unlimited (default). Lowering the limit

does not remove existing Objects.

0:namespace-index [UInt16]
Namespace for new entry BrowseNames and NodeIds (default: 0). Standard Properties and Methods remain in namespace 0.

0:refresh-interval [Double]
Directory refresh interval in milliseconds (default: 1000), at least 0.0001. 0 disables the periodic refresh, see UA_FileTransferDriver_refresh(). Vanished files with open handles remain until closed.

0:max-open-handles-per-session [UInt16]
Open-handle limit per Session in this driver (default: 64).

0:max-open-handles-per-file [UInt16]
Open-handle limit per file (default: 16). Either handle limit makes Open return Bad_ResourceUnavailable when reached.

0:activity-timeout [Double]
Close handles without a Method call for longer than this many milliseconds like Close, e.g. for the ActivityTimeout of a TrustList (Part 12, 7.8.2.1). The FileType Methods, openHandle and getHandleInfo count as calls. Idle handles are closed within twice the timeout. 0 disables it (default); otherwise at least 0.0001.

0:max-read-length [UInt32]
Maximum bytes per Read or Write (default, also when zero: 1048576). Published as MaxByteStringLength. Reads are capped at this length; larger writes fail with Bad_InvalidArgument.

Each refresh lists every mirrored directory with the server lock held, with max-nodes twice (removals first). For large or deep trees, set max-scan-depth and max-nodes and consider a longer refresh-interval, or 0 with UA_FileTransferDriver_refresh(). Skipped entries and trees cut off by these limits are logged at start and when their numbers change; unchanged results only at debug level.

```
/* Description of the root Object. Zero-initialized fields select defaults.
 * The description and its contents are borrowed only during construction. */
typedef struct {
    UA_NodeId nodeId;          /* Existing Object to reuse, or ID to create.
                                * Null generates a NodeId in namespace 1. */
    UA_NodeId parentNodeId;    /* Default: Objects folder */
    UA_NodeId referenceTypeId; /* Default: HasComponent */
    UA_QualifiedName browseName; /* Default: path basename (file), or 0:FileSystem */
    UA_NodeId typeDefinition;  /* Default: FileType or FileDirectoryType */
    UA_ObjectAttributes attributes; /* Default Object attributes; an empty
                                     * DisplayName uses the BrowseName */
} UA_FileTransferNodeDescription;

/* Construct a stopped driver for an existing backend file or directory.
 * newDirectory serves the empty path. description may be NULL for defaults.
 * A default file BrowseName uses namespace 1 and the path basename; for an
 * empty path, use getInfo's name or "File" if unnamed.
 *
 * An existing description->nodeId reuses a FileType/FileDirectoryType Object
 * (or subtype); other description fields are ignored. Children are matched
```

(continues on next page)

(continued from previous page)

```
 * by BrowseName text regardless of namespace. Ambiguous or incompatible
 * entries are rejected; unrelated existing nodes are untouched.
 *
 * Bound Object contexts point to the driver, including while stopped. Reused
 * roots are bound on first start. Freeing removes driver-created nodes and
 * restores reused Objects' contexts and Properties' values, sources and contexts.
 * Reused Objects also survive disappearance of their storage entry.
 * Delete and successful moves delete reused Objects at the source as well.
 * Before deleting an Object, its context from construction or reuse is restored
 * for the application lifecycle destructors.
 * While started, drivers supply the standard Methods, including shared
 * Namespace Zero Method nodes and copies inherited by subtypes. Other
 * Methods of subtypes stay with the application, see
 * UA_FileTransferDriver_getHandleInfo(). The drivers own these Method nodes:
 * the first start replaces callbacks set by the application, the last stop
 * removes them. FileType Objects of the application that are served without a
 * driver need their own Method nodes.
 *
 * On success, the backend struct is copied and its context becomes owned by
 * the driver. On failure, created nodes are removed, *outDriver is NULL and
 * the caller retains the backend. outDriver is required; outNodeId is optional
 * and receives an owned NodeId only on success.
 *
 * Registration is separate. Before successful UA_Server_addDriver(), the
 * caller must free the driver before deleting the server; afterwards the
 * server owns it. To free explicitly, stop and remove the driver first.
 * Start mirrors directory entries and the periodic refresh (unless
 * refresh-interval is 0) keeps them in sync.
 * Stop closes handles but preserves nodes and backend for restart. */
UA_THREADSAFE UA_StatusCode
UA_FileTransferDriver_newFile(UA_Server *server,
                              const UA_FileTransferFileBackend *backend,
                              const UA_String path,
                              const UA_FileTransferNodeDescription *description,
                              UA_NodeId *outNodeId, UA_Driver **outDriver);

UA_THREADSAFE UA_StatusCode
UA_FileTransferDriver_newDirectory(UA_Server *server,
                                   const UA_FileTransferBackend *backend,
                                   const UA_FileTransferNodeDescription *description,
                                   UA_NodeId *outNodeId, UA_Driver **outDriver);

/* Reconcile the directory tree of a started directory driver with its backend
 * now, e.g. with the periodic refresh disabled. Returns Bad_InvalidState for a
 * stopped driver and Bad_NotSupported for a file driver. */
UA_THREADSAFE UA_StatusCode
UA_FileTransferDriver_refresh(UA_Driver *driver);

/* FileType subtypes can define Methods on an open file, e.g. CloseAndUpdate of
```

(continues on next page)

(continued from previous page)

```
 * the PubSubConfiguration (Part 14). These Methods stay with the application,
 * which also implements the backend of the file. Their callbacks resolve the
 * client's FileHandle to the open mode and the backend handle, and may then
 * close it like the Close Method. Handles are bound to the Session and the file
 * Object that opened them; others return Bad_InvalidArgument. closeHandle
 * returns the result of the backend close. Both can be called from Method
 * callbacks. */
UA_THREADSAFE UA_StatusCode
UA_FileTransferDriver_getHandleInfo(UA_Driver *driver,
                                    const UA_NodeId fileNodeId,
                                    const UA_NodeId *sessionId,
                                    UA_UInt32 fileHandle, UA_Byte *mode,
                                    UA_UInt32 *backendHandle);

UA_THREADSAFE UA_StatusCode
UA_FileTransferDriver_closeHandle(UA_Driver *driver,
                                  const UA_NodeId fileNodeId,
                                  const UA_NodeId *sessionId,
                                  UA_UInt32 fileHandle);

/* Subtype Methods that open the file with their own arguments, e.g.
 * OpenWithMasks of a TrustList (Part 12), create the FileHandle with
 * openHandle. openContext reaches the backend open, e.g. for the masks. The
 * checks and results of the Open Method apply; a stopped driver returns
 * Bad_InvalidState and a node that is not a file of the driver
 * Bad_InvalidArgument. The handle belongs to the Session like one from Open:
 * it counts in OpenCount and the handle limits and closes with the Session.
 * Return fileHandle to the client, or close it with closeHandle if the Method
 * fails afterwards. backendHandle is optional. */
UA_THREADSAFE UA_StatusCode
UA_FileTransferDriver_openHandle(UA_Driver *driver,
                                 const UA_NodeId fileNodeId,
                                 const UA_NodeId *sessionId,
                                 void *sessionContext, UA_Byte mode,
                                 void *openContext, UA_UInt32 *fileHandle,
                                 UA_UInt32 *backendHandle);
```

15.4.3 Local Backends

```
/* Serve an existing local directory (UTF-8 path, copied on success).
 * out is written only on success. Reported rights follow the server process's
 * filesystem permissions, further restricted by the optional access callbacks.
 * Windows traversal checks are left to the filesystem operations.
 *
 * Paths with ".." segments are rejected, but links to regular files are followed
 * even outside rootPath: this is not a filesystem sandbox, including in
 * read-only mode. Directory symlinks and special files are skipped; directories
 * containing them cannot be deleted through the driver (Bad_InvalidState).
 * Windows rejects unrepresentable names. 32-bit POSIX positions are limited
```

(continues on next page)

(continued from previous page)

```
 * to 2 GiB. */
UA_StatusCode
UA_FileTransferBackend_localDirectory(const UA_String rootPath,
                                      UA_FileTransferBackend *out);

/* Serve one existing regular file using the same local I/O and permissions.
 * filePath is UTF-8 and copied; out is written only on success. Links to regular
 * files are followed. Only the empty backend path is accepted; getInfo reports
 * the basename and media type. Pass UA_STRING_NULL to newFile(). */
UA_StatusCode
UA_FileTransferFileBackend_localFile(const UA_String filePath,
                                     UA_FileTransferFileBackend *out);
```